



GE Medical Systems

Technical Publications

Direction 2143793

Revision 17

Signa® System Upgrades (SmartSpeed, HiSpeed, & EchoSpeed)

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Operating Documentation

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Rev. 08/15/2003



GE Medical Systems

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
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- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

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- 本维修手册仅存有英文本。
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- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
?	Feb. 6, 1995 . . .	Initial creation (copy over from 2123140)
A	Dec. 19, 1995 . . .	Preliminary Revision
0	Jan. 19, 1996 . . .	Initial Release
1	Apr. 15, 1996 . . .	Added instructions for upgrading 3.x and 4.x systems to Horizon. Also added changes, additions, and corrections to initial release instructions.
2	Jun. 7, 1996 . . .	General update with various changes, additions, and corrections.
2+	Oct. 4, 1996 . . .	Pre-production Release
3	Nov. 29, 1996 . . .	Added upgrade instructions for Horizon LX. Updated existing Horizon 5.5 upgrade instructions with changes, additions, and corrections received from Field Engineers, On-Line Center, and other sources.
4	Jun. 23, 1997 . . .	Changed Horizon 5.5 and LX references to Horizon 5.x and 8.x respectively throughout manual. Section 1 – added passive shimmming inspection instruction. Added optional keyboard catalogs and Operator Workspace options catalogs. Section 2 – added instructions for returning to GE consoles, computer, and system cabinet. Section 3 – updated flow chart for 8.x SV magnet passive shimmming. Section 4 – added info for alternate connections of Operator Workspace and GRAM cabinet to Compact PDU and 2-bay PDU on appropriate cable maps. Section 5 – revised Power On procedures. Added Supplier Performance Report. Added crimp tool ferrous Warning note. Tab 1 – revised cable connections for Run 041 and 042 at PDU. Added crimp tool ferrous Warning note. Tab 2 – added optional hex clamps for connecting RF drive cables to bulkhead. Added InSite cabinet magnet installation instructions. Tab 6 – Revise 8.x PDU instructions. Tab 13 – added Operator Workspace DASM jumper set-up. Included other changes, additions, or corrections to other parts of the manual.
5	Feb 10, 1998 . . .	Revised instructions throughout manual for upgrading to HiSpeed and EchoSpeed only with SGD Hi-Slew Gradient Cabinet. Section 1 – Revised catalog structure to reflect addition of SGD product. Section 3 – Revised mechanical installation to include SGD. Revised calibration flowcharts to reflect SGD changes. Added time and people graphics. Section 4 – Revised cable tables and maps to reflect SGD Changes. Section 5 – Made SGD changes. Tab 1 – Revised fiber optic, gradient output, and PDU power cable terminations and connections as related to SGD changes. Tab 2 – Corrected part number error in Magnishield section. Tab 6 – Revised connections to PDU Cabinet. Tab 8 – Added cable information for transferring of Fast Receiver module. Tab 9 – Changed instructions to installing new SGD Gradient cabinet. Tab 10 – Added RF/Pen II upgrade instructions. Tab 13 – Added instructions on installing new color monitors and wrapping cables to computer cabinet.
6	May 11, 1998 . . .	Revised instructions throughout manual for installation of Octane computer. Section 1 – Added catalogs to reflect additions of Octane computer. Section 3 – Updated flowcharts for Octane installation. Also revised calibration area of flowchart. Section 4 – Revised cable information for installation of Octane computer. Tab 6 – Changed required Compact PDU upgrade catalog from M1090MZ to M1090NZ. Tab 13 – Added Instructions for installing Octane computer, Direction 2210982.
7	Dec 4, 1998 . . .	Removed information on Indigo Computer installation through out manual. Section 1 – Revised catalog information. Section 3 – Added Oxford magnet prerequisite Service Note 60782 to installation flow chart. Added latest software installatinformation. Section 4 – Removed upgrade to Indigo information. Section 5 – Corrected reference information. Tab 1 – Revised Gradient output cable installation instructions. Tab 3 – Added BRM-D upgrade installation. Tab 13 – Revised Octane Computer installation. Removed Indigo installations. Revised DAT and MOD Drive installation instructions.
8	May 20, 1999 . . .	Section 1 – Added CVMR catalog information. Section 3 – REvised the flowcharts to correlate with the latest software upgrades. Also changed the CD-ROM menu selections in the flow chart boxes to relate with GUI improvements on the latest CD. Section 4 – Added cable maps for upgrading from Horizon 5.x to 8.x and keeping the existing Gradient/GRAM cabinets. Tab 1 – Added Direction 2245919 for installing BRM-D & CRM gradient output cables. Tab 2 – Added Direction 2245881, instructions for upgrading a system to CVMR. Tab 3 – Revised Direction 2223489 to include locating information for CRM Body Coil. Tab 13 – Updated the instructions on installing the Octane Operator Workspace.
9	Nov 17, 1999 . . .	Section 1 – Updated catalogs to reflect changes associated with the 8.3 software product. Section 3 – Updated flowcharts to reflect changes of installing 8.3 software and grafidy procedures. Section 4 – Updated cable information. Specifically Run 836 replaces 790 and 816 for Bit3 fiberoptics. Also made other minor corrections to cable map designators. Tab 1 – Revised fiber optic install for new Run 836. Revised power cable upgrade instructions for terminal strip connections (21, 15) of Run 047 for full-size PDU. This correlates with moving the OW Host circuit breaker to a new location. Section 4 – Added Oxford magnet ion pump Caution note. Added instructions for tightening pen panel mounting hardware. Tab 6 – Moved location of Operator Workspace (Host) circuit breaker to new location to resolve emergency stop CQA. Tab 8 – Revised System cabinet installation for new fiber optic Run 836. Tab 10 – Added Cable Extension kit (2105371) note. Tab 13 – Revised installation procedures to reflect new Run 836 connection

REVISION HISTORY (continued)

REV	DATE	PRIMARY REASON FOR CHANGE
10	Jan 12, 2001 . . .	Added SmartSpeed to the title page. Overview Tab: Section 1 – Updated catalogs for ACGD product release. Deleted upgrades to Horizon 5.x. Section 2 – Added full-size PDU removal. Deleted Signa 3.x removal procedures. Section 3 – Updated flowcharts for upgrading to ACGD. Section 4 – Updated cable removal and installation for ACGD. Section 5 – Removed references to Signa 3.x upgrade. Added ACGD information. Tab 1 – Added procedures for ACGD cables. Tab 3 – Added Water Chiller installation. Tab 5 – Removed tab and Console installation instructions. Tab 6 – Removed Horizon 5.x PDU upgrade information. Tab 7 – Removed tab and Computer installation instructions. Tab 8 – Removed Horizon 5.x System cabinet upgrade information. Tab 13 – Updated Operator Workspace installation instructions. Tab 15 – Added new ACGD/PDU Installation instructions. Tab 16 – Added new SRF Cabinet installation instructions.
11	Mar 19, 2001 . . .	Overview Tab: Section 1 – Updated catalog information. Section 3 – Revised flowcharts for mechanical upgrade to Dual Temperature Sensor. Removed references of black and white monitor installation. Section 4 – Revised cable removal and install tables for Run 749. Revised cable maps with latest product upgrade and corrected errors. Tab 2 – Revised enclosure upgrades for Magnishield, S-3 and newer, and CVMR for SRI module upgrade. Added Direction 2294109, Dual Temperature Sensor Upgrade, to end of tab. Tab 13 – Removed references to black & white monitor installation and removed installation of Runs 792 and 809. Tab 15 – Revised circuit breaker DIP switch setting instructions.
12	Apr 4, 2001 . . .	Tab 2 – Dual Temperature Sensor Upgrade instructions, Direction 2294109, were revised to reflect change from RTV adhesive to Oxybond epoxy for attaching cable to end bell.
13	Jul 1, 2002 . . .	The major change in this revision is the addition of information for upgrading to EXCITE Technology (MGD). Overview Tab: Section 1 – Revised product locator subsection. Revised catalog tables to flow charts and added EXCITE catalog information. Section 2 – Added PDU Runs 011 & 020 retermination instructions (MRlge68060). Section 3 – Added flow charts for upgrading to EXCITE. Revised existing flow charts with latest equipment room cabinet offerings. Section 4 – Added cable removal and installation information for upgrading to EXCITE. Added to cable information tables the following: Cable Type, Voltage Rating, Actual Voltage, Temp Rating, and Flame Rating. Section 5 – Revised material disposition subsection. Tab 1 – Added Direction 2338872. Tab 2 – Removed obsolete Directions 2147874, 2147875, & 2245881. Tab 3 – Added radial support installation instructions (MRlge59681) to Directions 2223489 & 2147882. Tab 4 – Added Direction 2147670. Tab 8 – Added Directions 2170004 & 2338502. Removed Tab 9. Removed Tab 12. Tab 13 – Revised Direction 2210982 for EXCITE installation, new keyboard, and other minor changes. Tab 15 – Added circuit breaker DIP switch settings for "I". Tab 16 – Added Direction 2346942.
14	Jun 16, 2003 . . .	Tab 4 – Added more information on attaching right angle heliax connectors to Pen Panel showing correct installation of O-rings (CQA 1029849). Tab 15 – Added neutral connection for SSM in PDU power cable connections table (SPR# MRlge82869). Revised wording on DIP switch settings. Tab 16 – Revised 1.0T and 1.5T designations for identifying Runs 236/748 and 235/745 (SPR # MRlge83424). Added note for location of UFI Label kit.
15	Jan 15, 2004 . . .	Renamed manual to Signa System Upgrades. The major change of this revision is the addition of EXCITE II upgrade and installation information. Overview Tab: Section 1 – Added catalog information for upgrading to EXCITE II. Also added information on upgrading WideOpen enclosures. Section 3 – Revised flowcharts to EXCITE II product. Removed references to MGD (EXCITE I). Section 4 – Added EXCITE II cable information for removal of obsolete cables and installation of new EXCITE II cables. Updated all cable maps and cable tables. Tab 1 – Added EXCITE II Fiber Optics and Times Microwave cable installation instructions. Tab 2 – Added EXCITE II enclosure upgrade instructions. Tab 4 – Added EXCITE I and II installation and upgrade instructions. Removed Tab 6 – PDU. Tab 8 – Added EXCITE II System cabinet installation and System Cabinet upgrade to EXCITE II instructions. Removed Tab 10 – RF/PEN. Tab 13 – Removed instructions on upgrading from Horizon 5.x to Octane. Added Linux OW installation instructions and OW upgrade to EXCITE II instructions. Tab 15 – Revised ACGD/PDU installation instructions and added GP Module upgrade for EXCITE instructions. Tab 16 – Revised SRFD2 information.
16	Aug 26, 2004 . . .	Overview Tab: Section 3 – Added SaveINFO and revised software load instructions in flowchart for upgrade to EXCITE 11.x. Section 4 – Corrected Run 1042 Remarks description in Magnet Room Run table. Added "N" (neutral) to ACGD end of Run 966 in cable maps. Tab 2 – Revised 8-channel module part number in Direction 2403142.
17	Dec 15, 2006 . . .	Overview Tab: Section 3 – Corrected IIQ Kit catalog number from M3090PE to M3090NE in three places (MRlhc19864).

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SECTION 1 – GETTING STARTED

1-1 UPGRADE INTRODUCTION

WARNING!

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT, CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

Magnetic Resonance (MR) system upgrades to Signa 1.5T LX With EXCITE, Signa LX with ACGD, Signa Horizon, and Signa CVMR are documented throughout this publication. Refer to the appropriate section listed below for overview of the upgrade content, refer to the remainder of this publication for upgrade requirements. System cable impacts are provided in Section 4.

- SECTION 1-10, UPGRADES TO 1.5T SIGNA LX WITH EXCITE
 - Signa Horizon LX SmartSpeed with EXCITE for 1.5T
 - Signa Horizon LX HiSpeed Plus with EXCITE for 1.5T
 - Signa Horizon LX EchoSpeed Plus with EXCITE for 1.5T
- SECTION 1-11, UPGRADES TO SIGNA LX WITH ACGD
 - Signa Horizon LX SmartSpeed with ACGD for 1.5T or 1.0T
 - Signa Horizon LX HiSpeed Plus with ACGD for 1.5T or 1.0T
 - Signa Horizon LX EchoSpeed Plus with ACGD for 1.5T
- SECTION 1-12, SIGNA HORIZON LX SYSTEM OPTIONS
- SECTION 1-13, UPGRADES TO SIGNA HORIZON LX WITH SGD SYSTEM

1-2 PASSIVE SHIM INSPECTION

Removal of the old body coil can disturb or displace passive shims. If magnet has passive shims glued to the bore, arrangements should be made to have the GE Field Engineer available to inspect, and if necessary repair the shims before installation of new epoxy-filled body coil.

1-3 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion. Refer to *Direction 2142460, Signa Horizon Upgrade Pre-installation*, Pre-installation Checklist tab.

Tools and test equipment to install and calibrate the Signa Horizon 8.x system are listed in *Direction 2142460, Signa Horizon Upgrade Pre-installation*, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit
- 2134776 Gradient Cable Termination kit

1-4 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

Note: The following equipment or material removed by the upgrade are ***required to be returned*** to GoldSeal:

- Octane Computer and Duplex with all associated cables and modules
- All parts, cables, modules, and equipment deinstalled for the EXCITE upgrade
- SGD or Gradient Cabinet
- System Cabinet
- Indigo Computer with all associated boards/cables (includes reflex board, cable kit, and audio cables)

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal.

The process for returning material is as follows:

1. Contact either of the following for disposition.
 - GoldSeal – Contact the MR Product Specialist at GoldSeal Logistics phone number 877-251-4468. If GoldSeal is accepting the returned equipment then all transportation arrangements and packaging will be processed through them. If GoldSeal is not accepting the return then they will advise you to contact GEMS Renewable Resources.
 - GEMS Renewable Resources – For coordination of transportation arrangements and packaging. Phone (414)-747-6997 or Dial Comm 8*579-6997; Fax (414)-747-6855; Field ASPEN 1-800-525-1516, Box #26041.
2. When calling, the following information is needed.
 - FDO (Future Delivery Order) number under which the upgrade was shipped.
 - Site name, address, and phone numbers.
 - Date of completed upgrade.
 - Name and Beeper number of Field Engineer that could be available when the equipment is picked up.
3. Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.
4. Process product locator information for all serialized components removed from the system according to policy.

1-5 PRE-DELIVERY FDO CHECKS FOR SIGNA LX

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa Horizon 8.x Upgrade Specialist (414-521-6161). Also verify the Surface coil compatibility issues have also been resolved. Refer to *Direction 2147446, Signa Horizon 1.5T & 1.0T UFI-Compatible Surface Coils Instructions*.

1-6 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The “Shipping Document” lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. The labels are color coded for different installation areas as follows:

- Green – Equipment Room
- Purple – Operator’s Room
- Orange – Magnet Room
- Yellow – Post-installation (Phantoms, Coils).

“Product Delivery Instructions” (PDI) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number (for example, PDI – M1090NC is for the Signa Horizon EchoSpeed Upgrade). Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with “Technical Publication” boxes for a list of delivered documents.

1-7 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following:
Product Locator Support Central Page – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-8 UPGRADE SCHEDULE CONSIDERATIONS

FIMs, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.



WARNING!

FERROUS MATERIAL HAZARD! THE CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 3 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

Alternate tasks are noted above the task block to set them aside from parallel tasks.

The general sequence of events is:

- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un-cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Cover replacement
- Complete final tasks and return site to customer.

1–10 UPGRADES TO 1.5T SIGNA LX WITH EXCITE

Illustration 1–19 shows the system's major equipment after a 1.5T Signa LX with EXCITE upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final Signa LX with EXCITE product configuration. The following sub-sections provide an overview of the various Signa LX with EXCITE upgrade hardware.

Note

At this time upgrades to EXCITE are not available for systems with 1.0T Magnets.



ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER TO DIRECTION 2142460 – POWER REQUIREMENTS.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER TO DIRECTION 2142460 – POWER REQUIREMENTS.

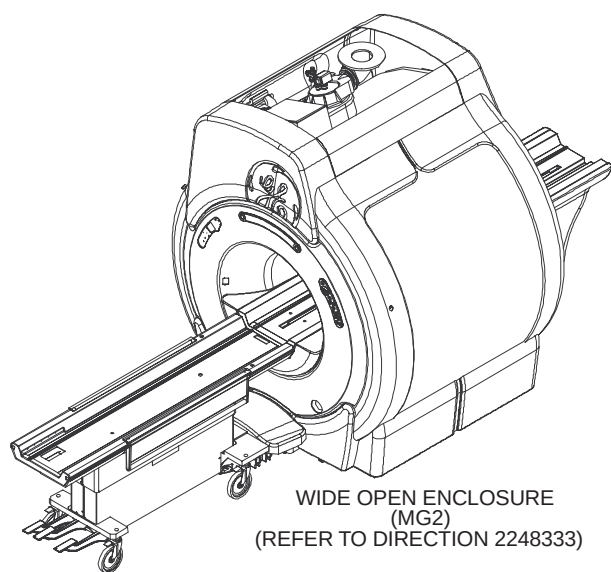
Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

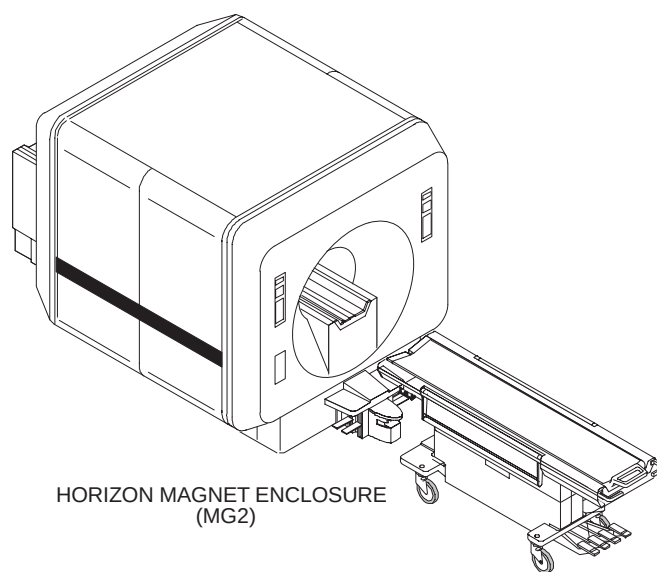
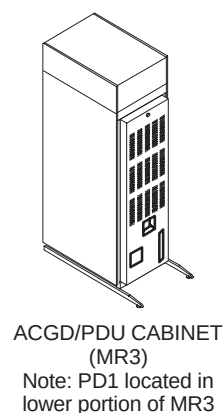
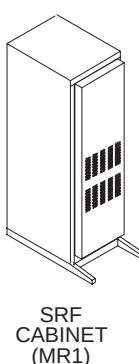
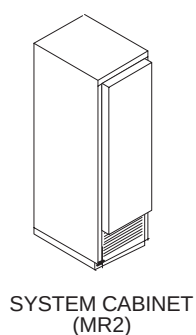
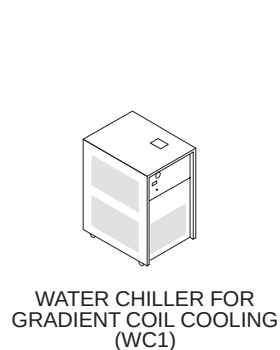
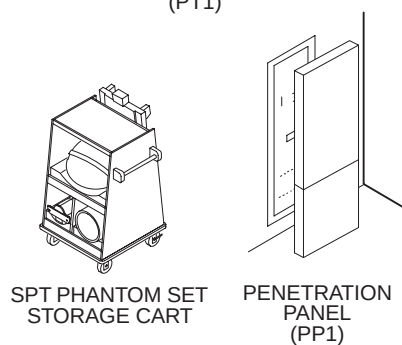
Note

FOR USA, FIXED SITE ONLY CONSIDERING MAGNET UPGRADE: Refer to *Direction 22461012 Pre-Sales/Site Survey For Upgrade To CXK4 Magnet With WideOpen Enclosure* for the many environmental factors and logistics that need to be understood to qualify a site for a magnet upgrade. The Magnet Upgrade Option requires the MR system to be operating at Release 8.X prior or coincident with the Magnet Upgrade option.

1-10 UPGRADES TO 1.5T SIGNA LX WITH EXCITE (Continued)

LIGHTWEIGHT
PATIENT TRANSPORT
(PT1)

OR

LIGHTWEIGHT
PATIENT TRANSPORT
(PT1)SIGNA LX WITH EXCITE TECHNOLOGY SYSTEM MAJOR EQUIPMENT
ILLUSTRATION 1-1

1–10–1 Signa CV/i & NV/i System

The flowchart shown in Illustration 1–2 lists the upgrade catalogs for a Signa CV/i or NV/i and current system installed magnet is an LCC Magnet upgrading to 1.5T Signa LX with EXCITE. Illustrations 1–3 and 1–4 show the cabinets replaced and modified by the catalogs listed in the flowchart.

Note

No magnet upgrades are allowed for Signa CV/i or Signa NV/i systems. If magnet upgrade is needed, new system must be ordered.

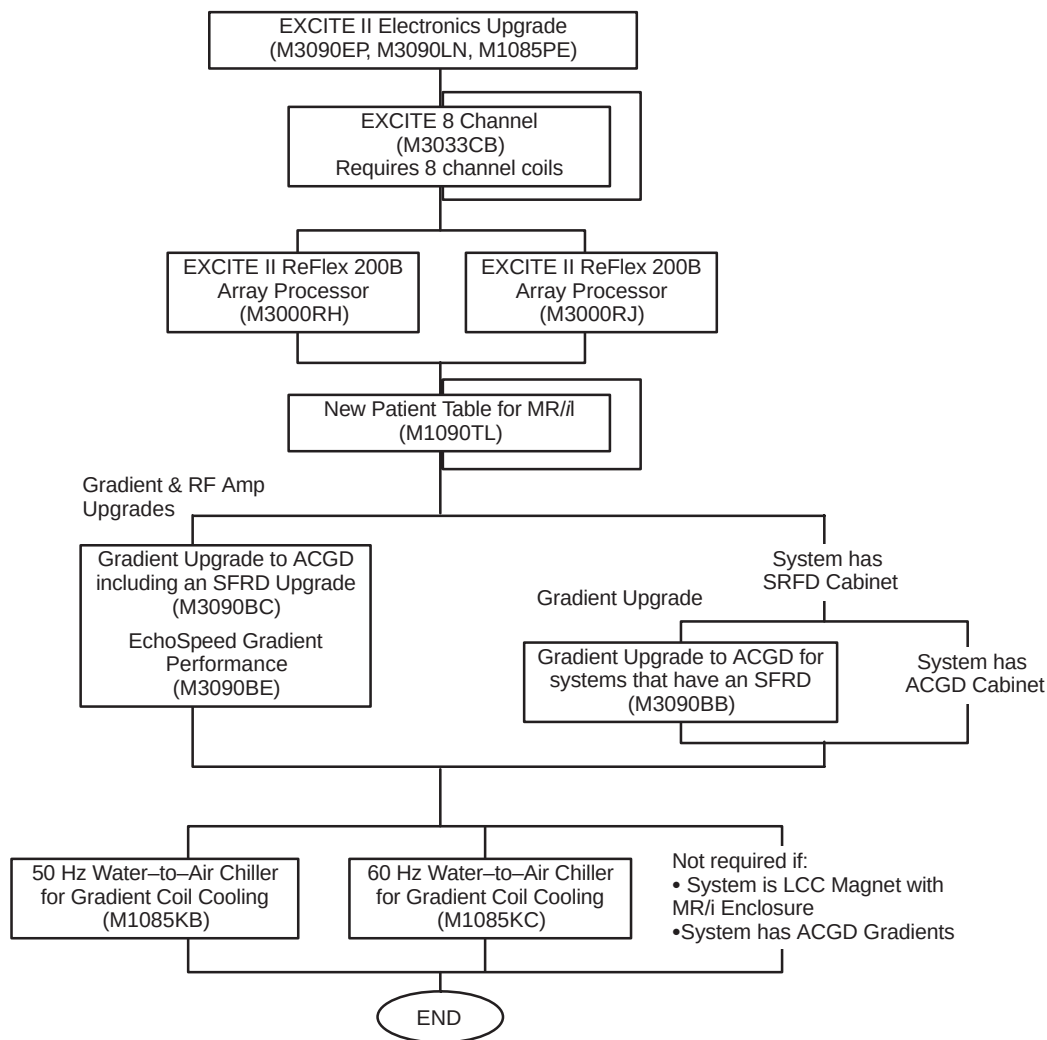
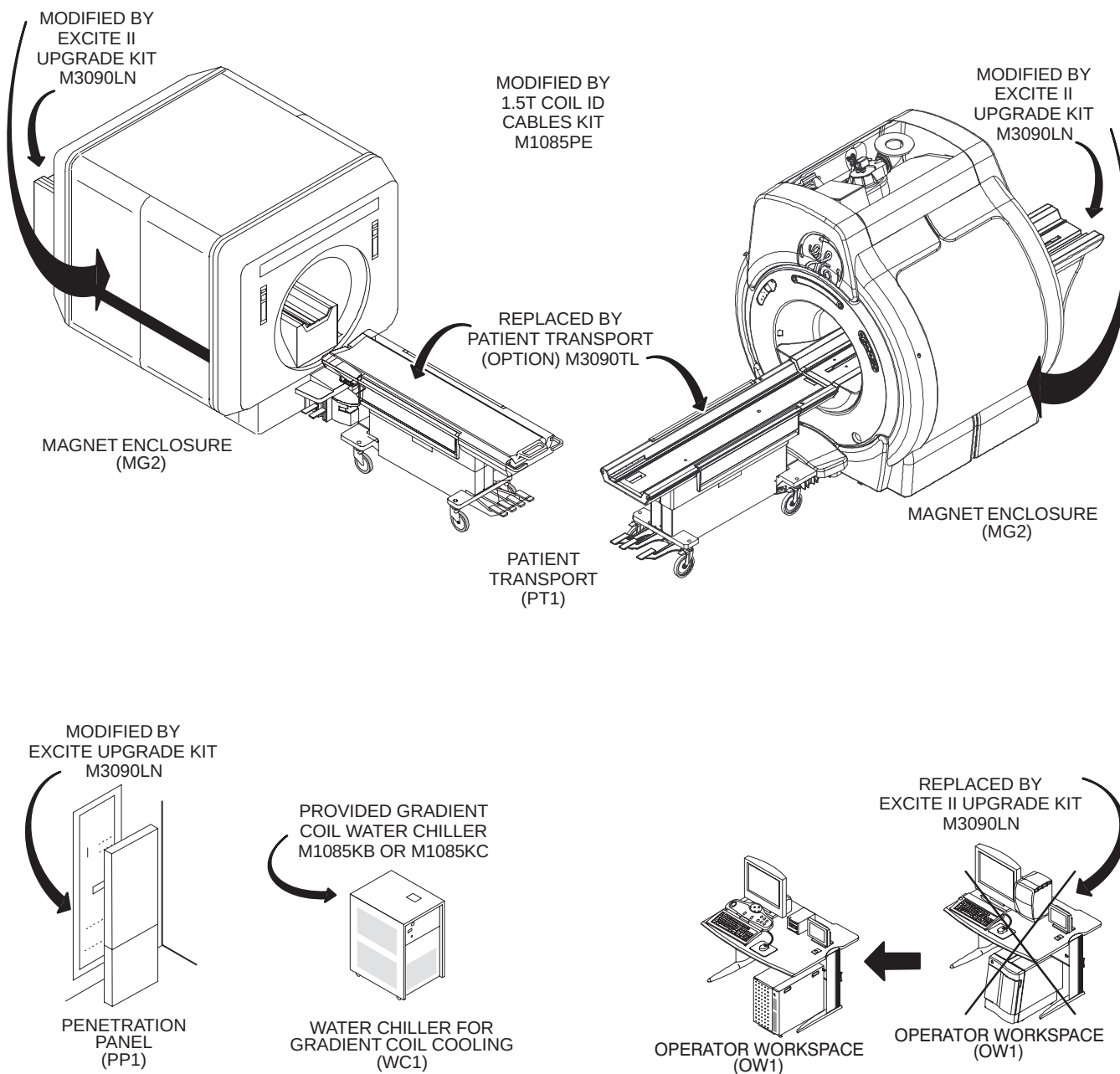
**SIGNA CV/i OR NV/i UPGRADE CATALOGS TO SIGNA 1.5T EXCITE**

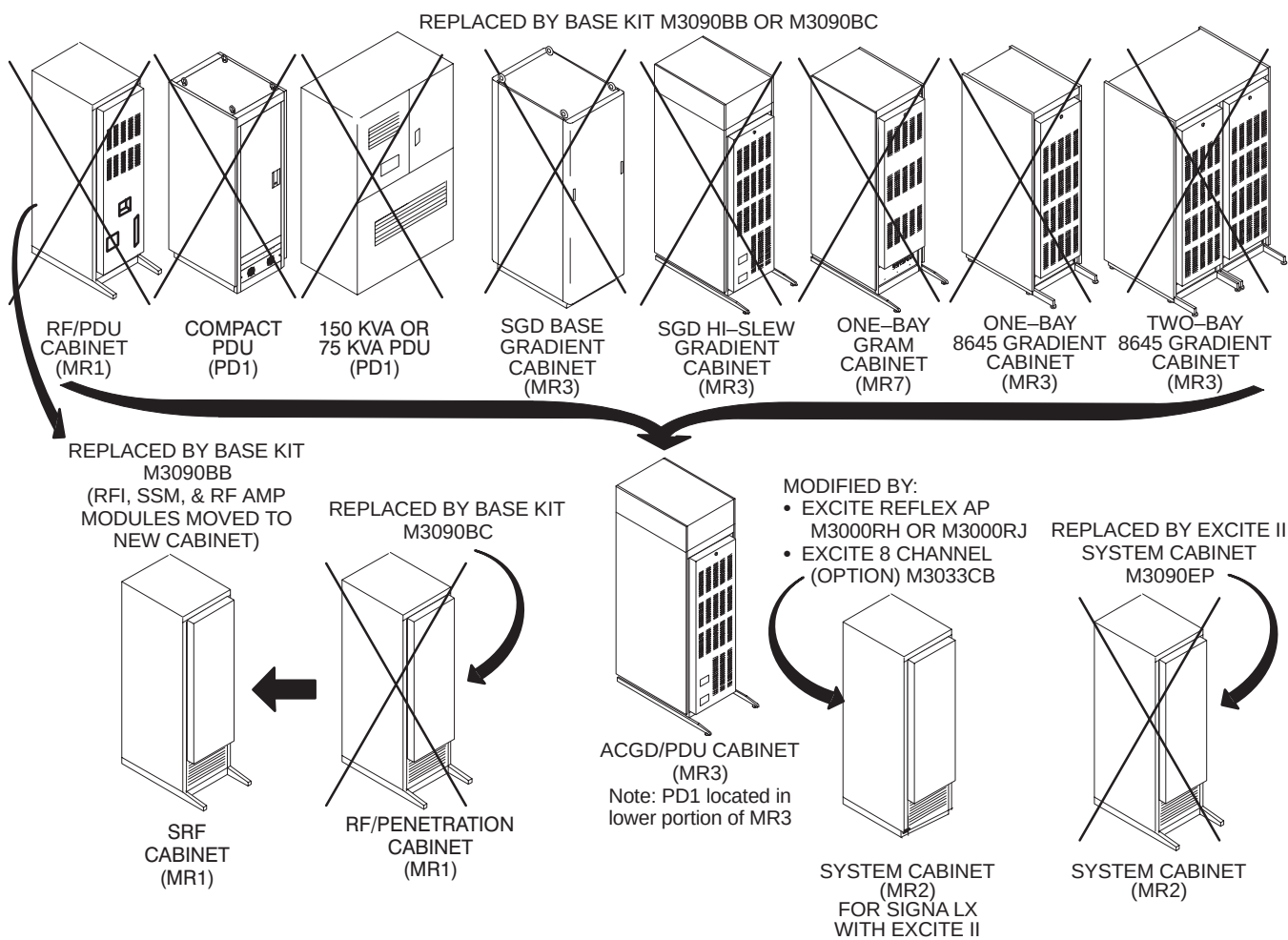
ILLUSTRATION 1–2

1-10-1 Signa CV/i & NV/i System (Continued)



SIGNA CV/i OR NV/i EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-3

1-10-1 Signa CV/i & NV/i System (Continued)



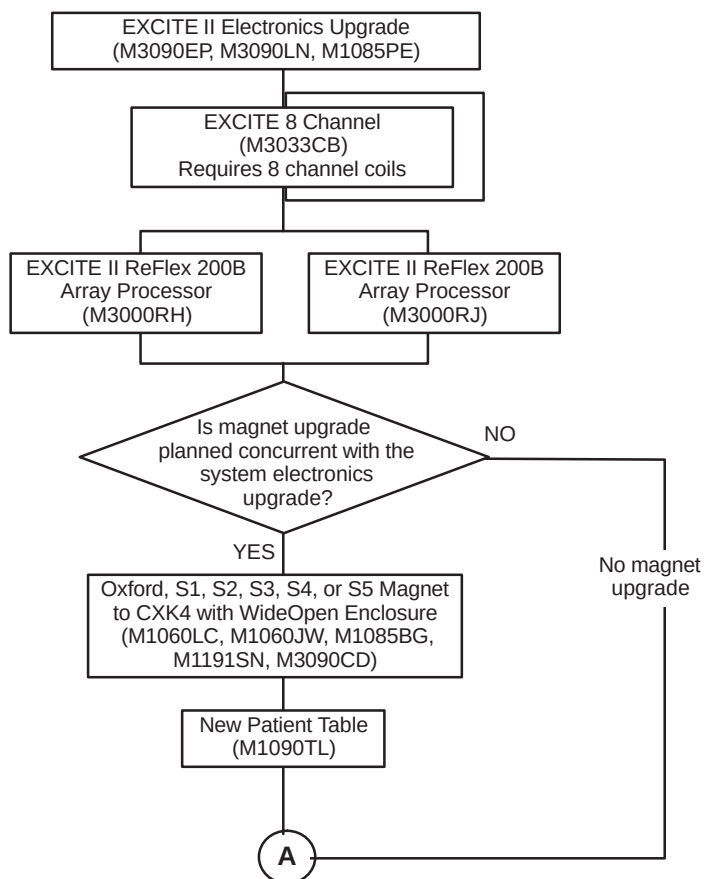
SIGNA CV/i OR NV/i EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-4

1-10-2 Signa LX System

Illustrations 1-5 and 1-6 lists the upgrade catalogs for Signa LX with ACGD or SGD and Signa MR/i system to Signa LX with EXCITE. Illustration 1-7 shows the cabinets replaced and modified by the listed catalogs.

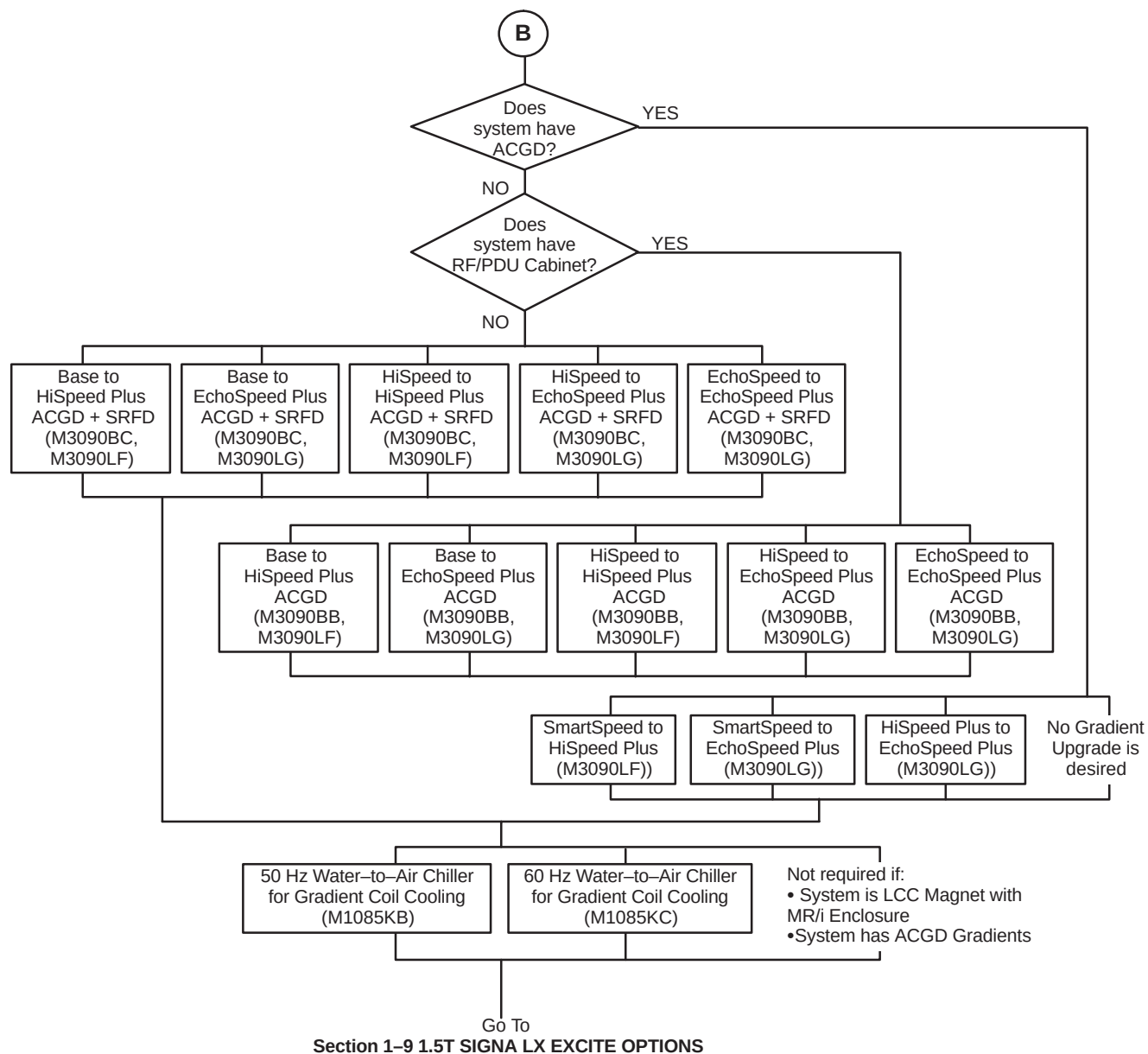
Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



1.5T SIGNA LX TO SIGNA LX WITH EXCITE CATALOGS
ILLUSTRATION 1-5

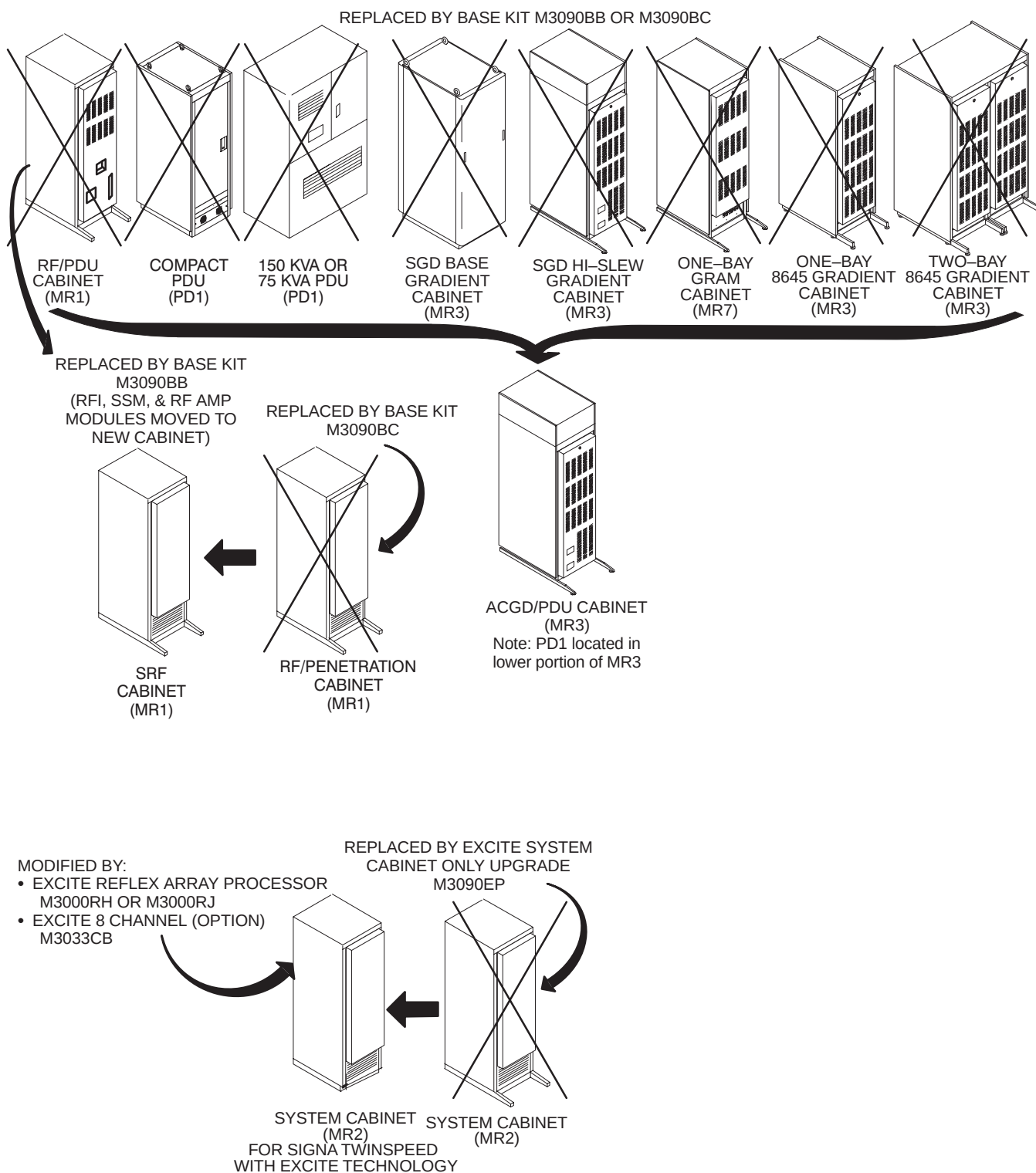
1-10-2 Signa LX System (Continued)



1.5T SIGNA LX TO SIGNA LX WITH EXCITE CATALOGS

ILLUSTRATION 1-6

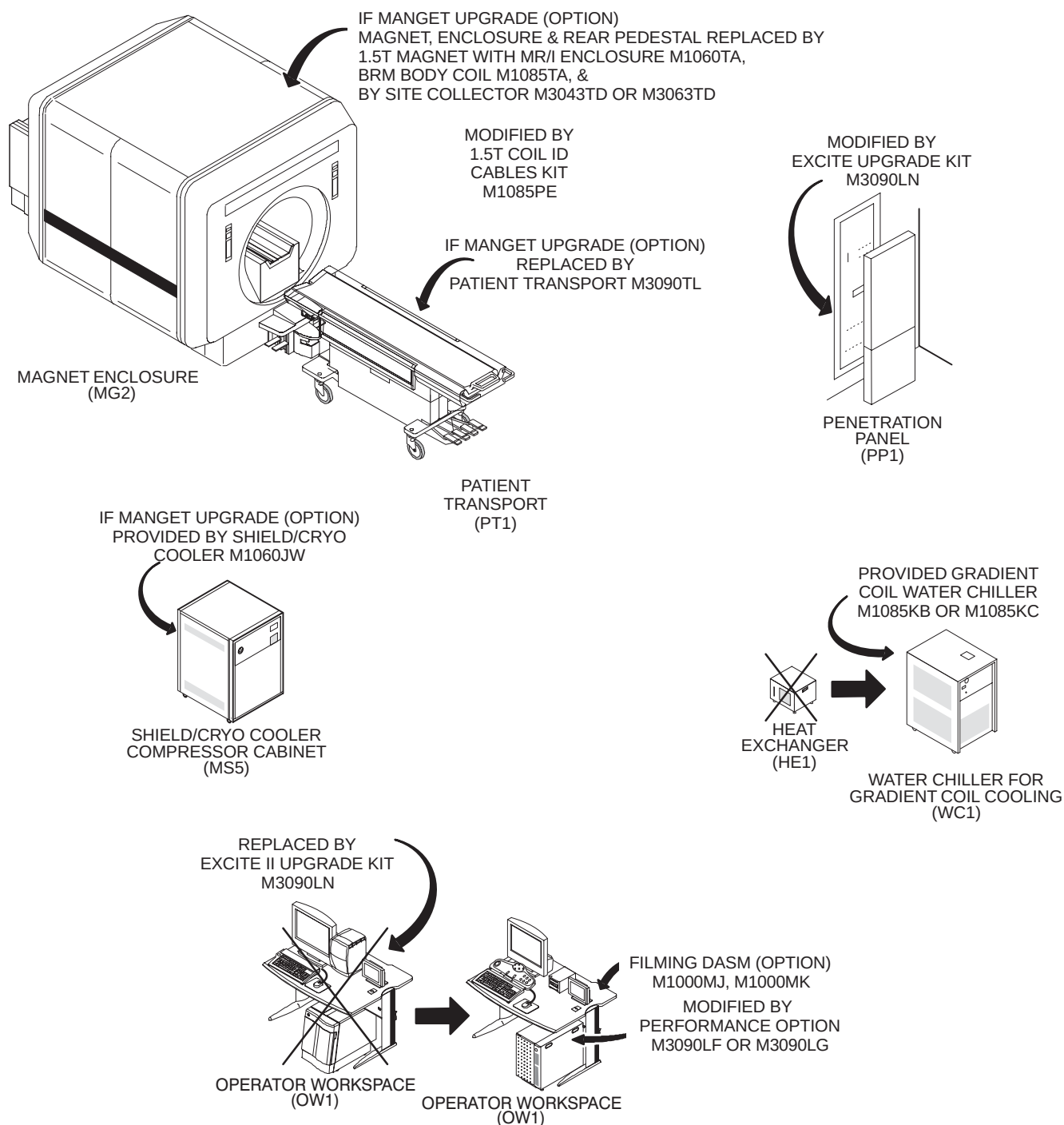
1-10-2 Signa LX System (Continued)



SIGNA HORIZON LX SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-7

1-10-2 Signa LX System (Continued)



SIGNA LX SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-8

1-10-3 Signa Horizon 5.X System

Illustrations 1-9 lists the upgrade catalogs for Signa Horizon 5.X system to Signa Horizon LX With EXCITE. Illustrations 1-10 and 1-11 how the cabinets replaced or modified and new equipment added by the listed catalogs.

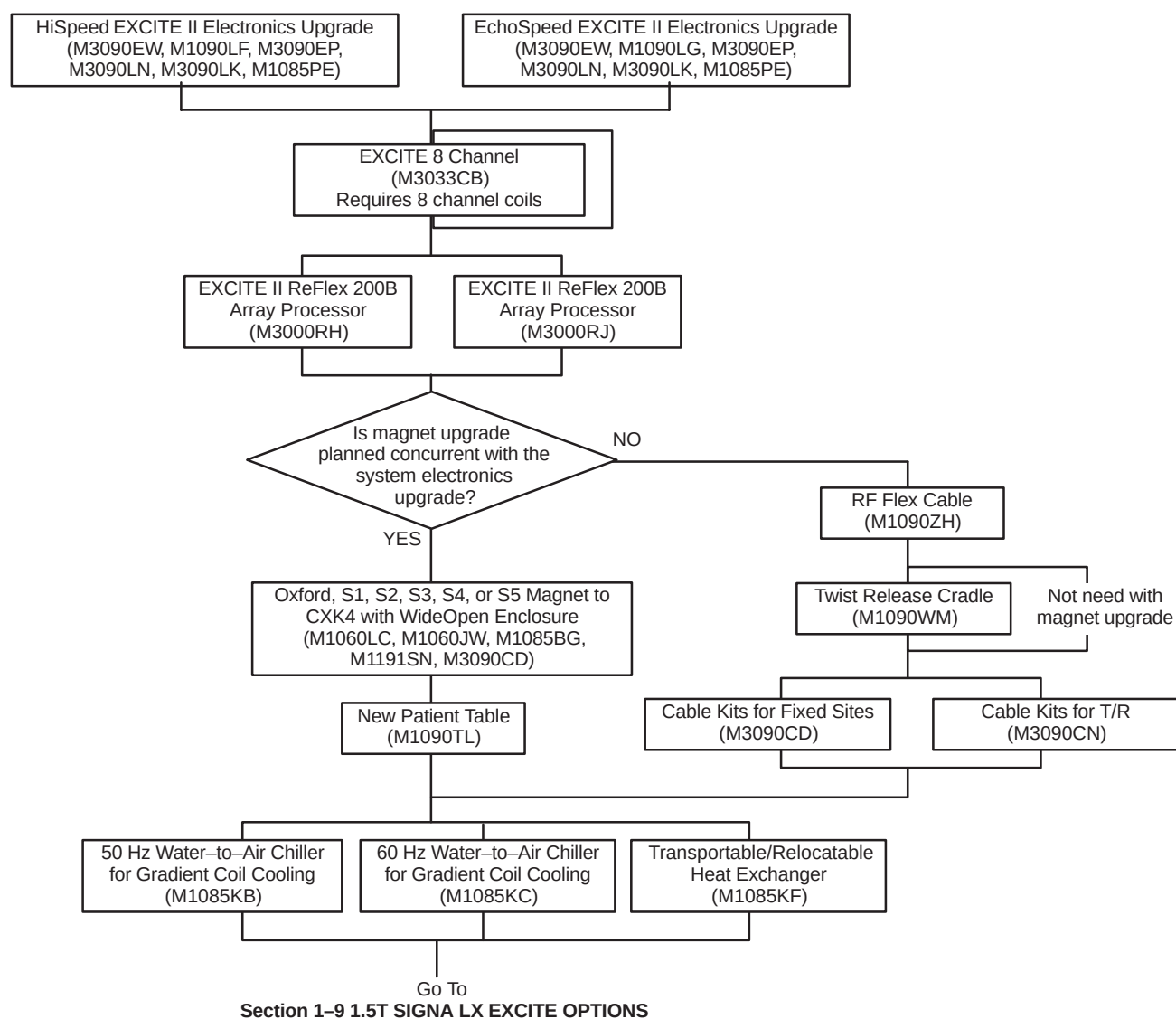
Refer to Section 1-10 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

Note

Signa Horizon 5.5 Software only systems have a 55 cm Body Coil and require an upgrade.

Note

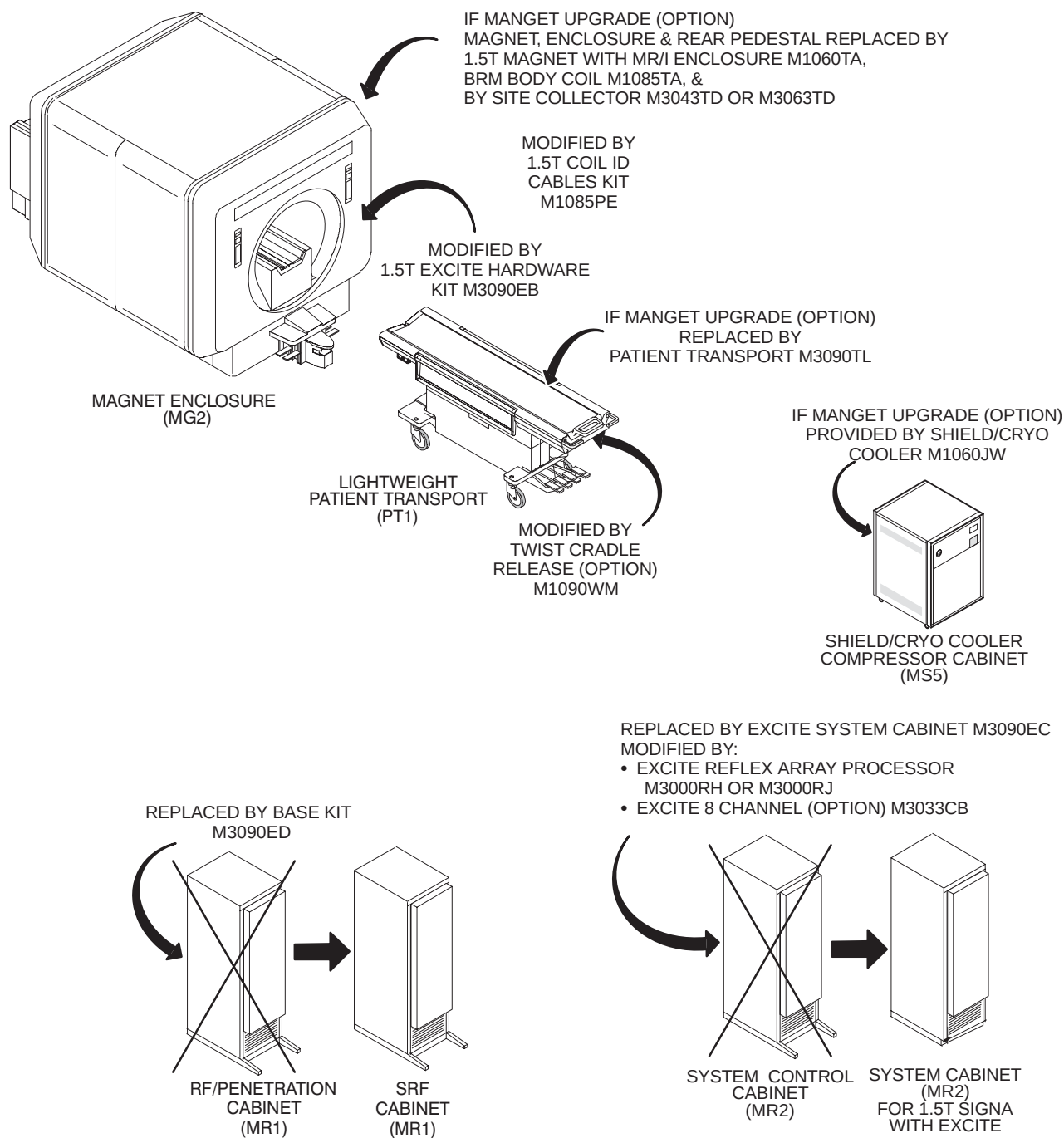
Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



1.5T SIGNA HORIZON TO SIGNA LX WITH EXCITE UPGRADE CATALOGS

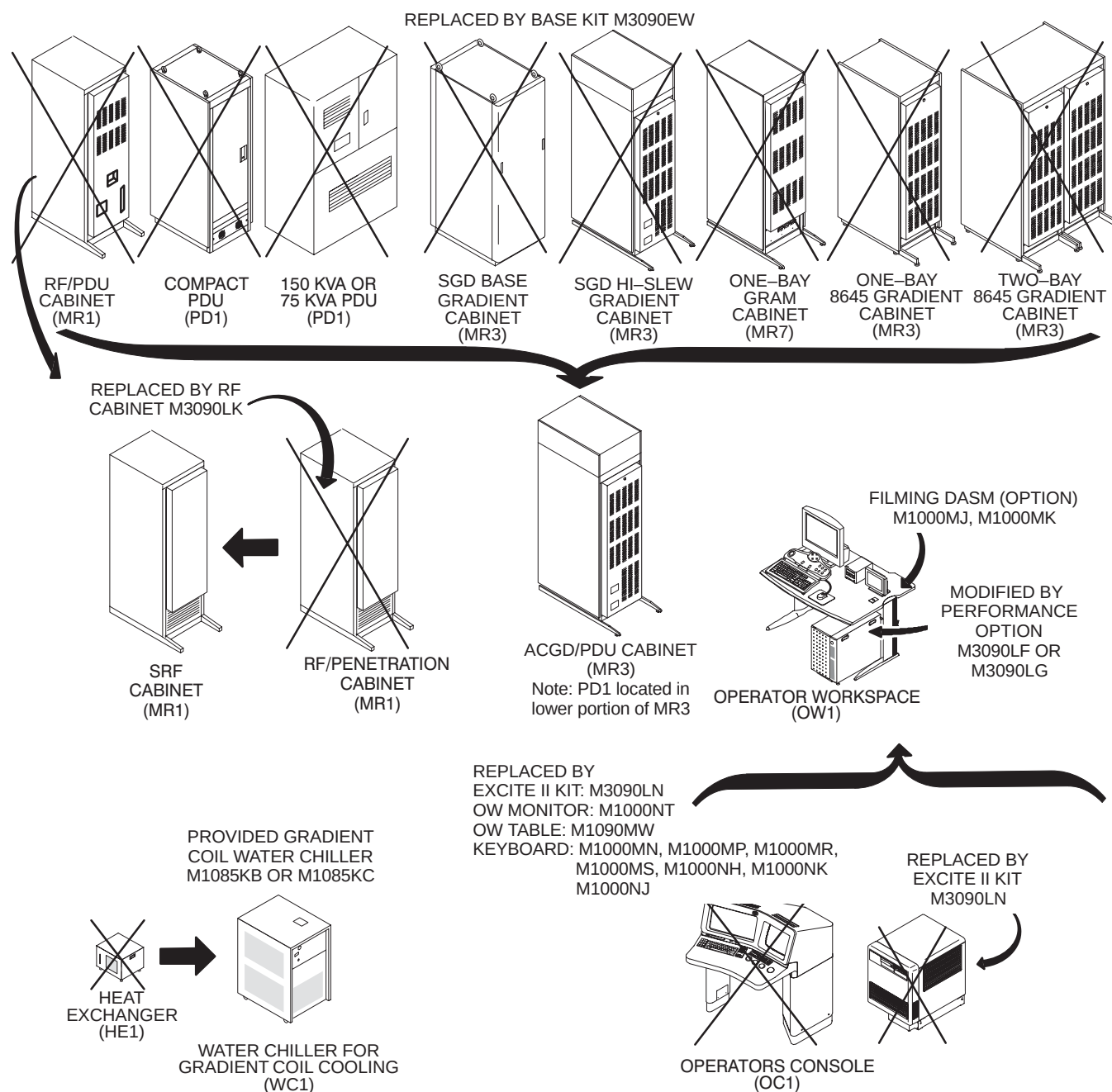
ILLUSTRATION 1-9

1-10-3 Signa Horizon 5.X System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-10

1-10-3 Signa Horizon 5.X System (Continued)



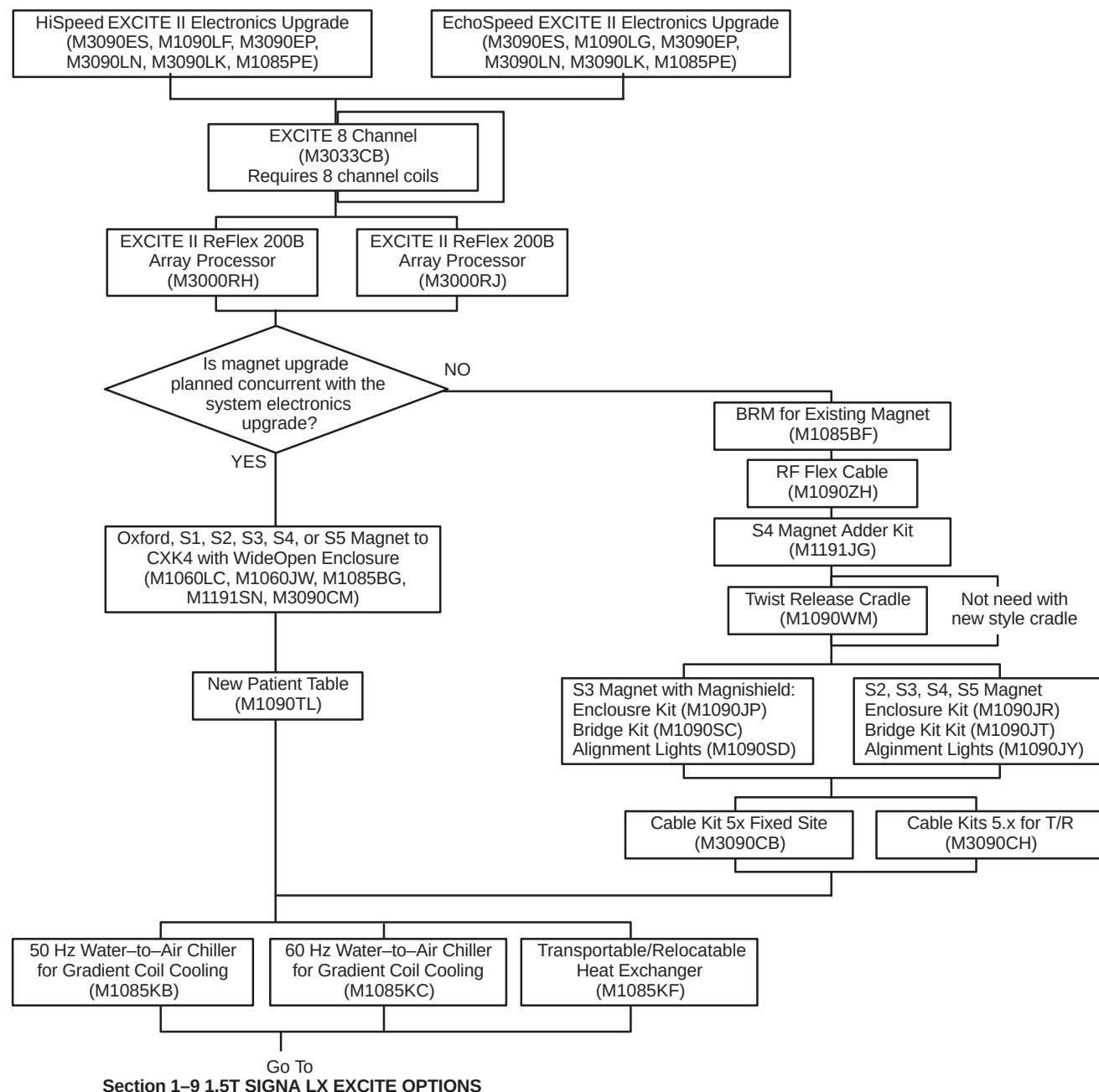
SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-11

1–10–4 Signa Advantage 5.x System

Illustration 1–12 lists the upgrade catalogs for 1.5T Signa 5.X to 1.5T Signa LX with EXCITE. Illustrations 1–13 and 1–14 show the cabinets replaced or modified and new equipment added by the listed catalogs.

Note

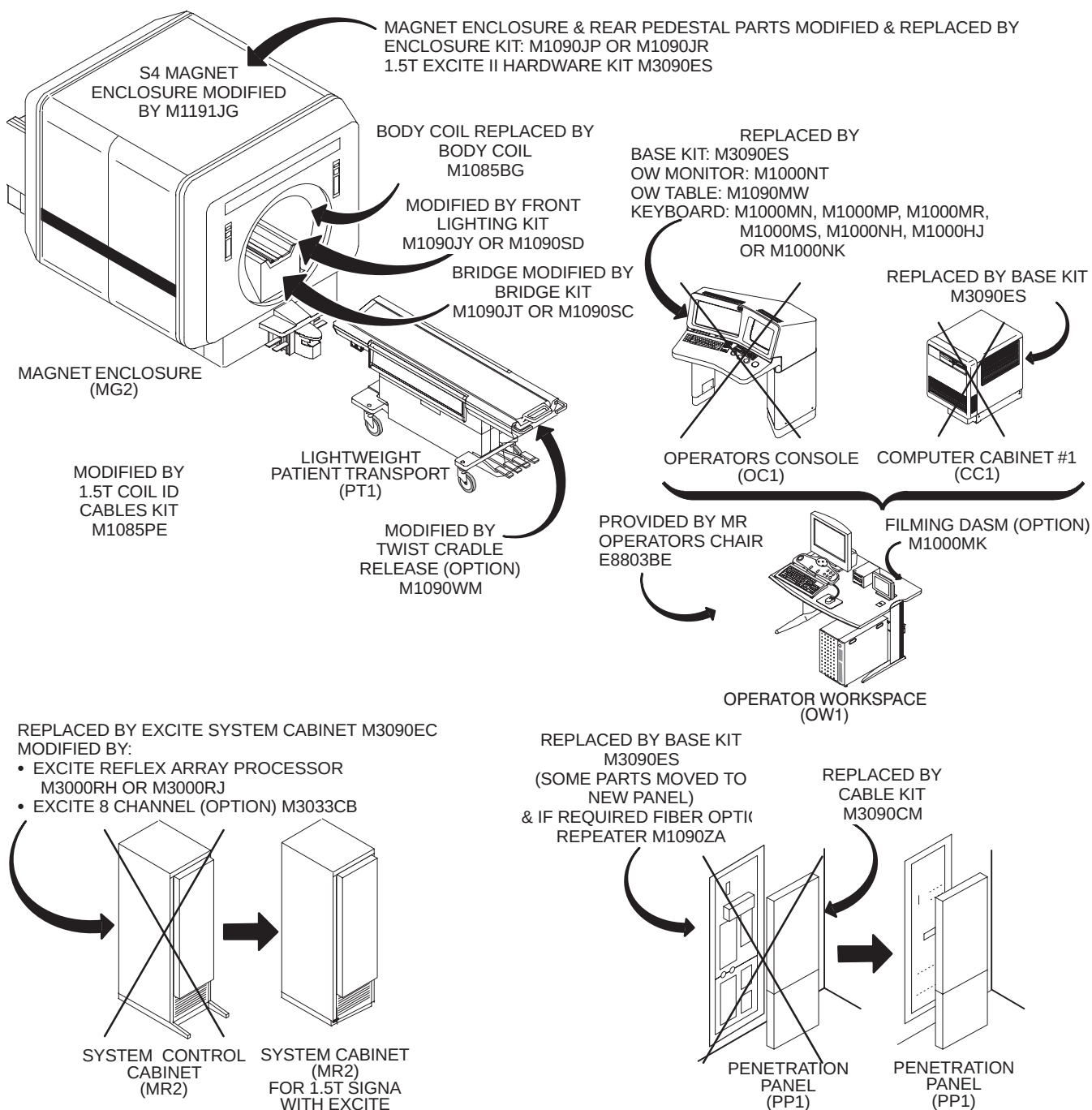
Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



1.5T SIGNA 5.X TO SIGNA LX WITH EXCITE UPGRADE CATALOGS

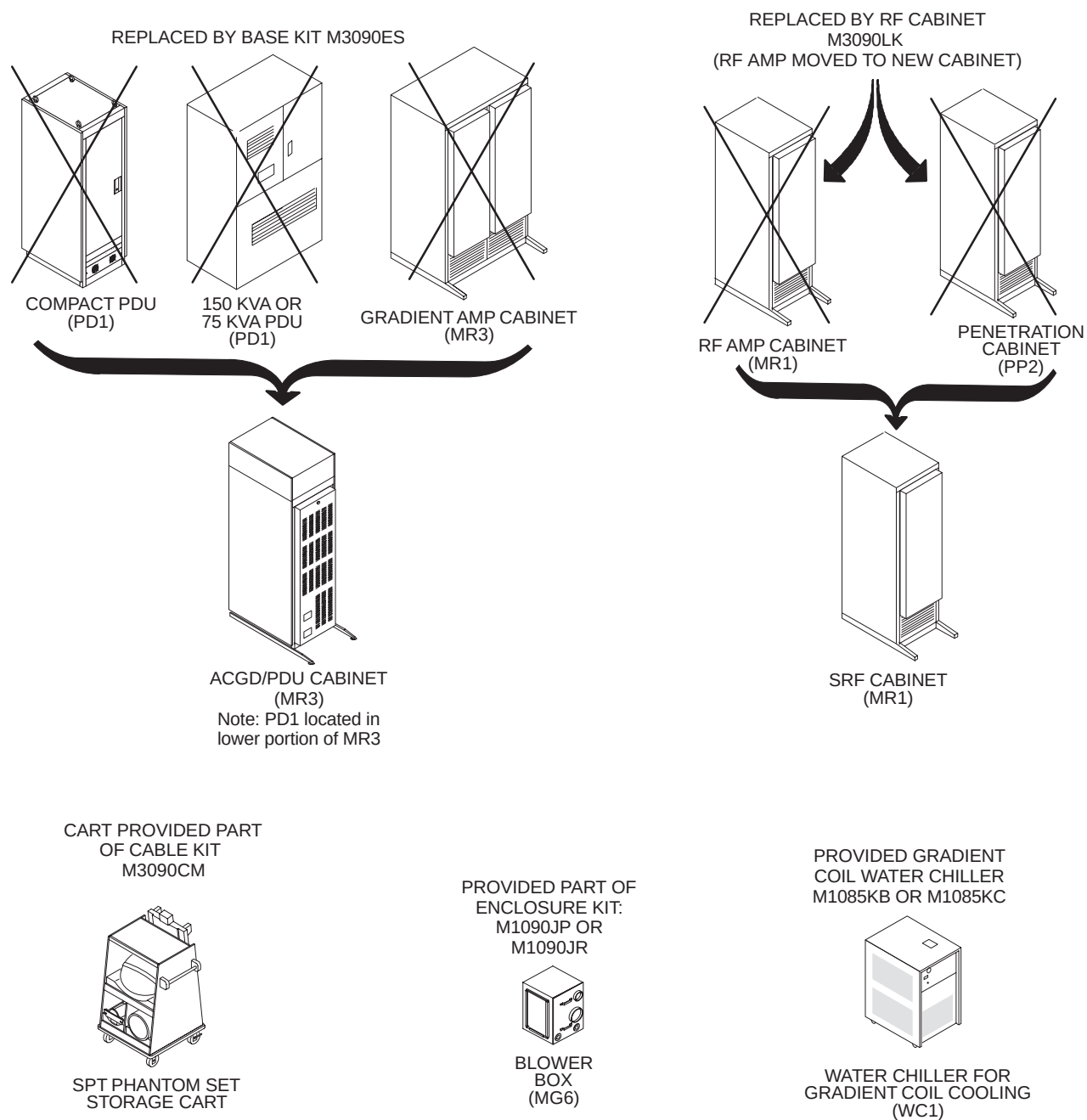
ILLUSTRATION 1 – 12

1-10-4 Signa Advantage 5.x (Continued)



SIGNA 5.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-13

1-10-4 Signa Advantage 5.x (Continued)



SIGNA 5.X EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-14

1–10–5 Signa Advantage 4.x

Illustration 1–15 lists the upgrade catalogs for 1.5T Signa 5.X to 1.5T Signa LX with EXCITE. Illustrations 1–16 and 1–17 show the cabinets replaced or modified and new equipment added by the listed catalogs.

Note

Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

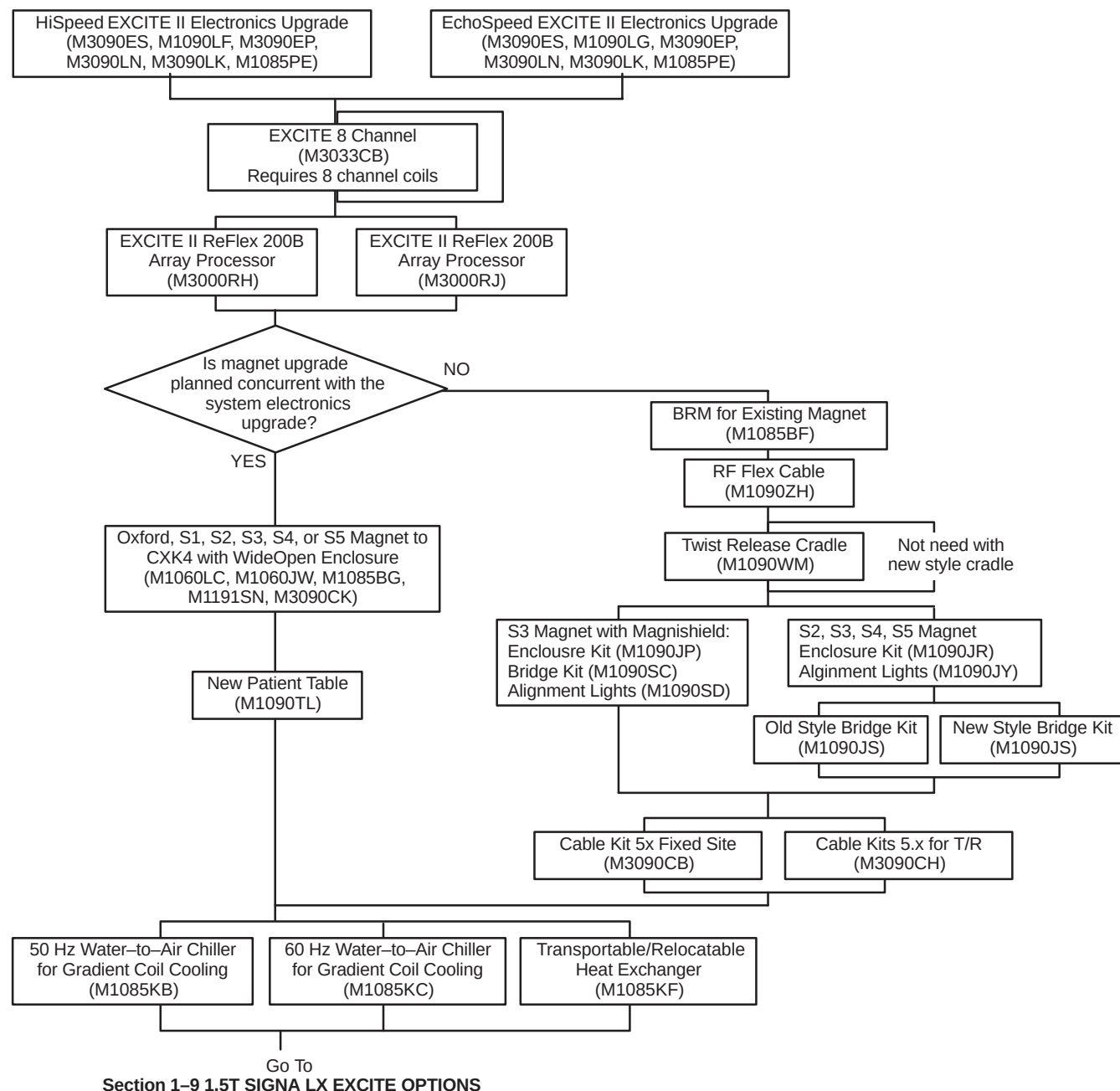
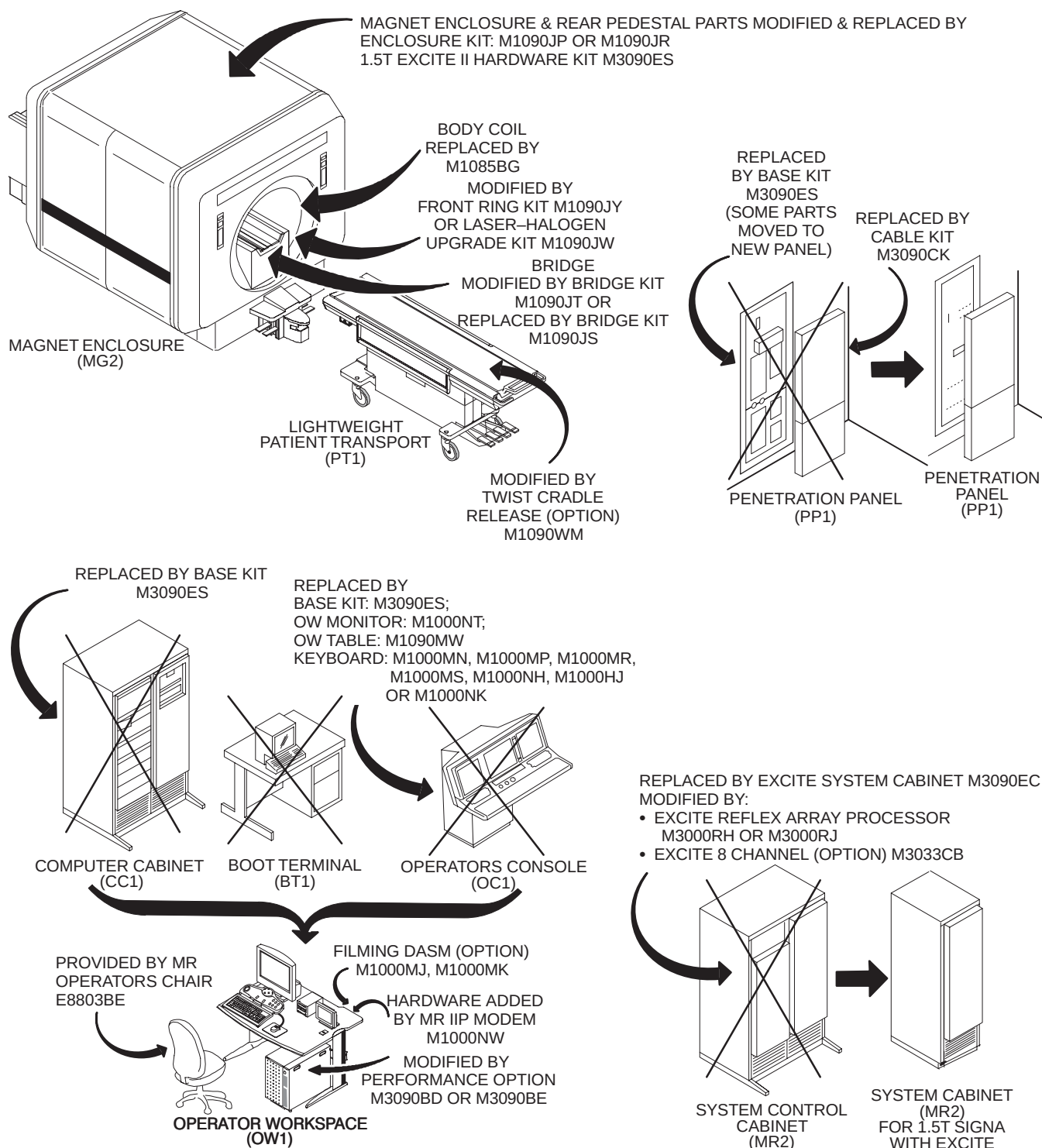
**1.5T SIGNA 4.X TO SIGNA LX WITH EXCITE UPGRADE CATALOGS**

ILLUSTRATION 1 – 15

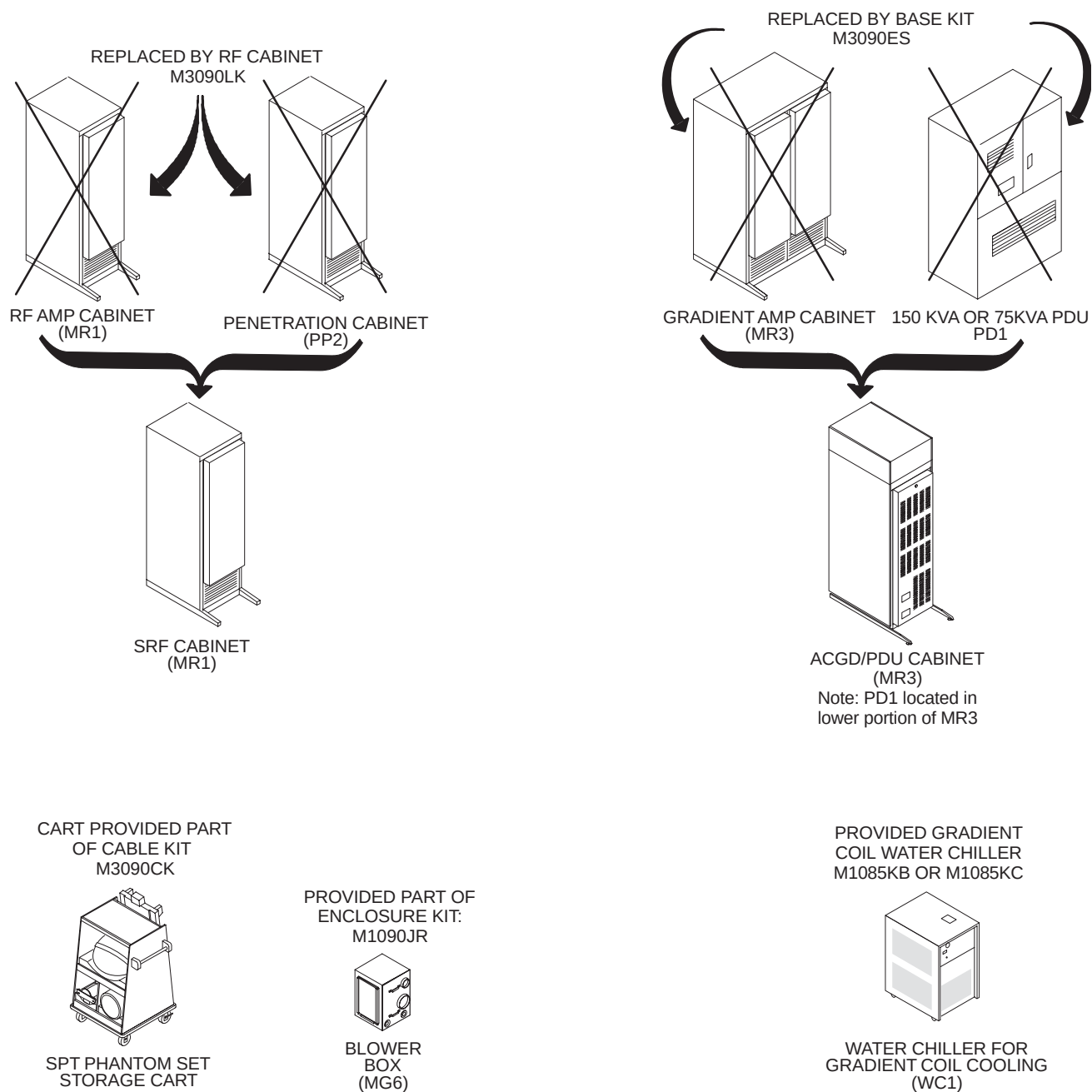
1-10-5 Signa Advantage 4.x (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-16

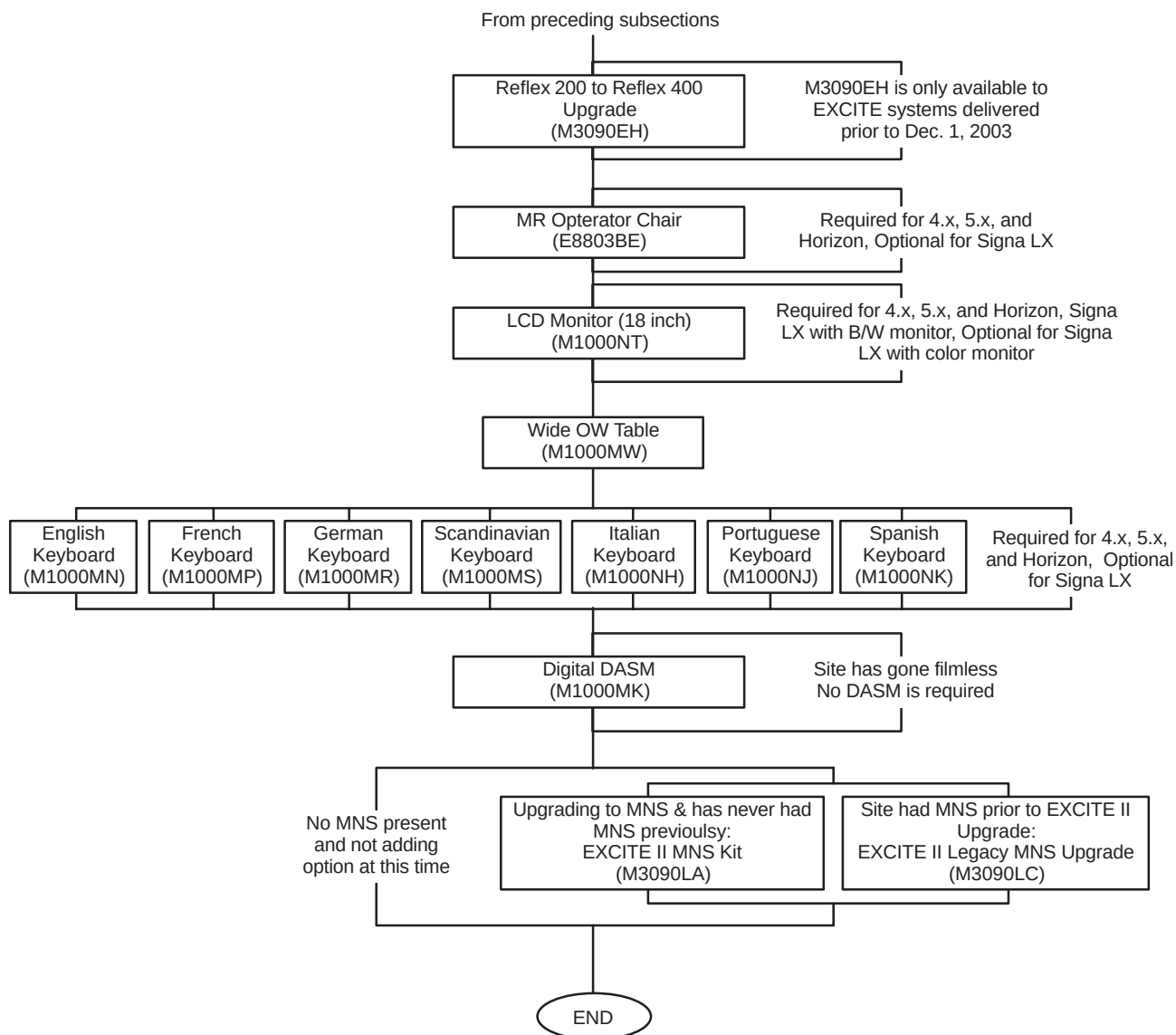
1-10-5 Signa Advantage 4.x (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-17

1-10-6 1.5T Signa LX with EXCITE Options

1.5T Signa LX EXCITE system options which may have site preparation impact are listed in Illustration 1-18. Surface Coils, software, training, and other options that do not provide MR system hardware requiring site preparation are not listed in this section.



SIGNA EXCITE 1.5T OPTIONS
ILLUSTRATION 1-18

1-11 UPGRADES TO SIGNA LX WITH ACGD

Illustration 1-19 shows the system's major equipment after a Signa LX with ACGD upgrade is complete. The upgrade content is dependent on the configuration of the system being upgraded and the final Signa LX with ACGD product configuration. The following sub-sections provide an overview of the various Signa LX with ACGD upgrade hardware.



WARNING!

ALL SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH THE ACGD/PDU CABINET NEED TO INVESTIGATE THAT THE FACILITY TRANSFORMER AND FEEDER WIRES ARE CORRECTLY SIZED. ALSO THE MAIN DISCONNECT CONTROL NEEDS TO BE INVESTIGATED FOR CORRECTLY SIZED WIRES AND RATED COMPONENTS TO MEET THE UPGRADE POWER REQUIREMENTS. REFER TO DIRECTION 2142460 – POWER REQUIREMENTS SECTION.

SYSTEMS UPGRADING TO SYSTEM CONFIGURATION WITH ACGD/PDU CABINET WITH AN UNINTERRUPTIBLE POWER SUPPLY (UPS) PROVIDING POWER TO THE ENTIRE SYSTEM NEED TO MAKE SURE THE UPS OPERATION PARAMETERS ARE COMPATIBLE WITH THE SIGNA SYSTEM WITH ACGD POWER AND REGULATION DEMANDS. REFER TO DIRECTION 2142460 – POWER REQUIREMENTS SECTION.

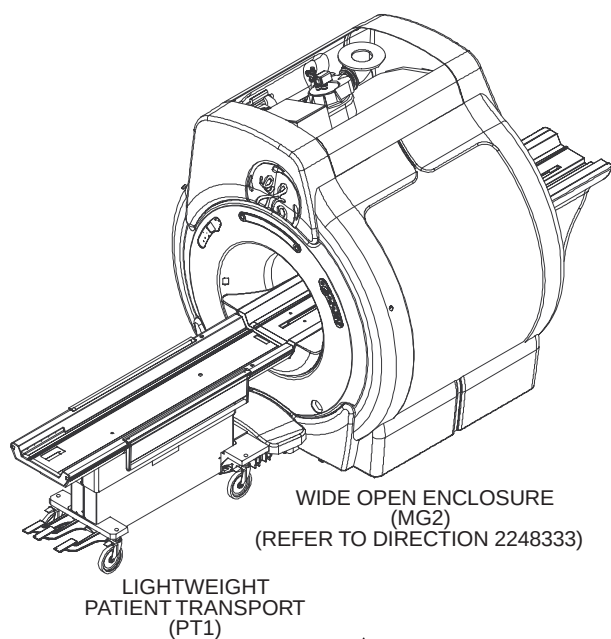
Note

Only catalogs which include hardware are documented in this publication. Software only catalogs are not included, existing system site parameters apply for software only upgrades.

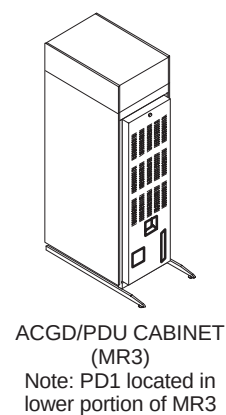
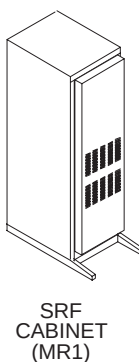
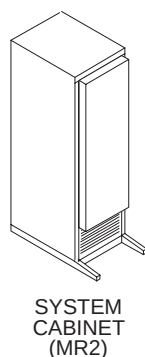
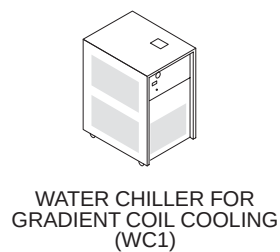
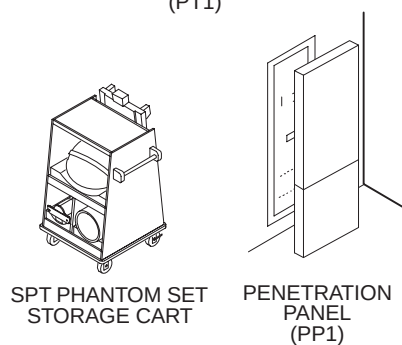
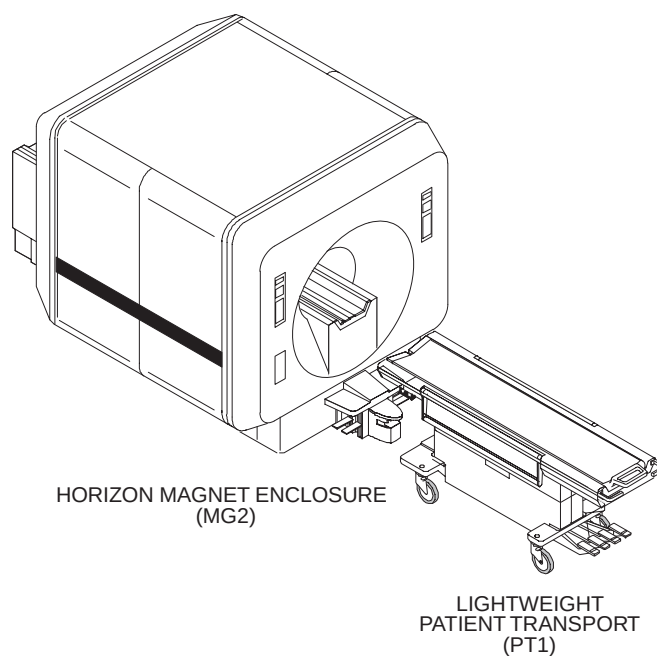
Note

FOR USA, FIXED SITE ONLY CONSIDERING MAGNET UPGRADE: Refer to *Direction 22461012 Pre-Sales/Site Survey For Upgrade To CXK4 Magnet With WideOpen Enclosure* for the many environmental factors and logistics that need to be understood to qualify a site for a magnet upgrade. The Magnet Upgrade Option requires the MR system to be operating a Release 8.X prior or coincident with the Magnet Upgrade option.

1-11 UPGRADES TO SIGNA LX WITH ACGD (Continued)



OR



SIGNA LX WITH ACGD SYSTEM MAJOR EQUIPMENT
ILLUSTRATION 1-19

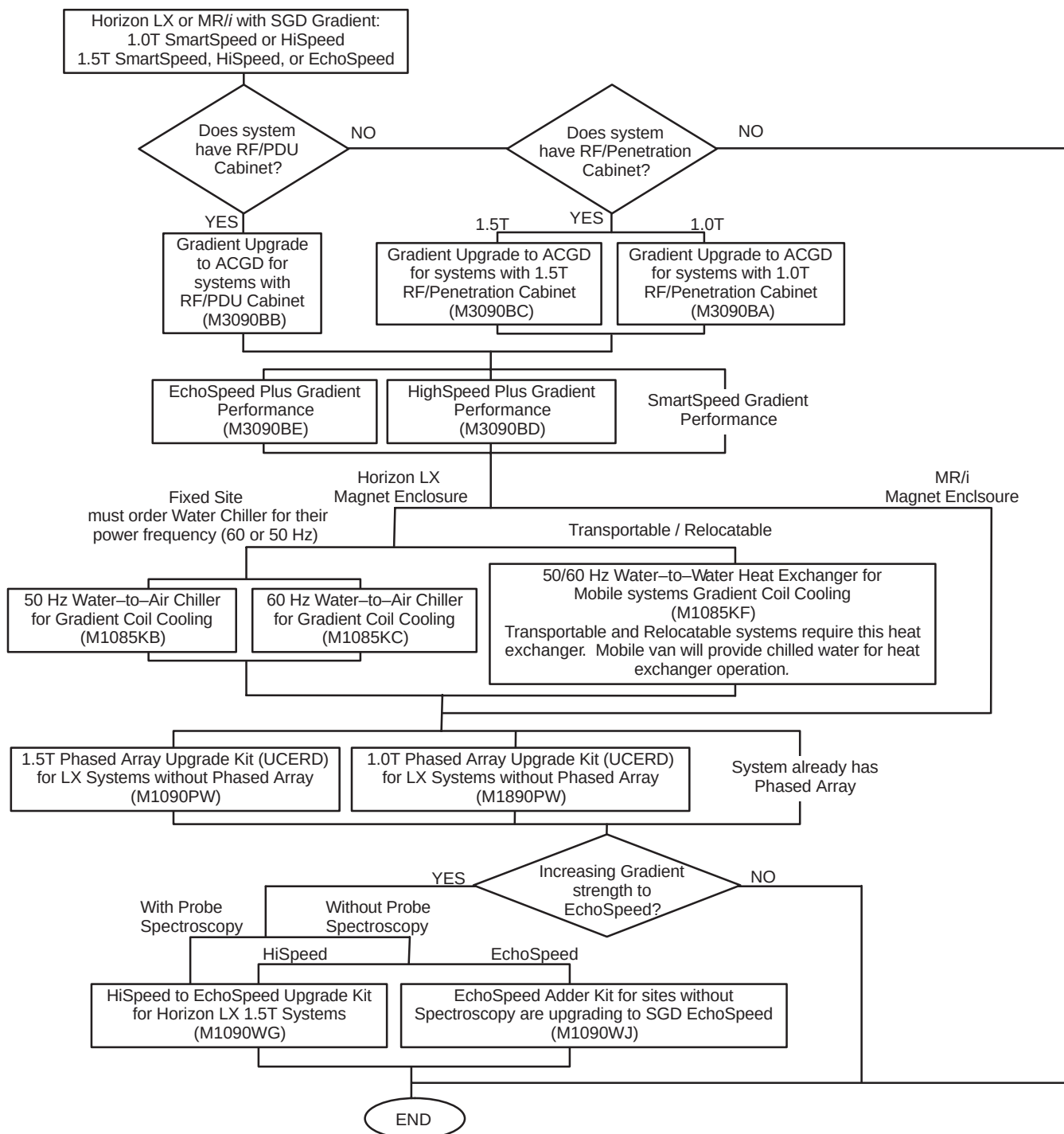
1-11-1 Signa Horizon LX With SGD & Signa MR/i System

Illustration 1-20 lists the upgrade catalogs for Signa Horizon LX with SGD and Signa MR/i with SGD system to Signa LX with ACGD. Illustration 1-21 shows the cabinets replaced and modified by the listed catalogs.

Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

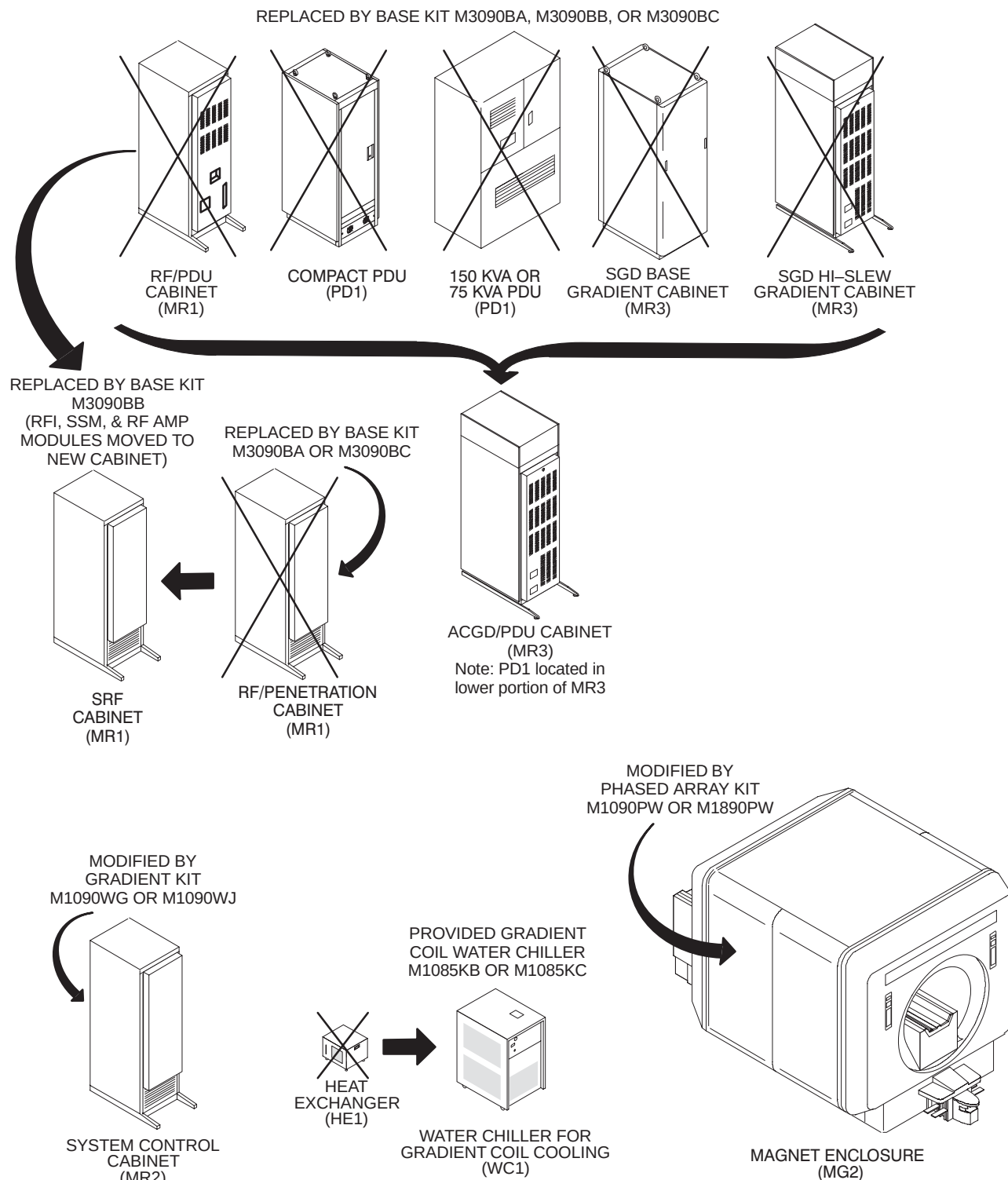
1-11-1 Signa Horizon LX With SGD & Signa MR/i System (Continued)



1.5T SIGNA LX OR SIGNA MR/i WITH SGD GRADIENTS TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-20

1-11-1 Signa Horizon LX With SGD & Signa MR/i System (Continued)



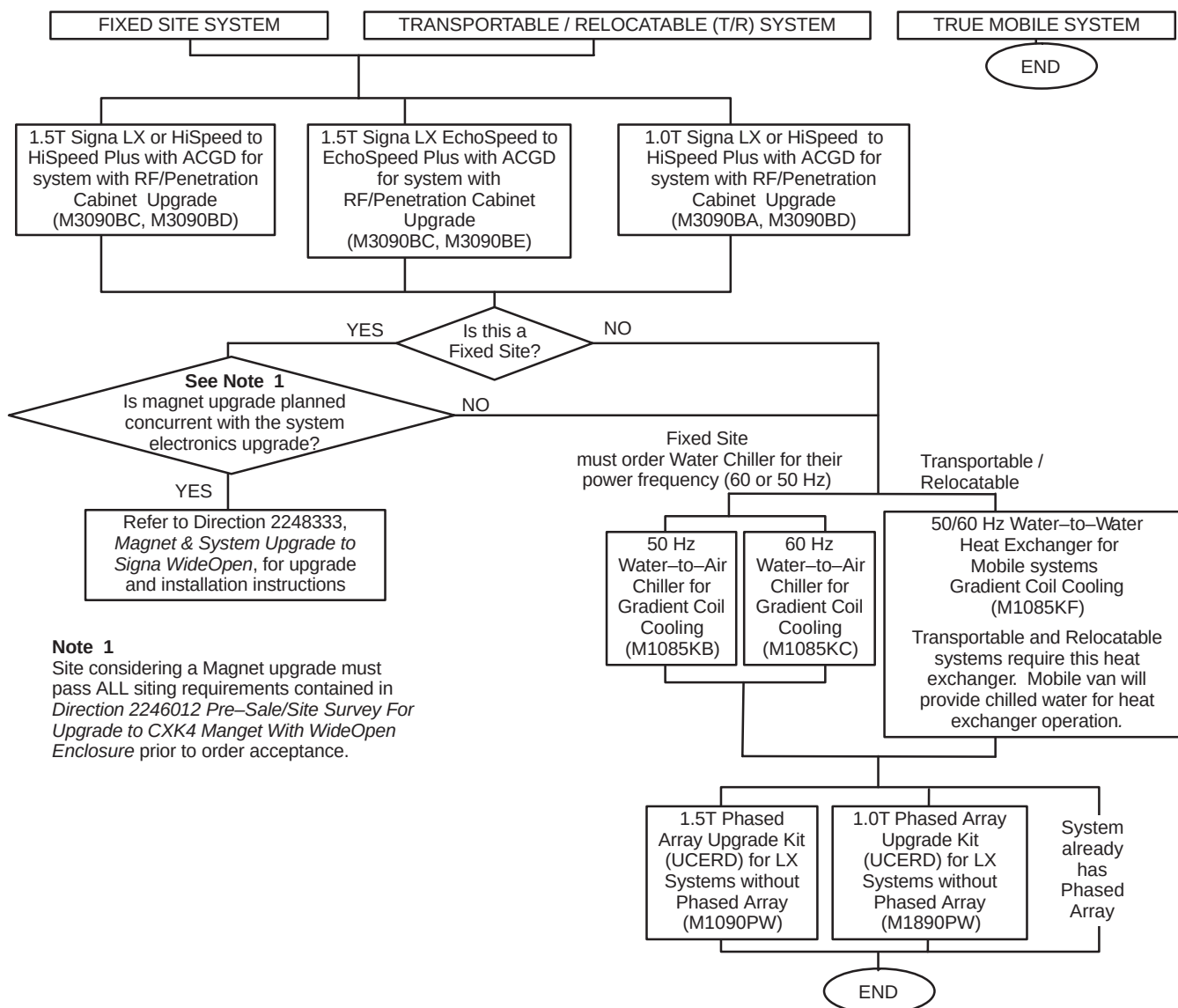
SIGNA HORIZON LX WITH SGD & SIGNA MR// SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-21

1-11-2 Signa Horizon LX With 8645 Gradients System

Illustration 1–22 lists the upgrade catalogs for Signa Horizon LX system with 8645 Gradients to Signa Horizon LX with ACGD. Illustration 1–23 shows the cabinets replaced and modified by the listed catalogs.

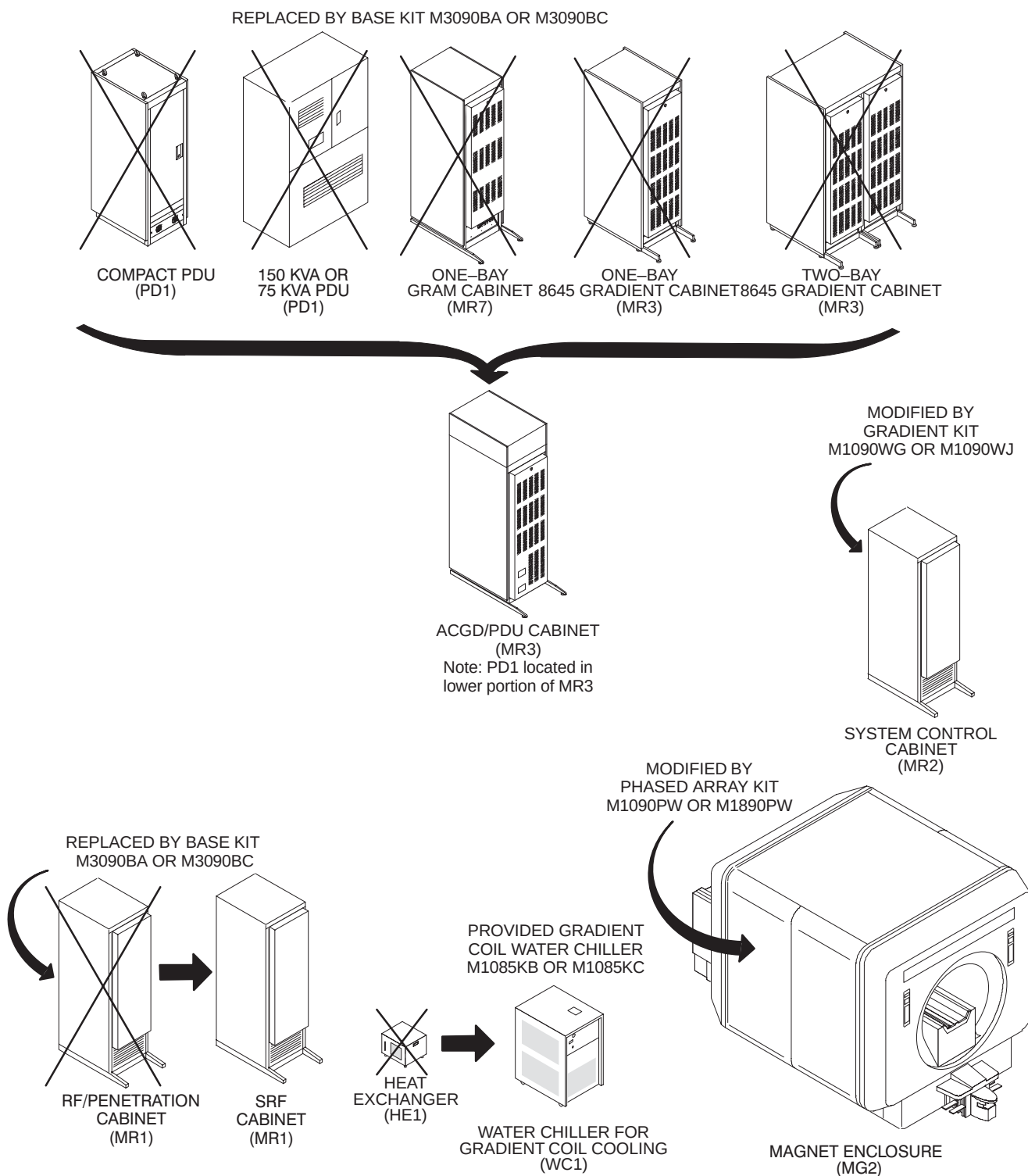
Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1-22

1-11-2 Signa Horizon LX With 8645 Gradients System (Continued)



SIGNA HORIZON LX SYSTEM UPGRADE MAJOR EQUIPMENT REPLACED & MODIFIED
ILLUSTRATION 1-23

1–11–3 Signa Horizon 5.x System

Illustrations 1–24 through 1–25 list the upgrade catalogs for Signa Horizon 5.X system to Signa Horizon LX with ACGD. Illustrations 1–20 and 1–21 show the cabinets replaced or modified and new equipment added by the listed catalogs.

Refer to Section 1–10 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

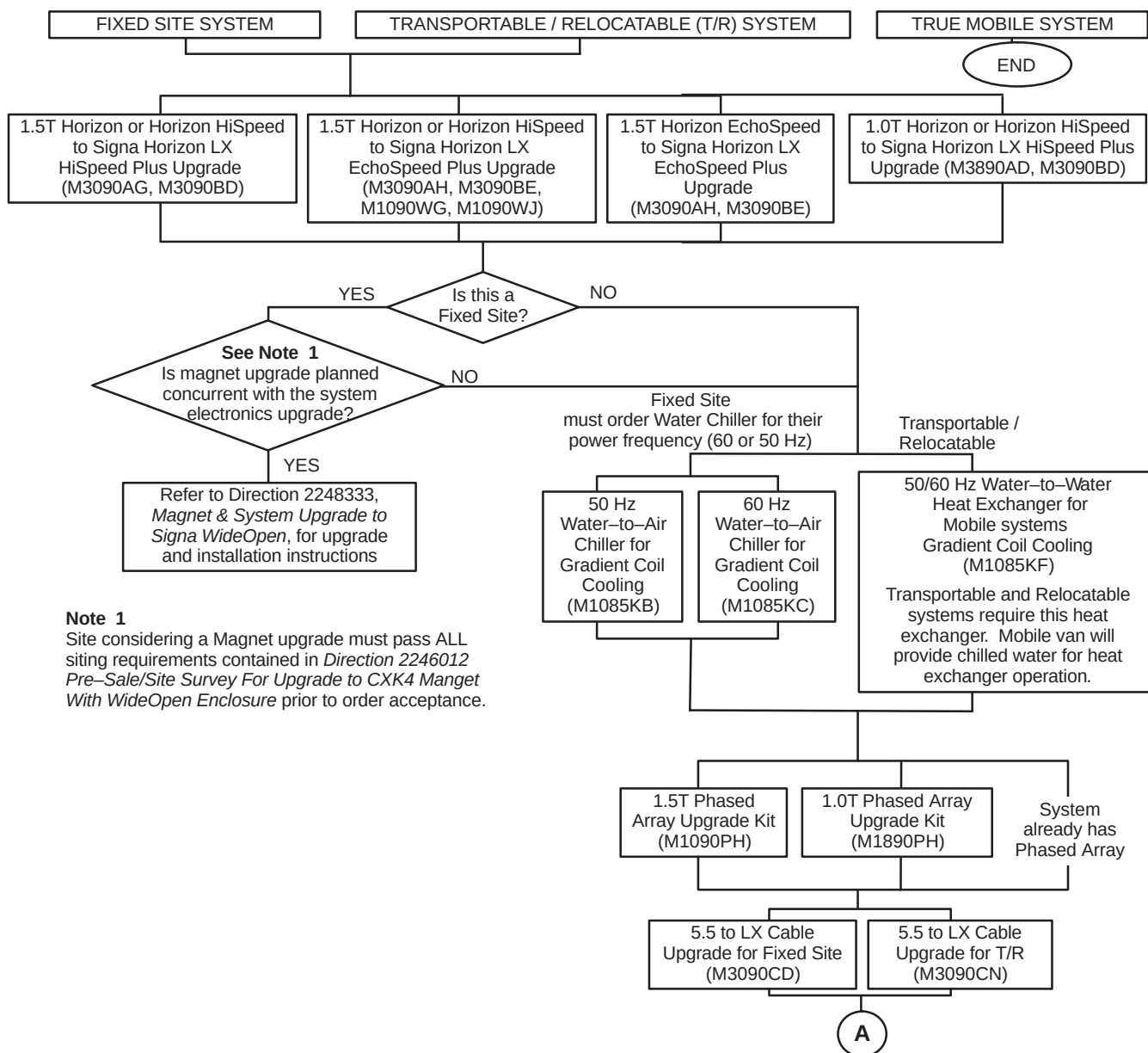
Note

Signa Horizon 5.5 Software only systems have a 55 cm Body Coil and require upgrade catalogs defined in Section 1–11–4, SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM UPGRADES.

Note

Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

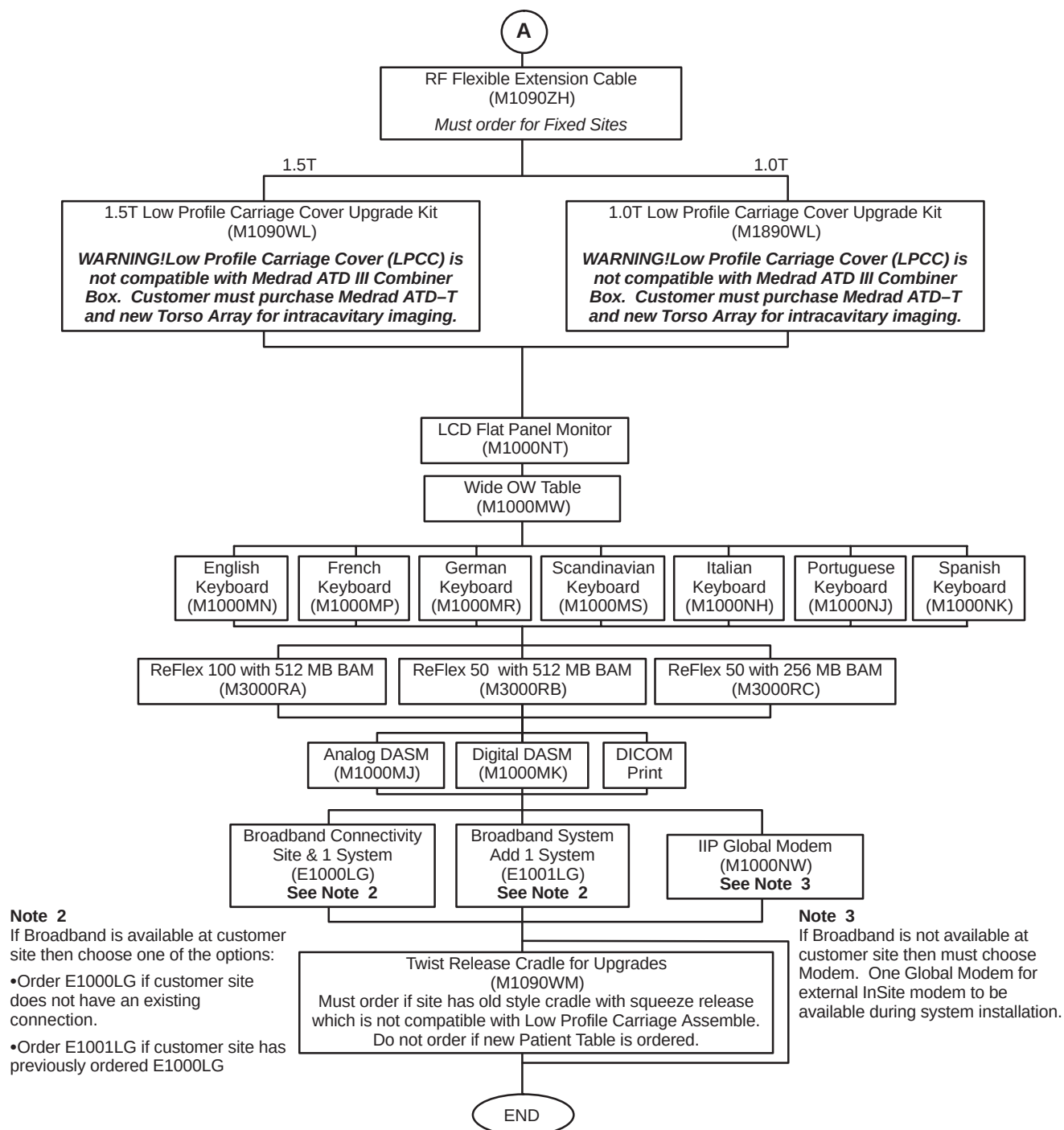
1-11-3 Signa Horizon 5.X System (Continued)



1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS

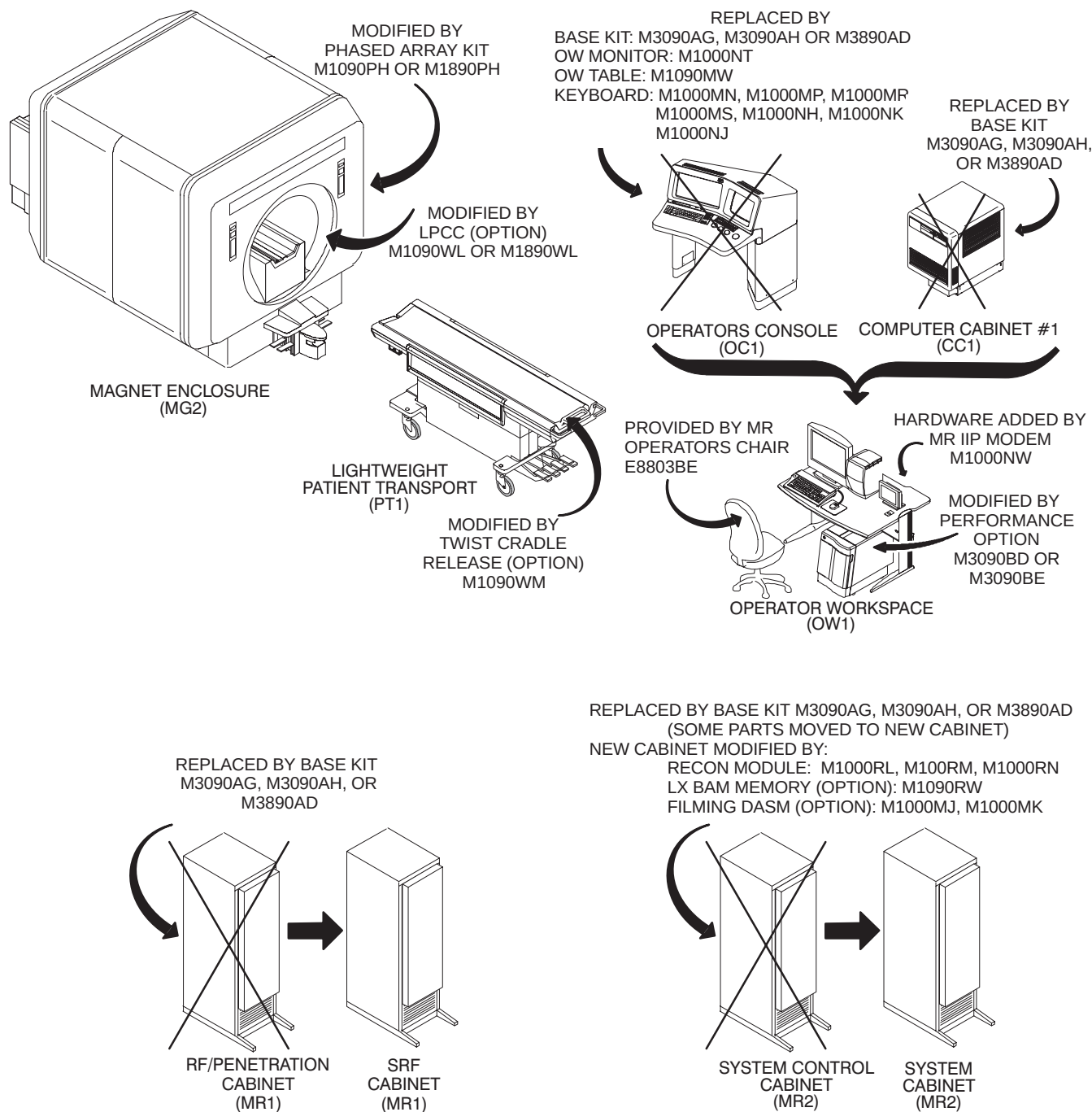
ILLUSTRATION 1-24

1-11-3 Signa Horizon 5.X System (Continued)



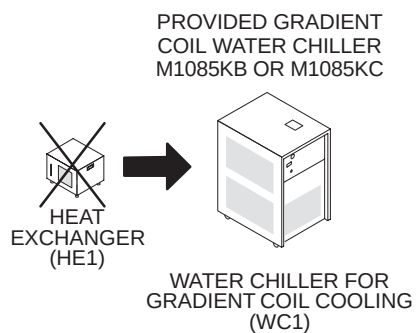
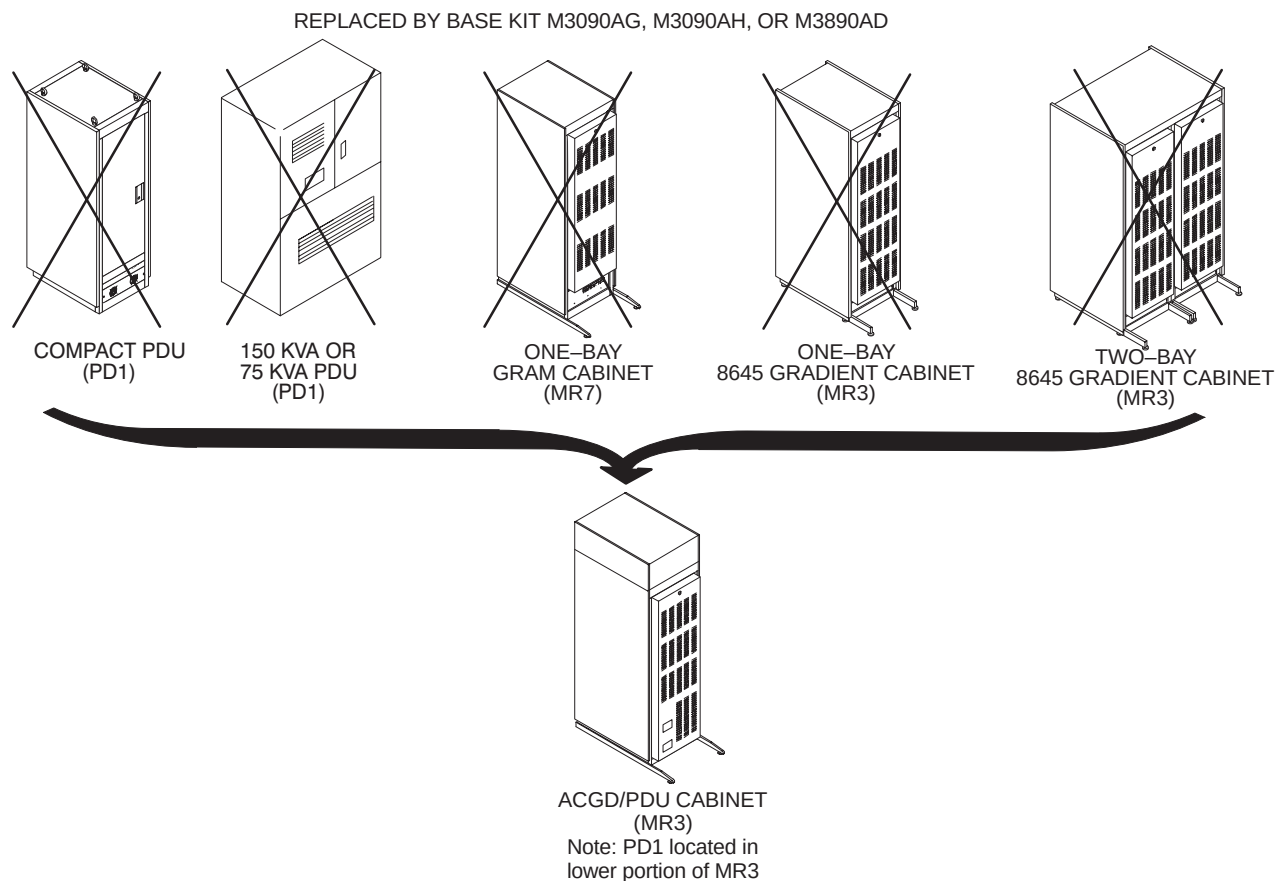
1.5T SIGNA HORIZON 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1-25

1-11-3 Signa Horizon 5.X System (Continued)



SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-26

1-11-3 Signa Horizon 5.X System (Continued)

SIGNA HORIZON 5.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-27

1–11–4 Signa Horizon 5.x Software Only & Signa Advantage 5.x (5.1–5.4) System

Illustrations 1–28 through 1–30 lists the upgrade catalogs for Signa Advantage 5.X hardware system and Signa Horizon 5.X software only system to Signa LX with ACGD. Illustrations 1–31 and 1–32 show the cabinets replaced or modified and new equipment added by the listed catalogs. Signa 5.1–5.4 hardware systems are the following configurations:

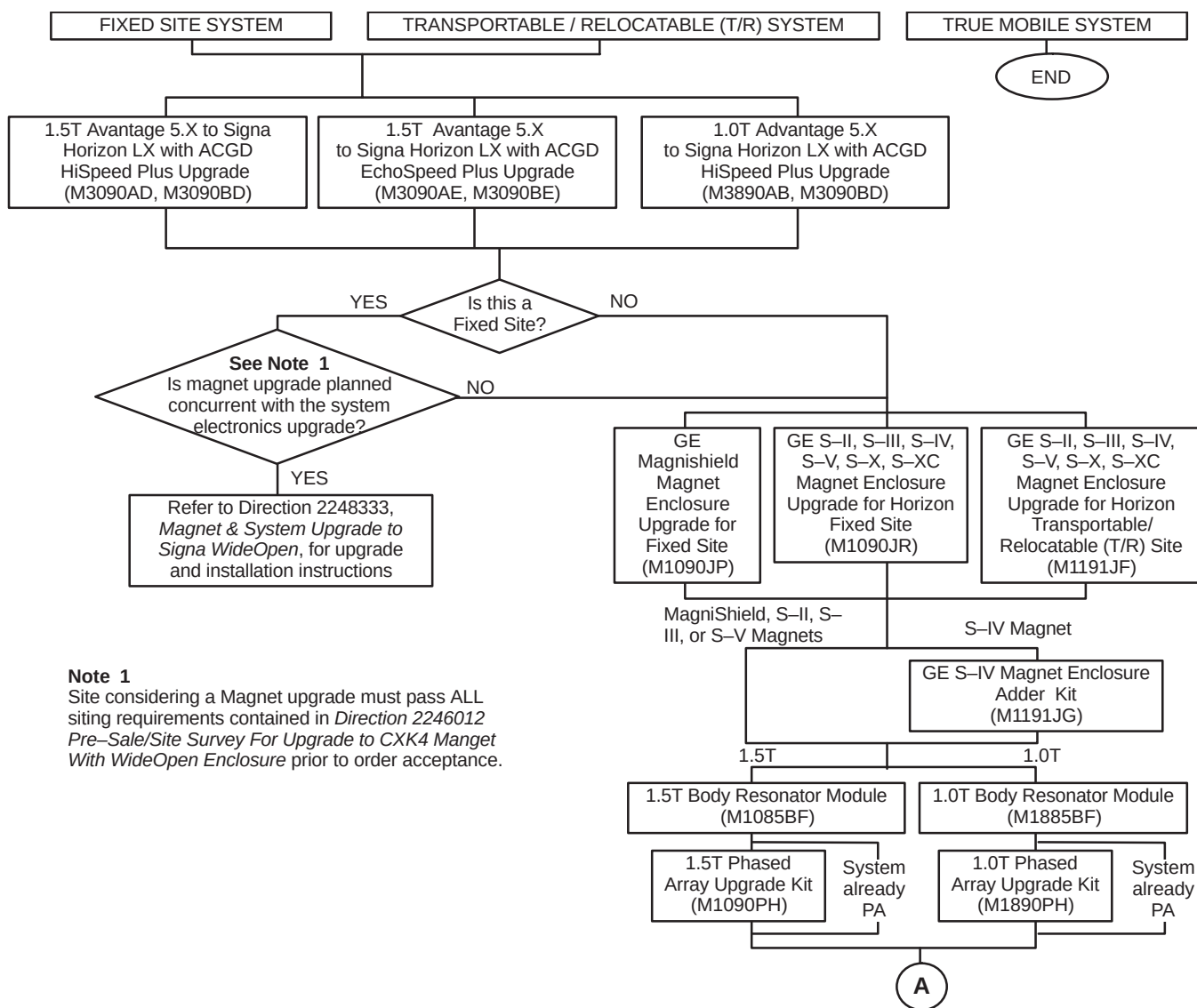
- Signa Horizon 5.X Software only systems which have a 55 cm Body Coil
- Signa Advantage 5.X systems identified by the Genesis Computer, 1-bay System Cabinet, and 8607 Gradient Cabinet.

Refer to Section 1–10 SIGNA HORIZON LX SYSTEM OPTIONS for additional option catalogs.

Note

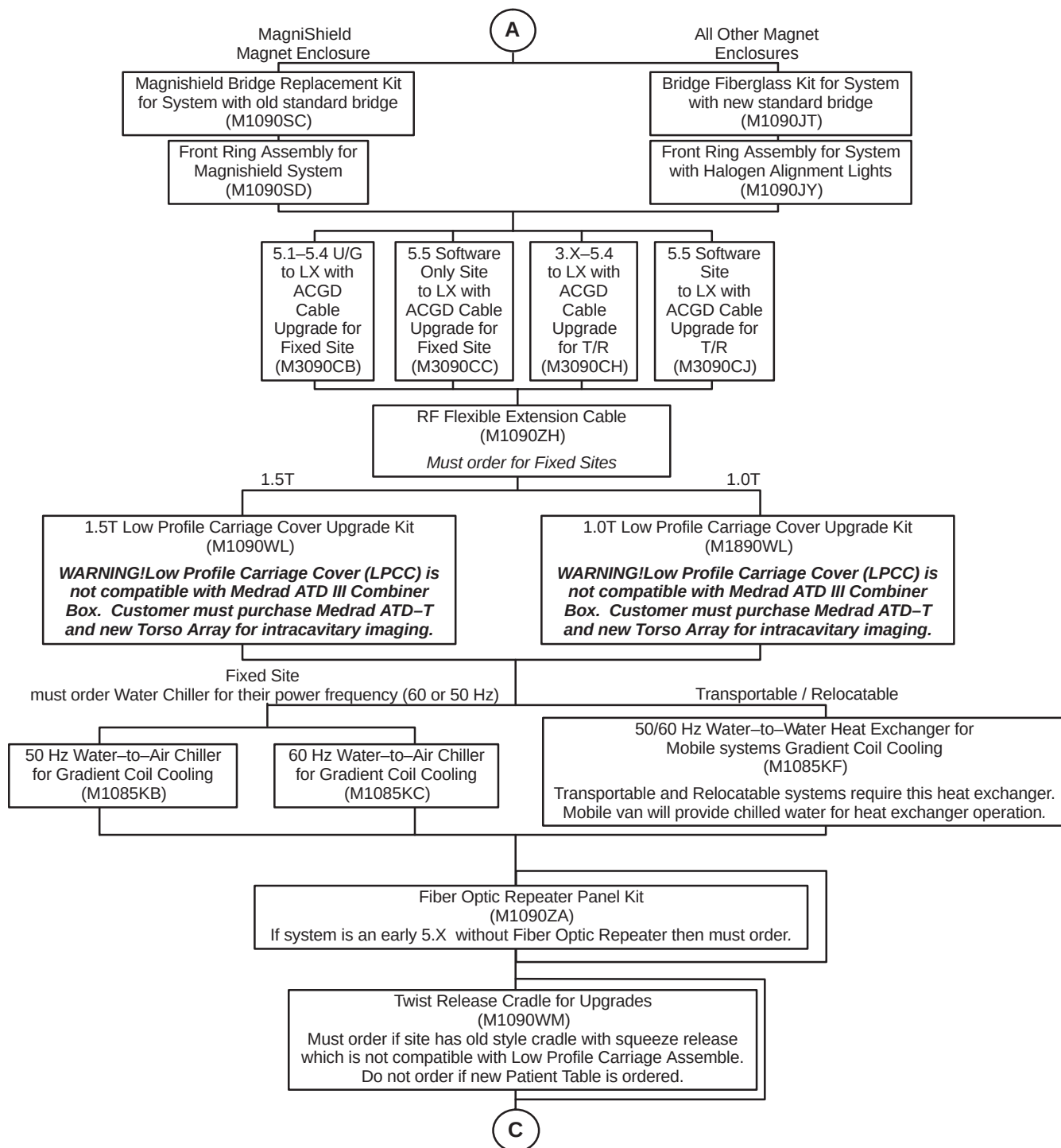
Systems with Oxford or S–1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.

1-11-4 Signa Horizon 5.X Software Only & Signa Advantage 5.X System – (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1 – 28

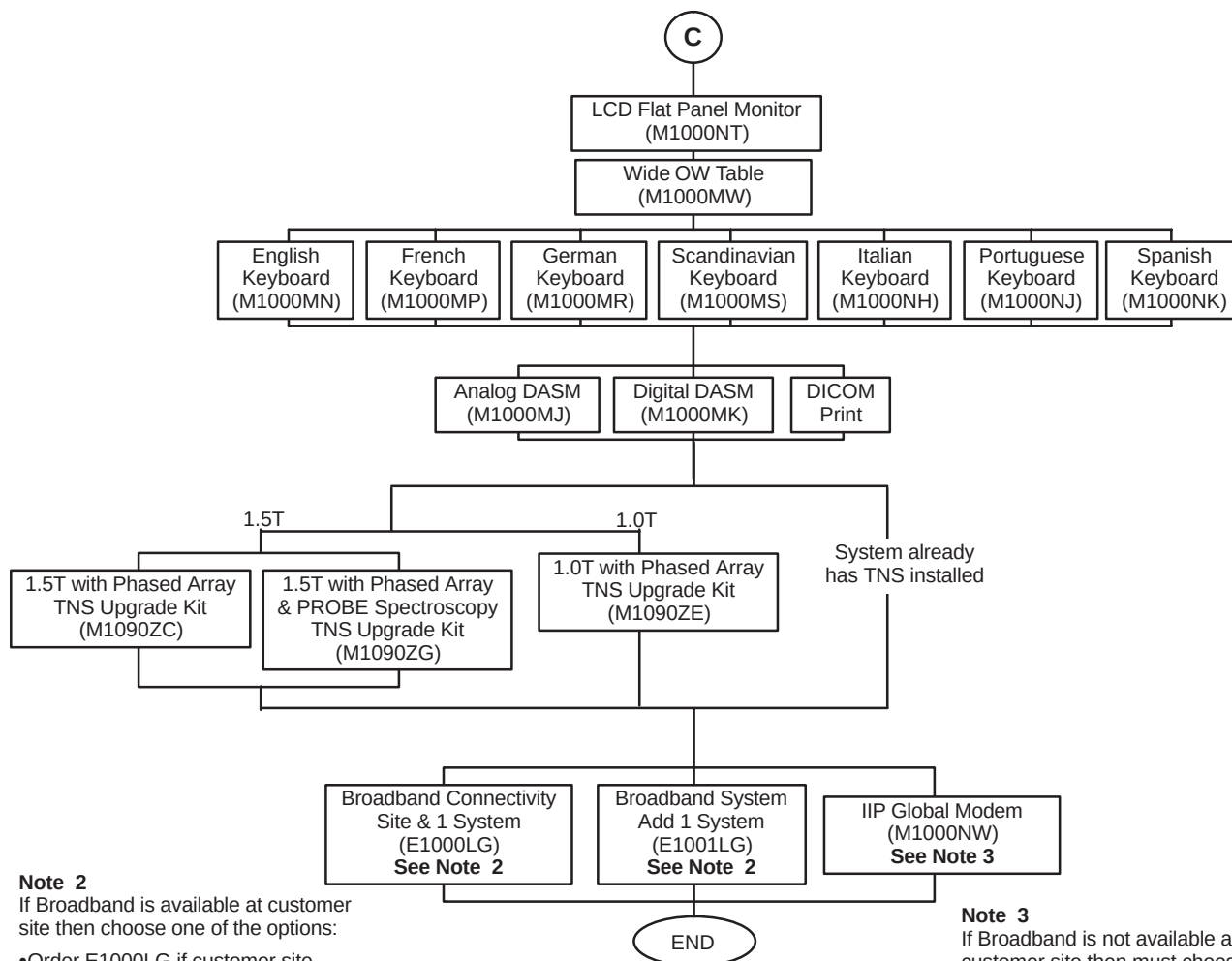
1-11-4 Signa Horizon 5.x Software Only & Signa Advantage 5.x System (Continued)



SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X TO SIGNA LX WITH ACGD CATALOGS

ILLUSTRATION 1-29

1-11-4 Signa Horizon 5.x Software Only & Signa Advantage 5.x System (Continued)

**Note 2**

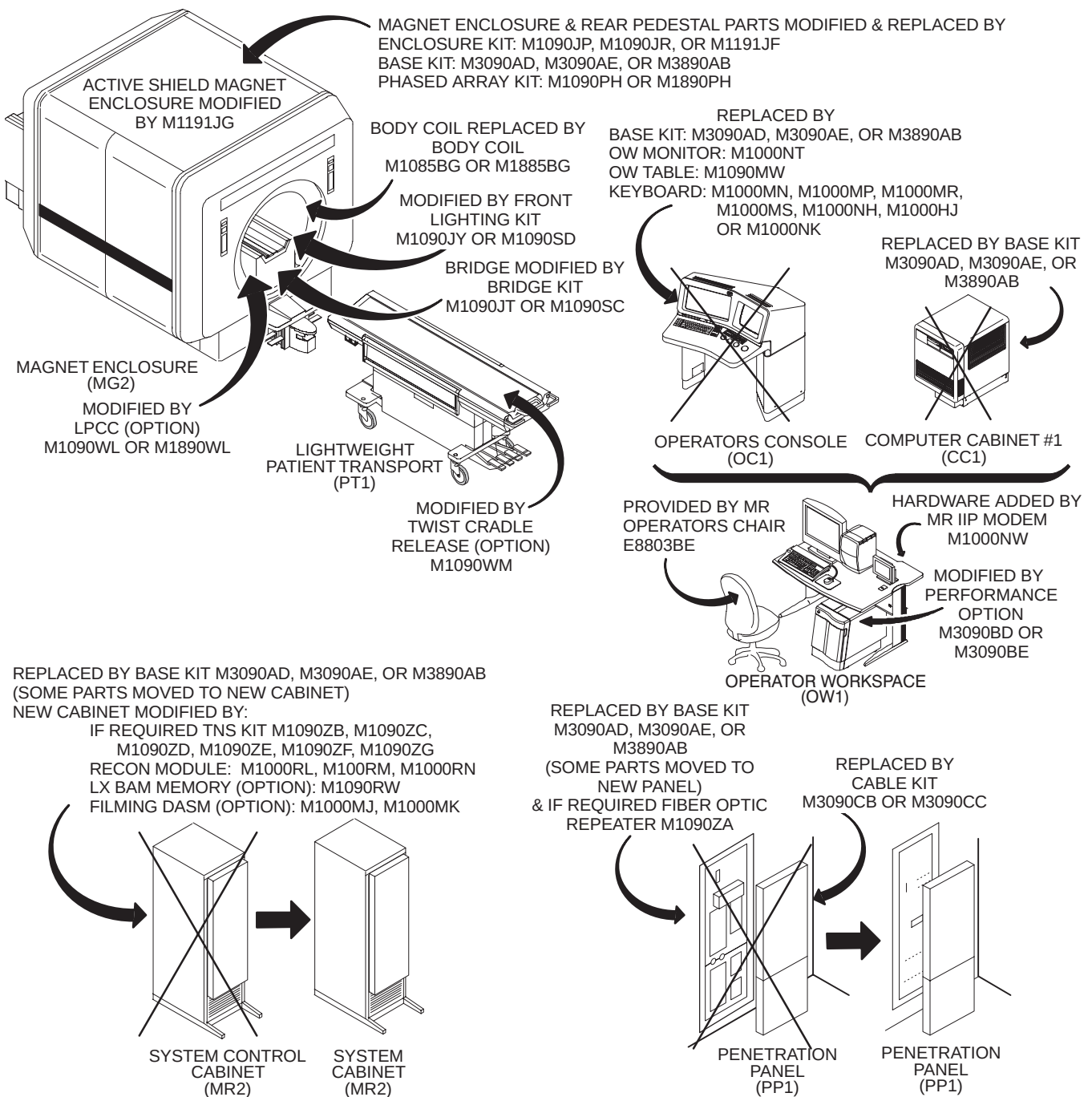
If Broadband is available at customer site then choose one of the options:

- Order E1000LG if customer site does not have an existing connection.
- Order E1001LG if customer site has previously ordered E1000LG

Note 3

If Broadband is not available at customer site then must choose Modem. One Global Modem for external InSite modem to be available during system installation.

1-11-4 Signa Horizon 5.x Software Only & Signa Advantage 5.x System (Continued)

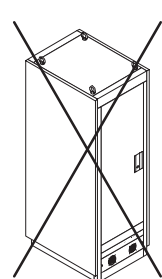
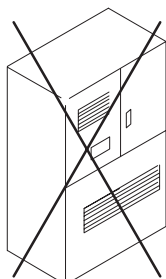
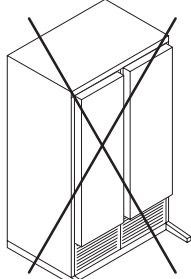
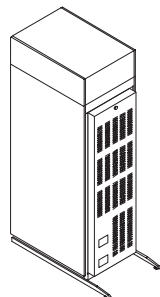
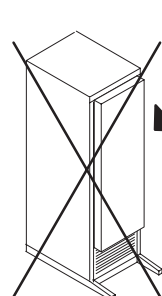
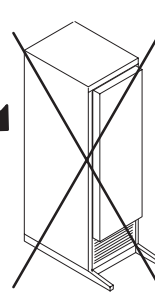
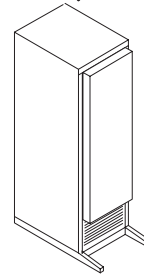
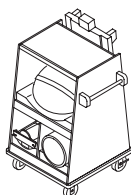
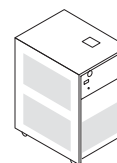


**SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED**

ILLUSTRATION 1-31

1-11-4 Signa Horizon 5.x Software Only & Signa Advantage 5.x System (Continued)

REPLACED BY BASE KIT M3090AD, M3090AE, OR M3890AB

COMPACT PDU
(PD1)150 KVA OR
75 KVA PDU
(PD1)GRADIENT AMP CABINET
(MR3)ACGD/PDU CABINET
(MR3)
Note: PD1 located in
lower portion of MR3REPLACED BY BASE KIT
M3090AD, M3090AE, OR M3890AB
(RF AMP MOVED TO NEW CABINET)RF AMP CABINET
(MR1)PENETRATION
CABINET
(PP2)SRF CABINET
(MR1)CART PROVIDED PART
OF CABLE KIT
M3090CB, M3090CC,
M3090CH, OR M3090CJSPT PHANTOM SET
STORAGE CARTPROVIDED PART OF
ENCLOSURE KIT:
M1090JP, M1090JR,
OR M1191JFBLOWER
BOX
(MG6)PROVIDED GRADIENT
COIL WATER CHILLER
M1085KB OR M1085KCWATER CHILLER FOR
GRADIENT COIL COOLING
(WC1)

SIGNA HORIZON 5.X SOFTWARE ONLY & SIGNA ADVANTAGE 5.X SYSTEM
EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED
 ILLUSTRATION 1-32

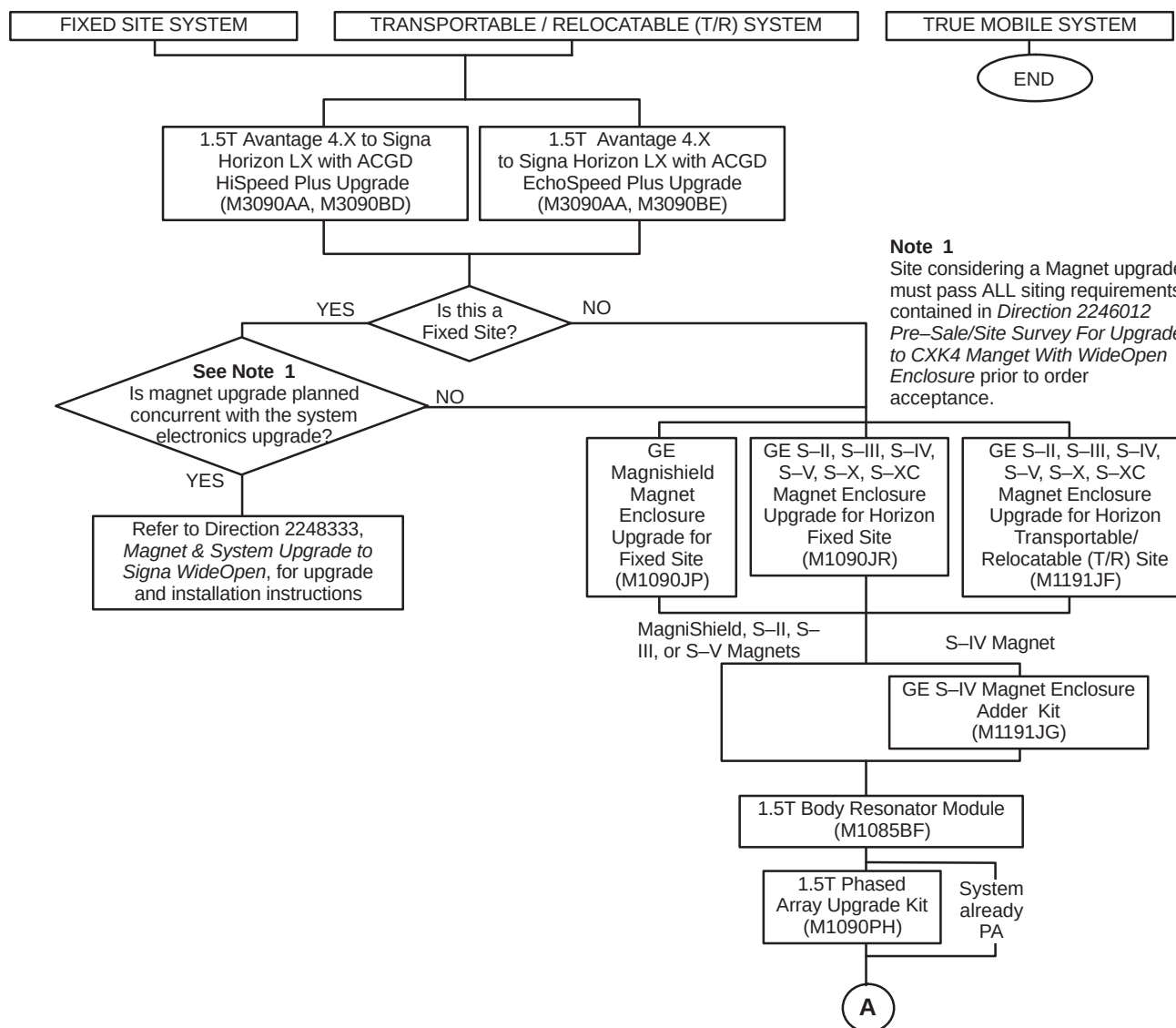
1-11-5 Signa Advantage 4.X System

Illustrations 1-33 through 1-35 lists the upgrade catalogs for Signa Advantage 4.X to Signa Horizon LX with ACGD. Signa Advantage 4.X hardware systems are Signa Advantage 4.5 to 4.8 which can be identified by RF Cabinet model number 46-306900 and Signa Advantage 4.0 to pre-4.5 which can be identified by RF Cabinet model number 46-282900. Illustrations 1-36 and 1-37 show the major equipment replaced or modified and new equipment added by the listed catalogs.

Refer to Section 1-10 SIGNA HORIZON LX SYSTEM OPTIONS for options catalogs.

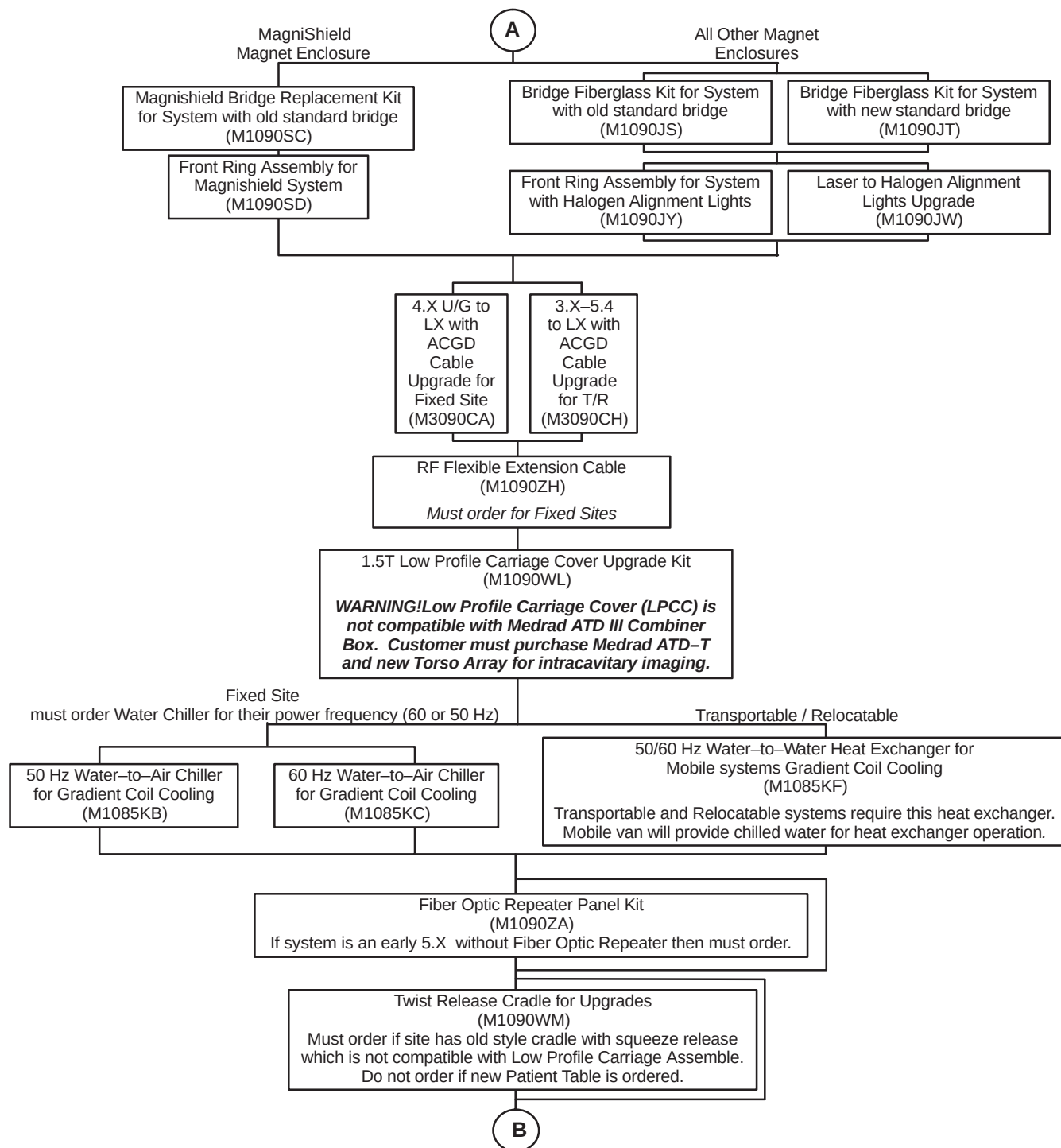
Note

Systems with Oxford or S-1 magnet must upgrade to the LCC magnet before or concurrent with system upgrade.



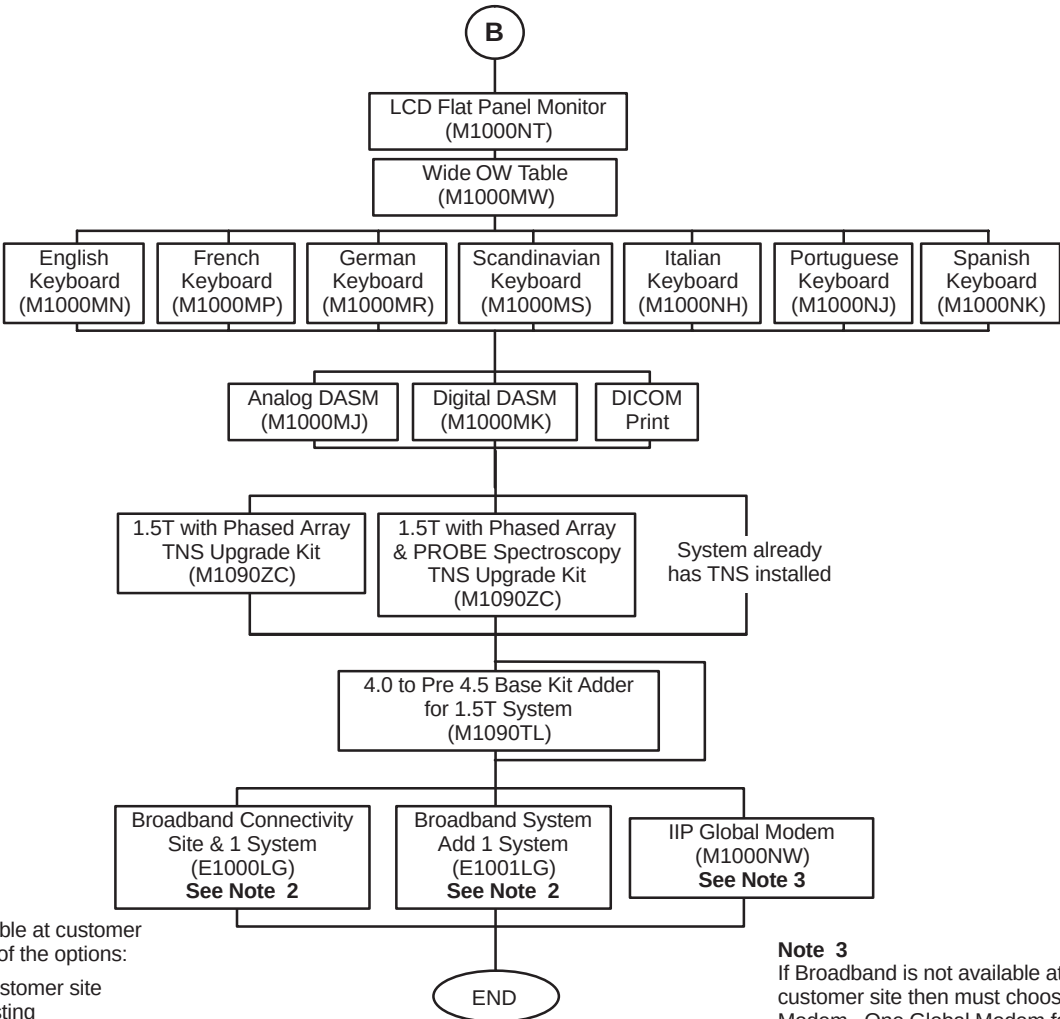
SIGNA ADVANTAGE 4.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1-33

1-11-5 Signa Advantage 4.X System (Continued)



SIGNA ADVANTAGE 4.X TO SIGNA LX WITH ACGD CATALOGS
ILLUSTRATION 1-34

1-11-5 Signa Advantage 4.x System (Continued)

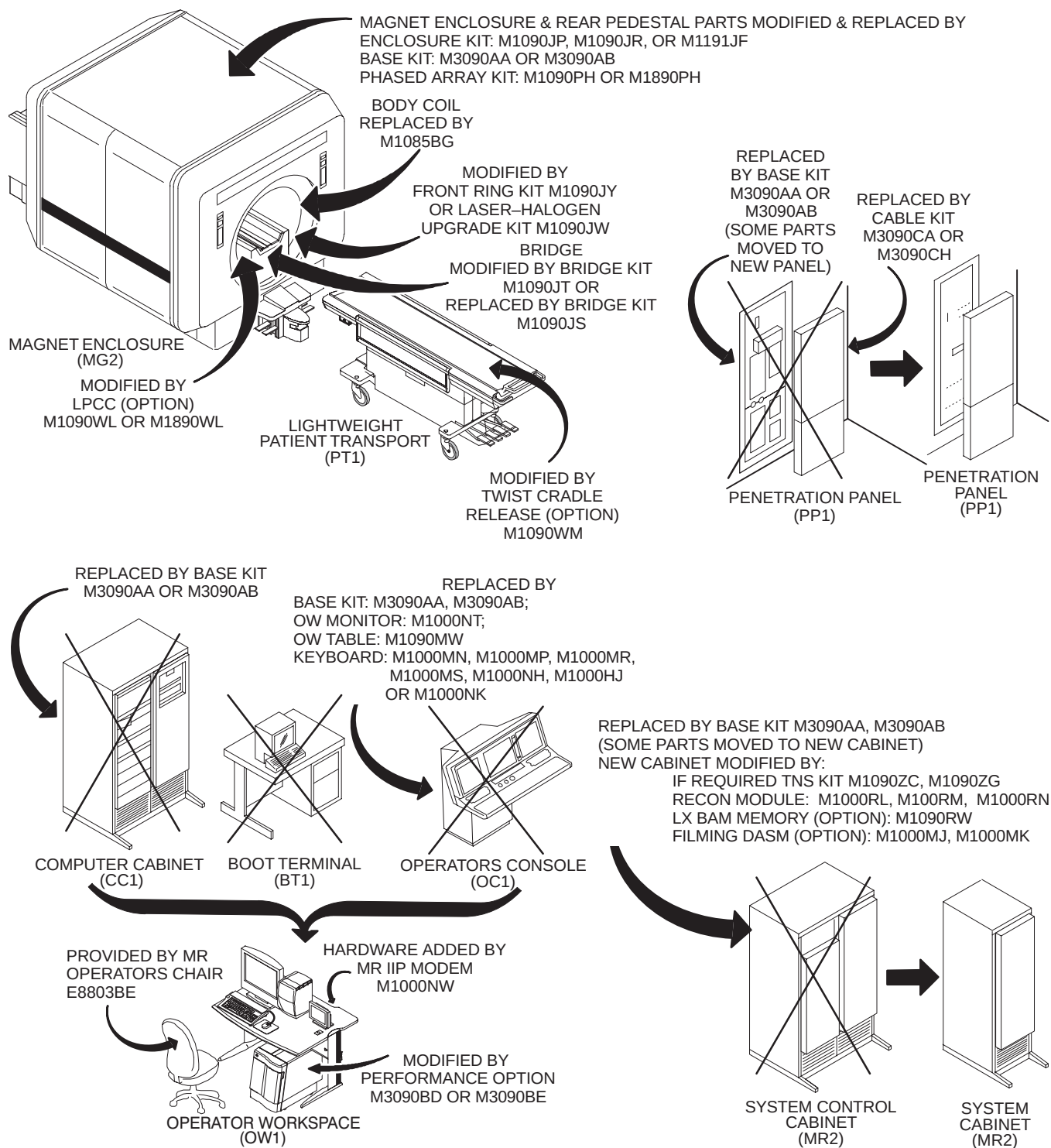


Note 2
If Broadband is available at customer site then choose one of the options:

- Order E1000LG if customer site does not have an existing connection.
- Order E1001LG if customer site has previously ordered E1000LG

Note 3
If Broadband is not available at customer site then must choose Modem. One Global Modem for external InSite modem to be available during system installation.

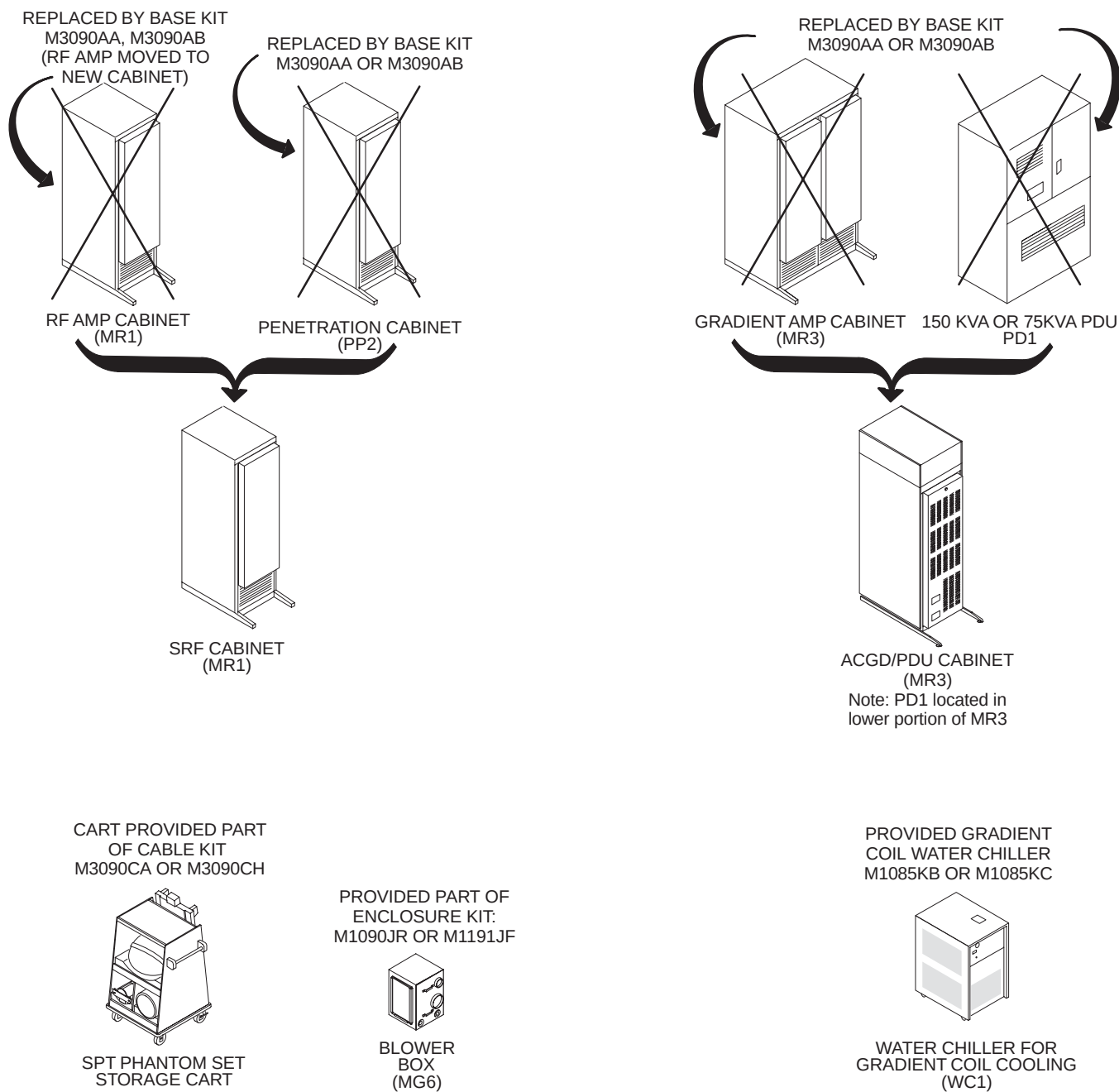
1-11-5 Signa Advantage 4.x System (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-36

1-11-5 Signa Advantage 4.x System (Continued)



SIGNA ADVANTAGE 4.X SYSTEM EXISTING HARDWARE REPLACED OR MODIFIED & NEW EQUIPMENT ADDED

ILLUSTRATION 1-37

1-12 SIGNA LX SYSTEM OPTIONS

Purchased options may also be installed concurrently with the system upgrade. System upgrade flow charts provide steering to the Option Installation Direction that is delivered with the option commonly delivered with the system upgrade.

- Refer to manual shipped with the option for installation of Advantage Window options and Camera Options.
- Refer to manual shipped with the option for installation of Surface Coil Options.
- Refer to installation manual shipped with the option for the following:

OPERATOR WORKSPACE OPTIONS

- M1000MJ – Analog DASM
- M1000MK – Digital DASM
- M1033K – Indigo to Octane Host Computer Upgrade
- M1090PM – Bar Code Reader
- M1090TZ – VCR Video Kit for Octane Computer
- M3000JB – 18Gb Disk Drive for Octane

SOFTWARE OPTIONS

- M1033JA – iDrive Pro
- M1033JB – SmartPrep 2000 (Includes SmartStep)
- M1033JD – Probe2000
- M1033JF – ScanTools 2000
- M1033JH – SmartPrep 2000 Update
- M1033JJ – Probe2000 Update (for customers that have Probe software)
- M1033JK – LX FuncToll 2000
- M1033JL – LX Tools
- M1033JZ – FuncTool 2000 for AW
- M1033MG –SAGE 7
- M1090CR – Cardiac
- M1090LP – EchoPlus Diffusion Weighted Imaging Software
- M1090LR – Single Shot – FSE
- M1090LT – SmartPrep IA
- M1090MW – 8.x Software Non-Z with Multi-slice
- M1090MZ – Clariview
- M1090NN – Functool
- M1090PL – Connectpro
- M1090PM – Bar Code Reader
- M3033JA – Probe 3D Bain
- M3033JB – Functool CSI
- M3033JC – Ultra-Short TR for 3D Fast GRE
- M3033JD – BodyPak
- M3033JE – ACGD Plus

1–12 SIGNA LX SYSTEM OPTIONS (continued)**PERFORMANCE UPGRADE OPTIONS**

- M1090RT – 1.0T/1.5T Base to Performance Upgrade, Sys. Cab. Kit (single i860 128MB to quad i860 128MB)
- M1090RW – 1.0T/1.5T Acquisition Memory Kit (128MB for Medcam Board)
- M1092RM – Quad i860 ReFlexPro to PPC ReFlex100 Upgrade
- M1092RN – Single i860 ReFlexPro to PPC ReFlex100 Upgrade
- M1092RP – i860 ReFlexPro to PPC ReFlex100 Upgrade
- M3033CA – EXCITE 8 Channel (for Signa 1.5T MR/i with EXCITE Technology Systems ONLY)
- M3000RD – EXCITE ReFlex 100 Array Processor
- M3000RE – EXCITE ReFlex 200 Array Processor
- M3000RF – EXCITE ReFlex 400 Array Processor

HARDWARE & SOFTWARE OPTIONS

- M1033BM BrainWave (1.5T only)
- M1033MD Multi-Nuclear Spectroscopy (1.5T only)

HARDWARE OPTIONS

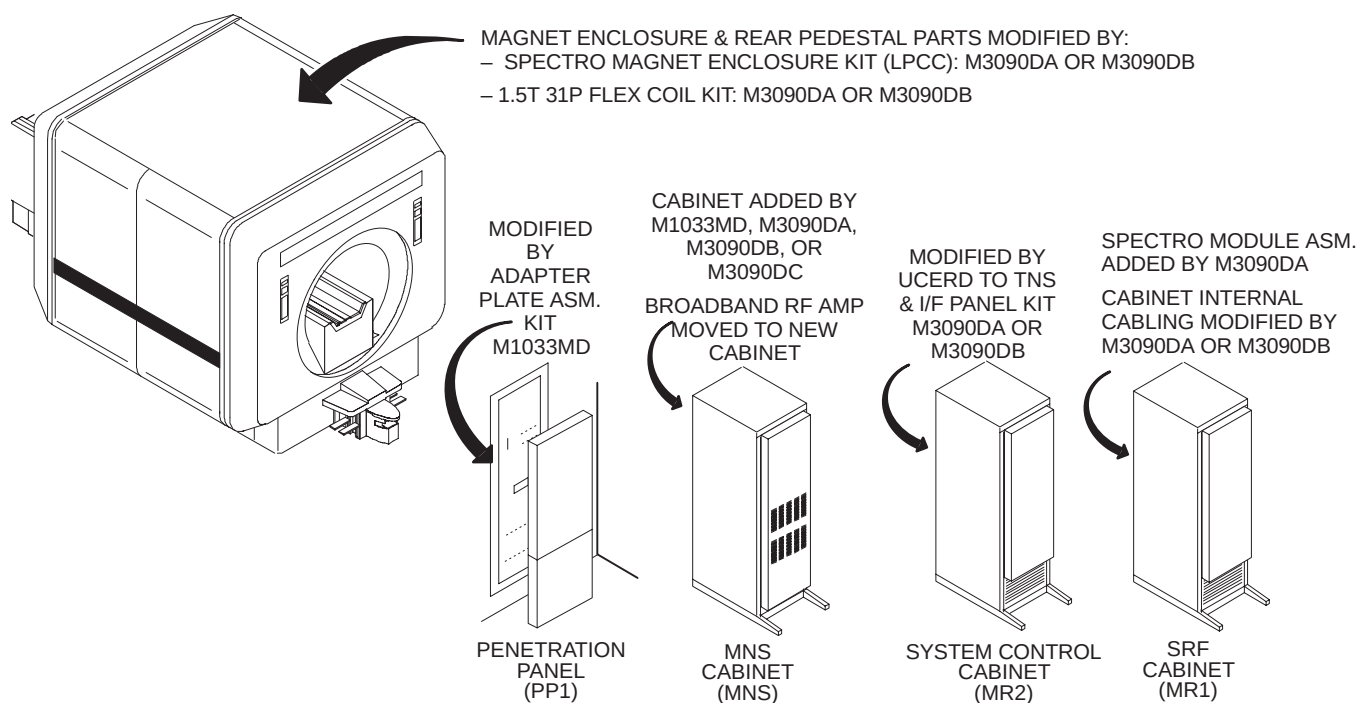
- M1000RM – Reflex50 PPC Recon Processor
- M1000RN – Reflex100 PPC Recon Processor
- M1090KM – Oxygen Monitor
- M1090WK – Low Profile Carriage Cover Upgrade for 1.5T Non-Phased Array Systems
- M1090WL – Low Profile Carriage Cover Upgrade for 1.5T Phased Array Systems
- M1090ZA – Fiber Optic Repeater
- M1090ZC – TNS Kit for 5X Upgrade to LX with Phased Array but without PROBE Spectroscopy
- M1090ZE – TNS Kit for 1.0T 5X Upgrade to LX with Phased Array
- M1090ZG – TNS Kit for 5X Upgrade to LX with Phased Array but with PROBE Spectroscopy
- M1890WK – Low Profile Carriage Cover Upgrade for 1.0T Non-Phased Array Systems
- M1890WL – Low Profile Carriage Cover Upgrade for 1.0T Phased Array Systems

PHASED ARRAY (MULTI-COIL) OPTIONS

- M1090PH – 1.5T 4 Channel Phased Array Upgrade
- M1090PW – 1.5T 4 Channel Phased Array Upgrade (UCERD)
- M1890PH – 1.0T 4 Channel Phased Array Upgrade
- M1890PW – 1.0T 4 Channel Phased Array Upgrade (UCERD)

1-12 SIGNA LX SYSTEM OPTIONS (continued)

- For 1.5T system ACGD/PDU Gradient Cabinet: The following Multi-Nuclear Spectroscopy (MNS) options are available (refer to Illustration 1-38):
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit For Signa LX Systems That Have RF/PDU Cabinet or ACGD Cabinet (M1033MD). Kit requires UCERD (can be ordered M1033MF), RF/PDU Cabinet or ACGD Cabinet.
 - For Signa Advantage 4.X or Signa Advantage 5.4 system upgrading to Signa LX: If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Advantage 4.X–5.4 (M3090DA).
 - For Signa Horizon 5.5 or later system upgrading to Signa LX: If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Horizon 5.5 And Later (M3090DB).
 - Signa LX with Spectroscopy pre-RF/PDU Cabinet to ACGD Upgrade (M3090DC).



1.5T SYSTEM WITH ACGD/PDU CABINET: MNS OPTIONS EXISTING HARDWARE MODIFIED & NEW EQUIPMENT ADDED
ILLUSTRATION 1-38

1-12 SIGNA LX SYSTEM OPTIONS (continued)

- For 1.5T system with SGD Cabinets: The following Multi-Nuclear Spectroscopy (MNS) options are available (refer to Illustration 1-38):
 - For Signa Advantage 4.X or Signa Advantage 5.4 system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Advantage 4.X–5.4 (M1033MA). Kit requires UCERD (can be ordered M1033MF), non-RF/PDU Cabinet, and non-ACGD Cabinet.
 - For Signa Horizon 5.5 or later system upgrading to Signa LX:
If system already has Multi-Nuclear Spectroscopy and have all the parts on site then can install 8.X MNS Upgrade For Signa Horizon 5.5 And Later (M1033MB). Kit requires UCERD (can be ordered M1033MF), non-RF/PDU Cabinet, and non-ACGD Cabinet.
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit for Signa LX Systems That Do Not Have RF/PDU Cabinet (M1033MC). Kit requires UCERD (can be ordered M1033MF), non-RF/PDU Cabinet, and non-ACGD Cabinet.
 - If system never had Multi-Nuclear Spectroscopy or did not retain all parts from the MNS Kit then need to install MNS Kit For Signa LX Systems That Have RF/PDU Cabinet or ACGD Cabinet (M1033MD). Kit requires UCERD (can be ordered M1033MF), RF/PDU Cabinet or ACGD Cabinet.

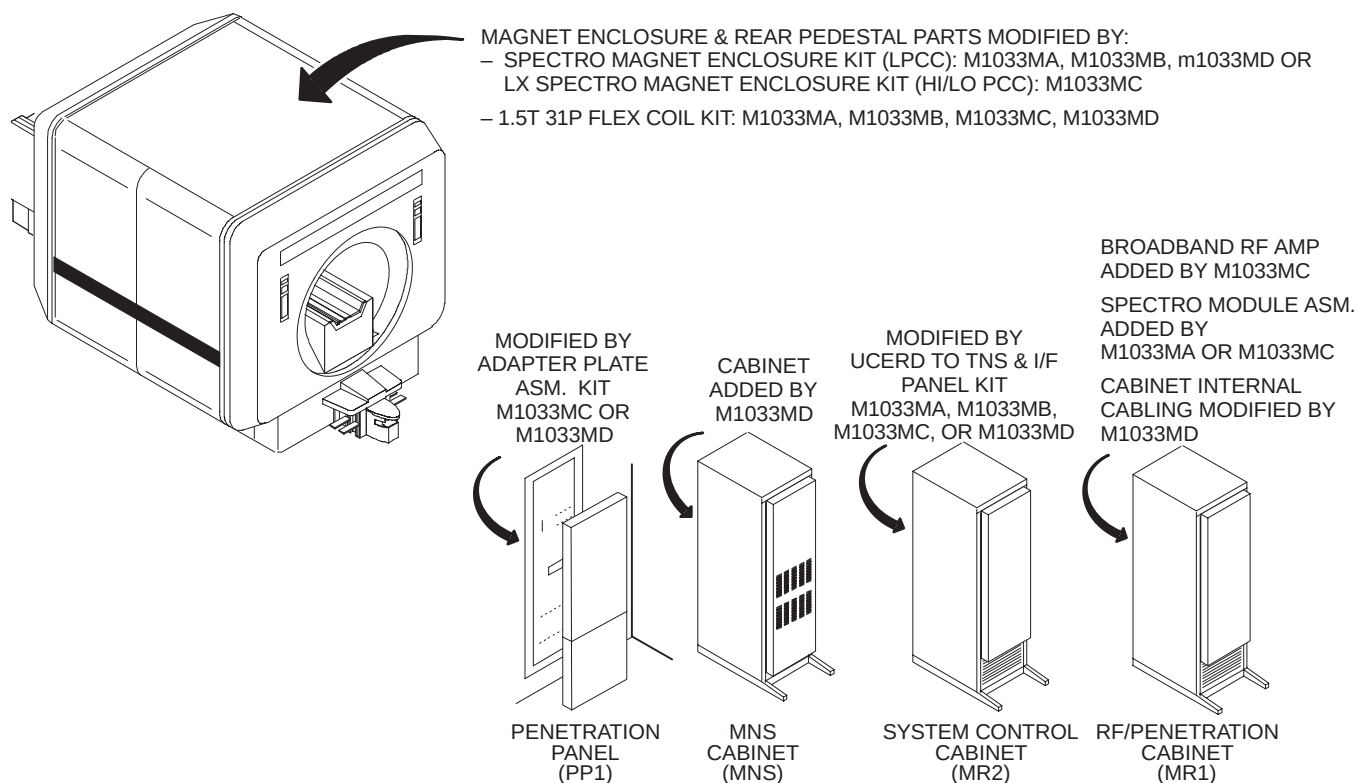
**1.5T SYSTEM WITH SGD CABINET: MNS OPTIONS EXISTING HARDWARE MODIFIED & NEW EQUIPMENT ADDED**

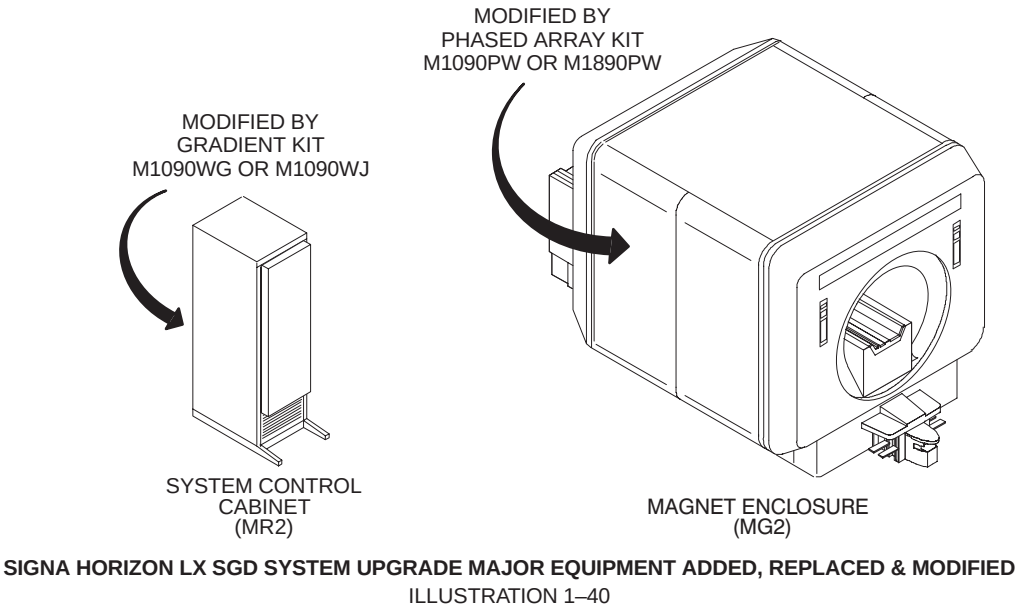
ILLUSTRATION 1-39

1-13 UPGRADES TO SIGNA LX WITH SGD SYSTEM

Table 1-1 lists the upgrade catalogs for Base Signa LX SGD or Signa LX SGD HiSpeed system within the Signa Horizon LX with SGD product line. Illustration 1-40 shows the cabinets added, replaced, and modified by the listed catalogs.

TABLE 1-1
SIGNA HORIZON LX SGD SYSTEM WITH SGD UPGRADE CATALOGS

SYSTEM HARDWARE CONFIG	UPGRADE KIT	HORIZON LX WITH SGD UPGRADES CATALOGS	
		HISPEED	ECHOSPEED
Base Horizon LX without Phased Array	Phased Array Kit	M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array M1890PW – 1.0T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array	M1090PW – 1.5T Phased Array Upgrade Kit (UCERD) for LX Systems without Phased Array
Horizon LX with SGD HiSpeed with Spectroscopy	Gradient Kit <i>Only required if increasing gradient strength</i>		M1090WG – HiSpeed SGD to EchoSpeed SGD Upgrade Kit for Horizon LX with SGD 1.5T Systems
Horizon LX with SGD without Spectroscopy	Gradient Kit <i>Only required if increasing gradient strength</i>		M1090WG – HiSpeed SGD to EchoSpeed SGD Upgrade Kit for Horizon LX with SGD 1.5T Systems M1090WJ – EchoSpeed Adder Kit for sites without Spectroscopy are upgrading to SGD EchoSpeed



1–14 TRANSPORTABLE/RELOCATABLE UPGRADE

Transportable and Relocatable MR system Horizon upgrade catalogs are listed in the previous upgrade sections. Note, the cable kits for Transportable/Relocatable include a complete set of system interconnect excluding magnet and option related cables.

In addition to the system, the van must also be upgraded to accommodate the Signa Horizon upgrades. The van/system upgrades can be preformed at the Van Manufacturer facility (preferred location) or at customer site. It is customer responsibility to arrange with the appropriate Van Manufacturer for any upgrade to be performed at site.

- Customer must contact Van Manufacturer to arrange for van/system upgrade at Van Manufacturer facility or customer site. Refer to Section 1–14, VAN MANUFACTURERS CONTACT INFORMATION.
- Customer is responsible for van upgrade costs with Van Manufacturer.
- Customer responsible for van transportation to and from Van Manufacturer facility for upgrades perform there.
- MR system check and calibration performance:
 - For van/system upgrades performed at Van Manufacturer facility, the installation will be checked by MR Manufacturing. Final calibrations will be performed by GE Field Service.
 - For van/system upgrades performed at customer site, the installation will be check and final calibrations will be performed by GE Field Service.

Note

Transportable or Relocatable with Oxford magnet can not be upgraded to Horizon configurations. The Oxford magnet can not be moved with the new body coil installed, serious magnet structural damage will result.

1–15 TRUE MOBILE UPGRADE

Note

Signa Horizon configurations are not available for 1.5T True Mobile systems due to structural limitations of the Oxford magnet in a mobile environment.

MR system Horizon upgrade catalogs for 1.0T True Mobiles are listed in the previous upgrade sections. Note, the cable kits for True Mobile include a complete set of system interconnect excluding magnet and option related cables.

In addition to the system, the van must also be upgraded to accommodate the Signa Horizon upgrades. The True Mobile van/system upgrade are preformed at the Van Manufacturer facility.

- System check and complete calibrations are performed by MR Manufacturing.

SECTION 2 – DE-INSTALLATION PROCESS

2-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Verify that System performance baseline measurements have been obtained. and center frequency should be recorded before system is shut down.
2. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
3. Press the FULL OFF button on the PDU front control panel.



ELECTRIC SHOCK HAZARD!! HIGH VOLTAGES ARE PRESENT ON INTERCONNECTING CABLES USED WITH SPECIAL FIELD INSTALLED ION-PUMPS USED WITH OXFORD MAGNETS. THE ION PUMP POWER SUPPLY IS USUALLY CONNECTED TO A POWER SOURCE THAT IS NOT CONTROLLED BY THE PDU AND HIGH VOLTAGE WILL BE PRESENT EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK, DISCONNECT POWER TO THE ION PUMP POWER SUPPLY, LOCK OUT, AND TAG BEFORE PERFORMING PROCEDURES IN THE FOLLOWING SECTIONS. ALSO CHECK FOR OTHER INTERCONNECTING EQUIPMENT WHICH WAS FIELD INSTALLED AND DISCONNECT POWER, LOCK OUT, AND TAG.

4. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
5. Locate and shut off main disconnect supplying power to the PDU.
6. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
7. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

2-2 RETURN MATERIAL INFORMATION FORMS

Fill in the information on “Attention: GEMS Recycling Center” form for Genesis Console (page 2-5), Genesis Computer (page 2-9), and Cabinets (page 2-13). Attach the appropriate sheet to each item.

2-3 REMOVING FULL-SIZE PDU

Sites that are upgrading to ACGD require that the full-size PDU or Compact PDU be de-installed.

If part of existing system, the shim and magnet power supply cables (Runs 011 and 020) need to be re-used for the new ACGD/PDU cabinet. **DO NOT DISCARD RUNS 011 AND 020.** Refer to Illustration 2-1 for instructions on removing and reterminating Runs 011 and 020 for use with new ACGD/PDU cabinet. Refer to Section 4 for other cable information.

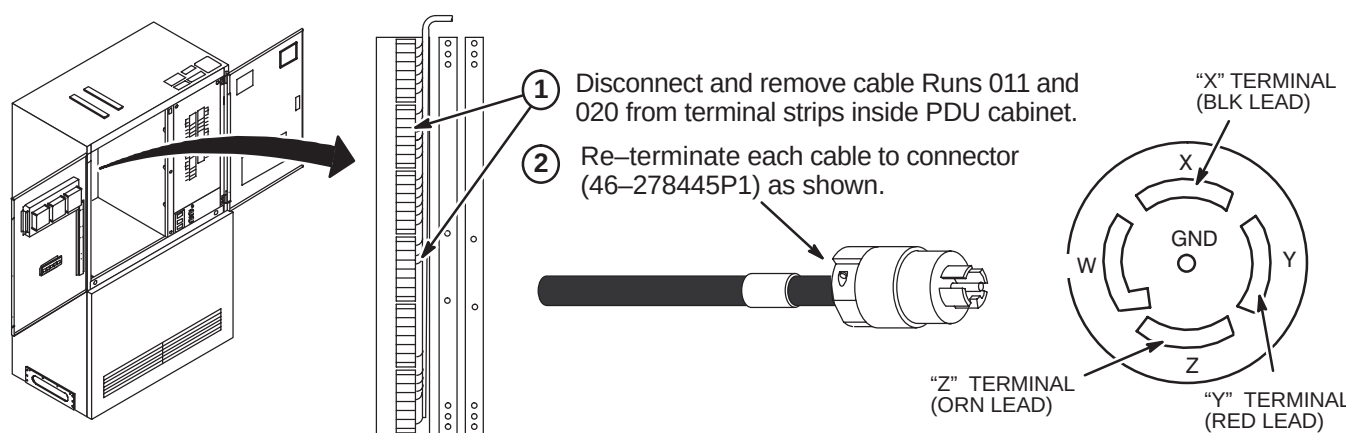


ILLUSTRATION 2-1
FULL-SIZE PDU WITH SHIPPING DOLLIES

The Compact PDU has built-in casters for ease of movement. The full-size PDU does not have casters and will have to be moved with two shipping dollies placed on each side of the PDU and clamped in place using straps with turnbuckles as shown in illustration below. If dollies are not provided, the Full-size PDU will have to be removed by riggers or other locally supplied equipment moving personnel. See Illustration 2-2.

CAUTION

PDU weight is about 1000 pounds (455 kilograms). Plywood or steel plates may be used to distribute weight and protect floor surface from overload damage.

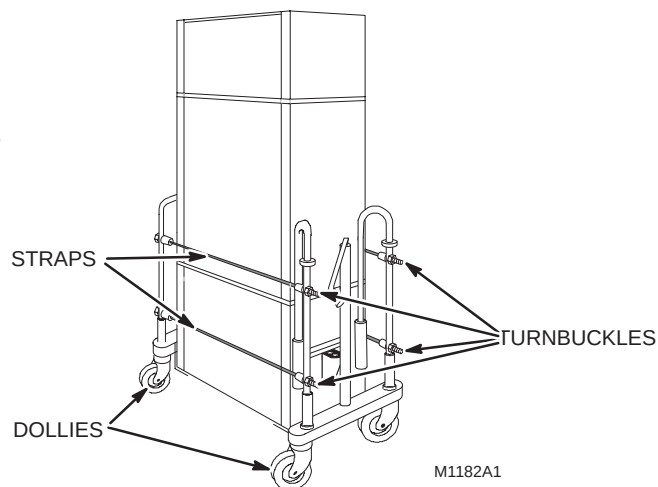


ILLUSTRATION 2-2
FULL-SIZE PDU WITH SHIPPING DOLLIES

2-4 5.x OPERATOR CONSOLE RETURN PROCEDURE

Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of 5.x Operator Console.

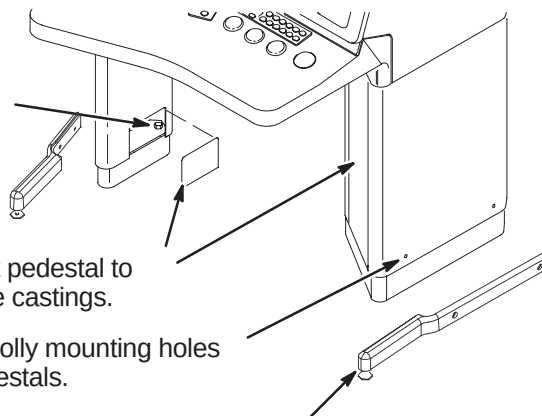
□ A1 – PREPARING OPERATOR CONSOLE REMOVAL

- ④ Loosen screws holding Console to pedestal risers.

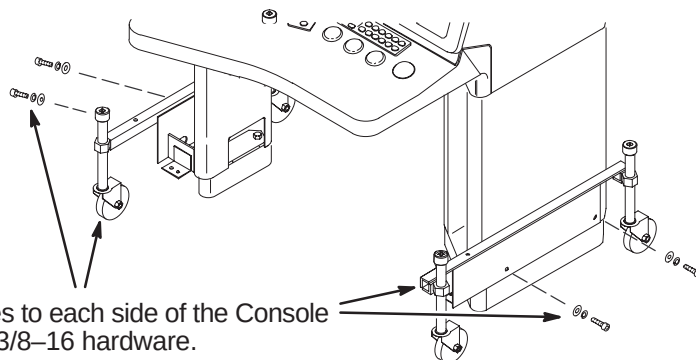
- ③ Remove covers on sides of support pedestal to access mounting holes on the Base castings.

- ② Remove Plug Buttons from dolly mounting holes on each side of Console pedestals.

- ① If installed, unfasten and remove legs from console.

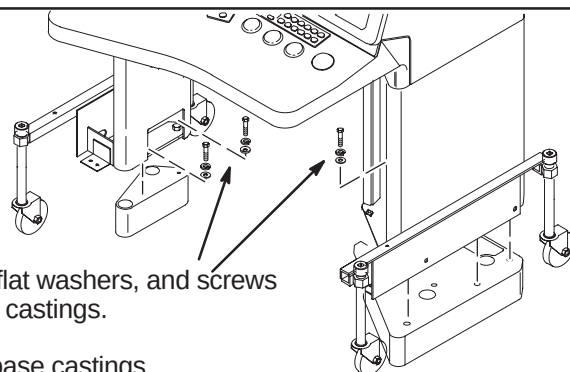
**□ A2 – ATTACHING DOLLY TO OPERATOR CONSOLE**

- ① Attach shipping dollies to each side of the Console legs using furnished 3/8-16 hardware.

**□ A3 – REMOVING OPERATOR CONSOLE**

- ① Remove lock washers, flat washers, and screws connecting legs to base castings.

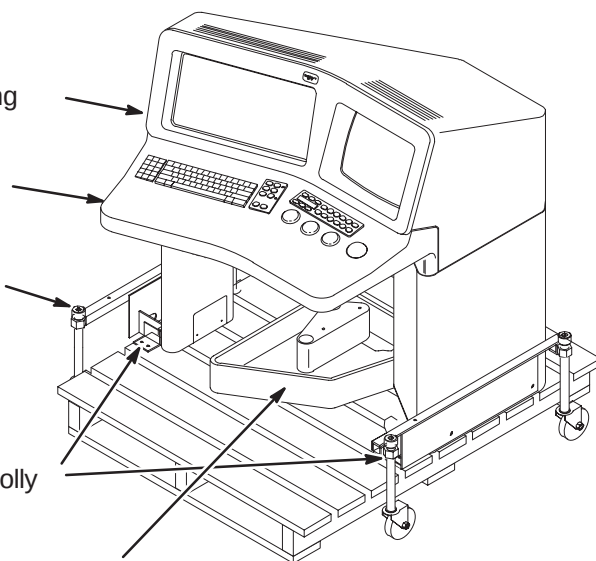
- ② Raise console up over base castings.



2-4 5.x OPERATOR CONSOLE RETURN PROCEDURE (Cont'd)

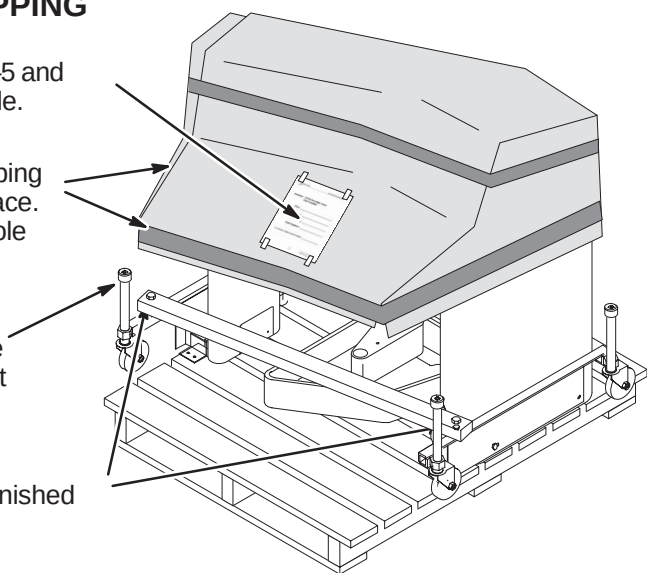
□ A4 – FASTENING OPERATOR CONSOLE TO SHIPPING PALLET

- ① Carefully roll Operator Console over an available shipping pallet.
- ② Position Console so that Keyboard Housing does not extend beyond shipping pallet.
- ③ Use a 1/2 inch socket wrench to turn the jack screws and lower the Operator Console onto pallet.
- ④ Locate (4) four provided 5/16 x 1-1/2 long lag bolts.
- ⑤ Install two of the lag bolts thru each shipping dolly bracket to fasten to shipping pallet.
- ⑥ Nest together Riser castings on pallet as shown.



□ A5 – FINAL PREPERATION BEFORE SHIPPING

- ④ Fill out *GEMS Recycling Form* on page 2-5 and attach to top of cardboard covering console.
- ③ Using any available cardboard (i.e. empty shipping boxes), cover Operator Console and tape in place. This is to prevent damage when shipping console back to manufacturer.
- ② Make sure that all four wheel assemblies are completely raised. If not, use 1/2 inch socket wrench to turn jack screws.
- ① Install provided dolly stiffener with furnished 1/4-20 x 1-1/2 long screws and nuts.



Return Console to GE Medical Systems with other system equipment that has been replaced during this upgrade.

**Attention: GEMS Recycling Center
Christine or Maxine**

FDO: _____

Site Name: _____

Removal Date: _____

Field Engineer: _____

Genesis Console Serial Number: _____

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2-5 5.x OPERATOR CONSOLE RETURN PROCEDURE

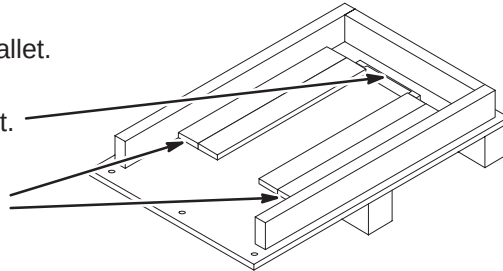
Confirm that all cables are disconnected from Computer. Perform the following procedure for return of 5.x Computer reusing the shipping pallet that the new Operator Workspace Cabinet arrived in.

□ A1 – PREPARING SHIPPING PALLET

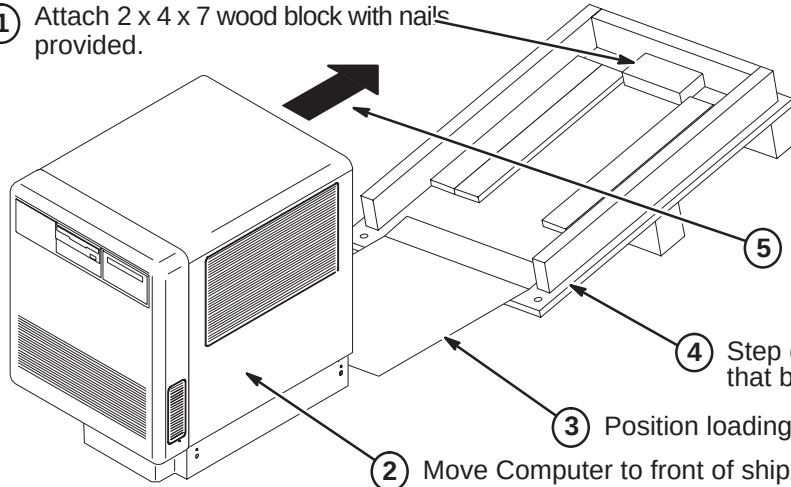
① Locate the Operator Workspace Cabinet shipping pallet.

② Remove existing bracket.

③ Re-position the protective foam on each side from a vertical to a horizontal position between the side of the pallet and the existing horizontal foam.

**□ A2 – LOADING COMPUTER ONTO SHIPPING PALLET**

① Attach 2 x 4 x 7 wood block with nails provided.

**Note:**

Pallet will return to horizontal position as Computer is rolled toward back of pallet.

⑤ With another person, carefully roll Computer onto shipping pallet so it will rest on the horizontal foam pieces.

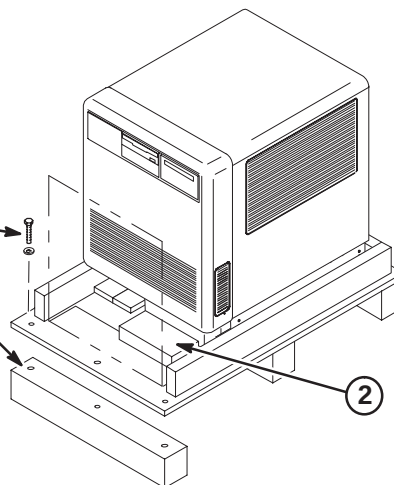
④ Step on front corner of shipping pallet so that bottom board of pallet touches floor.

③ Position loading board at front of pallet.

② Move Computer to front of shipping pallet.

□ A3 – REASSEMBLE SHIPPING PALLET

① Re-attach wood block to pallet using the three previously removed fasteners.



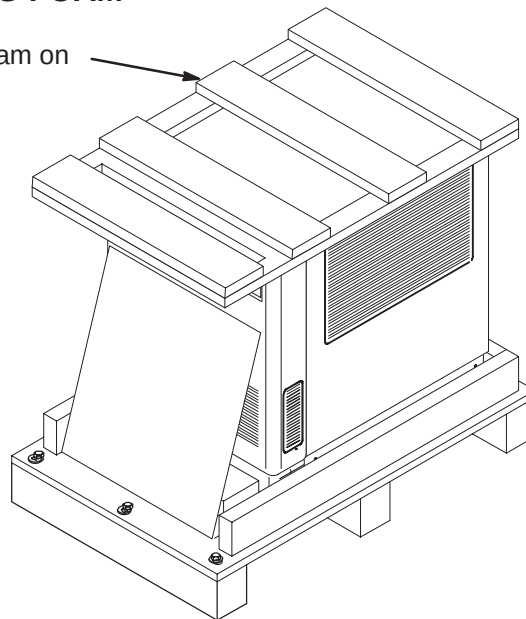
② Place 2 x 4 wood block against front right side of Computer and attach to pallet with nails provided.

2-5 5.x COMPUTER RETURN PROCEDURE (Cont'd)

□ A4 – INSTALLING COMPUTER TOP SHIPPING FOAM

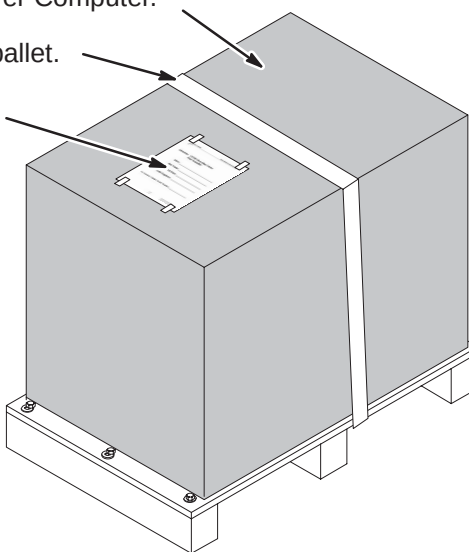
- ① Install the Operator Workspace Cabinet shipping foam on top of Computer.

Note: Shipping foam insert will not fit as well as it did with the new Operator Workspace Cabinet.



□ A5 – FINAL PREPERATION BEFORE SHIPPING

- ① Place Operator Workspace shipping box over Computer.
- ② Secure to shipping pallet.
- ③ Fill out *GEMS Recycling Form* on page 2-9 and attach to top of shipping box.



Return Computer to GE Medical Systems with other system equipment that has been replaced during this upgrade.

**Attention: GEMS Recycling Center
Christine or Maxine**

FDO: _____

Site Name: _____

Removal Date: _____

Field Engineer: _____

Genesis Computer Serial Number: _____

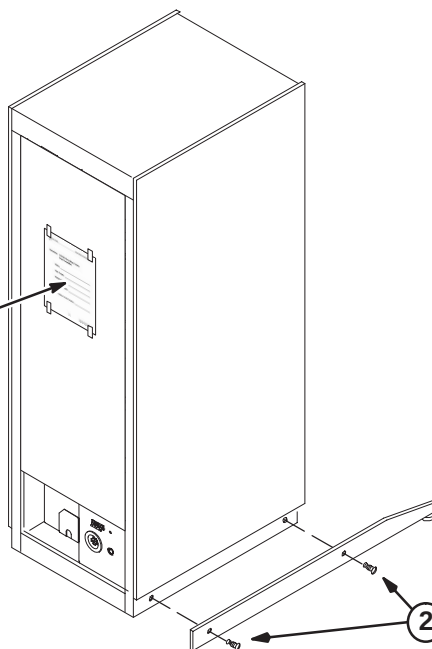
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2-6 CABINET RETURN PROCEDURE

Confirm that all cables are disconnected from applicable Cabinet. The System Cabinet is shown below. The following procedure should be used for return of any cabinet with anti-tip bars.

□ A1 – REMOVING ANTI-TIP BARS

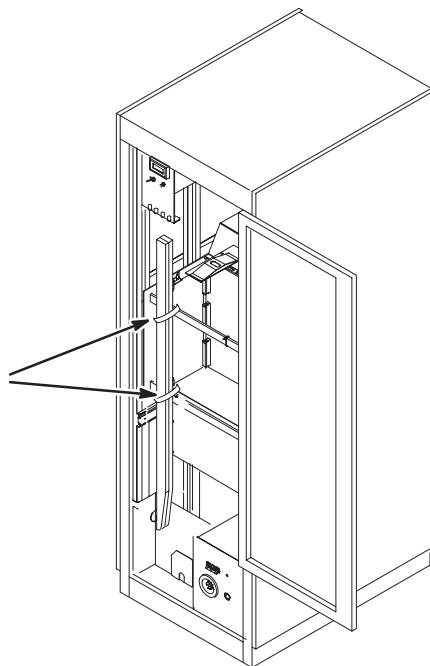
- ① Fill out *GEMS Recycling Form* on page 2-13 and attach to outside of cabinet door.



- ② Remove anti-tip bars from cabinet.

□ A2 – ATTACHING ANTI-TIP BARS AND FORM FOR RETURN SHIPMENT

- ① Fasten anti-tip bars to inside of cabinet for shipment back to GE Medical Systems.



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**Attention: GEMS Recycling Center
Christine or Maxine**

FDO: _____

Site Name: _____

Removal Date: _____

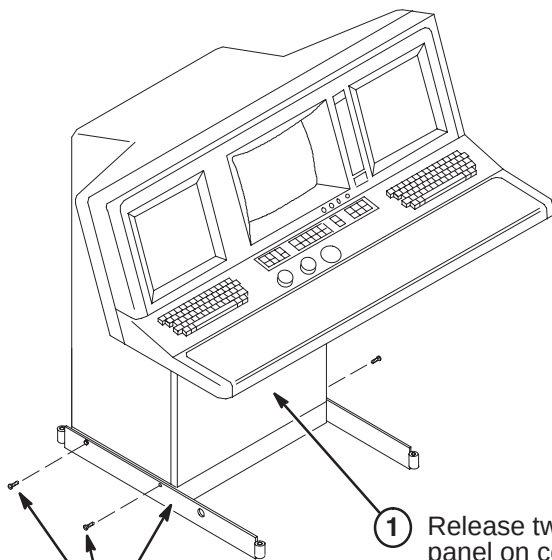
Field Engineer: _____

5.x Systems Cabinet Serial Number: _____

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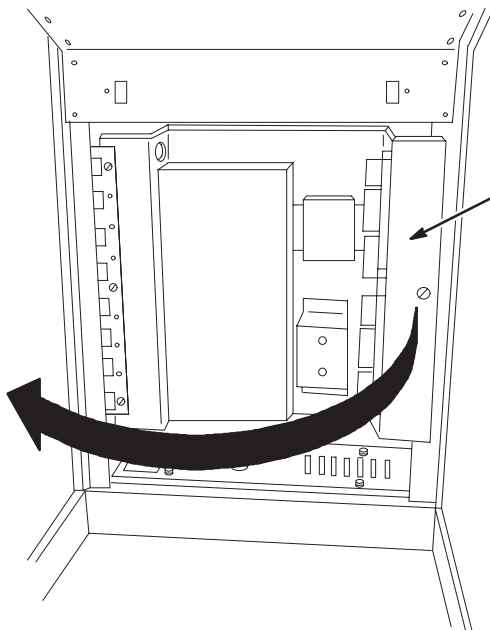
2-7 4.x OPERATOR CONSOLE RETURN PROCEDURE

Confirm that all cables are disconnected from Operator Console. Perform the following procedure for return of 3.x/4.x Operator Console.

□ A1 – INSTALL OPERATOR CONSOLE SHIPPING LEGS

- ① Fasten to each side of Console:
- (1) 46-225881G1 Anti-Tip Shipping Leg
 - (2) 46-208561P19 1/4-20x 3/4 Long Hex Head Screw

① Release two captive fasteners near top of lower front panel on console and remove panel.

A2 – OPERATOR CONSOLE REMOVAL

- ① Unlatch and swing open panel assembly.
- ② Remove hardware that attaches console to floor.
- ③ Close and latch panel assembly.
- ④ Re-attach Front Panel.

SECTION 3 – UPGRADE INSTALLATION PROCESS

3-1 UPGRADE PROCEDURE

The following upgrade flow charts contain information for upgrading to 1.0T and 1.5T Signa Horizon LX (9.x) or Signa 1.5T LX EXCITE (11.x). Follow the path of the flow chart that is appropriate for the upgrade you are installing.

**WARNING!**

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT, CRIMP TOOL, BLOWER BOX, AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

3-1 -1 Upgrade Flow

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

The applicable flow chart should be followed for an orderly and efficient installation of the Signa LX system. Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

Note

The flow chart references the Tab number and the PIP (Pictorial Installation Procedure) Direction number for the procedure. PIP Directions are similar to cable maps in that they are to be removed from this binder and used at the location where hardware installation is being performed. The PIP Direction may be returned to the binder for future reference upon completion of the installation. Other Directions and installation drawings delivered with kits are also referenced by the flowchart.

This installation flowchart has been developed assuming that all Signa LX equipment has been delivered together. Ensure that every part required is available before starting.

3-1 -2 Upgrade Flow Charts

The following flow charts are included in this section:

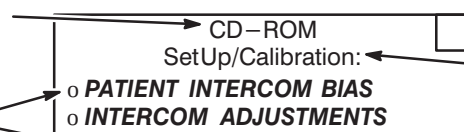
<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
3-2	SIGNA LX UPGRADE FLOWCHART EXPLANATION	3-3
	SYSTEM UPGRADE TO LX with EXCITE TECHNOLOGY FLOWCHARTS	
	Signa Horizon LX to Signa 1.5T LX EXCITE Upgrade Process	3-5
	Signa Horizon 5.x to Signa 1.5T LX EXCITE Upgrade Process	3-7
	Signa Advantage 5.x or 4.x to Signa 1.5T LX EXCITE Upgrade Process	3-15
	SYSTEM UPGRADE TO LX with ACGD FLOWCHARTS	
	Signa Horizon LX to Signa LX with with ACGD Upgrade Process	3-25
	Signa Horizon 5.x to Signa LX with with ACGD Upgrade Process	3-31
	Signa Advantage 5.x to Signa LX with ACGD Upgrade Process	3-37
	Signa Advantage 4.x to Signa LX with ACGD Upgrade Process	3-39

3-2 SIGNA LX UPGRADE FLOWCHART EXPLANATION

Shown below are three examples of the flow chart Installation Procedures and an explanation of the areas within the procedures.

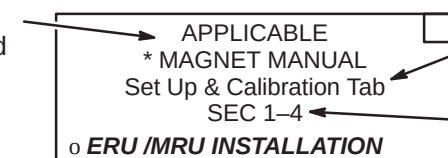
Use the CD-ROM that applies to the System being installed to locate the installation procedure listed below

Installation Procedure to be performed



Main menu selection

Use the correct Magnet Manual that applies to the System being installed to locate the installation procedure listed below

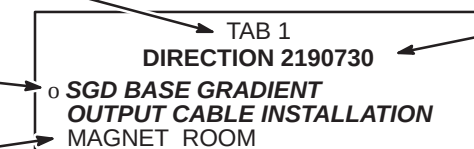


Tab within the manual where Installation Procedure is located

Section within Tab where Installation Procedure is located

Tab within this manual (Direction 2143793) where the Direction listed below is located

Installation Procedure to be performed



Direction within the Tab where the Installation Procedure listed below is located

Specific steps or room of the above Installation Procedure to be performed at this time

LEGEND FOR FOLLOWING INSTALLATION FLOWCHART



INDICATES MILESTONE COMPLETION GOAL IN DAYS



INDICATES AND OPTIONAL PATH DEPENDING ON SYSTEM CONFIGURATION



INDICATES GE PROPRIETARY PROCEDURE



INDICATES AN ADDITIONAL PROPRIETARY TEST OR ALTERNATIVE PROPRIETARY CALIBRATION PROCEDURE AVAILABLE FOR GE USE OR TO CUSTOMERS WITH AN ADVANCED SERVICE PACKAGE LIMITED LICENSE.



AS PART OF RE-CHECK AND AFTER QUICK SHIM (OR LV SHIM) IT MAY BE NECESSARY TO RE-PARK THE MAGNET.



MILESTONE MARKERS M5 – M8 ASSUME INSTALLATION OF AN S-V MAGNET REQUIRING PASSIVE SHIMMING. IF PASSIVE STEEL IS ADDED FOR X²-Y² AND XY, EDDY CURRENT COMP WILL NEED TO BE RE-CHECKED. IF SITE DOES NOT REQUIRE PASSIVE SHIMMING, ONE CYCLE DAY IS SAVED (ELIMINATE M5 AND CONVERT M6, M7, AND M8 TO M5, M6, AND M7).

NOTE 1: STANDARD CORRELATED NOISE CHECK NOT REQUIRED IF PROPRIETARY SPT FULL TEST MODE PROCEDURE PERFORMED EARLIER DURING INSTALLATION (HOWEVER, MULTICOIL AND FAST RECEIVER OPTION CORRELATED NOISE CHECKS IN PROCEDURE STILL REQUIRED).

NOTE 2: IF TROUBLESHOOTING NECESSARY.

NOTE 3: USE FINAL MAGNETOMETER READING FROM RAMPING OF MAGNET.

NOTE 4: A "saveINFO" SHOULD BE DONE BEFORE SPT IS RUN. IF YOU HAVE THE EPI OPTION, IT IS NECESSARY TO "saveINFO" AGAIN. IN THIS WAY, ANY DATA COLLECTED FOR CALIBRATION WILL NOT BE LOST IF A PROBLEM OCCURS.

NOTE 5: NOT USED.

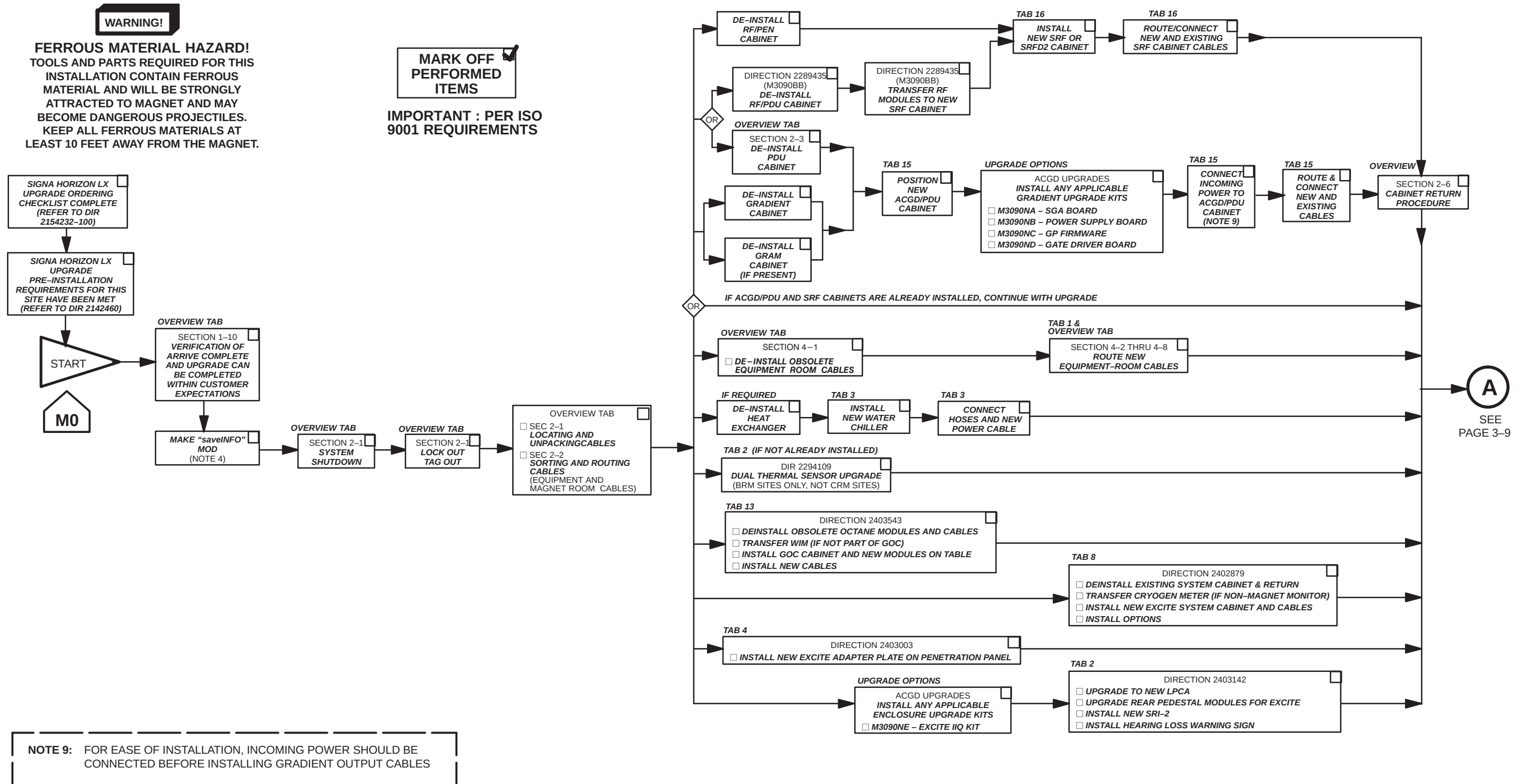
NOTE 6: VALUES OF EDDY CURRENT COMPENSATION ARE INDEPENDENT OF MAGNET SHIM.

NOTE 7: OVERNIGHT DIAGS MAY BE RUN AT END OF ANY DAY TO DETECT POSSIBLE ERRORS.

NOTE 8: RELEASE 5.x (5.5/5.6/5.7) SYSTEM CONFIG AND CALIBRATION DATA CAN NOT BE RESTORED ON AN LX SYSTEM USING "RESTORE INFO". THE CONFIG VALUES MUST BE MANUALLY RE-ENTERED ON THE 8.x (LX) SYSTEM. CALIBRATIONS FOR GRAFIDY (DUE TO DIFFERENT CAL FILE FORMAT) AND EPI (DUE TO NEW RECEIVERS) MUST BE RE-RUN ON LX.

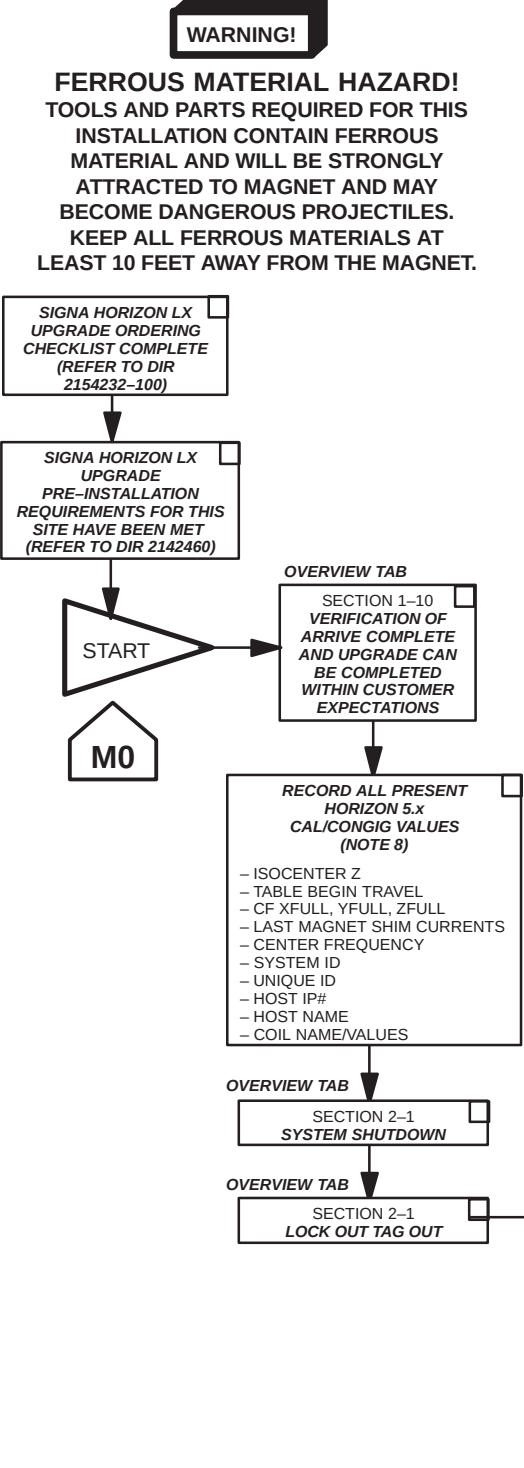
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HORIZON LX TO SIGMA LX with EXCITE TECHNOLOGY UPGRADE INSTALLATION FLOW CHART (FIXED-SITES ONLY)



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HORIZON 5.x TO SIGNA LX with EXCITE TECHNOLOGY UPGRADE
INSTALLATION FLOW CHART
(FIXED-SITES ONLY)

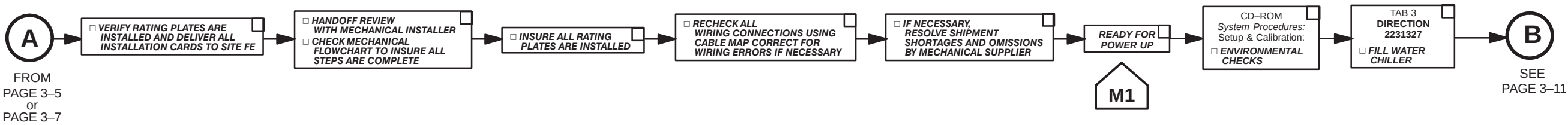


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HORIZON LX or 5.x TO SIGNA LX with EXCITE TECHNOLOGY (MGD) UPGRADE
INSTALLATION FLOW CHART
(FIXED-SITES ONLY)

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



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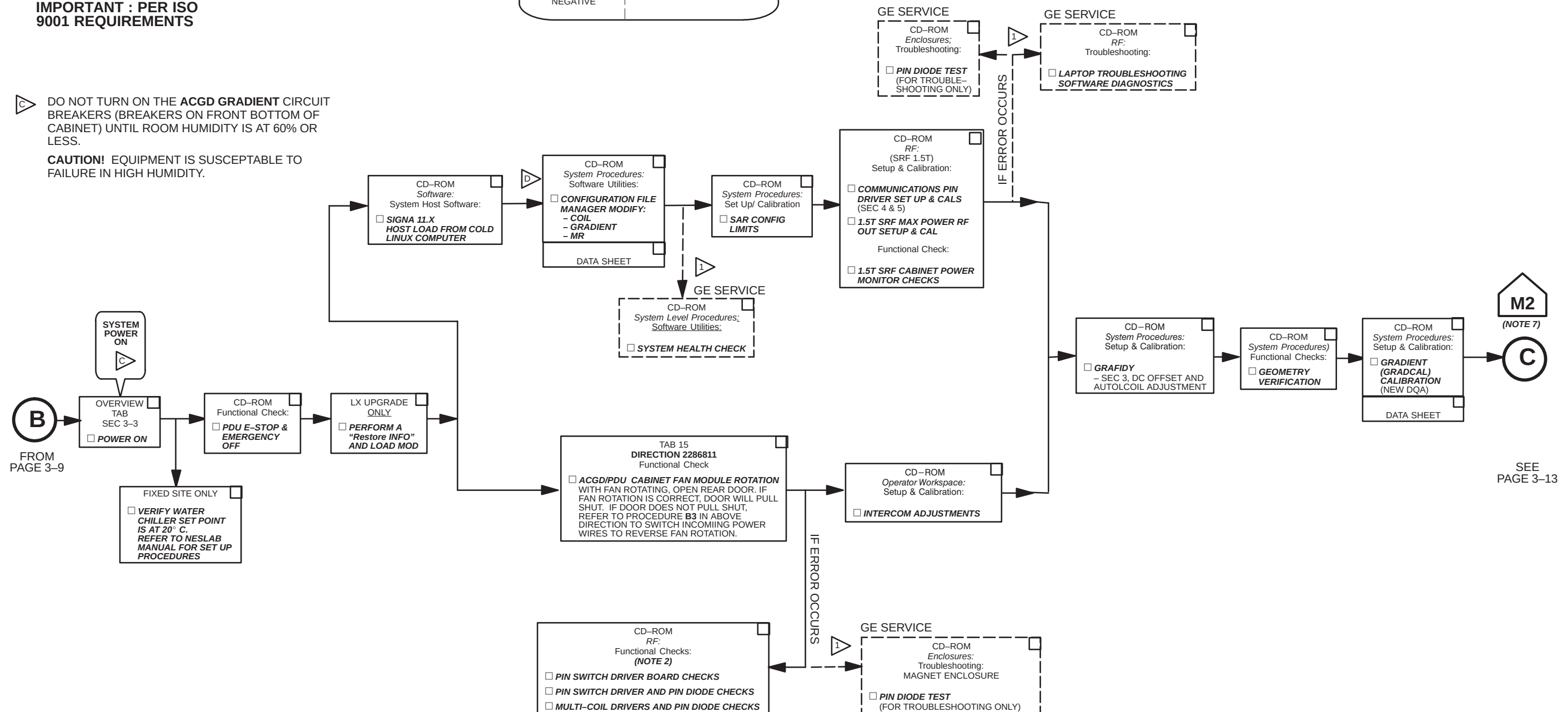
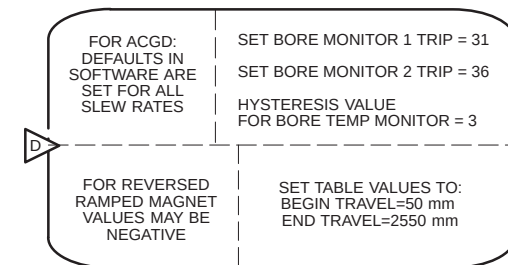
HORIZON LX or 5.x TO SIGNA LX with EXCITE TECHNOLOGY UPGRADE
SYSTEM CALIBRATION
(FIXED-SITES ONLY)

**MARK OFF
PERFORMED
ITEMS**

IMPORTANT : PER ISO 9001 REQUIREMENTS

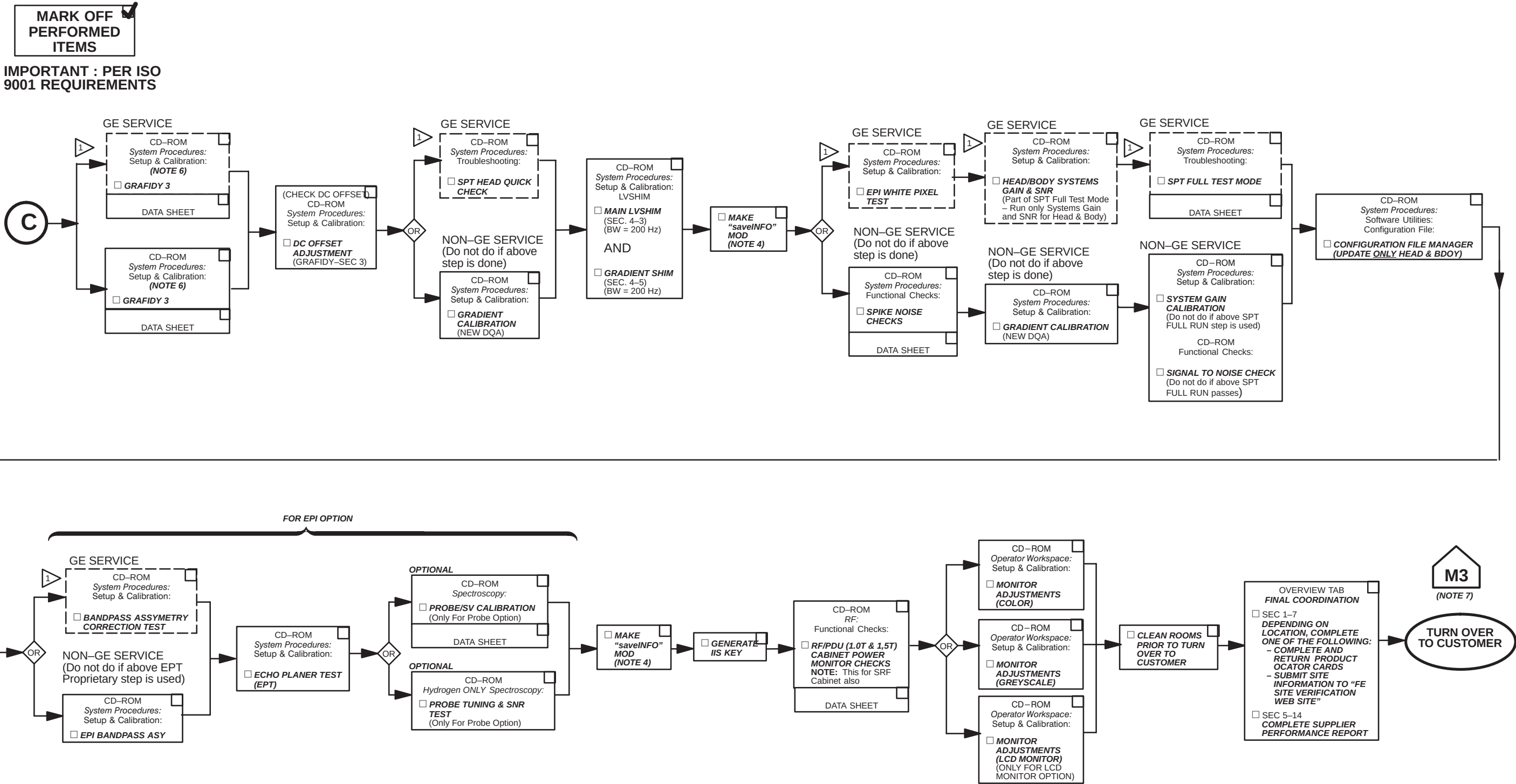
 DO NOT TURN ON THE **ACGD GRADIENT** CIRCUIT BREAKERS (BREAKERS ON FRONT BOTTOM OF CABINET) UNTIL ROOM HUMIDITY IS AT 60% OR LESS.

CAUTION! EQUIPMENT IS SUSCEPTABLE TO FAILURE IN HIGH HUMIDITY.



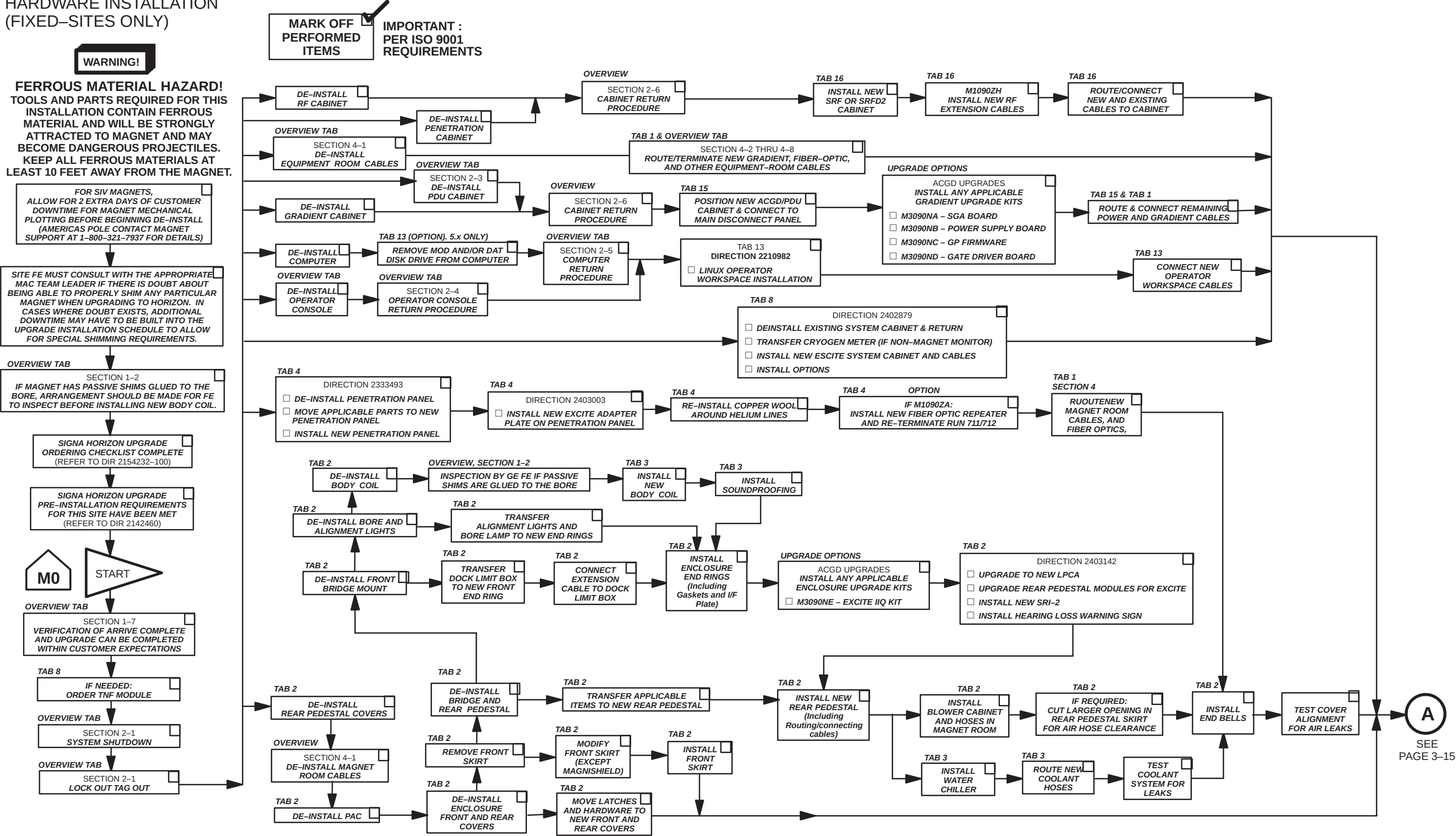
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HORIZON LX or 5.x TO SIGNA LX with EXCITE TECHNOLOGY UPGRADE
SYSTEM CALIBRATION
(FIXED-SITES ONLY)



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ADVANTAGE 5.x or 4.x TO SIGNA LX with EXCITE TECHNOLOGY UPGRADE
HARDWARE INSTALLATION
(FIXED-SITES ONLY)

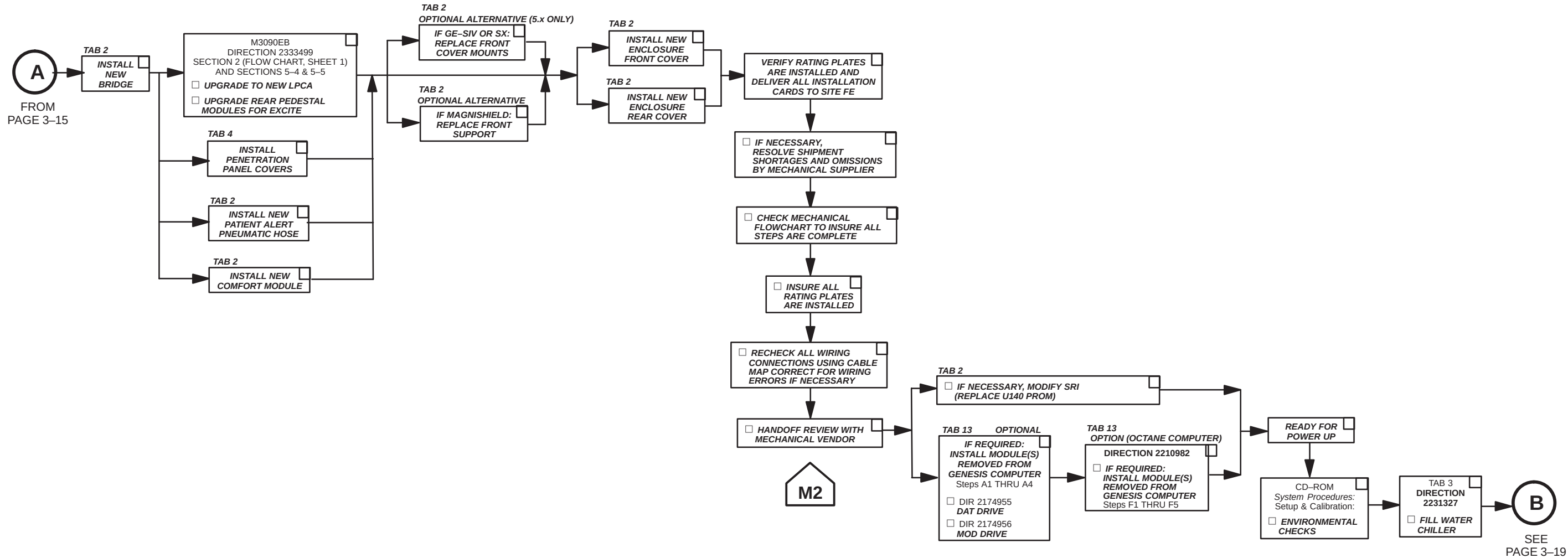


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ADVANTAGE 5.x or 4.x TO SIGNA LX with EXCITE TECHNOLOGY UPGRADE
HARDWARE INSTALLATION
(FIXED-SITES ONLY)

MARK OFF
PERFORMED
ITEMS

IMPORTANT :
PER ISO 9001
REQUIREMENTS

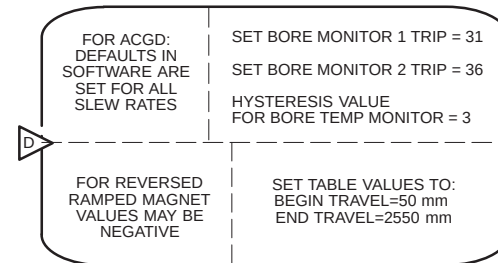


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ADVANTAGE 5.x or 4.x TO SIGNA LX with EXCITE TECHNOLOGY UPGRADE
HARDWARE INSTALLATION
(FIXED-SITES ONLY)

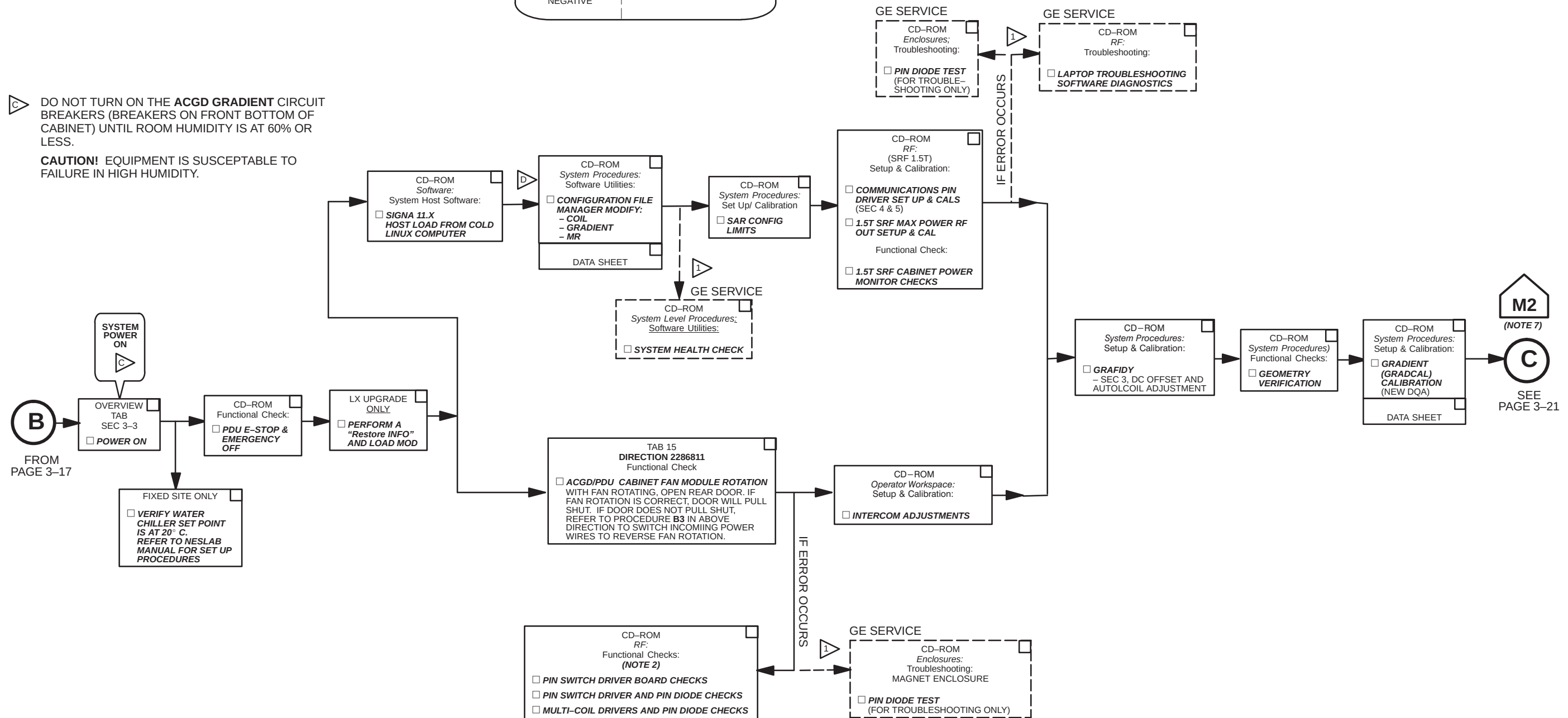
**MARK OFF
PERFORMED
ITEMS**

**IMPORTANT : PER ISO
9001 REQUIREMENTS**



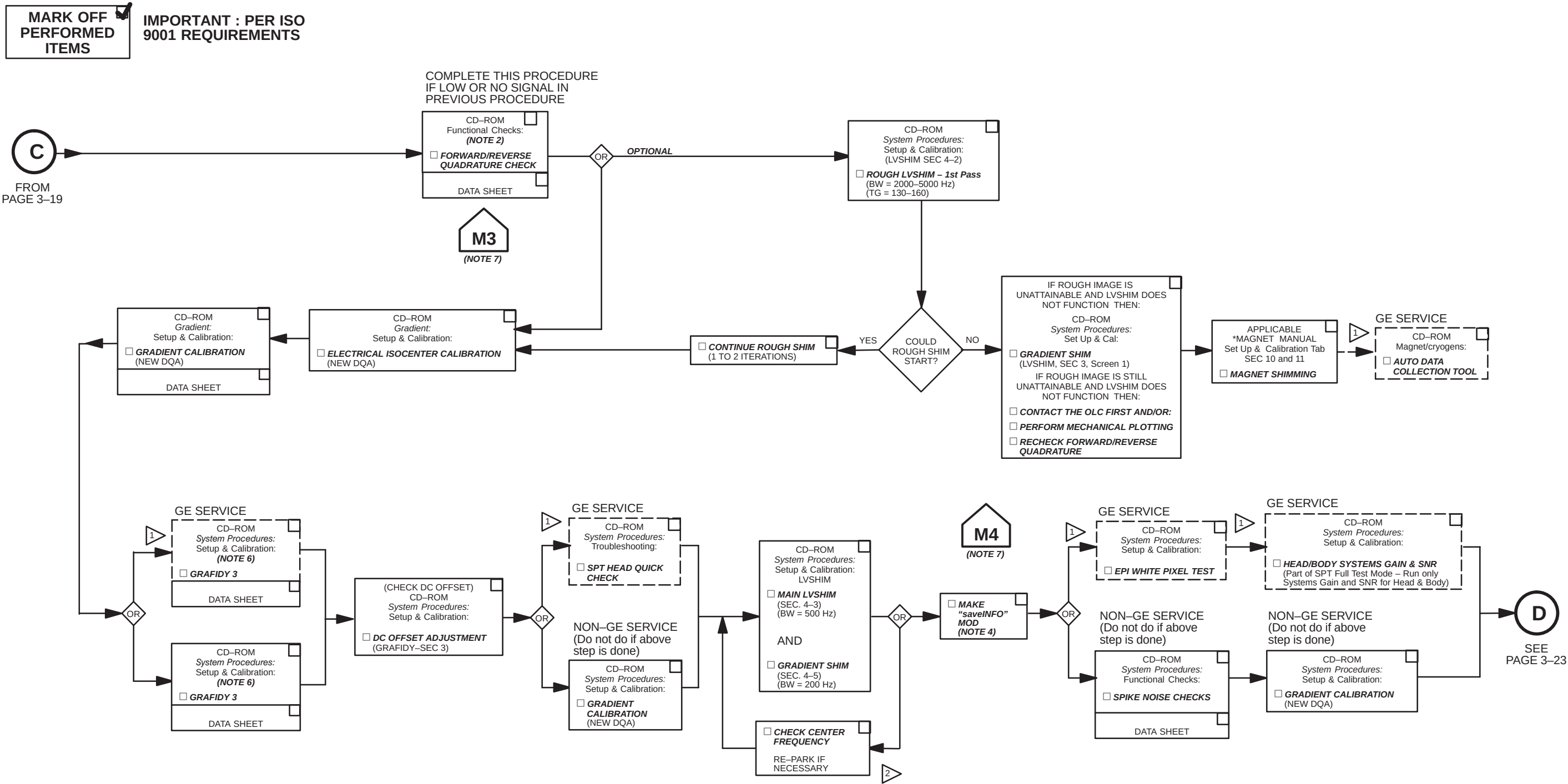
 DO NOT TURN ON THE **ACGD GRADIENT** CIRCUIT BREAKERS (BREAKERS ON FRONT BOTTOM OF CABINET) UNTIL ROOM HUMIDITY IS AT 60% OR LESS.

CAUTION! EQUIPMENT IS SUSCEPTABLE TO FAILURE IN HIGH HUMIDITY.



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ADVANTAGE 5.x or 4.x TO SIGNA LX with EXCITE TECHNOLOGY UPGRADE
HARDWARE INSTALLATION
(FIXED-SITES ONLY)

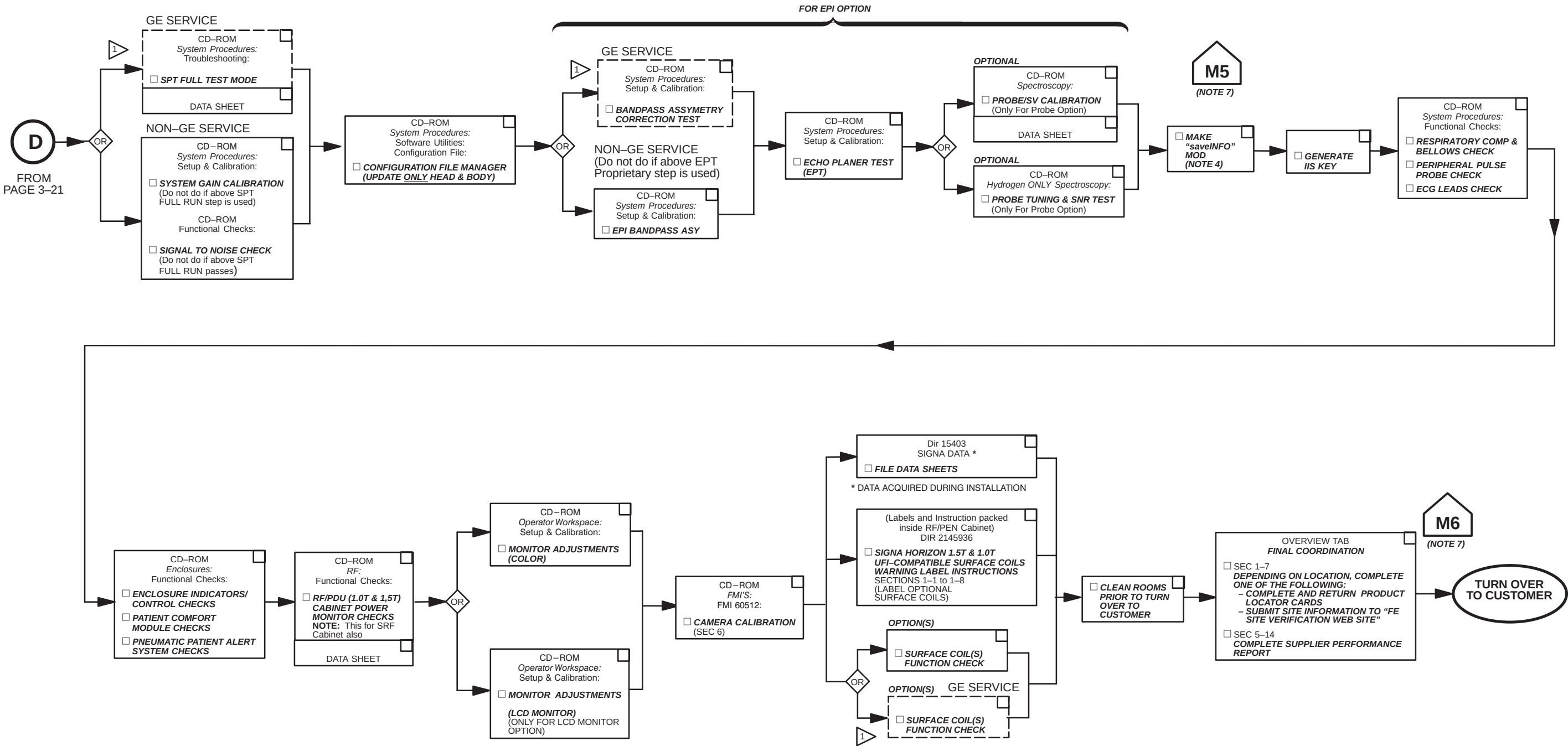


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ADVANTAGE 5.x or 4.x TO SIGNA LX with EXCITE TECHNOLOGY UPGRADE
HARDWARE INSTALLATION
(FIXED-SITES ONLY)

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



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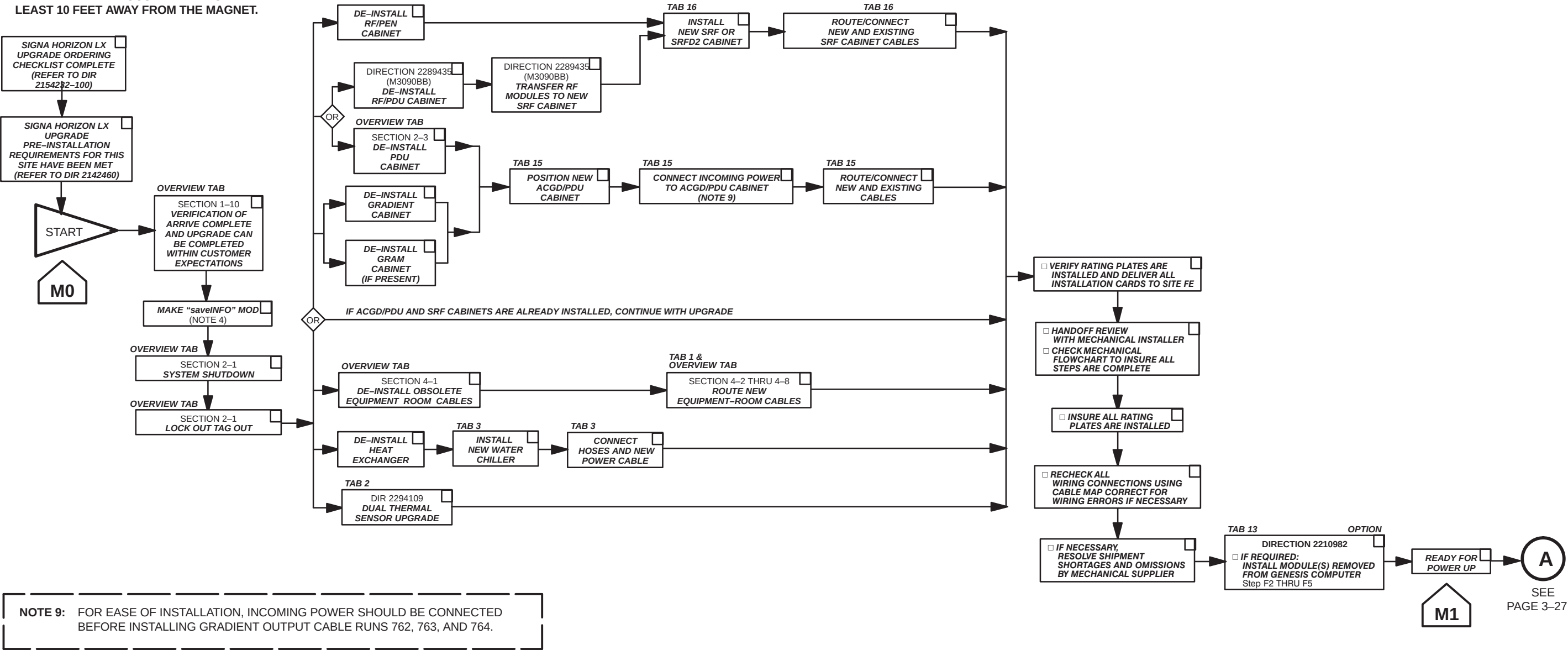
HORIZON LX TO SIGNA LX with ACGD UPGRADE
INSTALLATION FLOW CHART
(FIXED-SITES ONLY)

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

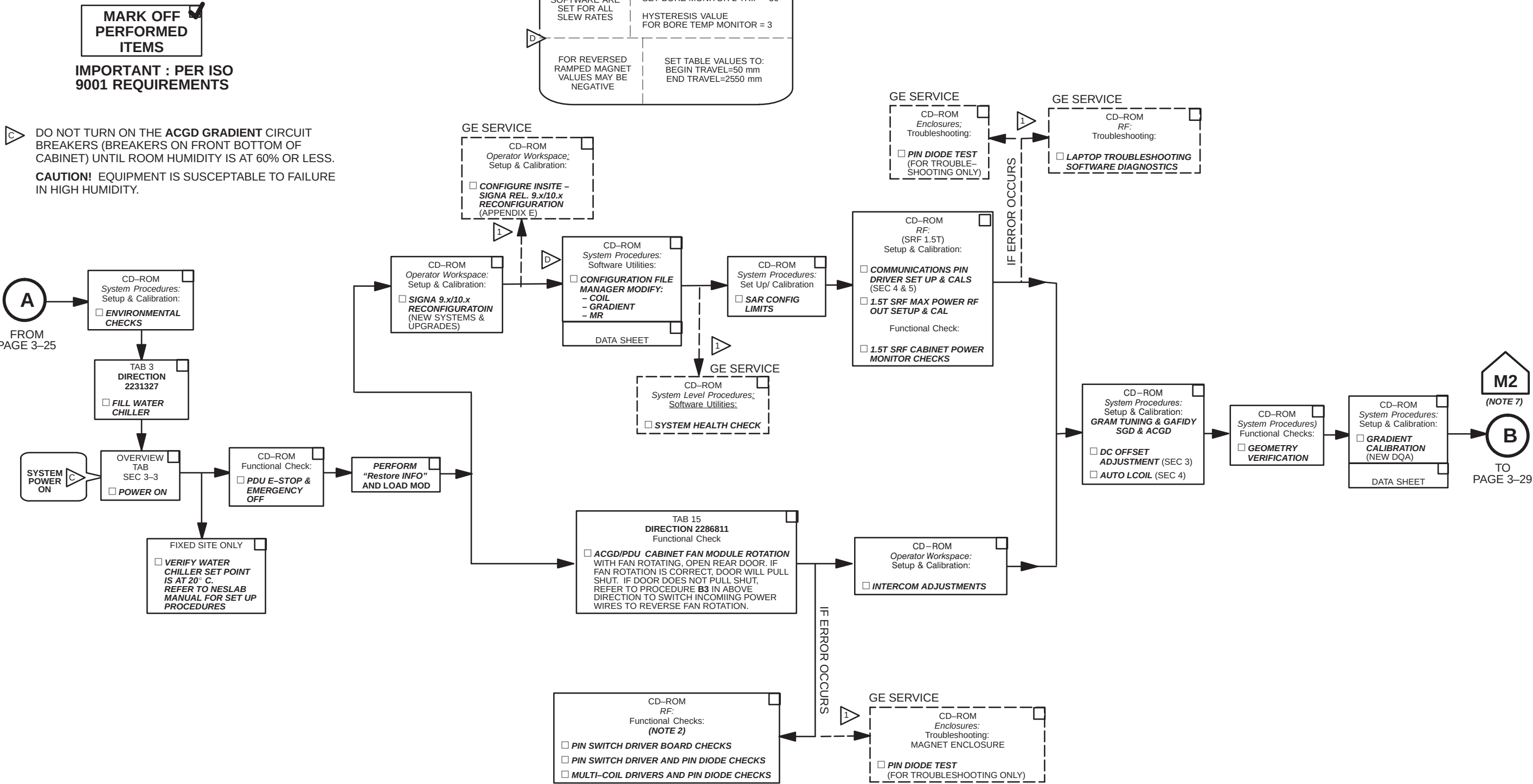
WARNING!

FERROUS MATERIAL HAZARD!
TOOLS AND PARTS REQUIRED FOR THIS
INSTALLATION CONTAIN FERROUS
MATERIAL AND WILL BE STRONGLY
ATTRACTED TO MAGNET AND MAY
BECOME DANGEROUS PROJECTILES.
KEEP ALL FERROUS MATERIALS AT
LEAST 10 FEET AWAY FROM THE MAGNET.



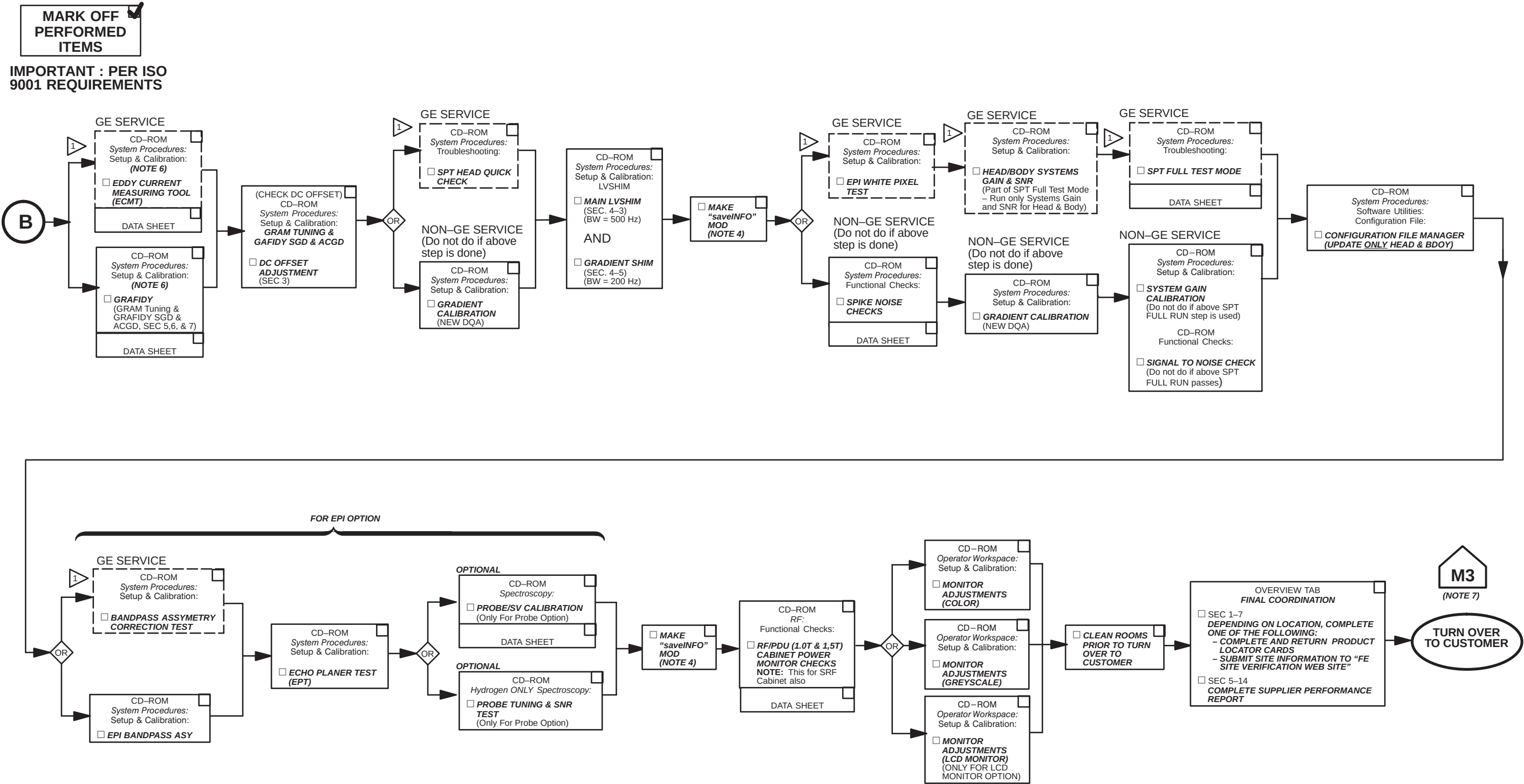
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HORIZON LX TO SIGNA LX with ACGD UPGRADE
SYSTEM CALIBRATION
(FIXED-SITES ONLY)



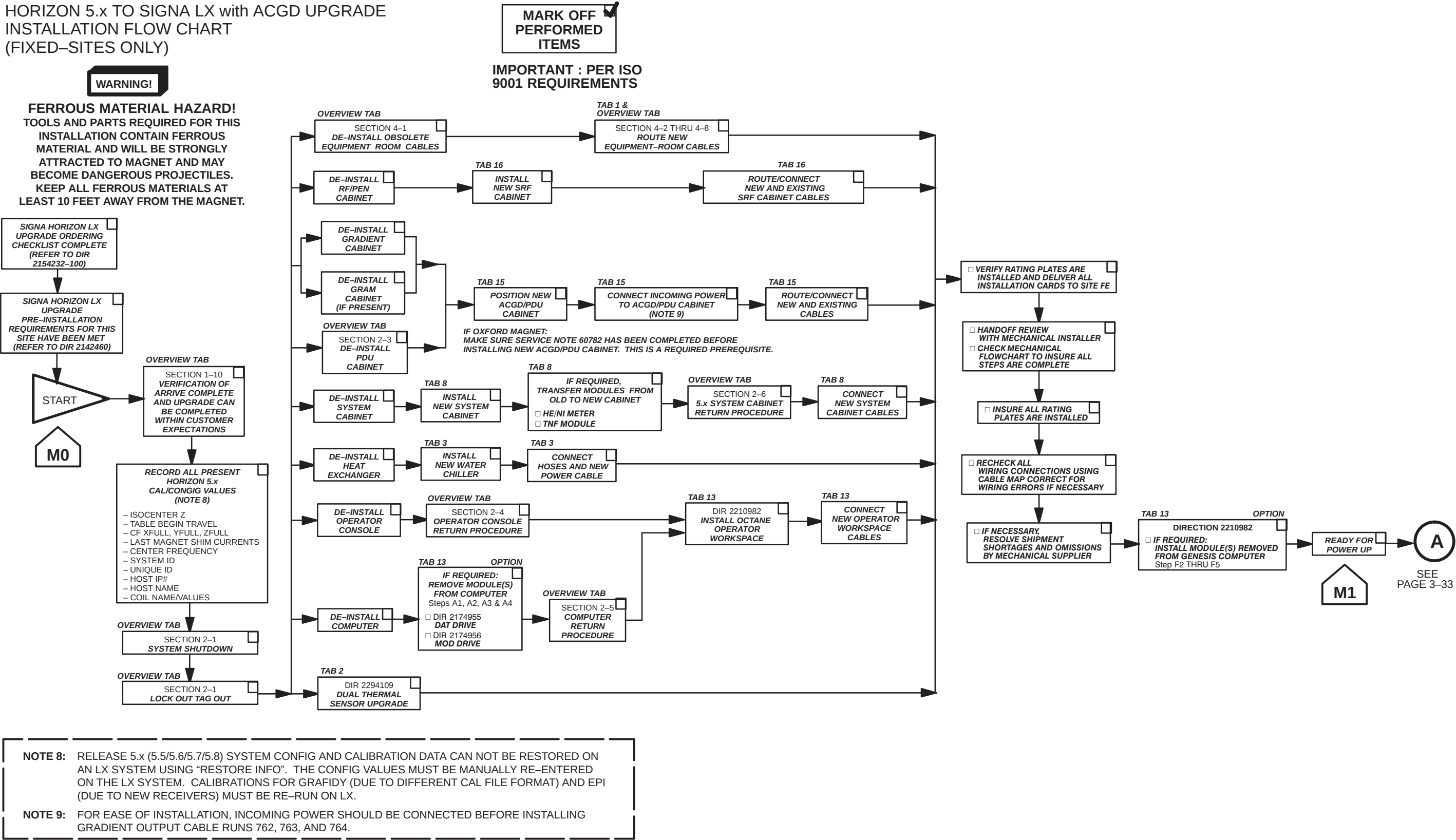
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HORIZON LX TO SIGNA LX with ACGD UPGRADE
SYSTEM CALIBRATION
(FIXED-SITES ONLY)



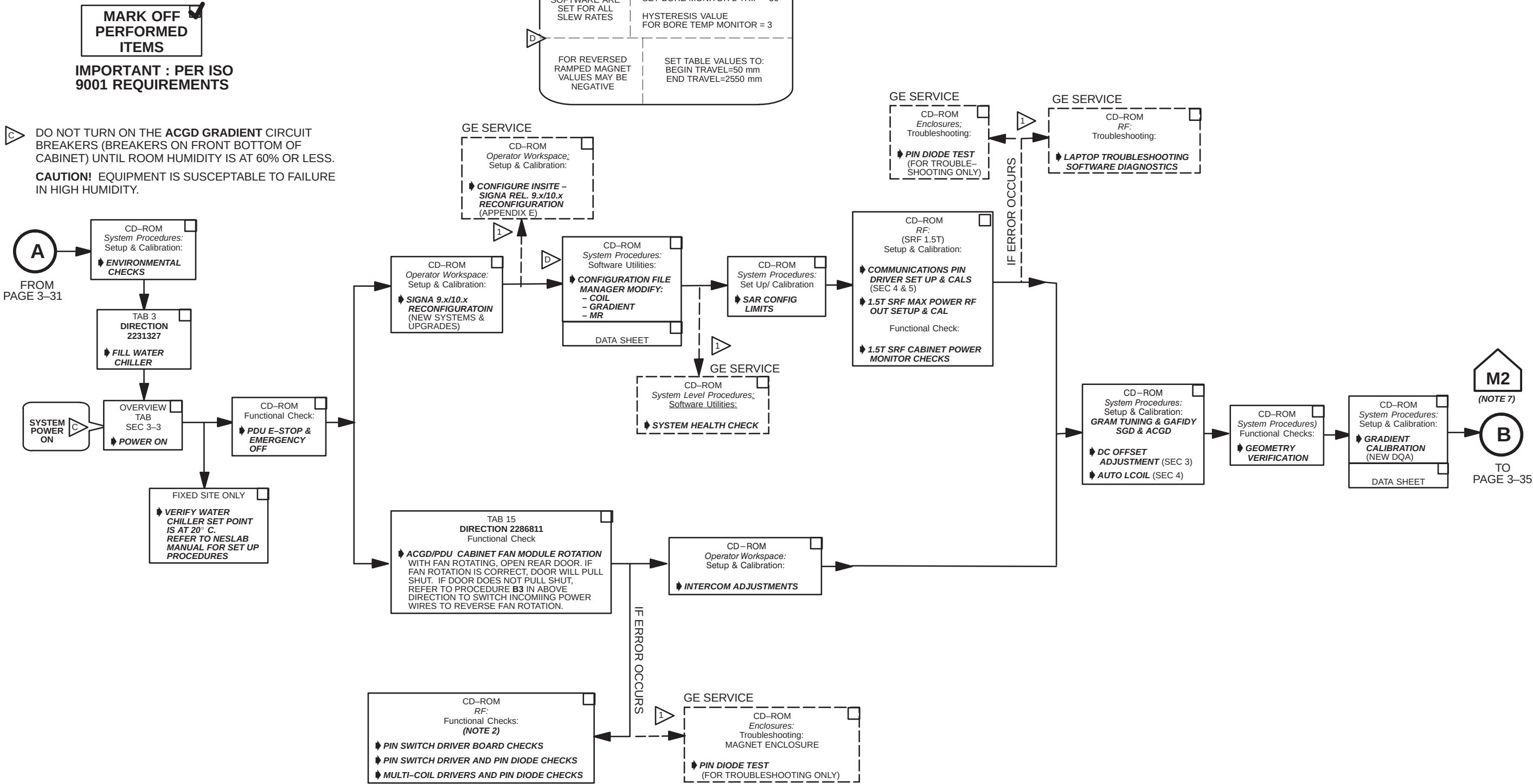
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HORIZON 5.x TO SIGNA LX with ACGD UPGRADE
INSTALLATION FLOW CHART
(FIXED-SITES ONLY)



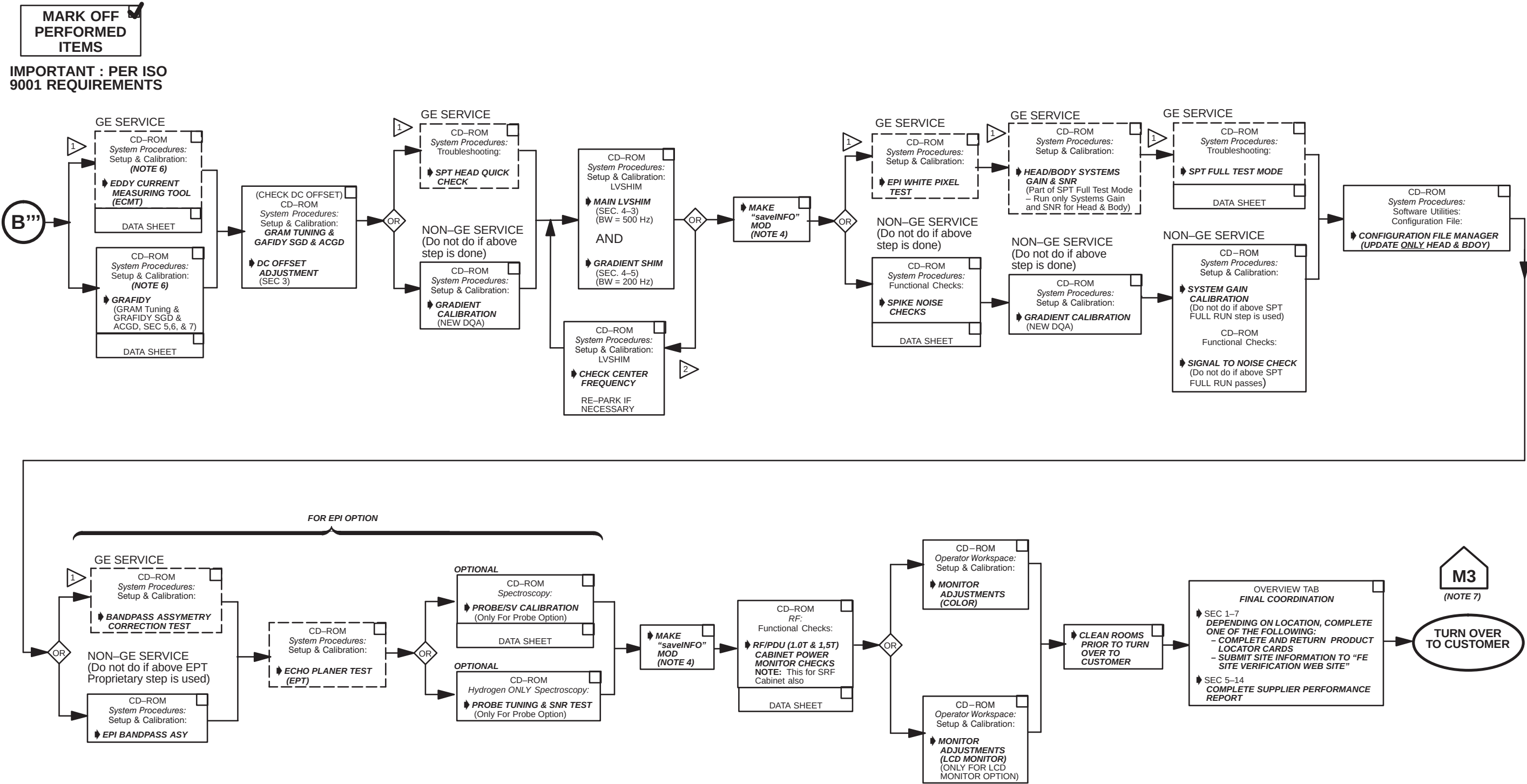
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HORIZON 5.x TO SIGNA LX with ACGD UPGRADE
SYSTEM CALIBRATION
(FIXED-SITES ONLY)



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HORIZON 5.x TO SIGNA LX with ACGD UPGRADE
SYSTEM CALIBRATION
(FIXED-SITES ONLY)



3-35

UPGRADE INSTALLATION PROCESS

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ADVANTAGE 5.x TO SIGNA LX with ACGD UPGRADE
HARDWARE INSTALLATION
(FIXED-SITES ONLY)

MARK OFF
PERFORMED
ITEMS

IMPORTANT :
PER ISO 9001
REQUIREMENTS

WARNING!

FERROUS MATERIAL HAZARD!
TOOLS AND PARTS REQUIRED FOR THIS
INSTALLATION CONTAIN FERROUS
MATERIAL AND WILL BE STRONGLY
ATTRACTED TO MAGNET AND MAY
BECOME DANGEROUS PROJECTILES.
KEEP ALL FERROUS MATERIALS AT
LEAST 10 FEET AWAY FROM THE MAGNET.

FOR SIV MAGNETS,
ALLOW FOR 2 EXTRA DAYS OF CUSTOMER
DOWNTIME FOR MAGNET MECHANICAL
PLOTING BEFORE BEGINNING DE-INSTALL
(AMERICAS POLE CONTACT MAGNET
SUPPORT AT 1-800-321-7937 FOR DETAILS)

SITE FE MUST CONSULT WITH THE APPROPRIATE
MAC TEAM LEADER IF THERE IS DOUBT ABOUT
BEING ABLE TO PROPERLY SHIM ANY PARTICULAR
MAGNET WHEN UPGRADING TO HORIZON. IN
CASES WHERE DOUBT EXISTS, ADDITIONAL
DOWNTIME MAY HAVE TO BE BUILT INTO THE
UPGRADE INSTALLATION SCHEDULE TO ALLOW
FOR SPECIAL SHIMMING REQUIREMENTS.

OVERVIEW TAB
SECTION 1-2
IF MAGNET HAS PASSIVE SHIMS GLUED TO THE
BORE, ARRANGEMENT SHOULD BE MADE FOR FE
TO INSPECT BEFORE INSTALLING NEW BODY COIL.

SIGNA HORIZON UPGRADE
ORDERING CHECKLIST COMPLETE
(REFER TO DIR 2154232-100)

SIGNA HORIZON UPGRADE
PRE-INSTALLATION REQUIREMENTS
FOR THIS SITE HAVE BEEN MET
(REFER TO DIR 2142460)



OVERVIEW TAB
SECTION 1-7
VERIFICATION OF ARRIVE COMPLETE
AND UPGRADE CAN BE COMPLETED
WITHIN CUSTOMER EXPECTATIONS

TAB 8
IF NEEDED:
ORDER TNF MODULE

OVERVIEW TAB
SECTION 2-1
SYSTEM SHUTDOWN

OVERVIEW TAB
SECTION 2-1
LOCK OUT TAG OUT

DE-INSTALL
RF CABINET

OVERVIEW TAB
SECTION 4-1
DE-INSTALL
EQUIPMENT ROOM CABLES

DE-INSTALL
GRADIENT CABINET

DE-INSTALL
COMPUTER

OVERVIEW TAB
DE-INSTALL
OPERATOR
CONSOLE

DE-INSTALL
5.4 SYSTEM CABINET

DE-INSTALL
PENETRATION
PANEL

TAB 2
DE-INSTALL
BODY COIL

TAB 2
DE-INSTALL BORE AND
ALIGNMENT LIGHTS

TAB 2
DE-INSTALL FRONT
BRIDGE MOUNT

TAB 2
DE-INSTALL REAR
PEDESTAL

TAB 2
DE-INSTALL
BRIDGE

TAB 2
REMOVE FRONT
SKIRT

TAB 2
DE-INSTALL
ENCLOSURE
FRONT AND REAR
COVERS

TAB 2
DE-INSTALL PAC

OVERVIEW
SECTION 4-1
DE-INSTALL MAGNET
ROOM CABLES

TAB 2
DE-INSTALL
REAR PEDESTAL
COVERS

DE-INSTALL
PENETRATION
CABINET

OVERVIEW TAB
SECTION 2-3
DE-INSTALL
PDU CABINET

TAB 13 (OPTION)
REMOVE MOD AND/OR DAT
DISK DRIVE FROM COMPUTER

OVERVIEW TAB
SECTION 2-5
COMPUTER
RETURN
PROCEDURE

OVERVIEW TAB
SECTION 2-4
OPERATOR
CONSOLE
RETURN PROCEDURE

OVERVIEW TAB
SECTION 2-6
5.x SYSTEM CABINET
RETURN PROCEDURE

TAB 4
MOVE
APPLICABLE PARTS
TO NEW PEN. PANEL

TAB 4
INSTALL NEW
PENETRATION
PANEL

TAB 3
INSTALL NEW
BODY COIL

TAB 2
TRANSFER
ALIGNMENT LIGHTS AND
BORE LAMP TO NEW END RINGS

TAB 2
TRANSFER DOCK LIMIT BOX
TO NEW FRONT END RING

TAB 2
TRANSFER APPLICABLE
ITEMS TO NEW REAR PEDESTAL

TAB 2
OPTIONAL
IF REQUIRED:
REPLACE OR CUT NOTCH IN
CARRIAGE BASE PLATE

TAB 2
MOVE CARRIAGE CABLE-TRAK AND
OTHER ITEMS TO NEW BRIDGE

TAB 2
MODIFY FRONT SKIRT
(EXCEPT MAGNISHIELD)

TAB 2
INSTALL FRONT
SKIRT

TAB 2
MOVE LATCHES
AND HARDWARE TO
NEW FRONT AND
REAR COVERS

TAB 2
DE-INSTALL MAGNET
ROOM CABLES

TAB 2
DE-INSTALL
REAR PEDESTAL
COVERS

TAB 2
DE-INSTALL
REAR PEDESTAL
COVERS

TAB 2
DE-INSTALL
REAR PEDESTAL
COVERS

TAB 2
DE-INSTALL
REAR PEDESTAL
COVERS

TAB 2
DE-INSTALL
REAR PEDESTAL
COVERS

TAB 1 & OVERVIEW TAB
SECTION 4-2 THRU 4-8
ROUTE/TERMINATE NEW GRADIENT, FIBER-OPTIC,
AND OTHER EQUIPMENT-ROOM CABLES

TAB 15
POSITION NEW ACGD/PDU CABINET &
CONNECT TO MAIN DISCONNECT PANEL

TAB 15 & TAB 1
ROUTE & CONNECT REMAINING
POWER AND GRADIENT CABLES

TAB 13
INSTALL
OPERATOR
WORKSPACE

TAB 13
CONNECT NEW
OPERATOR
WORKSPACE CABLES

OVERVIEW
SECTION 2-6
5.x SYSTEM CABINET
RETURN PROCEDURE

TAB 8
INSTALL
NEW SYSTEM
CABINET

TAB 8
CONNECT NEW
SYSTEM CABINET
CABLES

NOTE:
WHEN HANDLING TNF MODULE, THE INSTALLER MUST
WEAR A GROUNDING WRIST STRAP THAT IS PROPERLY
CONNECTED TO A GROUND TERMINAL.

TAB 8
TRANSFER CRYOGEN METER AND IF REQUIRED
TNF MODULE, FROM OLD TO NEW CABINET

TAB 4
OPTION
IF M1090ZA:
INSTALL NEW FIBER OPTIC REPEATER
AND RE-TERMINATE RUN 711/712

TAB 1
SECTION 4
RUN NEW
MAGNET ROOM CABLES,
AND FIBER OPTICS,

TAB 3
INSTALL
WATER
CHILLER

TAB 2
IF REQUIRED:
CUT LARGER OPENING IN
REAR PEDESTAL SKIRT
FOR AIR HOSE CLEARANCE

TAB 3
RUN NEW
COOLANT HOSES

TAB 3
TEST COOLANT
SYSTEM FOR
LEAKS

TAB 2
INSTALL ENCLOSURE END RINGS
(Including Gaskets and I/F Plate)

TAB 2
INSTALL NEW REAR PEDESTAL
(Including Routing/connecting cables)

TAB 2
INSTALL END BELLS

TAB 2
TEST COVER ALIGNMENT
FOR AIR LEAKS

TAB 2
INSTALL NEW BRIDGE

OPTIONAL ALTERNATIVE
TAB 2
IF GE-SIV OR SX: REPLACE
FRONT COVER MOUNTS

OPTIONAL ALTERNATIVE
TAB 2
IF MAGNISHIELD: REPLACE
FRONT SUPPORT

VERIFY RATING
PLATES ARE
INSTALLED AND
DELIVER ALL
INSTALLATION
CARDS TO SITE FE

A

SEE
PAGE 3-41

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ADVANTAGE 4.x TO SIGNA LX with ACGD UPGRADE
HARDWARE INSTALLATION
(FIXED-SITES ONLY)

MARK OFF PERFORMED ITEMS

IMPORTANT : PER ISO 9001 REQUIREMENTS

WARNING!

FERROUS MATERIAL HAZARD!
TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS MATERIALS AT LEAST 10 FEET AWAY FROM THE MAGNET.

SITE FE MUST CONSULT WITH THE APPROPRIATE MAC TEAM LEADER IF THERE IS DOUBT ABOUT BEING ABLE TO PROPERLY SHIM ANY PARTICULAR MAGNET WHEN UPGRADING TO HORIZON. IN CASES WHERE DOUBT EXISTS, ADDITIONAL DOWNTIME MAY HAVE TO BE BUILT INTO THE UPGRADE INSTALLATION SCHEDULE TO ALLOW FOR SPECIAL SHIMMING REQUIREMENTS.

OVERVIEW TAB
SECTION 1-2
IF MAGNET HAS PASSIVE SHIMS GLUED TO THE BORE, ARRANGEMENT SHOULD BE MADE FOR FE TO INSPECT BEFORE INSTALLING NEW BODY COIL.

SIGNA HORIZON UPGRADE ORDERING CHECKLIST COMPLETE (REFER TO DIR 2154232-100)

SIGNA HORIZON UPGRADE PRE-INSTALLATION REQUIREMENTS FOR THIS SITE HAVE BEEN MET (REFER TO DIR 2142460)

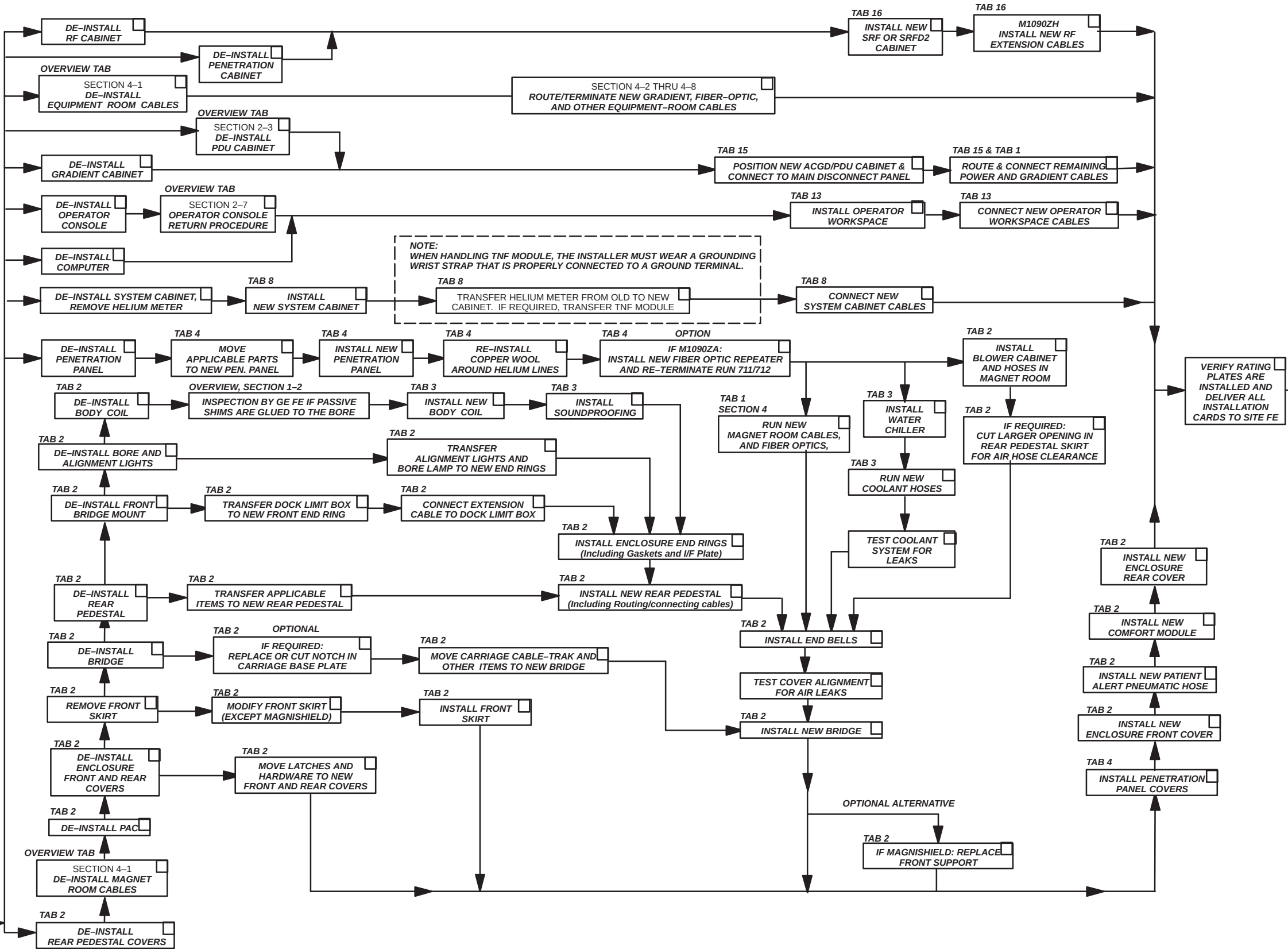


OVERVIEW TAB
SECTION 1-7
VERIFICATION OF ARRIVE COMPLETE AND UPGRADE CAN BE COMPLETED WITHIN CUSTOMER EXPECTATIONS

TAB 8
IF NEEDED:
ORDER TNF MODULE

OVERVIEW TAB
SECTION 2-1
SYSTEM SHUTDOWN

OVERVIEW TAB
SECTION 2-1
LOCK OUT TAG OUT



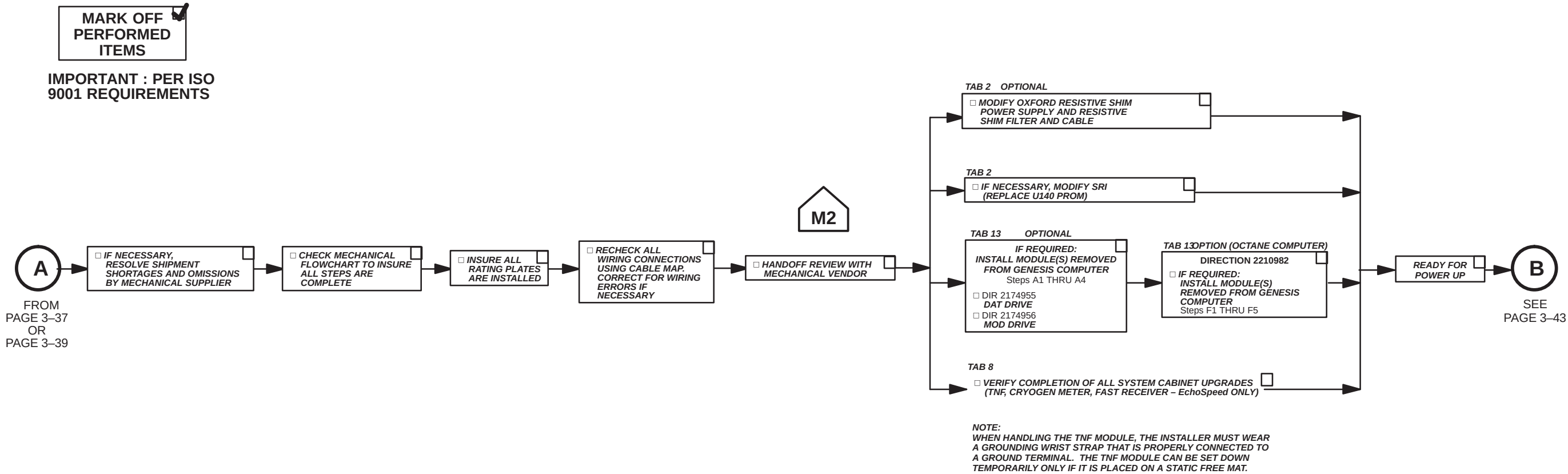
VERIFY RATING PLATES ARE INSTALLED AND DELIVER ALL INSTALLATION CARDS TO SITE FE

A

SEE PAGE 3-41

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ADVANTAGE 5.x or 4.x TO SIGNA LX with ACGD UPGRADE
PREPARATION FOR CALIBRATION
(FIXED-SITES ONLY)

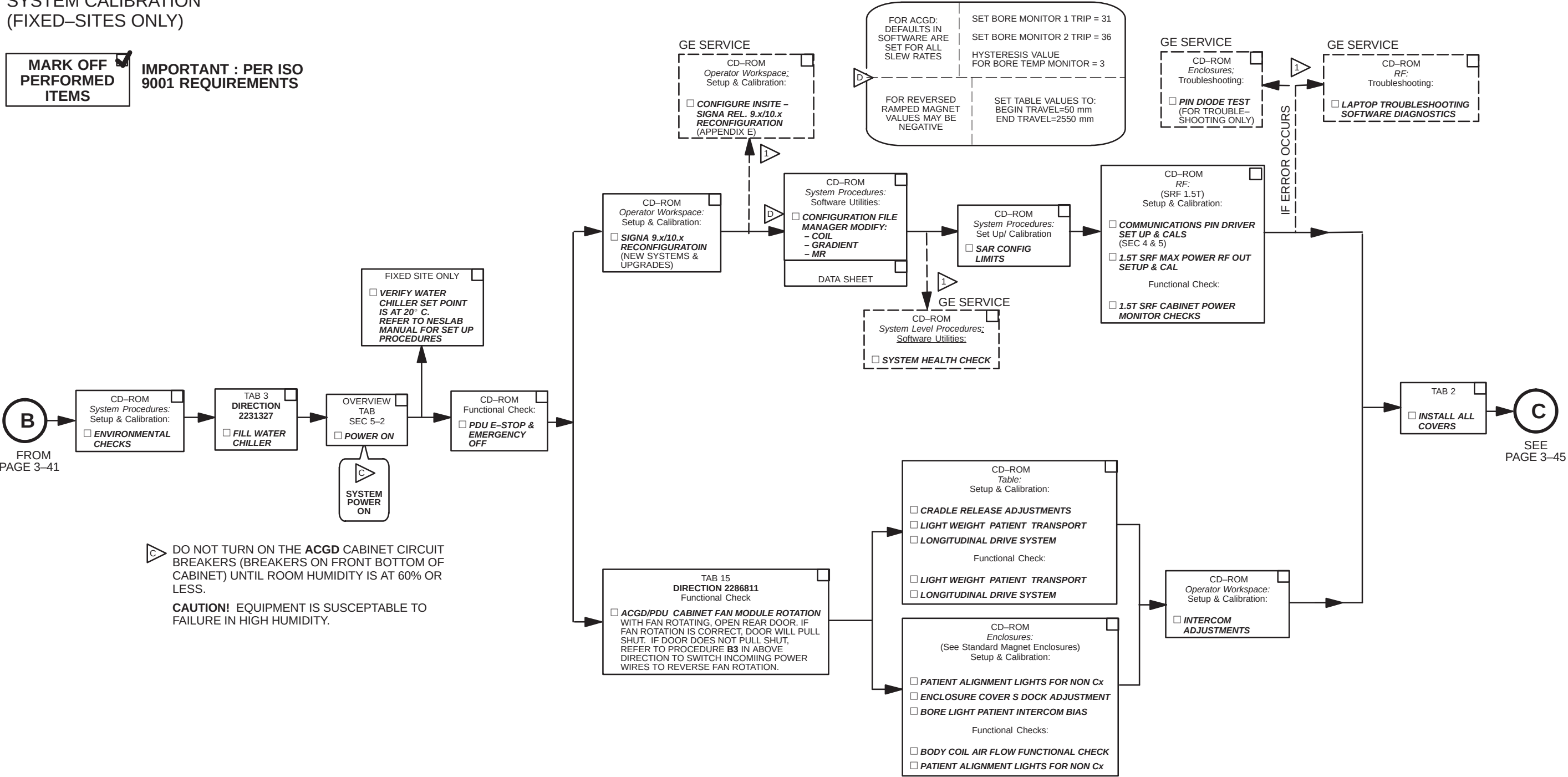


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ADVANTAGE 5.x or 4.x TO SIGNA LX with ACGD UPGRADE
SYSTEM CALIBRATION
(FIXED-SITES ONLY)

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

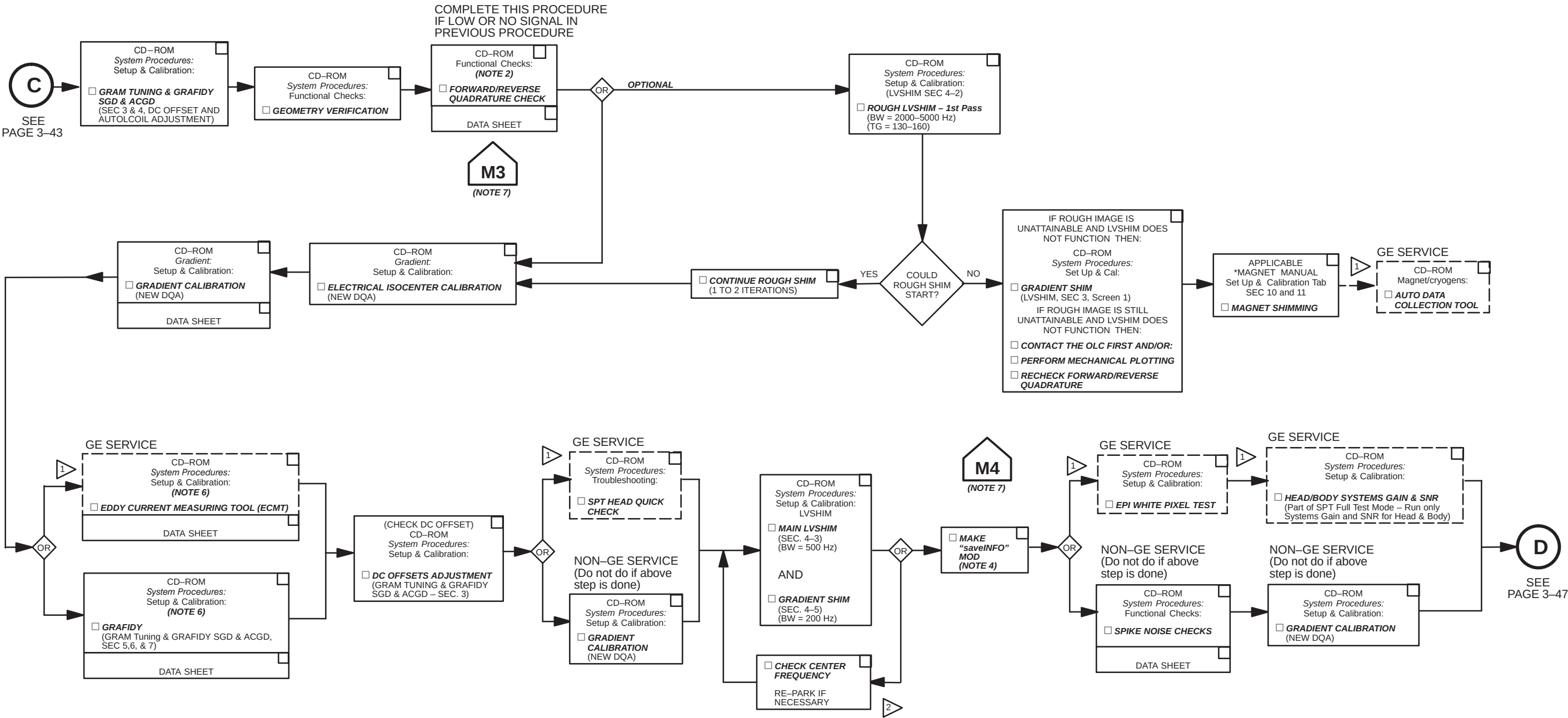


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ADVANTAGE 5.x or 4.x TO SIGNA LX with ACGD UPGRADE
SYSTEM CALIBRATION
(FIXED-SITES ONLY)

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

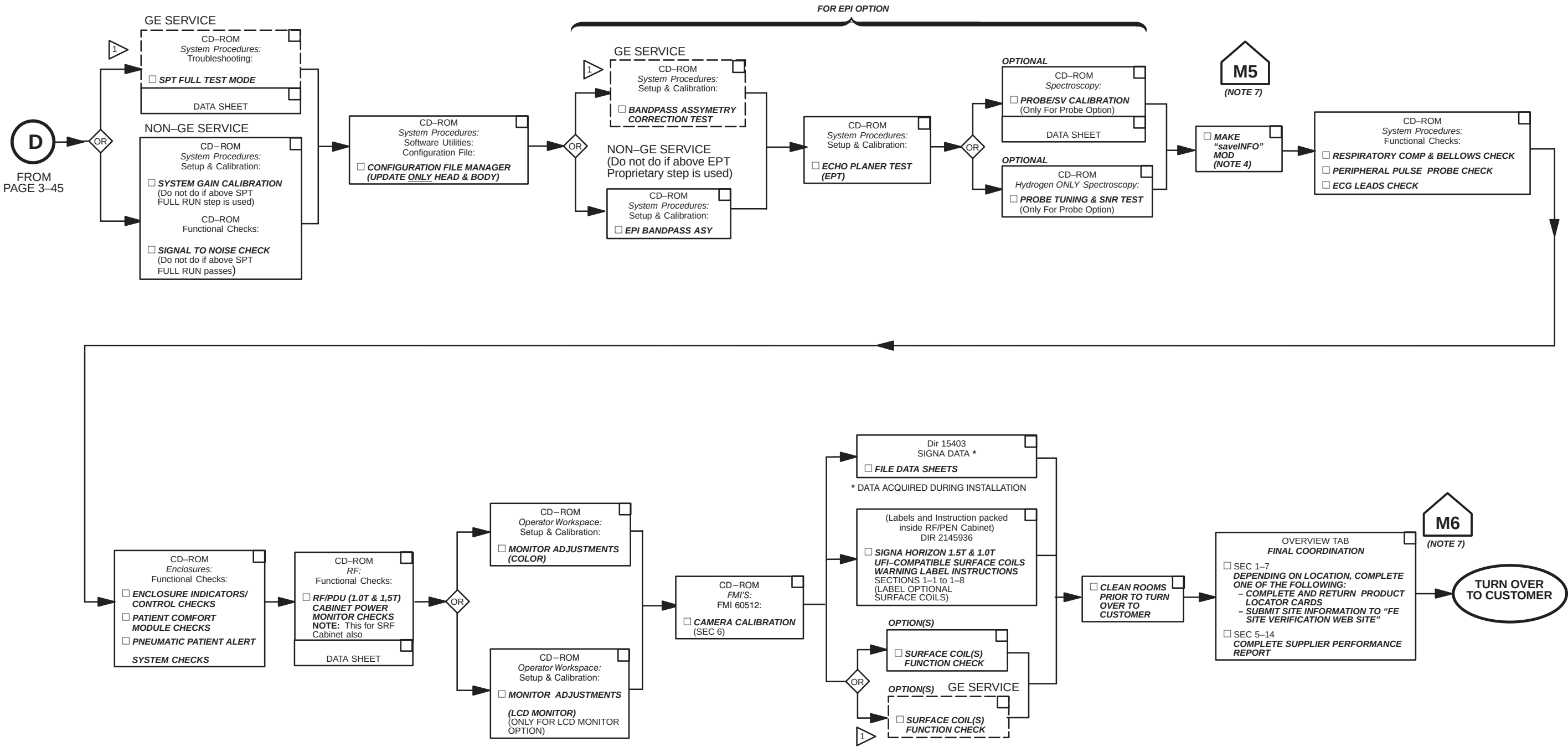


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ADVANTAGE 5.x or 4.x TO SIGNA LX with ACGD UPGRADE
SYSTEM CALIBRATION
(FIXED-SITES ONLY)

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS



SECTION 4 – SYSTEM CABLES

This section provides information on removing, relabeling and installing system cables for Signa LX (9.x) with ACGD and Signa LX 1.5T EXCITE (11.x) upgrades.

4-1 CABLE REMOVAL FOR UPGRADING TO SIGNA LX OR LX with EXCITE

Table 4-1 lists all cables to be removed from various early system configurations for upgrading to a Signa LX SmartSpeed, HiSpeed Plus or EchoSpeed Plus system or 1.5T Signa LX with EXCITE.

Signa Advantage 4.x systems that are being upgraded to Signa Horizon LX or EXCITE require many more cables to be removed or re-labeled than indicated on the cable removal maps on pages 4-9 thru 4-12. Table 4-1 lists all cables removed from various early system configurations, including Advantage 5.4 and Horizon 5.x, for upgrading to Signa LX SmartSpeed, HiSpeed Plus, or EchoSpeed Plus.

On the following table 4-1, the letters at the top of each column on the right side of the table are defined by the following cable removal explanations. An "X" in the box indicates that the cable should be removed.

If upgrading to 1.5T Signa LX with EXCITE Technology:

- A** – Existing system is Signa Horizon LX with Octane Computer (w/ACGD), upgrading to 1.5T Signa EXCITE
- B** – Existing system is Signa Horizon LX with Octane Computer (pre-ACGD), upgrading to 1.5T Signa EXCITE
- C** – Existing system is Signa Horizon LX with Indigo Computer, upgrading to 1.5T Signa EXCITE
- D** – Existing system is Signa Horizon 5.x, upgrading to 1.5T Signa EXCITE
- E** – Existing system is 5.4 Signa Advantage upgrading to 1.5T Signa LX with EXCITE
- F** – Existing system is 4.5 or later but pre 5.4 Signa Advantage upgrading to 1.5T Signa LX with EXCITE
- G** – Existing system is 4.x but pre 4.5 Signa Advantage upgrading to 1.5T Signa LX with EXCITE

If upgrading to Signa LX with ACGD:

- H** – Existing system is Signa Horizon 8.x pre-RF/PDU with SGD, upgrading to Signa LX with ACGD
- I** – Existing system is Signa Horizon 8.x pre-RF/PDU with Gradient/GRAM, upgrading to Signa LX w/ACGD
- J** – Existing system is Signa Horizon 5.x upgrading to Signa LX with ACGD
- K** – Existing system is 5.4 Signa Advantage upgrading to Signa LX with ACGD
- L** – Existing system is 4.5 or later but pre 5.4 Signa Advantage upgrading to Signa LX with ACGD
- M** – Existing system is 4.x but pre 4.5 Signa Advantage upgrading to Signa LX with ACGD

Note

If system does not have a Fiber Optic Repeater, Fiber Optic cables will have to be reterminated at Penetration Panel. Instructions to be found with M1090ZA, the Fiber Optic Repeater Option installation, which should have been ordered for this upgrade.

Note

Discard means to remove and return to GE for recycling

4-1 CABLE REMOVAL FOR UPGRADING TO SIGNA LX OR LX with EXCITE (continued)

TABLE 4-1
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA LX OR 1.5T EXCITE

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K	L	M
002	PD1-A2-TS1-32/NEUT/GND	MR2-A12-SERVICE						X	X					X	X
003	PD1-A2-TS1-1,2,3/NEUT/GND	MR3-A10-AC IN					X	X	X				X	X	X
007	PD1-FEEDER 15	PP2-A6-AC IN						X	X					X	X
008	PD1-FEEDER 16	PP2-A6-SERVICE						X	X					X	X
009	PD1-A2-TS1-10,11,12/NEUT/GND	MR1-A7-J1		X	X	X	X	X	X	X	X	X	X	X	X
012	PD1-A2-TS1-21/NEUT/GND	OC1-A8-J3 (AC IN)						X	X					X	X
013	PD1-A2-TS1-33/NEUT/GND	OC1-A8-J6 (SERVICE)						X	X					X	X
014	PD1-A2-TS1-34/NEUT/GND	CC1-A20-SERVICE						X	X					X	X
015	PD1-A2-TS1-13,14,15/NEUT/GND	CC1-A20-AC IN						X	X					X	X
016	PD1-A2-TS1-16,17,18/NEUT/GND	CC2-A20-AC IN						X	X					X	X
017	PD1-A2-TS1-34/NEUT/GND	CC2-A20-SERVICE						X	X					X	X
021	PD1-A2-GND	CC1-A20-GND						X	X					X	X
022	PD1-A2-GND	CC2-A20-GND						X	X					X	X
023	PD1-GND BUS	OC1-GND				X	X	X	X			X	X	X	X
027	PD1-A2-TS1-32/NEUT/GND	PP2-A15-J1-SERVICE						X	X					X	X
028	PD1-A2-TS1-31/NEUT/GND	PP2-A15-J2-AC IN						X	X					X	X
030	PD1-A2-TS1-16,17,18,NEUT/GND	MR2-A12-J1 AC IN								X	X				
035	PD1-A2-TS1-22,23,24/NEUT/GND	CC2-A20-AC IN						X	X					X	X
036	PD1-FEEDER 5-TS1-4,5,6	MR6					X	X	X				X	X	X
037	PD1-GND BUS	MR2-A11 GND													
038	PD1-GND BUS	MR1-A7 GND		X	X		X						X		
041	PD1-TS1-1,2,3,GND	MR3-CABI/FJ6			X	X				X	X	X			
042	PD1-TS1-4,5,6,GND	MR3-CABI/FJ7			X	X				X	X	X			
043*	MR7-A4-J7	PD1-A2				X					X	X			
044	PP1-GND	MR1-A7-GND													
045	PP1-GND	MG6-GND													
046	PD1-TS1-20,NEUT,GND	HE1			X	X				X	X	X			
047	PD1-TS1-X,X,NEUT,GND	OW1-A2-A5-J1								X	X				
048	PD1-GND BUS	OW1-A3-A5-GND			X										
049	OW1 A1 A1	OW1 A2 A5 J2	X	X	X										
050	OC1-AC-IN	PD1-TS1-13,14,15,NUE,GND				X	X					X	X		
064	MR2-A12-J1-AC OUT	CC1				X	X					X	X		
065	PD1-TS1-26,27,GND	WC1													
200	MR2-A11-J19	MR3-A9-J1						X	X					X	X
201	MR2-A11-J20	MR3-A9-J2						X	X					X	X
202	MR2-A11-J21	MR3-A9-J3						X	X					X	X
203	MR2-A11-J22	MR3-A9-J4						X	X					X	X
223	OC1-A8-J13	MR2-A1-J36						X	X					X	X
224	CC1-A21-J19	MR1-A7-J8						X	X					X	X
226	MR1-A7-J5	MR2-A11-J14						X	X					X	X
228	CC1-A21-J10	MR2-A11-J17						X	X					X	X

4-1 CABLE REMOVAL FOR UPGRADING TO SIGNA LX OR LX with EXCITE (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA LX OR 1.5T EXCITE

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K	L	M
229	MR2-A11-J1	MR1-A7-J3													
231	MR2-A11-J2	PP1-J8													
235	MR1-A7-J13	PP1-J44													
236	MR1-A7-J12	PP1-J4													
238	CC1-A21-J14	OC1-A8-J8						X	X					X	X
239	CC1-A21-J15	OC1-A8-J10						X	X					X	X
244	CC1-A21-J13	BTI-HOST						X						X	
248	CC2-A21-J1	OC1-A9-J1						X	X					X	X
249	CC2-A21-J22	OC1-A8-J12						X	X					X	X
256	OC1-A8-J11	PP1-J12						X	X					X	X
257	MG3-A3-J1	PP1-J44													
258	MG3-A17-J1	PP1-J4													
261	PP1-J6	MG2-A11-A1-J1							X						X
262	PP1-J8	MG3-A3-A5-OUTPUT													
274	CC1-A21-J20	MR1-A7-J7						X	X					X	X
280	PP2-A1-J2	PP1-J15						X	X					X	X
281	PP2-A1-J3	PP1-J11						X	X					X	X
282	MG3-A2-J8	PP1-J12													
287	CC2-A21-J15	CC1-A21-J2						X	X					X	X
296	PP1-J18	EO2													
297	PP1-J18	EO1													
300	PP1-J15	MG2-A15													
301	CC2-A21-J20	CC1-A21-J28						X	X					X	X
302	CC2-A21-J21	CC1-A21-J29						X	X					X	X
318	PP1-J11	MG3-A7-LONG DRIVE													
322	MR2-A11-J11	PP2-A6-J18						X	X					X	X
325	CC1-A21-J27	CC2-A21-J19						X	X					X	X
327	CC1-A21-J25	CC2-A21-J17						X	X					X	X
346	MR2-A11-J41	CM1-FRONT						X	X					X	X
347	MR2-A11-J42	CM1-BACK						X	X					X	X
354	CM1	ST1						X	X					X	X
370	MR1-A7-J19	PP1-A9-J46							X						X
384	PP2-A15-J17	PP1-J20					X	X	X				X	X	X
385	PP1-J20	MG2-A30-TS1,2,3													
386	PP2-A15-J21	PP1-J47					X	X	X				X	X	X
387	MG2-A29-P1	PP1-J47													
388	PP2-A15-J22	PP1-J11					X	X	X				X	X	X
389	PP2-A15-J20	PP1-J15					X	X	X				X	X	X
404	MG3 A17 J1	MG2 A16 A5 J6	X	X	X										
405	MR3-A9-TS1	MG3-TS1-1,2					X	X	X				X	X	X
406	MR3-A9-TS1	MG3-TS1-3,4					X	X	X				X	X	X

4-1 CABLE REMOVAL FOR UPGRADING TO SIGNA LX OR LX with EXCITE (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA LX OR 1.5T EXCITE

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K	L	M
407	MR3-A9-TS1	MG3-TS1-5,6					X	X	X				X	X	X
410	PP1-J14	PP2-A15-J23							X						X
411	PP1-J14	MG3-A13-J1							X						X
412	PP1-J51	MG3-A32-J1					X	X	X				X	X	X
413	PP1-J51	PP2-A15-J28(A16-J9)					X	X	X				X	X	X
414	MG2-A35-MIC	MG3-A2-J9													
415	PP1-A8-J45	MG3-A33-J7							X						X
416	MR1-A7-J21	PP1-A8-J45							X						X
417	PP2-A15-J19	MR2-A11-J32						X	X					X	X
418	MR2-A11-J39	PP2-A15-J							X						X
419	MR2-A11-J26	PP1-A10-J1						X	X					X	X
420	MG3-A33-J1	PP1-A10-J2						X	X					X	X
421	MG2-A33-J3	MG2-A31-J1													
422	MG2-A33-J2	MG3-A1-J4													
424	PP1-A10-J5/J6	MG2-A33-J11/J12						X	X					X	X
425	MR2-A11-J12	PP1-J6							X						X
426	MR1-A7-J9	MR3-A9-J5						X	X					X	X
427	MR2-A11-J17	CC1-A21-J13						X	X					X	X
428	MR2-A11-J44	CC1-A21-J35						X	X					X	X
429	BT1	CC1-A21-J25						X	X					X	X
430	MR2-A11-J24	CC1-A21-J26						X	X					X	X
431	MR2-A11-J25	CC1-A21-J27						X	X					X	X
433	OC1-A9-J1	CC1-A21-J10						X	X					X	X
434	CC1-A8-J12	CC1-A21-J29						X	X					X	X
435	MR2-A11-J13	PP1-A10-J3						X	X					X	X
436	MG2-A12-A1-J1	MG3-A33-J3					X	X	X				X	X	X
437	MG2-A12-A1-J2	MG3-A33-J4					X	X	X				X	X	X
439	MG3-A3-J4	MG3-A33-J1					X	X	X				X	X	X
440	MG3-A3-J3	MG3-A34-J2					X	X	X				X	X	X
441	MG3-A33-J7	MG3-A34-J8					X	X	X				X	X	X
446	PP1-A9-J46	MG2-A12-A13-J5							X						X
447	PP1-A9-J46	MG2-A12-A13-J3							X						X
448	MR2-A11-J44	CC1-A21-J1						X						X	
449	MR2-A11-J24	CC1-A21-J34						X						X	
450	MR2-A11-J25	CC1-A21-J35						X						X	
453	PP1-J50	MG2-A11-A1-J4						X	X					X	X
454	MR2-A11-J34	PP1-J50						X	X					X	X
470	MR1-A7-J17	PP1-A11-J71					X	X					X	X	
474	PP1-A11-J72	MG3-A12-A1-J3					X	X					X	X	
475	PP1-A11-J73	MG3-A12-A1-J4					X	X					X	X	
476	PP1-A11-J74	MG3-A33-J7					X	X					X	X	

4-1 CABLE REMOVAL FOR UPGRADING TO SIGNA LX OR LX with EXCITE (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA LX OR 1.5T EXCITE

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K	L	M
477	PP1-A11-J75	MG3-A12-A1-J5					X	X					X	X	
478	PP1-A11-J76	MG3-A12-A1-J6					X	X					X	X	
485	MG3 A17 J2	MG2 A16 A3 OUT	X	X	X										
486	PP1-A14-J85	MR1-A7-J22	X	X	X										
487	PP1-A15-J78	MR1-A7-J20	X	X	X										
488	PP1-J80	MR2-A11-J8	X	X	X										
489	PP1-J81	MR2-A11-J9	X	X	X										
490	PP1-J82	MR2-A11-J10	X	X	X										
491	PP1-J80	MG3-A36-J7	X	X	X										
492	PP1-J81	MG3-A36-J8	X	X	X										
493	PP1-J82	MG3-A36-J9	X	X	X										
494	PP1-A14-J87	MG3-A36-J6	X	X	X										
495	PP1-A15-J78	MG3-A36-J10	X	X	X										
499	PP1-A14-J84	MR2-A11-J5	X	X	X										
700	PP1 A14 J86	MG2/MG3 A17 J2	X	X	X										
701	MR2-J11-J15	RF DOOR SWITCH													
702	MR2-J11-J18	MR1-A7-J5													
704	PP1-J12	OC1-A10-J9				X	X					X	X		
706	MR2 A11 J14	PD1 A14 A1 J3 / MR1 PD1 J3	X	X	X										
707	MR2-A11-J26-ESTOP	OC1-J8				X	X					X	X		
708	MR2-A11-J16	MR1-A7-J27					X						X		
709	MR1-A7-J26	MR3-A9-J1					X						X		
710	MR2-A11-J17	MR3-A9-J2			X		X						X		
711	PP1-A17-J91	MG2-A33					X						X		
712	PP1-A17-J91	MG2-A11-A1					X						X		
711/ 712	PP1-A17-J91	MMR2-A11-J13					X						X		
715	PP1-J14	MG2-A33-J1													
716	PP1-J89	MG2-A11-J2													
718	MR2-A11-J29	OC1-A10-J4				X	X					X	X		
719	PP1-J6	MR2-A11-J4				X	X					X	X		
726	PP1-A16-J77	MR1-A7-J18	X	X	X										
727	PP1-A16-J77	MG3-A36-J5	X	X	X										
728	MG3 A36 J11	MG2 A16 OUT J1	X	X	X										
729	MG3 A36 J12	MG2 A16 OUT J2	X	X	X										
730	MG3 A36 J13	MG2 A16 OUT J3	X	X	X										
731	MG3 A36 J14	MG2 A16 OUT J4	X	X	X										
732	MG3 A36 J15	MG2 A16 OUT J5	X	X	X										
733	MG3 A36 J16	MG2 A16 OUT J6	X	X	X										
734	MG3 A36 J17	MG2 A16 OUT J7	X	X	X										
735	MG3 A36 J18	MG2 A16 OUT J8	X	X	X										

4-1 CABLE REMOVAL FOR UPGRADING TO SIGNA LX OR LX with EXCITE (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA LX OR 1.5T EXCITE

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K	L	M
736	MG3 A36 J19	MG2 A16 A8-J9	X	X	X										
743	MG2-A12-A2-J2	MG3-A3-J3													
744	MG2-A12-A2-J1	MG3-A3-J4													
749	MG2-A33-J10	MG2-A36	X	X	X	X	X	X	X	X	X	X	X	X	X
753*	MR7-A4-J1	MG3-A11-J5				X					X	X			
754*	MR7-A4-J2	MG3-A11-J4				X					X	X			
755*	MR7-A4-J3	MG3-A11-J3				X					X	X			
756*	MR7-A4-J4	MG3-A11-J8				X					X	X			
757*	MR7-A4-J5	MG3-A11-J7				X					X	X			
758*	MR7-A4-J6	MG3-A11-J6				X					X	X			
759*	MR3-11-J12	MR7-A4-J8				X					X	X			
760*	MR3-11-J11	MR7-A4-J9				X					X	X			
761*	MR3-11-J10	MR7-A4-J10				X					X	X			
762**	MR3CABI/FJ3	PP1-A7-1/2				X				X	X	X			
763**	MR3CABI/FJ4	PP1-A7-3/4				X				X	X	X			
764**	MR3CABI/FJ5	PP1-A7-5/6				X				X	X	X			
767	MR1-A7-J37	PP1-J51					X							X	
768	MR1-A7-J31	PP1-J20													
769	MR1-A7-J34	PP1-J15													
770	MR1-A7-J41	PP1-J11													
771	MR1-A7-J36	PP1-J47													
772	MR1-A7-J40	PP1-J14													
773	MR1-A7-J35	PP1-A17-J1													
774	MR1-A7-J32	MR2-A11-J25													
775	MR1-A7-J42	PP1-A11-J92													
776	MR1-A7-J44	PP1-A11-J93													
777	MR1-A7-J43	PP1-A11-J94													
778	MG2-A12-A2-J3	PP1-A11-J72													
779	MG2-A12-A2-J4	PP1-A11-J73													
780	MG3-A3-J6	PP1-A11-J74													
781	MG2-A12-A2-J5	PP1-A11-J75													
782	MG2-A12-A2-J6	PP1-A11-J76													
783	MR1-A7-J37	PP1-F5,F6													
784	MG6-J1	PP1-F5,F6													
785	MR1-A7-J47	PP1-F7,F8													
786	MG6-J2	PP1-F7,F8													
787	MR2-A11-J30	CC1-A1-A1-SERIAL A	X	X		X	X					X	X		
788	PP1-J12	OW1-A1-A4-J7													
789	MR2-A11-J26	OW1-A1-A2-J18													
790	OW1-A2-A6-XO,RO,X1,R1	MR2-A11-J21	X	X	X										
791	MR2-A11-J30	OW1-A2-A4-EXP1	X	X	X										

4-1 CABLE REMOVAL FOR UPGRADING TO SIGNA LX OR LX with EXCITE (continued)

TABLE 4-1 (CONT'D)
CABLES TO BE REMOVED AND DISCARDED FOR UPGRADING TO SIGNA LX OR 1.5T EXCITE

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K	L	M
792	OW1-A2-A1-MIC/AUDIO	OW1-A1-A4-J11,J13	X	X	X										
793	OW1-A2-A1-COM 1	OW1-A1-A4-J9			X										
794	OW1-A2-A3-MIC/AUDIO	OW1-A1-A4-J12,J14			X										
795	OW1-A2-A3-PWR	OW1-A1-A4-J17			X										
796	OW1-A2-A3-KBD/MS	OW1-A1-A4-J8			X										
797	OW1 A2 A1 PC	OW1 A3 LCD			X										
799	OW1-A15-MONITOR (OR 820)	OW1-A4 (BW) (OR 820)			X										
800	OC1-J1	CC1-A2-J1 (MUX)				X	X					X	X		
801	OC1-J10	CC1-A2-J2 (SECUR. KEY)				X	X					X	X		
802	OC1-J12	CC1-A1-J9 (PLASMA)				X	X					X	X		
803	OC1-J13	CC1-A1-A10 (VIDEO)				X	X					X	X		
804	OC1-J3	CC1-A1-A1 (BODY)				X	X					X	X		
805	OW1-A1-A4-J6	OW1-A7 (NON-SCIM KYBRD)	X	X	X										
806	OW1-A2-A3-SER 1	OW1-A13 (MODEM)			X										
807	OW1-A16	OW1-A5 (DASM)	X	X	X										
809	OW1-A15-MIC	OW1-A1-A4-J12,J14	X	X											
810	OW1-A15-MD,KYBD	OW1-A1-A4-J8	X	X											
811	OW1-A2-A4-PORT2	OW1-A15-ETHERNET	X	X											
813	OW1-A2-A1-COM2	OW1-A15-COM2	X	X											
814	OW1-A1-A4-J9	OW1-A2-A1-KBD,MOUSE	X	X											
815	OW1-A16-SCSI	OW1-A15-SCSI (PCI)	X	X											
816	OW1-A2-A6-J1	OW1-A15-BIT3	X	X	X										
817	OW1-A1-A2-J17	OW1-A2-A5-J3	X	X											
818	MR2-A11-J31	OW1-A16-PORT6,7,8	X	X											
819	OW1-A2-A7-SCSI	OW1-A15-SCSI	X	X											
820	OW1-A15-MONITOR	OW1-A4 (BW)													
836	MR2-A11-J21	OW1-A15-BIT3	X	X	X										
848	MG2-A4-J5	MG2-A11-J4													
861	MG2-A4-J4	MG2-A33-J5,6,7,8,9	X	X	X										
862	MG2-A4-J1	MG2-A5-J1													
863	MG2-A4-J2	MG2-A5-J2													
897	MG2-A4-J3	MG2-A33-J4													
966	MR1-A7-J1	MR3-PD1-7,8,9,GND													
967	MR1-A7-CB1	MR3-PD1-10,11,12,GND													
968	MR2-A11-J1	MR3-PD1-4,5,6,GND													
969	OW1-A2-A5-GND	MR3-PD1-18,19,GND,N													
970	WC1	MR3-PD1-16,17,GND													
971	MR2-A11-J23	MR3-PD1-J2													
972	MR2-A11-J14	MR3-PD1-J3													

Notes:

* These Runs are removed only if GRAM Cabinet is part of existing site being deinstalled.

** Remove only equipment room side of cable runs.

4-2 RE-LABELING & RE-ROUTING EXISTING LX CABLES TO SIGNA LX with EXCITE

Several existing LX cables need to be re-labeled and then re-routed per the new labels. Refer to Table 4-2.

TABLE 4-2
HORIZON LX CABLES RE-LABELED AND RE-LOCATED FOR UPGRADE TO LX with EXCITE

RUN	SYSTEM TYPE	ONE END		OTHER END	
		OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
047	Indigo	RUN 047 OW1-A2-A5-J1 46-307832P4	RUN 1081 OW1-A19-J1 2371515	RUN 047 PD1-A2-25,31,GND,N 46-307832P4	RUN 1081 MR3-PD1-18,19,GND,N 2371515
065	Octane with RF/PDU	RUN 065 PD1-26,27,GND 2223355	RUN 970 MR3-PD1-16,17,GND 2280289	RUN 065 WC1 2223355	RUN 970 WC1 2280289
969	Octane	RUN 969 OW1-A2-A5-J1 2280288	RUN 1081 OW1-A19-J1 2371515	RUN 969 MR3-PD1-18,19,GND,N 2280288	RUN 1081 MR3-PD1-18,19,GND,N 2371515
703	Indigo	RUN #703 PD1-A1-A2-J2 PD1-A14-A1-J2 46-258431G809	RUN #703/971 PD1-A1-A2-J2 PD1-A14-A1-J2 MR3-PD1-J2	RUN #703 MR2-A11-J23 46-228431G809	RUN #703/971 MR2-A11-J23 46-258431G941
706	Indigo	RUN #706 MR1-PD1-J3 PD1-A14-A1-J3 46-258431G808	RUN #706/972 PD1-A1-A3-J2 PD1-A14-A1-J3 MR3-PD1-J3	RUN #706 MR2-A11-J14 46-258431G808	RUN #706/972 MR2-A11-J14 46-258431G940
788	Indigo Octane	RUN 788 PP1-J12 46-328000G937	RUN 1085 PP1-J12 2363001-5	RUN 788 OW1-A1-A4-J7 46-328000G937	RUN 1085 OW1-A21-J7 2363001-5
789	Indigo Octane	RUN 789 MR2-A11-J26 46-328000G938	RUN 1084 MR2-A11-J26 2354601-5	RUN 789 OW1-A1-A4-J18 46-328000G938	RUN 1084 OW1-A21 (CONNECT TO RUN 1062) 2354601-5

4-3 RE-LABELING & RE-ROUTING EXISTING FIBER OPTIC CABLES TO SIGNA LX with EXCITE

Several existing LX fiber optic cables need to be re-labeled and some of them re-routed per the new labels. Blank write-on labels for marking cables (46-208781P3) are provided. Refer to Table 4-3 for information to write on the labels. Apply labels over existing labels at the appropriate ends of affected cables. Refer to cable maps for routing and connection of cables.

- Run 708 is re-labeled for Horizon 5.x or LX systems upgraded to Signa LX with EXCITE. The fiber optic connections to the 5.x and LX TYME board are different than the MGD STIF board. Refer to table 4-3. All pre-5.x system upgrades are provided a new cables for Run 708
- A new Run 1034 is provided which will replace Run 710. If existing Run 710 is retained, it will need to be re-labeled. Refer to table 4-3. The conduit may also need to be shortened to provide longer fiber optic wires required for routing and connecting within the System Cabinet. Refer to EXCITE System Cabinet Upgrade instructions in Tab 8.
- A new Run 1045 is provided which will replace Run 711/712 in the Equipment Room. If existing Run 711/712 is retained, it will need to be re-labeled. The fiber optic connections to the 5.x and LX TYME board are different than the MGD STIF board. Refer to table 4-3.

TABLE 4-3
HORIZON LX FIBER OPTIC CABLES RE-LABELED AND RE-LOCATED FOR UPGRADE TO LX with EXCITE

RUN	SYSTEM TYPE	ONE END		OTHER END	
		OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
708	Indigo Octane	RUN #708 MR2-A11-J16 (TR1/TX)	RUN 1033 MR2-A11-J16 2330480 (TR1/TX)	RUN #708 MR1-A7-TX (J19)	RUN 1033 MR1-A7-T 2330480X (MGD SLOT 13R-J21)
710	Indigo Octane	RUN #710 MR2-A11-J17 46-317085G603 -1 (J12) -2 (J13) -3 (J15) -4 (J17) -5 (J18)	RUN 1034 MR2-A11-J17 2331897 -1 (J14) -2 (J13) -3 (J12) -4 (J11) -5 (J20)	RUN #710 MR3-A11-J2 46-317085G603 -1 (J2) -2 (J3) -3 (J4) -4 (J1) -5 (J5)	RUN 1034 MR3-PD1-J2 2331897 -1 (J2) -2 (J3) -3 (J4) -4 (J1) -5 (J5)
711/712 Equip. Room	Indigo Octane	RUN #711/712 MR2-A11-J13 (J25,J27) (J23) (J20,J21)	RUN 1045 MR2-A11-J13 2331897 (MGD SLOT 13R-J15,J16) (MGD SLOT 13R-J17) (MGD SLOT 13R-J18,J19)	RUN #711/712 PP1-A17-A1	RUN 1045 PP1-A17-A1

4-4 RE-LABELING AND RE-ROUTING OF EXISTING PRE-HORIZON CABLES TO SIGNA LX

Several existing cables need to be re-labeled and then re-routed per the new labels. Refer to Table 4-4.

Runs 706, 772, 773, and 774 are only re-labeled for Advantage 5.4 systems upgraded to Signa LX with ACGD. All pre-5.4 system upgrades are provided new cables for Runs 706, 772, 773, and 774.

TABLE 4-4
PRE-HORIZON CABLES RE-LABELED AND RE-LOCATED FOR UPGRADE TO LX w/ACGD

OLD RUN	SYSTEM TYPE	ONE END		OTHER END	
		OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
207	4.x	RUN #207 MR2-A11-J6 46-243775G411	RUN #701 MR2-A11-J15 46-243775G807	RUN #207 DOOR SWITCH 46-243775G411	RUN #701 RF DOOR SWITCH 46-243775G807
227	4.x	RUN #227 MR2-A11-J16 46-243775G464	RUN #703/971 PD1-A1-A2-J2 PD1-A14-A1-J2 MR3-PD1-J2	RUN #227 PD1-A1-A2-J2 46-243775G464	RUN #703/971 MR2-A11-J23 46-258431G941
432	4.x	RUN #432 PD1-A1-A3-J2 46-243775G676	RUN #706/972 PD1-A1-A3-J2 PD1-A14-A1-J3 MR3-PD1-J3	RUN #432 MR2-A11-J45 46-243775G676	RUN #706/972 MR2-A11-J14 46-258431G940
705	5.x	RUN #705 MR2-A11-J25 46-243775G663	RUN #774 MR2-A11-J25 46-243775G811	RUN #705 PP2-A15-J19 46-243775G663	RUN #774 MR1-A7-A32 46-243775G811
713	5.x	RUN #713 PP2-A15-J23 46-243775G676	RUN #772 MR1-A7-J40 46-243775G808	RUN #713 PP1-J14 46-243775G676	RUN #772 PP1-J14 46-243775G808
714	5.x	RUN #714 PP2-A5-J27 46-243775G676	RUN #773 MR1-A7-J35 46-243775G808	RUN #714 PP1-J89 46-243775G676	RUN #773 PP1-J89 46-243775G808

4-5 RE-LABELING & RE-ROUTING EXISTING HORIZON 5.x OR 8.x CABLES TO SIGNA LX

Several existing Horizon 5.x and LX cables need to be re-labeled and then re-routed per the new labels. Refer to Table 4-5.

Runs 709 and 710 are re-labeled only for Horizon 5.x systems upgraded to Signa LX with ACGD. All pre-5.x system upgrades are provided new cables for Runs 709 and 710

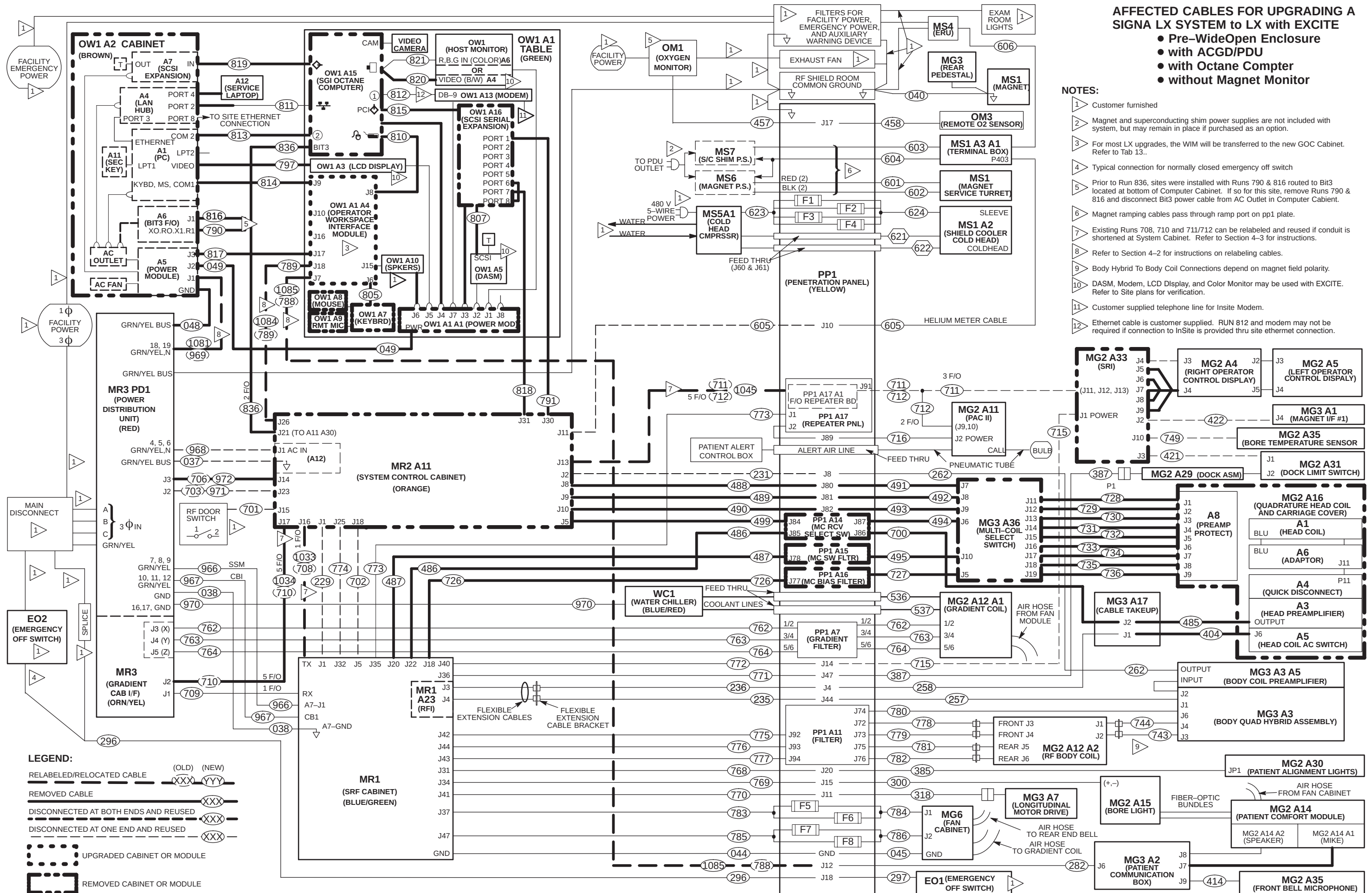
TABLE 4-5
HORIZON 5.x & 8.x CABLES RE-LABELED AND RE-LOCATED FOR UPGRADE TO LX w/ACGD

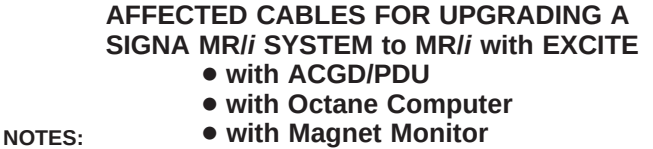
RUN	SYSTEM TYPE	ONE END		OTHER END	
		OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
703	5.x 8.x	RUN #703 PD1-A1-A2-J2 PD1-A14-A1-J2 46-258431G809	RUN #703/971 PD1-A1-A2-J2 PD1-A14-A1-J2 MR3-PD1-J2	RUN #703 MR2-A11-J23 46-228431G809	RUN #703/971 MR2-A11-J23 46-258431G941
706	5.x 8.x	RUN #706 MR1-PD1-J3 PD1-A14-A1-J3 46-258431G808	RUN #706/972 PD1-A1-A3-J2 PD1-A14-A1-J3 MR3-PD1-J3	RUN #706 MR2-A11-J14 46-258431G808	RUN #706/972 MR2-A11-J14 46-258431G940
709	5.x	RUN #709TR-1 MR3-A11-J1 46-317077G802	RUN #709TR-1 MR3CABI/FJ2 46-317077G803	RUN #709TR-1 MR1-A7-RX 46-317077G802	RUN #709TR-1 MR1-A7-RX 46-317077G803
710	5.x	RUN #710TR-1 MR3-A11-J2 46-317085G602	RUN #710TR-1 MR3CABI/FJ1 46-317085G603	RUN #710TR-1 MR2-A11-J17 46-317085G602	RUN #710TR-1 MR2-A11-J17 46-317085G603
710-1	5.x	RUN #710-1	RUN #710-1 MR3GIPJ2	NO CHANGE	NO CHANGE
710-2	5.x	RUN #710-2	RUN #710-2 MR3GIPJ3	NO CHANGE	NO CHANGE
710-3	5.x	RUN #710-3	RUN #710-3 MR3GIPJ4	NO CHANGE	NO CHANGE
710-4	5.x	RUN #710-4	RUN #710-4 MR3GIPJ1	NO CHANGE	NO CHANGE
710-5	5.x	RUN #710-5	RUN #710-5 MR3GIPJ5	NO CHANGE	NO CHANGE

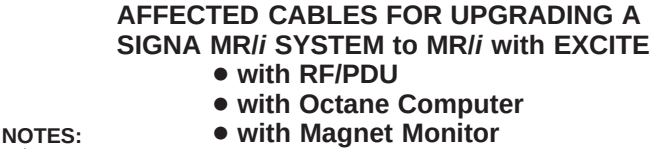
4-6 CABLE REMOVAL INTERCONNECT MAPS

The following “Cable Removal” interconnect maps refer only to cables removed, relabeled, and/or relocated when upgrading:

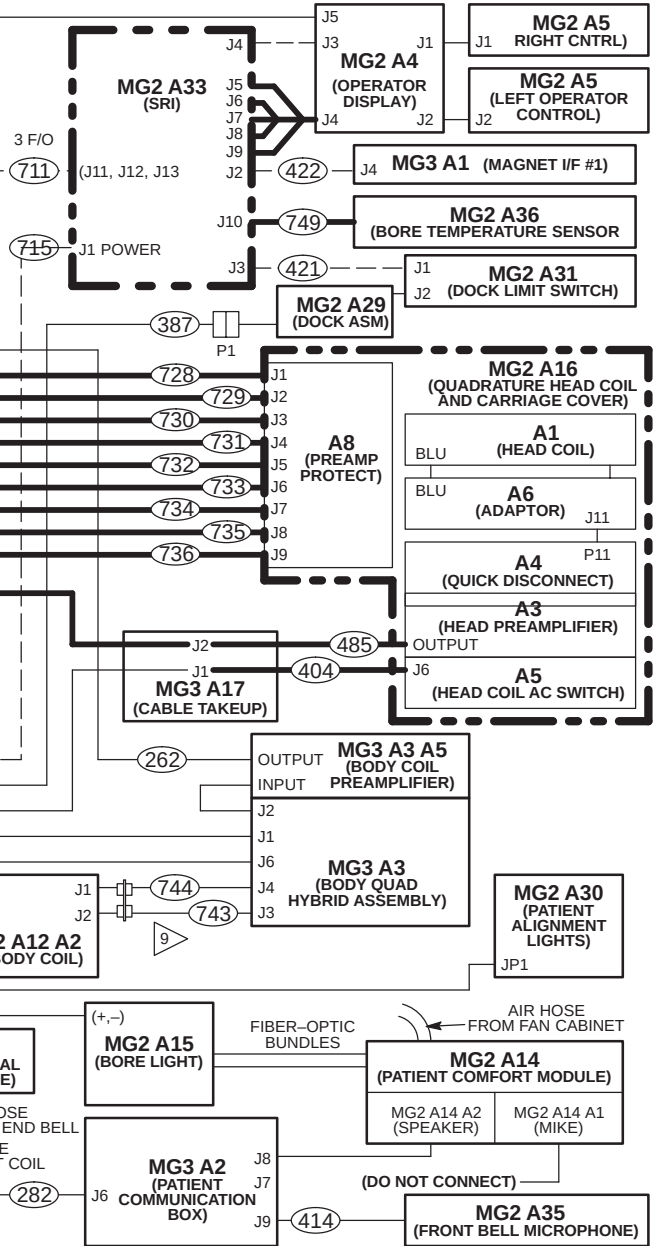
- Signa pre-WideOpen LX system with Octane, ACGD to Signa EXCITE Page 4-13
- Signa MR/i LX system with Octane, ACGD to Signa EXCITE Page 4-14
- Signa MR/i LX system with Octane, RF/PDU to Signa EXCITE Page 4-15
- Signa pre-WideOpen LX system with Octane, SGD, RF/PEN, PDU to Signa EXCITE Page 4-16
- Signa pre-WideOpen LX system with Indigo, SGD, RF/PEN, PDU to Signa EXCITE Page 4-17
- Signa Horizon 5.x system with Operator Console, Grad/GRAM, RF/PEN, PDU to EXCITE ... Page 4-18
- Signa Advantage 5.x and 4.x system to Signa EXCITE Page 4-19
- Signa LX system SGD, RF/PEN, PDU to Signa LX with ACGD Page 4-20
- Signa LX system Gradient/GRAM, RF/PEN, PDU to Signa LX with ACGD Page 4-21
- Signa Horizon 5.x system Gradient/GRAM, RF/PEN, PDU system to Signa LX with ACGD ... Page 4-22
- Signa Advantage 5.x and 4.x system to Signa LX with ACGD Page 4-23



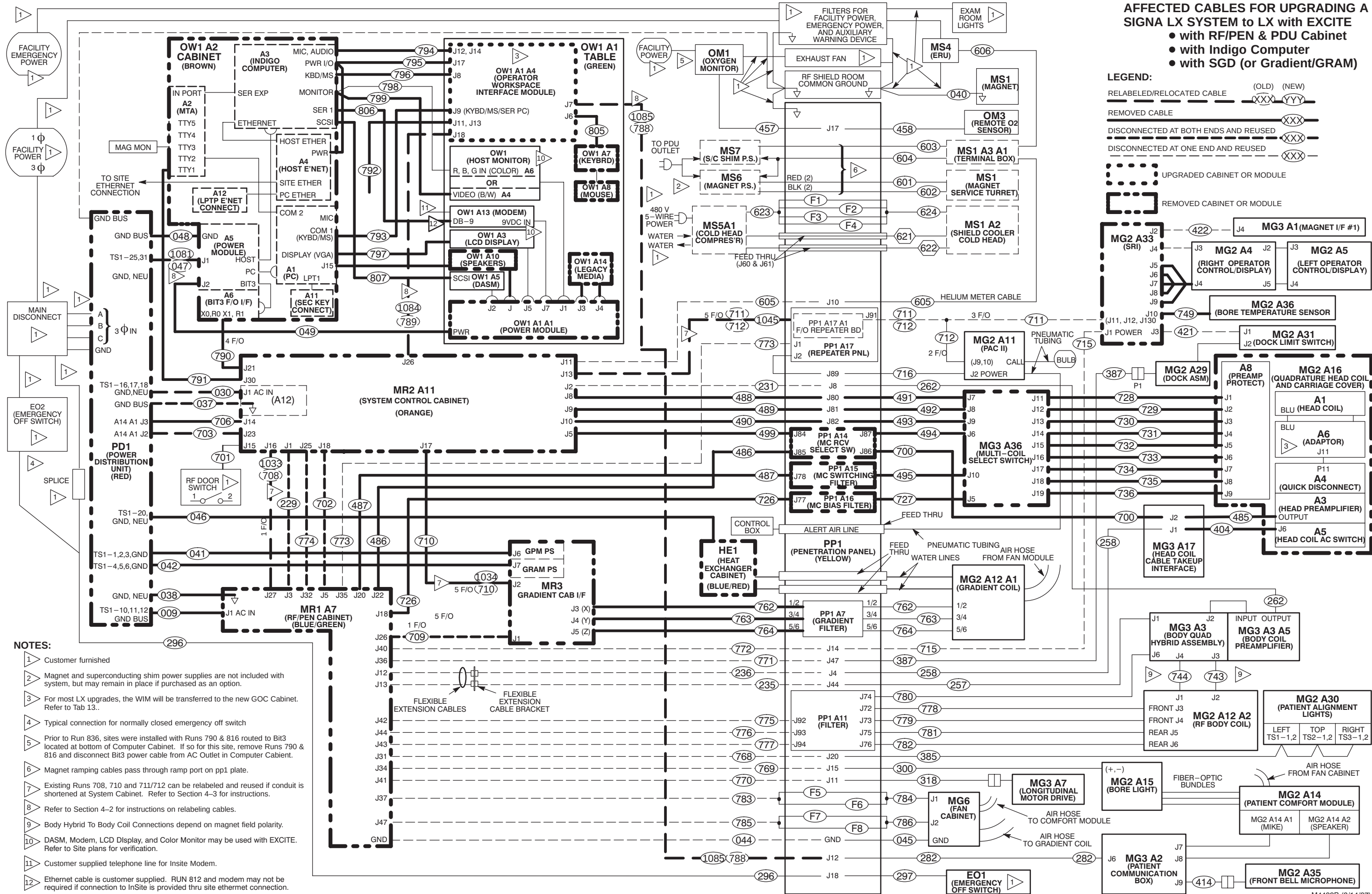


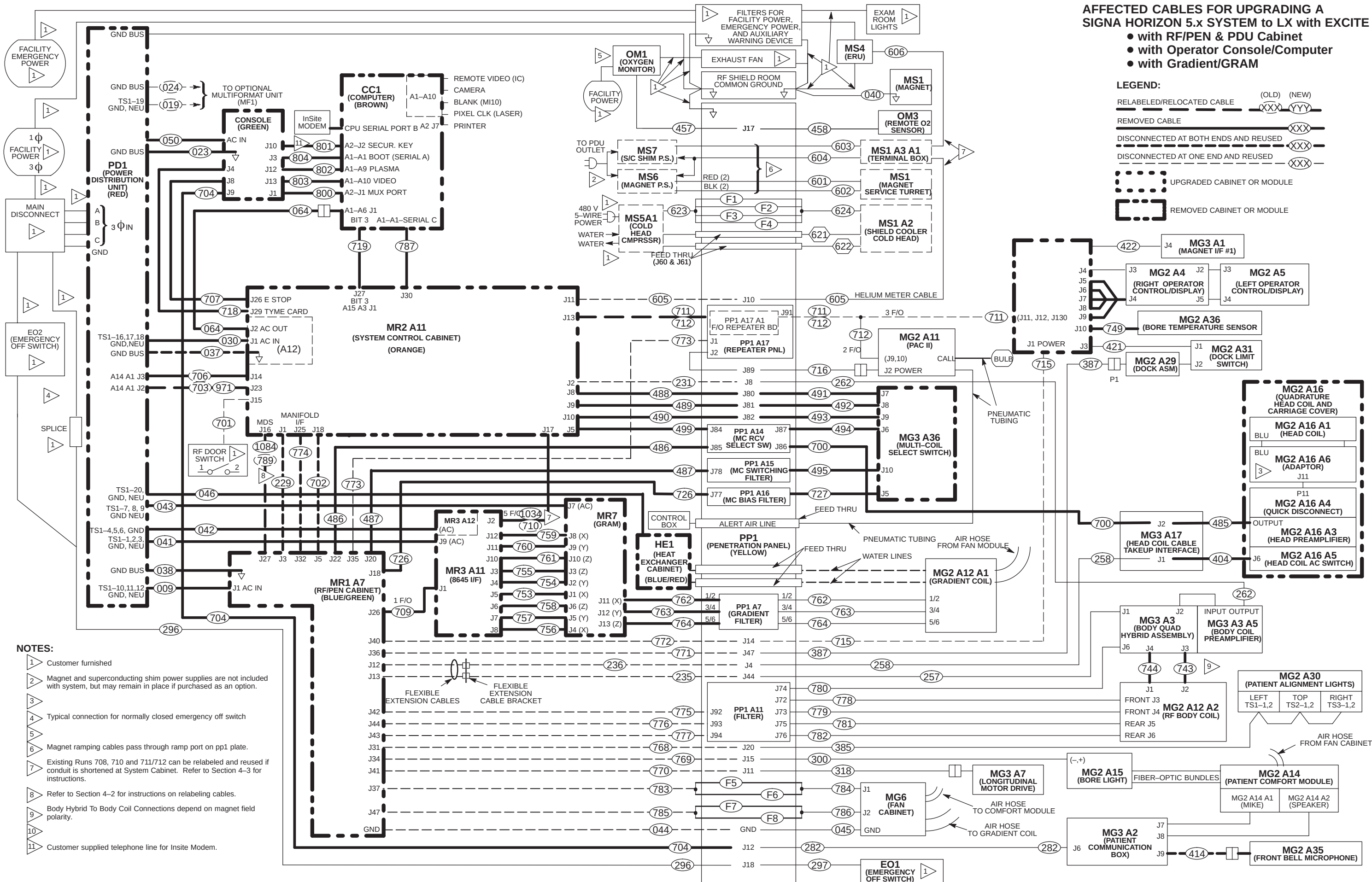


- 1 Customer furnished
- 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- 3 For most LX upgrades, the WIM will be transferred to the new GOC Cabinet. Refer to Tab 13..
- 4 Typical connection for normally closed emergency off switch
- 5 Prior to Run 836, sites were installed with Runs 790 & 816 routed to Bit3 located at bottom of Computer Cabinet. If so for this site, remove Runs 790 & 816 and disconnect Bit3 power cable from AC Outlet in Computer Cabinet.
- 6 Magnet ramping cables pass through ramp port on pp1 plate.
- 7 Existing Runs 708, 710 and 711/712 can be relabeled and reused if conduit is shortened at System Cabinet. Refer to Section 4-3 for instructions.
- 8 Refer to Section 4-2 for instructions on relabeling cables.
- 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 10 DASM, Modem, LCD Display, and Color Monitor may be used with EXCITE. Refer to Site plans for verification.
- 11 Customer supplied telephone line for Insite Modem.
- 12 Ethernet cable is customer supplied. Run 812 and modem may not be required if connection to InSite is provided thru site ethernet connection.



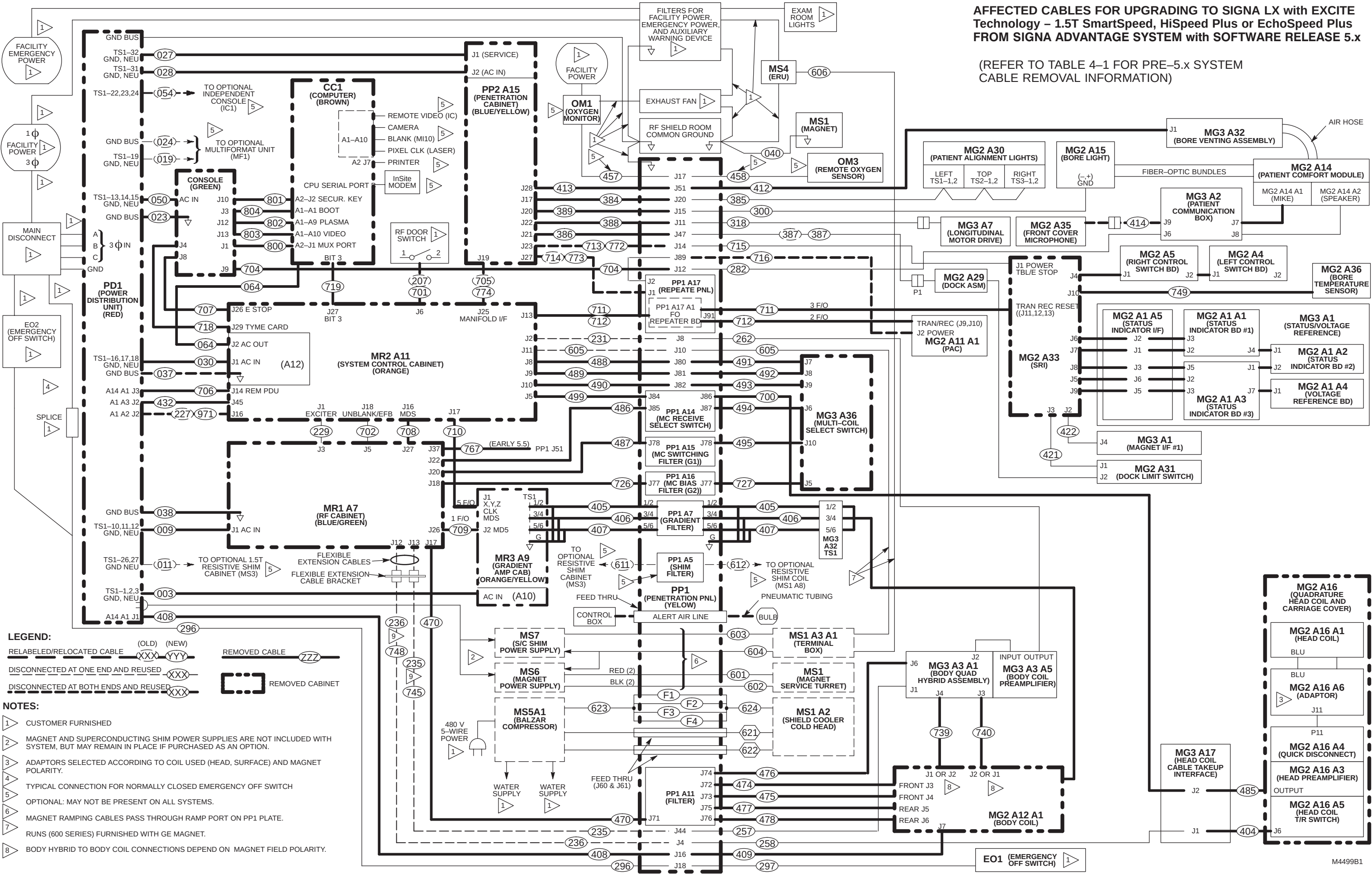


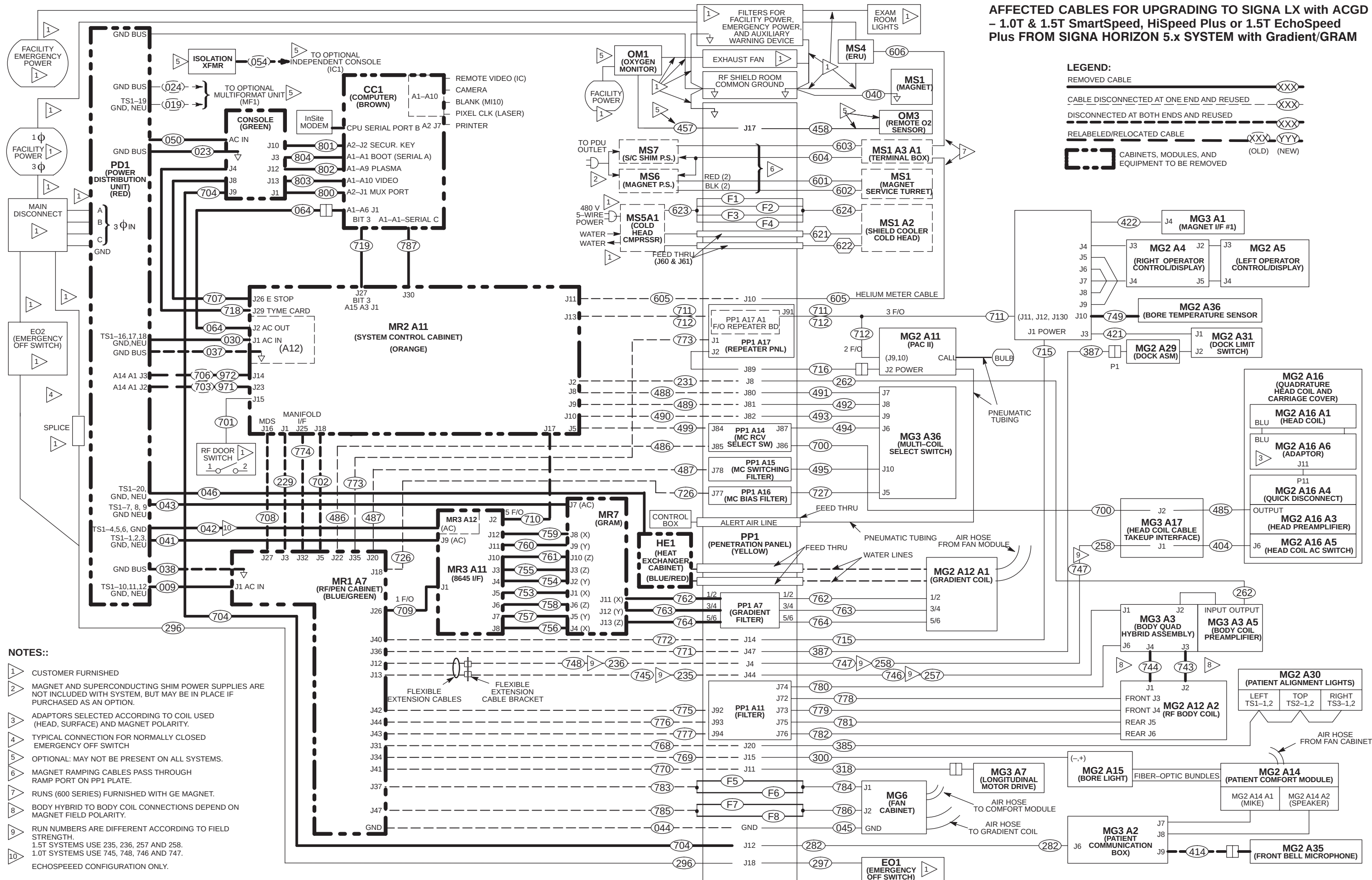




AFFECTED CABLES FOR UPGRADING TO SIGNA LX with EXCITE Technology – 1.5T SmartSpeed, HiSpeed Plus or EchoSpeed Plus FROM SIGNA ADVANTAGE SYSTEM with SOFTWARE RELEASE 5.x

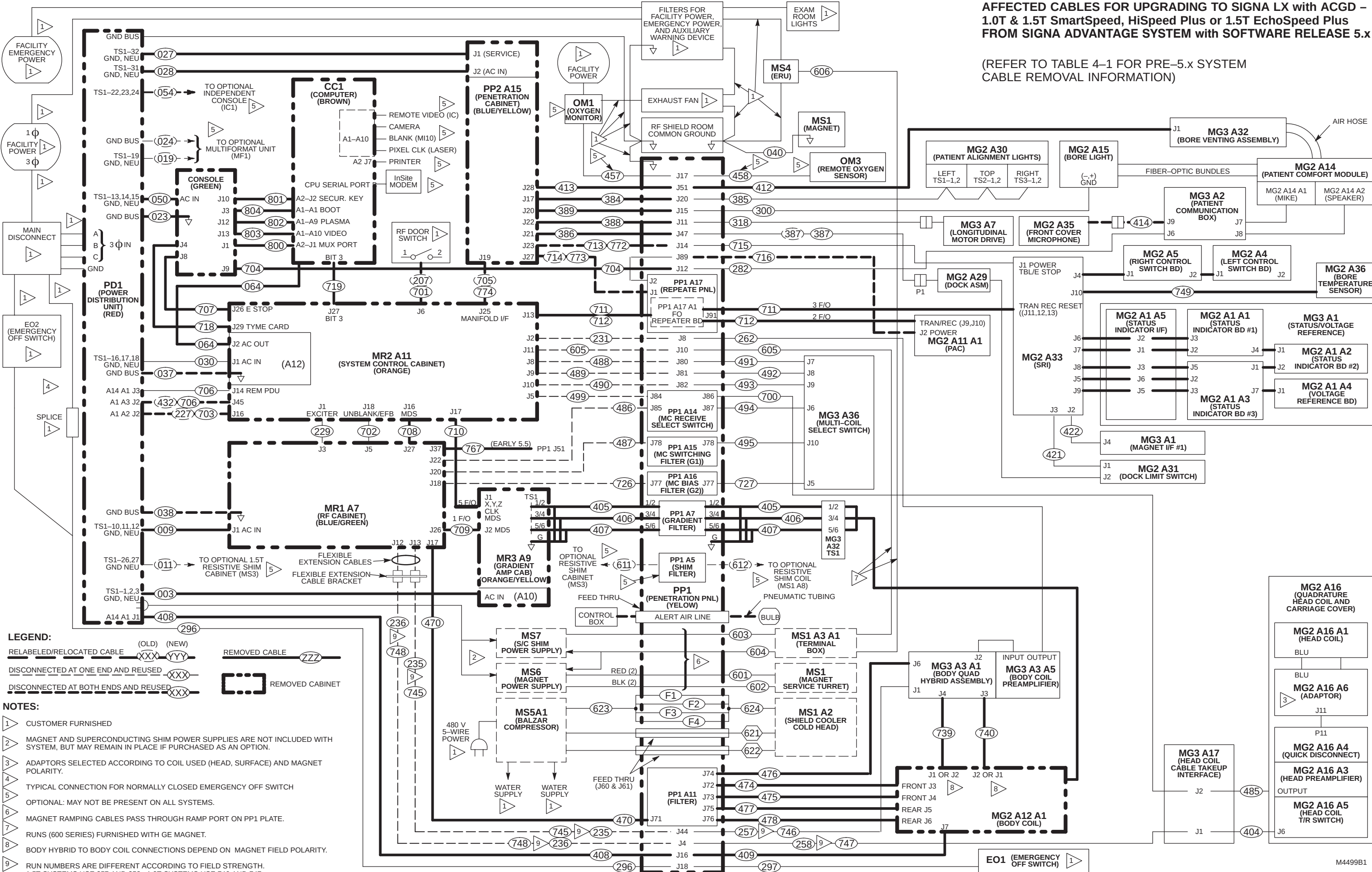
(REFER TO TABLE 4–1 FOR PRE-5.x SYSTEM CABLE REMOVAL INFORMATION)





AFFECTED CABLES FOR UPGRADING TO SIGNA LX with ACGD – 1.0T & 1.5T SmartSpeed, HiSpeed Plus or 1.5T EchoSpeed Plus FROM SIGNA ADVANTAGE SYSTEM with SOFTWARE RELEASE 5.x

(REFER TO TABLE 4–1 FOR PRE–5.x SYSTEM CABLE REMOVAL INFORMATION)



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4-7 IDENTIFYING NEW CABLES FOR SIGNA LX AND EXCITE

The procedures in this section assume Signa LX SmartSpeed, HiSpeed Plus, or EchoSpeed Plus with ACGD or LX 1.5T EXCITE upgrade components, appropriate Site Kit, and options are delivered complete with all cables present.

Cable shipment containers have color coded labels which identify the cable routing areas:

- Green – Equipment Room
- Purple – Operator's Room
- Orange – Magnet Room
- Yellow – Post-installation (Phantom's, Coils, etc.)

Table 4-6 lists all cables added for upgrading to Signa LX with ACGD SmartSpeed, HiSpeed Plus, or EchoSpeed Plus or LX 1.5T EXCITE from various early configurations. The letters on the right side of the table are defined as follows:

For sites that are upgrading to LX with 1.5T EXCITE, select from the following:

- A** – Cables installed if upgrading from ACGD/PDU Signa LX system to EXCITE
- B** – Cables installed if upgrading from RF/PDU Signa LX system to EXCITE
- C** – Cables installed if upgrading from Pre-RF/PDU Signa LX system to EXCITE
- D** – Cables installed if upgrading from Horizon 5.x system to EXCITE
- E** – Cables installed if upgrading from Advantage 5.4 or Horizon 5.x software only system to EXCITE
- F** – Cables installed if upgrading from Advantage 4.x system to EXCITE

For sites that are upgrading to LX, select from the following:

For sites upgrading from Indigo to Octane, the cables in column must be installed

- G** – Cables installed if upgrading from Indigo to Octane2 computer, M1033L. Instructions for this upgrade are delivered with this option.

Select either column C, D, E, or F for remaining cables to be installed.

- H** – Cables installed if upgrading from Pre-RF/PDU Signa LX system to LX
- I** – Cables installed if upgrading from Horizon 5.x system to LX
- J** – Cables installed if upgrading from Advantage 5.4 or Horizon 5.x software only system to LX
- K** – Cables installed if upgrading from Advantage 4.x system to LX



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

Many of the cables are common to both the LX and EXCITE System. The cables that are specific to LX and EXCITE installation are designated in the applicable cable maps and in the first column of the following tables as:

- (1) Cable used on LX installation only
- (2) Cable used on EXCITE installation only

Other general notes are indicated in the first column as follows:

- (3) See Section 4-7 table for connection to TYME boards within the LX System Cabinet.
- (4) For Mobile and T/R only.

4-7 IDENTIFYING NEW CABLES FOR SIGNA LX AND EXCITE (continued)

TABLE 4-6
CABLES ADDED FOR UPGRADING TO SIGNA LX OR LX with EXCITE

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K
038	MR1-A7-GND	PD1-GND BUS					X	X				X	X
044	MR1-A7-GND	PP1-GND					X	X				X	X
045	PP1-GND	MG6 GND (FAN)					X	X				X	X
048	OW1-A2-A5-GND	PD1-GND									X	X	X
404	MG3-A17-J1	MG2-A16-A5-J6	X	X	X	X	X	X					
488	PP1-J80	MR2-A11-J8	X	X	X	X	X	X					
489	PP1-J81	MR2-A11-J9	X	X	X	X	X	X					
490	PP1-J82	MR2-A11-J10	X	X	X	X	X	X					
491	PP1-J80	MG3-A36-J7	X	X	X	X	X	X					
492	PP1-J81	MG3-A36-J8	X	X	X	X	X	X					
493	PP1-J82	MG3-A36-J9	X	X	X	X	X	X					
499	PP1-A14-J84	MR2-A11-J5	X	X	X	X	X	X					
702	MR2-A11-J18	MR1-A7-J5						X					X
708 ⁽¹⁾	MR2-A11-J16	MR1-A7-TX										X	X
709	MR3-CAB-I/F-J2	MR1-A7-RX										X	X
710 ⁽³⁾	MR3-CAB-I/F-J1	MR2-A11-J17										X	X
711/ 712	PP1-A17-A1 F/O REPEATER BD	MG2-A33-J11,12,13 MG2-A11-A1-J9,10					X	X				X	X
711/ 712 (3)	MR2-A11-J13	PP1-A17-A1					X	X				X	X
715	MG2-A33-J1	PP1-J14						X					X
716	MG2-A11-A1-J2	PP1-J89						X					X
728	MG3-A36-J11	MG2-A16-OUT-J1	X	X	X	X	X	X					
729	MG3-A36-J12	MG2-A16-OUT-J2	X	X	X	X	X	X					
730	MG3-A36-J13	MG2-A16-OUT-J3	X	X	X	X	X	X					
731	MG3-A36-J14	MG2-A16-OUT-J4	X	X	X	X	X	X					
732	MG3-A36-J15	MG2-A16-OUT-J5	X	X	X	X	X	X					
733	MG3-A36-J16	MG2-A16-OUT-J6	X	X	X	X	X	X					
734	MG3-A36-J17	MG2-A16-OUT-J7	X	X	X	X	X	X					
735	MG3-A36-J18	MG2-A16-OUT-J8	X	X	X	X	X	X					
743	MG2-A12-A1-J1	MG3-A3-J4					X	X				X	X
744	MG2-A12-A1-J2	MG3-A3-J3					X	X				X	X
749	MG2-A33-J10	MG2-A35		X	X	X	X	X		X	X	X	X
762	MR3CABI/FJ3	MG3-A32-TS1/2			X	X	X	X		X	X	X	X
763	MR3CABI/FJ4	MG3-A32-TS3/4			X	X	X	X		X	X	X	X
764	MR3CABI/FJ5	MG3-A32-TS5/6			X	X	X	X		X	X	X	X
768	PP1-J20	MR1-A7-J31					X	X				X	X
769	PP1-J15	MR1-A7-J34					X	X				X	X
770	PP1-J11	MR1-A7-J41					X	X				X	X
771	PP1-J47	MR1-A7-J36					X	X				X	X
772	PP1-J14	MR1-A7-J40						X					X
773	PP1-A17-J1	MR1-A7-J35						X					X

4-7 IDENTIFYING NEW CABLES FOR SIGNA LX AND EXCITE (continued)

TABLE 4-6 (CONT'D)
CABLES ADDED FOR UPGRADING TO SIGNA LX OR LX with EXCITE

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K
774	MR2-A11-J25	MR1-A7-J32						X					X
775	MR1-A7-J42	PP1-A11-J92					X	X				X	X
776	MR1-A7-J44	PP1-A11-J93					X	X				X	X
777	MR1-A7-J43	PP1-A11-J94					X	X				X	X
778	MG2-A12-A13-J3	PP1-A11-J72					X	X				X	X
779	MG2-A12-A13-J4	PP1-A11-J73					X	X				X	X
780	MG3-A3-J6	PP1-A11-J74					X	X				X	X
781	MG2-A12-A12-J5	PP1-A11-J75					X	X				X	X
782	MG2-A12-A12-J6	PP1-A11-J76					X	X				X	X
783	MR1-A7-J37	PP1-F5/F6					X	X				X	X
784	PP1-F5/F6	MG6-J1 (FAN)					X	X				X	X
785	MR1-A7-J18	PP1-F7/F8					X	X				X	X
786	PP1-F7/F8	MG6-J2 (FAN)					X	X				X	X
788	PP1-J12	OW1-A1-A2-J7									X	X	X
789	MR2-A11-J26	OW1-A1-A2-J18									X	X	X
791	MR2-A11-J30	OW1-A2-A4-EXP1									X	X	X
797	OW1-A3 (LCD DISPLAY)	OW1-A2-A1 (VIDEO)							X				
805	OW1-A1-A4-J6	OW1-A7 (KEYBOARD)							X				
807	OW1-A16	OW1-A5 (DASM)							X				
810	OW1-A1-A2-J8	OW1-A15-KBD,MOUSE							X				
811	OW1-A2-A4-PORT2	OW1-A15-ETHERNET							X				
812	OW1-A13 (MODEM)	OW1-A15-COM1							X				
813	OW1-A2-A1-COM2	OW1-A15-COM2							X				
814	OW1-A1-A4-J9	OW1-A2-A1-KBD,MOUSE							X				
815	OW1-A16-SCSI	OW1-A15-SCSI (PCI)							X				
817	OW1-A1-A2-J17	OW1-A2-A5-J3							X				
818	MR2-A11-J31	OW1-A16-PORT6,7,8									X	X	X
819	OW1-A2-A7-SCSI	OW1-A15-SCSI							X				
820	OW1-A15-MONITOR	OW1-A4 (B/W)							X				
821	OW1-A15-MONITOR	OW1-A6-R,B,G (COLOR)							X				
836	OW1-A15-BIT3	MR2-A11-A30-BIT3									X	X	X
966	MR3-PD1-7,8,9,GND	MR1-A7-J1		X	X	X	X	X		X	X	X	X
967	MR3-PD1-10,11,12,GND	MR1-CB1		X	X	X	X	X		X	X	X	X
968	MR3-PD1-4,5,6,GND	MR2-A12-J1		X	X	X	X	X		X	X	X	X
969	MR3-PD1-18,19,GND	OW1-A2-A5-J1		X	X	X	X	X		X	X	X	X
970	MR3-PD1-16,17,GND	WC1		X	X	X	X	X		X	X	X	X
973 ⁽⁴⁾	MR3-PD1-16,17,GND	HE1		X	X	X	X	X		X	X	X	X
1011 ⁽²⁾	MG2-A33-J5	MG2-A4-J4	X	X	X	X	X	X					
1025 ⁽²⁾	MR2-A11-J40	PP1-J6	X	X	X	X	X	X					
1026 ⁽²⁾	MR2-A11-J41	PP1-J7	X	X	X	X	X	X					
1027 ⁽²⁾	MR2-A11-J42	PP1-J58	X	X	X	X	X	X					

4-7 IDENTIFYING NEW CABLES FOR SIGNA LX AND EXCITE (continued)

TABLE 4-6 (CONT'D)
CABLES ADDED FOR UPGRADING TO SIGNA LX OR LX with EXCITE

RUN#	FROM	TO	A	B	C	D	E	F	G	H	I	J	K
1028 ⁽²⁾	MR2-A11-J43	PP1-J59	X	X	X	X	X	X					
1029 ⁽²⁾	MR2-A11-J62	PP1-A16-J77	X	X	X	X	X	X					
1030 ⁽²⁾	MR2-A11-J61	PP1-A16-J78	X	X	X	X	X	X					
1033 ⁽²⁾	MR2-A11-J16	MR1-A7-TX	X	X	X	X	X	X					
1034 ⁽²⁾	MR2-A11-J17	MR3-A11-J1	X	X	X	X	X	X					
1035 ⁽²⁾	MG2-A16-A3/A6-J4	MG3-A36-J21	X	X	X	X	X	X					
1036 ⁽²⁾	PP1-J84	MG3-A36-J6	X	X	X	X	X	X					
1037 ⁽²⁾	PP1-J6	MG3-A36-J1	X	X	X	X	X	X					
1038 ⁽²⁾	PP1-J7	MG3-A36-J2	X	X	X	X	X	X					
1039 ⁽²⁾	PP1-J58	MG3-A36-J3	X	X	X	X	X	X					
1040 ⁽²⁾	PP1-J59	MG3-A36-J4	X	X	X	X	X	X					
1041 ⁽²⁾	PP1-A15-J78	MG3-A36-J10	X	X	X	X	X	X					
1042 ⁽²⁾	PP1-A16-J77	MG3-A36-J5	X	X	X	X	X	X					
1045 ⁽²⁾	MR2-A11-J13	PP1-A17-A1	X	X	X	X	X	X					
1049 ⁽²⁾	MG3-A36-J19	MG2-A16-A8-J9	X	X	X	X	X	X					
1053 ⁽²⁾	MR2-A11-J54	PP1-J85	X	X	X	X	X	X					
1054 ⁽²⁾	MG3-A36-J20	MG2-A33-J15	X	X	X	X	X	X					
1056 ⁽²⁾	OW1 A17 PRT6	OW1-A12 ETHRNET	X	X	X	X	X	X					
1057 ⁽²⁾	MG3-A8-J1	PP1-J85	X	X	X	X	X	X					
1059 ⁽²⁾	OW1 A17 PRT2	OW1-A15 ETHRNET	X	X	X	X	X	X					
1060 ⁽²⁾	OW1 A19 J3	OW1 A21 J17	X	X	X	X	X	X					
1061 ⁽²⁾	OW1 A15 KYBD/MS	OW1 A21 J8	X	X	X	X	X	X					
1062 ⁽²⁾	OW1 A20	OW1 A21 J6	X	X	X	X	X	X					
1063 ⁽²⁾	OW1 A15 Video 1	OW1 A6 HOST LCD	X	X	X	X	X	X					
1064 ⁽²⁾	OW1 A15 SCSI 1	OW1 A16 SCSI TOWER	X	X	X	X	X	X					
1065 ⁽²⁾	OW1 A11 J2	OW1 A6 Pwr	X	X	X	X	X	X					
1066 ⁽²⁾	OW1 A15	OW1 A13	X	X	X	X	X	X					
1067 ⁽²⁾	OW1 A15 SCSI 2	OW1 A5 DASM	X	X	X	X	X	X					
1068 ⁽²⁾	OW1 A11 J8	OW1 A3 LCD	X	X	X	X	X	X					
1069 ⁽²⁾	OW1 A11 J8	OW1 A3 Pwr	X	X	X	X	X	X					
1070 ⁽²⁾	OW1 A11 J11	OW1 A16 Pwr	X	X	X	X	X	X					
1071 ⁽²⁾	OW1 A11 J1	OW1 A15 Pwr	X	X	X	X	X	X					
1074 ⁽²⁾	OW1 A19 J2	OW1 A11	X	X	X	X	X	X					
1075 ⁽²⁾	OW1 A19 J4	OW1 A11	X	X	X	X	X	X					
1078 ⁽²⁾	MG3-A36-J29,J30	MG2-A33-J18,J19	X	X	X	X	X	X					
1081 ⁽²⁾	OW1 A19 J1	MR3 PD1	X	X	X	X	X	X					
1082 ⁽²⁾	OW1 A15 ETH SLOT 2	MR2 A11 J63	X	X	X	X	X	X					
1083 ⁽²⁾	MR2 A11 J22	OW1 A15 XMT1	X	X	X	X	X	X					
1084 ⁽²⁾	OW1 A21 J6 via Run 1062	MR2 A11 J26	X	X	X	X	X	X					
1085 ⁽²⁾	OW1 A21 J7	PP1 J12	X	X	X	X	X	X					

4-8 TYME BOARD FIBER OPTIC CABLE LABEL CONNECTION INFORMATION FOR LX

Runs 708, 710, 711, and 712 are used for both Horizon 5.x and LX. The fiber optic connections to the TYME board are different for 5.x and lx. Table 4-7 below indicates the connection labels as they should be marked on the TYME board end of the fiber optic cables. Use only the LX ONE END connections.

TABLE 4-7
SYSTEM CABINET FIBER OPTIC CONNECTIONS

RUN	SYSTEM CABINET ONE END		OTHER END	
	5.x ONE END	LX ONE END	5.x OTHER END	LX OTHER END
708-1	MR2-A23-A2-J5	MR2-A30-A23-J19	NO CHANGE	NO CHANGE
710-1	MR2-A23-A2-J13	MR2-A30-A23-J12	NO CHANGE	NO CHANGE
710-2	MR2-A23-A2-J15	MR2-A30-A23-J13	NO CHANGE	NO CHANGE
710-3	MR2-A23-A2-J17	MR2-A30-A23-J15	NO CHANGE	NO CHANGE
710-4	MR2-A23-A2-J14	MR2-A30-A23-J17	NO CHANGE	NO CHANGE
710-5	MR2-A23-A2-J4	MR2-A30-A23-J18	NO CHANGE	NO CHANGE
711-1	MR2-A23-A2-J6,J7	MR2-A30-A23-J25,J27	NO CHANGE	NO CHANGE
711-2	MR2-A23-A2-J8	MR2-A30-A23-J23	NO CHANGE	NO CHANGE
712-1	MR2-A23-A2-J9,J10	MR2-A30-A23-J20,J21	NO CHANGE	NO CHANGE

4-9 MODIFYING CABLE RUN 300

As indicated in table 4-8, cut off ground lead at the Bore Light connection.

TABLE 4-8
CABLE RUN 300 MODIFICATION

RUN	ONE END REWORK	OTHER END REWORK
300	PP1-J15 NO CHANGE	MG2-A15 CUT OFF GRN/BLK GND LEAD AT JACKET

4-10 LOCATING AND UNPACKING CABLES

- Unpack and move power and ground cable boxes to the PDU area where they will be routed. Set Run 044 and 045 aside. Run 038 may not be needed if existing Run 038 is of sufficient length for location of MR1.
- Unpack and move fixed site magnet room interconnect cables to magnet room.
 - Move Run 045 Ground Cable into magnet room .
 - Move applicable magnet room box of Fixed Site Cables into magnet room.
 - Locate Body Coil RF input Cables (ship with Body Coil) and move cables to magnet room.
 - Move gradient cables into magnet room. Gradient cables will be cut, re-terminated and routed according to *Direction 2291189 (BRM Body Coil)* or *Direction 2291190 (BRM-D & CRM Body Coil)*, Tab 1.
- Unpack and move equipment room system cables into routing location (cabinets, Operator Workspace, PDU). Note that labels are supplied to re-number cables 701, 703/971, 706/972, 709, 710, 772, 773, and 774.
- Move remainder of gradient cables to equipment room. Gradient cables will be cut, re-terminated and routed according to, refer to instructions in Tab 1.
- Unpack and move fiber optic cables into routing location.

4-11 SIGNA LX AND EXCITE CABLE INFORMATION

This section assumes Signa LX or EXCITE System and options are delivered complete with all cables present.

Many of the cables are common to both the 9.x and 10.x System. The cables that are specific to LX or EXCITE installation are designated in the applicable cable maps and in the first column of the following tables as:

- * Cable used on LX installation only
- ** Cable used on EXCITE installation only

Cables shipment containers have color coded labels which identify the cable routing areas:

- Green – Equipment Room
- Purple – Operator's Room
- Orange – Magnet Room
- Yellow – Post-installation (Phantom's, Coils, etc.)



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

The table below provides definitions for the following cable tables.

COLUMN HEADING	DESCRIPTION	SPECIAL NOTES
RUN	The run number as seen on the delivered cable	The vendor cable may not have run numbers.
"FROM" & "TO"	Labels the destination of the cable	
DESCRIPTION	Could include information on length, number of conductors, type of cable, and end plug type	Total length of cable delivered to site, type of cable specifies power from data.
CABLE TYPE	Type designation for the cable	
VOLTAGE RATING	The highest voltage that may be continuously applied to a wire	Specific to the core cable manufacturer given.
ACTUAL VOLTAGE	Nominal for continuous; maximum for variable voltage	If voltage varies during use, the maximum is given.
TEMP RATING	Maximum temperature insulating material may be used without loss of basic properties	Specific to the core cable manufacturer given.
FLAME RATING	Measure of material's ability to support combustion	Specific to the core cable manufacturer given.
REMARKS	Describe the cable's use in the MR system	

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)

1. Unpack magnet room cables from the GE Magnet accessory box(s) and move them into the magnet or equipment room for routing. The shield cooler runs should have been installed previously when magnet was delivered.
 - a. For magnet and equipment room standard runs delivered with GE Magnet, refer to Table 4-12.
2. Unpack and move power and ground cable boxes to the PDU area where they will be routed. Set Run 040, 044, and 045 aside. Power and ground cables are listed in Table 4-9.

TABLE 4-9
PDU POWER AND GROUND CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
030	PD1 or MR1 PD1	MR2 A12	35 ft Power (5-#8)	SO	600	208Y	75	FT-4	System Cabinet Power
037	PD1 GND	MR2 A12 GND	36 ft Ground Wire (#10)	SO	600	0	105	VW-1	MR2 "permanent" (UL) ground
038	PD1 A2 A2	MR1 A7 GND	31 ft Ground Wire (#10)	SO	600	0	105	VW-1	SRF Cabinet ground
040	MSI	RF COM GND	69 ft Ground Wire (#10)	SO	600	208Y	105	VW-1	Magnet Ground Cable
044	PP1 GND	MR1 A7 GND	31 ft Ground Wire (#10)	SO	600	0	105	VW-1	RF/Pen Cabinet Separate ground
045	PP1 GND	MG6 GND	41 ft Ground Wire (#10)	SO	600	0	105	VW-1	Blower Cabinet Separate ground
048	PD1 GND	OW1 A2 A5 GND	51 ft Ground Wire (#10/1)	SO	600	0	105	VW-1	Operator Workspace Cabinet ground
049	OW1 A1 A1	OW1 A2 A5 J2	Power Cord	STO	600	120	90	VW-1	Operator Workspace Table to OW Cabinet
966	MR1 A7 J1	MR3 PD1	35 ft Power (5-#8)	SO	600	208Y	75	FT-4	ACGD SSM Module Power
967	MR1 CB1	MR3 PD1	35.5 ft Power (4-#8)	SO	600	208	90	FT-4	ACGD SRF Cabinet Power
968	MR1 A12 J1	MR3 PD1	35 ft Power (5-#8)	SO	600	208Y	75	FT-4	ACGD System Cabinet Power
969	OW1 A2 A5 J1	MR3 PD1	65.5 ft Power (4-#10)	SO	600	120	75	FT-4	ACGD Operator Workspace Cabinet Power
970	WC1	MR3 PD1	82 ft Power (3-#10)	STO	600	208	90	FT-4	ACGD Water Chiller Power
973	HE1	MR3 PD1	82 ft Power (3-#12)	SO	600	208	75	FT-4	ACGD Heat Exchanger Power
1081**	OW1 A19 J1	MR3 PD1	65.5 ft Power (4-#10)	SO	600	120	75	FT-4	Operator Workspace Power
N/A	PD1 MAG/SHIM OUTLET	MAG/SHIM POWER SUPPLY	50 ft Power (5-#8) With 30A Plug and Receptacle						Cable for Compact PDU to GE Magnet Power Supply or SC Shim Supply Power

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)

3. Unpack and move fixed site magnet room interconnect cables to magnet room. Refer to Table 4-10.

TABLE 4-10
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
262	PP1 J8	MG3 A3 A5 OUT	45 ft. RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	Body Coil receive
282	PP1 J12	MG3 A2 J6	45 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Patient Intercom
297	PP1 J18	EO1	90 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
300	PP1 J15	MG2 A15	60 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Bore Light power
318	PP1 J11	MG3 A7	45 ft. 9 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Long Drive Motor power
385	PP1 J20	MG2 A30	67 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Patient Alignment Light power
387	PP1 J47	MG2 A29 J1	60 ft. 25 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Table Elevation motor power
404	MG3 A17 J1	MG2 A16 A5 J6	13.67 ft. RG58/U Coax	CI2	1900 VRMS	<30	80	VW-1	Head transmit cable take up
414	MG3 A35	MG3 A2 J9	30 ft. 9-pin Sub-D	CI3	300	<30	80	FT-4	Front Cover Microphone
421	MG2 A33 J1	MG3 A31 J1	15 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	Dock Limit
422	MG2 A33 J2	MG3 A1 J4	25 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	Long Drive Encoder/Limit Sw.
458	PP1 J17	OM3	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional Remote O2 Sensor
485*	MG3 A17 J2	MG2 A16 A3 OUT	13.67 ft. RG58/C Coax	CI2	1900 VRMS	<30	80	VW-1	Head receive cable take up
491	PP1-J80	MG3-A36-J7	80 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
492	PP1-J81	MG3-A36-J8	80 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC3 Receive
493	PP1-J82	MG3-A36-J9	80 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
494*	PP1-A14-J87	MG3-A36-J6	80 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC1 Receive
495*	PP1-A15-J78	MG3-A36-J10	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Multi-coil Switching
700*	PP1 A14 J86	MG2/MG3 A17 J2	45 ft. RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	Head/Surface Coil receive
715	PP1 J14	MG2 A33 J1	65 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	SRI Power
716	PP1 J89	MG2 A11 A1 J2	65 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	PAC Power
727*	PP1-A16-J77	MG3-A36-J5	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Multi-coil Bias
728	MG3 A36 J11	MG2 A16 OUT J1	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
729	MG3 A36 J12	MG2 A16 OUT J2	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
730	MG3 A36 J13	MG2 A16 OUT J3	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
731	MG3 A36 J14	MG2 A16 OUT J4	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
732	MG3 A36 J15	MG2 A16 OUT J5	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
733	MG3 A36 J16	MG2 A16 OUT J6	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
734	MG3 A36 J17	MG2 A16 OUT J7	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)

TABLE 4-10 (CONTINUED)
MAGNET ROOM RUNS (NOT CUT TO LENGTH)

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
735	MG3 A36 J18	MG2 A16 OUT J8	13.67 ft. BNC	CI2	500 VRMS	<30	200	VW-1	RF Cable
736*	MG3 A36 J19	MG2 A16 A8 J9	12 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	RF Cable
743	MG2 A12 A1 J2	MG3 A3 A1 J3	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
744	MG2 A12 A1 J1	MG3 A3 A1 J4	10 ft Coax w/M&F "N"	STO	2500 VDC	950	85	VW-1	Body Coil Drive
749	MG2 A33 J10	MG2 A36	35 ft 3 cond w/ 9-Sub D	CI3	300	<30	105	FT-4	Bore Temperature Sensor
778	PP1 A11 J72	MG2 A12 A2 J3	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
779	PP1 A11 J73	MG2 A12 A2 J4	70 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil frnt)
780	PP1 A11 J74	MG3 A3 A3 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Inductive Drive
781	PP1 A11 J75	MG2 A12 A2 J5	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
782	PP1 A11 J76	MG2 A12 A2 J6	60 ft. MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	Dynamic Disable (body coil rear)
784	PP1 F5, F6	MG6 J1	40 ft. 2 cond power cable	SO	600	120	80	FT-4	Patient cooling blower power
786	PP1 F7, F8	MG6 J2	40 ft. 2 cond power cable	SO	600	120	80	FT-4	Gradient cooling blower power
848	MG2 A11 J4	MG2 A4 J5	16 ft. 25 pin Sub-D/rt. angl	CI3	300	<30	80	FT-4	Operator Display to PAC II
861*	MG2 A4 J4	MG2 A33J5,6,7,8,9	20 ft 5/37 pin Sub-D/80 pin	CI2	300	<30	80	VW-1	Operator Display to SRI
862	MG2 A4 J1	MG2 A5 J1	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Oper Display to Rt. Op Cntrl
863	MG2 A4 J2	MG2 A5 J2	2 ft. 26 pin Rt. angle	CI2	300	<30	80	VW-1	Oper Display to Lt. Op Cntrl
893	MG3 A3 J2	MG3 A3 A5 INPUT	6.6 Inch BNC		1900 VRMS	<30	80	VW-1	Splitter to Preamp Cable
897	MG2 A4 J3	MG2 A33 J4	20 ft. 37 pin Sub-D/26 pin	CI3	300	<30	80	FT-4	Operator Display to SRI
1011**	MG2 A4 J4	MG2 A33 J5	20 ft. Data Cable (80-#28)	CI2	300	<30	80	VW-1	SRI Splitter to Oper Display
1035**	MG2 A16 A3/A5 J4	MG3 A36 J21	40 ft. RG58/C Coax	CI2	1900 VRMS	<30	80	VW-1	Head or Surface Coil Receive
1036**	PP1 J84	MG3 A36 J6	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
1037**	PP1 J6	MG3 A36 J1	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
1038**	PP1 J7	MG3 A36 J2	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
1039**	PP1 J58	MG3 A36 J3	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
1040**	PP1 J59	MG3 A36 J4	BNC	CMX	1900 VRMS	<30	80	VW-1	MC4 Receive
1041**	PP1 A15 J78	MG3 A36 J10	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	SRI Power
1042**	PP1 A16 J77	MG3 A36 J5	80 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Cross Point Sw Cntl/Pwr
1043**	MG3 A8 J11	MG3 A5 J1	0.5ft Disc. Wires (13-#22)	CI3	300	<30	60	VW-1	Head T/R Pwr/Coil ID LED
1049**	MG3 A36 J19	MG2 A16 A8 J9	12 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Multi-Coil Select Cable
1054**	MG3 A36 J20	MG2 A33 J15	Data Cable (9-#22)	CI3	300	<30	80	vw-1	MXA Signal Cable
1057**	PP1 J85	MG3 A8 J1	32 ft. R223/U Coax	CMX	1900 VRMS	<30	80	VW-1	UNTS Antenna

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)

4. Unpack and move fiber optic cables into routing location. Refer to Table 4-11.

TABLE 4-11
FIBER OPTIC CABLES DELIVERED WITH FIXED-SITE KIT

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
708*	MR2 A11 J16	MR1 A7 TX	20 ft. 1/2 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
709	MR3 A11 J2	MR1 A7 RX	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
710*	MR2 A11 J17	MR3 A11 J1	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic and one plastic fiber optic cables
711/ 712*	MR2 A11 J13	PP1-J91	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
711/ 712	PP1-J91	MG2 A33 and MG2 A11 A1	100 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	One duplex fiber optic cables and one simplex fiber optic cable and One duplex plastic fiber optic cable for PAC-II
836*	MR2 A11 J30 BIT3	OW1 A15 BIT3	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1033**	MR2 A11 J16	MR1 A7 TX	20 ft. 1/2 in. fiber optic conduit	N/A	N/A	0	85	N/A	Single simplex plastic fiber optic cable
1034**	MR2 A11 J17	MR3 A11 J1	20 ft. 3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Four glass fiber optic and one plastic fiber optic cables
1044**	MR2 A11 J22	OW1 A15 SLOT2	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares
1045**	MR2 A11 J13	PP1 A17 A1	3/4 in. fiber optic conduit	N/A	N/A	0	85	N/A	Two duplex fiber optic and one simplex fiber optic cables
1078**	MG2 A33 J30/J29	MG3 A36 J18/J19	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables, SRI to Cross Point Switch
1083**	MR2 A11 J22	OW1 A15 XMT1	72.2 ft. 3/8 in. jacketed cable	N/A	N/A	0	85	N/A	Two fiber optic cables with two spares

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)

TABLE 4-12
GE MAGNET CABLES

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
601	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Positive 4/0
602	MS8	MS1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp cables, Power Assembly Negative 4/0
603	MS7	MS1 A3 A1	Power Cable (#4/0)	SO	600	208	80	VW-1	Ramp and Shim Cable Assembly
604	MS7 or MS8	MS1 A3 A1	Data Cable (12-#22)	CI3	150	60 VDC	80	FT-4	Switch Heaters
605	MS1	MR2 A8	Data Cable (4-#22)	CI3	300	<30	105	FT-4	Instrumentation Cable, 80 feet
606	MS4	MS1 A3 A1	Data Cable (4-#22)	CI3	300	<30	105	FT-4	Emergency Run Down Unit
608	MS1	MR2 A8	Data Cable (4-#22)	CI3	300	<30	105	FT-4	Instrumentation Cable, 15 feet
609	MS7 OR MS8	PDU	Power Cable (5-#16)	SO	300	208Y	105	FT-4	Feeder Cables
621	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Supply Flexible Gas Line (PP1 J60 or J61)
622	MS5 A1	MS1 A2	7/8" Gas Flex Line	STO	N/A	N/A	85	VW-1	Return Flexible Gas Line (PP1 J60 or J61)
623	MS5 A1	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Compressor to Pen Panel Cable (equipment room)
624	MS1 A2	PP1 F1, F2, F3, F4	Power Cable (4-#18)	SO	600	200	105	VW-1	Cold Head Control Cable

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)

5. Unpack and move interconnect cables which are cut to length to the applicable room. Refer to Table 4-13.

TABLE 4-13
EQUIPMENT ROOM AND MAGNET ROOM CABLES CUT TO LENGTH

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
235	MR1 A7 J13	PP1 J44	7/8 in. HELIAX® Coax	STO	6000 VDC	<30	120	VW-1	1.5T Body Coil Run created from kit
236	MR1 A7 J12	PP1 J4	1/2 in. HELIAX® Coax	STO	4000 VDC	<30	120	VW-1	1.5T Head Coil Run created from kit
257	PP1 J44	MG3 A3 A1 J1	7/8 in. HELIAX® Coax	STO	6000 VDC	<30	120	VW-1	1.5T Body Coil Transmit
258	PP1 J4	MG3 A17 J1	1/2 in. HELIAX® Coax	STO	4000 VDC	<30	120	VW-1	1.5T Head Coil Transmit
745	MG3 A7 J13	PP1 J44	50 ft 1/2" Helix (HN)	STO	4000 VDC	950	120	VW-1	1.0T Body RF Transmit
746	PP1 J44	MG3 A3 J1	55 ft 1/2" Helix (HN)	STO	4000 VDC	950	120	VW-1	1.0T Body Coil Transmit
747	PP1 J4	MG3 A17 J1	50 ft 3/8" Helix (N)	STO	2500 VDC	950	85	VW-1	1.0T Head Coil Transmit
748	MG3 A7 J12	PP1 J4	50 ft 3/8" Helix (N)	STO	2500 VDC	950	85	VW-1	1.0T Head RF Transmit
536	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
537	WC1	MG2 A12 A1	Flexible Tubing	N/A	N/A	N/A	N/A	N/A	Gradient Coil Cooling Line
762	MR3 CABI/F J3	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT-4	X Gradient output
762	MG3 A32 1/2	PP1 A7 1, 2	2 Conductor #2	STO	1000	600	90	FT-4	X Gradient output
763	MR3 CABI/F J4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT-4	Y Gradient output
763	MG3 A32 3/4	PP1 A7 3, 4	2 Conductor #2	STO	1000	600	90	FT-4	Y Gradient output
764	MR3 CABI/F J5	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT-4	Z Gradient output
764	MG3 A32 5/6	PP1 A7 5, 6	2 Conductor #2	STO	1000	600	90	FT-4	Z Gradient output

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)

6. Verify magnet monitoring cables for GE Magnet are delivered (or if previously installed, remain in place). Refer to Table 4-14.

TABLE 4-14
MAGNET MONITORING CABLES DELIVERED WITH MAGNET

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
823	MSM1 A1 J10	MR2 A11 J24	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to System Cabinet I/F
824	PP1 J10	FJ1	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor Sensor I/F to Penetration Panel
825	PP1 J48	MSM1 A1 J8	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Magnet Monitor connector to Penetration Panel
826	MSM1 A1 J9	FJ3	Data Cable (15-#22)	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
827	FJ3	MS5 A5 A1 J1 / FJ4	Data Cable (8-#22) & Data Cable (4-#22)	CI3	300	<30	80	FT-4	Compressor I/F
828	PP1 J10	MA1 A3 A1 P403	Data Cable (9-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Sensor
829	PP1 J48	MS1 FJ2	Data Cable (25-#22)	CI3	300	<30	80	FT-4	Penetration Panel to Magnet I/F
830	MS1 FJ2	MS1 A2 J2/ MS1 A5 A2 J1/ MG3 TB1	3cond/3cond/25cond	CI3	300	<30	80	FT-4	Magnet I/F
831	FJ1	MSM1 A1 J7	Data Cable (8-#24)	CI3	300	<30	80	FT-4	Monitor to Sensor I/F
832	MS1 A2 J1	MS1 SLEEVE/ MS1 COLDHEAD	Data Cable (8-#22)	CM	300	<30	80	FT-4	Buffer Amp to Cooling Sleeves
833	FJ4	MS5 A1 A6 JR	Data Cable (6-#24)	CM	300	<30	80	FT-4	Compressor to Compressor I/F
940	MSM3 RS232	MSM1 J1	6 ft. Data Cable (9-#22)	CI3	300	<30	80	FT-4	Monitor to Modem
	MR2 A11 J24	MR2 IPG J9/ MR2 PDU J5							System Cabinet I/F

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)

7. Unpack and move equipment room system cables into routing location (for cabinets, Operator Workstation, PDU etc.). Refer to Table 4-15.

TABLE 4-15
EQUIPMENT ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
229	MR2 A11 J1	MR1 A7 J3	20 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	Exciter RF output
231	MR2 A11 J2	PP1 J8	40 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	Body Coil receive
296	FACILITY DISCNCT	PP1 J18	45 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Emergency off switch
457	OM1	PP1 J17	9 pin Sub-D	CI3	300	<30	80	FT-4	Optional O2 Monitor cable
486*	PP1-A14-J85	MR1-A7-J22	30 ft. 9 Sub-D	CI3	300	<30	80	FT-4	Head/MC1 Receive Select Switch
487*	PP1-A15-J78	MR1-A7-J20	40 ft.15 Sub-D	CI3	300	<30	80	FT-4	Multi-coil Switching
488	PP1-J80	MR2-A11-J8	40 ft. BNC	CMX	300	<30	80	VW-1	MC2 Receive
489	PP1-J81	MR2-A11-J9	40 ft. BNC	CMX	300	<30	80	VW-1	MC3 Receive
490	PP1-J82	MR2-A11-J10	40 ft. BNC	CMX	300	<30	80	VW-1	MC4 Receive
499/233	MR2 A11 J5	PP1 J7	40 ft RG223/U Coax	CMX	300	<30	80	VW-1	Head and Surface Coil receive
701	MR2 A11 J15	RF Door switch	70 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	RF Door switch
702	MR2 A11 J18	MR1 A7 J5	20 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Unblank/EFB
703	MR2 A11 J23	PD1 A14 A1 J2/ MR1 PD1 J2	40 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	PDU Status
706*	MR2 A11 J14	PD1 A14 A1 J3/ MR1 PD1 J3	40 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Remote PDU
726*	PP1-A16-J77	MR1-A7-J18	30 ft. 15 Sub-D	CI3	300	<30	80	FT-4	Multi-coil Bias
768	MR1 A7 J31	PP1 J20	30 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	Patient Alignment Light power
769	MR1 A7 J34	PP1 J15	30 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	Bore Light power
770	MR1 A7 J41	PP1 J11	30 ft 9 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Long Drive Motor power
771	MR1 A7 J36	PP1 J47	30 ft 25 pin Sub-D	CI3	300	38.5 VDC	80	FT-4	Dock Power
772	MR1 A7 J40	PP1 J14	30 ft 37 pin Sub-D	CI3	300	<30	80	FT-4	SRI Power
773	MR1 A7 J35	PP1 A17 J1	30 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	PAC & Fiber Optic I/F Power
774	MR2 A11 J25	MR1 A7 J32	20 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	Hydraulic Manifold Interface
775	MR1 A7 J42	PP1 A11 J92	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	100V Dynamic Disable
776	MR1 A7 J44	PP1 A11 J93	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	100V Dynamic Disable
777	MR1 A7 J43	PP1 A11 J94	30 ft MHV Coax	CI2	1900 VRMS	1000 VDC	80	VW-1	100V Dynamic Disable
783	MR1 A7 J37	PP1 F5/F6	30 ft 2-#8 AWG	SO	600	120	80	FT-4	Blower Cabinet Power
785	MR1 A7 J18	PP1 F7/F8	30 ft 2-#8 AWG	SO	600	120	80	FT-4	Blower Cabinet Power
788*	OW1 A1 A4 J7	PP1 J12	40 ft Data (25-#22)	CI3	300	<30	80	FT-4	Intercom
789*	OW1 A1 A4 J18	MR2 A11 J26	51 ft Data (9-#22)	CI3	300	<30	80	FT-4	E-stop

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)

TABLE 4-15 (CONT'D)
EQUIPMENT ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
791*	OW1 A16 PORT1	MR2 A11 J30	51 ft. Data (25-#22)	CI3	300	<30	80	FT-4	TNF monitoring (Octane)
797*	OW1 A2 A1 PC	OW1 A3 LCD	2 line; 3.5 mm jacks	CI3	300	<30	80	FT-4	LCD Display
798*	OW1 A2 A3 HOST	OW1 A6	10.5 ft. Data (3-#24)	CI3	300	<30	80	FT-4	Color Monitor cable
805*	OW1 A1 A4 J6	OW1 A7	3.3 ft. Data	CI3	300	<30	80	FT-4	Keyboard cable
807*	OW1 A2 A3 HOST	OW1 A5	9.8 ft. Data	CI3	300	<30	80	FT-4	DASM SCSI cable
810*	OW1 A1 A2 J8	OW1 A15 KBD,MS	9 ft. 25 pin Sub-D	CI3	300	<30	80	UL-1581	Keyboard/Mouse cable (Octane)
811*	OW1 A2 A4	OW1 A15	12 ft. Ethernet	CI3	300	<30	80	FT-4	Communication
812*	OW1 A15	OW1 A13	12 ft. 25/9 pin Sub-D	CM	300	<30	80	UL-1581	Modem
813*	OW1 A2 A1 COM2	OW1 A415 PORT7	12 ft. 9 pin Sub-D	CI3	600	<30	105	VW-1	Communication
814*	OW1 A2 A1 KBD,MS	OW1 A1 A4 J9	9 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Keyboard/Mouse cable (PC)
815*	OW1 A16	OW1 A15 SCSI	6 ft. 68/50 pin	CI3	300	<30	80	FT-4	SCSI Cable (Octane)
817*	OW1 A2 A5 J3	OW1 A1 A2 J17	8 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	Workspace I/F Module
818*	OW1 A16 J6,7,8	MR2 A11 J31	70 ft. 25/3-25 pin	CI3	300	<30	80	FT-4	TNF monitoring
819*	OW1 A2 A7 SCSI	OW1 A15	10 ft. 68/50 pin	CI3	300	<30	80	FT-4	SCSI Cable
821*	OW1 A15 MON-ITOR	OW1 A6 R,B,G	10.5 ft. 3 BNC/SubD	CI3	300	<30	80	FT-4	Color Monitor cable
971	MR2 A11 J23	MR3 PD1 J2	40 ft 15 pin Sub-D	CI3	300	<30	80	FT-4	PDU Status (ACGD)
972*	MR2 A11 J14	MR3 PD1 J3	40 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	Remote PDU (ACGD)
1025**	MR2 A11 J40	PP1 J6	40 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
1026**	MR2 A11 J41	PP1 J7	40 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
1027**	MR2 A11 J42	PP1 J58	40 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
1028**	MR2 A11 J43	PP1 J59	40 ft. BNC	CMX	1900 VRMS	<30	80	VW-1	MC2 Receive
1029**	MR2 A11 J62	PP1 A16 J77	60 ft. 15 pin Sub-D	CI3	300	<30	80	FT-4	Monitor to Compressor I/F
1030**	MR2 A11 J61	PP1 A16 J78	30 ft. 37 pin Sub-D	CI3	300	<30	80	FT-4	SRI Power
1032	MR2-A11-J31	OW1-A16-FJ100	65 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Serial Communication
1046	FJ100	OW1-A16-PORT 1,6,7,8	4.5 ft. 25 pin Sub-D	CI3	300	<30	80	FT-4	Serial Communication
1053**	PP1 J85	MR2 A11 J54	34.5 ft RG223/U Coax	CMX	1900 VRMS	<30	80	VW-1	UNTS Antenna
1056**	OW1 A17 PRT6	OW1-A12 ETHRNET	3.3 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Laptop Access cable

4-11 SIGNA LX AND EXCITE CABLE INFORMATION (continued)TABLE 4-15 (CONT'D)
EQUIPMENT ROOM RUNS

RUN	"FROM"	"TO"	DESCRIPTION	CABLE TYPE	VOLTAGE RATING	ACTUAL VOLTAGE	TEMP RATING	FLAME RATING	REMARKS
1059**	OW1 A17 PRT2	OW1-A15 ETHRNET	5.9 ft. Ethernet Cat 5	CI3	30	<30	80	FT-4	Lan Hub cable
1060**	OW1 A19 J3	OW1 A21 J17	3.9 ft. 9 pin Sub-D	CI3	300	<30	80	FT-4	WIM Power cord
1061**	OW1 A15 KYBD/MS	OW1 A21 J8	2.9 ft. 25 pin Sub-D 2 PS/2s	CI3	300	<30	90	CL-2	Keyboard/Mouse cable
1062**	OW1 A20	OW1 A21 J6	13.2 ft. 50 pin SCSI, 9 pin Sub-D	CI3	300	<30	80	VW-1	SCIM cable
1063**	OW1 A15 Video 1	OW1 A6 HOST LCD	9.9 ft. 29 pin DVI, 15 pin VGA	CI3	300	<30	80	VW-1	Host LCD Display Video cable
1064**	OW1 A15 SCSI 1	OW1 A16 SCSI TOWER	9.9 ft. SCSI cable 50	CI3	300	<30	80	VW-1	SCSI Tower Data cable
1065**	OW1 A11 J2	OW1 A6 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host LCD Display Power cable
1066**	OW1 A15	OW1 A13	3.9 ft. 25/9 pin Sub-D	CM	300	<30	80	FT-4	Modem I/O cable
1067**	OW1 A15 SCSI 2	OW1 A5 DASM	5.9 ft. SCSI cable 50	CI3	30	<30	80	VW-1	DASM I/O cable
1068**	OW1 A15 Video2	OW1 A3 LCD	9.9 ft. Data (15-#28)	CI3	300	<30	80	VW-1	LCD Display Video cable
1069**	OW1 A11 J8	OW1 A3 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	LCD Display Power cable
1070**	OW1 A11 J11	OW1 A16 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	SCSI Tower Power cable
1071**	OW1 A11 J1	OW1 A15 Pwr	9.9 ft. Power (3-#18)	SVT	300	120	80	VW-1	Host Computer Power cable
1074**	OW1 A19 J2	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1075**	OW1 A19 J4	OW1 A11	3.3 ft. Power (3-#14)	SVT	600	120	105	VW-1	Power Strip cord
1082**	OW1 A15 ETH SLOT 2	MR2 A11 J63	75 ft. Ethernet Cat 5	CI3	300	<30	80	FT-4	EXCITE Ethernet C'cable
1084**	OW1 A21 J6 via Run 1062	MR2 A11 J26	60 ft 9 pin Sub-D	CI3	300	<30	80	FT-4	E-stop
1085**	OW1 A21 J7	PP1 J12	65 ft 25 pin Sub-D	CI3	300	<30	80	FT-4	Intercom

8. Unpack additional equipment room cables for Signa LX Options:

- a. Refer to the service/installation manual shipped with options.

4-12 SORTING AND ROUTING CABLES



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

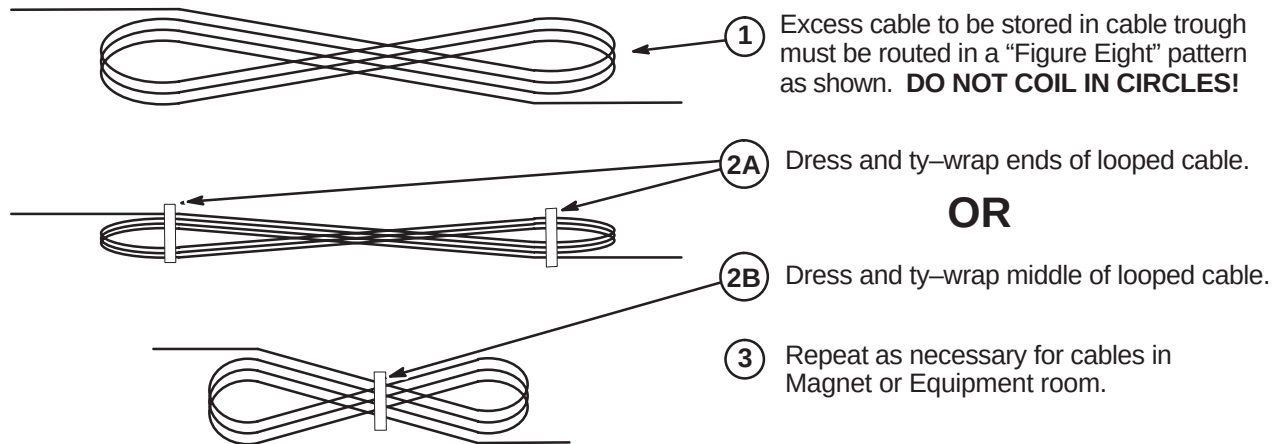
System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note

Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 2-1. Some cables are usually cut to length such as gradient cables, power cables, Helix, and Fiber-optic cables.



PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 2-1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.

4-13 SIGNA LX AND EXCITE CABLE INTERCONNECT MAPS

For Signa LX and EXCITE cable installation, fold-out Cable Interconnect Maps are provided on pages 4-43 thru 4-46. These maps show the final system interconnects for LX and EXCITE on WideOpen and pre-WideOpen systems.

Pre-5.4 Signa Advantage systems upgraded to LX or EXCITE will have many additional new cables to install and re-label, which would be very difficult to show in a cable map form. Refer to table 4-1 for information on cable removal.

- For EXCITE, refer to tables 4-2 and 4-3 for affected cables to be relabeled and the appropriate column in table 4-6 for new cables to be installed.
- For LX, refer to tables 4-4 and 4-5 for affected cables to be relabeled and the appropriate column in table 4-6 for new cables to be installed.

From any early configuration, the cable maps listed below illustrate the final configuration of various magnet systems after the completion of the Signa LX with ACGD or EXCITE upgrade. The cable maps illustrate the Signa LX or EXCITE system interconnect for all supplied cables and interface with customer supplied wiring.

The Signa 1.5T EXCITE Upgrade Interconnect Cable Maps can be found on the following pages:

- Page 4-42 Signa EXCITE System Interconnect – 1.5T SmartSpeed, HiSpeed, or EchoSpeed – Non-WideOpen Enclosure without Magnet Monitor
- Page 4-43 Signa EXCITE MR/i System Interconnect – 1.5T SmartSpeed, HiSpeed, or EchoSpeed – WideOpen Enclosure with Magnet Monitor

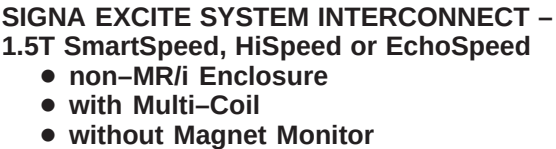
The Signa LX with ACGD Upgrade Interconnect Cable Maps can be found on the following pages:

- Page 4-44 Signa LX with ACGD System Interconnect – 1.5T SmartSpeed, HiSpeed, or EchoSpeed – Non-WideOpen Enclosure without Magnet Monitor
- Page 4-45 Signa LX with ACGD MR/i System Interconnect – 1.5T SmartSpeed, HiSpeed, or EchoSpeed – WideOpen Enclosure with Magnet Monitor

Use only the applicable Upgrade Interconnect Cable Map that applies to the Signa LX or 1.5T EXCITE system being upgraded. Tape map in a convenient location to check off routed cable. The fold-out map should be returned later to the binder for future reference.

Interconnects for additional cables supplied with other options (Spectroscopy, etc.) are illustrated in the installation manual shipped with the option.

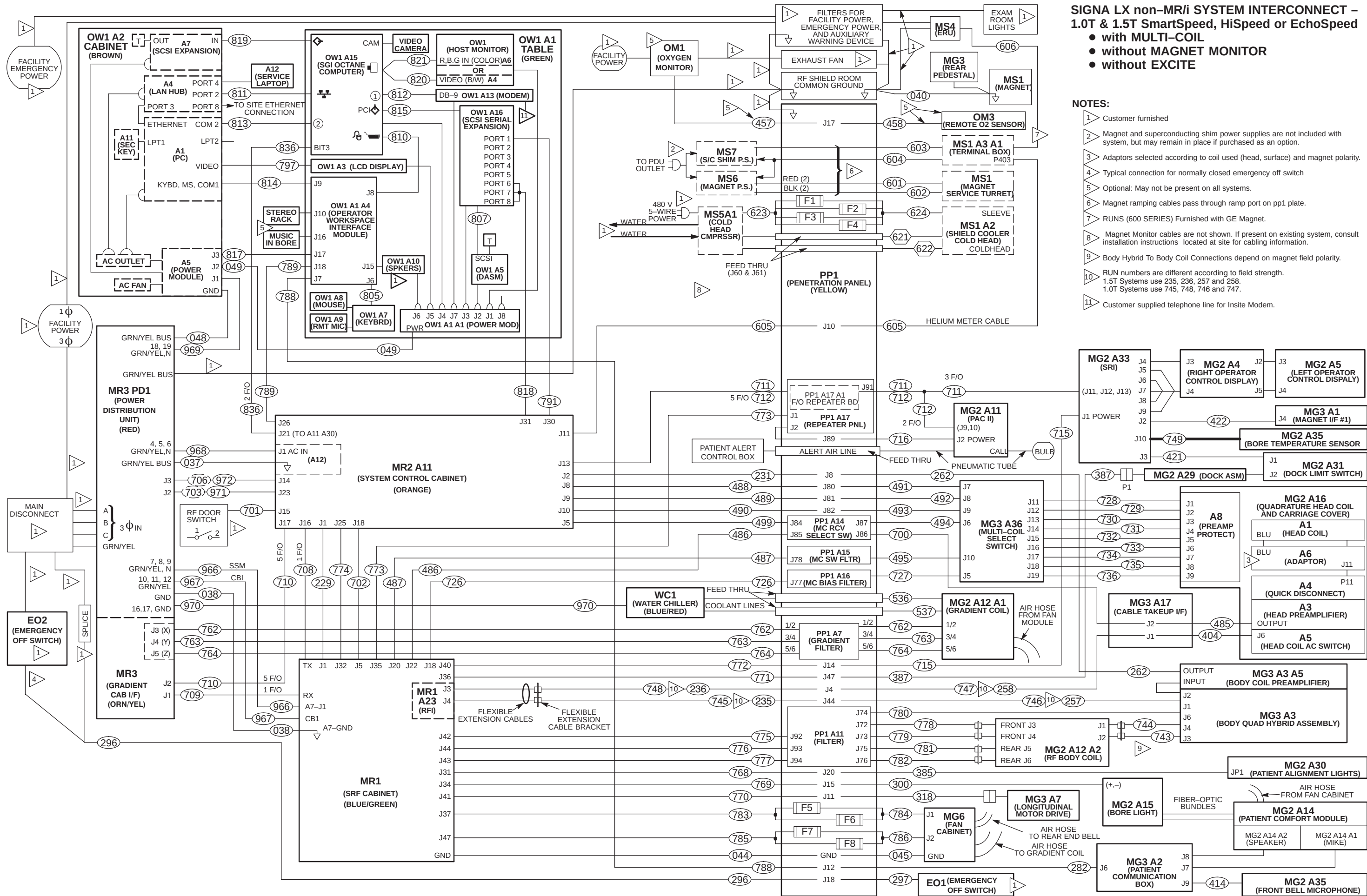
Procedures for connections made at the Magnet Enclosure, Operator Workspace, cabinets, etc. are detailed in the respective installation instructions (PIP Direction) within this manual within the appropriate Tab. Refer to upgrade flowcharts in Section 3.

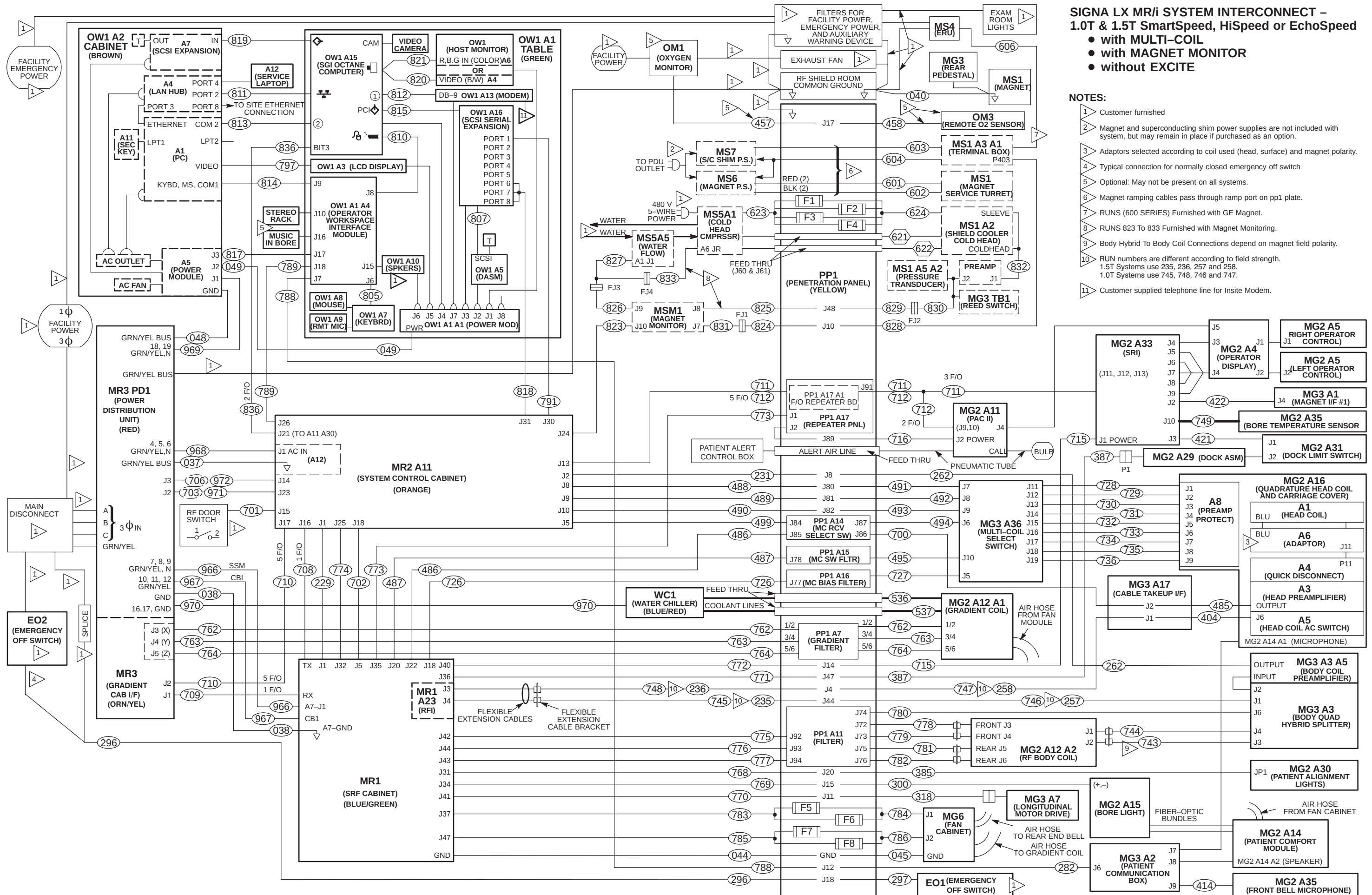


- 1 Customer furnished
- 2 Magnet and superconducting shim power supplies are not included with system, but may remain in place if purchased as an option.
- 3 Adaptors selected according to coil used (head, surface) and magnet polarity.
- 4 Typical connection for normally closed emergency off switch
- 5 Optional: May not be present on all systems.
- 6 Magnet ramping cables pass through ramp port on pp1 plate.
- 7 RUNS (600 SERIES) Furnished with GE Magnet.
- 8 RUNS 823 To 833 Furnished with Magnet Monitoring.
- 9 Body Hybrid To Body Coil Connections depend on magnet field polarity.
- 11 Customer supplied telephone line for Insite Modem.
- 15 Ethernet cable is customer supplied. RUN 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection.



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SECTION 5 – COMPLETION

5-1 PHANTOMS AND SERVICE KITS

1. Locate SPT Phantom Cart Kit.
2. Unpack the phantoms carefully to reduce risk of damage, breakage, or chemical spill.
3. The MSDS babel, in eight different languages, is provided with the phantoms. Select the appropriate language label for the site and apply to phantom.
4. Place service phantoms into SPT Phantom Cart or in a service storage cabinet to be organized and ready for be used during applicable checks and calibrations. The DQA Phantom is to be stored in a customer designated location.
5. Locate TPS RF I/F Kit (if applicable) and Signa Spares Kit. Place kits into service storage cabinet for ready use.

5-2 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Set-Up & Calibration, and perform the following:
 - PDU Voltage Checks Compact
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the ACGD Gradient power modules, switch all cabinet circuit breakers to ON. The ACGD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. Make sure the UPS circuit breaker is switched to the ON position. This is required to feed power to the Operator Workspace.

5-3 CALIBRATION AND SYSTEM CHECKS

Refer to Installation Flow Chart, Section 3, to perform:

- Software load and Configuration file edit.
- Functional checks
- Set-up and calibration
- System performance checks

5-4 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 5-5. Recycling Operation will route them to appropriate location.

5-5 MATERIAL DISPOSITION GUIDANCE

The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

Note

Disposed material includes items in Sections 5-6 through 5-10, MATERIAL RETURN CHECKLIST.

Note

The following equipment or material removed by the upgrade are ***required to be returned*** to GoldSeal:

- Octane Computer and Duplex with all associated cables and modules
- All parts, cables, modules, and equipment deinstalled for the EXCITE upgrade
- SGD or Gradient Cabinet
- System Cabinet
- Indigo Computer with all associated boards/cables (includes reflex board, cable kit, and audio cables)

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal.

The process for returning material is as follows:

1. Contact either of the following for disposition.
 - GoldSeal – Contact the MR Product Specialist at GoldSeal Logistics phone number 877-251-4468. If GoldSeal is accepting the returned equipment then all transportation arrangements and packaging will be processed through them. If GoldSeal is not accepting the return then they will advise you to contact GEMS Renewable Resources.
 - GEMS Renewable Resources – For coordination of transportation arrangements and packaging. Phone (414)-747-6997 or Dial Comm 8*579-6997; Fax (414)-747-6855; Field ASPEN 1-800-525-1516, Box #26041.
2. When calling, the following information is needed.
 - FDO (Future Delivery Order) number under which the upgrade was shipped.
 - Site name, address, and phone numbers.
 - Date of completed upgrade.
 - Name and Beeper number of Field Engineer that could be available when the equipment is picked up.
3. Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.
4. Process product locator information for all serialized components removed from the system according to policy.

5-6 MATERIAL RETURN CHECKLIST, SIGNA ADVANTAGE 4.x TO SIGNA LX or EXCITE UPGRADE

The following items are designated to be put on return pile during the upgrade. Refer to Section 5-5, MATERIAL DISPOSITION GUIDANCE.

- ☐ 1. PDU Cabinet
- ☐ 2. Penetration Cabinet
- ☐ 3. RF Cabinet less RF Amplifier
- ☐ 4. Gradient Cabinet
- ☐ 5. System Cabinet
- ☐ 6. Boot Terminal monitor and keyboard
- ☐ 7. Computer Cabinet(s)
- ☐ 8. T3 Exchange Boards
- ☐ 9. Computer Cabinet
- ☐ 10. Cardiac Monitor
- ☐ 11. Operator Console
- ☐ 12. Gated Studies/PAC Asm
- ☐ 13. Carriage (early version)
- ☐ 14. Carriage Cover
- ☐ 15. SRI Module (if applicable)
- ☐ 16. Rear Pedestal
- ☐ 17. Patient Communication Box
- ☐ 18. Penetration Panel
- ☐ 19. Select Switch
- ☐ 20. Comfort Module
- ☐ 21. Body Coil On Cart
- ☐ 22. Magnet Enclosure Front Cover mounted to dolly
- ☐ 23. Magnet Enclosure Back Cover mounted to dolly
- ☐ 24. Left Rear Side Cover
- ☐ 25. Left and Right Lasers (early version)
- ☐ 26. Bridge
- ☐ 27. Front Bridge Mount
- ☐ 28. Front Mid-support Plate
- ☐ 29. Rear Bridge Mount (early version)
- ☐ 30. Rear Mid-Support (early version)
- ☐ 31. Rear Support (early version)
- ☐ 32. Cable Take-up Track, Pulley, Spring (early version)
- ☐ 33. Obsolete Cables
- ☐ 34. DSIM/SRI

5-7 MATERIAL RETURN CHECKLIST, SIGNA ADVANTAGE 5.x TO SIGNA LX OR EXCITE UPGRADE

The following items are designated to be put on return pile during the upgrade. Refer to Section 5-5, MATERIAL DISPOSITION GUIDANCE.

- ☐ 1. PDU Cabinet
- ☐ 2. Penetration Cabinet
- ☐ 3. RF Cabinet less RF Amplifier
- ☐ 4. Gradient Cabinet
- ☐ 5. System Cabinet
- ☐ 6. Computer Cabinet
- ☐ 7. Operator Console
- ☐ 8. Rear Pedestal
- ☐ 9. Penetration Panel
- ☐ 10. LPCC or Carriage Cover Assembly
- ☐ 11. SRI Module (if applicable)
- ☐ 12. Select Switch
- ☐ 13. Comfort Module
- ☐ 14. PAC Asm
- ☐ 15. Body Coil On Cart
- ☐ 16. S-IV Swing Front Cover side weld mounts
- ☐ 17. Magnet Enclosure Front Cover mounted to dolly
- ☐ 18. Magnet Enclosure Back Cover mounted to dolly
- ☐ 19. Bridge
- ☐ 20. Front Bridge Mount
- ☐ 21. Front Mid-support Plate
- ☐ 22. Obsolete Cables

5-8 MATERIAL RETURN CHECKLIST, SIGNA HORIZON 5.x TO SIGNA LX with ACGD UPGRADE

The following items are designated to be put on return pile during the upgrade. Refer to Section 5-5, MATERIAL DISPOSITION GUIDANCE.

- ☐ 1. PDU Cabinet
- ☐ 2. RF/PEN Cabinet
- ☐ 3. System Cabinet
- ☐ 4. Computer Cabinet
- ☐ 5. LPCC or Carriage Cover Assembly
- ☐ 6. Select Switch
- ☐ 7. SRI Module (if applicable)
- ☐ 8. Multi-Coil Panel on Penetration Panel
- ☐ 9. Operator Console
- ☐ 10. Heat Exchanger
- ☐ 11. SGD Cabinet OR Gradient & GRAM (if applicable) Cabinet
- ☐ 12. Obsolete Cables

5-9 MATERIAL RETURN CHECKLIST, SIGNA HORIZON LX TO SIGNA LX with ACGD UPGRADE

The following items are designated to be put on return pile during the upgrade. Refer to Section 5-5, MATERIAL DISPOSITION GUIDANCE.

- ☐ 1. PDU Cabinet
- ☐ 2. RF/PEN Cabinet
- ☐ 3. Heat Exchangeer
- ☐ 4. SGD Cabinet OR Gradient & GRAM (if applicable) Cabinet
- ☐ 5. Obsolete Cables

5-10 MATERIAL RETURN CHECKLIST, SIGNA HORIZON LX TO SIGNA LX with EXCITE UPGRADE

- ☐ 1. PDU Cabinet (if applicable)
- ☐ 2. RF/PEN Cabinet (if applicable)
- ☐ 3. RF/PDU Cabinet (if applicable)
- ☐ 4. SGD Cabinet OR Gradient & GRAM (if applicable) Cabinet
- ☐ 5. System Cabinet (if applicable)
- ☐ 6. Octane or Indigo Computer and related OW modules
- ☐ 7. LPCC or Carriage Cover Assembly
- ☐ 8. Select Switch or Cross Point Swtich
- ☐ 9. SRI Module
- ☐ 10. Multi-Coil Panel on Penetration Panel
- ☐ 11. Heat Exchangeer
- ☐ 12. SGD Cabinet OR Gradient & GRAM (if applicable) Cabinet
- ☐ 13. Obsolete Cables

5-11 PHANTOM REPLACEMENT FOR SIGNA ADVANTAGE 4.x

A new TLT Head Phantom and a new TLT Body Phantom are delivered with this upgrade. These new phantoms contain Nickle Chloride solution which has been found to be more temperature stable than the Copper Sulfate solution in the existing phantoms.



NEW TLT PHANTOMS CONTAIN NICKEL – A SUSPECTED CARCINOGEN. DO NOT INGEST. DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO REGULATIONS.



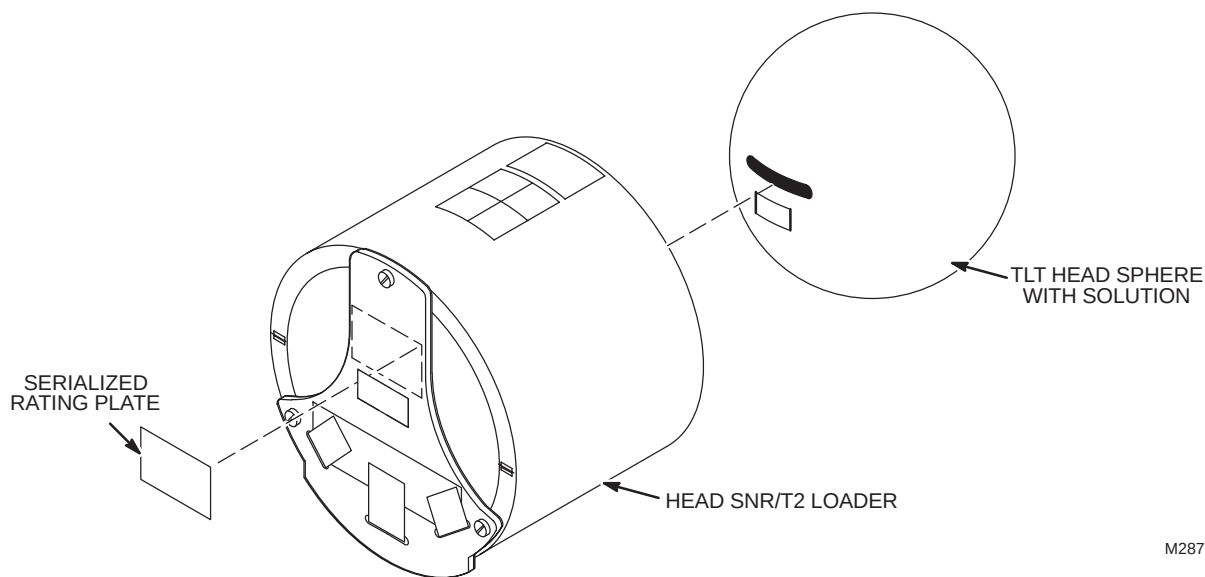
Old phantoms are filled with a solution of Copper Sulfate. Copper Sulfate is considered an environmentally hazardous material and must be disposed of properly. Regulations governing the disposal of this material vary from location to location and from country to country. In order to insure compliance with all local, state or national regulations, these old phantoms must be returned to one of the locations listed at the end of this section. Failure to comply will put you at personal risk of violating environmental regulations.

1. Locate model 46-265826G6 Head TLT Sphere; Return Label, Rating plate, and Product locator card set.
2. Locate model 46-265635G6 Body TLT Sphere; Return Label, Rating plate, and Product locator card set.
3. Unpack the phantoms carefully to enable re-use of packing material to return old phantoms.
4. Locate Material Safety Data Sheets (MSDS) packed with new phantoms, they must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
5. Remove old 46-265826G5 Head TLT Sphere(Copper Sulfate Solution) from service storage cabinet. Pack it in carton that new Nickel Chloride TLT Head Sphere was shipped in.
6. Remove old 46-265635G5 Body TLT Sphere(Copper Sulfate Solution) from service storage cabinet. Pack it in carton that new Nickel Chloride TLT Body Sphere was shipped in.
7. Seal each phantom box and place boxes on hazardous material return pile.

5-12 PHANTOM REMANUFACTURE FOR SIGNA ADVANTAGE 4.x

Because these phantoms are being replaced, it is necessary to change the rating plates on the Head Loader Assembly and the Body Loader Assembly and process Product Locator Remanufacture Cards.

1. Remove Head SNR/T2 Loader from service storage cabinet. Record the serial number from the Head Loader Assembly rating plate in the appropriate place on Product Locator Remanufacture Card (Illustration 5-3).
2. Remove old Model 46-287900G1 rating plate from end. See Illustration 5-1. Attach new Model 46-287900G2 rating plate in same location (above "Property of General Electric" label. Seal the new rating plate with clear acrylic aerosol spray to help insure that label remains securely attached.
3. Fill out furnished Product Locator Remanufacture Card for Head Loader (Illustration 5-3). Attach old 46-287900G1 Head Loader rating plate to card.
4. Check serial number on Installation Card and rating plate. Fill out Product Locator Installation Card for Head Loader.
5. Record the serial number from the new rating plate in the appropriate place on Product Locator Remanufacture Card (Illustration 5-3).
6. Sign and date the Product Locator Remanufacture Card. Return both cards to Product Locator.

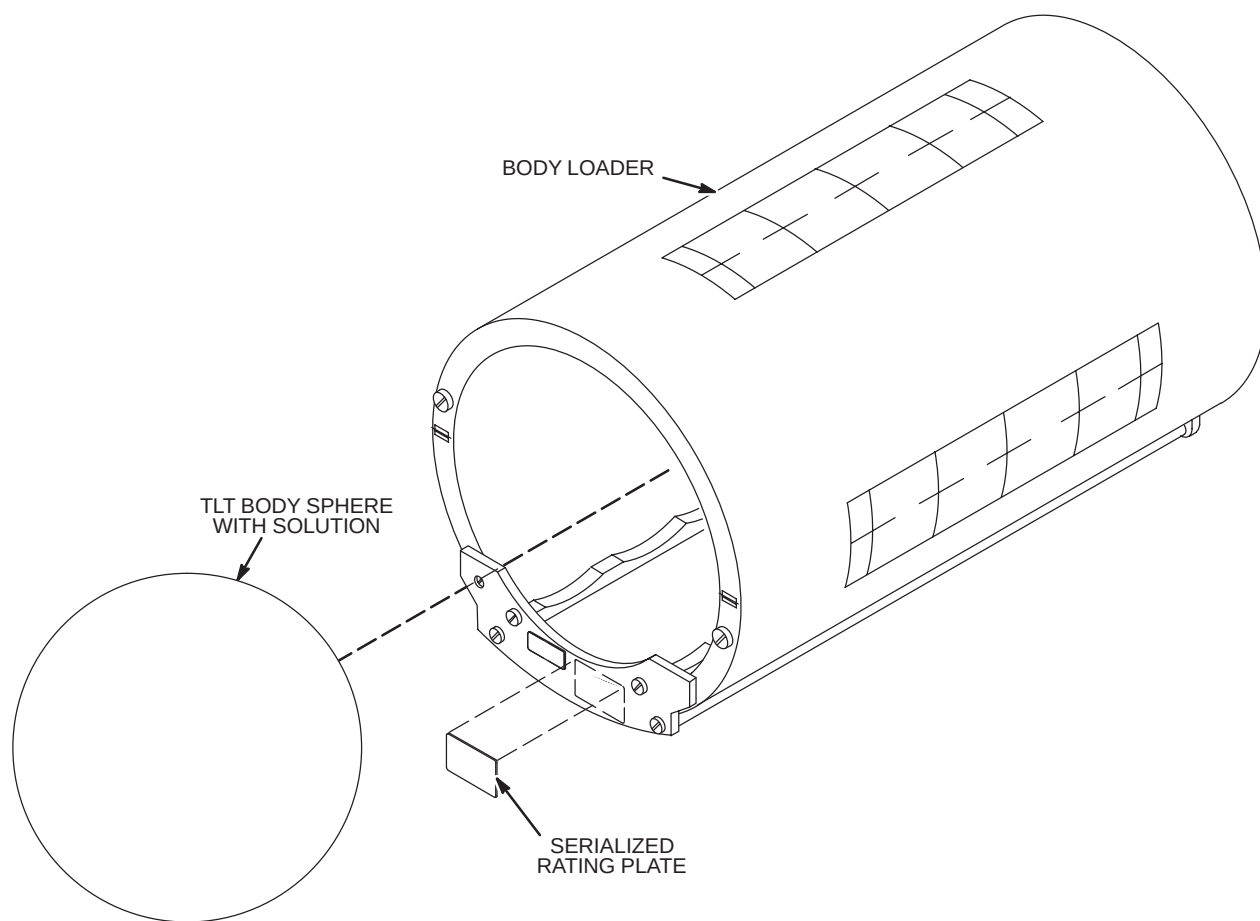


M287900A

HEAD TLT PHANTOM
ILLUSTRATION 5-1

5-12 PHANTOM REMANUFACTURE FOR SIGNA ADVANTAGE 4.x (continued)

7. Remove Body SNR/T2 Loader from service storage cabinet. Record the serial number from the Head Loader Assembly rating plate in the appropriate place on Product Locator Remanufacture Card (Illustration 5-2).
8. Remove old 46-287903G1 rating plate from Body Loader. See Illustration 5-2. Attach new 46-287903G2 rating plate. Seal the new rating plate with clear acrylic aerosol spray to help insure that label remains securely attached.
9. Fill out furnished Product Locator Remanufacture Card for Body Loader (Illustration 5-4). Attach old 46-287903G1 Head Loader rating plate to card.
10. Check serial number on Installation Card and rating plate. Fill out Product Locator Installation Card for Body Loader.
11. Record the serial number from the new rating plate in the appropriate place on Product Locator Remanufacture Card (Illustration 5-2).
12. Sign and date the Product Locator Remanufacture Card. Return both cards to Product Locator.
13. Place Head and Body Loaders, and new Head and Body TLT spheres back into service storage cabinet.



M287903A

BODY TLT PHANTOM
ILLUSTRATION 5-2

5-12 PHANTOM REMANUFACTURE FOR SIGNA ADVANTAGE 4.x (continued)

14. Fill in the four shaded boxes on each of the two Product Locator Remanufacture Cards and return to:

**GE Medical Systems
Product Locator File
P.O. Box 414, W-523
Milwaukee, WI 53201-0414**

TC 180 REMANUFACTURE	NEW PRODUCT IDENTIFICATION	LOT	HEAD LOADER DESCRIPTION	46-287900G2 MODEL	SERIAL
	OLD PRODUCT IDENTIFICATION	LOT	HEAD LOADER DESCRIPTION	46-287900G1 MODEL	SERIAL
	 GE Medical Systems Product Locator File P.O. Box 414, W-523, Milwaukee, WI 53201-0414			SIGNATURE	DATE
				46-265826G5, Head TLT Phantom was removed and replaced by 46-265826G6, Head TLT Phantom.	
			The above product was remanufactured per the above description.		

HEAD TLT PHANTOM REMANUFACTURE CARD

ILLUSTRATION 5-3

TC 180 REMANUFACTURE	NEW PRODUCT IDENTIFICATION	LOT	BODY LOADER DESCRIPTION	46-287903G2 MODEL	SERIAL
	OLD PRODUCT IDENTIFICATION	LOT	BODY LOADER DESCRIPTION	46-287903G1 MODEL	SERIAL
	 GE Medical Systems Product Locator File P.O. Box 414, W-523, Milwaukee, WI 53201-0414			SIGNATURE	DATE
				46-265635G5, Body TLT Phantom was removed and replaced by 46-265635G6, Body TLT Phantom.	
			The above product was remanufactured per the above description.		

BODY TLT PHANTOM REMANUFACTURE CARD

ILLUSTRATION 5-4

5-13 PHANTOM RETURN GUIDANCE

The “hazardous material return pile” consists of phantoms that contain Copper Sulfate. Regulations governing the disposal of this material vary from location to location and from country to country. In order to insure compliance with all local, state or national regulations, these old phantoms must be returned to one of the locations listed below:

If you are performing this upgrade on a system **within the Americas**, return the old Copper Sulfate filled phantoms to the following address:

**G.E. Medical Systems W-853
3200 N. Grandview Blvd.
Waukesha, Wisconsin 53188
Attn: CHEM LAB**

NOTE

Unfortunately, the U.S. Post Office does NOT deliver to the above address. Return phantoms via a commercial carrier such as RPS, Federal Express etc.

If you are performing this upgrade on a system within the area supported by **GE Medical Systems – Europe**, return the old Copper Sulfate filled phantoms to the following address:

**GE Medical Systems – Europe
198 Avenue du President Wilson
93210 La Plaine Saint Denis, FRANCE**

5-13 PHANTOM RETURN GUIDANCE (continued)

If you are performing this upgrade on a system within the area supported by **YMS/GEMSA** (Asia and Pacific), return the old Copper Sulfate filled phantoms to one of the the following:

Hong Kong:	G.E. Medical Systems Attn: Service Manager 37F Shun Tak Centre West Tower 200 Connaught Road Central Hong Kong
Singapore:	G.E. Medical Systems Attn: Regional Tech Support Manager 16th Floor, The Gateway, West Tower 150 Beach Road, Singapore
India:	Vice President, Customer Service Wipro GE A-1 Block, 1st Floor Golden Enclave, Corporate Powers Airport Road, Bangalore 560017 India
Tiawan:	Service Manager GETSCO – Medical Systems 7th Floor, Formosia Plastics Building 201 Tun-Hau North Road Taipei Tiawan, Republic of China
Korea:	Samsung Medical Systems Attn: Service Manager 15th Floor, Seo Woo Building 837-12 Yeoksan-dong Kangnam-gu Seoul, Korea

If none of the above work, Field Engineers in Asia should contact YMS Legal Console, at 0425-85-5055 for assistance.

5-14 FINAL COORDINATION

1. Complete assembly of PPG cable and probe head.
2. Consult with customer as to location of PPG Cable and install PPG cable hooks on frontor side of Magnet Enclosure front cover.
3. Consult with customer as to location of Operator Alert squeeze bulb. Install bulb hook on frontor side of Magnet Enclosure front cover. Route, terminate, and connect tubing at PAC Assembly connection. Verify squeeze bulb will operate alarm.
4. Replace cabinet covers that have not been previously replaced.
5. Replace Magnet Enclosure covers that have not been previously replaced. Close and latch front and rear covers.
6. Replace Rear Pedestal covers.
7. Dock Patient Transport in position at front of bore.
8. Record and enter applicable data into applicable site configuration files and records.
9. Complete Product Locator information for all serialized components, new or updated, that were installed via either:
 - **(U.S. Only) FE Site Verification Web Site**
or
 - Process and return product locator installation cards for all serialized components to:
Product Locator File, P.O. Box 414, W-523, Milwaukee, WI 53201-0414

Refer to Section 1 for details on submitting Product Locator information.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

10. Process the **Supplier Performance Report** located in Common Forms. A reference copy of the form is provided at the end of this section.
11. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals. Since considerable hardware remains from initial installation, (PDU, Magnet Enclosure etc.) retain *Direction 15020 Signa Renewal Parts*, *Direction 15304, Signa Advantage Renewal Parts*, and/or *Direction 15404, Signa Advantage 1.5T, 1.0T, & .05T Renewal Parts*.
12. Locate Material Safety Data Sheets (MSDS) packed with phantoms and gradient coil coolant. They must be retained on site. Customer is to be informed that material with MSDS was brought into site and customer should know/decide where MSDS should be retained at site.
13. Verify that coordination of application support for instruction of site users on Software Release 8.x has been made before returning system to the users. Turn site over to applications who will instruct users.

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GE Medical Systems Field Service

SUPPLIER PERFORMANCE REPORT

Send completed forms to the 4 addresses at the end of this form.

Supplier Name: _____ Date: _____

Work Performed / Evaluated: _____

Commodity / Item Purchased: _____

GE Customer Name: _____

FDO Number: _____ Test Equipment Bar-code No; _____

GEMS Evaluator Name: _____

Please rate suppliers as 1 to 5 on each of the 15 questions below:

1 = Unacceptable 2 = Poor 3 = Average 4 = Excellent 5 = Outstanding

There are 5 groups with three related questions in each group. Groups can be rated either as a group or 3 separate questions.

Any rating below 3 triggers a discrepancy review with the supplier.

Corrective actions will be faster and more effective when you provide examples and/or detail information.

Supplier schedules well and:					
1	starts on time?	1	2	3	4 5
2	finishes on time?	1	2	3	4 5
3	avoids disruptions?	1	2	3	4 5
Supplier is flexible and:					
4	responds to GEMS/customer requests?	1	2	3	4 5
5	cooperates with changes in schedule?	1	2	3	4 5
6	deals with unexpected problems?	1	2	3	4 5
Supplier tools and equipment are:					
7	appropriate for contracted work?	1	2	3	4 5
8	adequate for contracted work?	1	2	3	4 5
9	calibrated and in good working order?	1	2	3	4 5
Supplier meets GEMS quality standards with employees who:					
10	are trained, skilled and technically competent?	1	2	3	4 5
11	always keep job sites clean and free of debris?	1	2	3	4 5
12	ensure both function and appearance are as expected?	1	2	3	4 5
Supplier employees enhance GEMS image by their:					
13	appearance?	1	2	3	4 5
14	behavior?	1	2	3	4 5
15	attitude?	1	2	3	4 5

Examples / Details: _____

Any rating below 3 triggers a discrepancy review with the supplier.

SEND COMPLETED FORMS TO:

- Senior Operations Specialist for your LCT
- The contractor who performed the work
- ISS (already in default distribution)
- Sourcing Supplier Quality (already in default distribution)

SYSTEM UPGRADE INSTRUCTIONS

Tab 1, System, contains upgrade instructions for the following configurations:

- *DIRECTION 2398342* – Signa EXCITE II Fiber Optic Cable Installation Instructions
- *DIRECTION 2338872* – Signa EXCITE I Fiber Optic Cable Installation Instructions
- *DIRECTION 2287106* – Signa 8.x ACGD Fiber Optic Cable Installation Instructions
- *DIRECTION 2287109* – Signa 8.x ACGD Power Cable Installation Instructions
- *DIRECTION 2291189* – Signa 8.x ACGD Gradient Cable Upgrade Instructions for BRM Body Coil
- *DIRECTION 2291190* – Signa 8.x ACGD Gradient Cable Upgrade Instructions for BRM-D & CRM Body Coil
- *DIRECTION 2389819* – 1.5T Times Microwave RF Cable Installation Instructions

Use only the applicable System Upgrade Instruction Directions for the site being upgraded

INSTALLATION STEPS SUMMARY

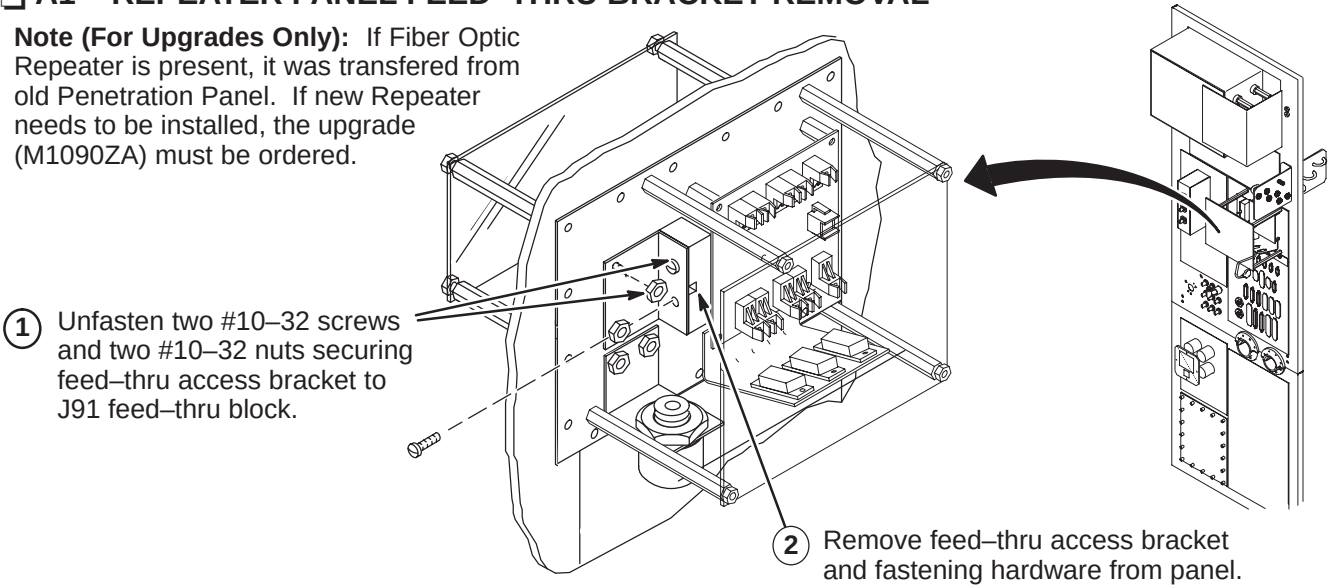
- A1 – Repeater Panel Feed-thru Bracket Removal
- A2 – Routing Run 711/712 in Magnet Room
- A3 – Connecting Magnet Room Run 711/712 to Repeater Panel
- A4 – Route/Connect Equipment Room Run 1045 to Repeater Panel
- A5 – Cable Map of Fiber Optic Runs 709, 711/712, 1032, 1033, 1034, 1044, and 1045

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

A1 – REPEATER PANEL FEED-THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transferred from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.



A2 – ROUTING RUN 711/712 IN MAGNET ROOM

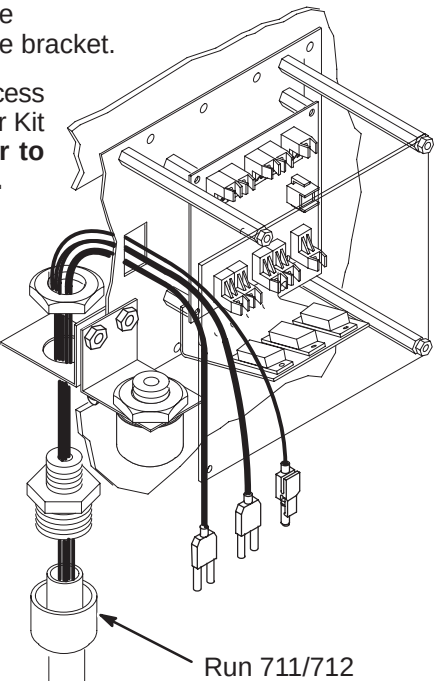
- 1 Move Run 711/712 (46-328079G9) to the Magnet Room side of the Penetration Panel. The “PP1-J91” end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

- 2 Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
- 3 Route ends of individual fiber-optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
- 4 Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

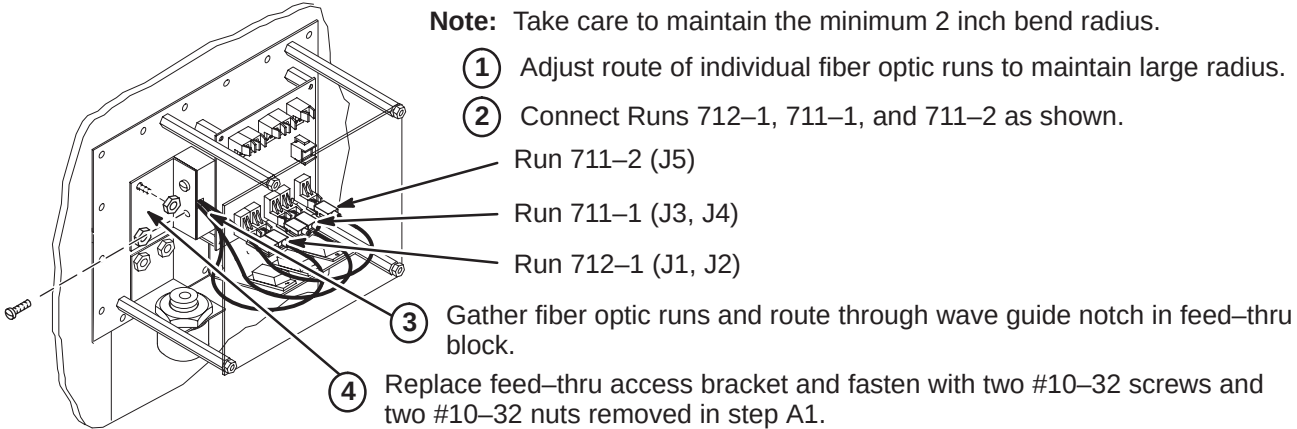
Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC-II.

- 5 Route Run 711/712 to the front right side of the magnet (SRI location).
- 6 Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC-II.



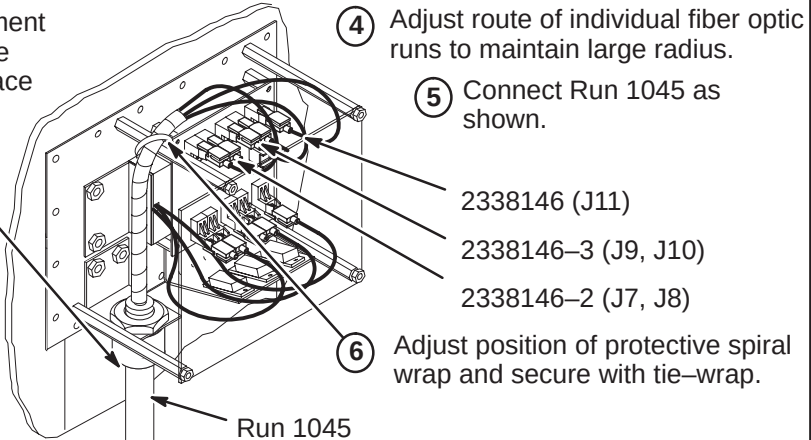
A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

Note: Take care to maintain the minimum 2 inch bend radius.



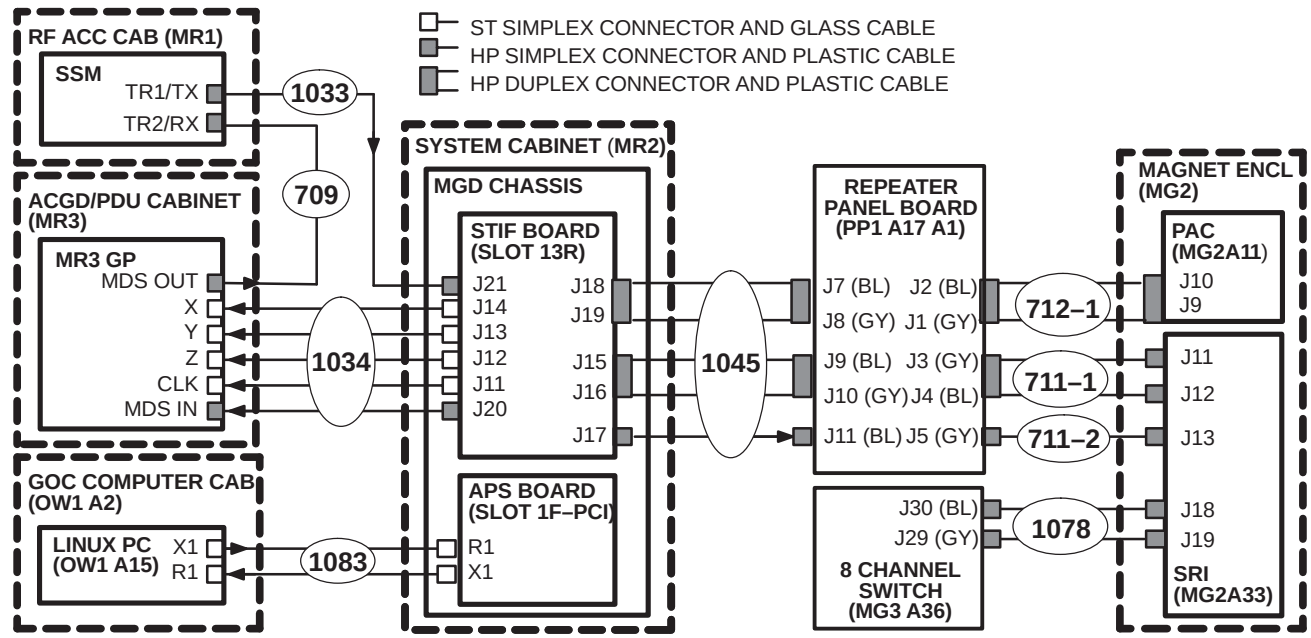
A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 1045 TO REPEATER PANEL

- 1 Move Run 1045 (2338146) to the Equipment Room side of the Penetration Panel. The “PP1-A17” end will be attached to interface bracket.
- 2 Attach Run 1045 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
- 3 Route other end of Run 1045 to System Cabinet. Refer to Direction 2338502 (Tab 8) for connecting fiber optics in System Cabinet.



A5 – CABLE MAP OF F/O RUNS 709, 1033, 1034, 1045, 1078, AND 1083

- 1 Unroll coiled fiber optic conduits for Runs 709, 1033, 1034, 1045, 1078, and 1083 along intended routes. Connection of Runs is shown in diagram below. Plan for storage of excess Run 1034 conduit of as it cannot be reterminated.



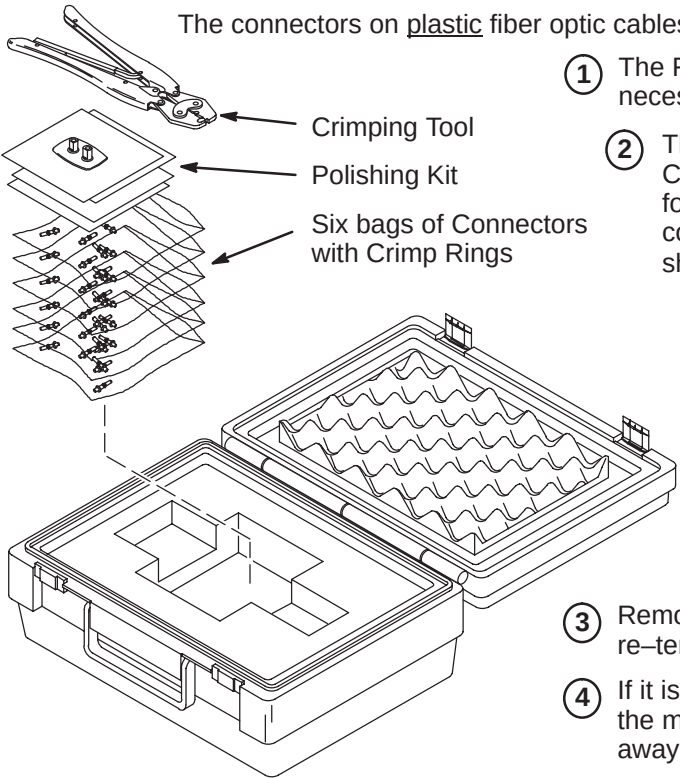
INSTALLATION STEPS SUMMARY

- ❑ B1 – Fiber Optic Cable Termination Kit
- ❑ B2 – Stripping Fiber Optic Outer Jacket
- ❑ B3 – Crimping Connector to Fiber Optic Cable
- ❑ B4 – Trimming Excess Fiber Optic from Connector
- ❑ B5 – Polishing Fiber Optic End

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

❑ B1 – FIBER OPTIC CABLE TERMINATION KIT



The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.

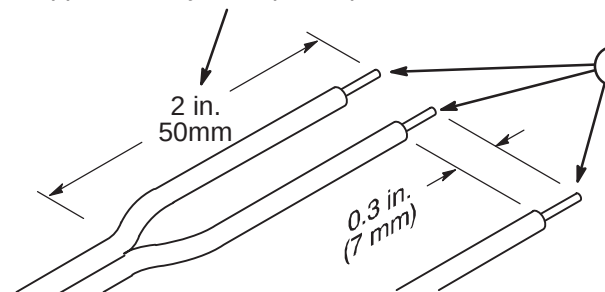
- 1 The Fiber Optic Repair Kit, 46-301450G1, contains all the necessary materials to replace a damaged connector.
- 2 The "Spares Kit", delivered with the Shipping Collector, contains a 46-320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.

- 3 Remove the end of fiber optic cable to be re-terminated from magnet room if possible.
- 4 If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
- 5 Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

- 1 For Duplex Cables, separate the two fibers approximately 2 in. (50mm) back from the ends.



Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

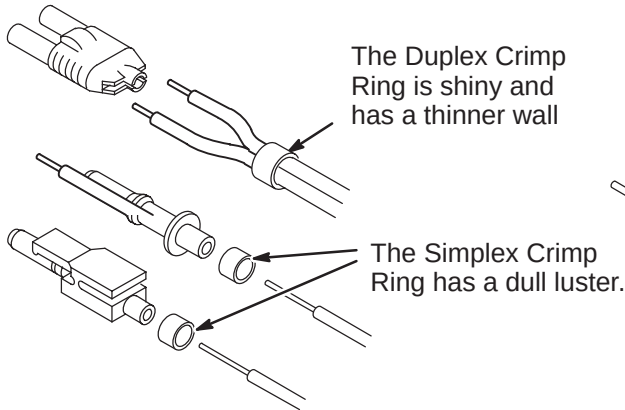
- 2 Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.

❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

- 1 Place the Crimp Ring and Connector over the end of the cable.

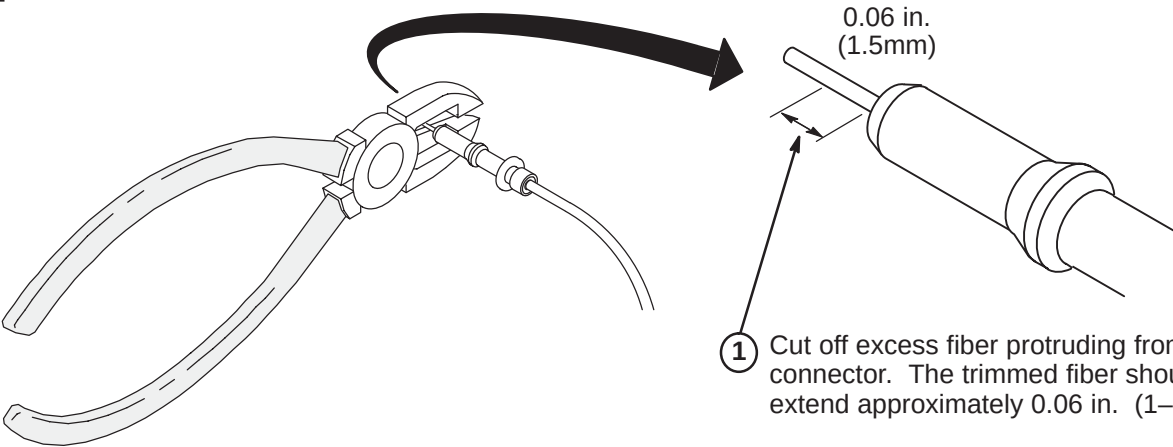
Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

- 2 The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



- 3 Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR



- 1 Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1-1/2mm).

❑ B5 – POLISHING FIBER OPTIC END

- 1 Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
- 2 Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
- 3 Wipe the connector and fixture with a clean cloth or tissue.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

- 4 Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
- 5 The fiber end should be flat, smooth and clean. The cable end can now be connected.

INSTALLATION STEPS SUMMARY

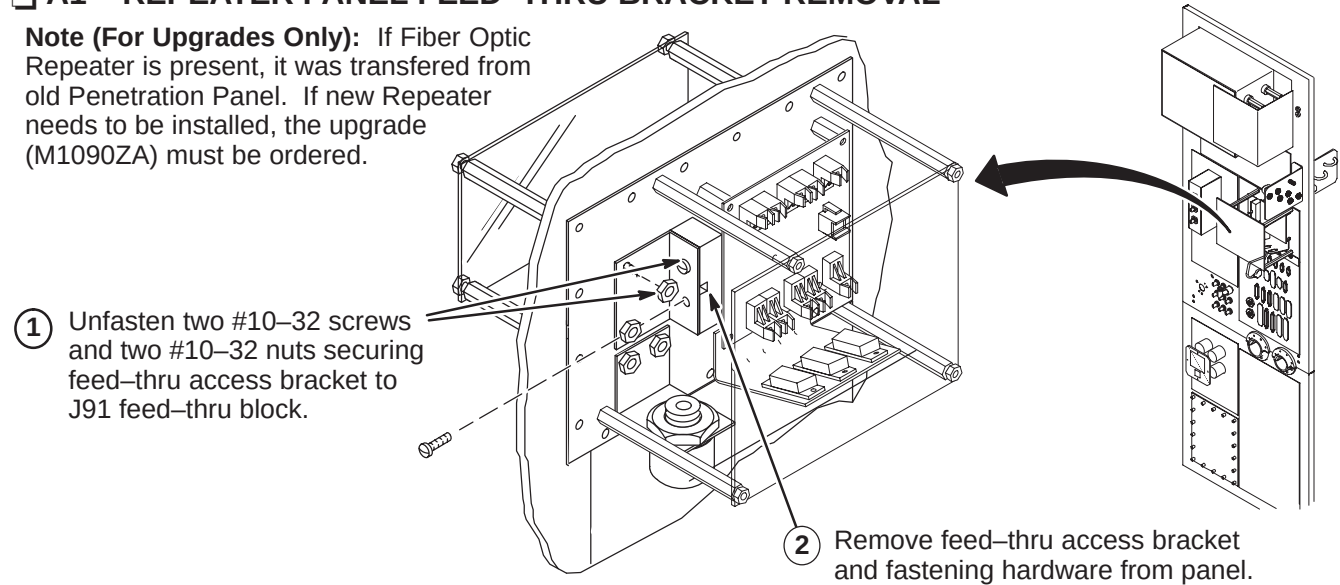
- A1 – Repeater Panel Feed-thru Bracket Removal
- A2 – Routing Run 711/712 in Magnet Room
- A3 – Connecting Magnet Room Run 711/712 to Repeater Panel
- A4 – Route/Connect Equipment Room Run 1045 to Repeater Panel
- A5 – Cable Map of Fiber Optic Runs 709, 711/712, 1032, 1033, 1034, 1044, and 1045

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

A1 – REPEATER PANEL FEED-THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transferred from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.



A2 – ROUTING RUN 711/712 IN MAGNET ROOM

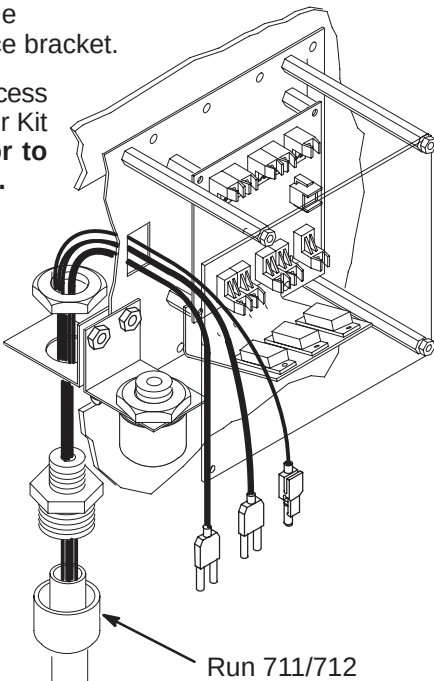
- 1 Move Run 711/712 (46-328079G9) to the Magnet Room side of the Penetration Panel. The “PP1-J91” end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

- 2 Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
- 3 Route ends of individual fiber-optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
- 4 Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

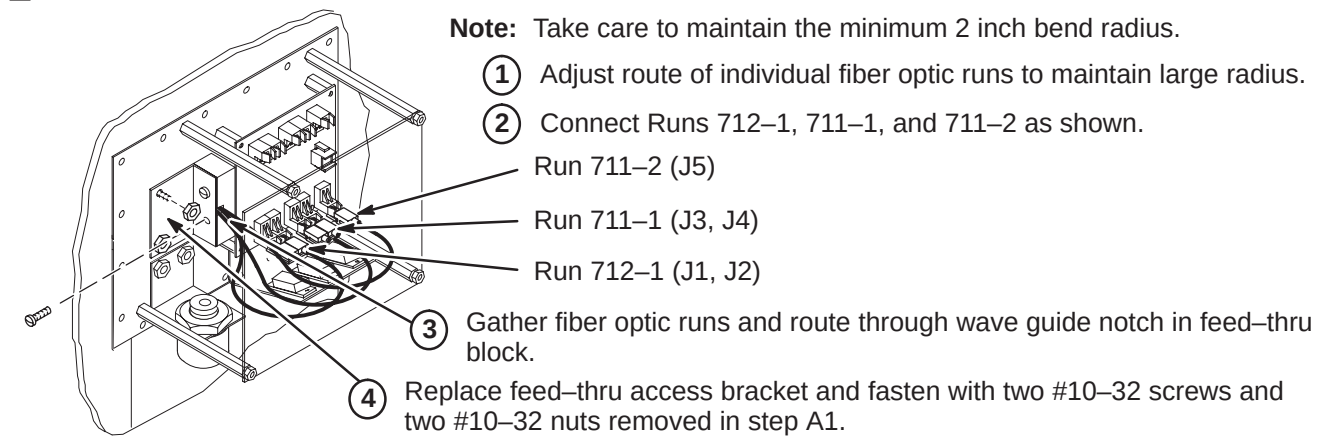
Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC-II.

- 5 Route Run 711/712 to the front right side of the magnet (SRI location).
- 6 Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC-II.



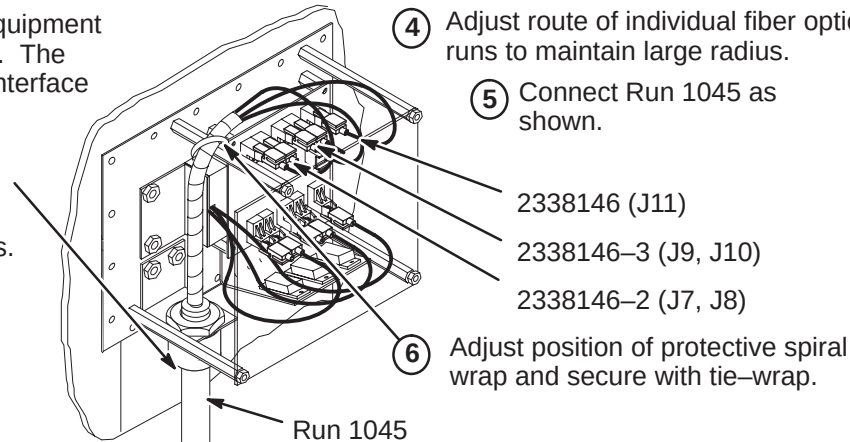
A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

Note: Take care to maintain the minimum 2 inch bend radius.



A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 1045 TO REPEATER PANEL

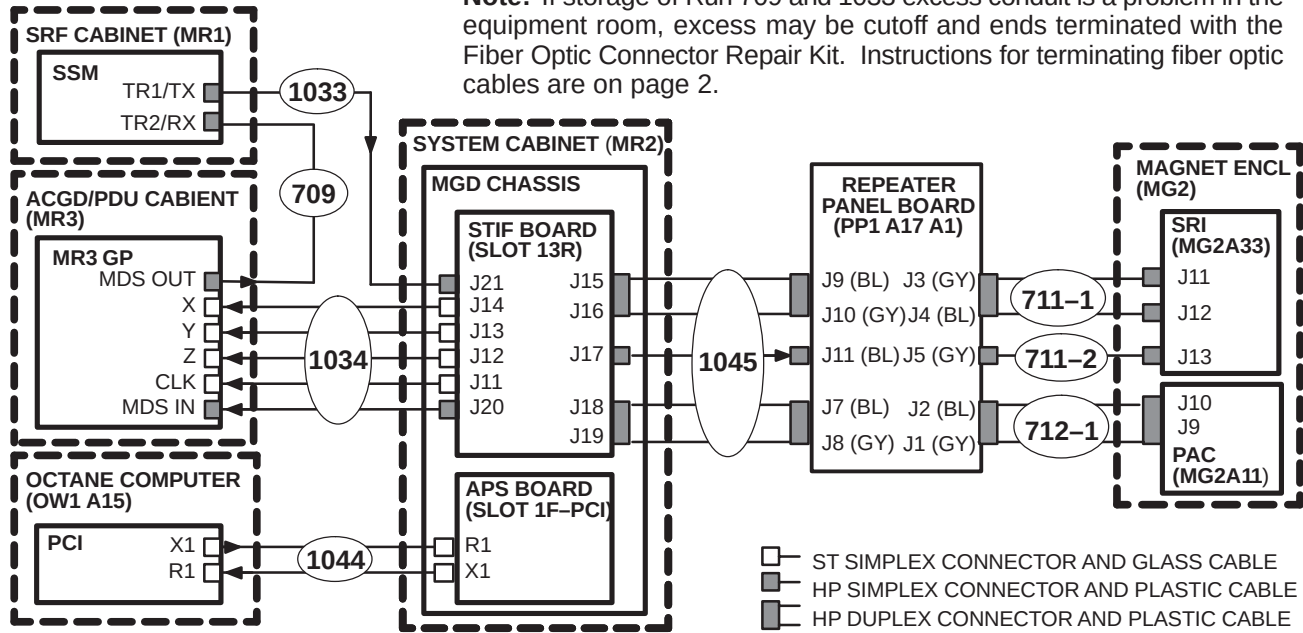
- 1 Move Run 1045 (2331846) to the Equipment Room side of the Penetration Panel. The “PP1-A17” end will be attached to interface bracket.
- 2 Attach Run 1045 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
- 3 Route other end of Run 1045 to System Cabinet. Refer to Direction 2338502 (Tab 8) for connecting fiber optics in System Cabinet.
- 4 Adjust route of individual fiber optic runs to maintain large radius.
- 5 Connect Run 1045 as shown.
- 6 Adjust position of protective spiral wrap and secure with tie-wrap.



A5 – CABLE MAP OF F/O RUNS 709, 711/712, 1032, 1033, 1034, 1044, AND 1045

- 1 Unroll coiled fiber optic conduits for Runs 709, 1033, and 1034 along intended route. Connection of Runs is shown in diagram below. Plan for storage of excess Run 1034 conduit of as it cannot be reterminated.

Note: If storage of Run 709 and 1033 excess conduit is a problem in the equipment room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit. Instructions for terminating fiber optic cables are on page 2.



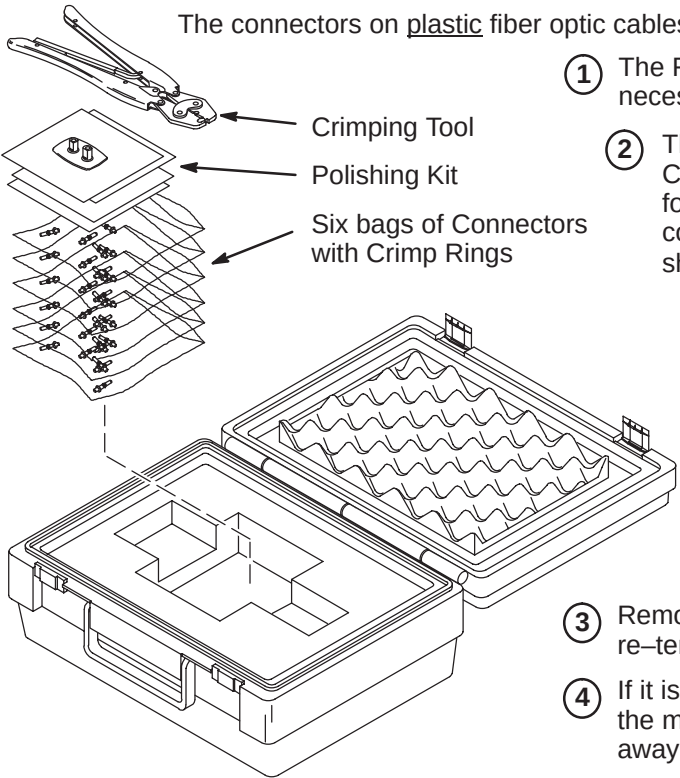
INSTALLATION STEPS SUMMARY

- ❑ B1 – Fiber Optic Cable Termination Kit
- ❑ B2 – Stripping Fiber Optic Outer Jacket
- ❑ B3 – Crimping Connector to Fiber Optic Cable
- ❑ B4 – Trimming Excess Fiber Optic from Connector
- ❑ B5 – Polishing Fiber Optic End

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

❑ B1 – FIBER OPTIC CABLE TERMINATION KIT



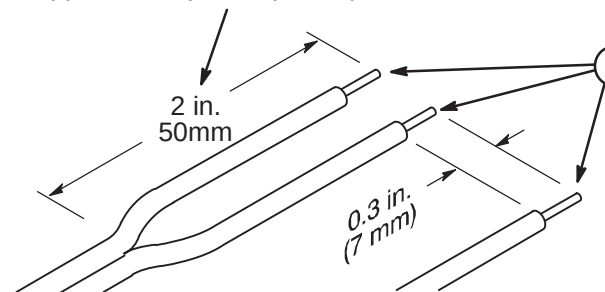
- 1 The Fiber Optic Repair Kit, 46-301450G1, contains all the necessary materials to replace a damaged connector.
- 2 The "Spares Kit", delivered with the Shipping Collector, contains a 46-320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.

- 3 Remove the end of fiber optic cable to be re-terminated from magnet room if possible.
- 4 If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
- 5 Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

- 1 For Duplex Cables, separate the two fibers approximately 2 in. (50mm) back from the ends.



Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

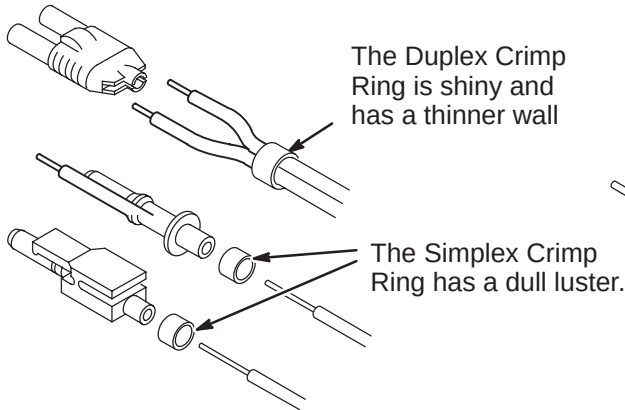
- 2 Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.

❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

- 1 Place the Crimp Ring and Connector over the end of the cable.

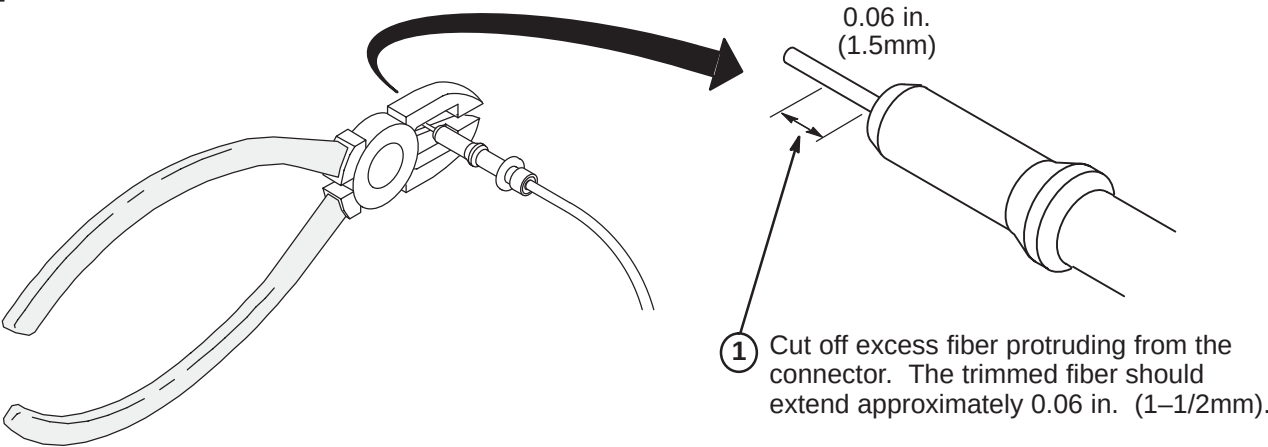
Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

- 2 The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



- 3 Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR



- 1 Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1-1/2mm).

❑ B5 – POLISHING FIBER OPTIC END

- 1 Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.
- 2 Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.
- 3 Wipe the connector and fixture with a clean cloth or tissue.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

- 4 Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.
- 5 The fiber end should be flat, smooth and clean. The cable end can now be connected.

INSTALLATION STEPS SUMMARY

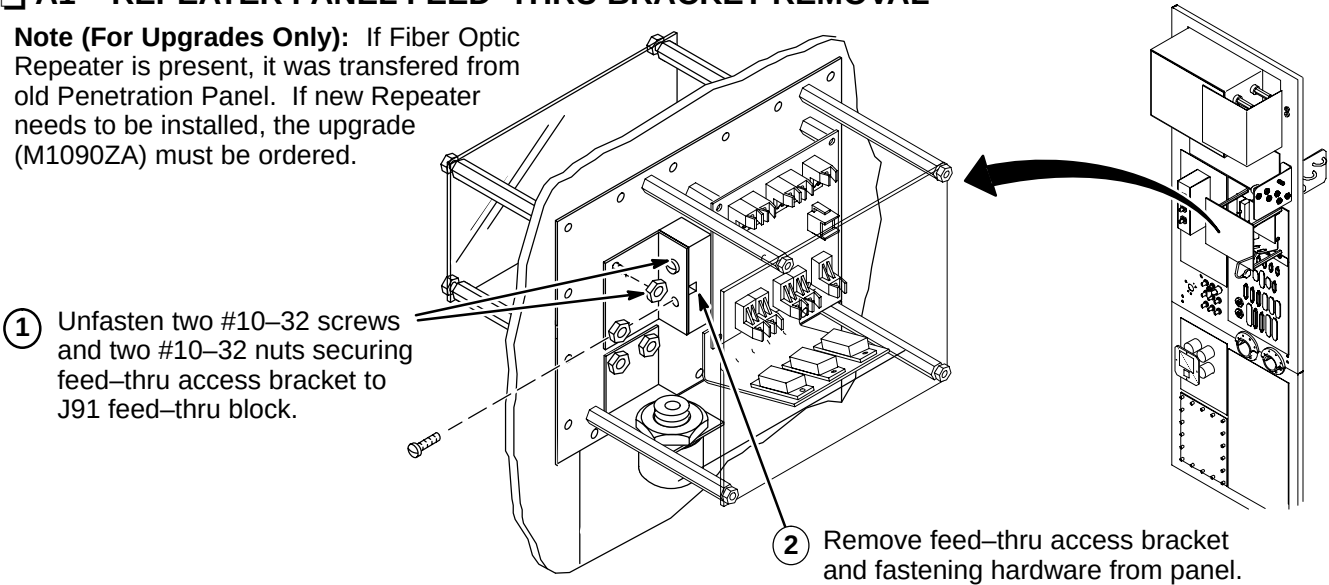
- A1 – Repeater Panel Feed-thru Bracket Removal
- A2 – Routing Run 711/712 in Magnet Room
- A3 – Connecting Magnet Room Run 711/712 to Repeater Panel
- A4 – Route/Connect Equipment Room Run 711/712 to Repeater Panel
- A5 – Route/Connect Fiber Optic Equipment Room Runs 708, 709, 710, 711/712, and 836

CAUTION

Fiber optic cables are easily damaged! Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend fiber optic cables to radius smaller than two inches.

A1 – REPEATER PANEL FEED-THRU BRACKET REMOVAL

Note (For Upgrades Only): If Fiber Optic Repeater is present, it was transferred from old Penetration Panel. If new Repeater needs to be installed, the upgrade (M1090ZA) must be ordered.



A2 – ROUTING RUN 711/712 IN MAGNET ROOM

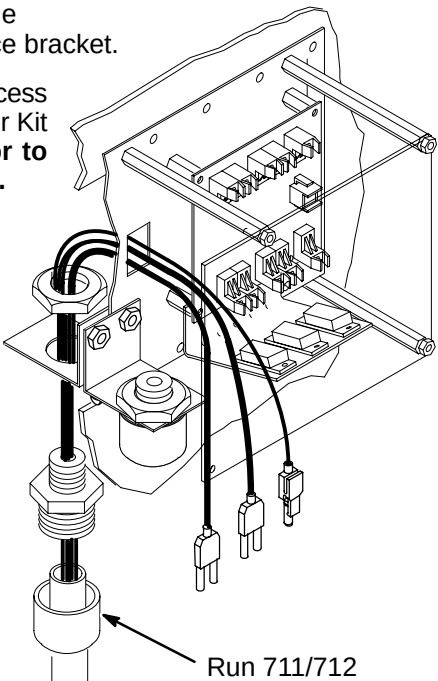
- 1 Move Run 711/712 (46-328079G9) to the Magnet Room side of the Penetration Panel. The “PP1-J91” end will be attached to interface bracket.

Note: If Run 711/712 excess conduit is a problem in the magnet room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit per page 2. **If cables need to be shortened, mark each cable prior to cutting. This will aid in identifying the cables when reterminating.**

- 2 Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connection installation.
- 3 Route ends of individual fiber-optic Runs through J91 opening to the Repeater Board (PP1 A17 A1)
- 4 Route Run 711/712 to Magnet Enclosure Rear Pedestal and underneath magnet.

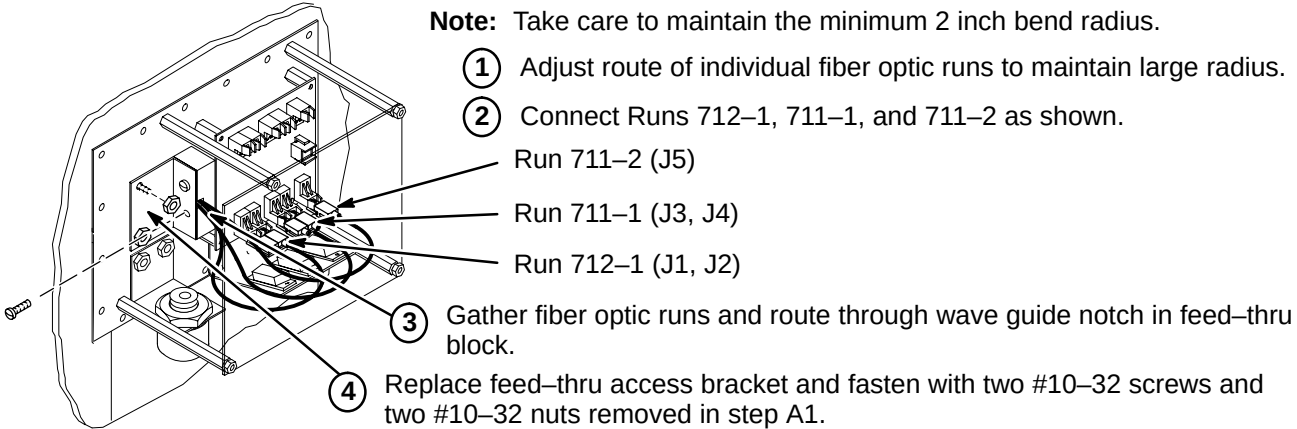
Note: For the next two steps, refer to Magnet Enclosure Installation (Tab 2) for connecting Run 711 to SRI Module and Run 712 to PAC-II.

- 5 Route Run 711/712 to the front right side of the magnet (SRI location).
- 6 Route Run 712 into the magnet enclosure front cover, across the top and down the right side to the PAC-II.



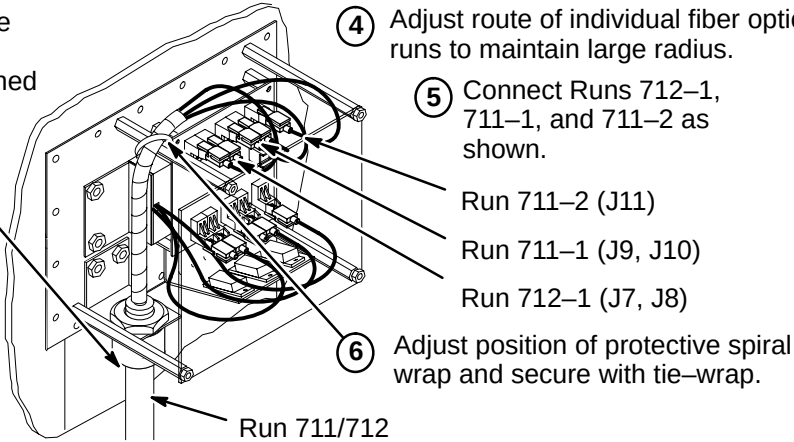
A3 – CONNECTING MAGNET ROOM RUN 711/712 TO REPEATER PANEL

- Note:** Take care to maintain the minimum 2 inch bend radius.
- 1 Adjust route of individual fiber optic runs to maintain large radius.
 - 2 Connect Runs 712-1, 711-1, and 711-2 as shown.



A4 – ROUTE/CONNECT EQUIPMENT ROOM RUN 711/712 TO REPEATER PANEL

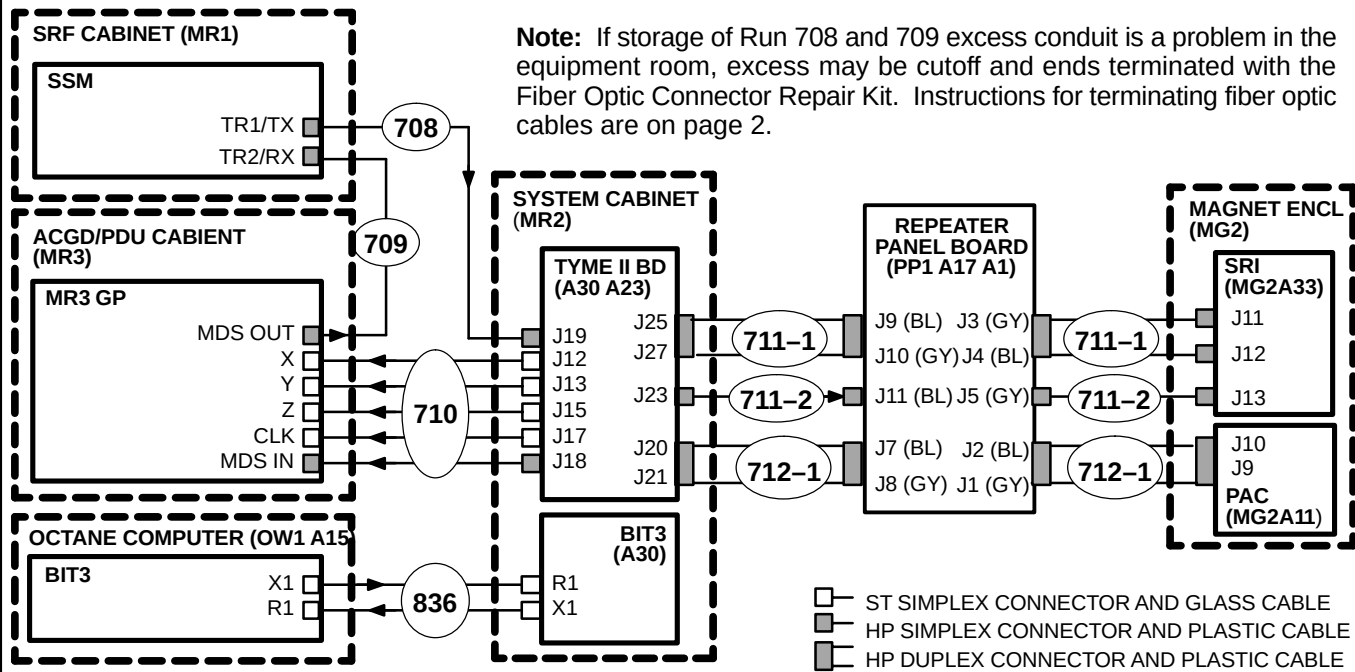
- 1 Move Run 711/712 (46-328078G5) to the Equipment Room side of the Penetration Panel. The “PP1-A17” end will be attached to interface bracket.
- 2 Attach Run 711/712 to access bracket. Refer to Direction 2123149 (packed with bag of FO Conduit connectors) for connector installation instructions.
- 3 Route other end of Run 711/712 to System Cabinet. Refer to Direction 2170004 (Tab 8) for connecting fiber optics in System Cabinet.
- 4 Adjust route of individual fiber optic runs to maintain large radius.
- 5 Connect Runs 712-1, 711-1, and 711-2 as shown.
- 6 Adjust position of protective spiral wrap and secure with tie-wrap.



A5 – ROUTE/CONNECT F/O EQUIPMENT ROOM RUNS 708, 709, 710, 711/712 & 836

- 1 Unroll coiled fiber optic conduits for Runs 708, 709, and 710 along intended route. Connection of Runs is shown in diagram below. Plan for storage of excess Run 710 conduit of as it cannot be reterminated.

Note: If storage of Run 708 and 709 excess conduit is a problem in the equipment room, excess may be cutoff and ends terminated with the Fiber Optic Connector Repair Kit. Instructions for terminating fiber optic cables are on page 2.



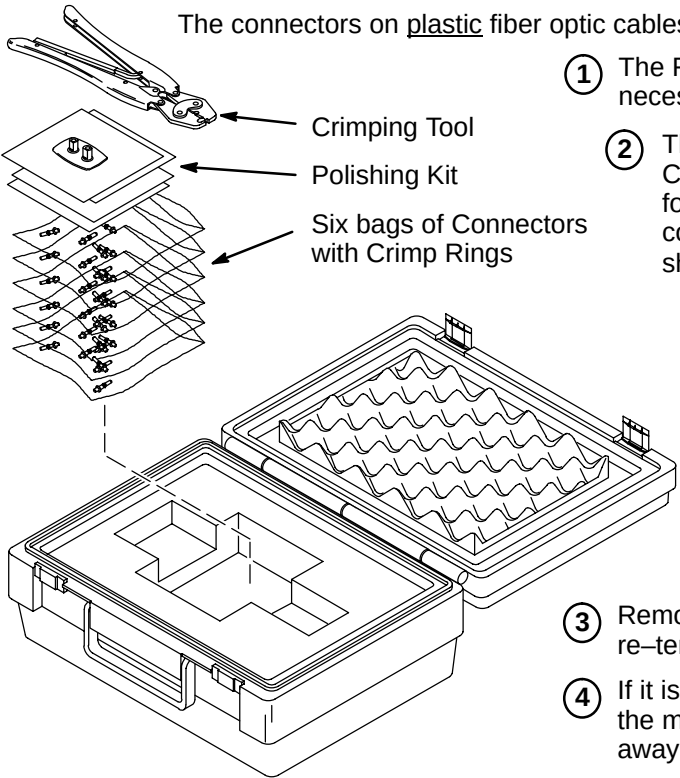
INSTALLATION STEPS SUMMARY

- ❑ B1 – Fiber Optic Cable Termination Kit
- ❑ B2 – Stripping Fiber Optic Outer Jacket
- ❑ B3 – Crimping Connector to Fiber Optic Cable
- ❑ B4 – Trimming Excess Fiber Optic from Connector
- ❑ B5 – Polishing Fiber Optic End

WARNING!

FERROUS MATERIAL HAZARD! CRIMP TOOL AND OTHER TOOLS REQUIRED FOR THIS PROCEDURE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

❑ B1 – FIBER OPTIC CABLE TERMINATION KIT



The connectors on plastic fiber optic cables can be replaced if damaged or cut off to shorten.

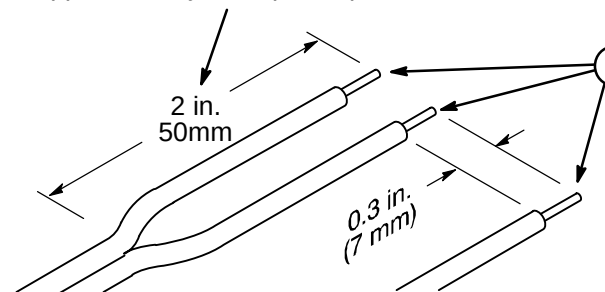
- 1 The Fiber Optic Repair Kit, 46-301450G1, contains all the necessary materials to replace a damaged connector.
- 2 The "Spares Kit", delivered with the Shipping Collector, contains a 46-320119P1 Termination Kit for Fiber Optic Cables. This kit consists of consumables such as terminals and polishing sheets used to terminate cables during installation.

Note: The connectors on glass fiber optic cables **cannot** be replaced using the F/O Repair Kit. At this time, there is no kit available to repair glass fiber optic cable connectors.

- 3 Remove the end of fiber optic cable to be re-terminated from magnet room if possible.
- 4 If it is not possible to remove the end to be repaired from the magnet room, the cable must be at least 10 feet away from the magnet before it can be worked on.
- 5 Cut cable to desired length.

❑ B2 – STRIPPING FIBER OPTIC OUTER JACKET

- 1 For Duplex Cables, separate the two fibers approximately 2 in. (50mm) back from the ends.



Note: The separated duplex cable must be stripped to equal lengths on each side of the cable. This allows proper seating of the cable into the connector.

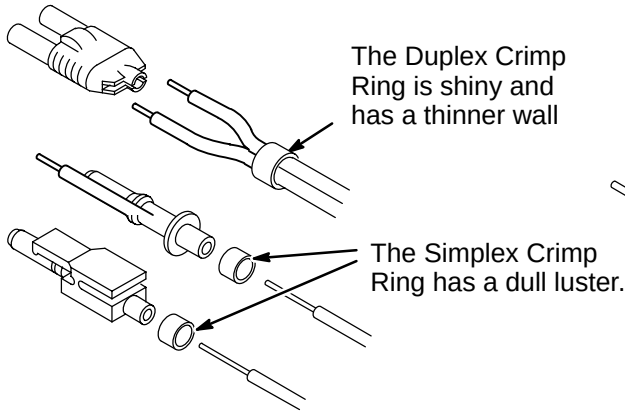
- 2 Strip off approximately 0.3 in. (7 mm) of the outer jacket with the 16 gauge wire strippers. Excess webbing on duplex cable may have to be trimmed to allow the connector to slide over the cable.

❑ B3 – CRIMPING CONNECTOR TO FIBER OPTIC CABLE

- 1 Place the Crimp Ring and Connector over the end of the cable.

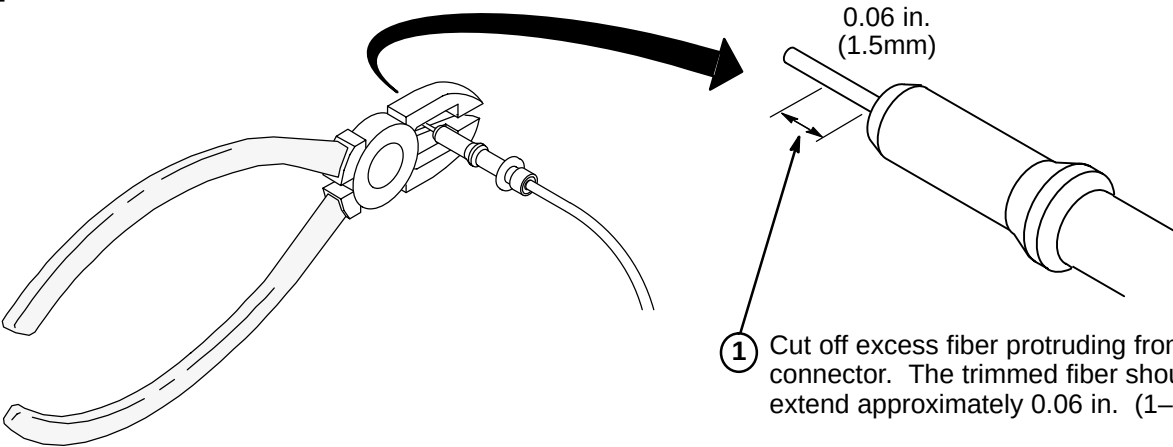
Note: The Connector Crimp Rings cannot be interchanged between simplex and duplex connectors.

- 2 The fiber should extend approximately 0.12 in. (3mm) through the end of the connector.



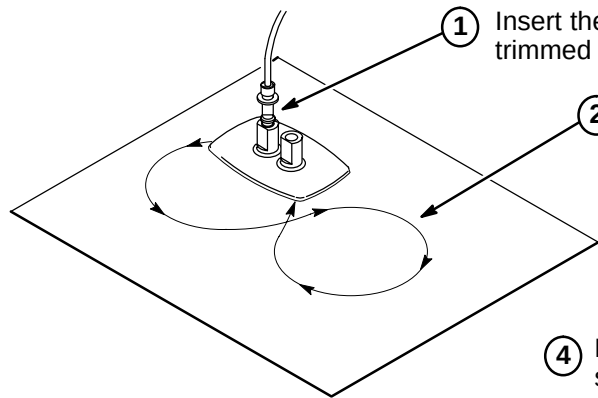
- 3 Carefully position the ring so that it is entirely on the connector and then crimp the ring in place with the crimping tool.

❑ B4 – TRIMMING EXCESS FIBER FROM CONNECTOR



- 1 Cut off excess fiber protruding from the connector. The trimmed fiber should extend approximately 0.06 in. (1-1/2mm).

❑ B5 – POLISHING FIBER OPTIC END



- 1 Insert the connector fully into the polishing fixture with the trimmed fiber end protruding from the bottom of the fixture.

- 2 Place the 600 grit abrasive paper on a flat smooth surface. Pressing down on the indicator, polish the fiber and the connector using a figure eight motion until the connector is flush with the end of the polishing fixture.

- 3 Wipe the connector and fixture with a clean cloth or tissue.

- 4 Place the flush connector and polishing fixture on the dull side of the 3 micron pink lapping film and continue to polish the fiber and connector for approximately 25 strokes.

Note: The four dots on the bottom of the polishing fixture are wear indicators. Replace the polishing fixture when any dot is no longer visible.

- 5 The fiber end should be flat, smooth and clean. The cable end can now be connected.

INSTALLATION STEPS SUMMARY

- A – System Power Cable Installation
- B – Power Cable Re-termination
- C – Power Cable Pre-connect checks
- D – Power Cable Connections Details

A – SYSTEM POWER CABLE INSTALLATION

1. Verify that all power and ground cables for the Signa System with ACGD are present. Refer to Section 2–1, LOCATING AND UNPACKING CABLES.
2. Lengths of PDU Power cables for:

	FIXED SITE	MOBILE	T/R
Run 966, SSM	35.0 ft. (10.7M)	35.0 ft. (10.7M)	48.0 ft. (14.6M)
Run 967, SRF Cabinet	35.5 ft. (11.1M)	29.5 ft. (9.0M)	49.0 ft. (15.0M)
Run 968, System Cabinet	35.0 ft. (10.7M)	25.0 ft. (7.6M)	45.0 ft. (13.7M)
Run 969, Operator Workspace Computer	65.5 ft. (20.0M)	33.0 ft. (10.0M)	33.0 ft. (10.0M)
Run 970, Water Chiller (Fixed–Site) (Note 1)	82.0 ft. (25.0M)		
Run 973, Heat Exchanger (Mob & T/R) (Note 1)		82.0 ft. (25.0M)	82.0 ft. (25.0M)
Run 989, Twin Accessory Cabinet (Note 1)	25.0 ft. (7.62M)	25.0 ft. (7.62M)	25.0 ft. (7.62M)

Note 1: Runs 970 and 973 are for Non–TwinSpeed systems, Run 989 is only used on TwinSpeed.

- If insufficient room exists to store excess cables:
Plan for storage location if cables are not being cut to length. Proceed to Step C.
- If insufficient room exists to store excess cables:
They can be cut shorter and re-terminated at the PDU. Complete Step B. Then proceed to Step C.

3. Route all power cables per applicable System Interconnect Diagram. Unroll cables to lay without twists or kinks from cabinet to PDU through provided troughs, conduit, or ducts.

B – POWER CABLE RE-TERMINATION

1. Cut power and ground cables to length as necessary. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans.
2. Cut and trim insulation for connection of power and ground cable conductors to connectors at PDU.
3. Transfer color code tape, run-number labels, and applicable wire marking labels as each cable is cut to size.

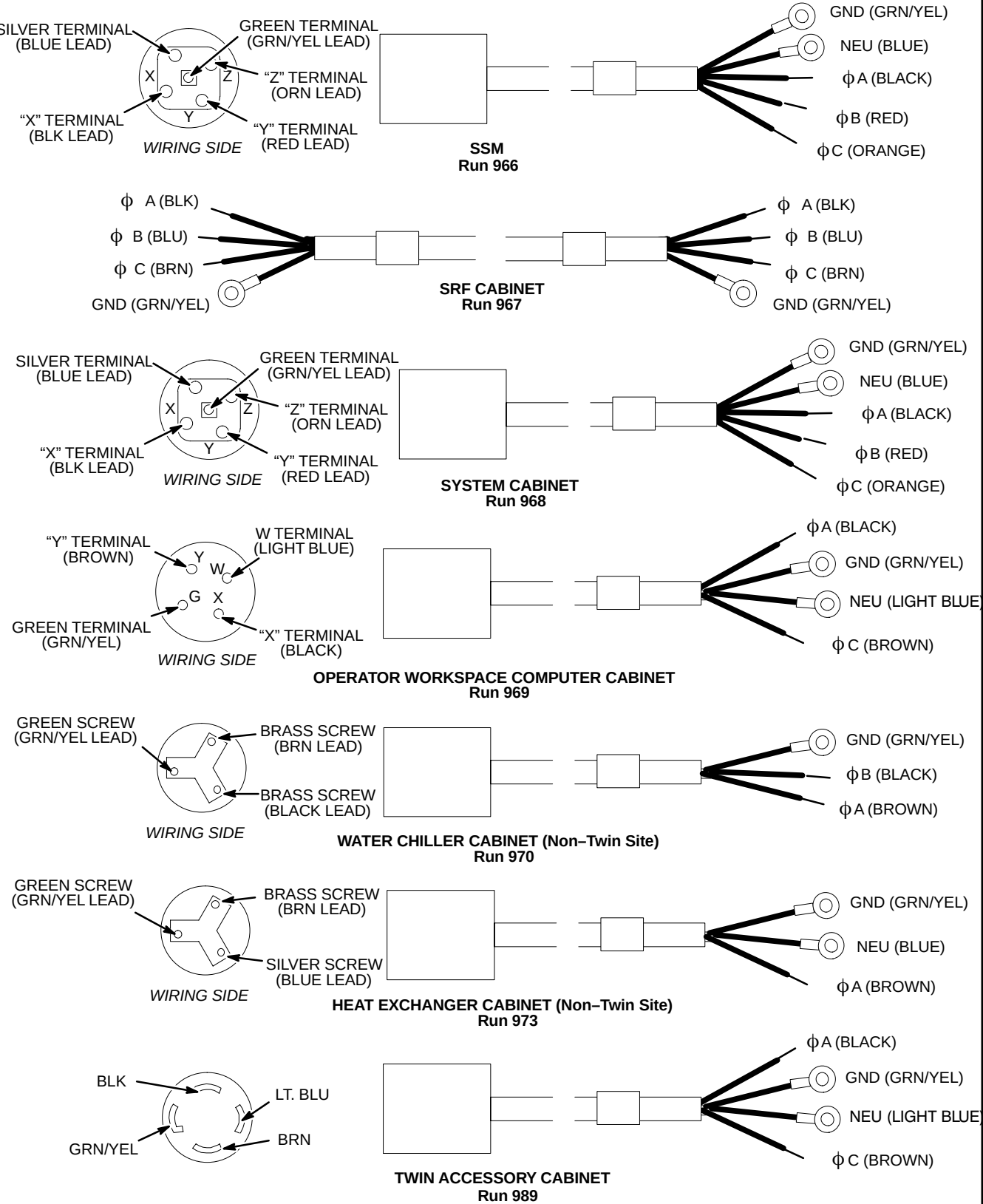
C – POWER CABLE PRE-CONNECT CHECKS

Checks for opens, shorts, and crossed connections are made for field re-terminated power cables prior to connection to PDU and cabinets. This does not provide ground resistance checks for cabinets. Ground resistance checks are in MR CD–ROM, Set Up and Calibration: System, GROUNDING AND LEAKAGE CURRENT CHECKS.

1. Use the ohmmeter and the cable plug connector Illustration to check each power cable for the following:
 - Open circuit between cable neutral and cable ground.
 - Open circuit between cable neutral and each line on the cable.
 - Open circuit between cable ground and each line on the cable.
 - Less than 1 ohm between cable neutral and the light blue (or white) conductor.
 - Less than 1 ohm between cable ground and the green/yellow conductor.
2. Any out-of-spec conditions must be immediately corrected. See Procedure D for plug connection details. After all power cables have been satisfactorily checked out, continue installation.

D – POWER CABLE CONNECTIONS DETAILS

Note
After performing cable pre-connect checks, avoid potential problems by taping up exposed ACGD Gradient Cabinet cables or immediately install ACGD/PDU Gradient Cabinet and connect terminals to power modules before PDU power is turned on.



INSTALLATION STEPS SUMMARY

- A – Cut cables to length and route in Equipment and Magnet Rooms
- B – Terminate Runs 762, 763, and 764 for ACGD/PDU Cabinet
- C – Terminate Runs 762, 763, and 764 in Magnet Room
- D – Connect Cables at Penetration Panel
- E – Connection of Body Coil Gradient Cables to Runs 762, 763 & 764

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

A – CUT CABLES TO LENGTH AND ROUTE IN EQUIPMENT AND MAGNET ROOMS

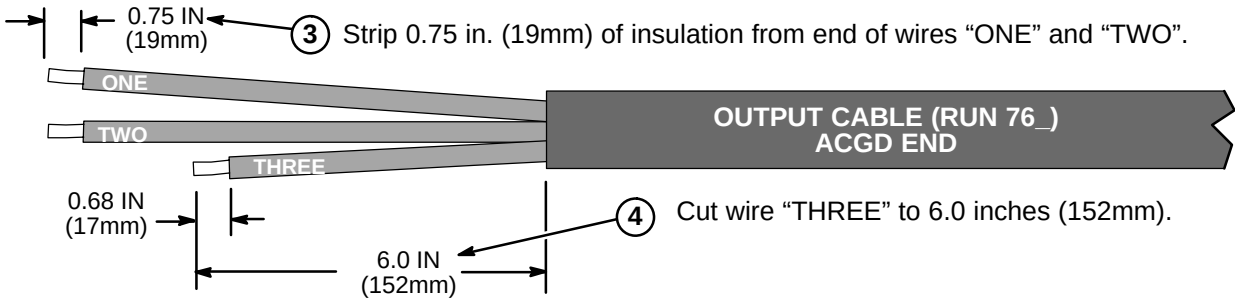
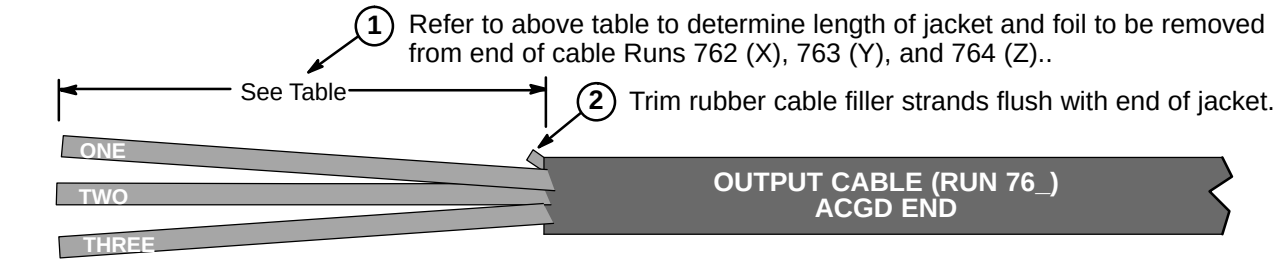
- Verify that 100 ft. (30.5 M) length of supplied Run 762, 763, and 764 cable is sufficient to route from ACGD/PDU Cabinet to Penetration Panel and Penetration Panel to Body Coil Gradient Cables at rear of magnet.
Each cable has identical terminations at each end for connection to both sides of the Penetration Panel Gradient Filter after completing Steps 2 and 3 below.
- Unroll each cable. Route cables from equipment room side of Penetration Panel to ACGD/PDU Cabinet. The end of cable with terminals will be connected to Penetration Panel. Make sure enough slack is allowed for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
- Move cable to magnet room. Route cables from Gradient Filter of Penetration Panel to Body Coil gradient cables at rear of magnet.
- Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. Do not cut Body Coil gradient cables.
- Locate furnished bag of parts and attach terminal and labels to each cable which was cut to length. See Procedures B and C.
- Route Run 762, 763, and 764 from equipment room side of Penetration Panel to ACGD/PDU cabinet.
- See Procedures D for connection to Penetration Panel.
- Route Run 762, 763, and 764 from magnet room side of Penetration Panel to Body Coil gradient cables.
- See Procedure D for connection to Penetration Panel.
- See Procedure E for connection of Runs 762, 763, and 764 to Body Coil gradient cables.
- See ACGD/PDU Cabinet installation instructions (DIRECTION 2286811 IN TAB 15) for connection of Runs 762, 763, and 764 to ACGD/PDU cabinet.

TOOLS REQUIRED: 2135776 – Cable Crimper/Stripper Kit

B – TERMINATE RUNS 762, 763, AND 764 FOR ACGD/PDU CABINET

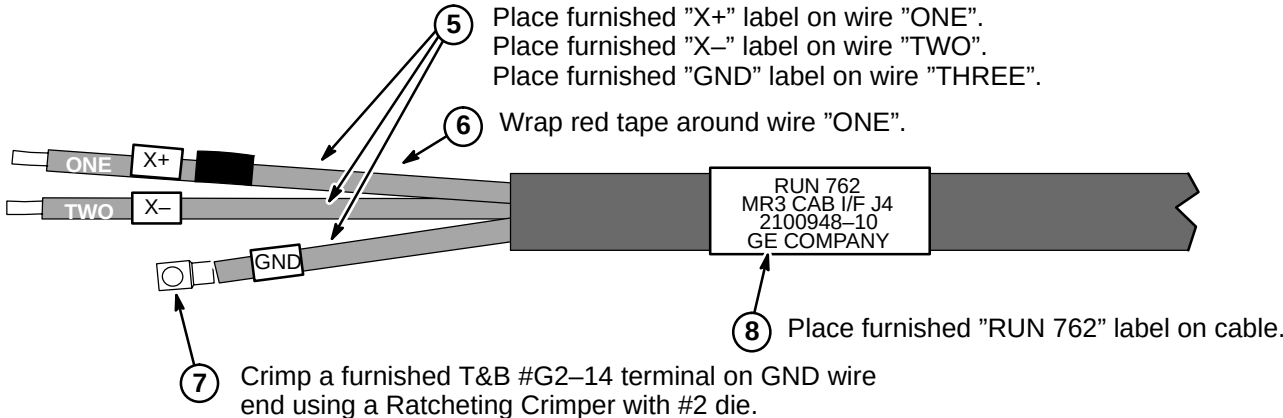
Note: The following steps are required to terminate the end of the cable that was cut in Step A4. The other end of the cable is connected to the Penetration Panel.

	Run 762	Run 763	Run 764
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)



Note: Steps in illustration below show completion of Run 762. Repeat same process for Runs 763 (Y+, Y-) and 764 (Z+, Z-).

Red tape, Y+ (RUN 763) and Z+ (RUN 764) labels are attached to wire "ONE".



C – TERMINATE RUNS 762, 763, AND 764 FOR MAGNET ROOM

Note: The following steps are required to terminate the end of the cable that was cut in Step A2. The other end of the cable is connected to the Penetration Panel.

1 Remove 84 in. (2130 mm) of jacket from end of cable Run 762.

2 Trim rubber cable filler strands flush with end of jacket.

3 Cut and remove GROUND wire.

4 Strip 0.68 in. (17 mm) of insulation from end of each wire and crimp a furnished T&B #G2–38 terminal on each wire end using a Ratcheting Crimper with #2 die.

5 Wrap red tape (from Pen Panel end of cable) around wire "ONE".

6 Place furnished "X+" label on wire "ONE". Place furnished "X-" label on wire "TWO".

7 Place furnished "Run 762" label on cable.

8 Repeat above process for Runs 763 and 764.

Note: Red tape, Y+ (RUN 763) and Z+ (RUN 764) labels are attached to wire "ONE".

D – CONNECT CABLES AT PENETRATION PANEL

1 Remove gradient filter terminal covers.

2 Route Run 762, 763, and 764 to gradient filter on Penetration Panel.

4 Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft-lbs.

Note: Red tape indicates positive (+) wires #1, #3, and #5.

3 Secure each cable Run to strain-relief bracket with U-bolts.

5 Secure GND lead wire under clamp nut for each cable Run.

6 Repeat procedure and connect cables to Gradient Filter on other side of Penetration Panel.

7 Replace gradient filter terminal covers.

EQUIPMENT ROOM MAGNET ROOM

RUN 764 RUN 763 RUN 762 RUN 762 RUN 763 RUN 764

E – CONNECT GRADIENT CABLES AT TERMINAL BLOCK AT REAR OF MAGNET

1 Route Runs 762, 763, & 764 to Terminal Block as marked below.

2 Fasten Gradient Cables to terminal block using screws removed above. Torque to 15 foot-pounds..

3 Perform a "Tug Test" to Gradient Cable connections to verify each terminal is securely crimped to the wire.

4 Install strain relief.

5 Install terminal block cover.

X1 (+) X2 (-) Y1 (+) Y2 (-) Z1 (+) Z2 (-)

F – SECURE GRADIENT OUTPUT CABLES

1 Tape cables together as shown with electrical tape, approximately 100mm (4 in.) apart, all the way back to cable jacket

INSTALLATION STEPS SUMMARY

- ☐ A – Cut cables to length and route in Equipment and Magnet Rooms

☐ B – Terminate Runs 762, 763, and 764 for ACGD/PDU Cabinet

☐ C – Terminate Runs 762, 763, and 764 in Magnet Room

☐ D – Connect Cables at Penetration Panel

☐ E1 – Active Shield Magnet Gradient Cable Connection Preperation

☐ E2 – NON-Active Shield Magnet Gradient Cable Connection Preperation

} PERFORM E1 OR E2

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

IF MAGNET IS AT FULL FIELD – CUT GRADIENT CABLE TO LENGTH AND REMOVE FROM MAGNET ROOM TO PERFORM CABLE TERMINATION PROCEDURES.

ALSO

FOR NON-ACTIVELY SHIELDED MAGNETS AT FULL FIELD – REFER TO TAB 3, BODY COIL, SIGNA HORIZON BRM-D & CRM BODY COIL IN MAGNET UPGRADE INSTRUCTIONS PROCEDURE FOR ATTACHING THE 2119600 SERIES EXTENSION CABLES TO THE BODY COIL GRADIENT CABLES.

A – CUT CABLES TO LENGTH AND ROUTE IN EQUIPMENT AND MAGNET ROOMS

- 1

Verify that 100 ft. (30.5 M) length of supplied Run 762, 763, and 764 cable is sufficient to route from ACGD/PDU Cabinet to Penetration Panel and Penetration Panel to Body Coil Gradient Cables at rear of magnet.
Each cable has identical terminations at each end for connection to both sides of the Penetration Panel Gradient Filter after completing Steps 2 and 3 below.
- 2

Unroll each cable. Route cables from equipment room side of Penetration Panel to ACGD/PDU Cabinet. The end of cable with terminals will be connected to Penetration Panel. Make sure enough slack is allowed for termination and connection to cabinet. Mark location for cutting. Cut cables to length.
- 3

Move cable to magnet room. Route cables from Gradient Filter of Penetration Panel to Body Coil gradient cables at rear of magnet.
- 4

Verify sufficient slack of at least two feet (610mm) is allowed for termination and connection. Mark location for cutting. Cut cables to length and label cut ends with applicable run number. Do not cut Body Coil gradient cables.
- 5

Locate furnished bag of parts and attach terminal and labels to each cable which was cut to length. See Procedures B and C.
- 6

Route Run 762, 763, and 764 from equipment room side of Penetration Panel to ACGD/PDU cabinet.
- 7

See Procedures D for connection to Penetration Panel.
- 8

Route Run 762, 763, and 764 from magnet room side of Penetration Panel to Body Coil gradient cables.
- 9

See Procedure D for connection to Penetration Panel.
- 10

See Procedure E for connection of Runs 762, 763, and 764 to Body Coil gradient cables.
- 11

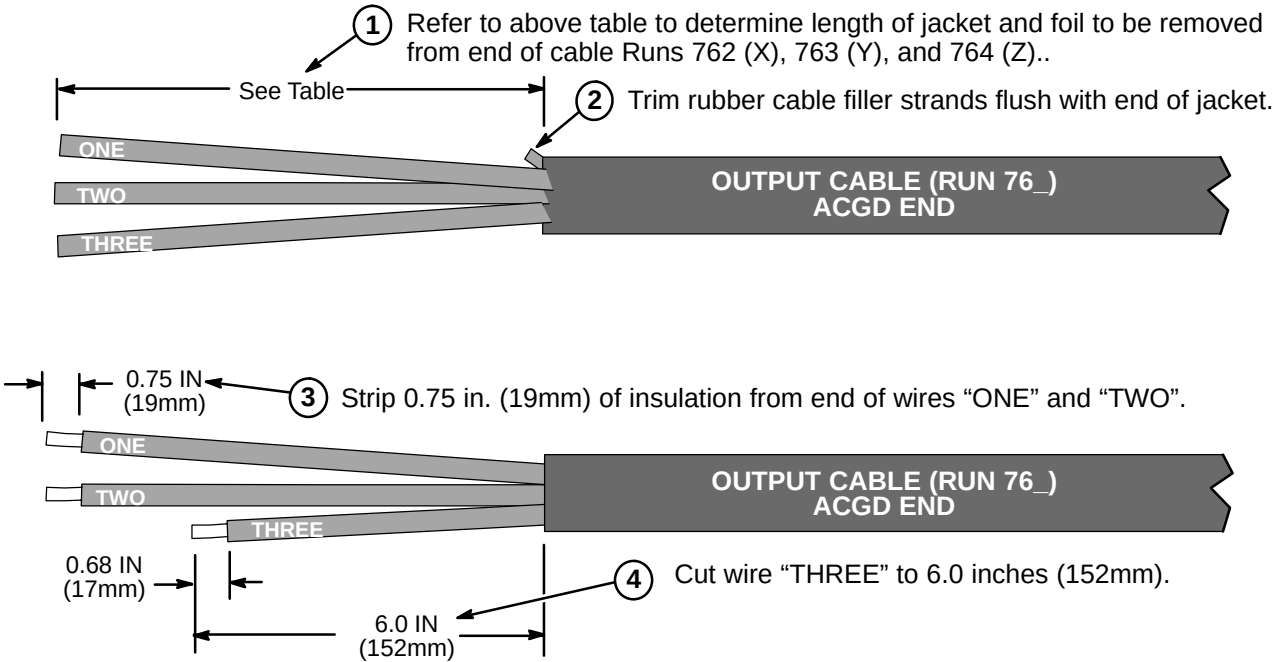
See ACGD/PDU Cabinet installation instructions (DIRECTION 2286811 IN TAB 15) for connection of Runs 762, 763, and 764 to ACGD/PDU cabinet.

TOOLS REQUIRED: 2135776 – Cable Crimper/Stripper Kit

B – TERMINATE RUNS 762, 763, AND 764 FOR ACGD/PDU CABINET

Note: The following steps are required to terminate the end of the cable that was cut in Step A3. The other end of the cable is connected to the Penetration Panel.

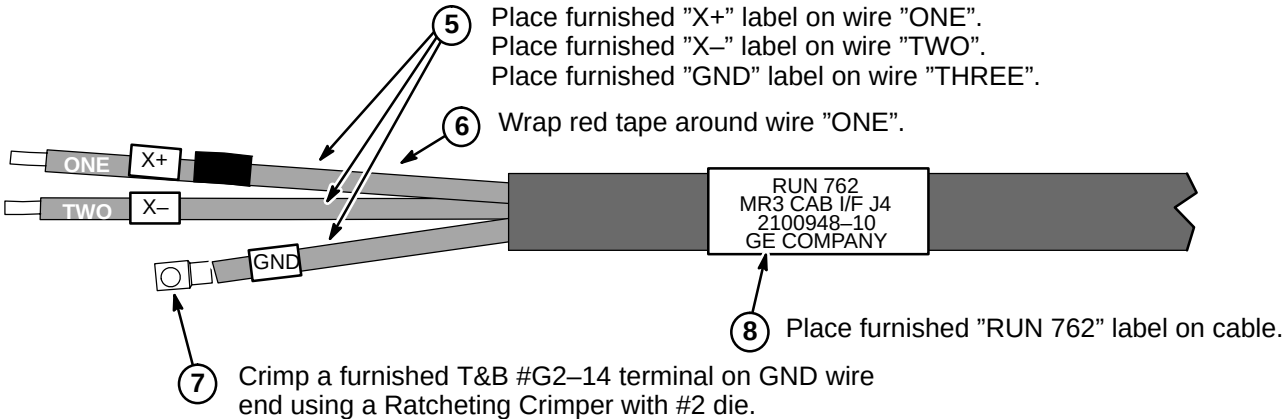
	Run 762	Run 763	Run 764
Jacket Length to be Removed	8.0 IN (203mm)	11.0 IN (280mm)	14.0 IN (356mm)



Note:

Steps in illustration below show completion of Run 762. Repeat same process for Runs 763 (Y+, Y-) and 764 (Z+, Z-).

Red tape, Y+ (RUN 763) and Z+ (RUN 764) labels are attached to wire "ONE".



ONE

TWO

GND

24 IN
(288mm)

1

Remove 24 in. (288 mm) of jacket from end of cable Run 762.

GRADIENT CABLE (MAGNET ROOM)
(RUN 762)

2

Trim rubber cable filler strands flush with end of jacket.

ONE

TWO

GND

3

Cut and remove GROUND wire.

4

Strip 0.75 in. (19 mm) of insulation from end of each wire.

5

Wrap red tape (from Pen Panel end of cable) around wire "ONE".

X+

ONE

X-

TWO

6

Place furnished "X+" label on wire "ONE".
Place furnished "X-" label on wire "TWO".

7

Place furnished "Run 762" label on cable.

8

Repeat above process for Runs 763 and 764.
Note: Red tape, Y+ (RUN 763) and Z+ (RUN 764) labels are attached to wire "ONE".

RUN 762
MG3-A32-TS1/2
2100948
GE COMPANY

1

Remove gradient filter terminal covers.

2

Route Run 762, 763, and 764 to gradient filter on Penetration Panel.

4

Connect wires to gradient filter terminals as marked. Align terminals to obtain maximum spacing from other terminals and conductive metal objects. Tighten nuts with a torque wrench to 35 ft-lbs.
Note: Red tape indicates positive (+) wires #1, #3, and #5.

3

Secure each cable Run to strain-relief bracket with U-bolts.

5

Secure GND lead wire on Ground Stud for each cable Run.

6

Repeat procedure and connect cables to Gradient Filter on other side of Penetration Panel.

7

Replace gradient filter terminal covers.

EQUIPMENT ROOM

MAGNET ROOM

1

2

3

4

5

6

2

1

4

3

6

5

RU

RUN

RUN

7621

763

764

RUN

RUN

RUN

764

763

762

1

Route Gradient cables as shown for butt splice connection in cable trough.

2

Refer to **CONNECTION OF BODY COIL GRADIENT CABLES TO RUNS 762, 763, & 764** Procedure in **Direction 2223489 of Tab 3** for instructions on connecting Gradient cables from Body Coil to cables coming from Penetration Panel.

Gradient Output Cables from Penetration Panel

Gradient Cables from Body Coil

Note: This illustration is typical for an Active Shield magnet.

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

IF MAGNET IS NON-ACTIVE SHIELD AND IS AT FULL FIELD – MOVE END OF BODY COIL EXTENSION CABLES TO FARTHEST POINT AWAY FROM MAGNET TO PERFORM BUTT SPLICE CONNECTION TO RUNS 762, 763, AND 764 COMING FROM PENETRATION PANEL.

1

If possible, make butt splice connection of Gradient cables prior to mounting Rear Pedestal. This would give you more distance from magnet for performing butt splice.

- Make sure cables from Penetration Panel have been routed and cut to proper length.
- Make sure cables are spliced on side of magnet where cables are routed to Penetration Panel.

2

Refer to **CONNECTION OF BODY COIL GRADIENT CABLES TO RUNS 762, 763, & 764** Procedure in **Direction 2223489 of Tab 3** for instructions on connecting Gradient cables from Body Coil to cables coming from Penetration Panel.

3

After completion of butt splices, route cables in final position and install Rear Pedestal.

Gradient Output Cables from Penetration Panel

Gradient Output Cables from Body Coil

Unmounted Rear Pedestal shown

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1-2 LAYOUT OF CABLES	1-3
1-3 TERMINATE AND CONNECT RF BODY CABLES AT PENETRATION PANEL	1-4
1-4 TERMINATE AND CONNECT RF HEAD COIL CABLES	1-5

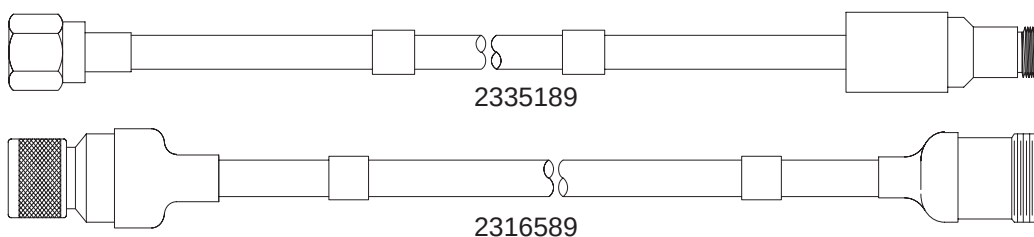
1-1 TIMES MICROWAVE RF CABLES INFORMATION

Four cables and other associated parts to be installed are provided for this procedure.

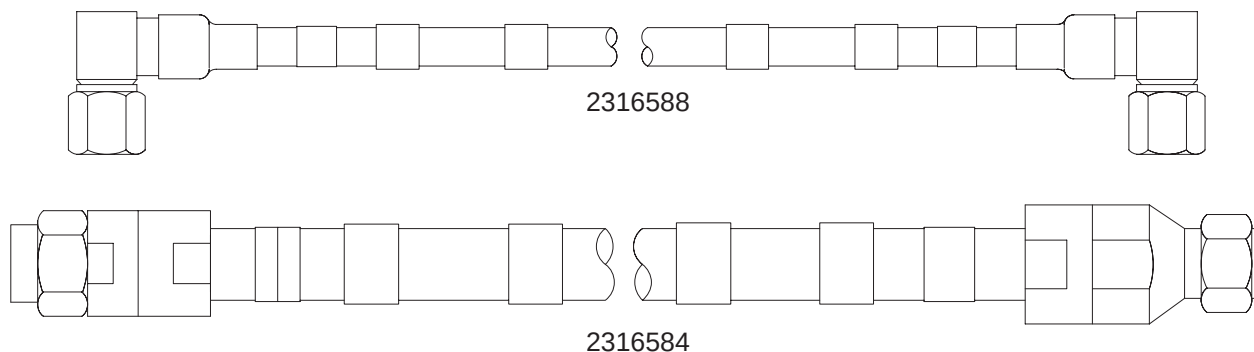


Do not subject LMR-600FR cable (Head) to a bend radius less than 1.5 inches (138mm). Do not subject LMR-900FR cable (Body) to a bend radius less than 3 inches (75mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

The two short cables, 2335189 (Head) and 2316589 (Body) are connected to the SRFD2 cabinet.



The two long cables, 2316588 (Head, Runs 236 & 258) and 2316584 (Body, Runs 235 & 257), will be cut to length and terminated for routing in the equipment and scan rooms.

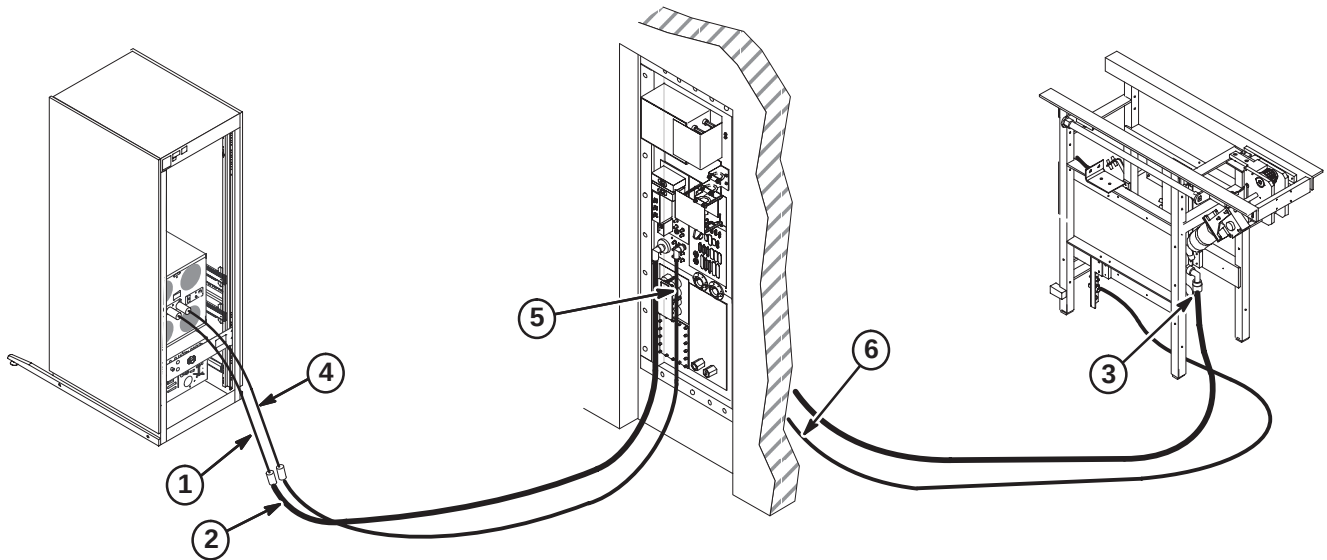


The remaining connectors, fasteners, and labels will be installed as directed in the following steps.

1-2 LAYOUT OF CABLES

RF cables will be routed and connected as indicated in the following steps.

1. Connect extension cable 2316589 to J4 Body Output at rear of RF module in RF cabinet.
2. Connect end of Times RF Body cable (2316584) marked "(RUN 235) FJ5" to Body extension cable. Route to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1-3 for termination procedure of cut end.
3. Move remainder of RF Body cable to Scan Room and connect other end of cable (2316584) marked "(RUN 257) MG3A3-J1" to Rear Pedestal as shown below. Route to J44 at Penetration Panel. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1-3 for termination procedure of cut end.
4. Connect extension cable 2335189 to J3 Head Output at rear of RF module in RF cabinet.
5. Connect end of Times RF Head cable (2316588) marked "(RUN 236) PP1-J4" to J4 at Penetration Panel. Route to RF Head extension cable at rear of RF cabinet. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1-4 for termination procedure of cut end.
6. Move remainder of RF Head cable to Scan Room and connect other end of cable marked (RUN 258) PP1-J4" to Penetration Panel at J4. Route to Rear Pedestal Takeup bracket J1. Determine if any take-up slack is needed and cut to length. Refer to *Times Connector Installation Procedures* provided with cables for cutting information. Refer to Section 1-4 for termination procedure of cut end.



1-3 TERMINATE AND CONNECT RF BODY CABLES AT PENETRATION PANEL

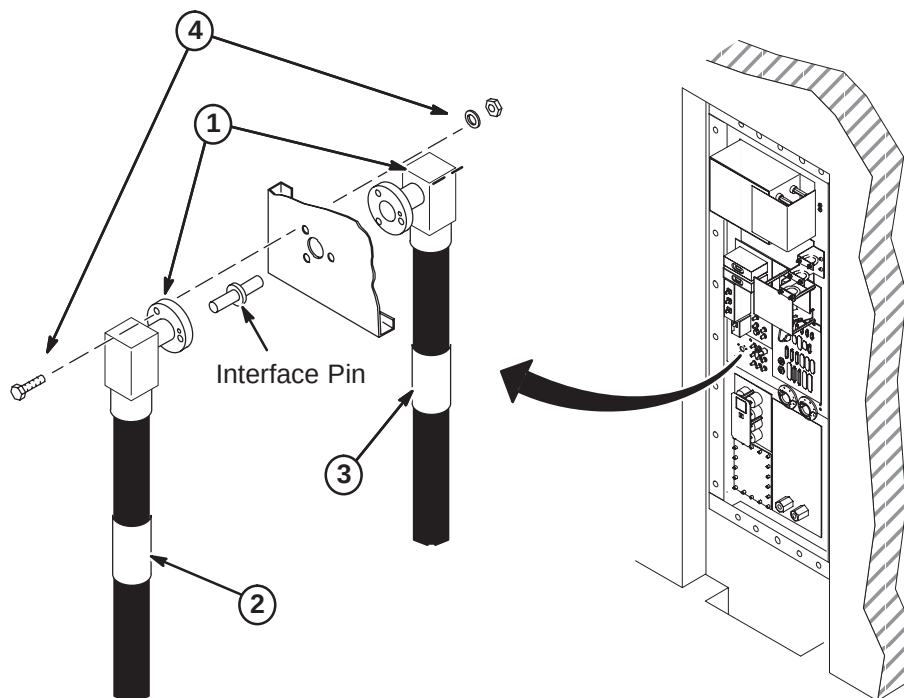
Perform the following steps for terminating and connecting the RF Body cables at the Penetration Panel J44.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an EIA Right Angle Connector (2319986) on each end of the RF cables. Use provided LMR900 Stripping Tool (2335190).

Note

Make sure the fastener holes in the connector flanges align with the holes in the Penetration Panel so the cables will route toward floor when installed.

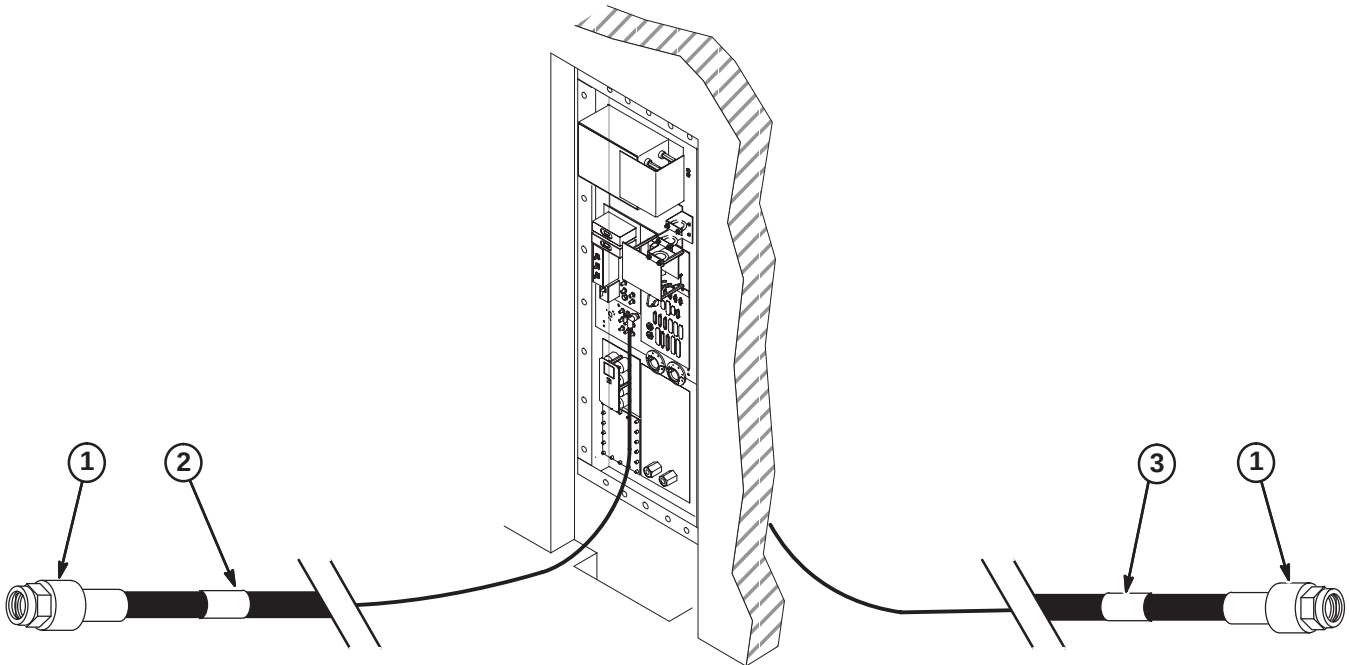
2. Attach label "RUN 235, PP1-J44" (2357230) to cable as shown on Equipment room side of Penetration Panel.
3. Attach label "RUN 257, PP1-J44" (2357230-2) to cable as shown on Scan room side of Penetration Panel.
4. Insert Interface Pin into one of the connectors. At J44, align connectors on each side of Penetration Panel and push together. Fasten together using three of each of the following:
 - 1/4-20 x 1.25 Hex Head Bolt (brass)
 - 1/4 Screw Size Washer (phos bronze)
 - 1/4-20, 7/16" Hex x 3/16" thk Nut (brass)



1-4 TERMINATE AND CONNECT RF HEAD COIL CABLES

Perform the following steps for terminating and connecting the RF Head cables in the Equipment and Scan rooms.

1. Refer to *Times Connector Installation Procedures* provided with cables for information on installing an Straight Male N Connector (2352195) on each end of the RF cable. Use provided LMR600 Stripping Tool (2352193).
 - Connect Equipment room end to Extension cable at FJ6.
 - Connect Scan room end to MG3A17-J1 at Interface bracket on Rear Pedestal.
2. Attach label "RUN 236 FJ6" (2357230-3) to cable as shown in Equipment room.
3. Attach label "RUN 258 MG3A17-J1" (2357230-4) to cable as shown in Scan room.



MAGNET ENCLOSURE UPGRADE INSTRUCTIONS

Tab 2, Magnet Enclosure, contains upgrade instructions for the following configurations:

- *DIRECTION 2147876* – Magnishield Magnet Enclosure Upgrade
- *DIRECTION 2147878* – Fixed–Site GE Magnet Enclosure Upgrade
- *DIRECTION 2294109* – Dual Temperature Sensor Upgrade
- *DIRECTION 2403142* – Signa Magnet Enclosure Upgrade for EXCITE II

Use only the applicable Magnet Enclosure Upgrade Instruction Direction for the site being upgraded

GENERAL ENCLOSURE NOTES:

- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.

INSTALLATION STEPS SUMMARY

- ☐ A1 – Remove Enclosure Covers, Bridge, Bridge Supports, DSIM/SRI and PAC Asm
- ☐ A2 – Remove Alignment Lights
- ☐ A3 – Remove Body Coil, Rear Pedestal, Bore Light and Comfort Module

☐ A1 – REMOVE COVERS, BRIDGE, BRIDGE SUPPORT, DSIM/SRI AND PAC ASM

③ Discard Rear cover. **Save all latches and all mounting hardware.**

① Remove and discard Pedestal Rear Cover. **Save mounting hardware for reuse.**

② Remove and **save Rear Pedestal Side Covers and hardware. These will be reused.**

④ For 3.x or 4.x, Remove and discard DSIM/SRI Module.

⑤ Discard Front Cover, Displays and Operator's Controls. **Save all latches and all mounting hardware.**

⑥ Remove and discard PAC Asm and Fiber Optic Runs 711/712

⑦ Remove Bridge Asm and **save all parts removed from the bridge.** Discard fiberglass Bridge.

⑧ Remove from Bridge Support Asm and **save Dock Control Box Asm and attaching hardware.**

⑨ Remove and discard Bridge Support Asm.

⑩ See Steps F1 and F2 if Carriage base plate does not have a notch for Cable Trak clearance. Notch is required for Upgrade procedure.

☐ A2 – REMOVE ALIGNMENT LIGHTS

② Remove cable clamp and terminal strip screws.

③ **Save Light assemblies with wires and cables attached to Terminal Strips.**

④ **Remove and save Light Asm mounting hardware.**

⑤ Discard alignment light base plates and stand-offs.

① Remove Alignment Light Asm, stand-offs, and brackets. Leave Run 385 attached to Terminal Strips.

☐ A3 – REMOVE BODY COIL, REAR PEDESTAL, BORE LIGHT, & COMFORT MODULE

① Remove Body Coil.

② **Remove and save Bore Light electronics and hardware for transfer to new bracket.** Discard bracket.

③ Remove and discard Comfort Module.

****CAUTION**
MAGNETIC MATERIAL**

④ Unfasten Rear Pedestal and remove from Magnet Room. **Many of the components of the removed Rear Pedestal HAVE to be transferred to the new Rear Pedestal.** Refer to procedure D1 on page 4 for details.

⑤ Discard old Rear Pedestal frame after transfer of components.

GENERAL ENCLOSURE NOTES:

- A 1-ounce tube of Permatex "Anti-Seize" lubricant is provided. Follow instructions on back of tube. "Anti-Seize" lubricant is to be applied to threads of stainless-steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.

INSTALLATION STEPS SUMMARY

- ❑ B1 – Install Front Ring Asm and Vertical Upright
- ❑ B2 – Route and Connect Runs 778 and 779
- ❑ B3 – Install Front Gasket and Mount End Bell
- ❑ B4 – Install/Connect Alignment Lights & Microphone Bracket

❑ B1 – INSTALL FRONT RING ASM AND VERTICAL UPRIGHT

1 Remove and discard Left Front Upright and left side bracket. Save mounting hardware.

2 Install new Left Front Upright (2120127) using existing hardware. **Note reversed orientation.**

3 Install new Left Front Upright bracket (2142631) using existing hardware. Tighten all hardware attaching Upright and bracket.

4 Apply Permatex "Anti-Seize" Lubricant (2119594) to :
(12) 2116325 3/8-16 x 55mm SST Stud
Install studs into Front Ring holes.

5 Place Front Ring Asm (46-307710G7) on studs.

6 Mount Upper Alignment Light Bracket (2129348-2) using existing hardware.

7 Place Dock Control Box ring segment (2129109) on studs.

8 Transfer Dock Control Box and attaching hardware from previously removed Bridge Support to Ring.

9 Place End Ring Section Spacer (2129823) on studs.

❑ B2 – ROUTE AND CONNECT RUNS 778 AND 779

1 Remove Air Plenum from Front End Bell to access bottom mounting studs.

2 Position Front End Bell Asm (46-317004G2) on floor in front of Magnet

3 Route Microphone Cable through hole in lower right side of Front End Bell Asm.

4 Route Runs 778 and 779 through hole in lower left side of Front End Bell Asm.

5 Connect Body Coil cables to Runs 778 and 779 using Bulkhead Asm. (Furnished with Body Coil)

Cable J3/J5

Run 778

Run 779

Run 749

❑ B3 – INSTALL FRONT GASKET AND MOUNT END BELL

1 Position around End Ring segments:
(6) 2115369-2 1.5mm thick Rubber Seal w/Adhesive

2 Move Front End Bell Asm into position on studs. Adjust slack of Runs 778 and 779. Fasten End Bell with:
(12) 1000904P447 25/64 Flat Washer
(12) 46-220196P6 3/8 SST Lock Washer
(12) 46-208948P2 3/8-16 SST Hex Nut

CAUTION: The Bushing in Step 3 is a special design for this application. Do not substitute or cable damage will occur.

3 Place Bushings over Microphone Cable and Run 749 on right side and Runs 778 and 779 on left side and secure into holes in End Bell Asm.

4 Install previously removed Air Plenum.

❑ B4 – INSTALL/CONNECT ALIGNMENT LIGHTS & MICROPHONE BRACKET

Note: If Halogen lights and terminal strips are still attached to Run 385, mount the Halogen lights and terminal strips as required during the routing of Run 385.

1 Route and secure Run 385 through rubber lined clamps around End Ring towards Right Alignment Light.

2 Mount the three previously removed alignment lights using original mounting hardware.

3 Connect Alignment Light wires to terminal strip.

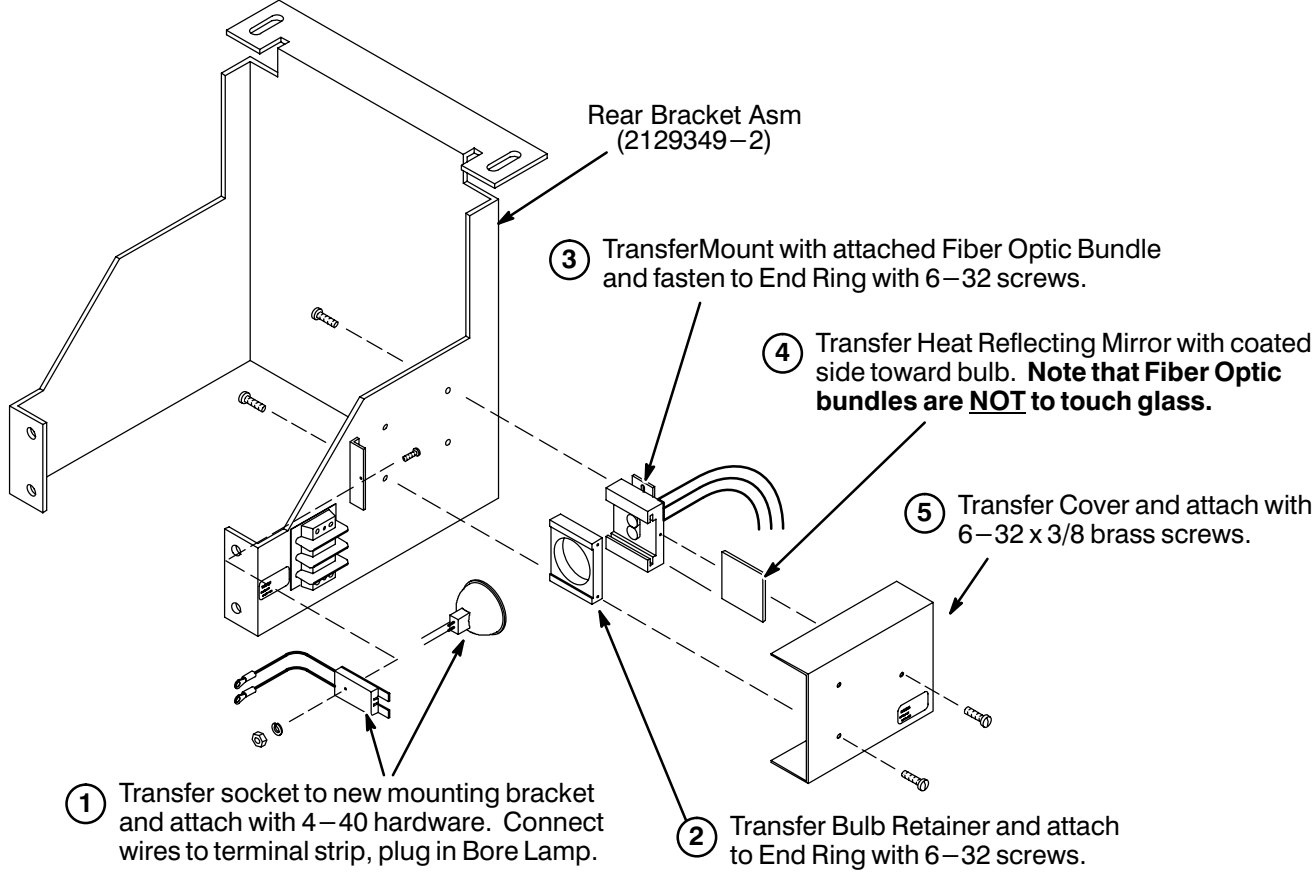
4 From the Hardware Box, install Front Microphone Connector Bracket (2109027) with existing hardware. Insert (2) Cable Mounts (46-265067P1) and connect Microphone cable from End Bell to Run 414 at this bracket.

Note: Secure electrical ground required. Insure low resistance connection and tighten fasteners securely.

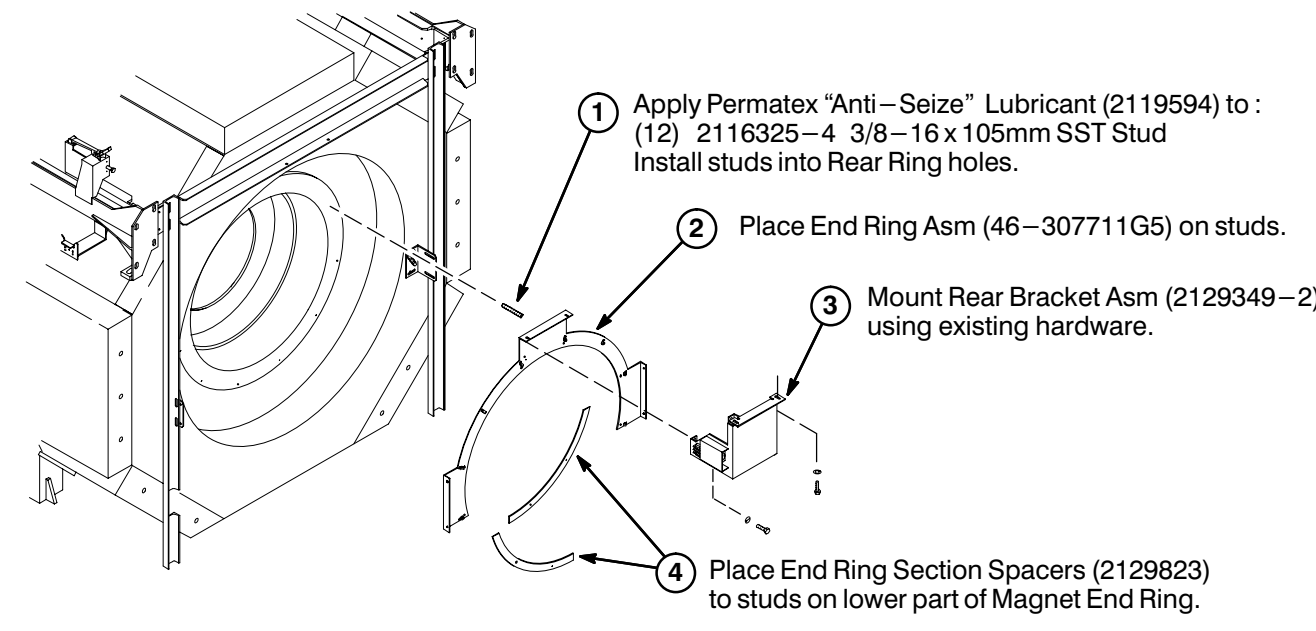
INSTALLATION STEPS SUMMARY

- C1 – Transfer & Assemble Bore Light to New Rear Bracket Asm
- C2 – Install Rear End Rings and Bracket Asm
- C3 – Formula for Determining Rear Spacers
- C4 – Install Rear Spacers and Gradient Terminal Block

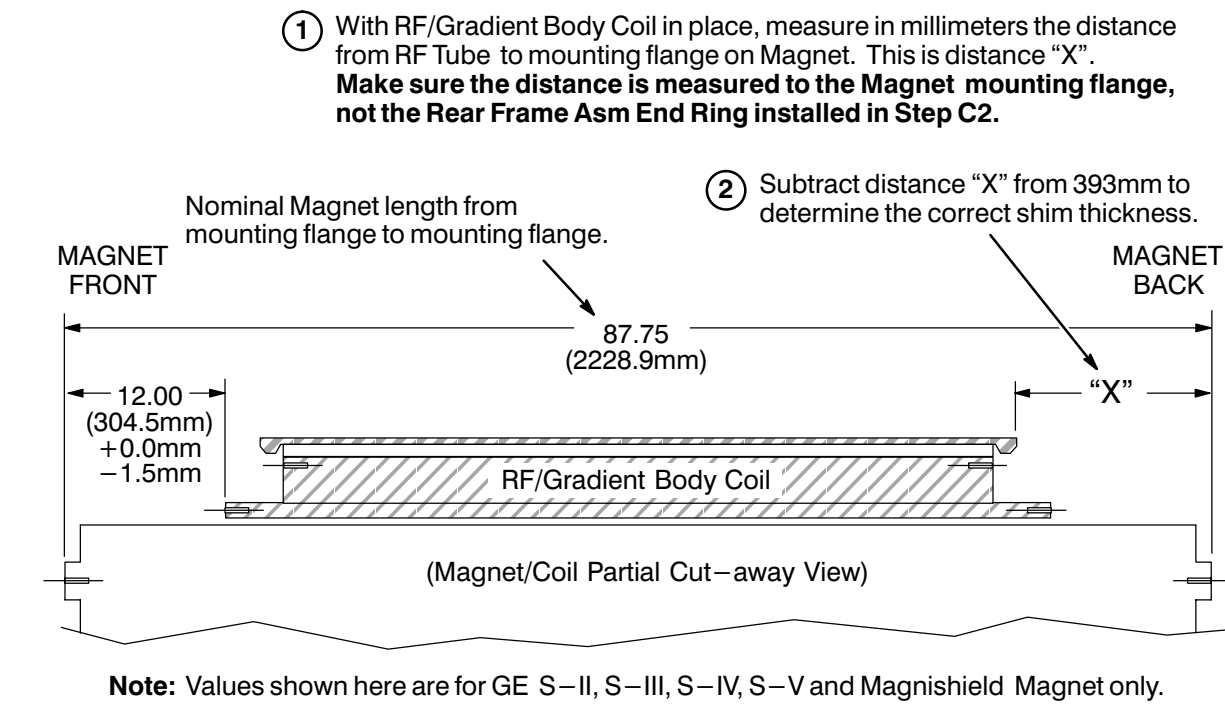
C1 – TRANSFER & ASSEMBLE BORE LIGHT TO NEW REAR BRACKET ASM



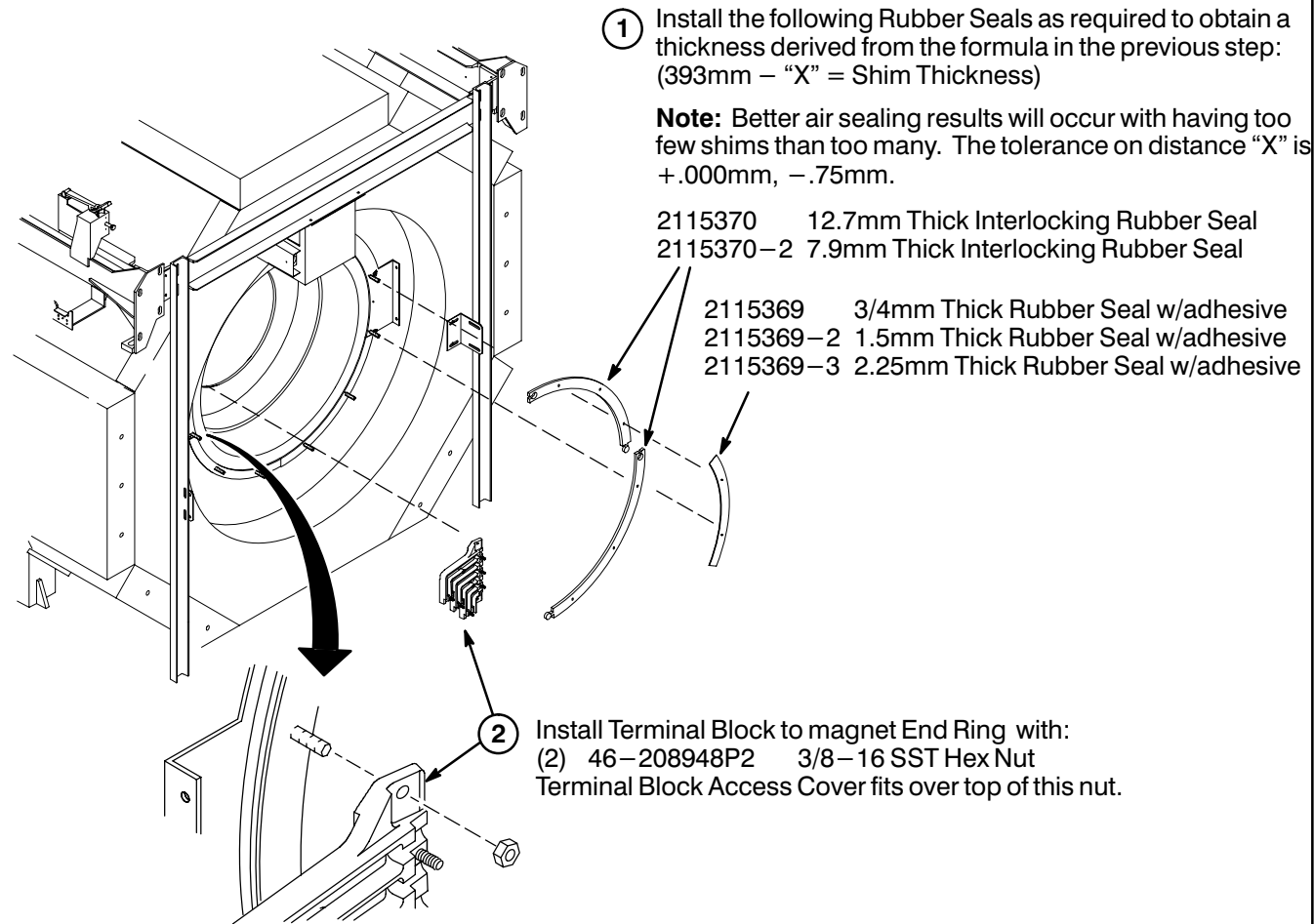
C2 – INSTALL REAR END RINGS AND BRACKET ASM



C3 – FORMULA FOR DETERMINING REAR SPACERS



C4 – INSTALL REAR SPACERS AND GRADIENT TERMINAL BLOCK



INSTALLATION STEPS SUMMARY

- D1 – Transfer Existing Components to New Rear Pedestal Frame
- D2 – Install Rear Bulkhead and Rear Pedestal
- D3 – Connect Body Coil Gradient Cables to Terminal Block
- D4 – Route and Connect Bulkhead Cables and Hoses

□ D1 – TRANSFER EXISTING COMPONENTS TO NEW REAR PEDESTAL FRAME

Note: Because of possible damage to smaller pieces of mounting hardware during transfer of components to new Rear Frame, a bag of #6, #8, and #10 screws, nuts, and lock washers is provided as spares.

- ③ Transfer 1.5T Body Preamp and Rigid Coax Cable to new Body Hybrid Splitter (46–321312G3). Install Body Splitter.
- ② Remove and re–install Magnet Interface #1 box.
- ④ If upgrading from 5.x, remove and re–install Patient Communication Box. OR If upgrading from 3.x or 4.x, install new Patient Communication Box (46–317185G1).
- ⑤ If Multi–coil, transfer Multi–coil Select Switch to new Rear Pedestal.
- ⑥ Transfer Coil Cable Take–up Interface Assembly to new Rear Pedestal.
- ① Remove from existing Rear Pedestal the Long Drive Asm and re–install into new Rear Pedestal (46–328137G15).

□ D2 – INSTALL REAR BULKHEAD AND REAR PEDESTAL

Note: If desired, End Bell installation procedures D3 & D4 can be performed after completion of Step ③ below and prior to starting Step ④, installing Rear Pedestal.

- ③ Place Bulkhead Asm (2129410) on two lower studs of End Ring Asm.
- ② Verify Support Straps are fastened to rear legs
- ④ Install (2) Hex Spacers (2124040) on two lower studs.
- ⑤ Install Rear Pedestal and fasten to lower End Ring studs with:
 - (2) 46–220196P6 3/8 Lock Washer
 - (2) 46–208948P2 3/8–16 x 3/4 SST Hex Screw
- ⑥ Place Level on lower frame channel; adjust leveling screws on rear legs as needed.
- ① Verify/re–assemble front and rear legs in vertical orientation.

□ D3 – CONNECT BODY COIL GRADIENT CABLES TO TERMINAL BLOCK

- ① Connect Gradient Cables as marked to studs in Terminal Block.
- ② Tape cables together as shown after connections are made in Steps ①.

□ D4 – ROUTE AND CONNECT BULKHEAD CABLES AND HOSES

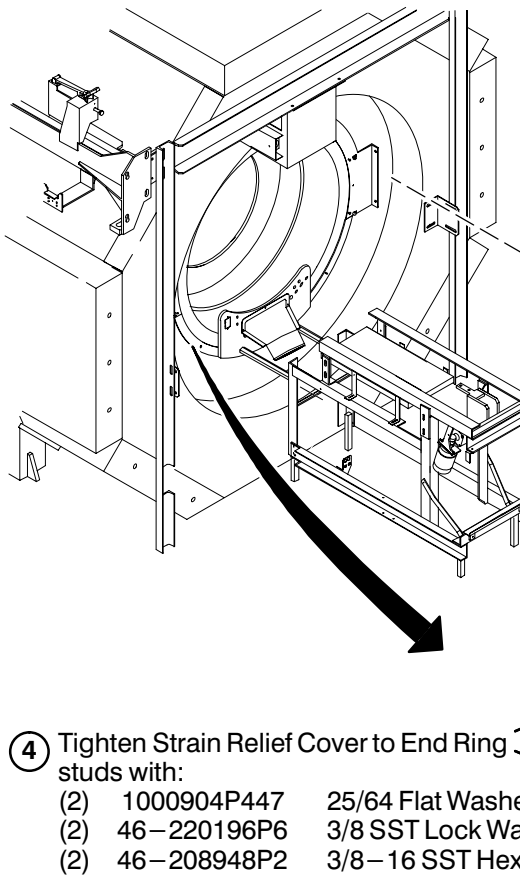
- ① Route Body Coil water lines from Penetration Panel, thru Bulkhead, and connect to Body Coil.
 - CAUTION**
To prevent pinching of hoses and blue cables when Rear End Bell is installed, do not coil or loop hoses, between bulkhead and body coil.
 - Note:** Fragile connections to RF Body Coil. Do not kink or bend sharply.
 - ② Install as marked, without twisting, Body Coil Drive Cables to J1 and J2 on Bulkhead.
 - ④ Connect as marked Dynamic Disable Drive Cables to J5 and J6 on Bulkhead.
 - ③ If available, position within the upper and lower boundaries left and right Hex Clamps (2179135 & 2179135–2) around Drive Cable hex fittings. Remove protective cover from adhesive and attach Hex Clamps to bulkhead.
- Upper Boundary
Lower Boundary
Left and Right Hex Clamps
(Rear View of Bulkhead)

INSTALLATION STEPS SUMMARY

- E1 – Install Rear End Bell
- E2 – Connect Runs 257, 258, 262, 282, 318, 414, 422, 700, AND 780 at Rear Pedestal
- E3 – Route/Connect Runs 743, 744, 781, and 782
- E4 – Connect Runs 421 and Dock Cable & Install Skirt Upgrade Kit

E1 – INSTALL REAR END BELL

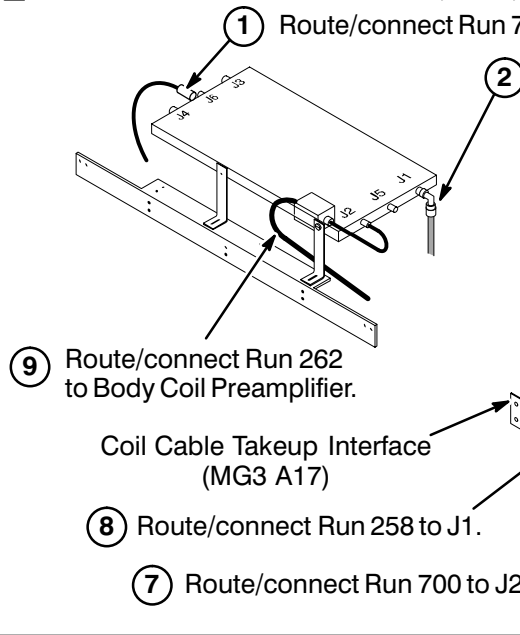
Note: A good seal is required to prevent air leaks. Any air leaks between RF Tube (Body Coil) and End Bell joint can cause a loud humming noise during patient scanning. If possible, conduct an air leakage test prior to Bridge installation. This would require connecting power to the Blower Box and connecting 6" blower hose from Blower Box to Bulkhead.



- 1 Install Rear End Bell (2113076) and fasten to studs with:
 - (8) 1000904P447 25/64 Flat Washer
 - (8) 46-220196P6 3/8 SST Lock Washer
 - (8) 46-208948P2 3/8-16 SST Hex Nut
- 2 Install Rear End Bell to Bulkhead Asm with:
 - (3) 2109878-3 M6 SST Flat Washer
 - (3) 2109866-13 M6 SST Hex Screw
- 3 Install Strain Relief Cover to Terminal Block using hardware provided with assembly.
- 4 Tighten Strain Relief Cover to End Ring studs with:
 - (2) 1000904P447 25/64 Flat Washer
 - (2) 46-220196P6 3/8 SST Lock Washer
 - (2) 46-208948P2 3/8-16 SST Hex Nut

Strain Relief Cover

E2 – CONNECT RUNS 257, 258, 262, 282, 318, 414, 422, 700, and 780



- 1 Route/connect Run 780 to J6 on RF Splitter.
- 2 Route/connect Run 257 Heliax® to J1 on RF Splitter.
- 3 Route/connect Run 282 to J6 on Patient Communications Box (MG3 A2).
- 4 Route/connect Run 414 to J9.
- 5 Route/connect Run 422 to J4 on Magnet Interface #1 (MG3 A1).
- 6 Route/connect Run 318 to short cable on Longitudinal Motor Drive (MG3 A7).
- 7 Route/connect Run 700 to J2.
- 8 Route/connect Run 258 to J1.
- 9 Route/connect Run 262 to Body Coil Preamplifier.

Coil Cable Takeup Interface (MG3 A17)

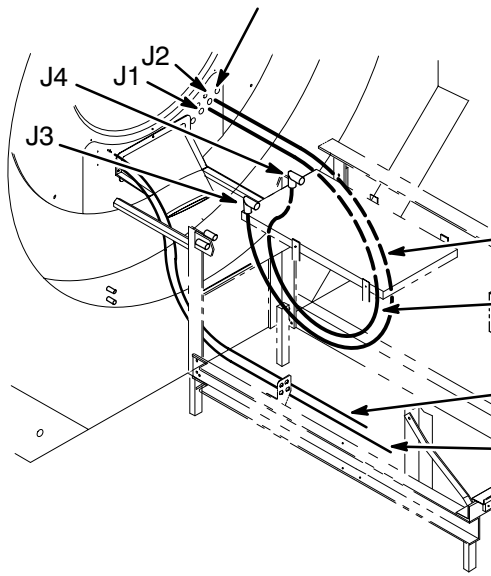
Magnet Interface #1 (MG3 A1)

E3 – ROUTE/CONNECT RUNS 743, 744, 781, AND 782



The RF Drive Input cables, Runs 743 and 744, utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

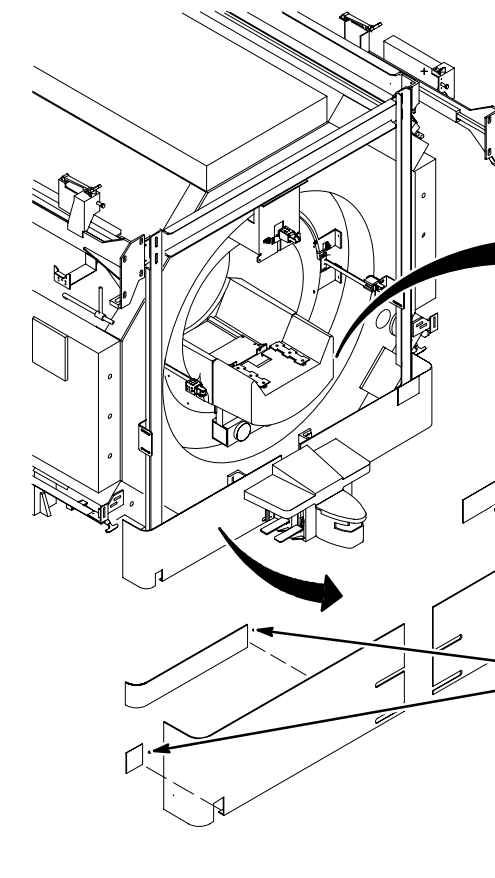
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



Note: Runs 743 and 744, Body Coil to RF Splitter cable connections may be reversed depending on magnet field polarity. Consult local Field Engineer for proper routing.

- 1 Route and connect Run 744 (2105691-5) from J1 on the Bulkhead to J4 on the RF Splitter.
- 2 Route and connect Run 743 (2105691-4) from J2 on the Bulkhead to J3 on the RF Splitter.
- 3 Route and connect Run 781 (46-328000G919) from J5 on the Bulkhead to J75 on the Penetration Panel.
- 4 Route and connect Run 782 (46-328000G920) from J6 on the Bulkhead to J76 on the Penetration Panel.

E4 – CONNECT RUNS 421 AND DOCK CABLE & INSTALL SKIRT UPGRADE KIT

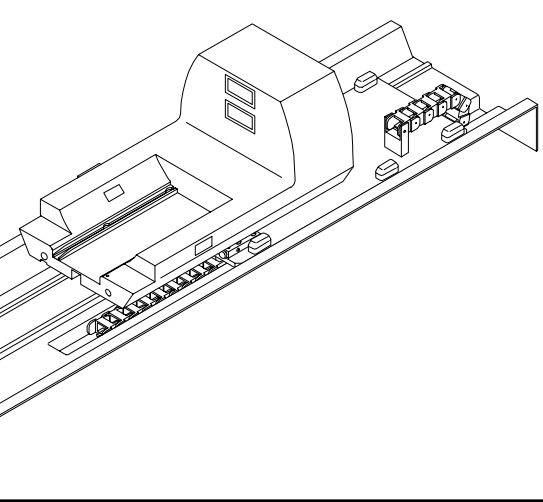


- 1 Attach Cable Extension (2143072) to Dock Cable and connect to J2 on Dock Limit Module (MG3 A31).
- 2 Route/connect Run 421 from SRI Module to J1 on Dock Limit Module (MG3 A31).
- 3 Install Front Skirt Upgrade Kit (2139555). Remove protective coating from following parts and install as shown:
 - (1) 2139555-2 Vacuum Break Port Access
 - (1) 2139555-3 Trim for Front Skirt
 - (1) 2139555-4 Front Skirt Wraparound
 - (1) 2139555-5 Front Skirt Top Plate
 - (1) 2139555-6 Front Skirt Corner Trim

INSTALLATION STEPS SUMMARY

- ☐ F1 – Remove Carriage from Bridge for Notch Modification (If Needed)
- ☐ F2 – Cutting Notch in Carriage (If Needed)

☐ F1 – REMOVE CARRIAGE FROM BRIDGE FOR NOTCH MODIFICATION (If Needed)

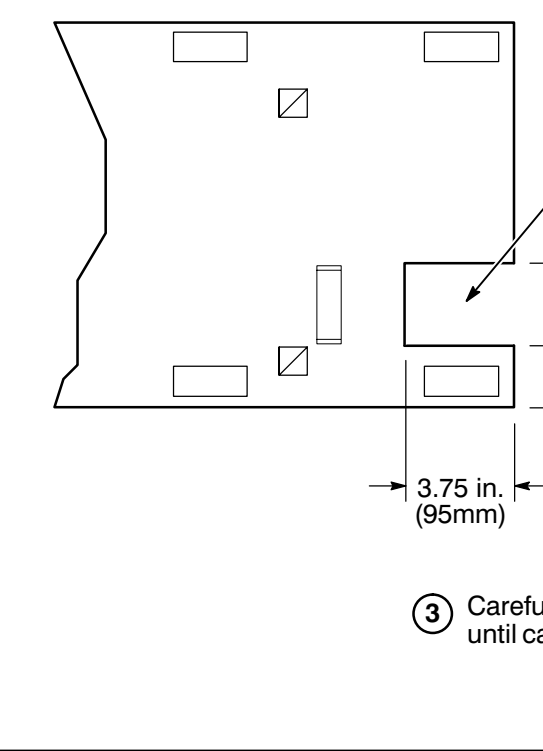


Note:
Cable track mount is easily broken.
Pry evenly with two screwdrivers.

① Detach Cable Track with two screwdrivers.

② Remove Carriage from Magnet Room to cut required notch.

☐ F2 – CUTTING NOTCH IN CARRIAGE (If Needed)



① Cut Notch according to dimensions.
Remove sharp edges.

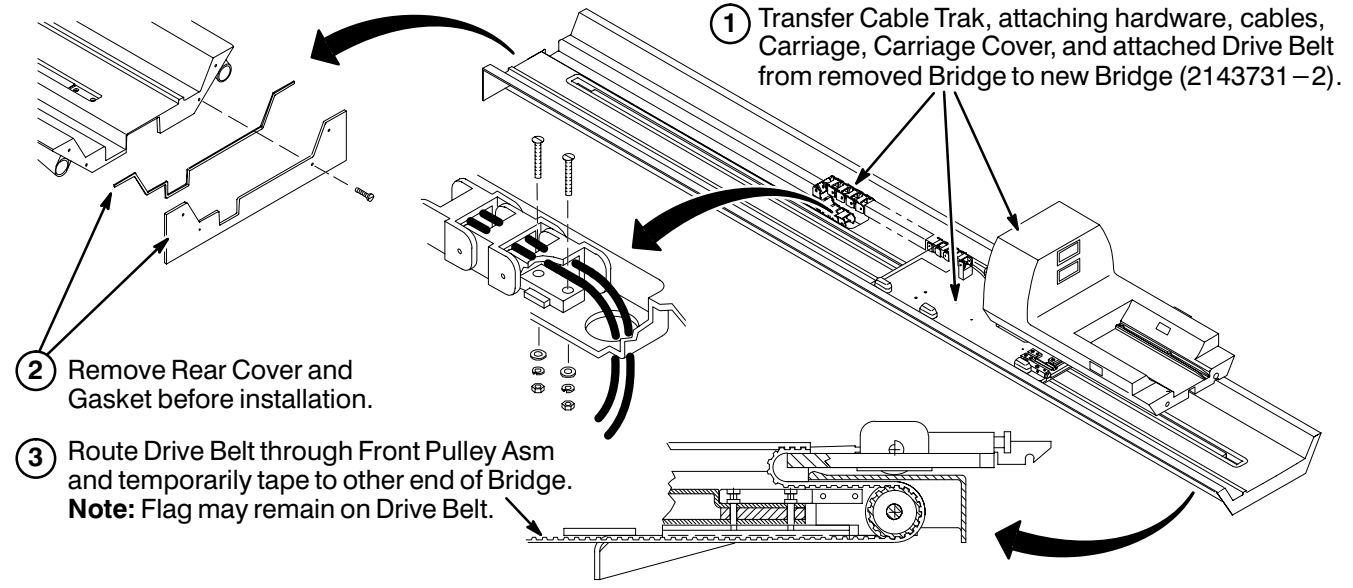
② Move Carriage back to Bridge.

③ Carefully spread tabs with screwdrivers until cable track drops into place.

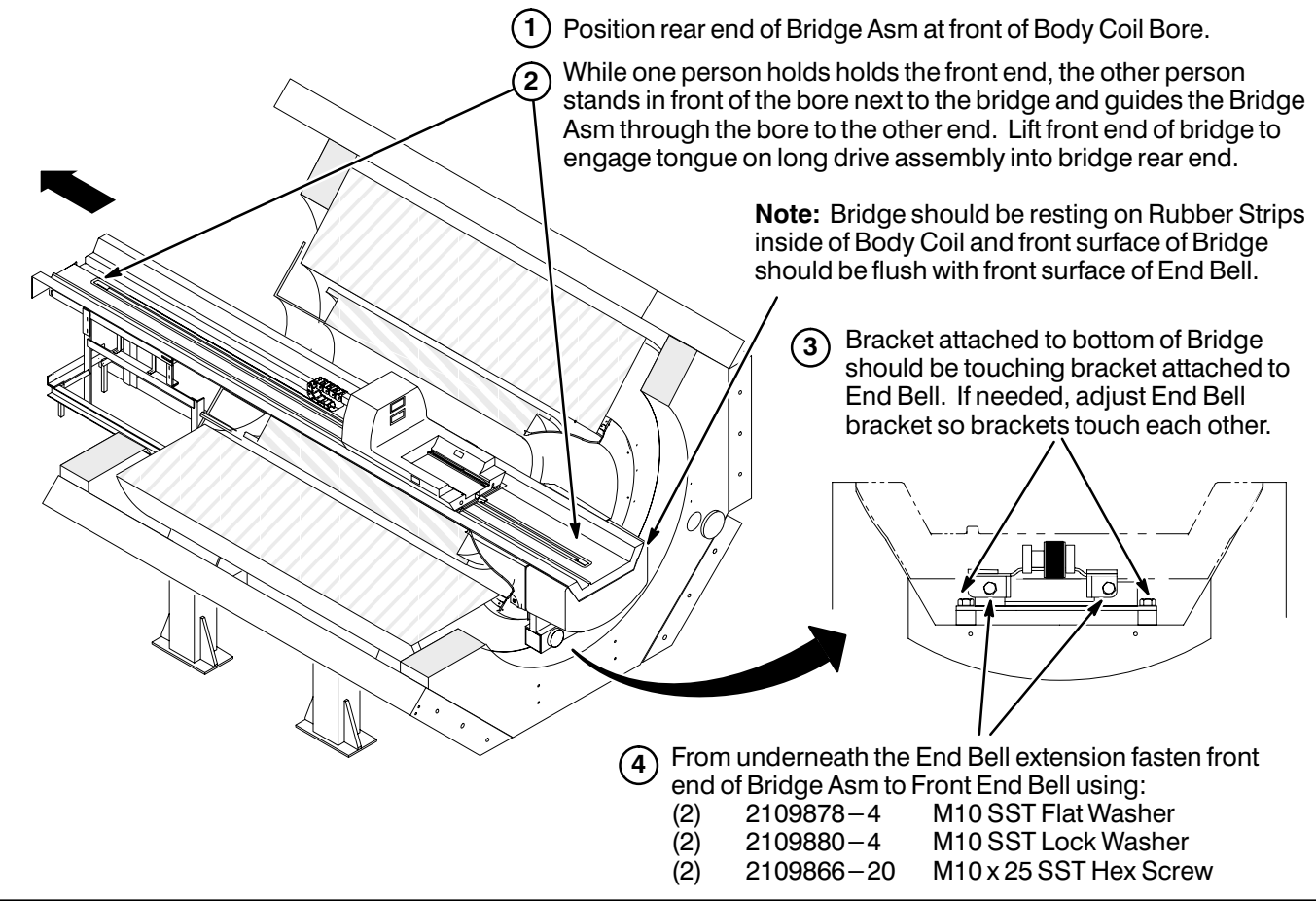
INSTALLATION STEPS SUMMARY

- G1 – Bridge Assembly Preparation Before Installing in Magnet
- G2 – Securing Front of Bridge
- G3 – Leveling Bridge
- G4 – Connect Photo Sensor & Bridge End Cover

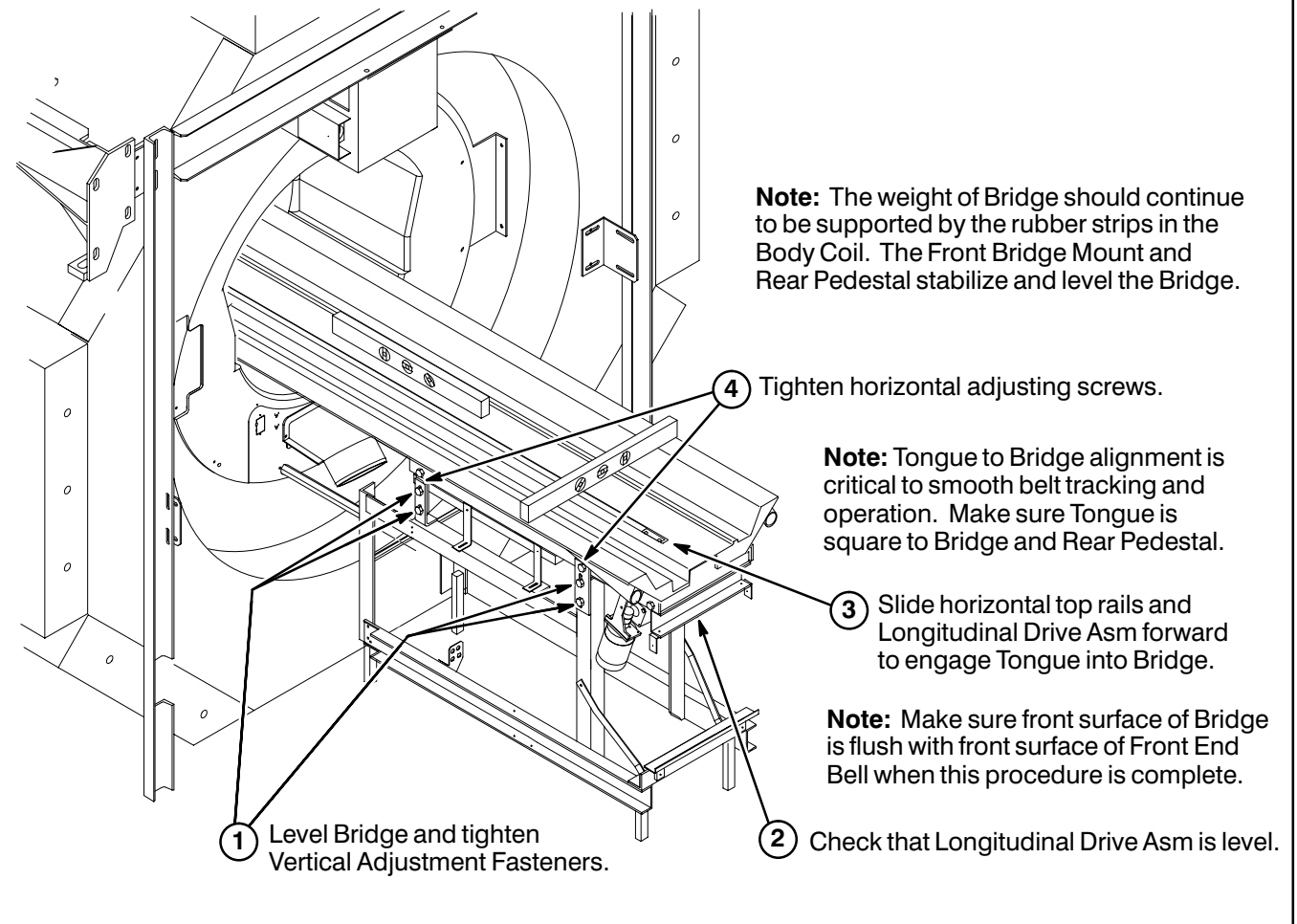
□ G1 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET



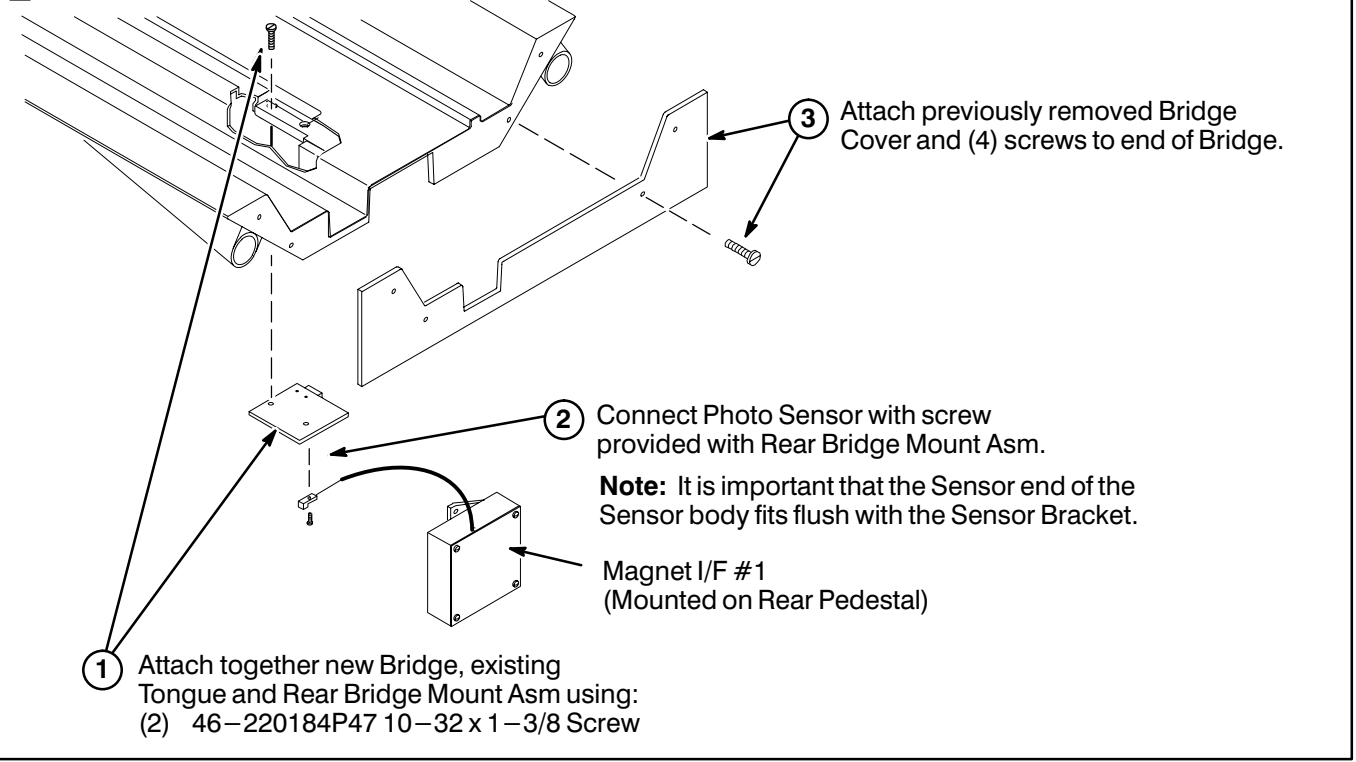
□ G2 – SECURING FRONT OF BRIDGE



□ G3 – LEVELING BRIDGE



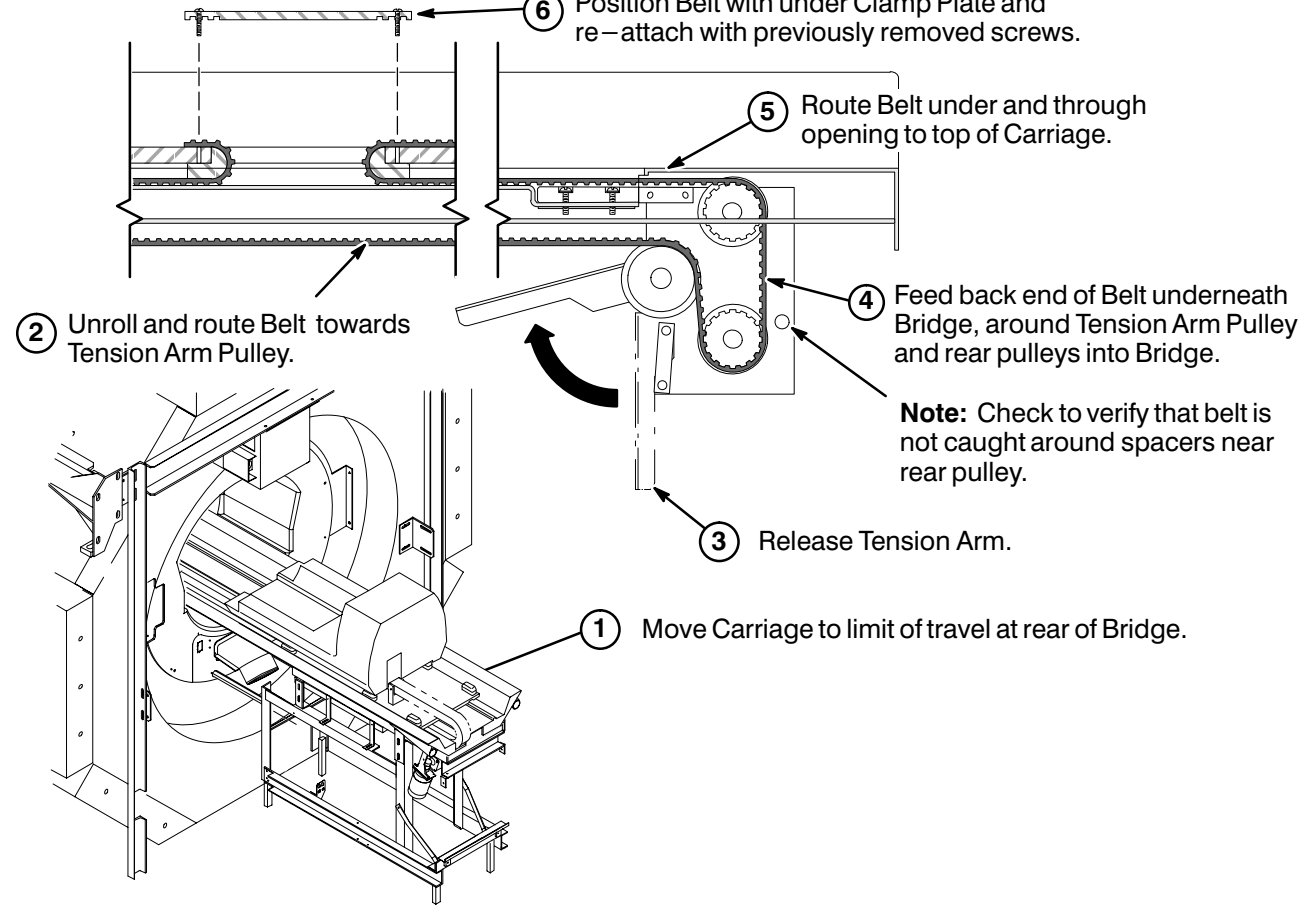
□ G4 – CONNECT PHOTO SENSOR & BRIDGE END COVER



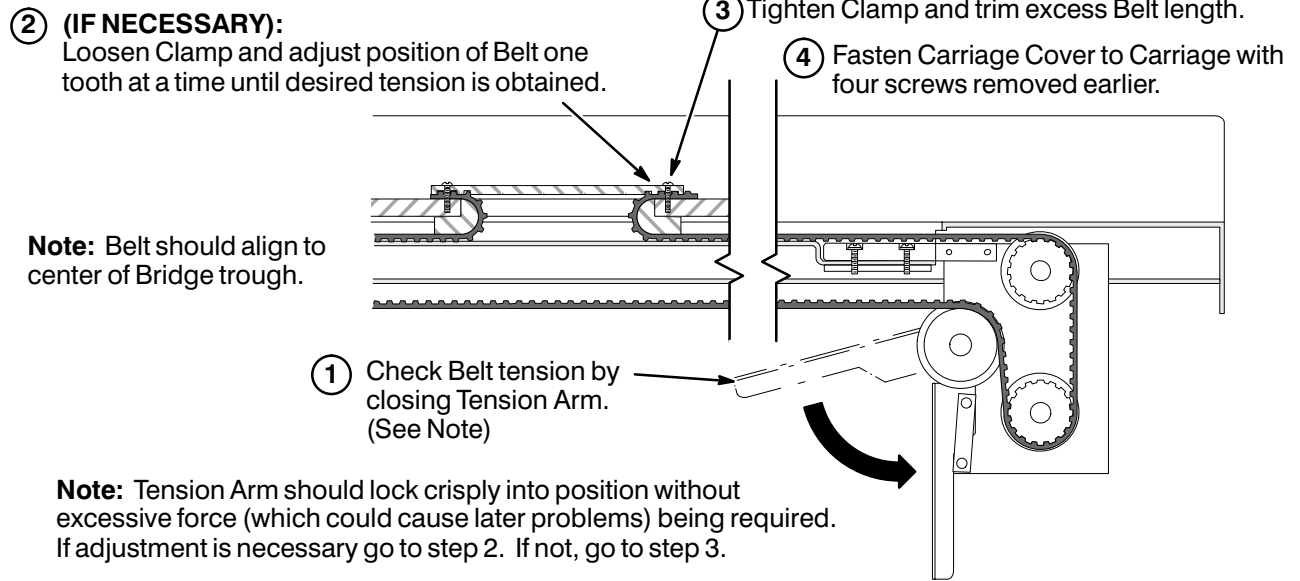
INSTALLATION STEPS SUMMARY

- ☐ H1 – Routing Drive Belt
- ☐ H2 – Final Adjustment of Drive Belt
- ☐ H3 – Installation of Sensor Flag
- ☐ H4 – Route Runs 404 and 485 (and Multi-coil Cables if required)

H1 – ROUTING DRIVE BELT



H2 – FINAL ADJUSTMENT OF DRIVE BELT

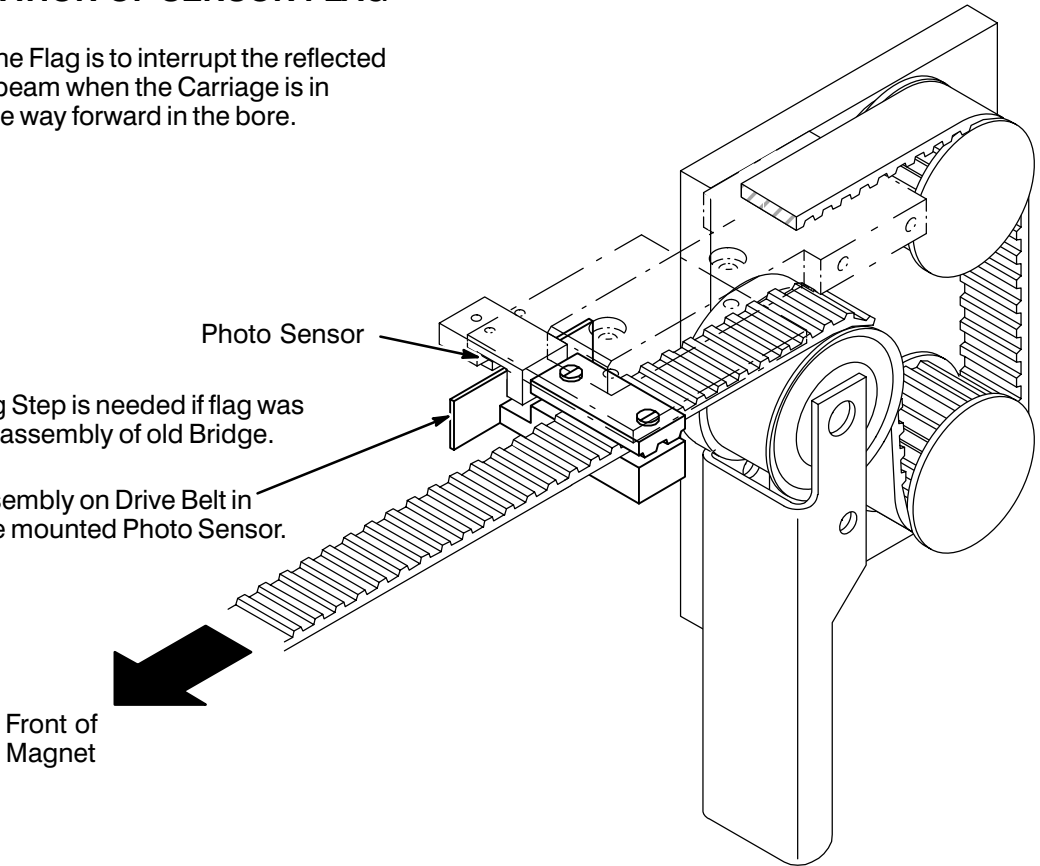


H3 – INSTALLATION OF SENSOR FLAG

Note: Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

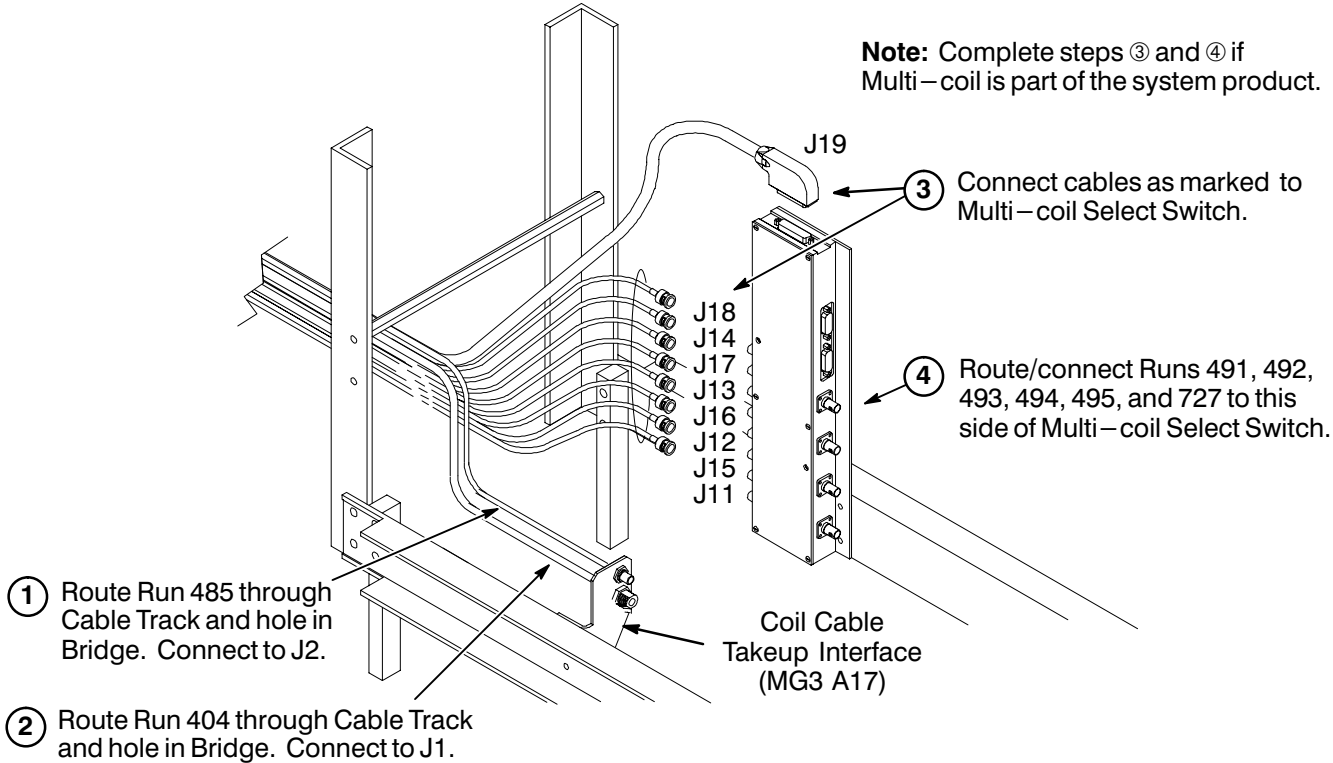
Note: The following Step is needed if flag was removed during disassembly of old Bridge.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



H4 – ROUTE RUNS 404 AND 485 (AND MULTI-COIL CABLES IF REQUIRED)

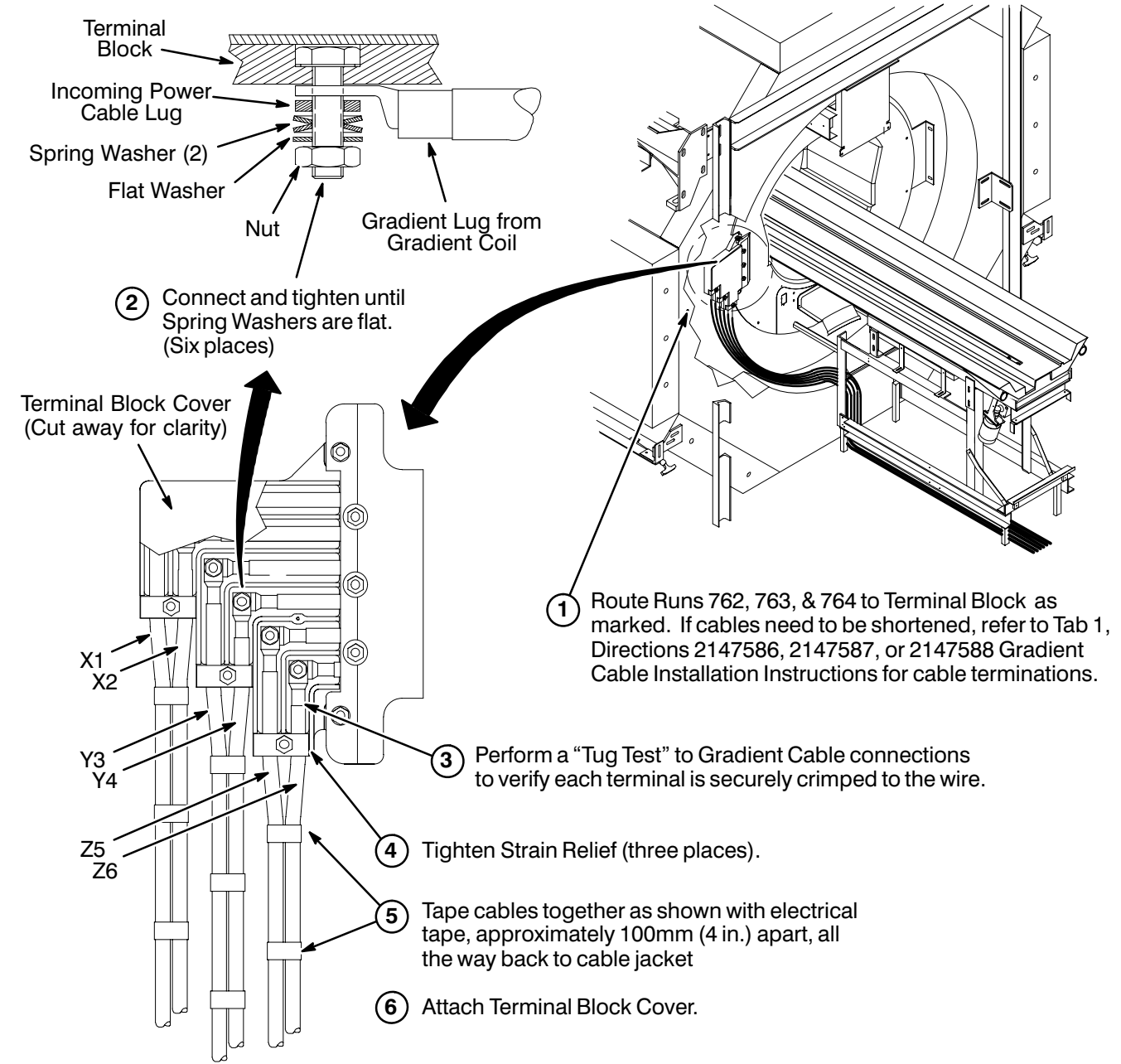
Note: Complete steps ③ and ④ if Multi-coil is part of the system product.



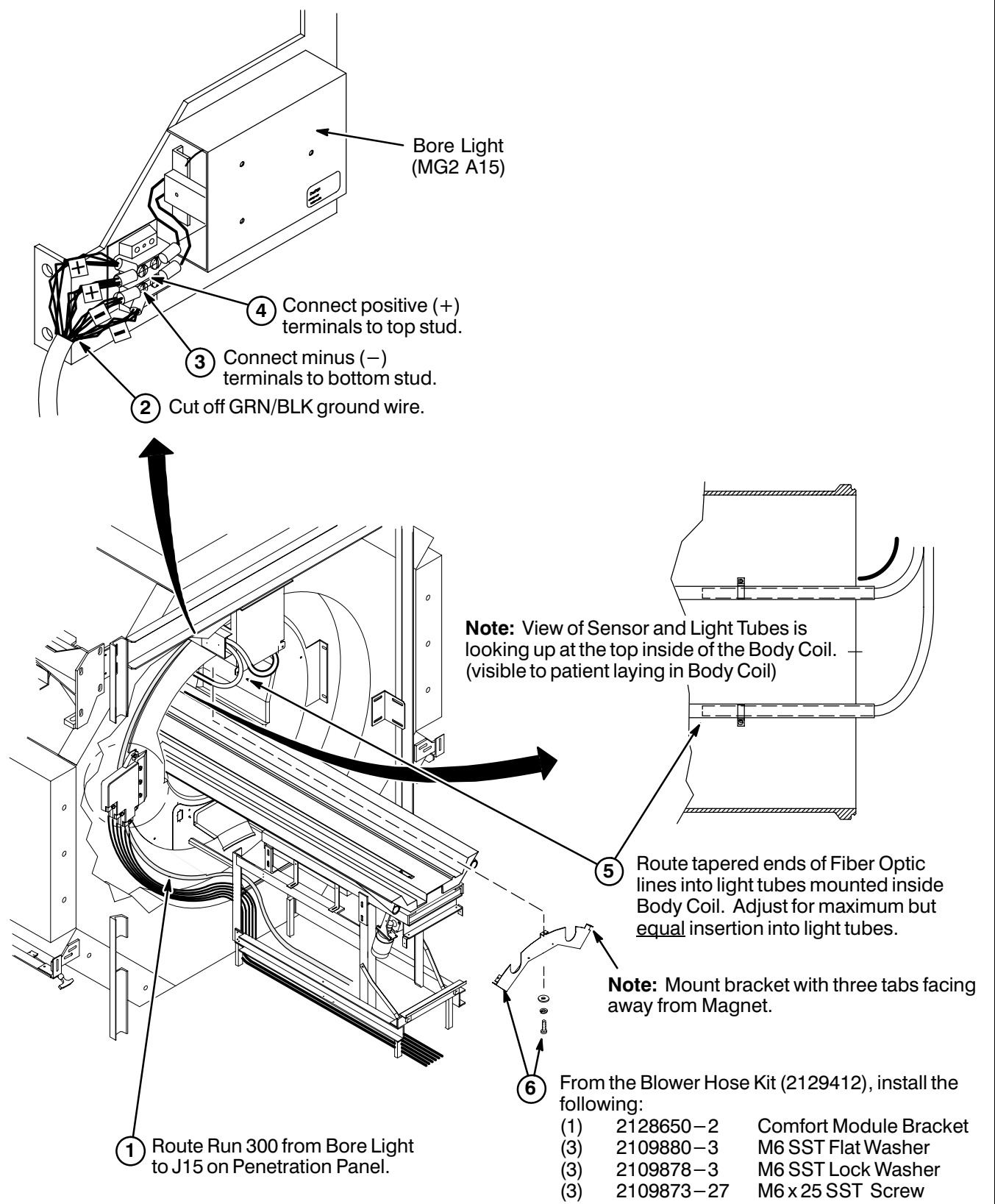
INSTALLATION STEPS SUMMARY

- J1 – Route/Connect Gradient Cables
- J2 – Route/Connect Run 300 and Bore Light Fiber Optic Lines

J1 – ROUTE/CONNECT GRADIENT CABLES



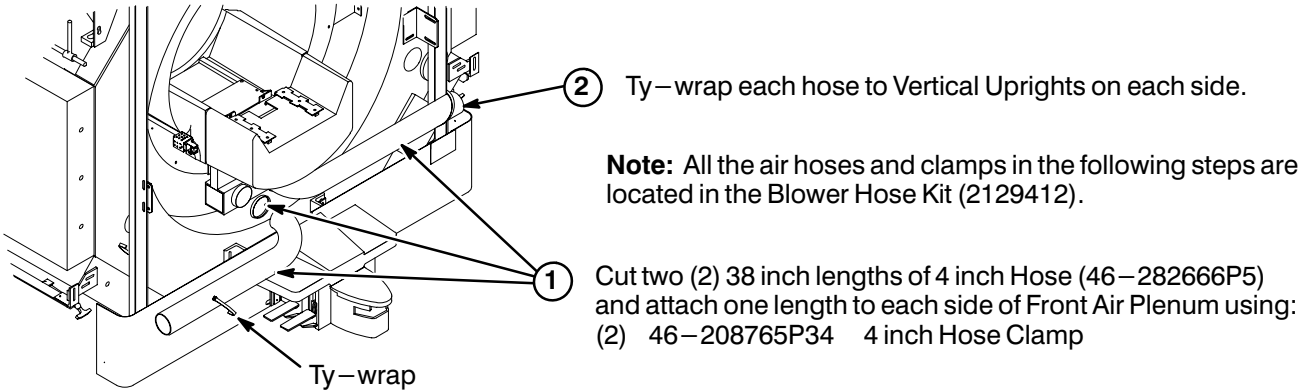
J2 – ROUTE/CONNECT RUN 300 AND BORE LIGHT FIBER OPTIC LINES



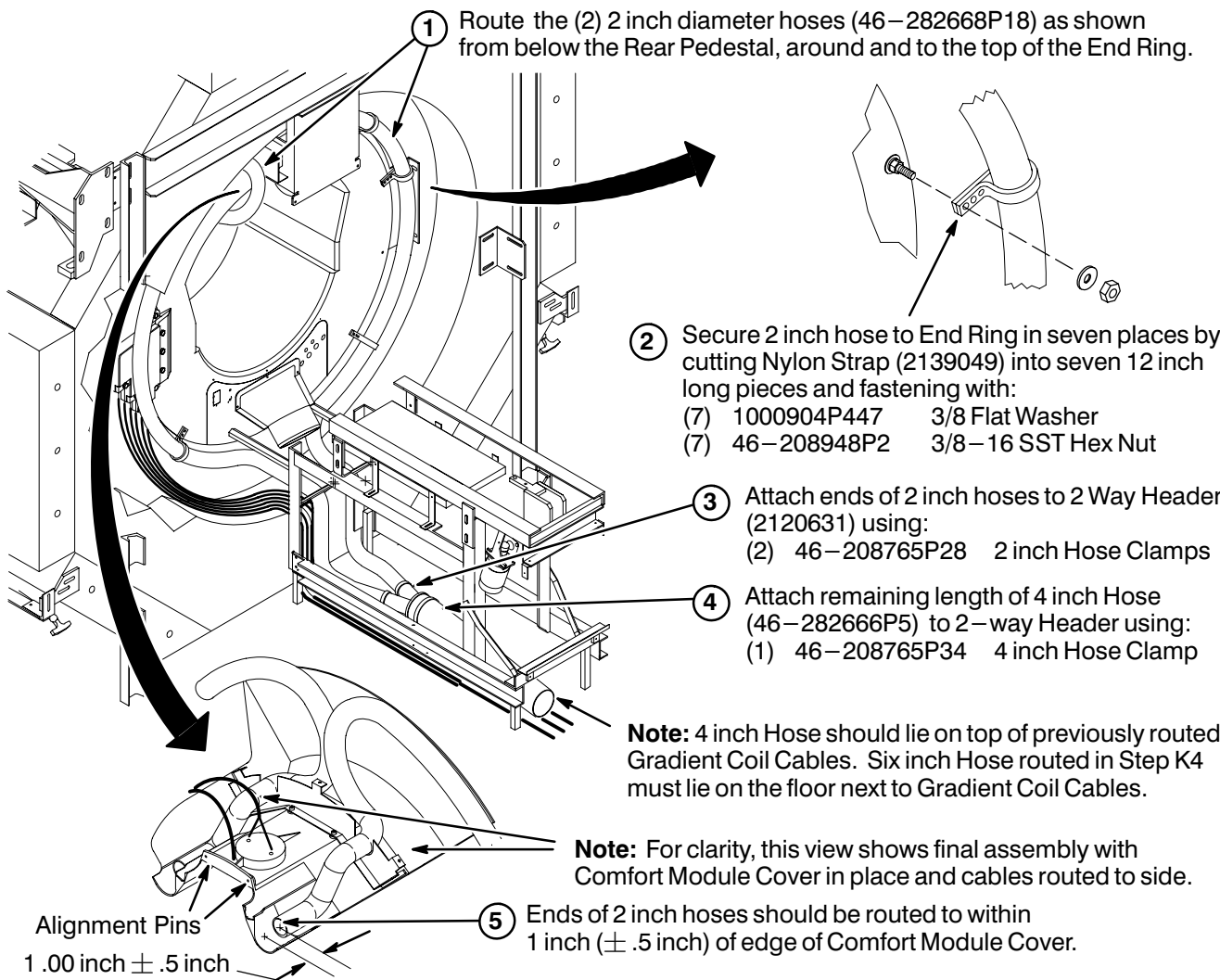
INSTALLATION STEPS SUMMARY

- ❑ K1 – Connecting Front Air Plenum Hoses (2129412)
- ❑ K2 – Routing/Connection of Patient Comfort Air Hoses
- ❑ K3 – Attaching Patient Comfort Module
- ❑ K4 – Route/Connect Patient Comfort Cables and Air Hoses

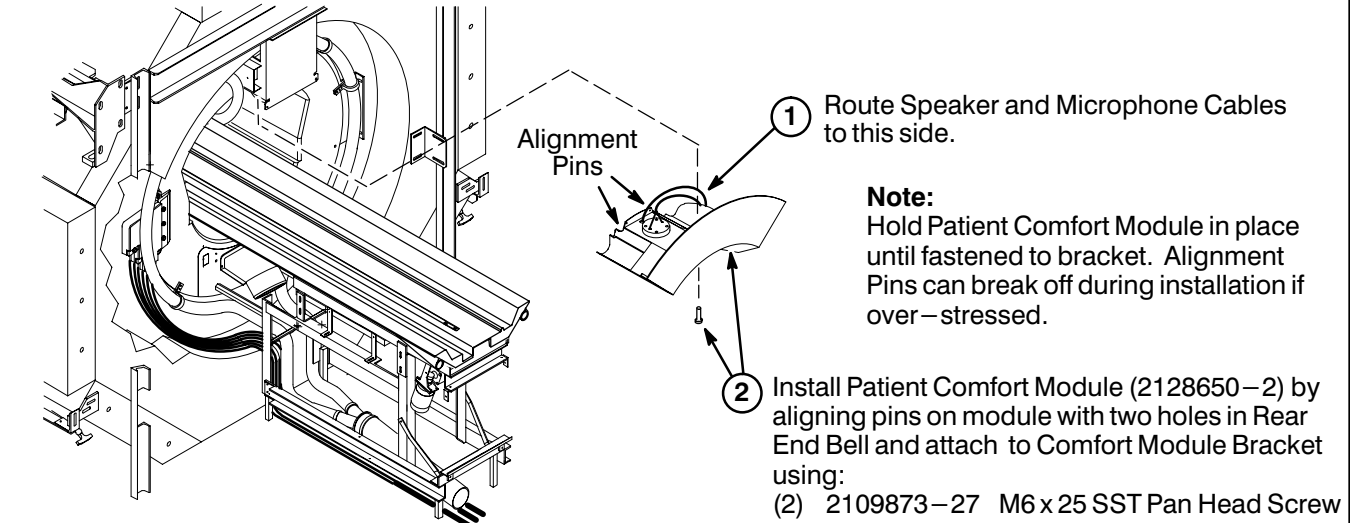
❑ K1 – CONNECTING FRONT AIR PLENUM HOSES (2129412)



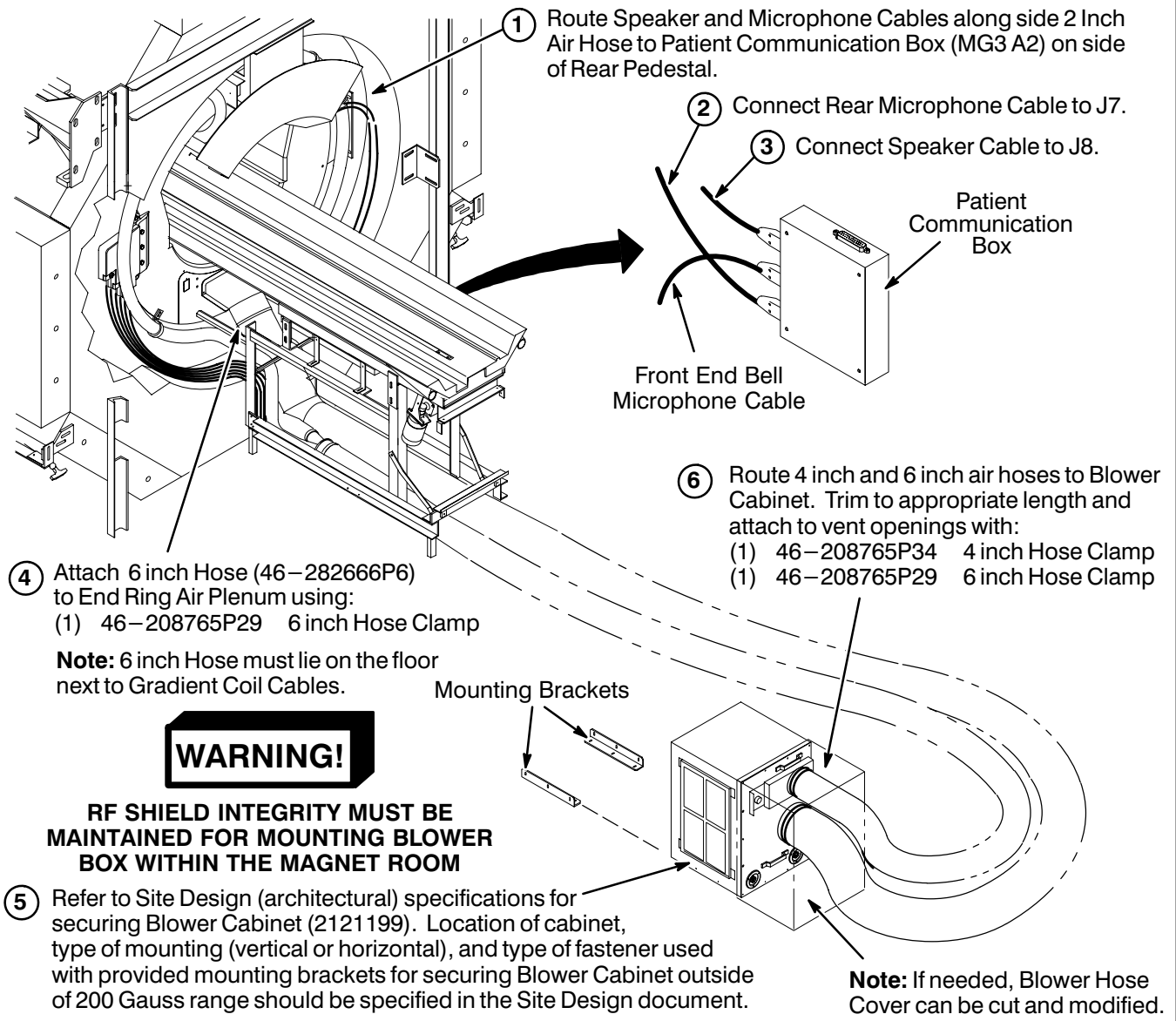
❑ K2 – ROUTING/CONNECTION OF PATIENT COMFORT AIR HOSES



❑ K3 – ATTACHING PATIENT COMFORT MODULE



❑ K4 – ROUTE/CONNECT PATIENT COMFORT CABLES AND AIR HOSES



INSTALLATION STEPS SUMMARY

- ☐ L1 – Install Front Cover (2120889–2)
- ☐ L2 – Disassemble Cover Dolly
- ☐ L3 – Remove Cover Dolly

DANGER!!
FERROUS MATERIAL HAZARD! THE COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE FRONT COVER FROM DOLLY AND CARRY COVER INTO POSITION AT FRONT OF MAGNET.

☐ L1 – INSTALL FRONT COVER (2120889–2)

- 1 Move Shipping Dolly carrying Front Cover (2120889–2) to front of Magnet.
- 2 Fully extend both Front Cover Slides.
- 3 Lock both Slides with Pins hanging from chains.

- 4 Move Cover Dolly to position cover studs into slots on extended slides.

Note: Cover Studs have slotted ends to enable adjustment of length.

Note: Leave the following nuts and washers loose. They will be tightened after Cover is adjusted later in procedure.

- 5 Mount new Cover to both front slides by fastening four cover studs to each slide assembly with previously removed hardware.

☐ L2 – DISASSEMBLE COVER DOLLY

- 1 Remove bolts from each side of dolly that were used for attaching cover to dolly.
- 2 Remove three screws from each end of cross brace and remove brace.
- 3 Remove screws holding dolly side section. Side section will slide down when last bolt is removed. Repeat above for removal of opposite side section of dolly.

☐ L3 – REMOVE COVER DOLLY

- 6 Place remaining dolly parts on recycle pile.
- 5 Place the four bracket and caster assemblies in pre-addressed box provided with dolly for shipment.
- 4 With casters removed, slide frame away from magnet. **Use care not to damage floor when sliding and removing dolly.**
- 3 With end lifted, remove caster assemblies from dolly deck. Repeat process for other end of dolly deck.

- 2 Lift this end of dolly deck.
- 1 Remove screws from each side of bracket holding caster assembly to dolly deck. Repeat process for other three corners.

INSTALLATION STEPS SUMMARY

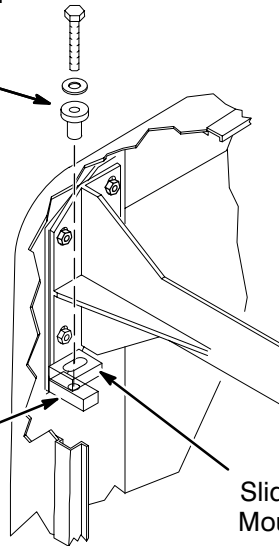
- ❑ M1 – Install Front Cover Alignment Hardware
- ❑ M2 – Transfer Lower Latch Assemblies to New Front & Rear Covers
- ❑ M3 – Adjust Front Cover Alignment
- ❑ M4 – Align Front Cover
- ❑ M5 – Install Rear Cover (2118419)

❑ M1 – INSTALL FRONT COVER ALIGNMENT HARDWARE

❶ Transfer and insert the Alignment Hardware from the old Front Cover into the oval hole on the Slide Assembly Mounting Block

❷ Engage threads in Cover Mounting Block.

❸ Repeat process for other side of Front Cover.

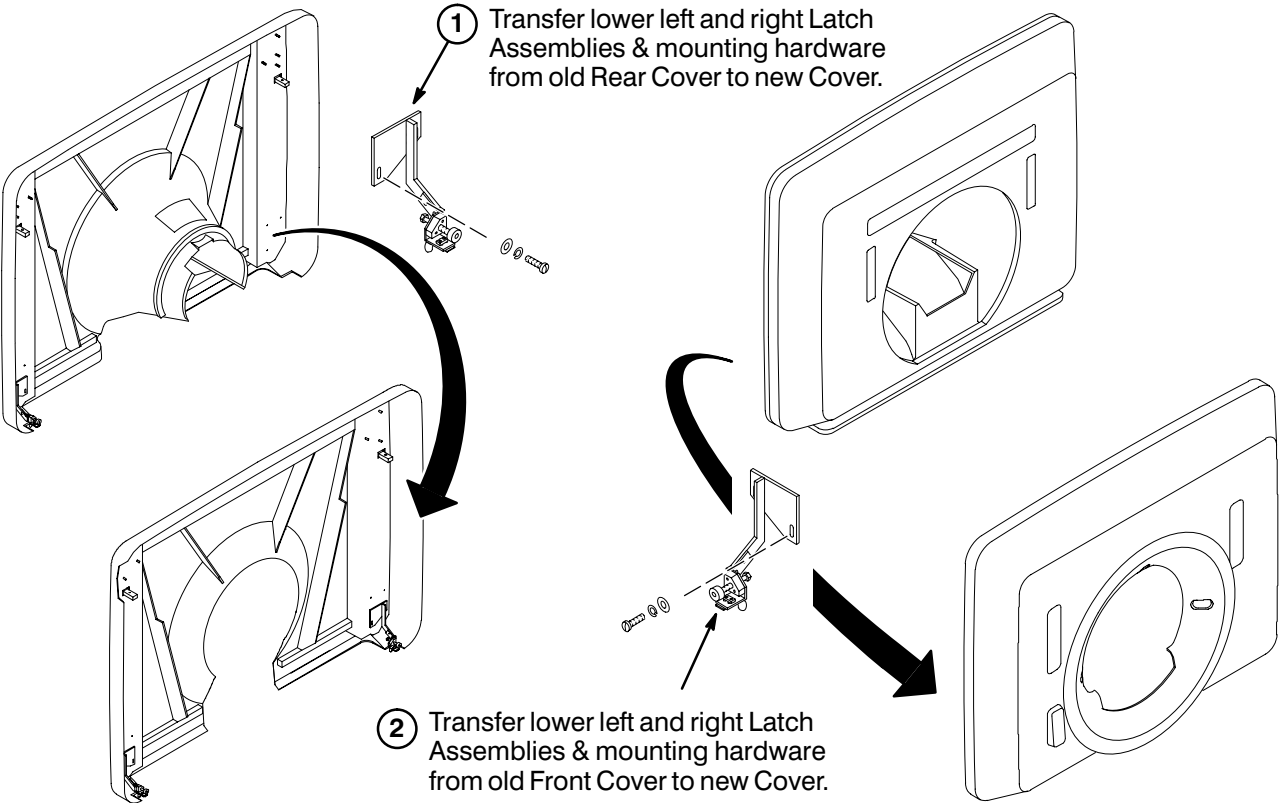


Slide Assembly Mounting Block

❑ M2 – TRANSFER LOWER LATCH ASSEMBLIES TO NEW FRONT & REAR COVERS

❶ Transfer lower left and right Latch Assemblies & mounting hardware from old Rear Cover to new Cover.

❷ Transfer lower left and right Latch Assemblies & mounting hardware from old Front Cover to new Cover.



❑ M3 – ADJUST FRONT COVER ALIGNMENT

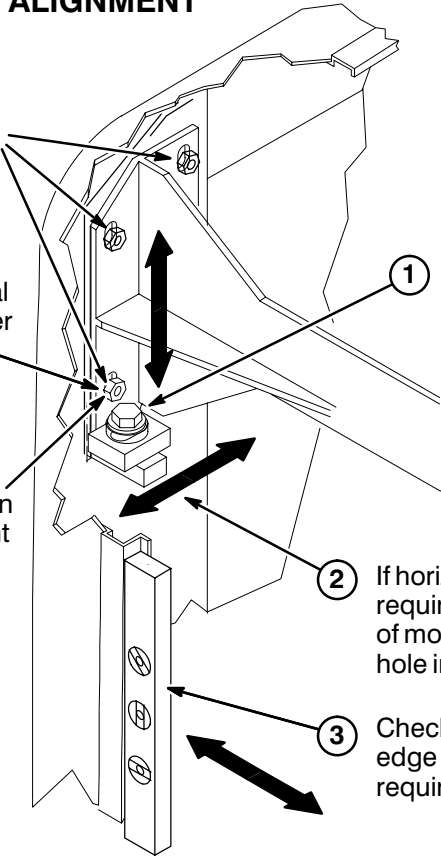
❶ Tighten all hardware when alignment is completed.

❷ Shim at this stud only to obtain vertical alignment. Place shim between Cover and Slide front face. Repeat process for Rear Cover if required.

❸ If vertical alignment is required, loosen hex nut at bottom stud on left and right sides. Cut shims from excess rubber seals:
3/4mm Rubber Seal 2115369
1.5mm Rubber Seal 2115369-2
2.25mm Rubber Seal 2115369-3

❹ If horizontal level adjustment is required, slide cover sideways. Length of movement is determined by slotted hole in Slide Assembly Mounting Block.

❺ Check vertical alignment (plumb) at edge of Front Cover. If alignment is required, proceed to Steps ❹ and ❺.



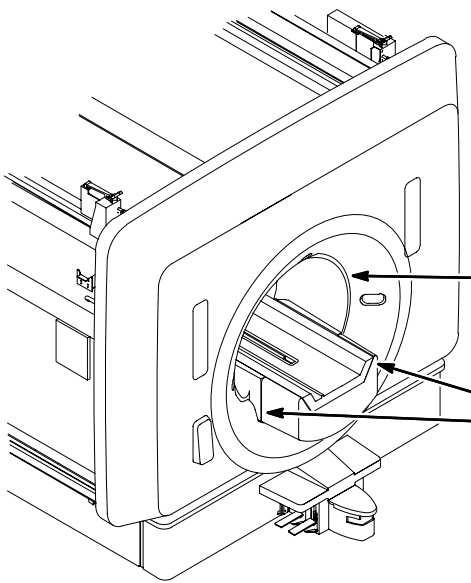
❑ M4 – ALIGN FRONT COVER

❶ Alignment checks:

Cover should just touch and be centered on End Bell when T-Handle Flex Latches are secured and brackets contact bumpers at each corner.

Lower part of Front Cover opening should be equally spaced around Bridge.

Note: Refer to previous step for alignment adjustment procedure.



❑ M5 – INSTALL REAR COVER (2118419)

THE STEPS ARE THE SAME AS THE JUST COMPLETED FRONT COVER INSTALLATION PROCEDURE
Note: Experience has shown that the Rear Cover does not require shimming for vertical levelness.

DANGER!!

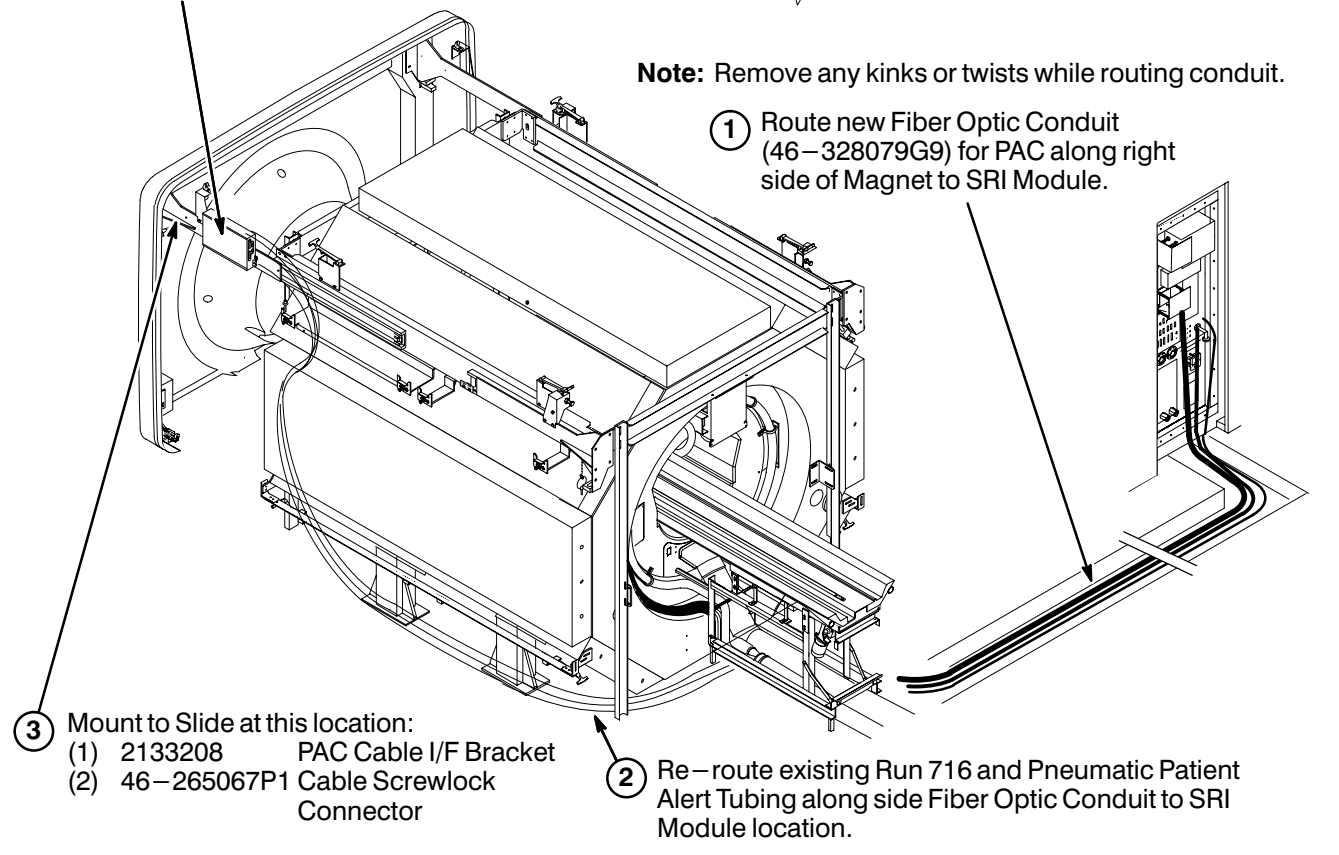
FERROUS MATERIAL HAZARD! THE REAR COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE COVER FROM DOLLY AND CARRY COVER INTO POSITION AT REAR OF MAGNET.

INSTALLATION STEPS SUMMARY

- ❑ N1 – Route/Connect Runs 711/712, 716, and Patient Alert Tubing
- ❑ N2 – Connect Run 715, Front Cover Cables and Install Spiral Wrap
- ❑ N3 – Adjust Spiral Wrapped Cables

❑ N1 – ROUTE/CONNECT RUNS 711/712, 716, AND PATIENT ALERT TUBING

- ⑥ Route Run 716 along side existing cables, over top of SRI Module, thru clamp, to I/F bracket on slide. Connect to PAC II cable (46–224712G20) from Front Cover using (2) Cable Mounts.
- Note:** A secure electrical ground is required. Insure low resistance connection and tighten fasteners securely.
- ⑦ Route fiber optic cable Run 712 and Patient Alert Tubing to PAC II Asm on Front Cover. **Do not clamp.**
- ⑧ Connect fiber optic cable Run 711 to circuit board as marked. Leave a small loop in fiber optic cable and secure to clamp. (Remove fiber optic cable guides and pass fiber optic cables through if necessary) Close SRI cover and tighten captive screws.
- Note:** Fiber optic cables loops must not be smaller than two inch radius.
- ④A For **Horizon 5.x and LX**, SRI Board within module is replaced. Refer to Direction 2294109, Dual Thermal Sensor Upgrade at the end of this Tab. Section 1–13 contains the information for replacing the SRI board within the module.
- OR**
- ④B For **Advantage 4.x & 5.x**, install new SRI Module..
- ⑤ Route fiber optic conduit to within 16 inches of Slide Assembly. Group with existing routed cables.

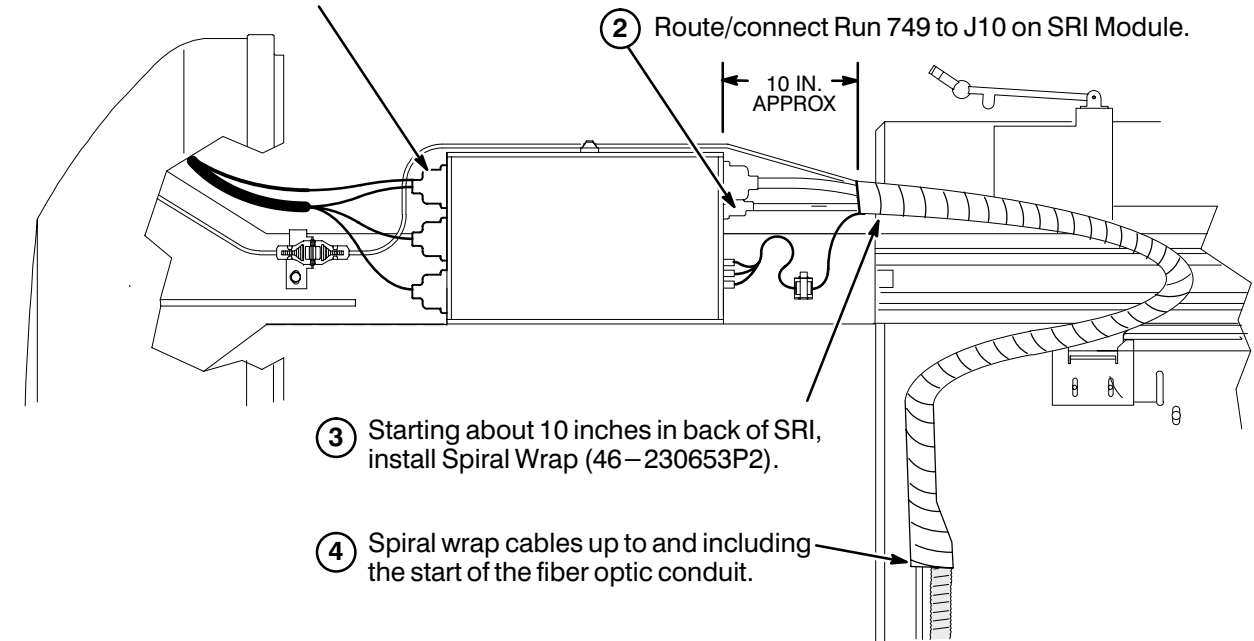


CAUTION

EQUIPMENT DAMAGE POSSIBILITY. Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend to radius smaller than two inches. Do not pull connector from cable. Avoid scratching connector ends. Keep ends protected until ready to connect.

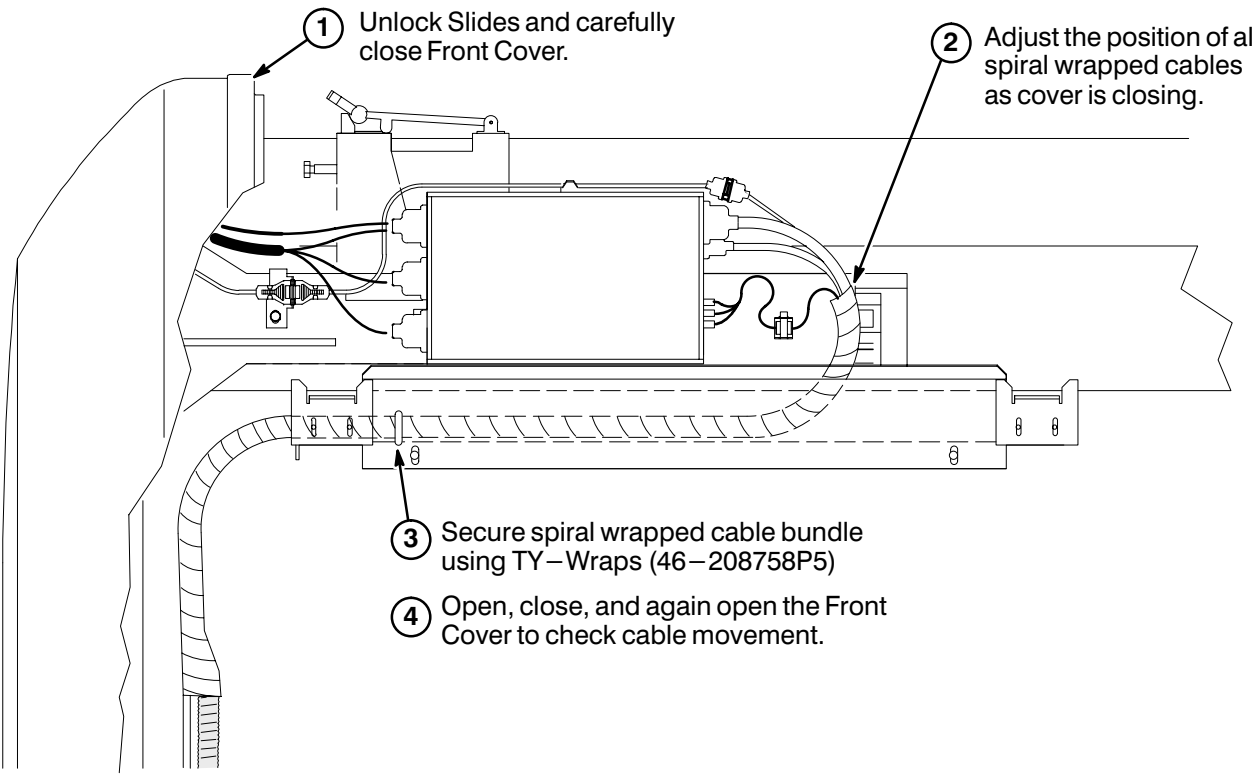
❑ N2 – CONNECT RUN 715, FRONT COVER CABLES AND INSTALL SPIRAL WRAP

- ① Route ends of six remaining cables from Front Cover to the SRI and connect as marked.
- ② Route/connect Run 749 to J10 on SRI Module.
- ③ Starting about 10 inches in back of SRI, install Spiral Wrap (46–230653P2).
- ④ Spiral wrap cables up to and including the start of the fiber optic conduit.



❑ N3 – ADJUST SPIRAL WRAPPED CABLES

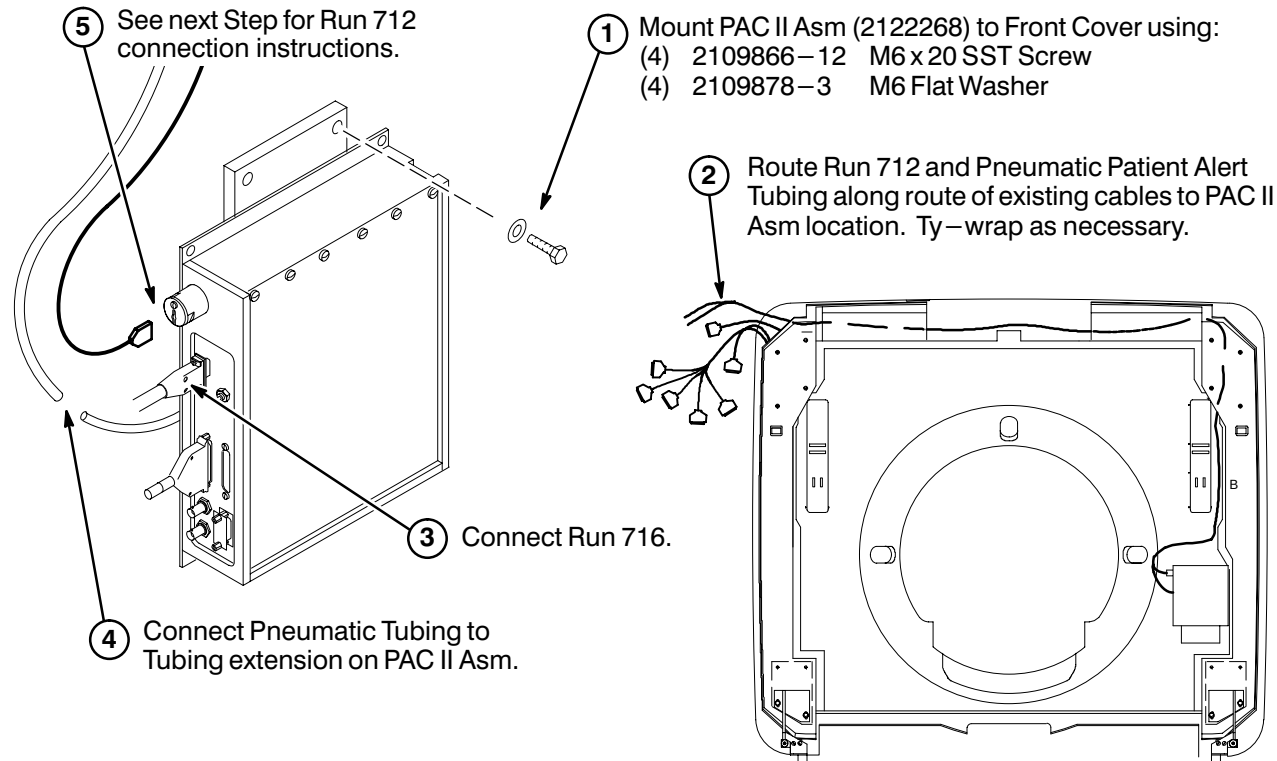
- ① Unlock Slides and carefully close Front Cover.
- ② Adjust the position of all spiral wrapped cables as cover is closing.
- ③ Secure spiral wrapped cable bundle using TY–Wraps (46–208758P5)
- ④ Open, close, and again open the Front Cover to check cable movement.



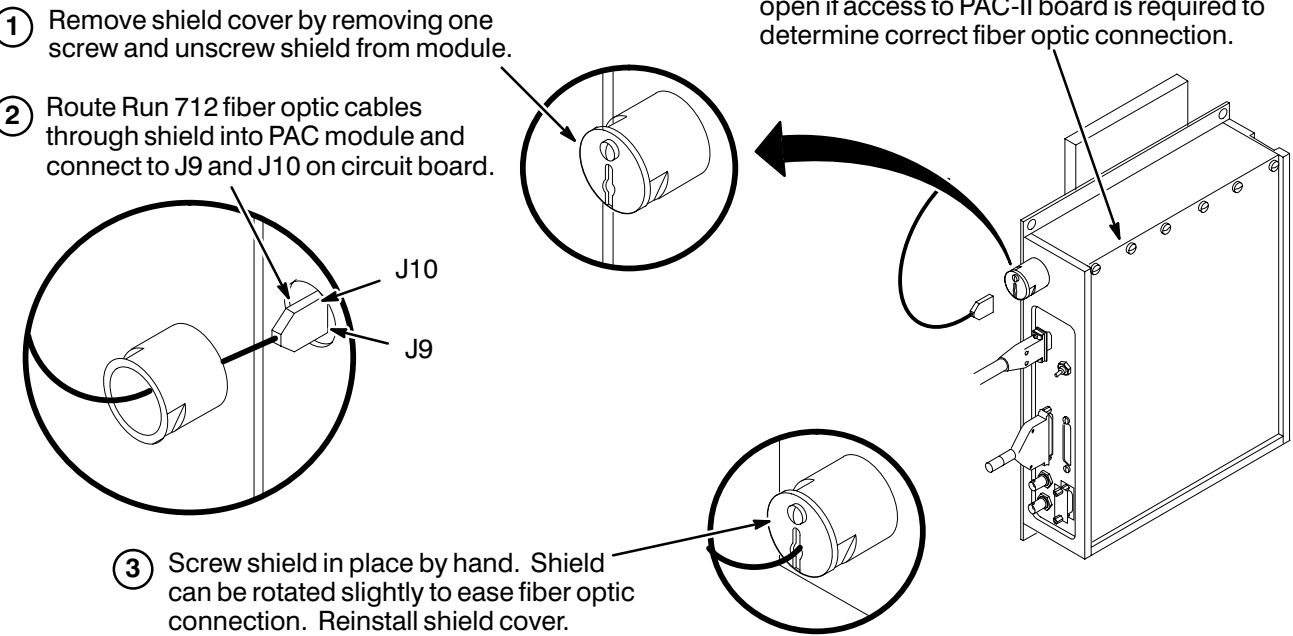
INSTALLATION STEPS SUMMARY

- ☐ P1 – Connect Run 712, 716, and Pneumatic Patient Alert Tubing
- ☐ P2 – Run 712 Connection
- ☐ P3 – Install Drip Shield Plate Kit (2143475)

P1 – CONNECT RUN 712, 716, & PNEUMATIC PATIENT ALERT TUBING

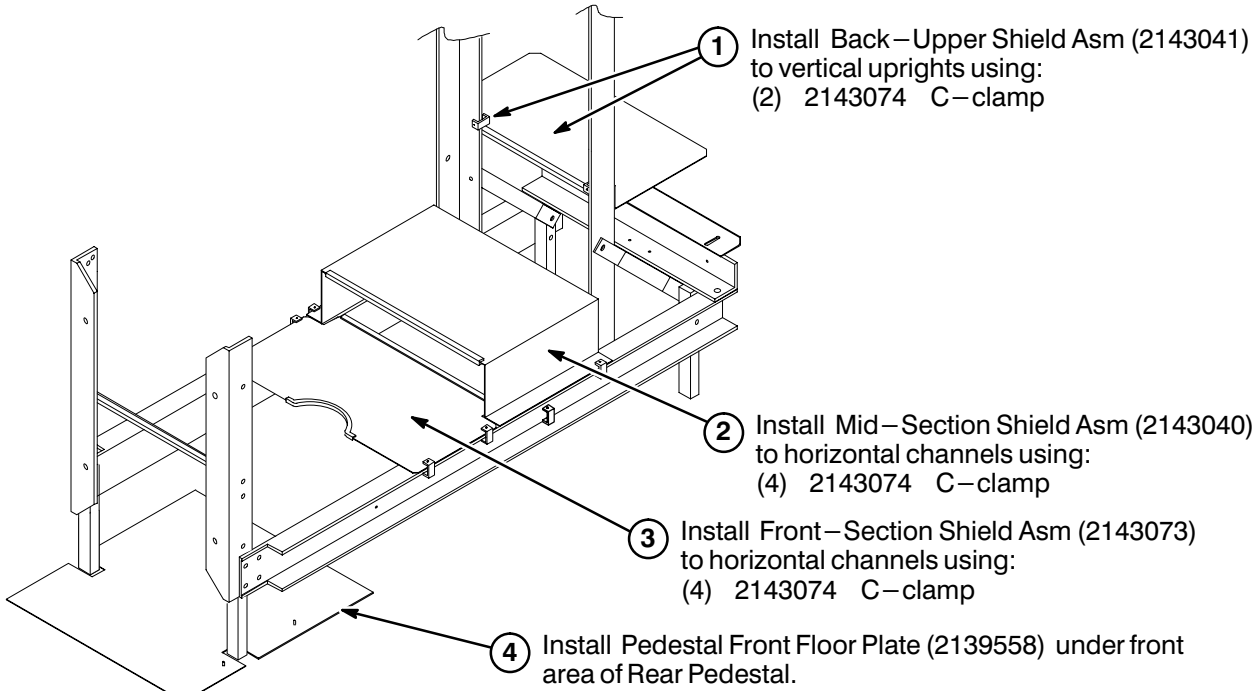


P2 – RUN 712 CONNECTION



P3 – INSTALL DRIP SHIELD PLATE KIT (2143475), IF SUPPLIED

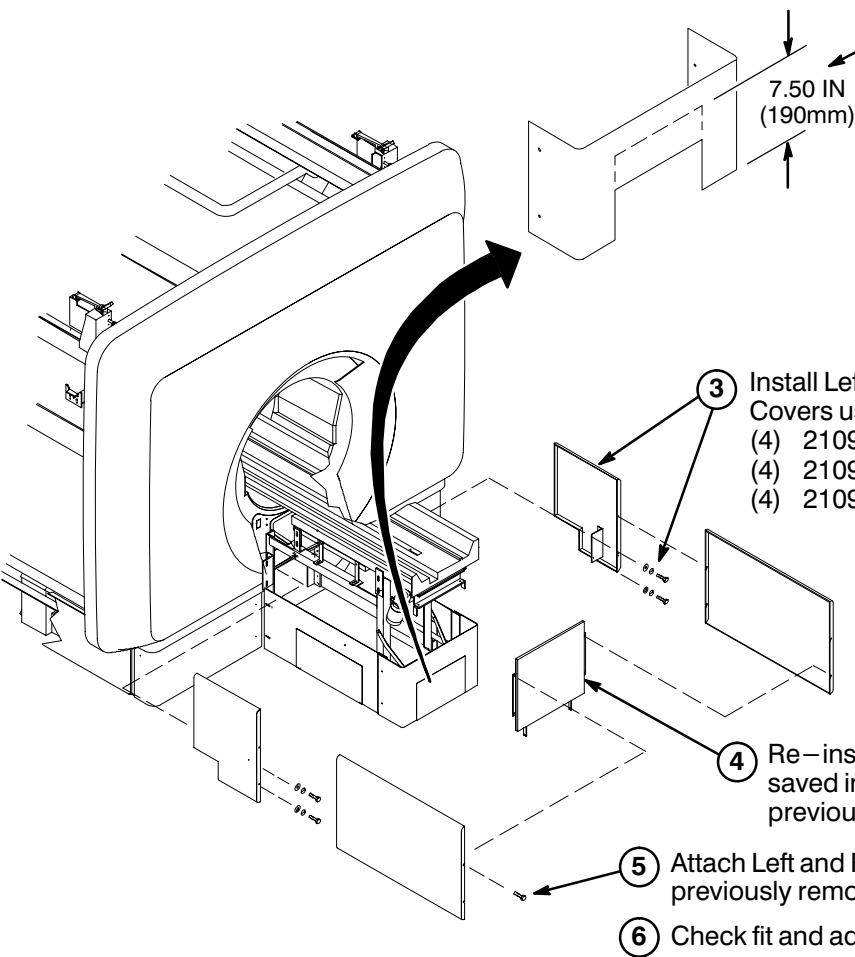
Note: Drip Shield Plate Kit installation is required to meet UL requirements.



INSTALLATION STEPS SUMMARY

- Q1 – Install Rear Pedestal Covers (2131492–2)
- Q2 – Install Enclosure Side Covers
- Q3 – Install PPG Cable Hooks & Rating Plate on Front Cover

Q1 – INSTALL REAR PEDESTAL COVERS (2131492–2)



7.50 IN (190mm)

1 Remove Rear Trim Cover from room and cut opening to height indicated. Remove sharp edges.

2 Re-install Base Trim Covers with saved hardware.

3 Install Left (2131044) and Right (2131045) Side Covers using:

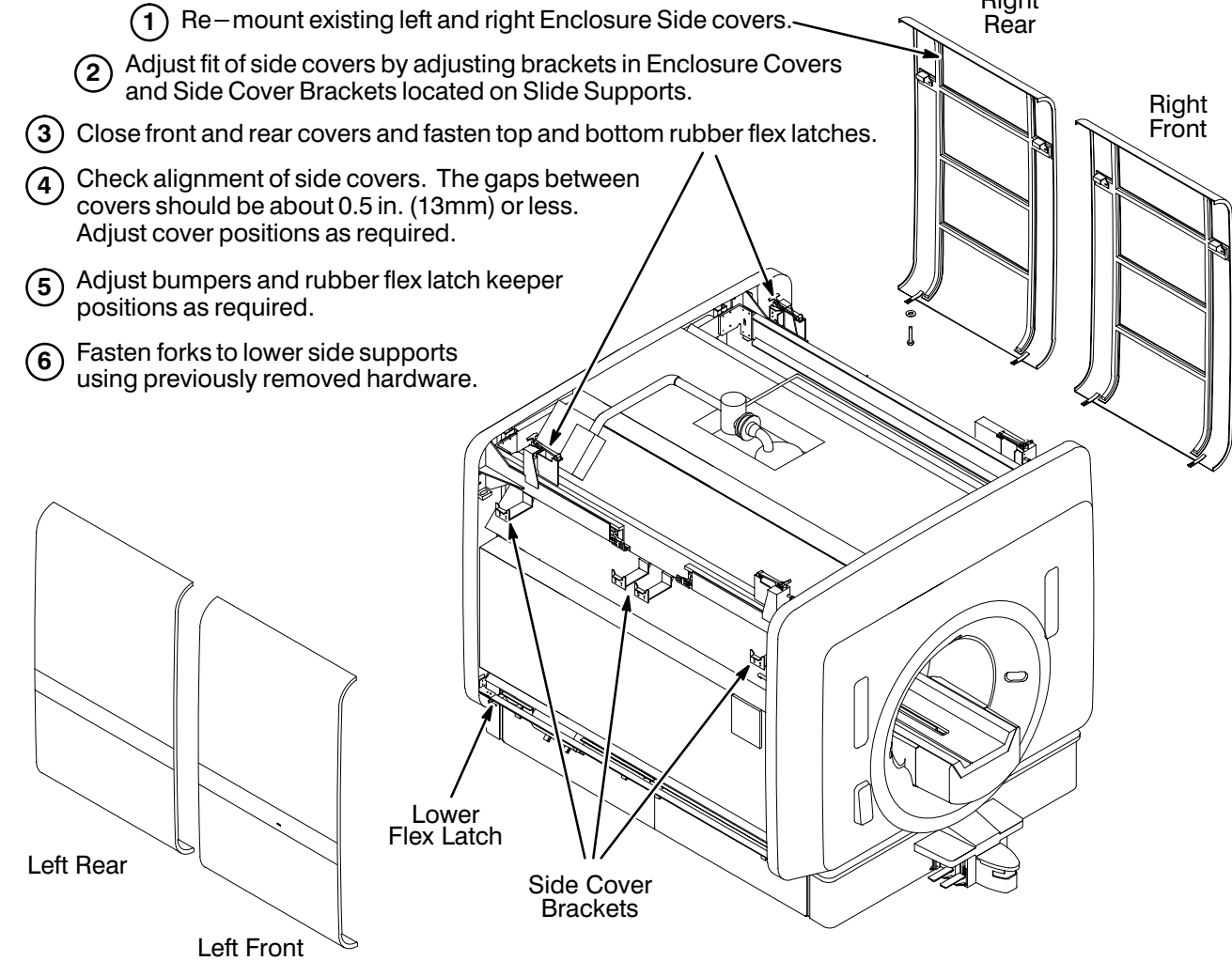
(4) 2109878–3	M6 SST Flat Washer
(4) 2109880–3	M6 SST Lock Washer
(4) 2109866–13	M6 x 25 SST Hex Screw

4 Re-install Rear Pedestal Back Cover Asm saved in A1 – Step 1 to Rear Pedestal with previously removed hardware.

5 Attach Left and Right Rear Pedestal Side Covers with previously removed hardware.

6 Check fit and adjust as required.

Q2 – INSTALL ENCLOSURE SIDE COVERS



1 Re-mount existing left and right Enclosure Side covers.

2 Adjust fit of side covers by adjusting brackets in Enclosure Covers and Side Cover Brackets located on Slide Supports.

3 Close front and rear covers and fasten top and bottom rubber flex latches.

4 Check alignment of side covers. The gaps between covers should be about 0.5 in. (13mm) or less. Adjust cover positions as required.

5 Adjust bumpers and rubber flex latch keeper positions as required.

6 Fasten forks to lower side supports using previously removed hardware.

Right Rear

Right Front

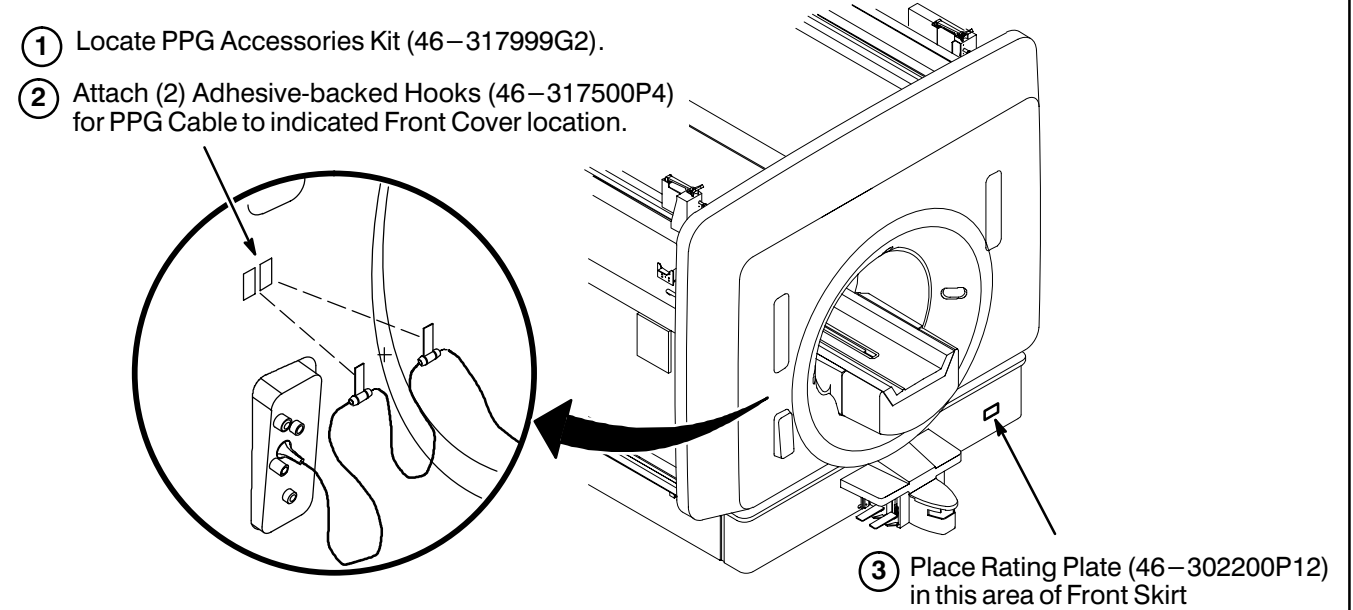
Left Rear

Left Front

Lower Flex Latch

Side Cover Brackets

Q3 – INSTALL PPG CABLE HOOKS & RATING PLATE ON FRONT COVER



1 Locate PPG Accessories Kit (46–317999G2).

2 Attach (2) Adhesive-backed Hooks (46–317500P4) for PPG Cable to indicated Front Cover location.

3 Place Rating Plate (46–302200P12) in this area of Front Skirt

GENERAL ENCLOSURE NOTES:

- Front, rear, left, and right, as used in this section, are defined as follows: the patient entrance end of the magnet is the front; the opposite end is the rear; left and right are in respect to a person's left and right while facing the patient entrance end of the magnet.

INSTALLATION STEPS SUMMARY

- ❑ A1 – Remove Enclosure Covers, Bridge, Bridge Supports, DSIM/SRI and PAC Asm
- ❑ A2 – Remove Alignment Lights
- ❑ A3 – Remove Body Coil, Rear Pedestal, Bore Light and Comfort Module

❑ A1 – REMOVE COVERS, BRIDGE, BRIDGE SUPPORTS, DSIM/SRI AND PAC ASM

1 Remove and discard Pedestal Rear Cover. Save mounting hardware for reuse.

2 Remove and save Rear Pedestal Side Covers and hardware. These will be reused.

3 Discard Rear cover. Save all latches and all mounting hardware.

4 For 3.x or 4.x, Remove and discard DSIM/SRI Module.

5 **Note:** See special removal instructions, steps B1 and B2, if Front Cover is an S-IV Swing-up design. Discard Front Cover, Displays, and Operator's Controls. Save all latches and all mounting hardware.

6 Remove and discard PAC Asm and Fiber Optic Runs 711/712

7 Remove Bridge Asm and save all parts removed from the bridge. Discard fiberglass Bridge.

8 Remove from Bridge Support Asm and save Dock Control Box Asm and attaching hardware.

9 Remove and discard Bridge Support Asm.

10 See Steps G3 and G4 if Carriage base plate does not have a notch for Cable Trak clearance. Notch is required for Upgrade procedure.

❑ A2 – REMOVE ALIGNMENT LIGHTS

1 Unfasten Dock and remove Front Trim Skirt from Magnet room. Cut notch at top of Trim Panel according to dimensions below.

2 Re-install Front Trim Skirt and Dock to Magnet

3 Remove Alignment Light Asm and brackets. Leave Run 385 attached to Terminal Strips.

4 Remove cable clamp and terminal strip screws.

5 Save Light assemblies with wires and cables attached to Terminal Strips.

6 Remove and save Light Asm mounting hardware.

7 Discard alignment light base plates on recycle pile.

❑ A3 – REMOVE BODY COIL, REAR PEDESTAL, BORE LIGHT, & COMFORT MODULE

1 Remove Body Coil.

2 Discard Bore Light brackets. Save Bore Light and hardware for transfer to new Rear Ring.

3 Remove asnd discard Comfort Module.

4 Unfasten Rear Pedestal and remove from Magnet Room. **Many of the components of the removed Rear Pedestal HAVE to be transferred to the new Rear Pedestal.** Refer to procedure E3 on page 5 for details.

5 Discard old Rear Pedestal frame after transfer of components.

INSTALLATION STEPS SUMMARY (S-IV & S-X FRONT COVER REMOVAL, IF NEEDED))

- ☐ B1 – Remove Front Cover Gas Spring (S–IV & S–X)
- ☐ B2 – Remove Front Cover, Install New Weld Mounts (S–IV & S–X)

CAUTION

Magnet Material Hazard! The cover dolly was not designed for cover installation with magnet at field. The caster assemblies and fastening hardware are magnetic. If magnet is at field, three installers are required to remove front cover from dolly and carry cover into position at front of magnet.

B1 – REMOVE FRONT COVER GAS SPRING (S-IV & S-X)

-
- 1 Fully extend Front Cover/Slides and lock both slides with Pins hanging from chains.
- 2 Remove screw from left and right side Front Cover Hinge assemblies. This will allow Front Cover to swing up.
- DANGER** – Do not remove this screw unless cover is attached. Hinge plate will fly open with dangerous speed from Gas Spring force.
- 3 Swing Front Cover up.
- 4 Insert Locking Pin in bracket hole for both left and right sides after cover is swung up. This will hold Front Cover in place for removal of gas spring.
- WARNING!**
- GAS SPRINGS CONTAIN FERROUS MATERIAL. HANDLE WITH CARE DURING REMOVAL.**
- 5 Remove safety clip at each end of Gas Spring for both left and right sides.
- 6 Remove Gas Spring for both left and right sides.
- Note:** Because of weight and size, at least two persons are required to swing Front Cover down and lock into place.
- 7 Push up bottom of Front Cover. Remove Locking Pin inserted in step 4.
- 8 Slowly swing the Front Cover down into the vertical position.

❑ B2 – REMOVE FRONT COVER, INSTALL NEW WELD MOUNTS (S-IV & S-X)

-
- The diagram illustrates the process of replacing the front cover and side weld mounts of a microwave oven. It shows the internal chassis with the door removed, and the front cover being detached. The steps are numbered 1 through 5.
- 1 Remove Front Cover. Save all Latches and mounting hardware.
 - 2 Remove SRI electronics and hardware from existing front weld mount.
 - 3 Remove and discard left and right front weld mounts.
 - 4 Install new right side weld mount (from M1191JG Kit) with existing hardware. Transfer electronics and attaching hardware.
 - 5 Install new left side weld mount with existing hardware.

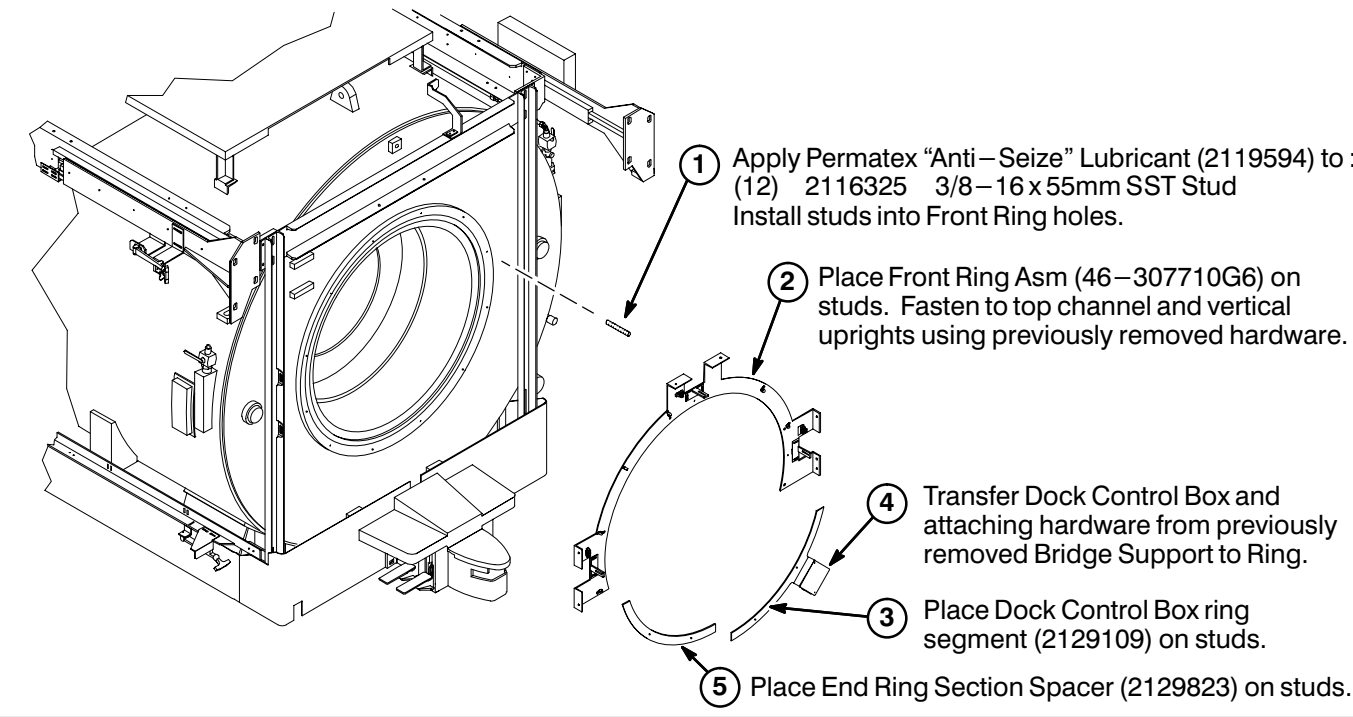
GENERAL ENCLOSURE NOTES:

- A 1-ounce tube of Permatex “Anti-Seize” lubricant is provided. Follow instructions on back of tube. “Anti-Seize” lubricant is to be applied to threads of stainless-steel screws before installing into tapped holes of magnet to prevent galling and seizing of cap screw.

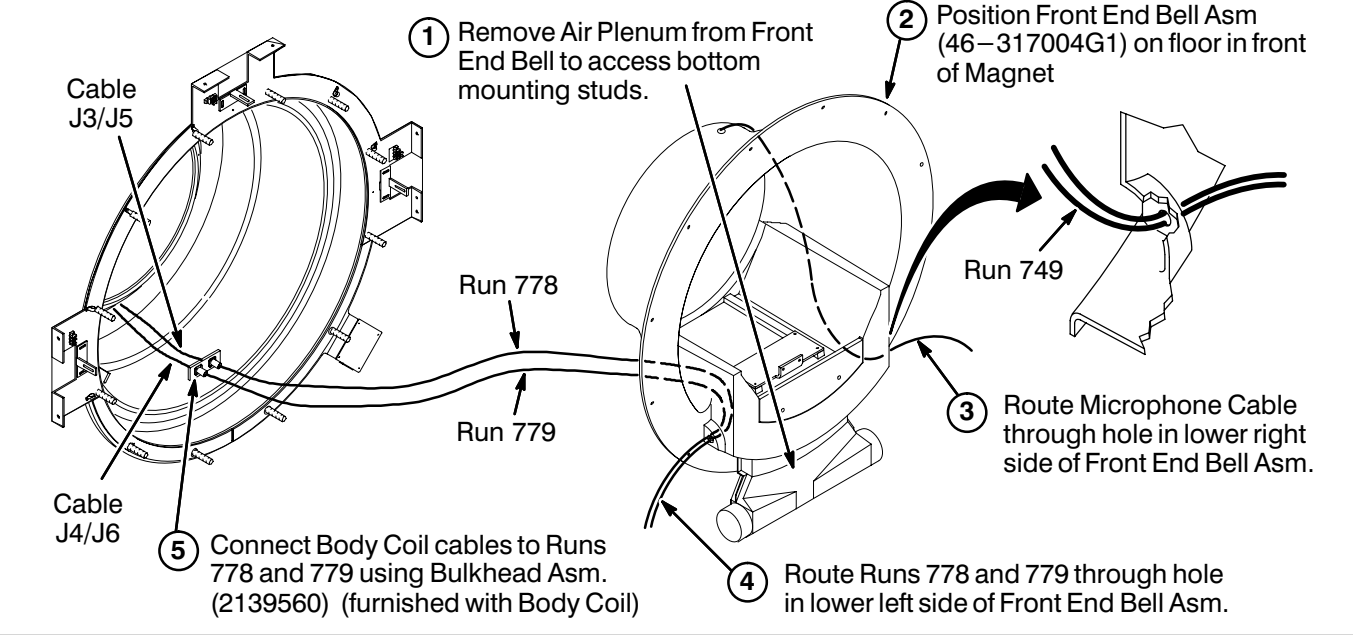
INSTALLATION STEPS SUMMARY

- C1 – Install Front Frame Asm
- C2 – Route and Connect Runs 778 and 779
- C3 – Install Front Gasket and Mount End Bell
- C4 – Install and Connect Alignment Lights

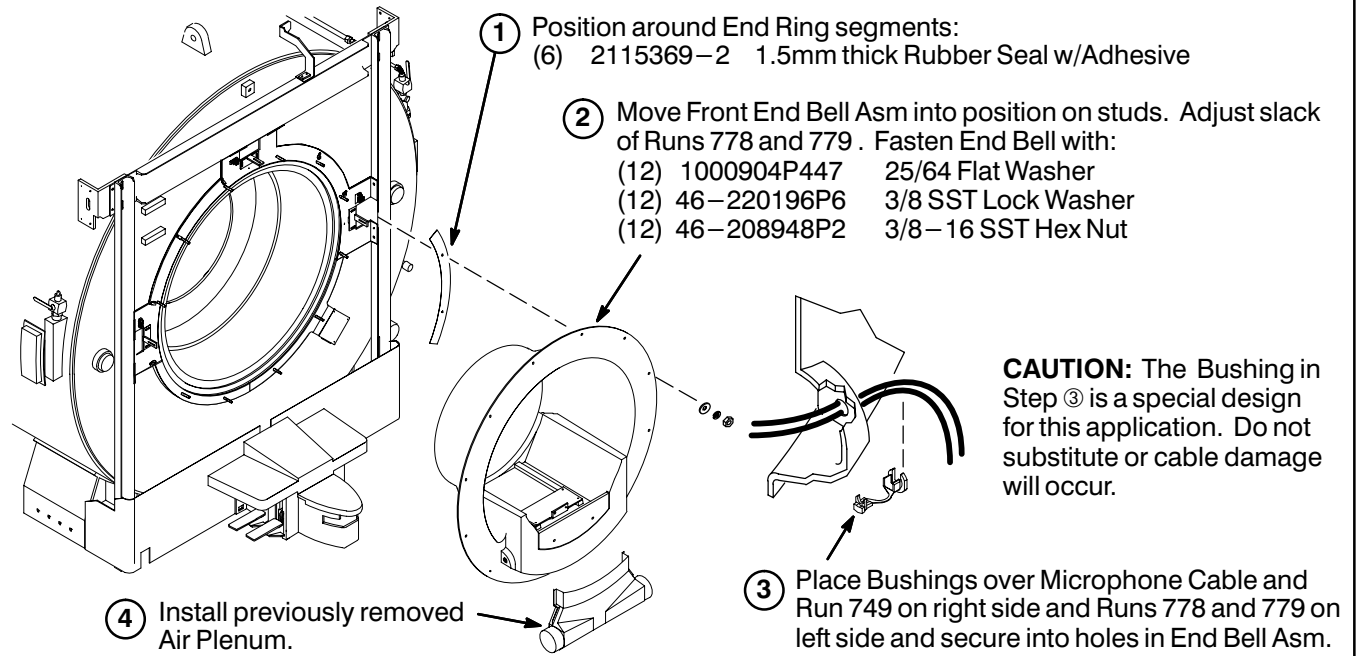
□ C1 – INSTALL FRONT FRAME ASM



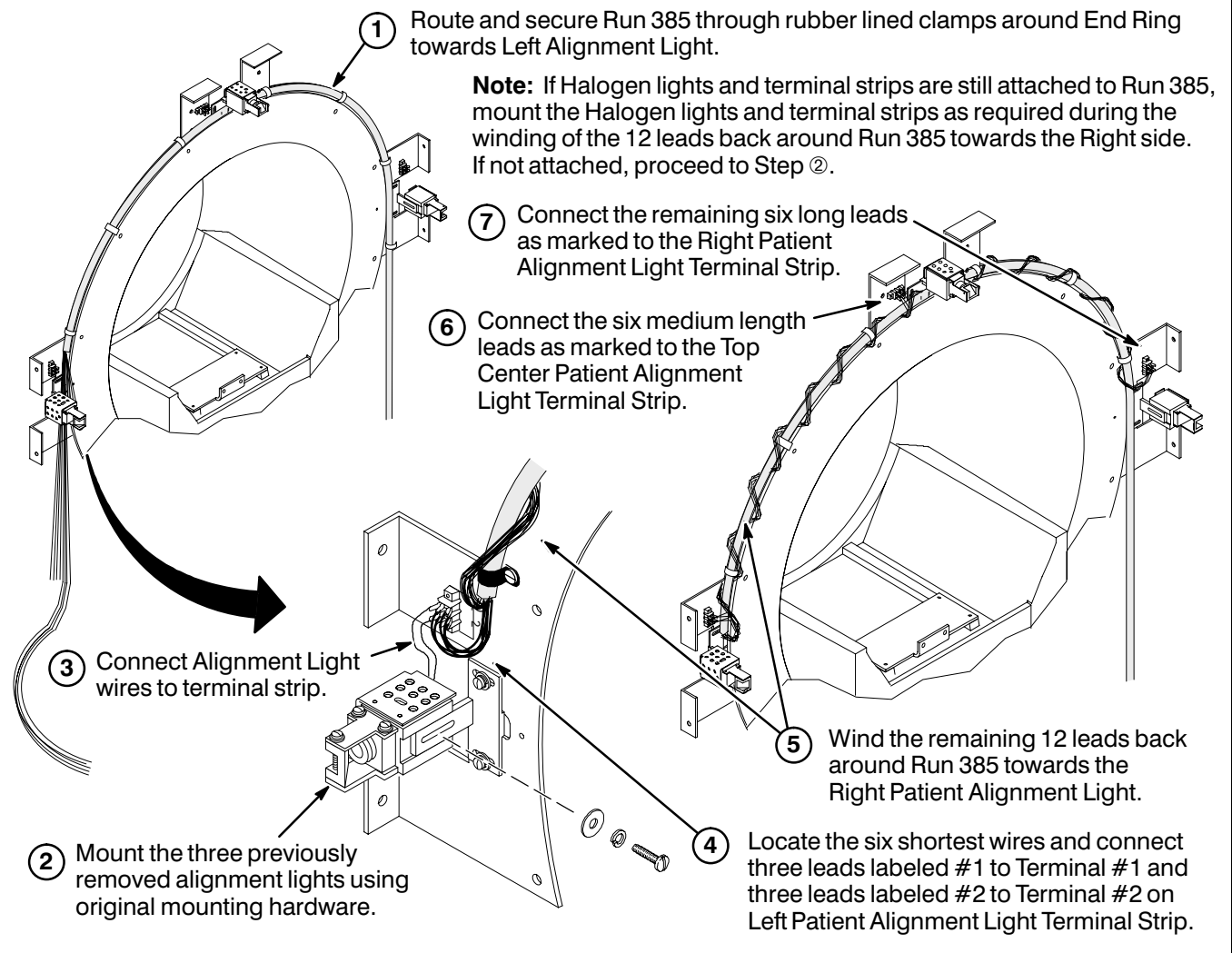
□ C2 – ROUTE AND CONNECT RUNS 778 AND 779



□ C3 – INSTALL FRONT GASKET AND MOUNT END BELL



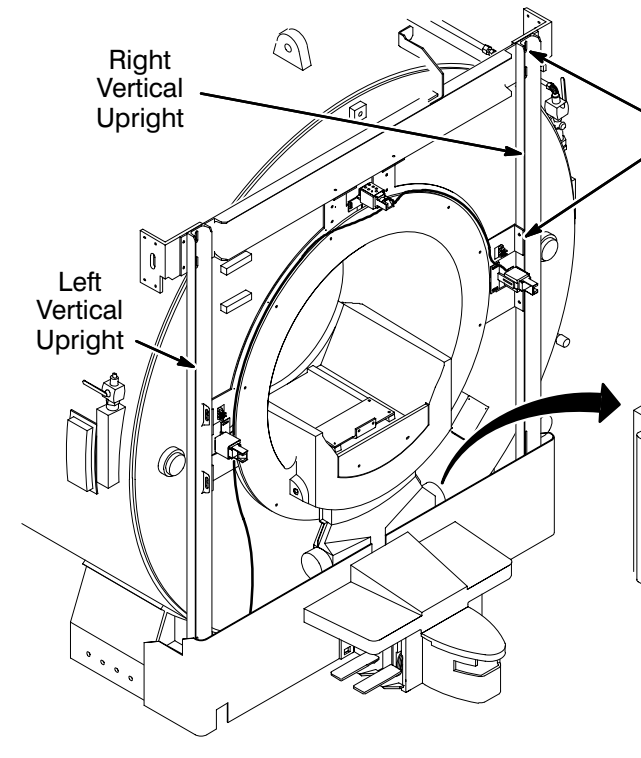
□ C4 – INSTALL AND CONNECT ALIGNMENT LIGHTS



INSTALLATION STEPS SUMMARY

- ☐ D1 – Adjusting Front Vertical Uprights & Installing Microphone Bracket
- ☐ D2 – Transfer & Assemble Bore Light to New Rear End Ring
- ☐ D3 – Install Rear End Rings
- ☐ D4 – Adjusting Rear Vertical Uprights

☐ D1 – ADJUST FRONT VERTICAL UPRIGHTS & INSTALLING MICROPHONE BRACKET



Right Vertical Upright

Left Vertical Upright

1 Verify that vertical uprights contact the floor. If adjustment is needed, proceed to Step 2. If not, proceed to Step 5.

2 Loosen four fasteners attaching Upright to Top Channel and Front Ring Asm.

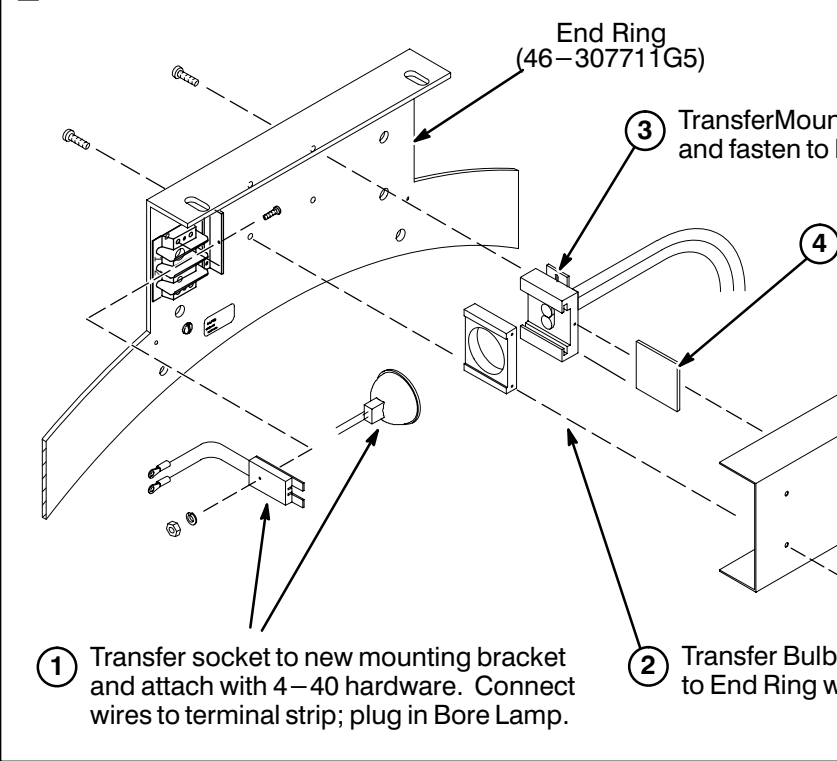
3 Adjust Vertical Upright to make firm contact with the floor. Tighten fasteners.

4 Repeat Steps 2 and 3 for adjustment of Left Vertical Upright.

5 From the Hardware Box, install Front Microphone Connector Bracket (2109027) with existing hardware. Insert (2) Cable Mounts (46-265067P1) and connect Microphone cable from End Bell to Run 414 at this bracket.

Note: Secure electrical ground required. Insure low resistance connection and tighten fasteners securely.

☐ D2 – TRANSFER & ASSEMBLE BORE LIGHT TO NEW REAR END RING



End Ring (46-307711G5)

1 Transfer socket to new mounting bracket and attach with 4-40 hardware. Connect wires to terminal strip; plug in Bore Lamp.

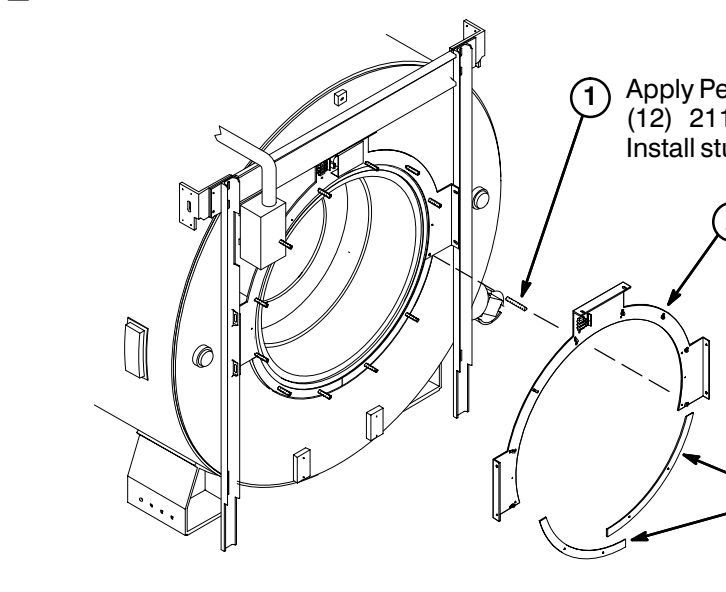
2 Transfer Bulb Retainer and attach to End Ring with 6-32 screws.

3 Transfer Mount with attached Fiber Optic Bundle and fasten to End Ring with 6-32 screws.

4 Transfer Heat Reflecting Mirror with coated side toward bulb. **Note that Fiber Optic bundles are NOT to touch glass.**

5 Transfer Cover and attach with 6-32 x 3/8 brass screws.

☐ D3 – INSTALL REAR END RINGS

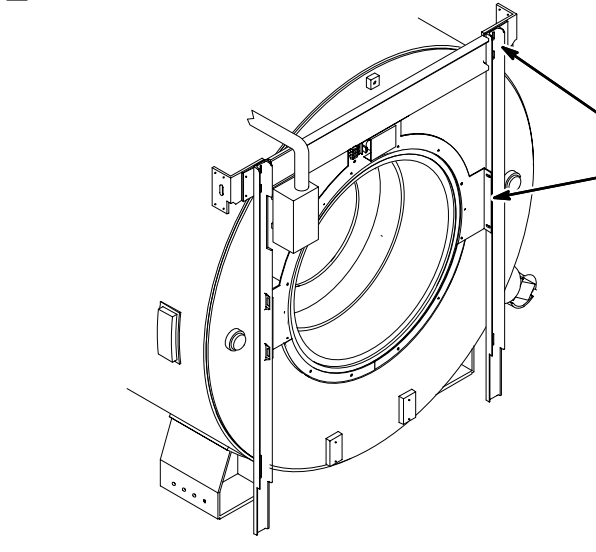


1 Apply Permatex "Anti-Seize" Lubricant (2119594) to : (12) 2116325-4 3/8-16 x 105mm SST Stud. Install studs into Rear Ring holes.

2 Install End Ring Asm (46-307711G5) on studs.

3 Install End Ring Section Spacers (2129823) to studs on lower part of Magnet End Ring.

☐ D4 – ADJUSTING REAR VERTICAL UPRIGHTS



1 Verify that vertical uprights contact the floor. If adjustment is needed, proceed to Step 2. If not, proceed to Step 4.

2 Loosen four fasteners attaching Upright to Top Channel and Front Ring Asm.

3 Adjust Vertical Upright to make firm contact with the floor. Tighten fasteners.

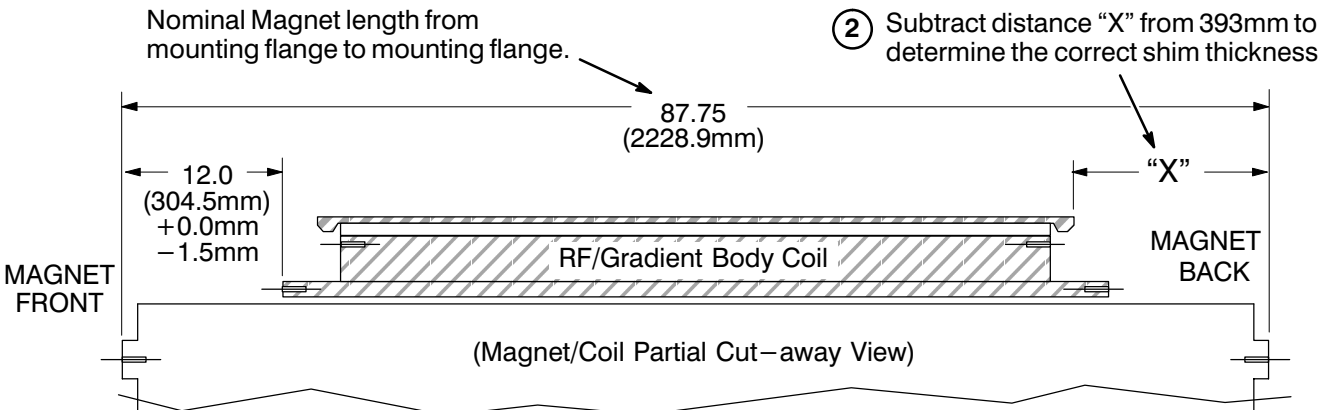
4 Repeat Steps 2 and 3 if adjustment is needed for Right Rear Vertical Upright.

INSTALLATION STEPS SUMMARY

- E1 – Formula for Determining Rear Spacers
- E2 – Install Rear Spacers and Terminal Blocks
- E3 – Transfer Existing Components to New Rear Pedestal Frame
- E4 – Install Rear Bulkhead and Rear Pedestal

E1 – FORMULA FOR DETERMINING REAR SPACERS

- 1 With RF/Gradient Body Coil in place, measure in millimeters the distance from RF Tube to mounting flange on Magnet. This is distance “X”.
Make sure the distance is measured to the Magnet mounting flange, not the Rear Frame Asm End Ring installed in Step D3.

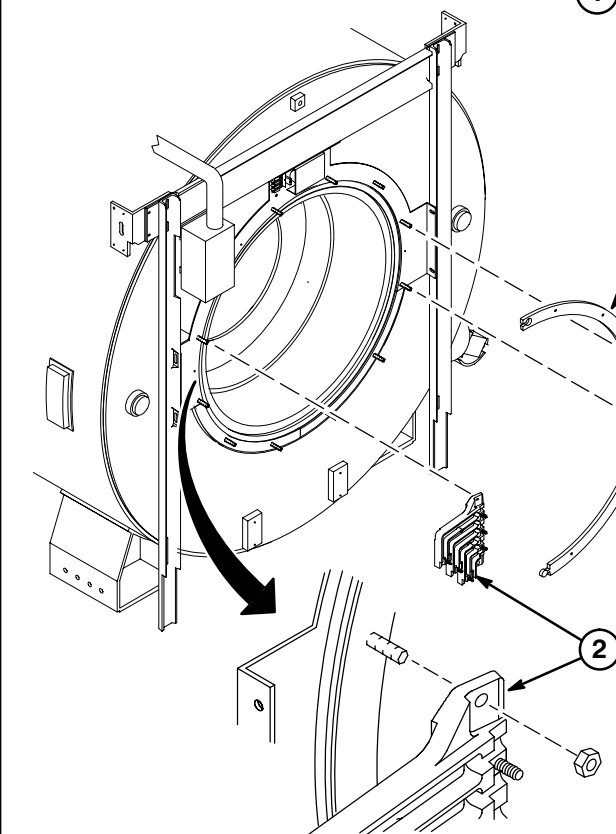


E2 – INSTALL REAR SPACERS AND GRADIENT TERMINAL BLOCK

- 1 Install the following Rubber Seals as required to obtain a thickness derived from the formula in the previous step: (393mm – “X” = Shim Thickness)

Note: Better air sealing results will occur with having too few shims than too many. The tolerance on distance “X” is +.000mm, –.75mm.

- 2115370 12.7mm thick Interlocking Rubber Seal
- 2115370-2 7.9mm thick Interlocking Rubber Seal
- 2115369 3/4mm thick Rubber Seal w/adhesive
- 2115369-2 1.5mm thick Rubber Seal w/adhesive
- 2115369-3 2.25mm thick Rubber Seal w/adhesive

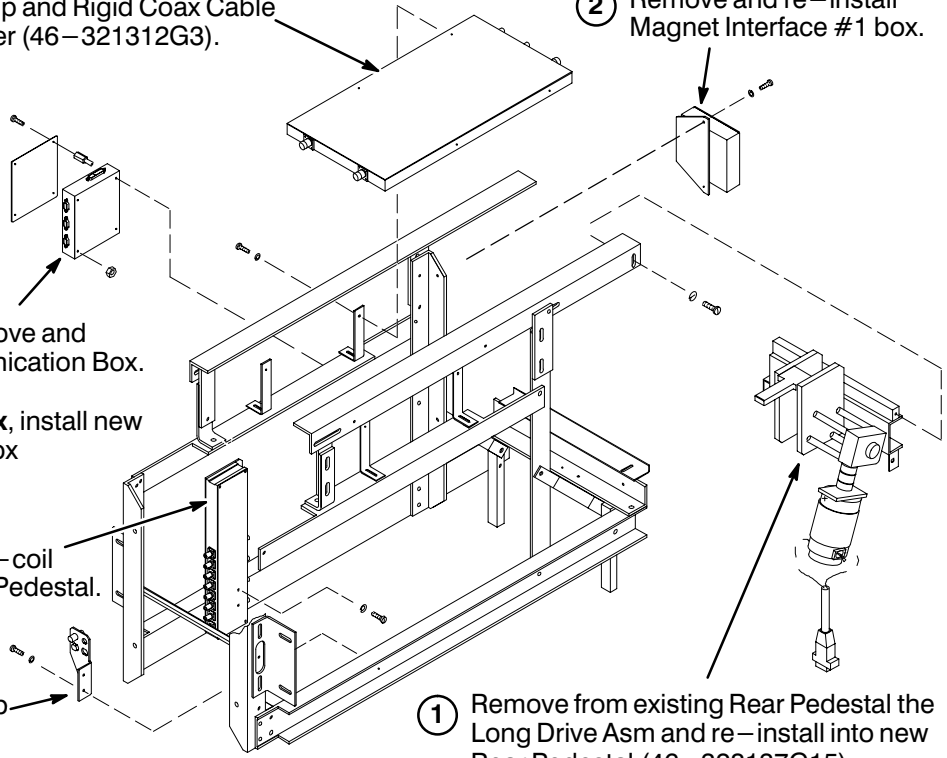


- 2 Install Terminal Block to magnet End Ring with:
(2) 46-208948P2 3/8-16 SST Hex Nut
Terminal Block Access Cover fits over top of this nut.

E3 – TRANSFER EXISTING COMPONENTS TO NEW REAR PEDESTAL FRAME

Note: Because of possible damage to smaller pieces of mounting hardware during transfer of components to new Rear Frame, a bag of #6, #8, and #10 screws, nuts, and lock washers is provided as spares.

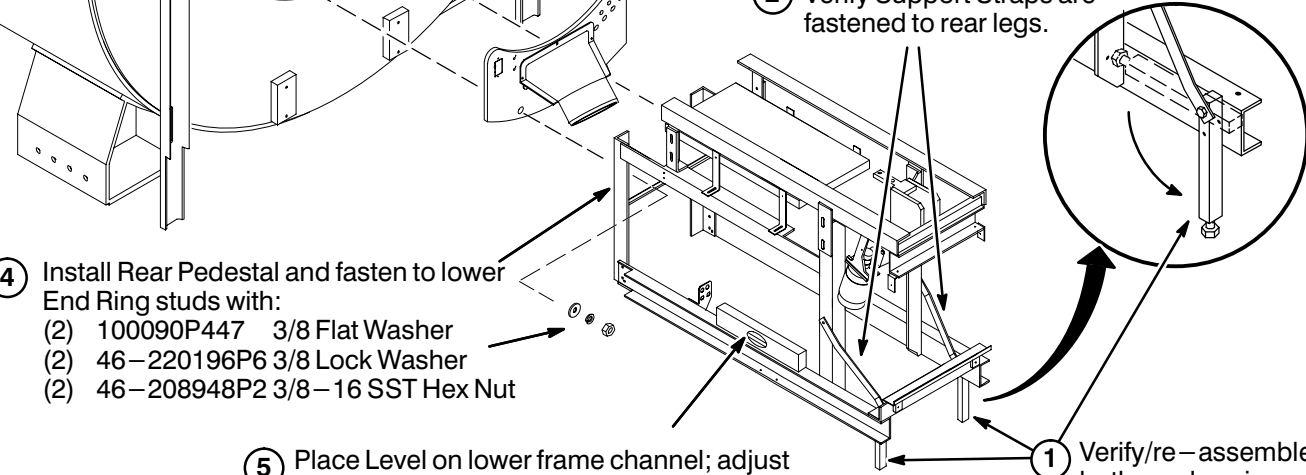
- 3 Transfer 1.5T Body Preamp and Rigid Coax Cable to new Body Hybrid Splitter (46-321312G3). Install Body Splitter.
- 2 Remove and re-install Magnet Interface #1 box.
- 4 If upgrading from 5.x, remove and re-install Patient Communication Box. OR
If upgrading from 3.x or 4.x, install new Patient Communication Box (46-317185G1).
- 5 If Multi-coil, transfer Multi-coil Select Switch to new Rear Pedestal.
- 6 Transfer Coil Cable Take-up Interface Assembly to new Rear Pedestal.
- 1 Remove from existing Rear Pedestal the Long Drive Asm and re-install into new Rear Pedestal (46-328137G15).



E4 – INSTALL REAR BULKHEAD AND REAR PEDESTAL

Note: If desired, End Bell installation procedures F1 & F2 can be performed after completion of Step 3 below and prior to starting Step 4, installing Rear Pedestal.

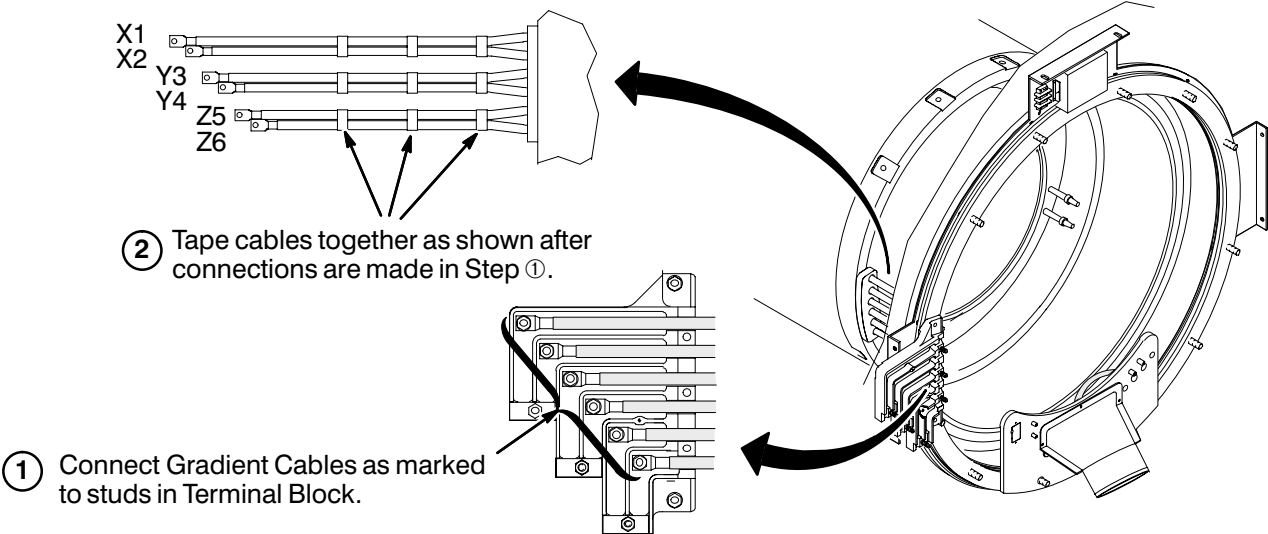
- 3 Place Bulkhead Asm (2129410) on two lower studs of End Ring Asm.
- 2 Verify Support Straps are fastened to rear legs.
- 4 Install Rear Pedestal and fasten to lower End Ring studs with:
(2) 100090P447 3/8 Flat Washer
(2) 46-220196P6 3/8 Lock Washer
(2) 46-208948P2 3/8-16 SST Hex Nut
- 5 Place Level on lower frame channel; adjust leveling screws on rear legs as needed.
- 1 Verify/re-assemble both rear legs in vertical orientation.



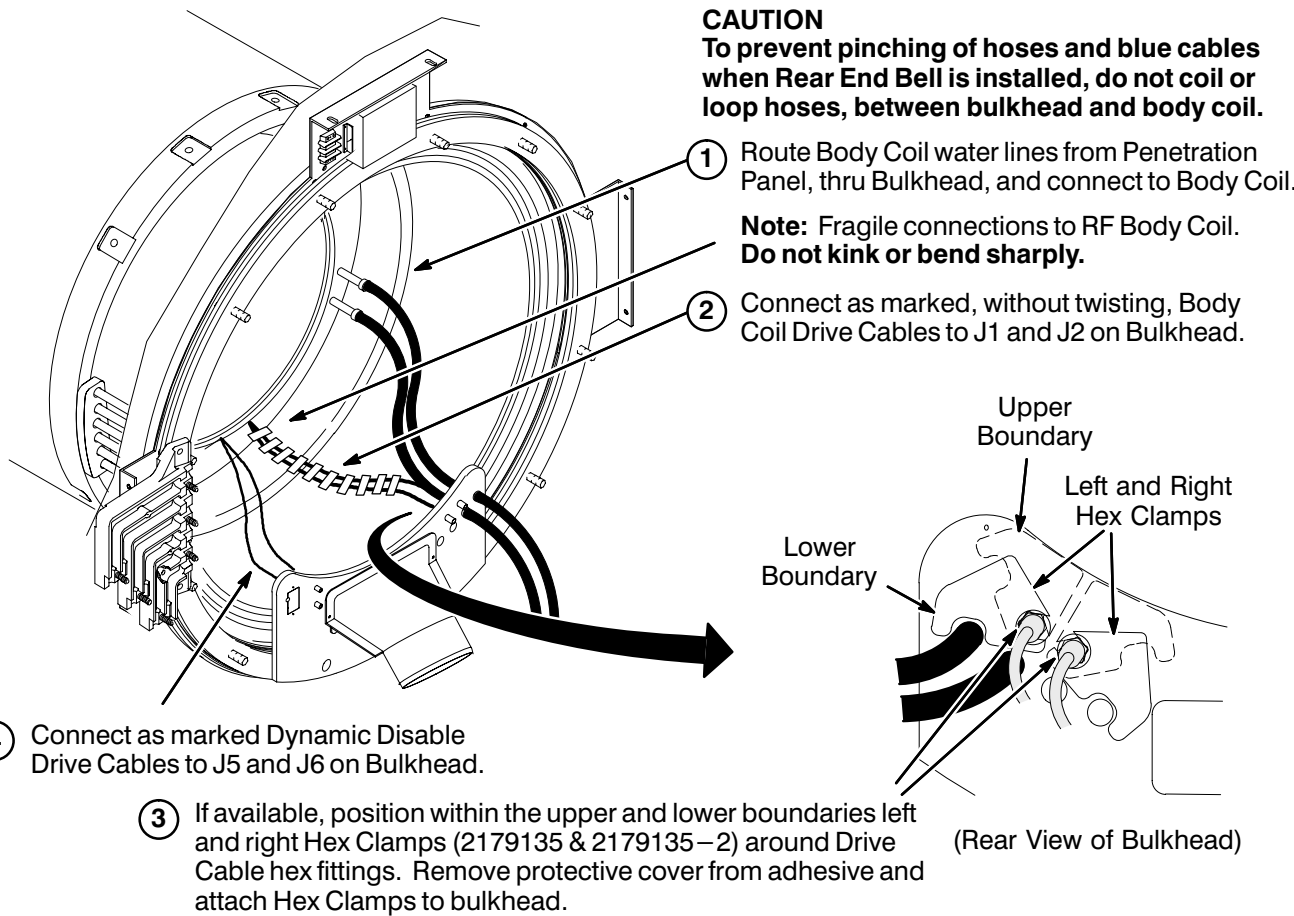
INSTALLATION STEPS SUMMARY

- F1 – Connect Body Coil Gradient Cables to Terminal Block
- F2 – Route and Connect Bulkhead Cables and Hoses
- F3 – Install Rear End Bell
- F4 – Connect Runs 257/746, 258/747, 262, 282, 318, 414, 422, 700 and 780 at Rear Pedestal

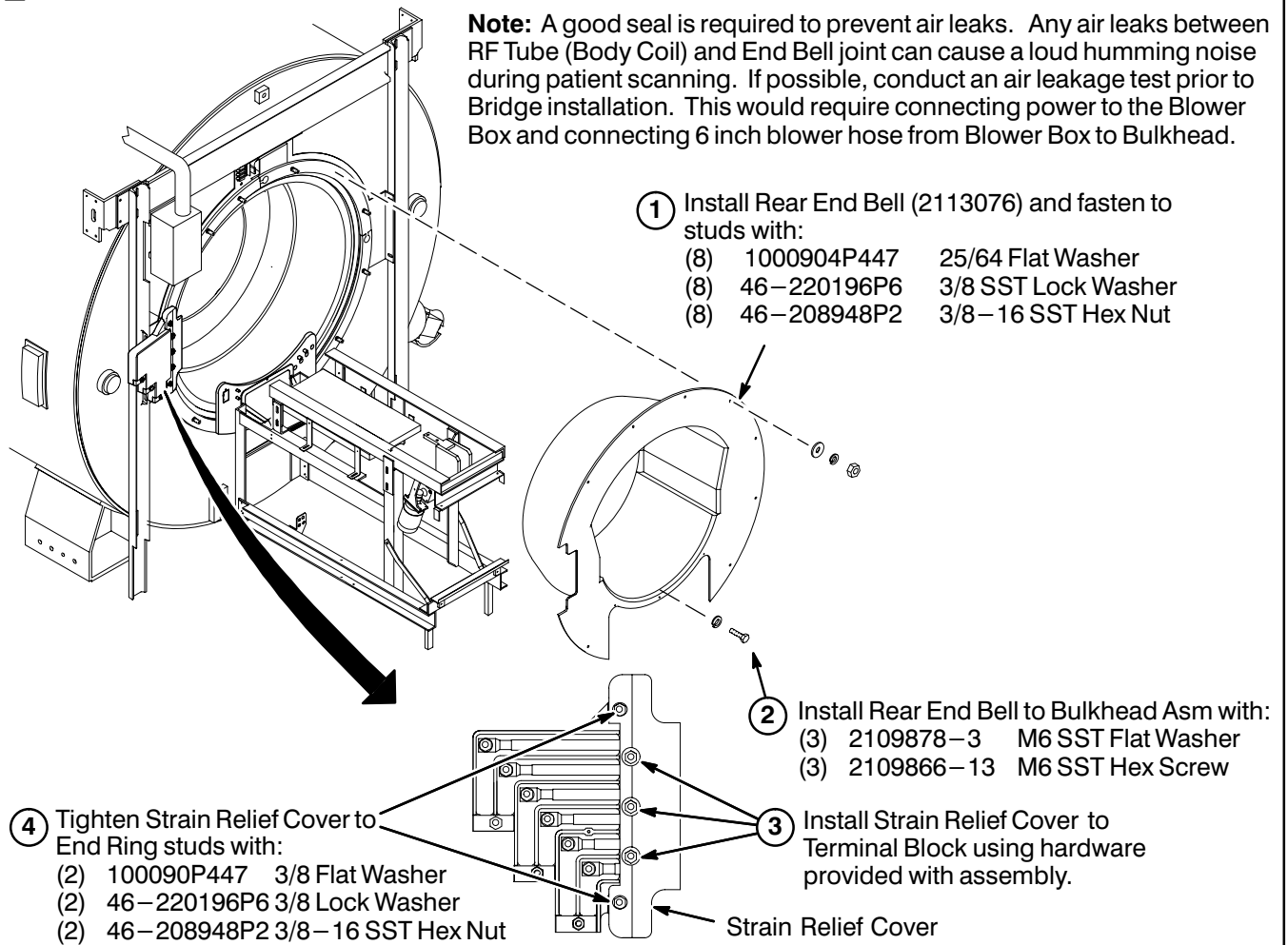
F1 – CONNECT BODY COIL GRADIENT CABLES TO TERMINAL BLOCK



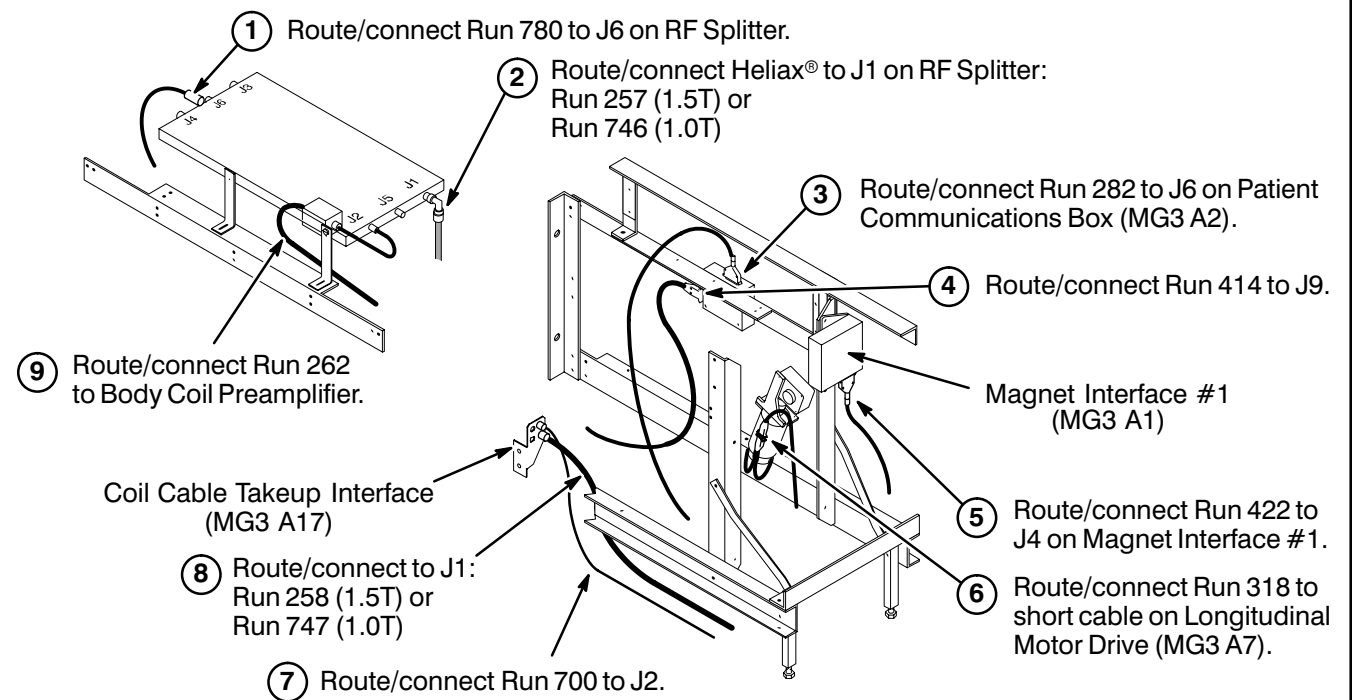
F2 – ROUTE AND CONNECT BULKHEAD CABLES AND HOSES



F3 – INSTALL REAR END BELL



F4 – CONNECT RUNS 257/746, 258/747, 262, 282, 318, 414, 422, 700, and 780



INSTALLATION STEPS SUMMARY

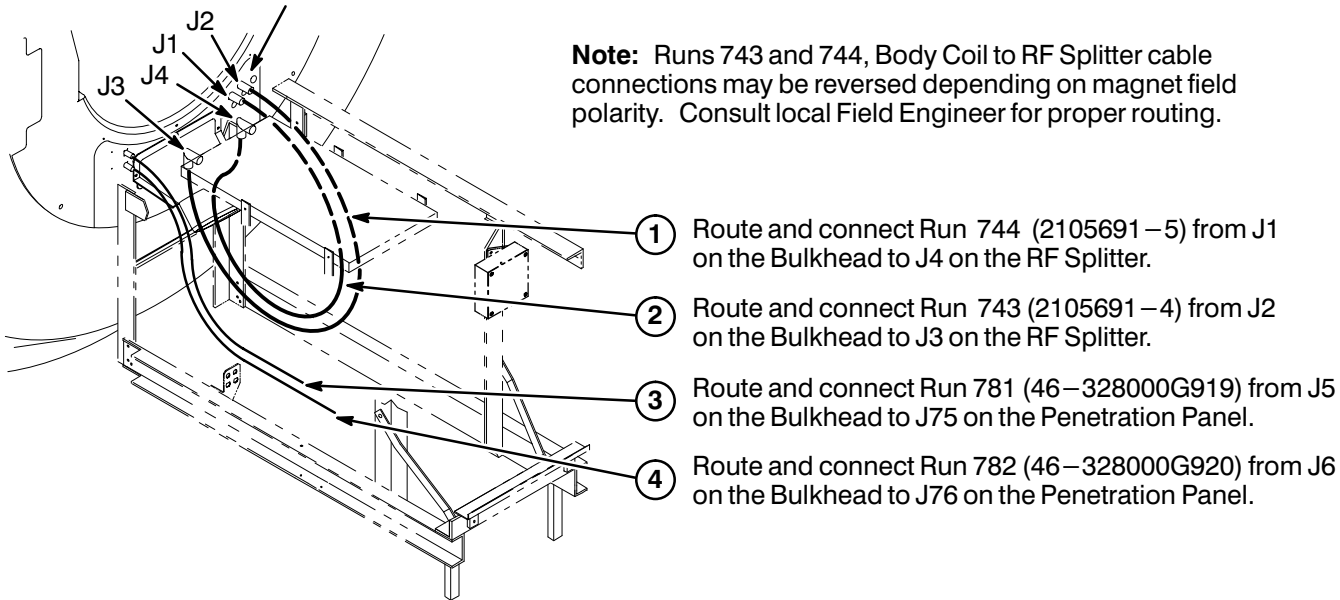
- ☐ G1 – Route/Connect Runs 743, 744, 781, and 782
- ☐ G2 – Route/Connect Runs 421 and Dock Cable
- ☐ G3 – Remove Carriage from Bridge for Notch Modification (If Needed)
- ☐ G4 – Cutting Notch in Carriage (If Needed)

☐ G1 – ROUTE/CONNECT RUNS 743, 744, 781, AND 782

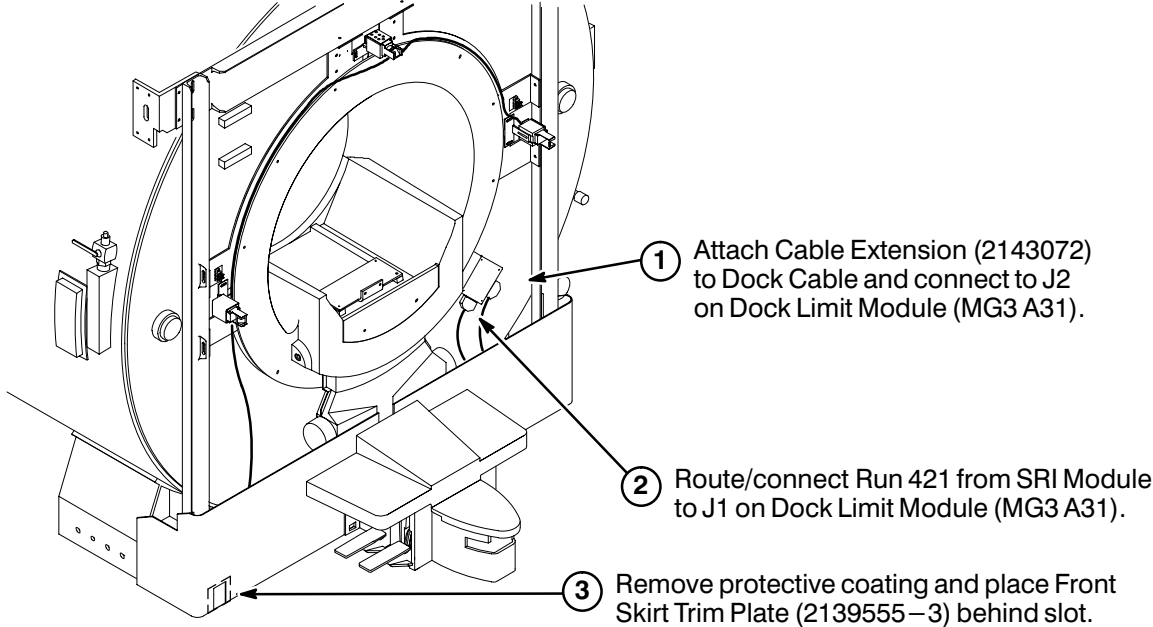
CAUTION

The RF Drive Input cables, Runs 743 and 744, utilize a corrugated solid RF shield. If mishandled, the cable can be kinked and subsequently broken.

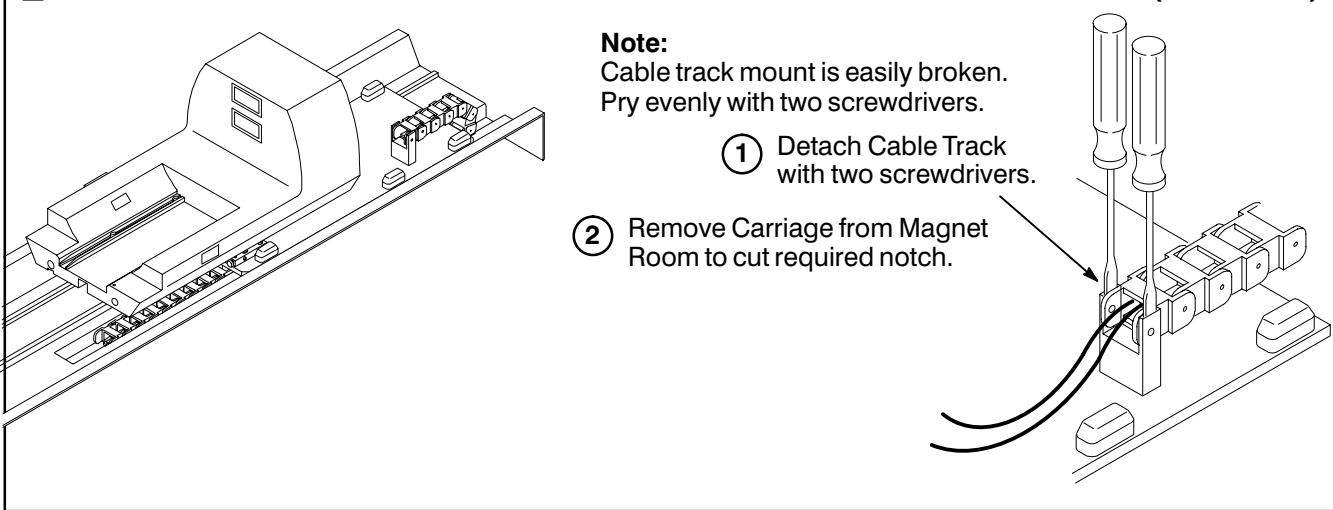
Also, use care when connecting these cable to the bulkhead connectors J1 and J2 and RF Splitter connectors J3 and J4. Cross threading or inadequate engagement of the center pin can cause arcing and damage the cables.



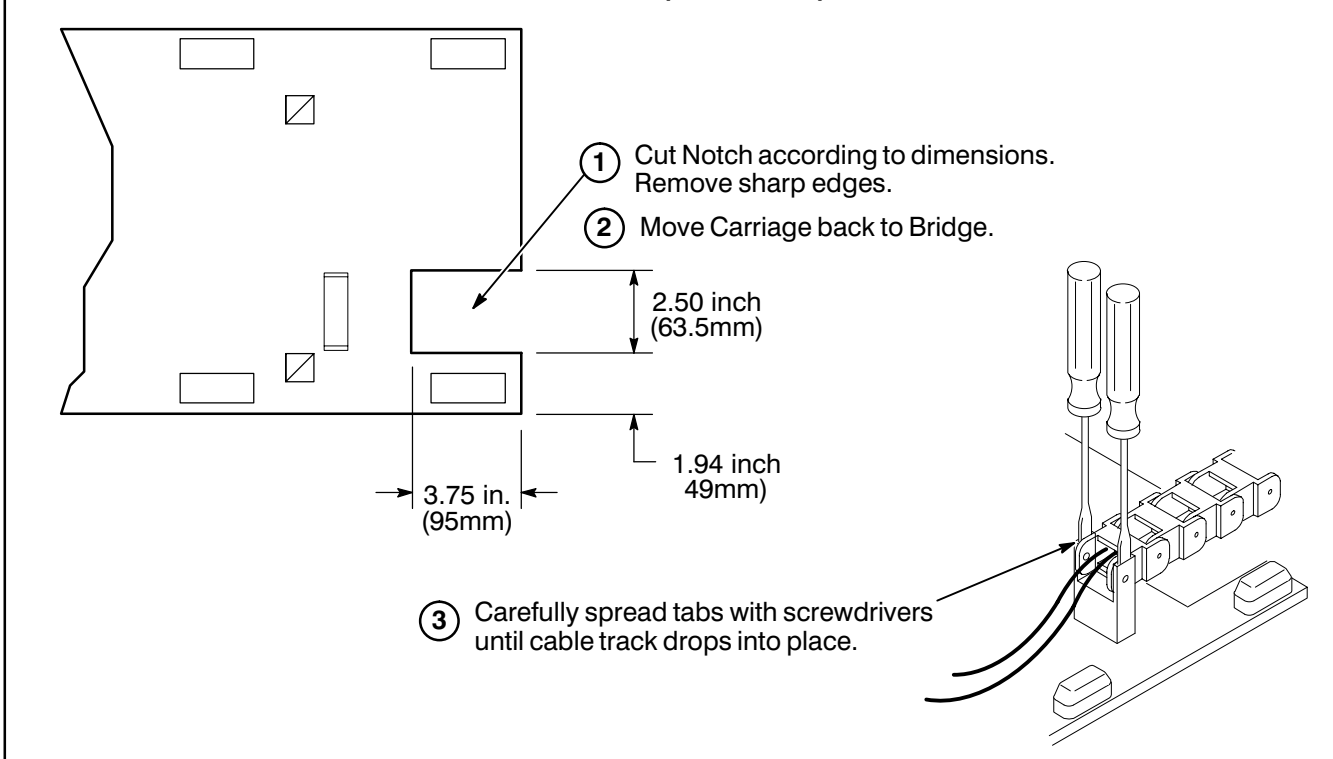
☐ G2 – ROUTE/CONNECT RUNS 421 AND DOCK CABLE



☐ G3 – REMOVE CARRIAGE FROM BRIDGE FOR NOTCH MODIFICATION (If Needed)



☐ G4 – CUTTING NOTCH IN CARRIAGE (If Needed)



INSTALLATION STEPS SUMMARY

☐ H1 – Bridge Assembly Preparation Before Installing in Magnet

☐ H2 – Securing Front of Bridge

☐ H3 – Leveling Bridge

☐ H4 – Connect Photo Sensor & Bridge End Cover

H1 – BRIDGE ASSEMBLY PREPARATION BEFORE INSTALLING IN MAGNET

1

Transfer Cable Trak, attaching hardware, cables, Carriage, Carriage Cover, and attached Drive Belt from removed Bridge to new Bridge (2119595–7).

2

Remove Rear Cover before installation.

3

Route Drive Belt through Front Pulley Asm and temporarily tape to other end of Bridge.
Note: Flag may remain on Drive Belt.

H2 – SECURING FRONT OF BRIDGE

1

Position rear end of Bridge Asm at front of Body Coil.

2

While one person holds the front end, the other person stands in front of the bore next to the bridge and guides the Bridge Asm through the bore to the other end. Lift front end of bridge to engage tongue on long drive assembly into bridge rear end.
Note: Bridge should be resting on Rubber Strips inside of Body Coil and front surface of Bridge should be flush with front surface of End Bell.

3

Bracket attached to bottom of Bridge should be touching bracket attached to End Bell. If needed, adjust End Bell bracket so brackets touch each other.

4

Fasten front end of Bridge Asm to Front End Bell using:
(2) 2109878–4 M10 SST Flat Washer
(2) 2109880–4 M10 SST Lock Washer
(2) 2109866–20 M10 SST Hex Screw
Note: Bridge and End Bell front surfaces should still be flush.

5

Install previously removed End Bell Panel with:
(2) 2109873–27 M6 x 25 Pan Screw

H3 – LEVELING BRIDGE

1

Level Bridge and tighten Vertical Adjustment Fasteners. Check that Longitudinal Drive Asm is level.

2

Slide horizontal top rails and Longitudinal Drive Asm forward to engage Tongue into Bridge.
Note: Tongue to Bridge alignment is critical to smooth belt tracking and operation. Make sure Tongue is square to Bridge and Rear Pedestal.

3

Tighten horizontal adjusting screws.
Note: Make sure front surface of Bridge is flush with front surface of Front End Bell when this procedure is complete.

H4 – CONNECT PHOTO SENSOR & BRIDGE END COVER

1

Attach together new Bridge, existing Tongue and Rear Bridge Mount Asm using:
(2) 46–220184P47 10–32 x 1–1/2 Screw

2

Connect Photo Sensor with screw provided with Rear Bridge Mount Asm.
Note: It is important that the Sensor end of the Sensor body fits flush with the Sensor Bracket.

Magnet I/F #1
(Mounted on Rear Pedestal)

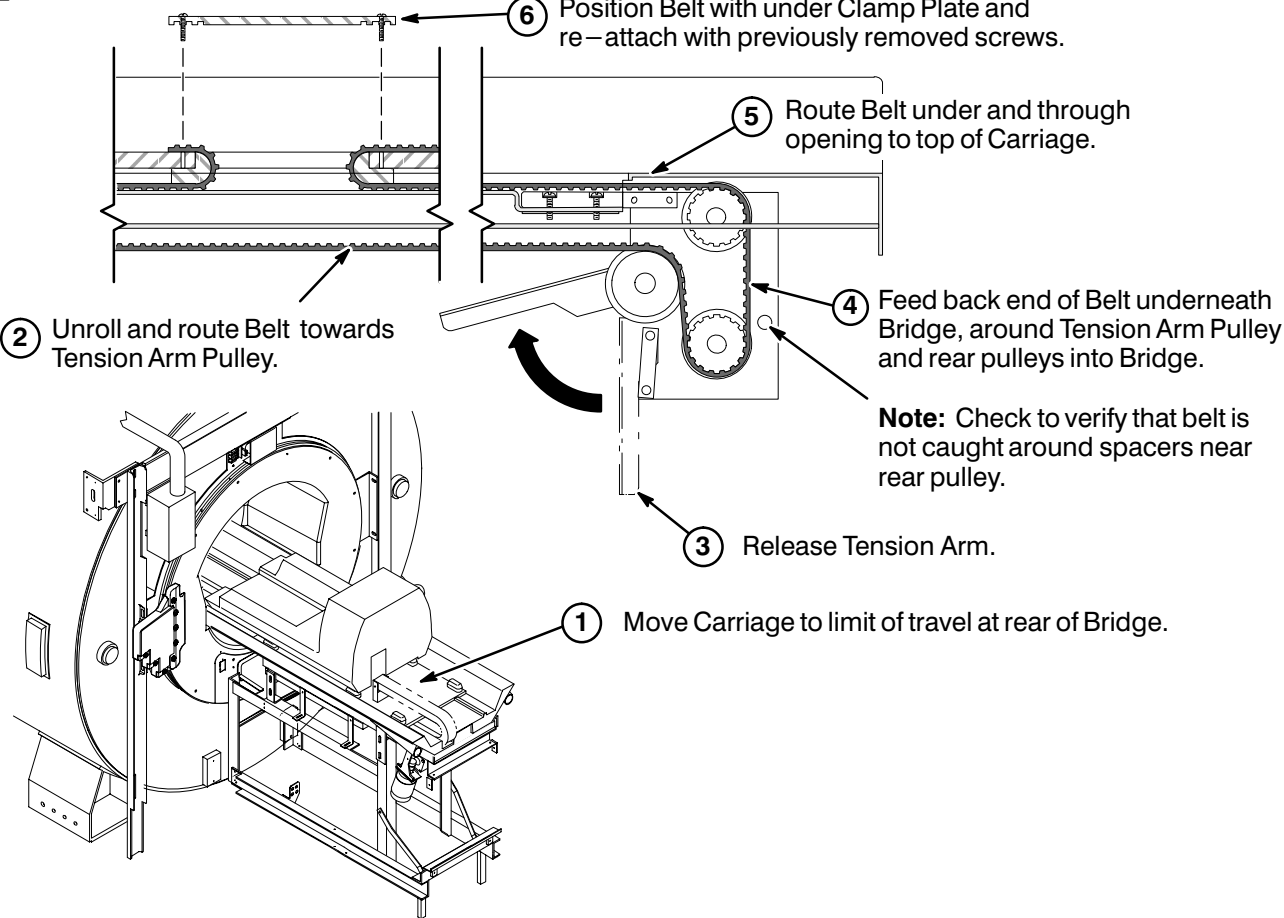
3

Attach previously removed Bridge Cover and (4) screws to end of Bridge.

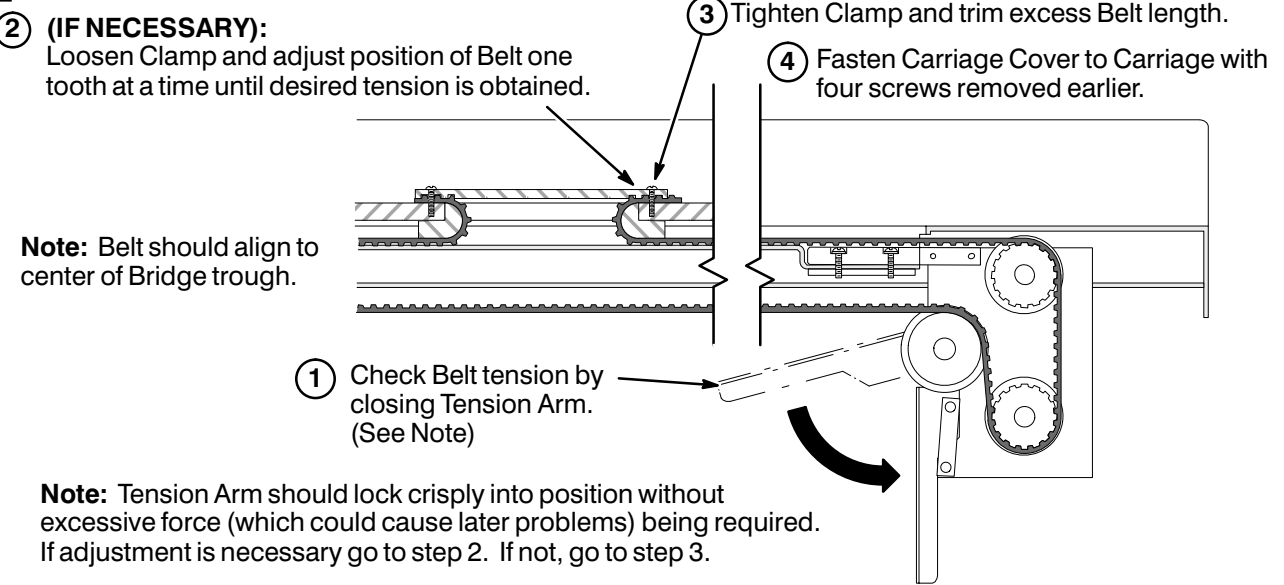
INSTALLATION STEPS SUMMARY

- J1 – Routing Drive Belt
- J2 – Final Adjustment of Drive Belt
- J3 – Installation of Sensor Flag
- J4 – Route Runs 404 and 485 (and Multi-coil Cables if required)

J1 – ROUTING DRIVE BELT



J2 – FINAL ADJUSTMENT OF DRIVE BELT

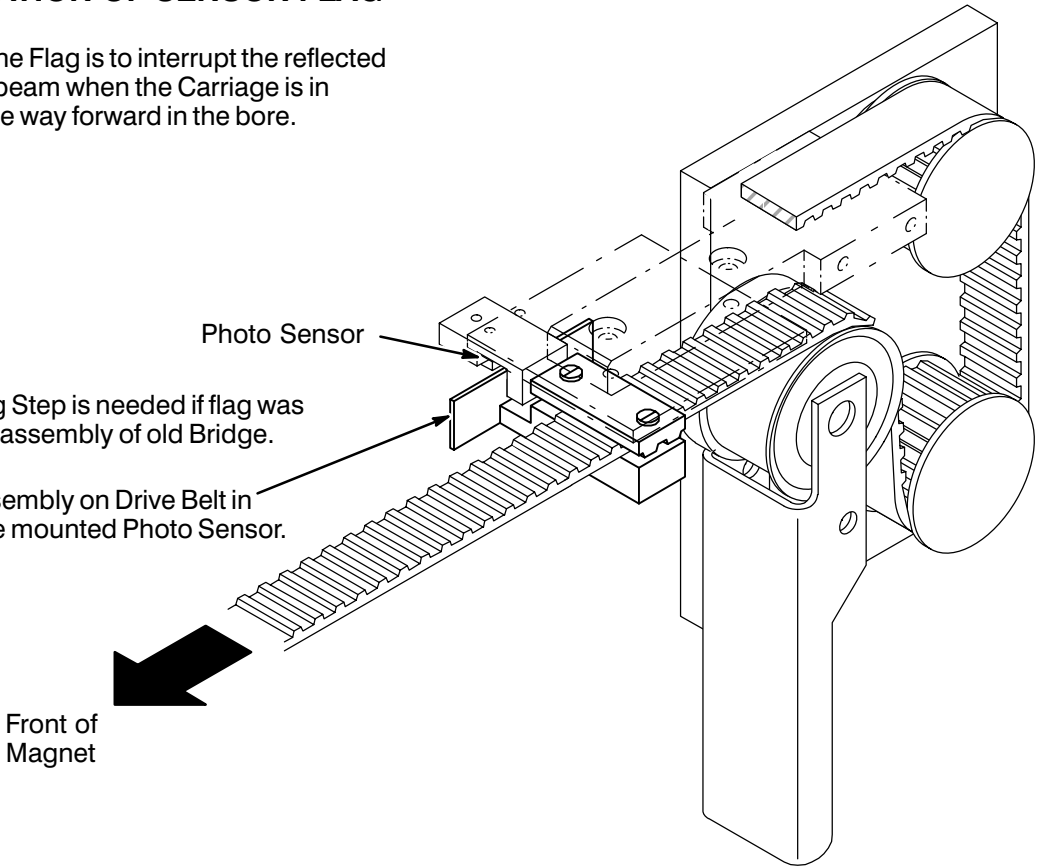


J3 – INSTALLATION OF SENSOR FLAG

Note: Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

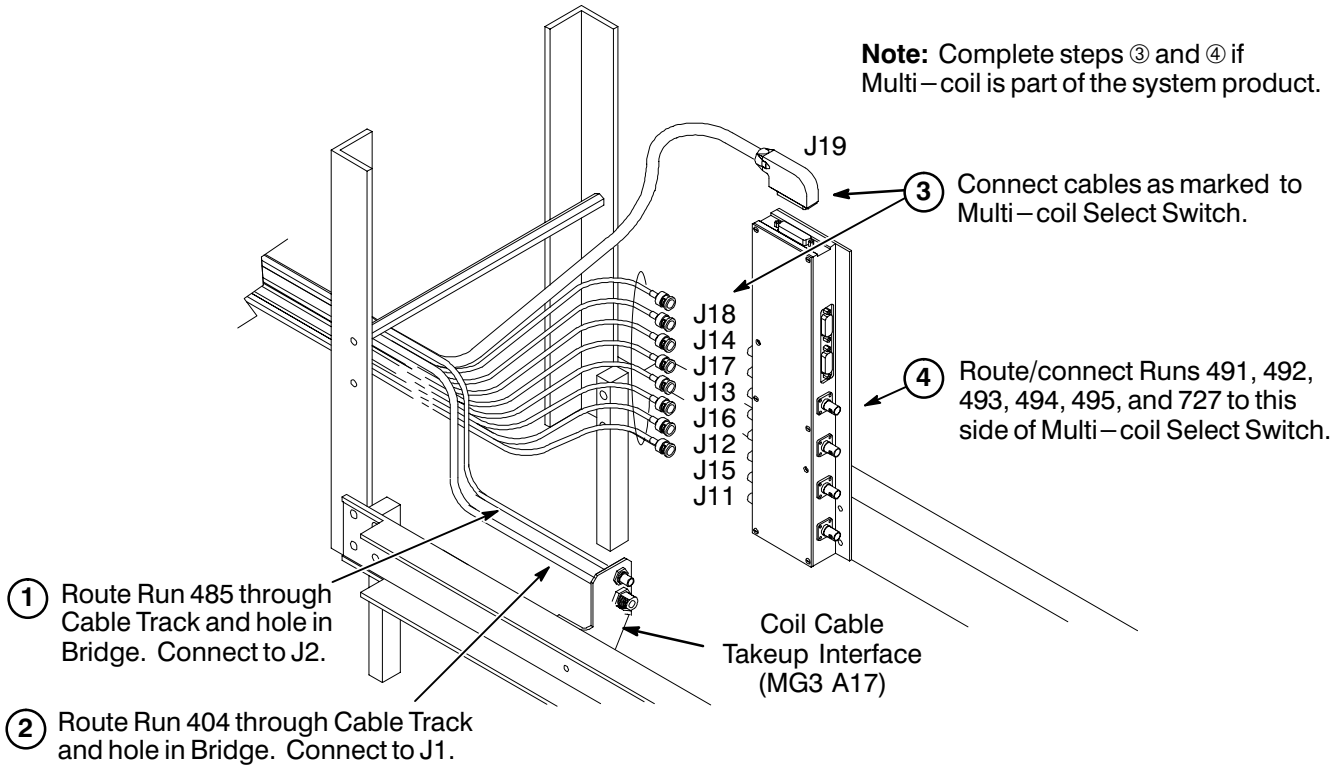
Note: The following Step is needed if flag was removed during disassembly of old Bridge.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



J4 – ROUTE RUNS 404 AND 485 (AND MULTI-COIL CABLES IF REQUIRED)

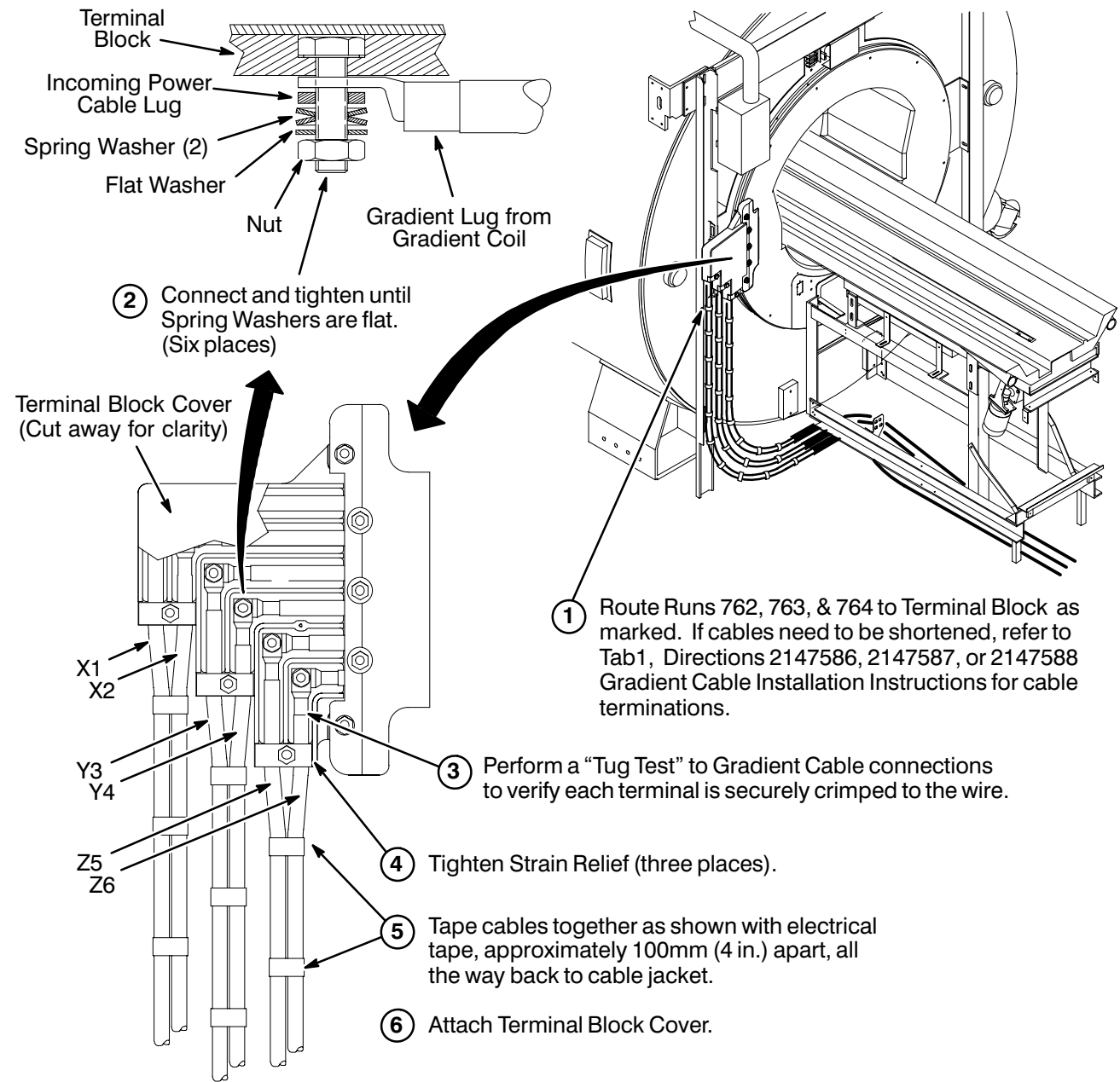
Note: Complete steps ③ and ④ if Multi-coil is part of the system product.



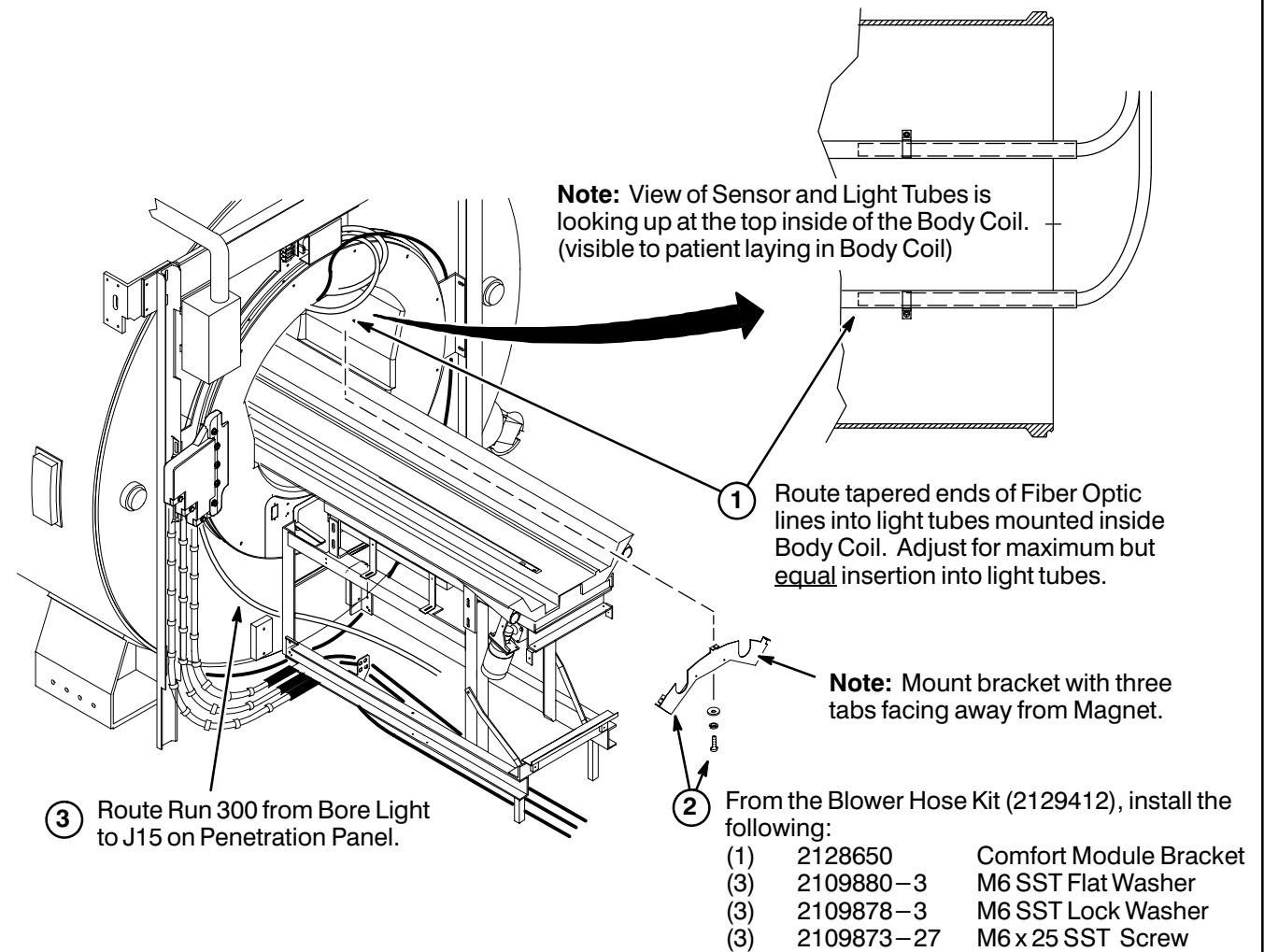
INSTALLATION STEPS SUMMARY

- ☐ K1 – Route/Connect Gradient Cables
- ☐ K2 – Route Run 300, Run 749, and Install Light Tube Fiber Optics
- ☐ K3 – Connect Run 300 to Bore Light

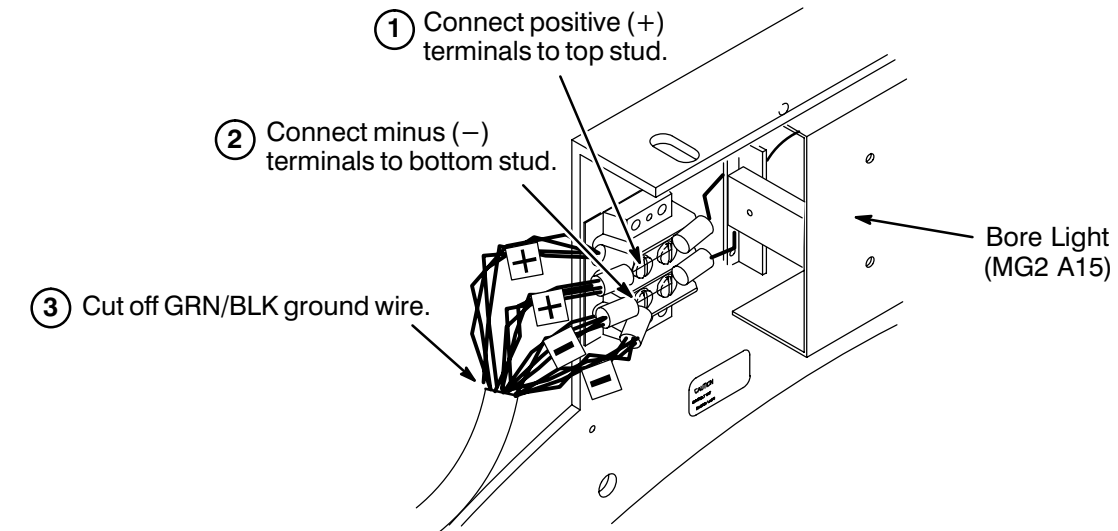
☐ K1 – ROUTE/CONNECT GRADIENT CABLES



☐ K2 – ROUTE RUN 300 AND INSTALL LIGHT TUBE FIBER OPTICS



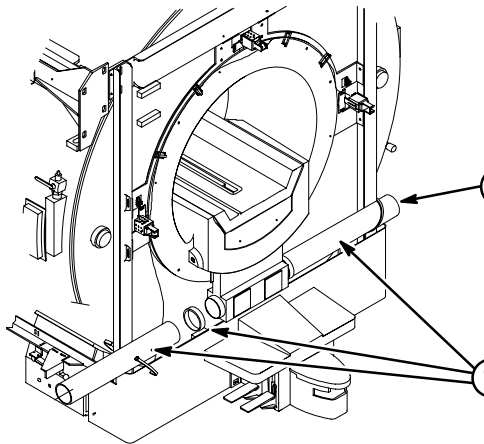
☐ K3 – CONNECT RUN 300 TO BORE LIGHT



INSTALLATION STEPS SUMMARY

- ❑ L1 – Connecting Front Air Plenum Hoses
- ❑ L2 – Routing/Connection of Patient Comfort Air Hoses
- ❑ L3 – Attaching Patient Comfort Module
- ❑ L4 – Route/Connect Patient Comfort Cables and Air Hoses

❑ L1 – CONNECTING FRONT AIR PLENUM HOSES

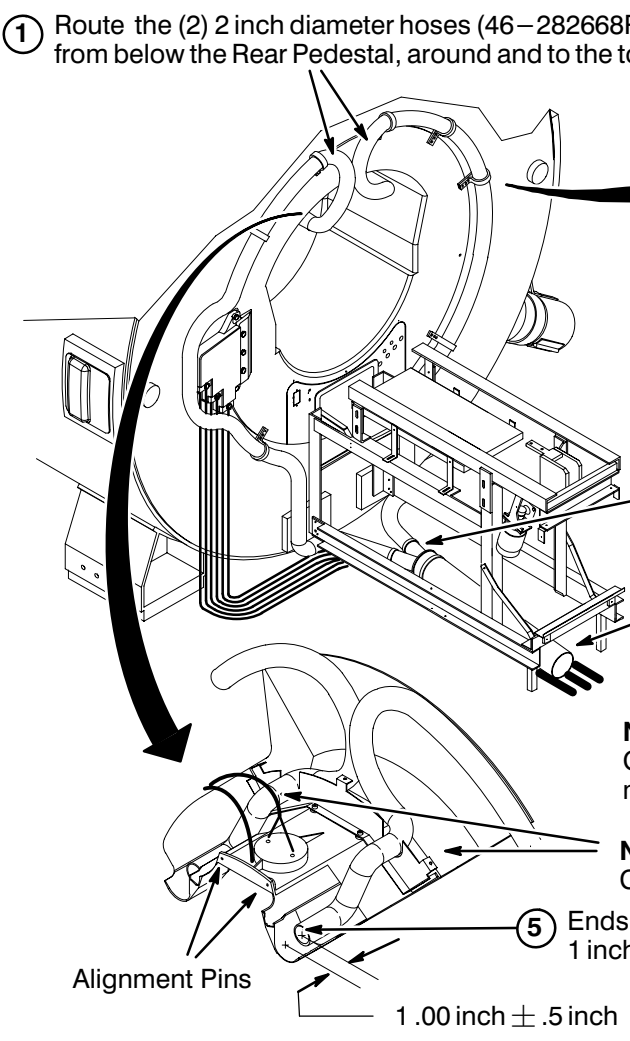


❶ Ty-wrap each hose to Vertical Uprights on each side.

Note: All the air hoses and clamps in the following steps are located in the Blower Hose Kit (2129412).

❷ Cut two (2) 26 inch lengths of 4 Inch Hose (46-282666P5) and attach one length to each side of Front Air Plenum using:
(2) 46-208765P34 4 inch Hose Clamp

❑ L2 – ROUTING/CONNECTION OF PATIENT COMFORT AIR HOSES



❶ Route the (2) 2 inch diameter hoses (46-282668P18) as shown from below the Rear Pedestal, around and to the top of the End Ring.

❷ Secure 2 inch hose to End Ring in seven places by cutting Nylon Strap (2139049) into seven 12 inch long pieces and fastening with:
(7) 1000904P447 3/8 Flat Washer
(7) 46-208948P2 3/8-16 SST Hex Nut

❸ Attach ends of 2 inch hoses to 2 Way Header (2120631) using:
(2) 46-208765P28 2 inch Hose Clamps

❹ Attach remaining length of 4 inch Hose (46-282666P5) to 2-way Header using:
(1) 46-208765P34 4 inch Hose Clamp

Note: 4 Inch Hose should lie on top of previously routed Gradient Coil Cables. Six Inch Hose routed in Step L4 must lie on the floor next to Gradient Coil Cables.

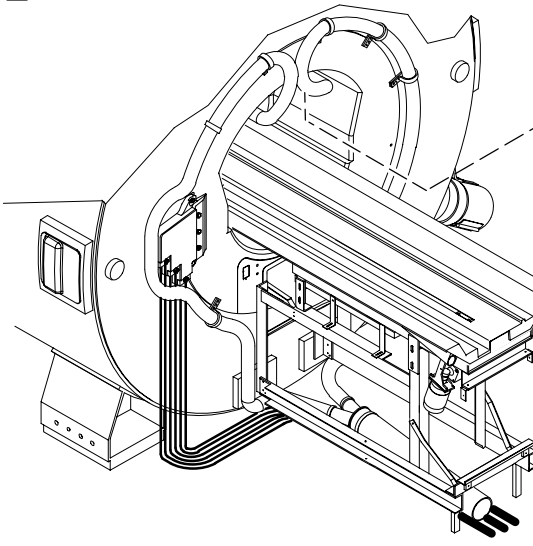
Note: For clarity, this view shows final assembly with Comfort Module Cover in place and cables routed to side.

❺ Ends of 2 inch hoses should be routed to within 1 inch (± .5 inch) of edge of Comfort Module Cover.

Alignment Pins

1.00 inch ± .5 inch

❑ L3 – ATTACHING PATIENT COMFORT MODULE

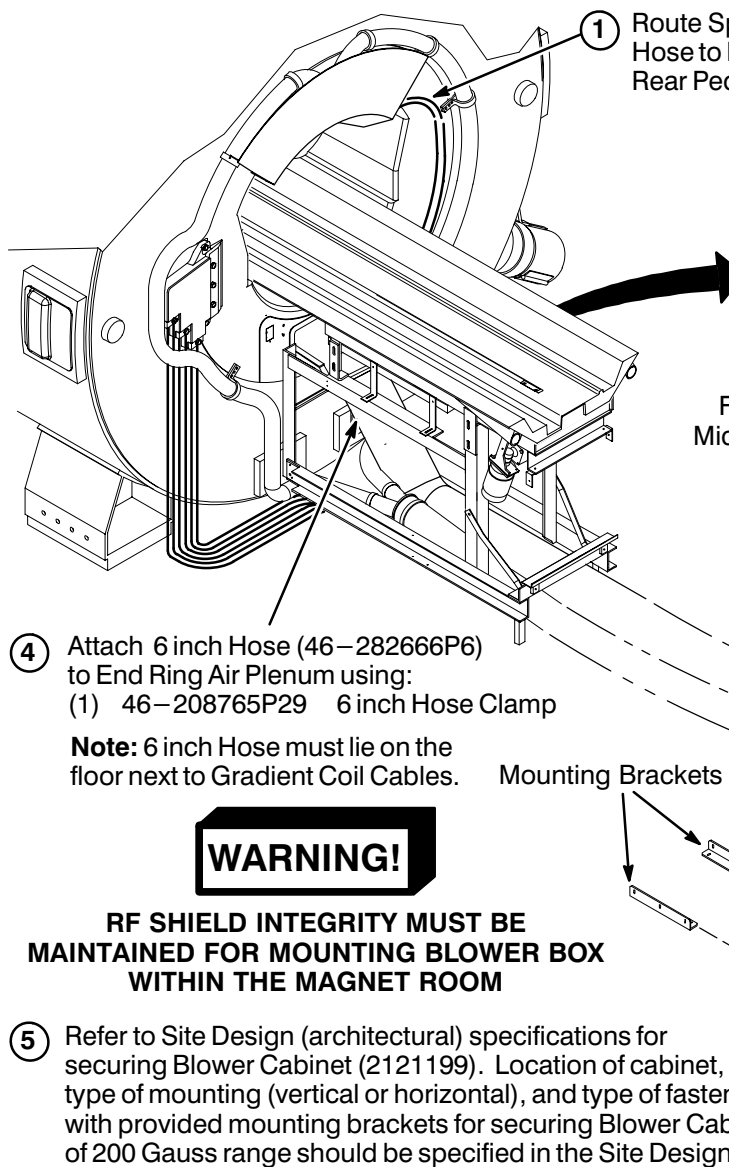


❶ Route Speaker and Microphone Cables to this side.

Note: Hold Patient Comfort Module in place until fastened to bracket. Alignment Pins can break off during installation if over-stressed.

❷ Install Patient Comfort Module (2128650) by aligning pins on module with two holes in Rear End Bell and attach to Comfort Module Bracket using:
(2) 2109873-27 M6 x 25 SST Pan Head Screw

❑ L4 – ROUTE/CONNECT PATIENT COMFORT CABLES AND AIR HOSES



❶ Route Speaker and Microphone Cables along side 2 Inch Air Hose to Patient Communication Box (MG3 A2) on side of Rear Pedestal.

❷ Connect Rear Microphone Cable to J7.

❸ Connect Speaker Cable to J8.

❹ Attach 6 inch Hose (46-282666P6) to End Ring Air Plenum using:
(1) 46-208765P29 6 inch Hose Clamp

Note: 6 inch Hose must lie on the floor next to Gradient Coil Cables.

❺ Refer to Site Design (architectural) specifications for securing Blower Cabinet (2121199). Location of cabinet, type of mounting (vertical or horizontal), and type of fastener used with provided mounting brackets for securing Blower Cabinet outside of 200 Gauss range should be specified in the Site Design document.

❻ Route 4 inch and 6 inch air hoses to Blower Cabinet. Trim to appropriate length and attach to vent openings with:
(1) 46-208765P34 4 inch Hose Clamp
(1) 46-208765P29 6 inch Hose Clamp

WARNING!
RF SHIELD INTEGRITY MUST BE MAINTAINED FOR MOUNTING BLOWER BOX WITHIN THE MAGNET ROOM

Mounting Brackets

Note: If needed, Blower Hose Cover can be cut and modified.

INSTALLATION STEPS SUMMARY

- ☐ M1 – Install Front Cover (2120888-2)
- ☐ M2 – Disassemble Cover Dolly
- ☐ M3 – Remove Cover Dolly

DANGER!!

FERROUS MATERIAL HAZARD! THE COVER DOLLY WAS NOT DESIGNED FOR COVER INSTALLATION WITH MAGNET AT FIELD. THE CASTER ASSEMBLIES AND FASTENING HARDWARE ARE MAGNETIC. IF MAGNET IS AT FIELD, THREE INSTALLERS ARE REQUIRED TO REMOVE FRONT COVER FROM DOLLY AND CARRY COVER INTO POSITION AT FRONT OF MAGNET.

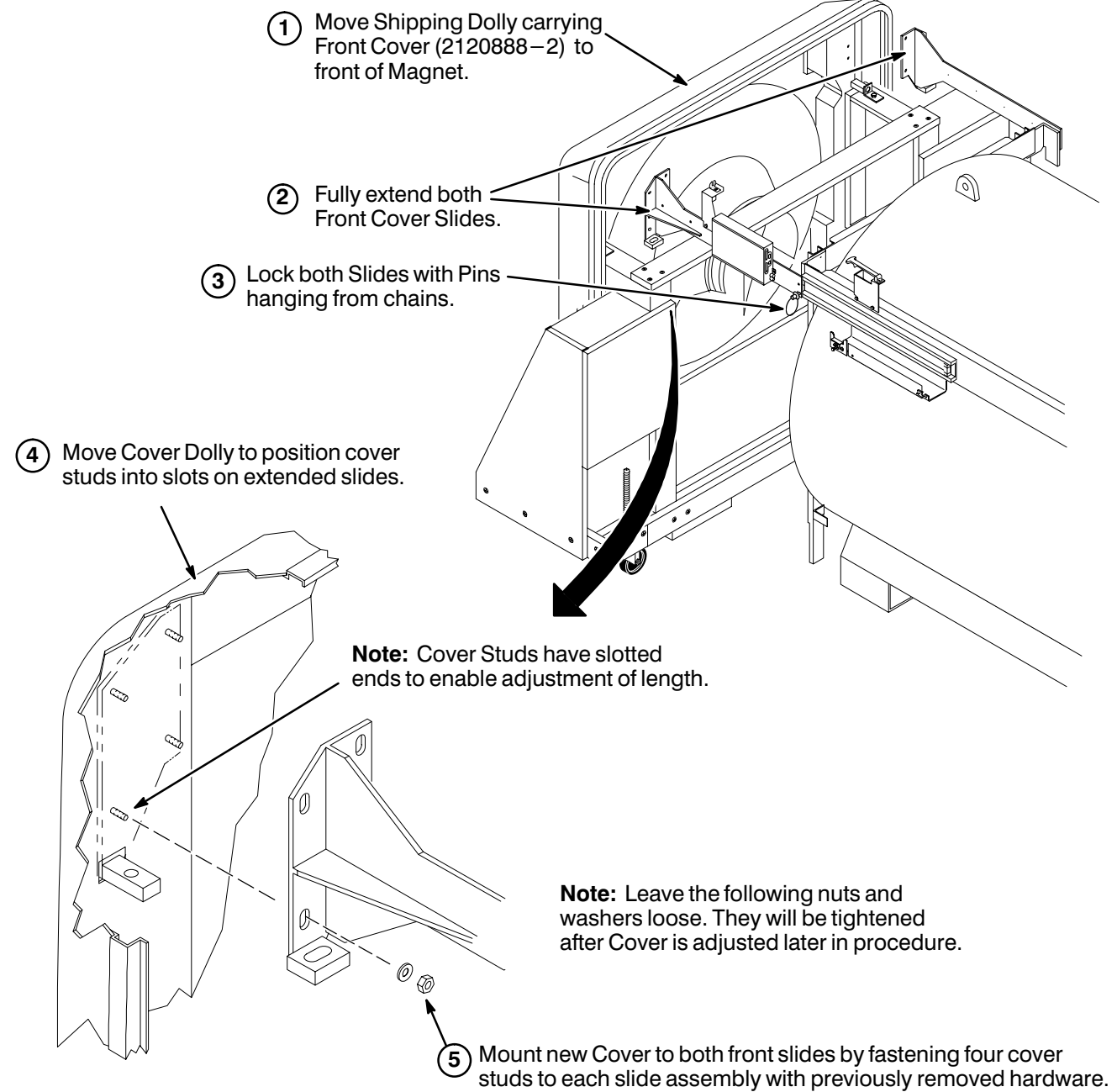
☐ M1 – INSTALL FRONT COVER (2120888-2)

1 Move Shipping Dolly carrying Front Cover (2120888-2) to front of Magnet.

2 Fully extend both Front Cover Slides.

3 Lock both Slides with Pins hanging from chains.

4 Move Cover Dolly to position cover studs into slots on extended slides.



Note: Cover Studs have slotted ends to enable adjustment of length.

Note: Leave the following nuts and washers loose. They will be tightened after Cover is adjusted later in procedure.

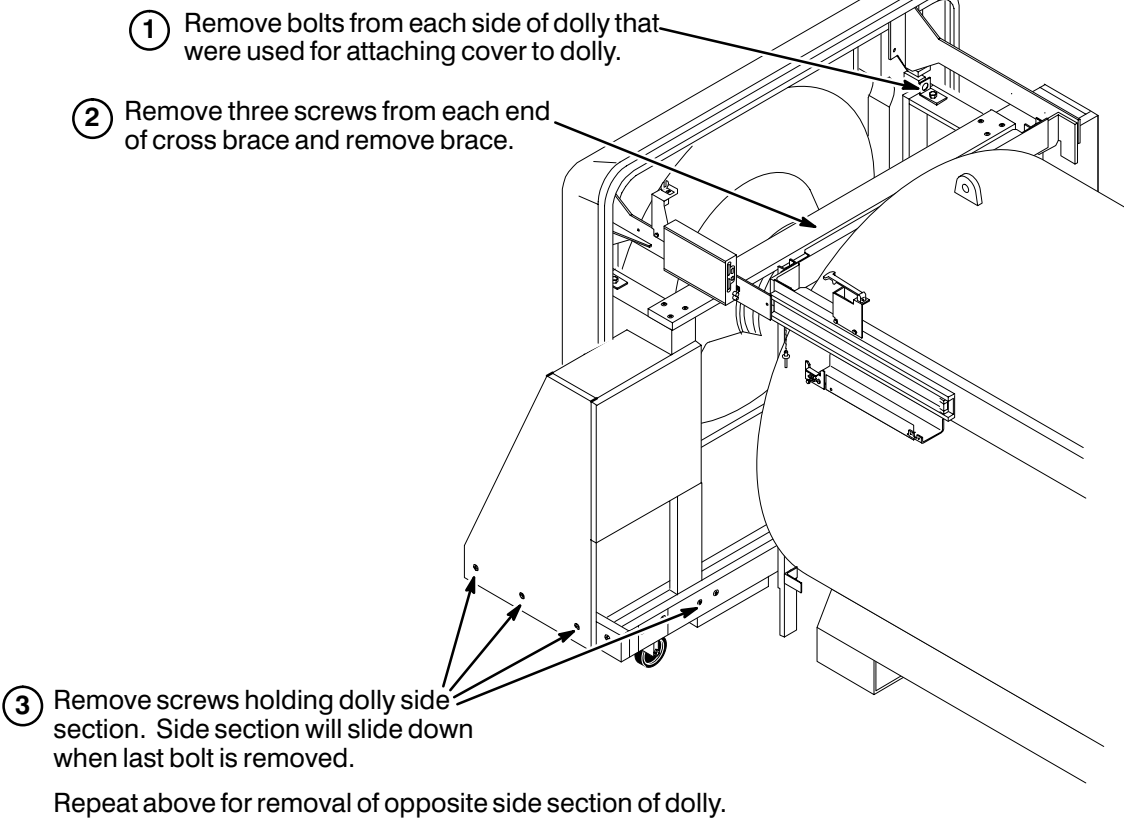
5 Mount new Cover to both front slides by fastening four cover studs to each slide assembly with previously removed hardware.

☐ M2 – DISASSEMBLE COVER DOLLY

1 Remove bolts from each side of dolly that were used for attaching cover to dolly.

2 Remove three screws from each end of cross brace and remove brace.

3 Remove screws holding dolly side section. Side section will slide down when last bolt is removed.



Repeat above for removal of opposite side section of dolly.

☐ M3 – REMOVE COVER DOLLY

6 Place remaining dolly parts on recycle pile.

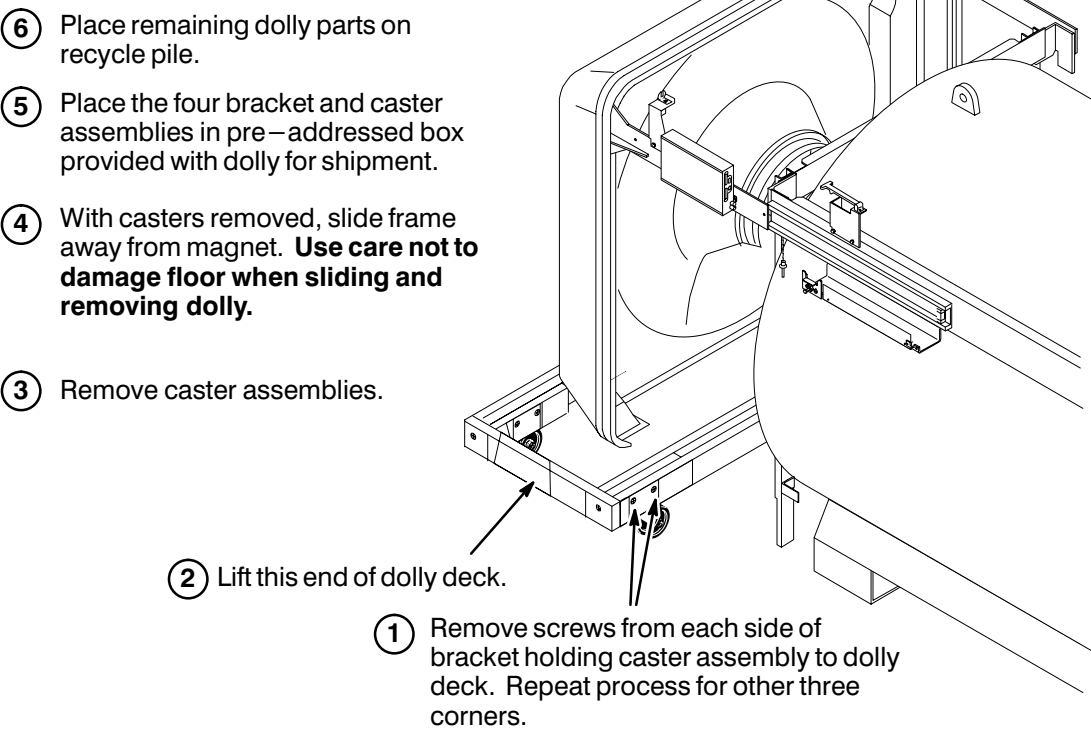
5 Place the four bracket and caster assemblies in pre-addressed box provided with dolly for shipment.

4 With casters removed, slide frame away from magnet. **Use care not to damage floor when sliding and removing dolly.**

3 Remove caster assemblies.

2 Lift this end of dolly deck.

1 Remove screws from each side of bracket holding caster assembly to dolly deck. Repeat process for other three corners.



INSTALLATION STEPS SUMMARY

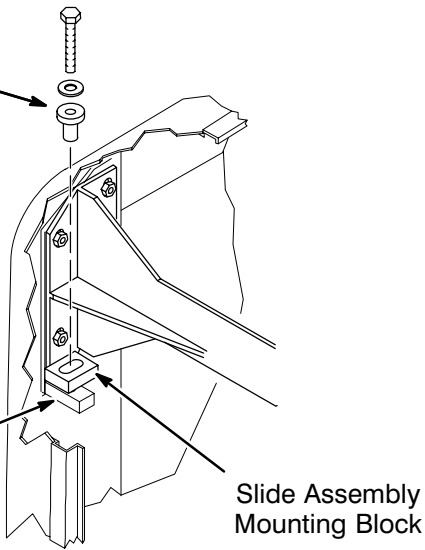
- ❑ N1 – Install Front Cover Alignment Hardware
- ❑ N2 – Transfer Lower Latch Assemblies to New Front & Rear Covers
- ❑ N3 – Adjust Front Cover Alignment
- ❑ N4 – Align Front Cover
- ❑ N5 – Install Rear Cover (46-320517P1)

❑ N1 – INSTALL FRONT COVER ALIGNMENT HARDWARE

❶ Transfer and insert the Alignment Hardware from the old Front Cover into the oval hole on the Slide Assembly Mounting Block

❷ Engage threads in Cover Mounting Block.

❸ Repeat process for other side of Front Cover.

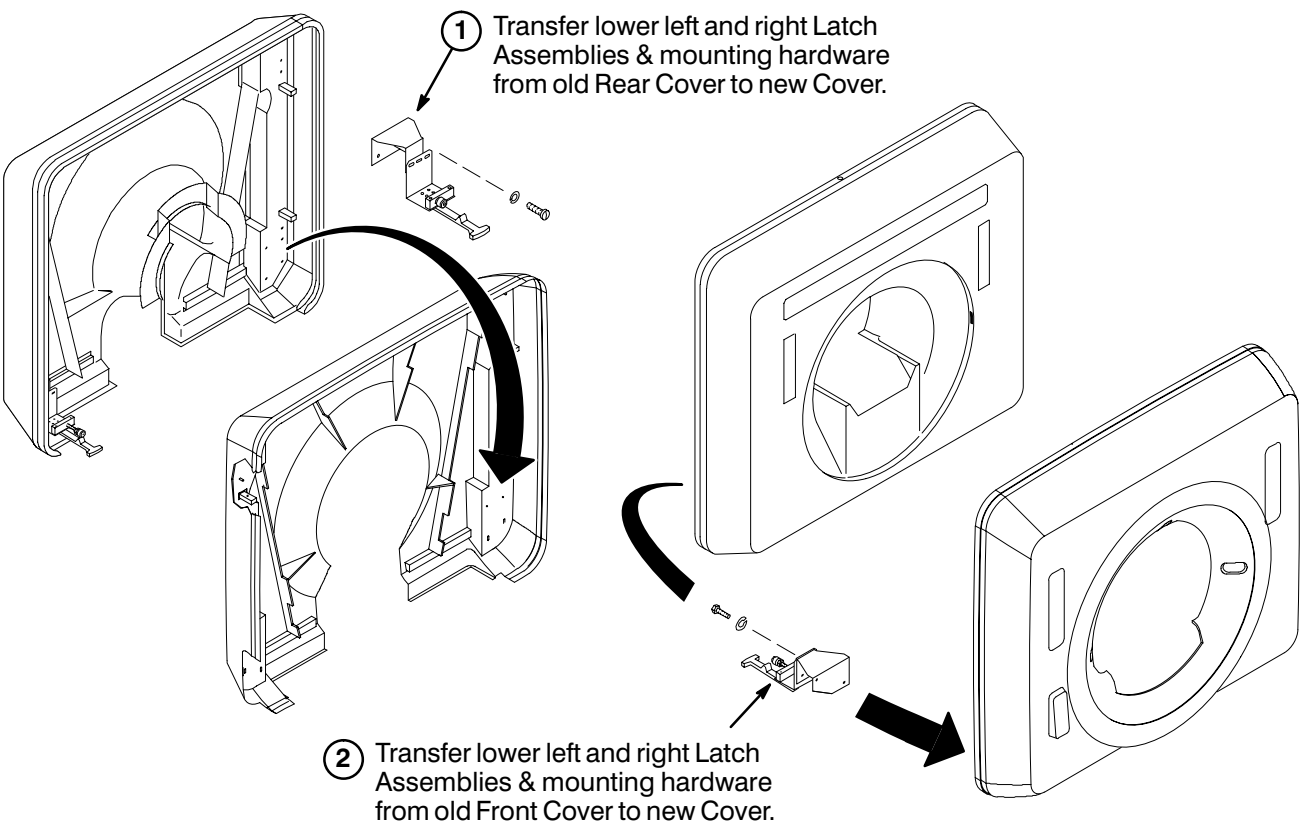


Slide Assembly Mounting Block

❑ N2 – TRANSFER LOWER LATCH ASSEMBLIES TO NEW FRONT & REAR COVERS

❶ Transfer lower left and right Latch Assemblies & mounting hardware from old Rear Cover to new Cover.

❷ Transfer lower left and right Latch Assemblies & mounting hardware from old Front Cover to new Cover.



❑ N3 – ADJUST FRONT COVER ALIGNMENT

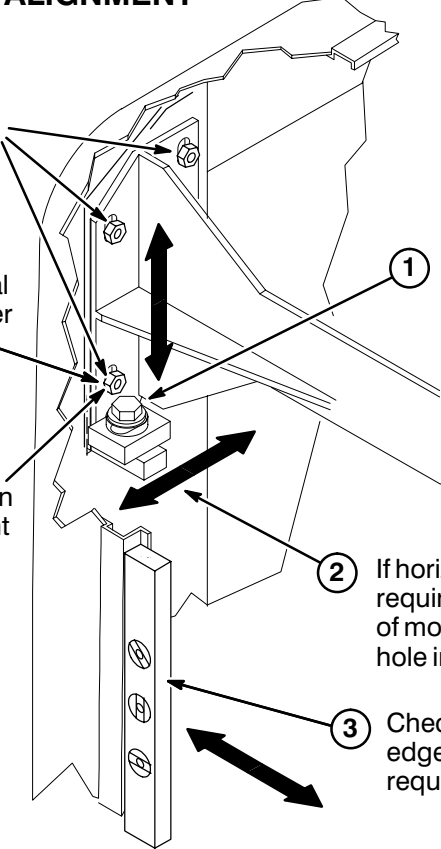
❶ Tighten all hardware when alignment is completed.

❷ Shim at this stud only to obtain vertical alignment. Place shim between Cover and Slide front face. Repeat process for Rear Cover if required.

❸ If vertical alignment is required, loosen hex nut at bottom stud on left and right sides. Cut shims from excess rubber seals:
3/4mm Rubber Seal 2115369
1.5mm Rubber Seal 2115369-2
2.25mm Rubber Seal 2115369-3

❹ If horizontal level adjustment is required, slide cover sideways. Length of movement is determined by slotted hole in Slide Assembly Mounting Block.

❺ Check vertical alignment (plumb) at edge of Front Cover. If alignment is required, proceed to Steps ❹ and ❺.

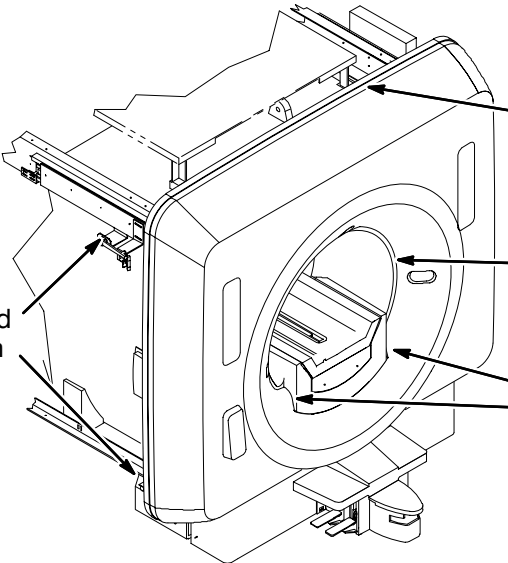


❑ N4 – ALIGN FRONT COVER

❶ Close Front Cover, turn key to lock.

❷ Alignment checks:
Cover should just touch and be centered on End Bell when T-Handle Flex Latches are secured and brackets contact bumpers at each corner.
Lower part of Front Cover opening should be equally spaced around Bridge.

Note: Refer to previous step for alignment adjustment procedure.



T-Handled Flex Latch

❑ N5 – INSTALL REAR COVER (46-320517P1)

THE STEPS ARE THE SAME AS THE JUST COMPLETED FRONT COVER INSTALLATION PROCEDURE
Note: Experience has shown that the Rear Cover does not require shimming for vertical levelness.

DANGER!!

FERROUS MATERIAL HAZARD!! The rear cover dolly was not designed for cover installation with magnet at field. The caster assemblies and fastening hardware are magnetic. If magnet is at field, three installers are required to remove cover from dolly and carry cover into position at rear of magnet.

INSTALLATION STEPS SUMMARY

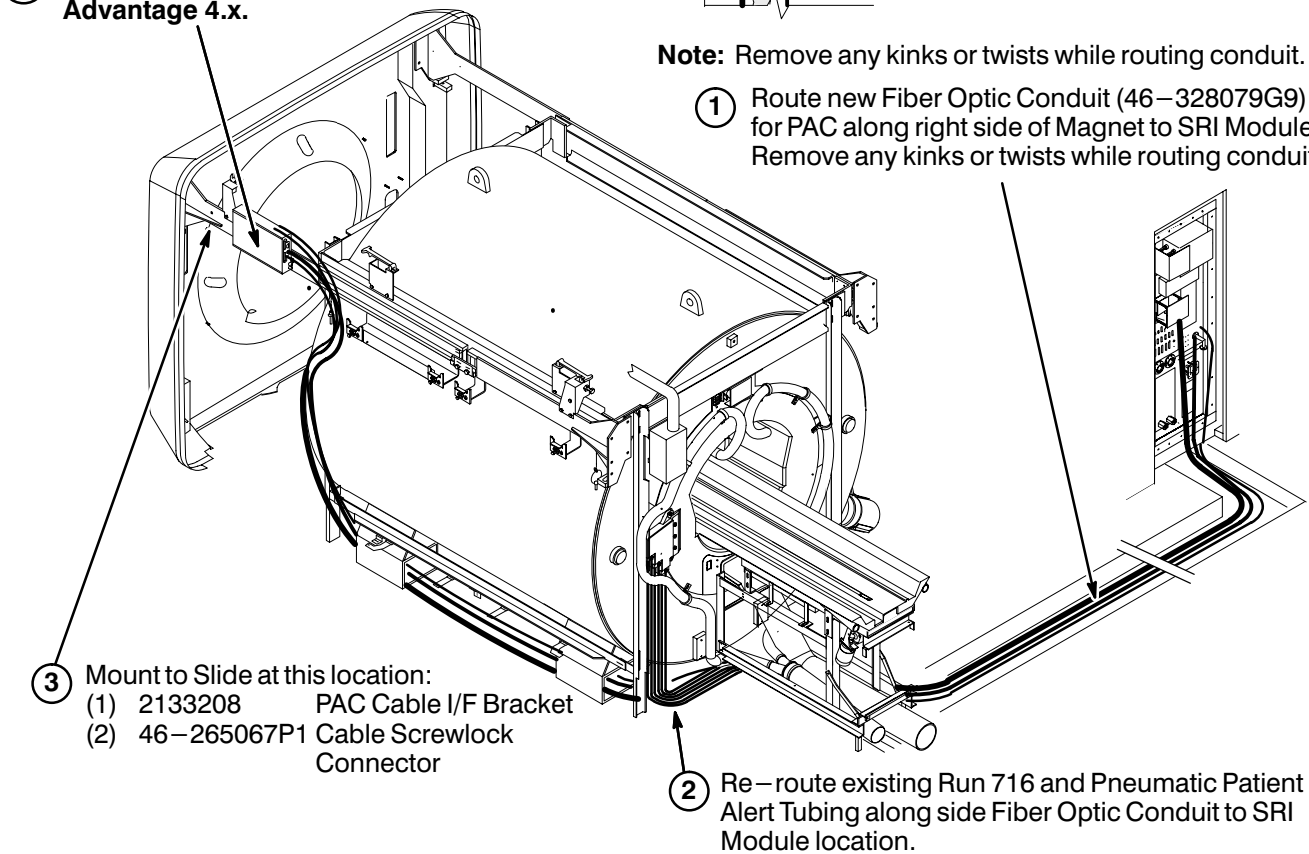
- ☐ P1 – Route/Connect Runs 711/712, 716, and Patient Alert Tubing
- ☐ P2 – Connect Run 715, 749, Front Cover Cables and Install Spiral Wrap
- ☐ P3 – Adjust Spiral Wrapped Cables

☐ P1 – ROUTE/CONNECT RUNS 711/712, 716, AND PATIENT ALERT TUBING

- ⑥ Route Run 716 along side existing cables, over top of SRI Module, thru clamp, to bracket on slide. Connect to PAC II cable (46–224712G20) from Front Cover.
- Note:** A secure electrical ground is required. Insure low resistance connection and tighten fasteners securely.
- ⑦ Route fiber optic cable Run 712 and Patient Alert Tubing to PAC II Asm on Front Cover. **Do not clamp.**
- ⑧ Connect fiber optic cable Run 711 to circuit board as marked. Leave a small loop in fiber optic cable and secure to clamp. (Remove fiber optic cable guides and pass fiber optic cables through if necessary) Close SRI cover and tighten captive screws.
- Note:** Fiber optic cables loops must not be smaller than two inch radius.
- ④A For **Advantage 5.x** SRI Board within module is replaced. Refer to Direction 2294109, Dual Thermal Sensor Upgrade at the end of this Tab. Section 1–13 contains the information for replacing the SRI board within the module.
- OR**
- ④B Install new SRI Module for upgrading from **Advantage 4.x.**
- ⑤ Route fiber optic conduit to within 16 inches of Slide Assembly. Group with existing routed cables.

Note: Remove any kinks or twists while routing conduit.

- ① Route new Fiber Optic Conduit (46–328079G9) for PAC along right side of Magnet to SRI Module. Remove any kinks or twists while routing conduit.

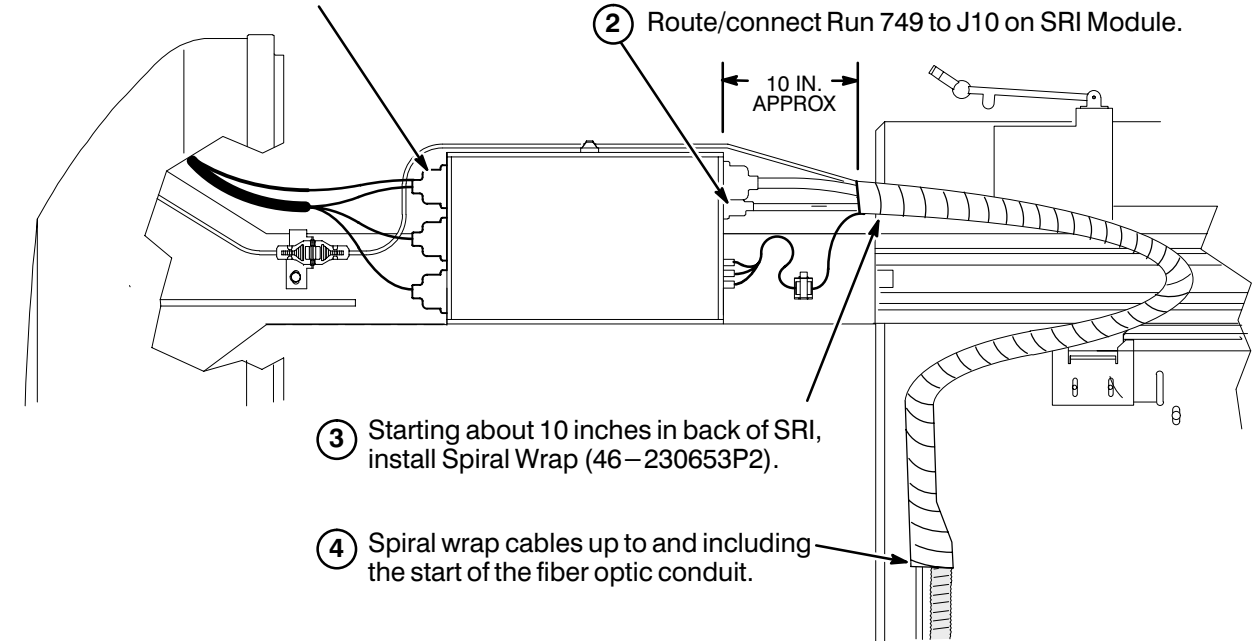


CAUTION

EQUIPMENT DAMAGE POSSIBILITY. Handle fiber optic cables very carefully. Failure to do so may cause intermittent problems difficult to isolate. Do not bend to radius smaller than two inches. Do not pull connector from cable. Avoid scratching connector ends. Keep ends protected until ready to connect.

☐ P2 – CONNECT RUN 715, 749, FRONT COVER CABLES AND INSTALL SPIRAL WRAP

- ① Route ends of six remaining cables from Front Cover to the SRI and connect as marked.
- ② Route/connect Run 749 to J10 on SRI Module.



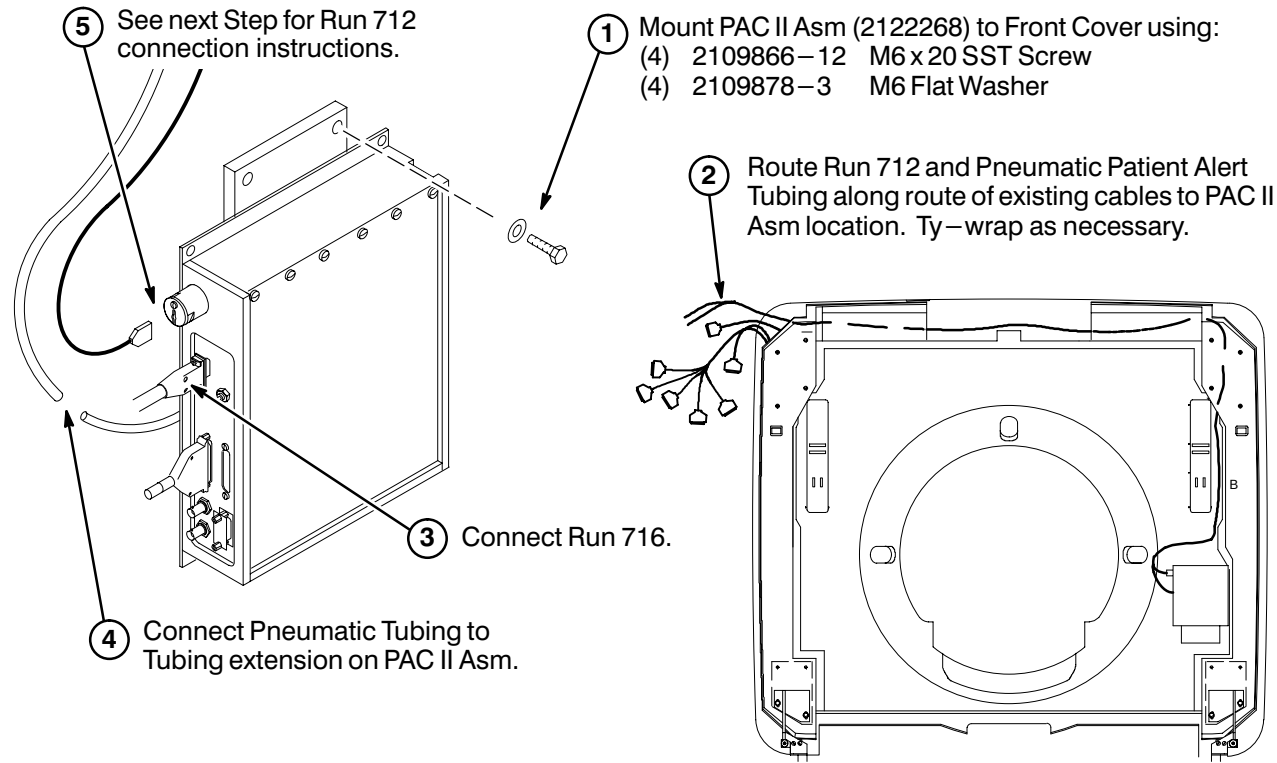
☐ P3 – ADJUST SPIRAL WRAPPED CABLES

- ① Unlock Slides and carefully close Front Cover.
- ② Adjust the position of all spiral wrapped cables as cover is closing.
- ③ Secure spiral wrapped cable bundle using Ty-wraps (46–208758P5)
- ④ Open, close, and again open the Front Cover to check cable movement.

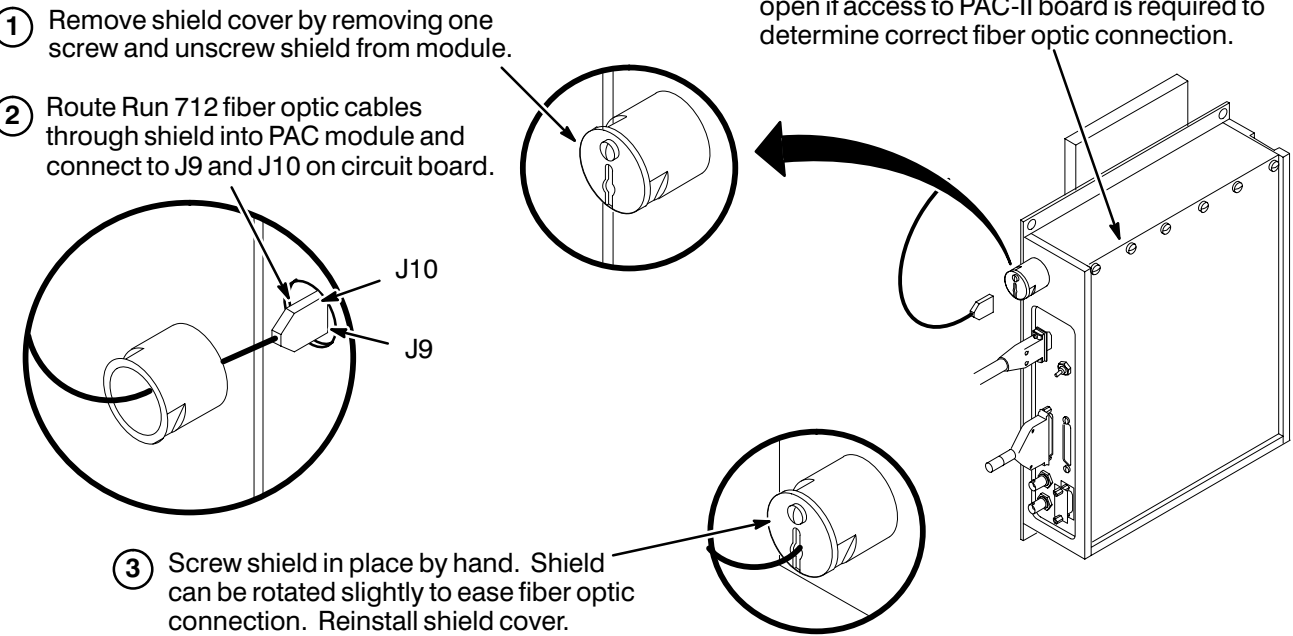
INSTALLATION STEPS SUMMARY

- Q1 – Connect Run 712, 716, and Pneumatic Patient Alert Tubing
- Q2 – Run 712 Connection
- Q3 – Install Drip Shield Plate Kit (2143475), If Supplied

Q1 – CONNECT RUN 712, 716, & PNEUMATIC PATIENT ALERT TUBING

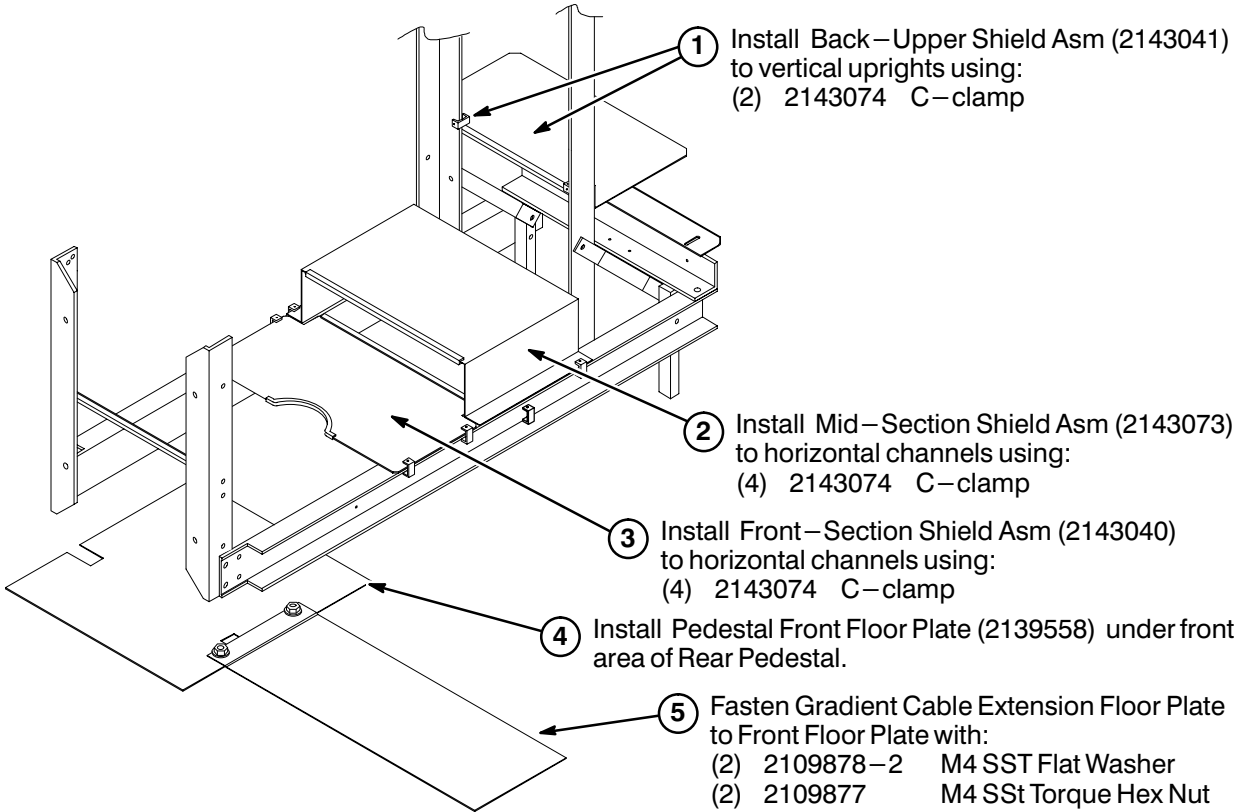


Q2 – RUN 712 CONNECTION



Q3 – INSTALL DRIP SHIELD PLATE KIT (2143475), IF SUPPLIED

Note: Drip Shield Plate Kit installation is required to meet UL requirements.



INSTALLATION STEPS SUMMARY

- ❑ R1 – Install Rear Pedestal Covers (2131492-3)
- ❑ R2 – Install Enclosure Side Covers
- ❑ R3 – Install PPG Cable Hooks & Rating Plate on Front Cover

❑ R1 – INSTALL REAR PEDESTAL COVERS (2131492-3)

1 Remove Rear Trim Cover from room and cut opening to height indicated. Remove sharp edges.

2 Re-install Base Trim Covers with saved hardware.

3 Install Left (2131044) and Right (2131045) Side Covers using:

- (4) 2109878-3 M6 SST Flat Washer
- (4) 2109880-3 M6 SST Lock Washer
- (4) 2109866-13 M6 x 25 SST Hex Screw

4 Re-install Rear Pedestal Back Cover Asm saved in A1-Step 1 to Rear Pedestal with previously removed hardware.

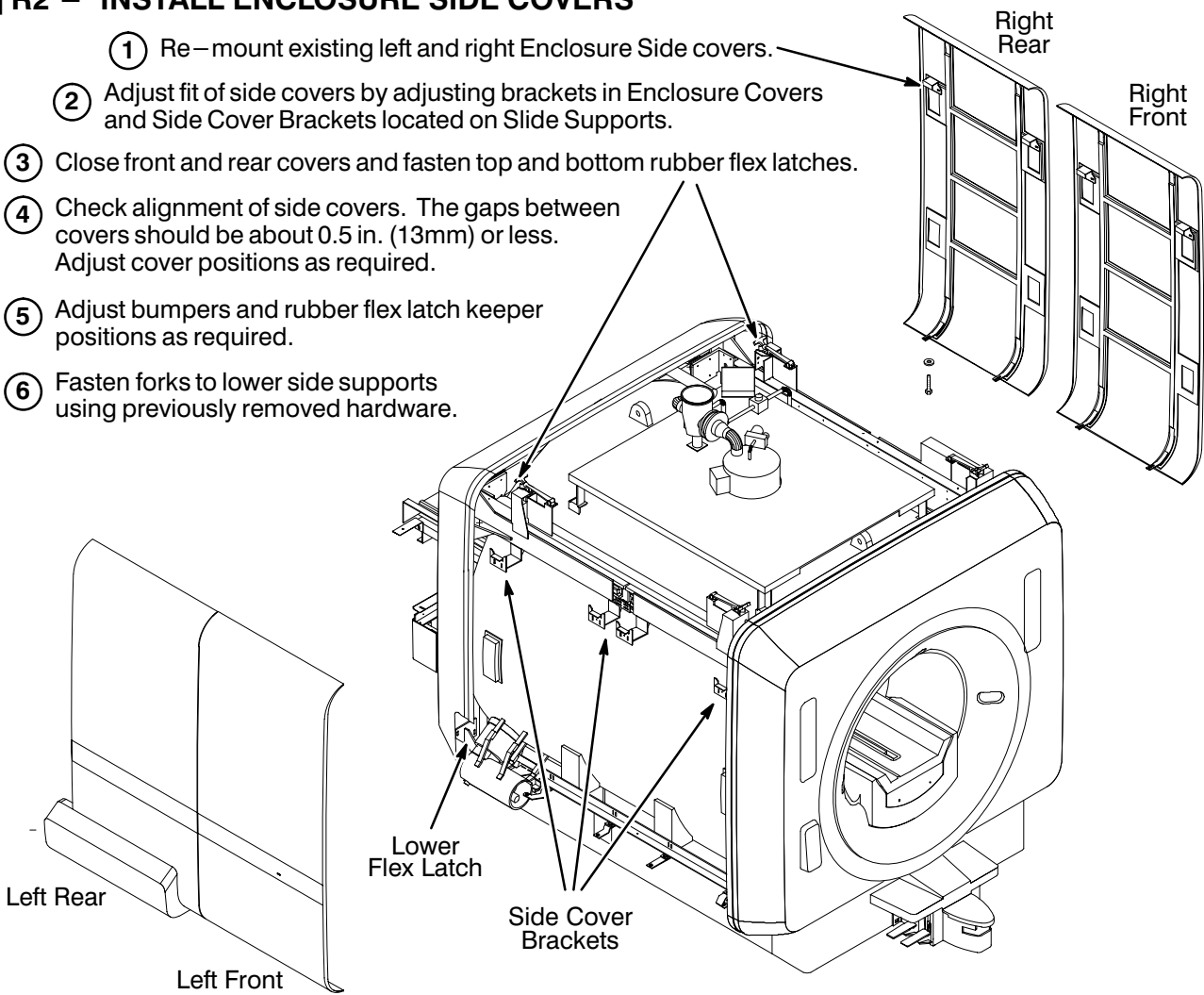
5 Attach Left and Right Rear Pedestal Side Covers with previously remove hardware.

6 Check fit and adjust as required.

7.50 IN (190mm)

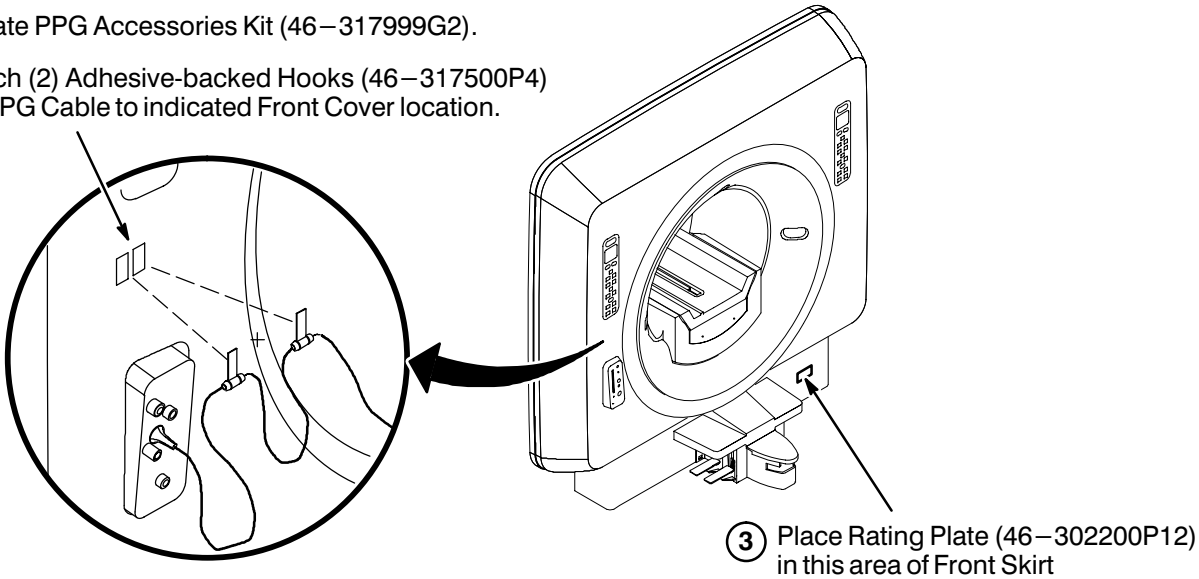
❑ R2 – INSTALL ENCLOSURE SIDE COVERS

- 1 Re-mount existing left and right Enclosure Side covers.
- 2 Adjust fit of side covers by adjusting brackets in Enclosure Covers and Side Cover Brackets located on Slide Supports.
- 3 Close front and rear covers and fasten top and bottom rubber flex latches.
- 4 Check alignment of side covers. The gaps between covers should be about 0.5 in. (13mm) or less. Adjust cover positions as required.
- 5 Adjust bumpers and rubber flex latch keeper positions as required.
- 6 Fasten forks to lower side supports using previously removed hardware.



❑ R3 – INSTALL PPG CABLE HOOKS & RATING PLATE ON FRONT COVER

- 1 Locate PPG Accessories Kit (46-317999G2).
- 2 Attach (2) Adhesive-backed Hooks (46-317500P4) for PPG Cable to indicated Front Cover location.



DUAL TEMPERATURE SENSOR UPGRADE

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1-1 INTRODUCTION

The Thermal Sensor Upgrade, #2287391, improves bore temperature sensing reliability by adding a second temperature sensor.

This update applies to the Signa LX (8.x) magnet enclosures only. The following illustrations show an S3 magnet. The instructions apply to any LX magnet enclosure

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!

1-2 TOOLS AND EQUIPMENT

The following tools and equipment are required to perform this procedure:

- Drill bit, 7 mm or 5/16 inch
- Drill
- Dykes
- Digital Multimeter
- Lockout/Tagout equipment
- Tape, electrical or duct
- Permanent marker, fine-tip
- Slot-head screwdriver, 5/16" blade
- Nut driver, 5/16"
- Static-grounding wrist strap
- Copper tape (for replacement only)

The following items are provided in the Thermal Sensor Upgrade Kit, P/N 2287391

- 2285729 Run 749, Dual-head sensor cable
- 46-288016G9 SRI board with dual-sensor daughter board
- 46-208747P1 Cable ties
- 46-208747P2
- 46-282264P1 Epoxy adhesive
- 2298033 Applicator
- 46-320187P1 Nozzle

Manpower Requirements: Two or three people are required to place the board under the bridge and to remove the end bell.

1-3 POWER DOWN PROCEDURE

If complete system is required to be powered down, refer to OVERVIEW Tab, Section 2, SYSTEM POWER SHUTDOWN.

1. Press the **[In Fast]** button on the magnet enclosure to move the carriage all the way to the rear of the magnet.

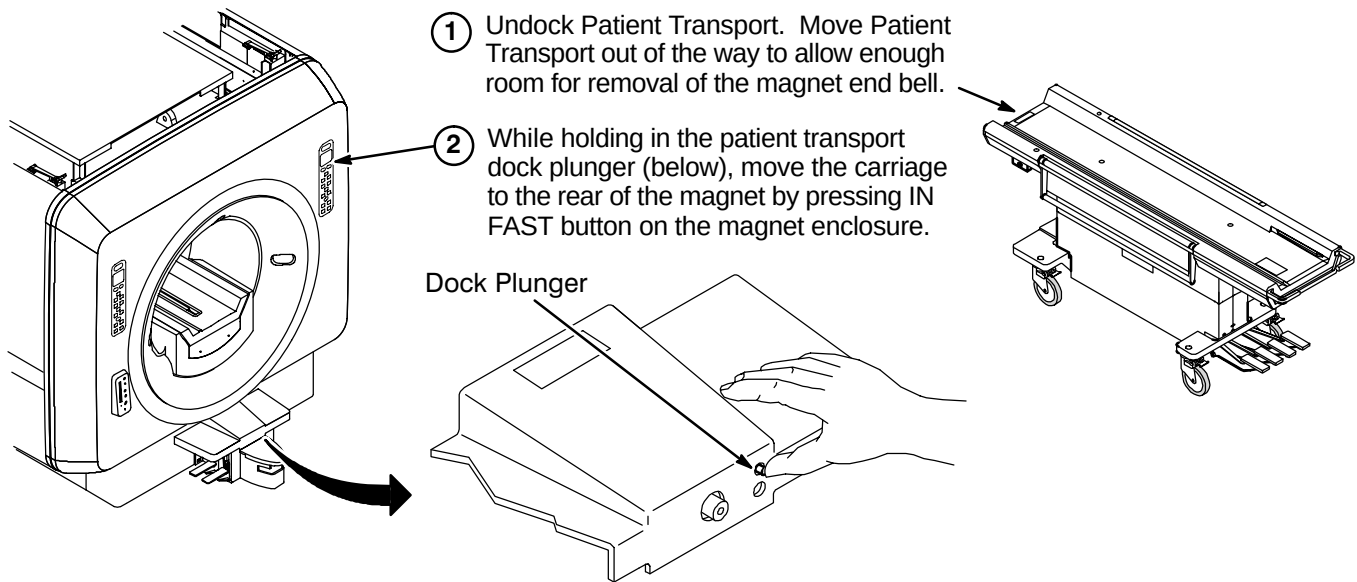
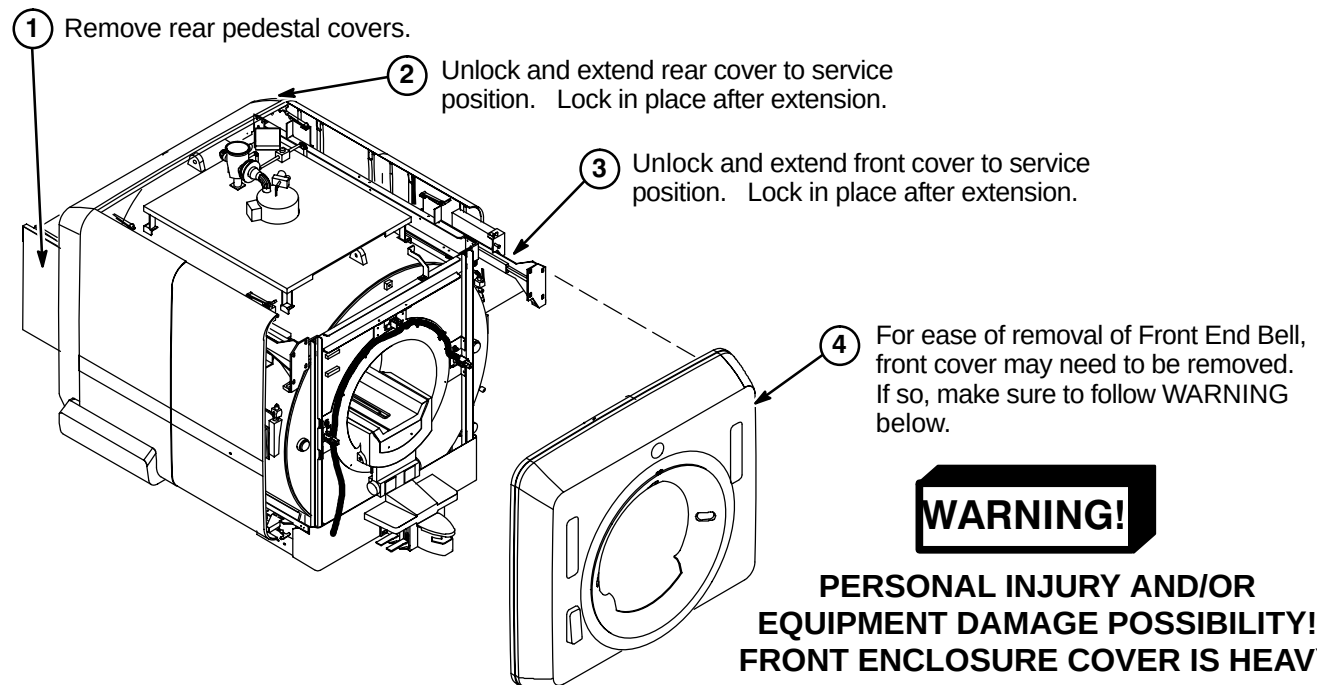


2. Click the  (Service Tools) icon. From the Service Desktop Manager, click , then . The system will begin the shutdown process. After about 30 seconds, the CPU will shut down.

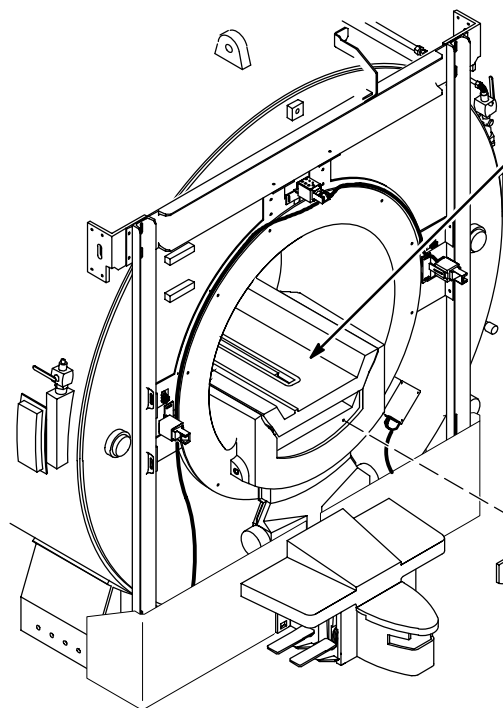
3. On the system PC, click , **[Shut Down...]**, **[Yes]**. The PC will shut down.
4. For RF/Pen Cabinet: Lock out and tag out circuit breaker MR1 A14 CB6 at rear of RF/Pen Cabinet (MR1) on magnet enclosure power supply (MEPS) module.
 - Verify that voltage between MR1 A14 TP1 (+) and TP14 (–) is 0 Vdc.
 - Verify that voltage between MR1 A14 TP4 (+) and TP13 (–) is 0 Vdc.

For RF/Pen II and RF/PDU Cabinet: Lock out and tag out circuit breaker MR1 A20 CB4 at rear of RF/Pen II and RF/PDU Cabinet on System Support Module (SSM).

- Verify that voltage between MR1 A20 TP4 (+) (SRI+24V) and TP13 (–) (SRI RTN) is 0 Vdc.
- Verify voltage between MR1 A20 TP1 (+) (Dock) and TP14 (–) (Dock RTN) is 0 Vdc.

1-4 TABLE REMOVAL AND CARRIAGE POSITIONING**1-5 REMOVAL OF ENCLOSURE COVERS****WARNING!**

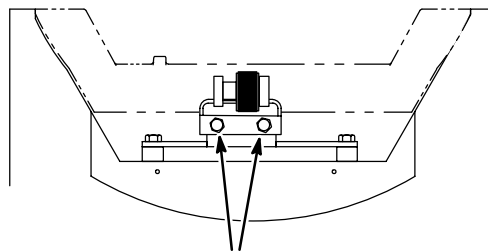
PERSONAL INJURY AND/OR EQUIPMENT DAMAGE POSSIBILITY! FRONT ENCLOSURE COVER IS HEAVY AND CUMBERSOME. AT LEAST TWO PEOPLE ARE REQUIRED TO LIFT AND MOVE IT; THREE IS PREFERABLE

1-6 REMOVAL OF FRONT END BELL PANEL

③ For removal of Front End Bell, Bridge must be moved out of the way either by lifting the front end or complete removal.

Refer to Section for procedure to lift the Bridge.

Refer to Section for removal of Bridge.



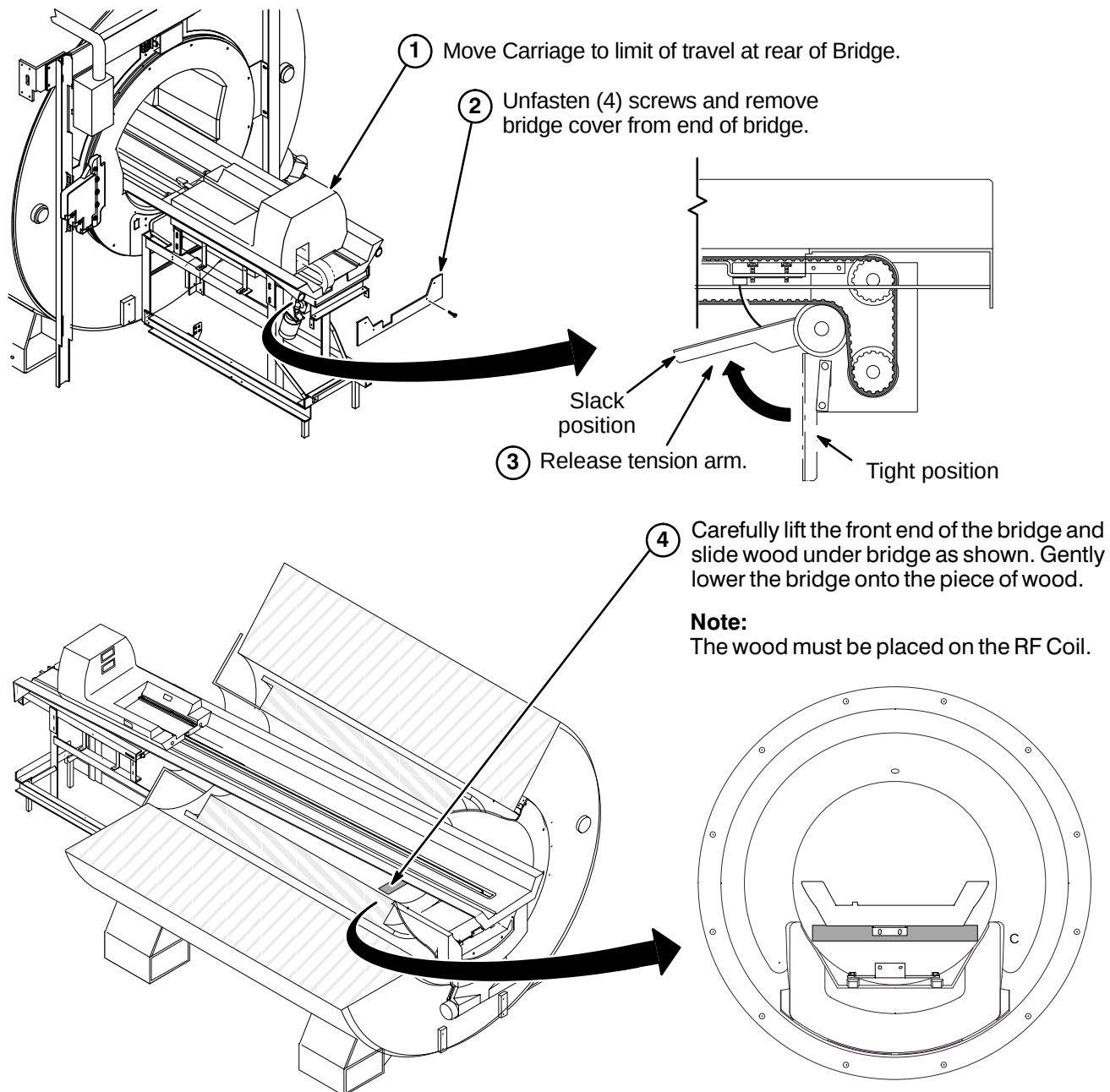
② Unfasten M10 flat washers, lock washers, and screws attaching front end of bridge asm to front end bell.

① Unfasten (2) M6 x 25 pan screws and remove end bell panel.

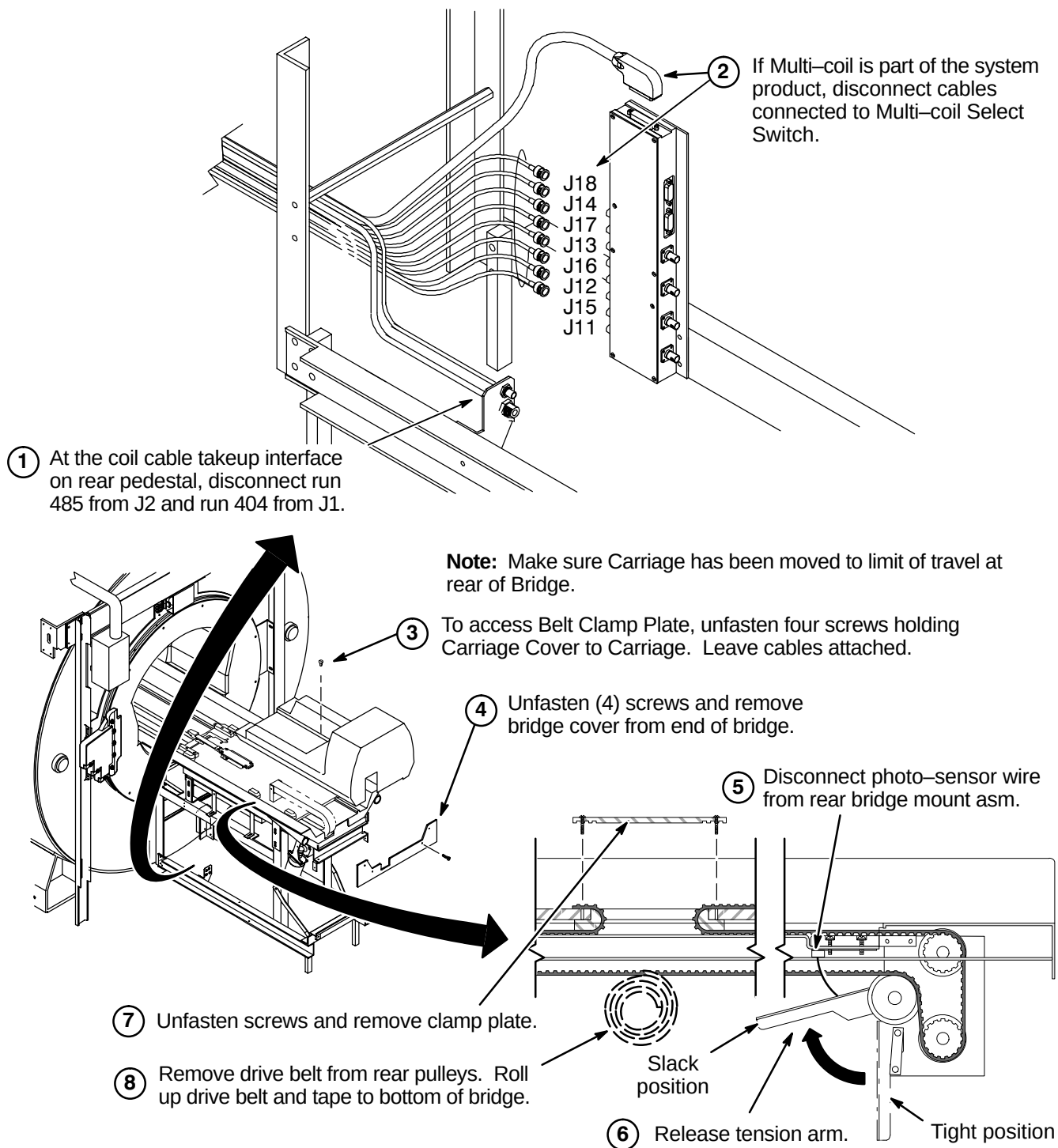
1-7 BRIDGE HANDLING OPTIONS PRIOR TO REMOVAL OF FRONT END BELL

The Bridge needs to be repositioned or removed prior to removal of Front End Bell. Review sections 1-7-1 and 1-7-2. Determine which method would best fit the needs of your site.

1-7-1 Lifting Front End Of Bridge

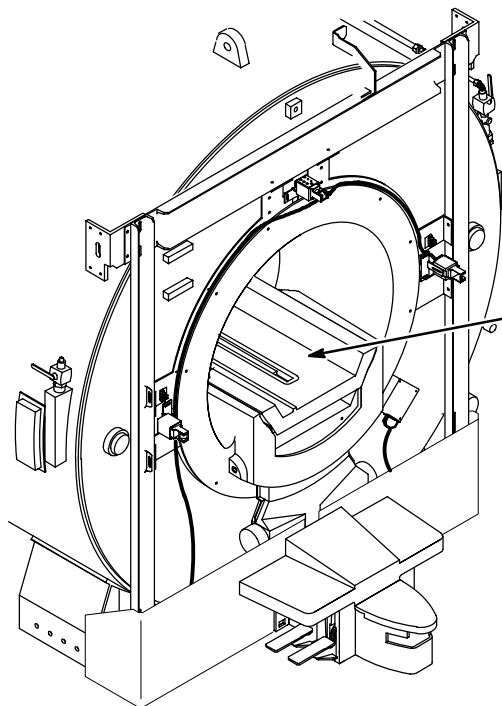


1-7-2 Removal of Bridge



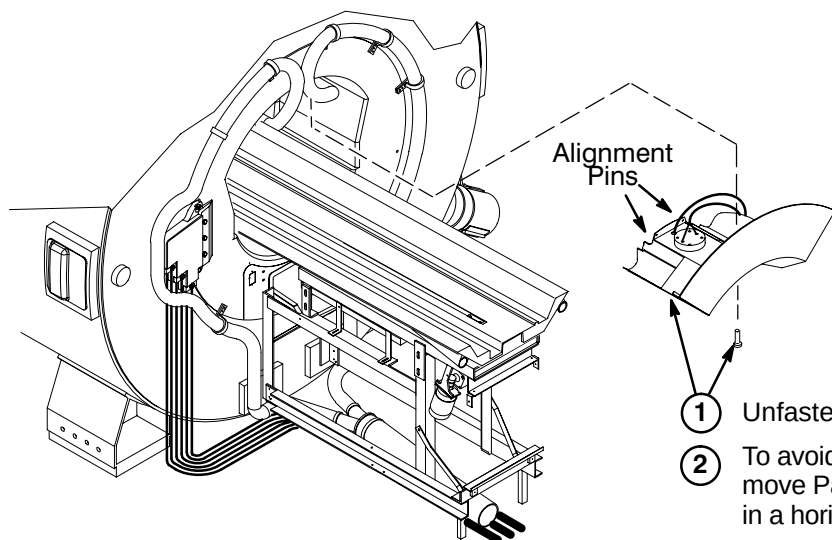
1-7-2 Removal of Bridge (continued)**CAUTION**

Possible equipment damage! Be careful not to damage the home flag on the drive belt when removing or replacing the bridge. The home flag is fragile and can easily be damaged.



- 1 To remove bridge, one person holds the front end, lifts slightly, and carefully pulls the bridge forward. The other person stands in front of the magnet next to the bridge to guide the bridge asm through the bore. Place bridge aside in a safe area to prevent damage.

L

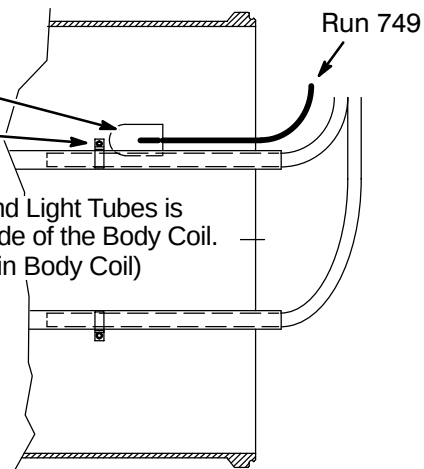
1-8 REMOVE EXISTING RUN 749, TEMPERATURE SENSOR CABLE**Note:**

Hold Patient Comfort Module in place until mounting screws in Step 1 are removed. Alignment Pins can break off during removal if over-stressed.

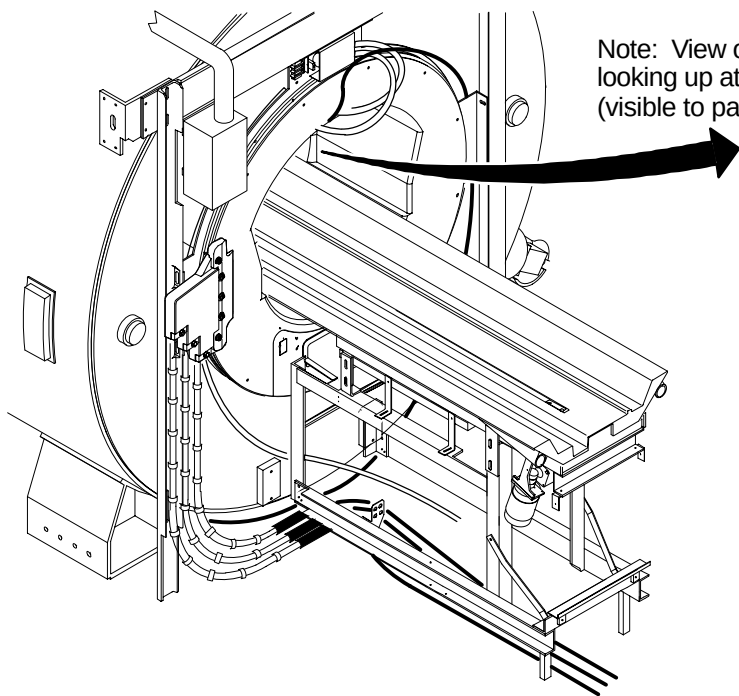
- ① Unfasten two M6 x 25 SST Pan Head Screws.
- ② To avoid breaking alignment pins during removal, move Patient Comfort Module away from End Bell in a horizontal path.

- ③ Unfasten existing Temperature Sensor Cable (Run 749) from top inside of Body Coil. It may be attached with:
adhesive fastener (as shown)
OR
with the Light and Cable clamp

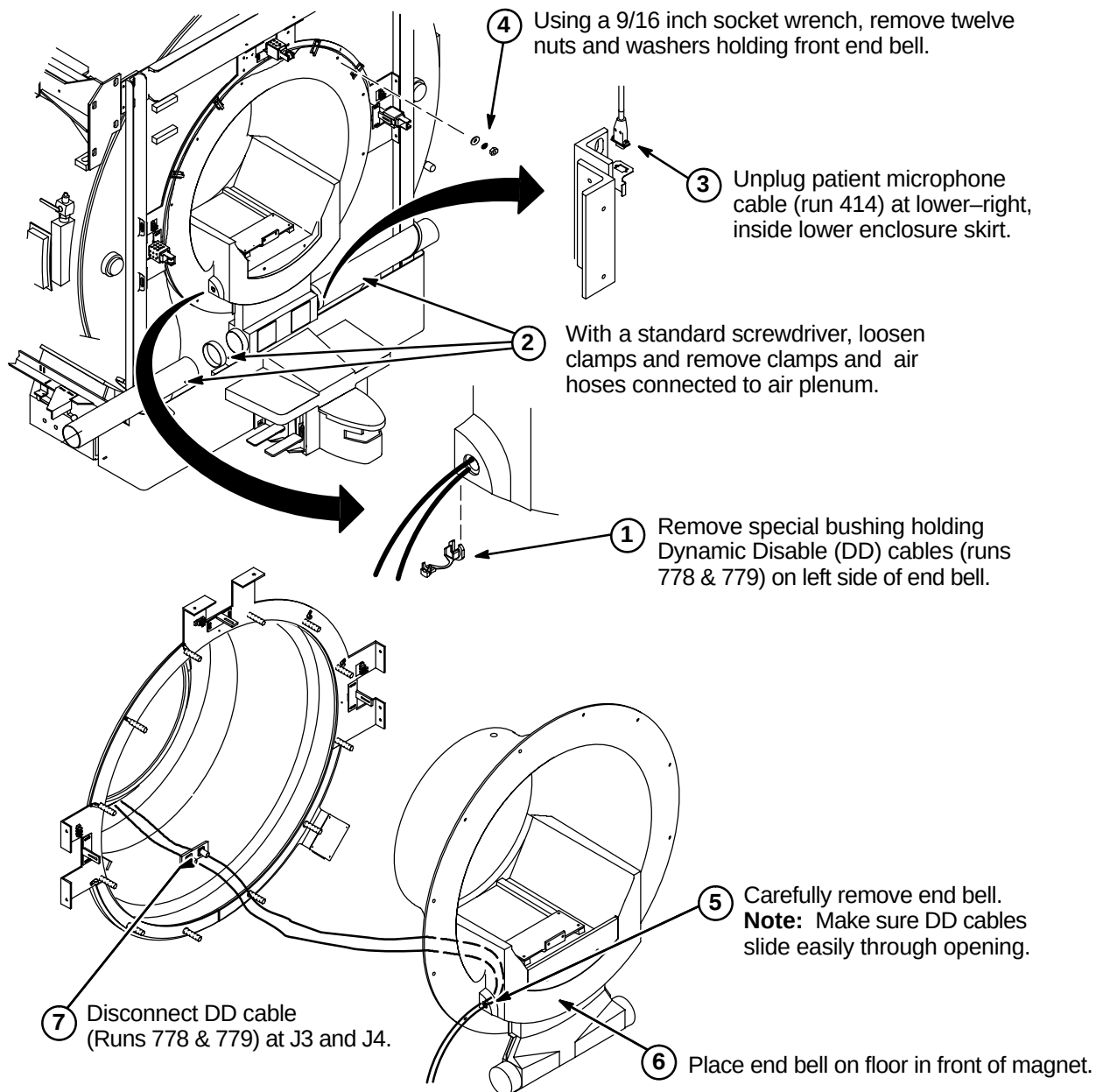
Note: View of Sensor and Light Tubes is looking up at the top inside of the Body Coil. (visible to patient laying in Body Coil)



- ④ Disconnect other end of Run 749 from J10 on SRI module.
- ⑤ Remove and discard cable.



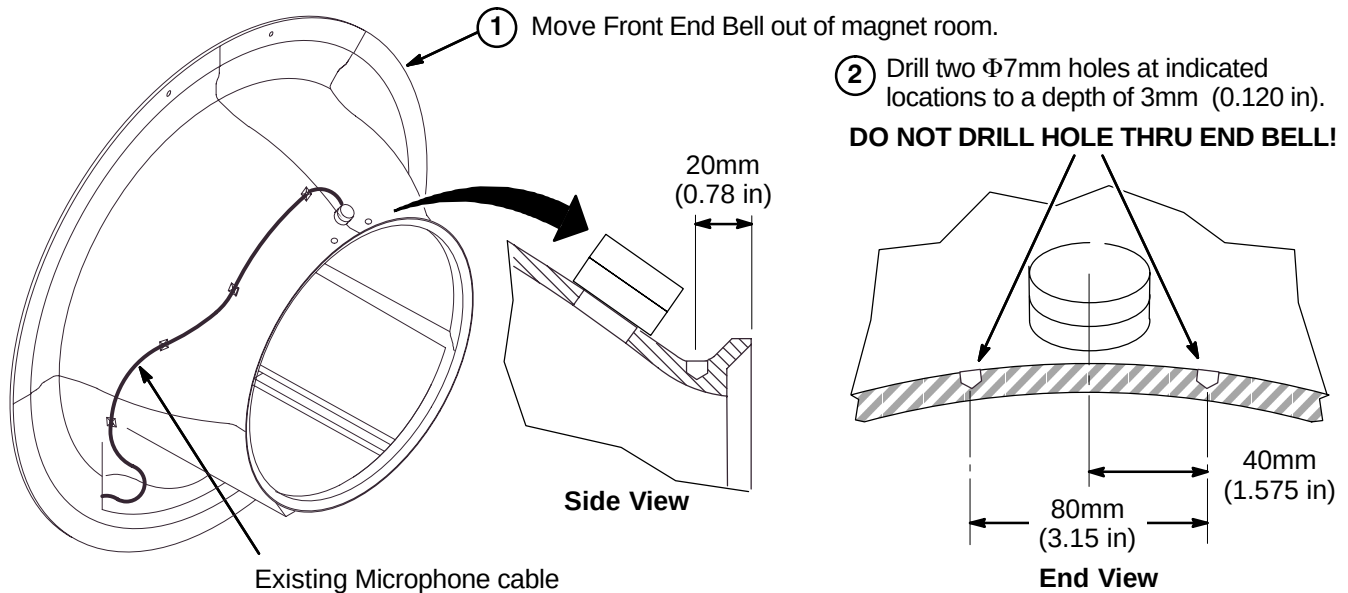
1-9 FRONT END BELL REMOVAL



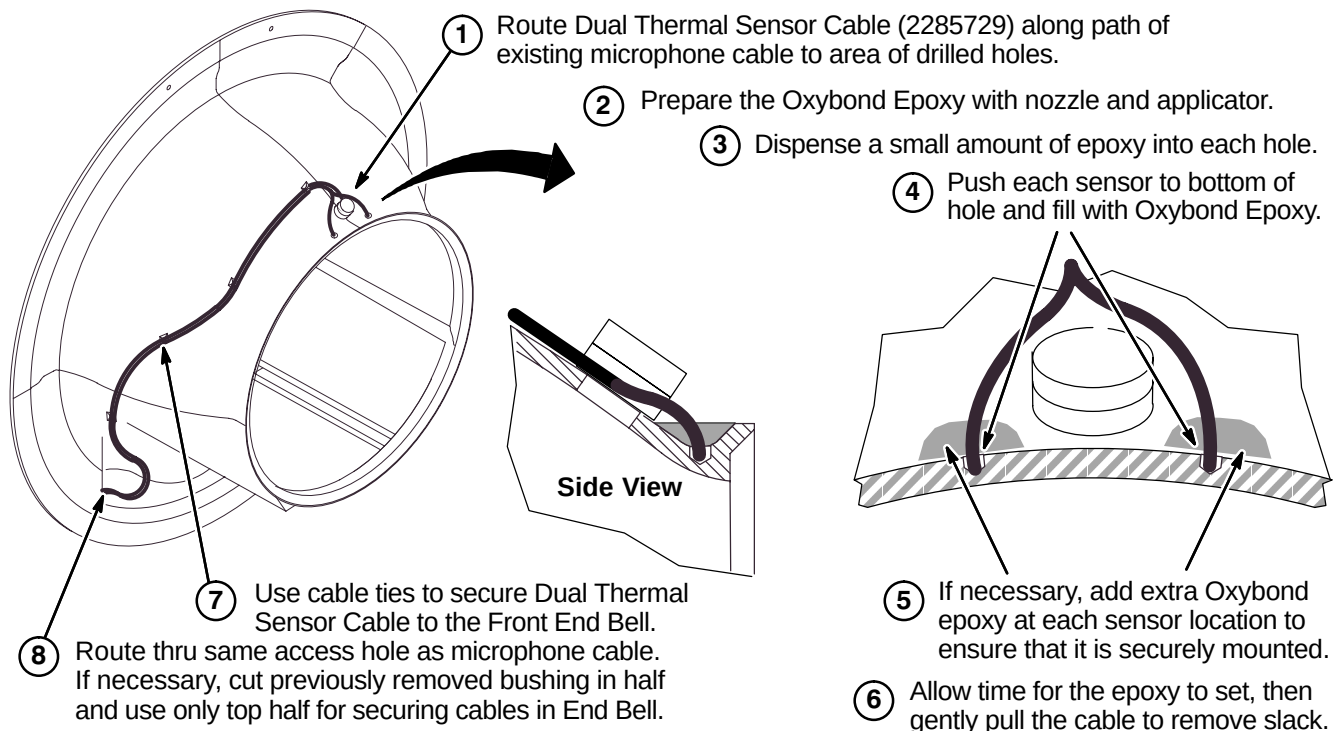
1-10 PREPARATION OF END BELL FOR INSTALLATION OF DUAL THERMAL SENSOR CABLE

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!



1-11 INSTALL DUAL THERMAL SENSOR CABLE



1-12 REASSEMBLY OF MAGNET ENCLOSURE**1-12-1 Front End Bell Replacement**

1. Reverse the Front End Bell Removal procedure (Section 1-9) to reinstall the end bell.

1-12-2 Bridge Replacement

1. Depending on which method was used, Lifting the Front of Bridge (Section 1-7-1) or Removal of Bridge (Section 1-7-2), reverse the procedure to reinstall bridge.
2. Make sure the tension arm is moved to the "Tight Position" at completion.
3. Install covers on rear pedestal.

1-13 MODIFYING THE SRI MODULE

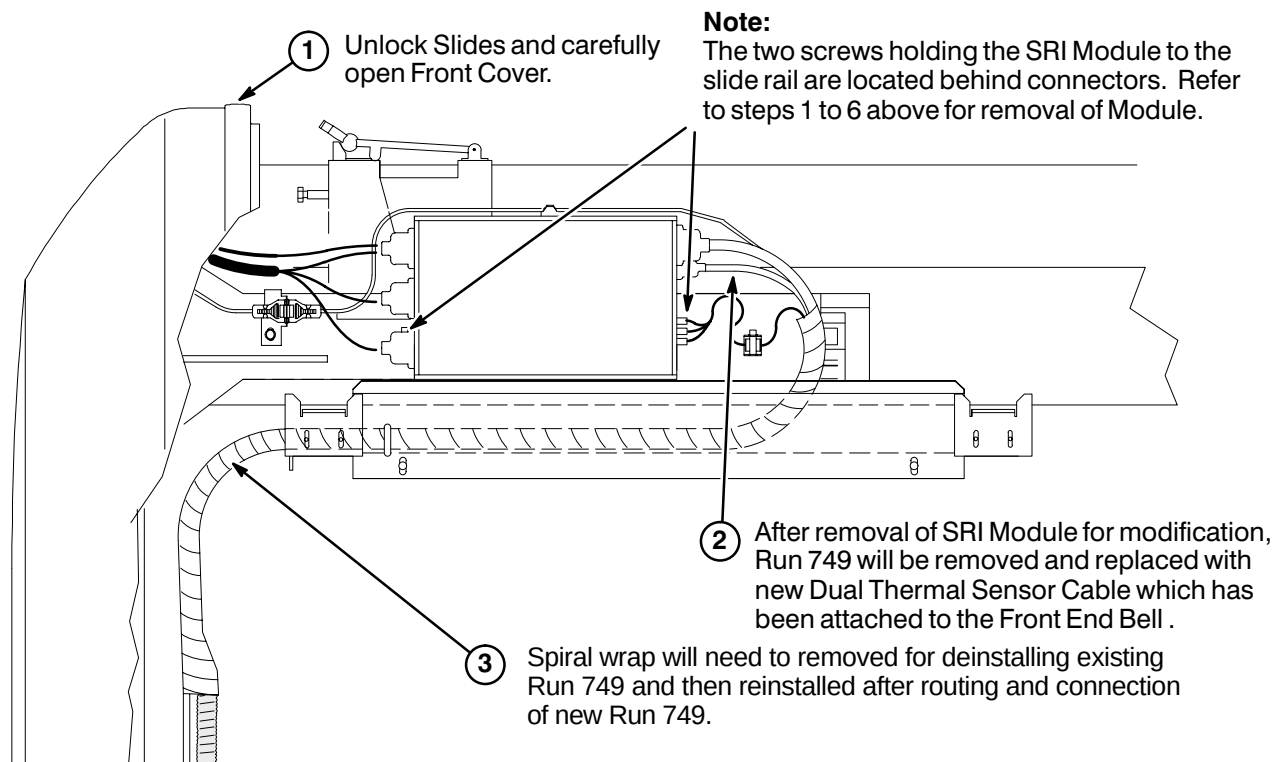
This section contains instructions for replacing the circuit board in the SRI module. The SRI module must be removed from the magnet room before replacing the circuit board. Refer to illustrations 1-1 and 1-2 for locations of SRI Module on standard and Cx magnets.

1-13-1 Removing the SRI Module

1. If copper tape is present on the SRI module cover, note all the locations where the tape was applied, then remove the tape.
2. Loosen the six captive screws securing the top cover of the SRI module enclosure, then slide the cover off.
3. Carefully release the three fiber optic cables from their connectors on the SRI PCB and pull the cables free from the SRI module.
4. Disconnect the remaining cables attached to the SRI.
5. Remove two screws holding SRI to rail or bracket. See Illustrations .
6. Lift SRI off tray and remove.



Ferrous material present! Remove the SRI module from the magnet room before replacing the SRI module circuit board.



SRI MODULE LOCATION FOR STANDARD MAGNET ENCLOSURES

ILLUSTRATION 1-1

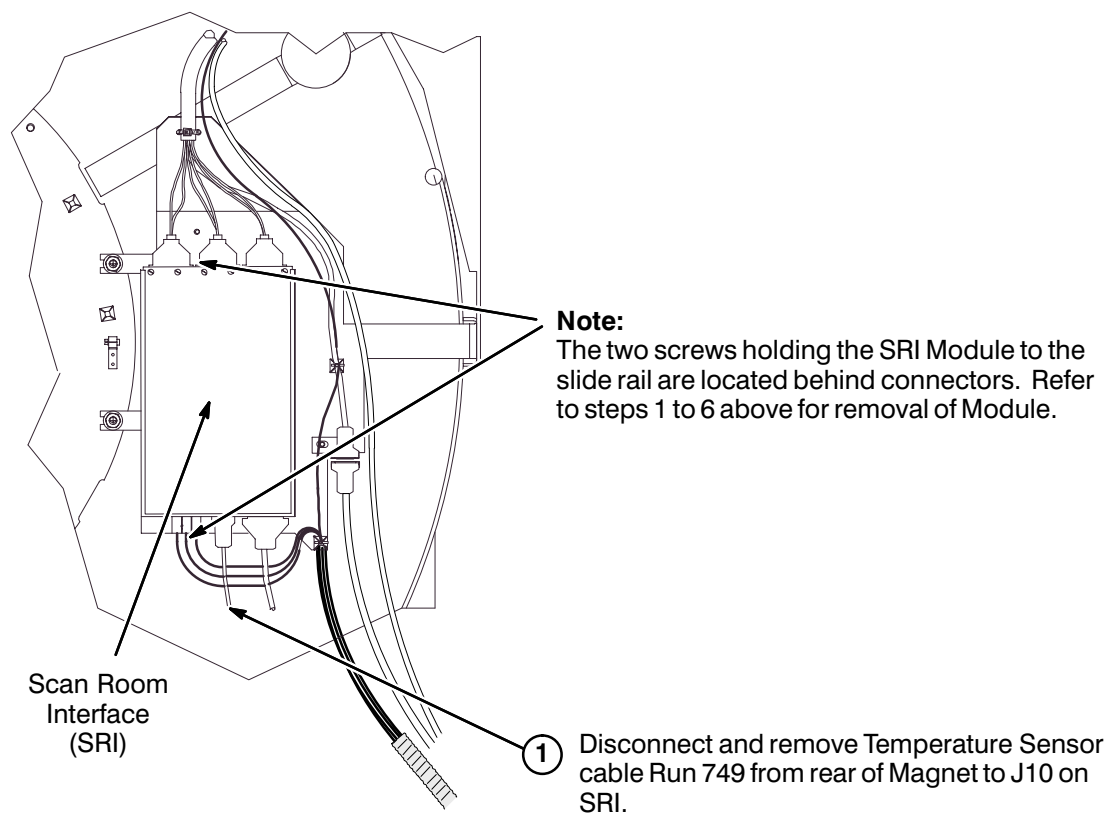
1-13-1 Removing the SRI Module (continued)**SRI MODULE LOCATION FOR CX MAGNET ENCLOSURES**

ILLUSTRATION 1-2

7. Replace the existing SRI module circuit board with the replacement circuit board. Refer to Section 1-13-2 "Replacing the SRI Module Circuit Board" in this procedure.

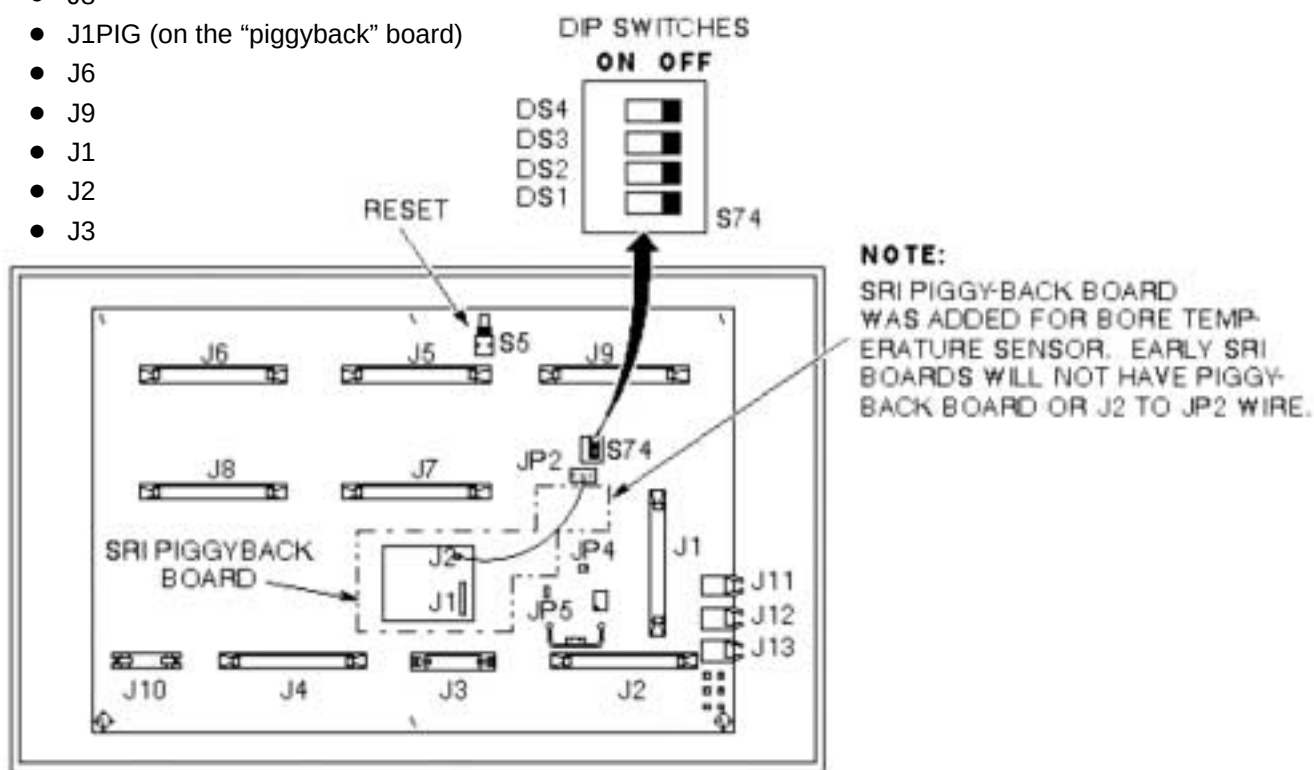
1-13-2 Replacing the SRI Module Circuit Board

CAUTION

Potential equipment damage! Wear a static-grounding wrist strap to prevent ESD (electrostatic discharge) damage to the circuit board.

1. Disconnect all ribbon cables from the SRI circuit board in the following sequence:

- J5
- J7
- J4
- J8
- J1PIG (on the "piggyback" board)
- J6
- J9
- J1
- J2
- J3



SRI BOARD CONNECTION LOCATIONS

ILLUSTRATION 4-1

2. Make sure the free ends of the ribbon cables will not interfere with the removal of the circuit board. If necessary, use masking tape to temporarily secure the free ends of the cables to the outside of the SRI module enclosure.
3. Remove six hex nuts and six washers securing the SRI circuit board to the SRI module enclosure mounting studs.
4. Carefully lift the board off the studs (w/ standoffs) and remove it from the SRI module enclosure. Place the board on an anti-static mat. Do not remove the fish paper underneath the board.
5. Position the replacement board over the six mounting studs.
6. Secure the board using the six hex nuts and six washers removed in step 3
7. Connect the ribbon cables to the board in the reverse sequence listed in step 1.

1-13-2 Replacing the SRI Module Circuit Board (Cont'd)

8. Refer to Tables 4-1 and 4-2 to verify proper jumper and switch settings on replacement SRI board.

TABLE 4-1
SRI BOARD SWITCH INFORMATION

Switch	Schematic Location	Position	Description	Notes
S5	2-B4	Open	BOARD RESET	
S74-SW1	2-H9	Open	(TEST USE)	
S74-SW2	2-H9	Open	(TEST USE)	
S74-SW3	2-H9	Open	(TEST USE)	
S74-SW4	2-H9	Open	(TEST USE)	

TABLE 4-2
SRI BOARD JUMPER INFORMATION

Jumper	Schem. Location	Position	Description	Notes
JP2	2-F6	B	POLARITY SOURCE (SRI without Piggy-back board)	Selects source for LCDPOLAR signal. Position A selects the 8OC196 as source; position B selects off-board source.
JP2	2-C5	Default-Installed	XTAL TEST	Provided for testability
JP5	2-E9	Default-Installed	MEMORY SELECT	Configures memory accessibility. Jumper not installed allows stand-alone diagnostics shell to be accessed. Jumper installed allows normal operation.

1-13-3 Installing The Modified SRI Module

1. Place new SRI module on rail or bracket.
2. Secure SRI module to tray using two mounting screws.
3. Connect all previously removed cables to SRI module.

Note

The replacement SRI circuit board may have small protective plugs in the fiber optic connectors. If so, remove the plugs before connecting the fiber optic cables.

4. Replace the SRI module cover and secure it with the captive screws.
5. If copper tape was removed from the SRI module for this procedure, replace it with new copper tape in the same locations.
6. Close front cover and replace enclosure side panels.

1-14 RESTORING POWER

1. Refer to Overview, Section 5, POWER ON, for procedures on restoring power.

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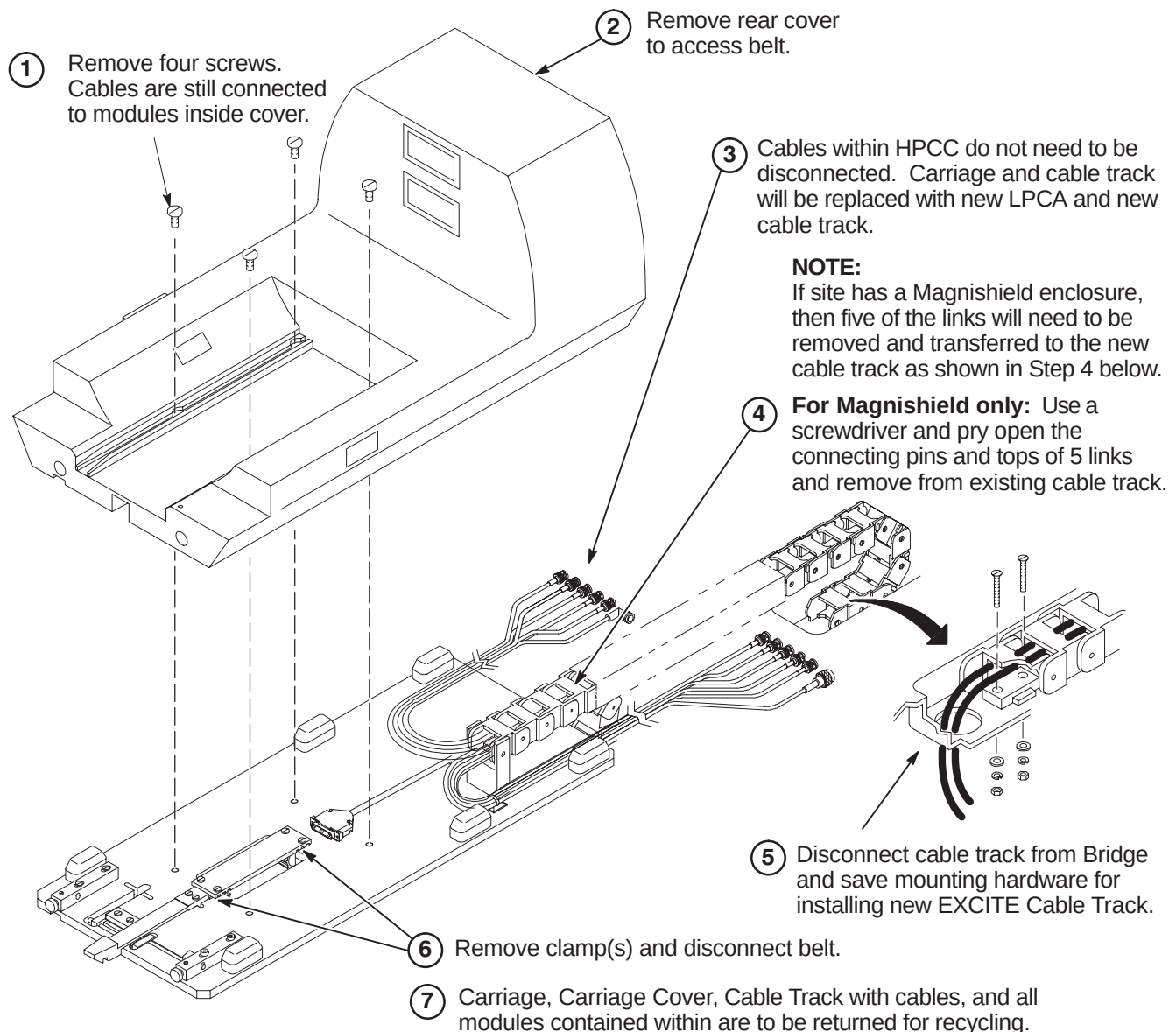
1-1 LPCA (LOW PROFILE CARRIAGE ASSEMBLY)

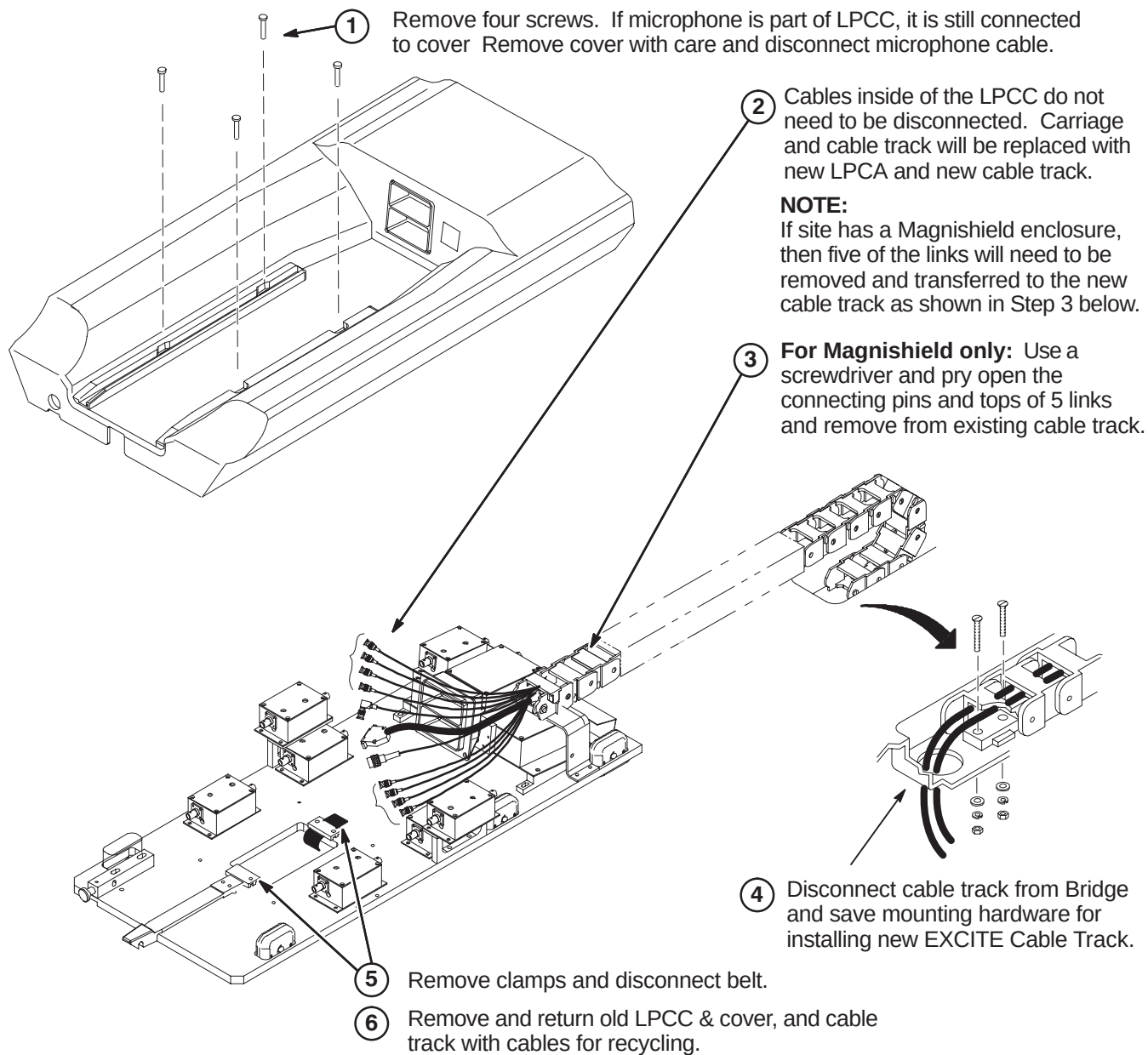
The site being upgraded may have either a High Profile Carriage Cover or a Low Profile Carriage Cover that will need to be replaced with the new LPCA.

If site has a Magnishield magnet, five of the links from the existing cable track will need to be reused until the upgrade kit is revised to include a 15 link extension rather than 10 link. The instructions for upgrading the existing cable track are found at Procedure D2a.

Refer to sections 1-1-1 or 1-1-2 for the applicable removal procedure.

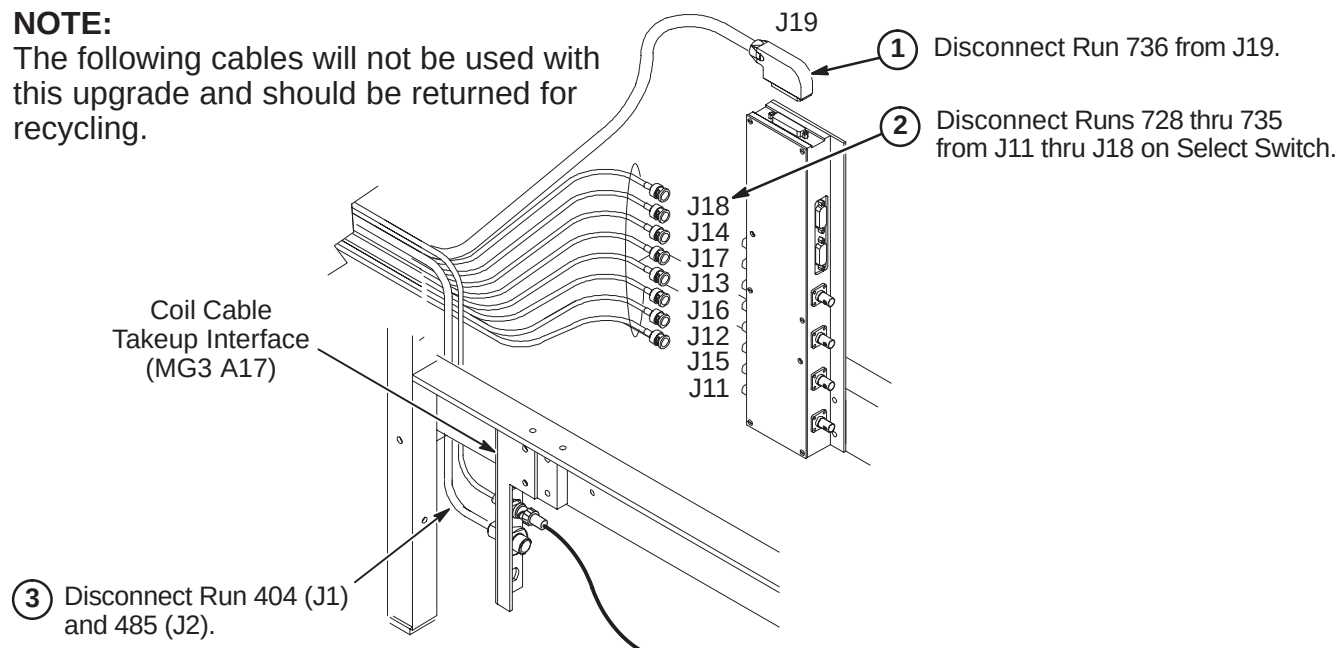
1-1-1 REMOVE EXISTING HIGH PROFILE CARRIAGE COVER & CABLE TRACK



1-1-2 REMOVE EXISTING LOW PROFILE CARRIAGE COVER & CABLE TRACK

1-1-3 DISCONNECT CARRIAGE COVER CABLES IN REAR PEDESTAL**NOTE:**

The following cables will not be used with this upgrade and should be returned for recycling.

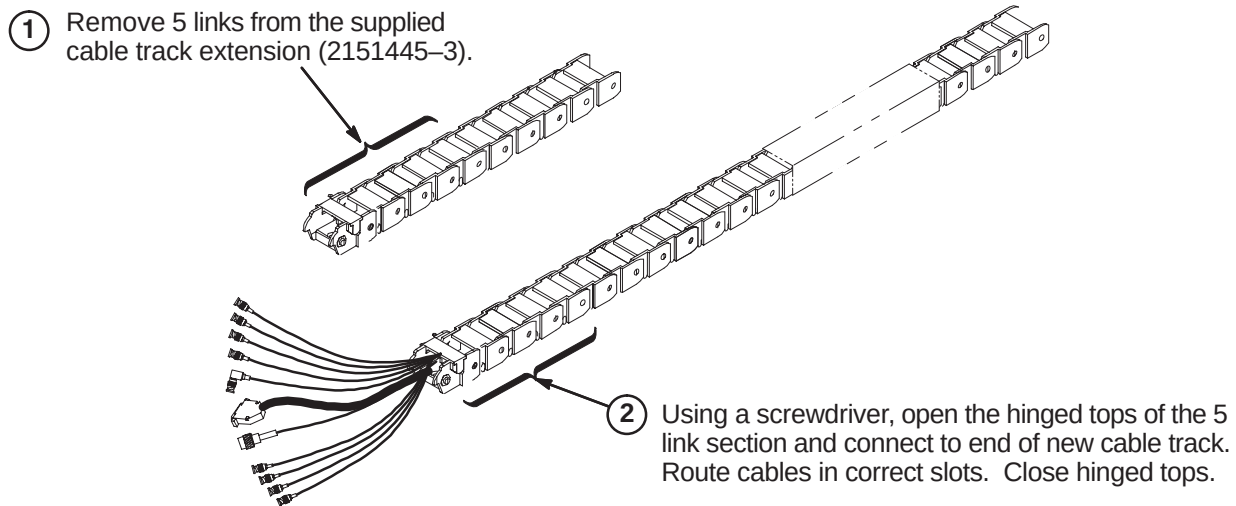


1-1-4 MODIFY CABLE TRACKS FOR S-SERIES AND MAGNISHIELD ENCLOSURES

The S-Series and Magnishield magnet enclosures require longer cable tracks than the provided 35-link track that is part of this upgrade. The S-Series uses a 40-link cable track and the Magnishield uses a 50-link cable track. Modifications are required as follows:

- For the S-Series magnet, five links need to be added to new 35-link cable track supplied with upgrade to extend it to the required 40 links. Ten extra links are provided for this extension. Only five are required. Discard the extra five links that are not used. Refer to section **1-1-4-1**.
- For Magnishield, fifteen links need to be added to the new 35-link cable track to create the required 50-link track. The ten extra links provided with this upgrade along with 5 links removed from the existing cable track will be required. Refer to section **1-1-4-2**.

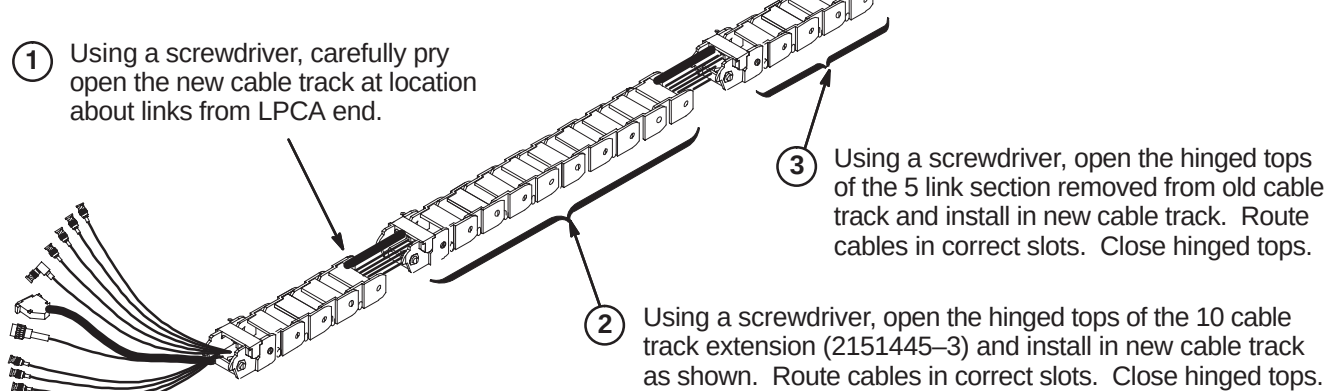
1-1-4-1 S-SERIES CABLE TRACK MODIFICATION



1-1-4-2 MAGNISHIELD CABLE TRACK MODIFICATION

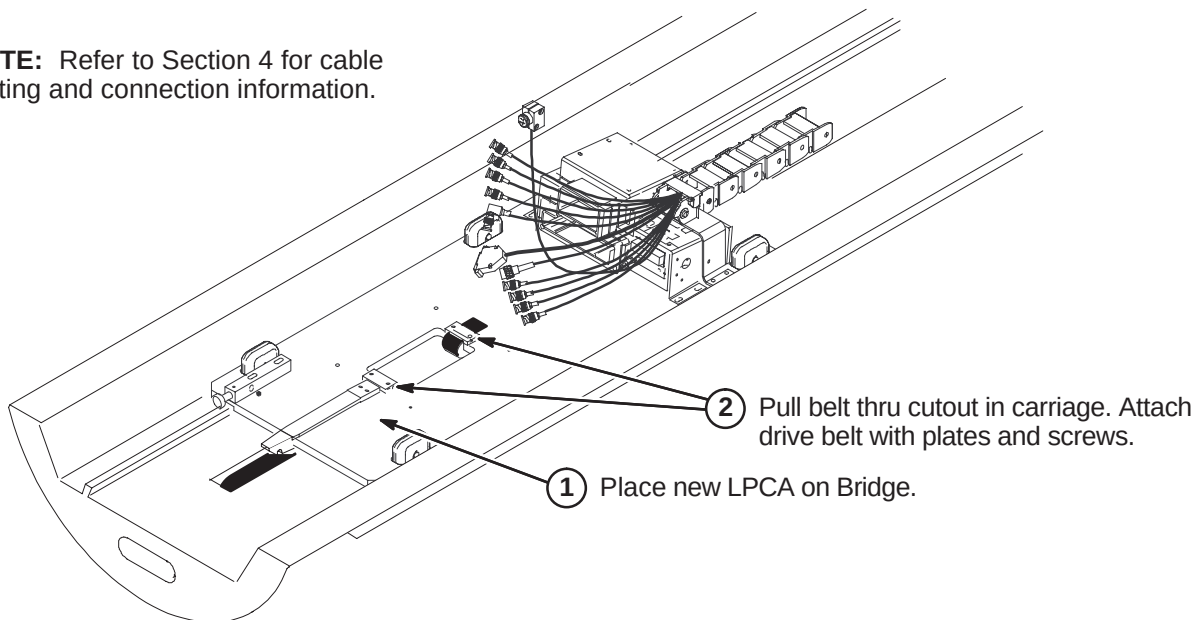
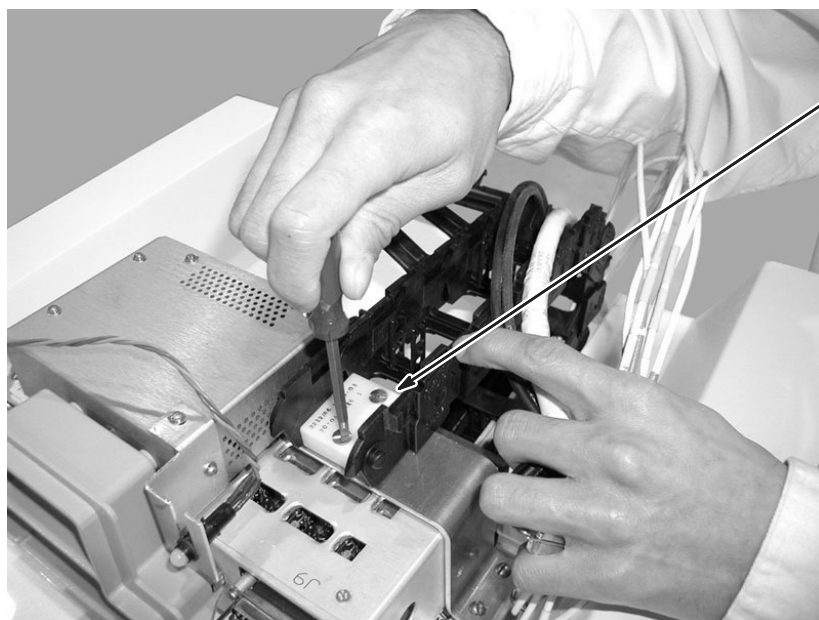
NOTE:

The links of the new cable track are of a slightly different design than the existing cable track for connecting to the LPCA. Therefore, it is required that the 5-link addition from the old cable track not be installed at the end of the cable track.

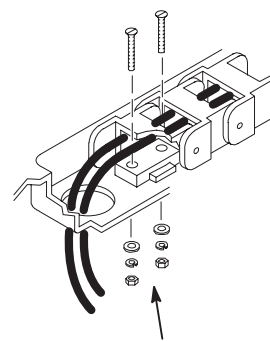


1-1-5 INSTALL NEW LPCA

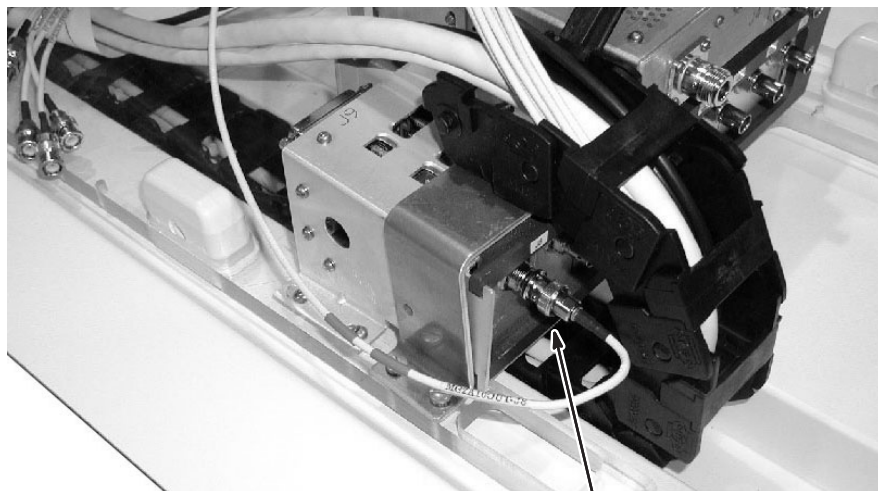
NOTE: Refer to Section 4 for cable routing and connection information.

**1-1-6 INSTALL CABLE TRACK**

① Attach cable track to bracket with cable track clamp (46-282169P1) and screws (46-208913P12).

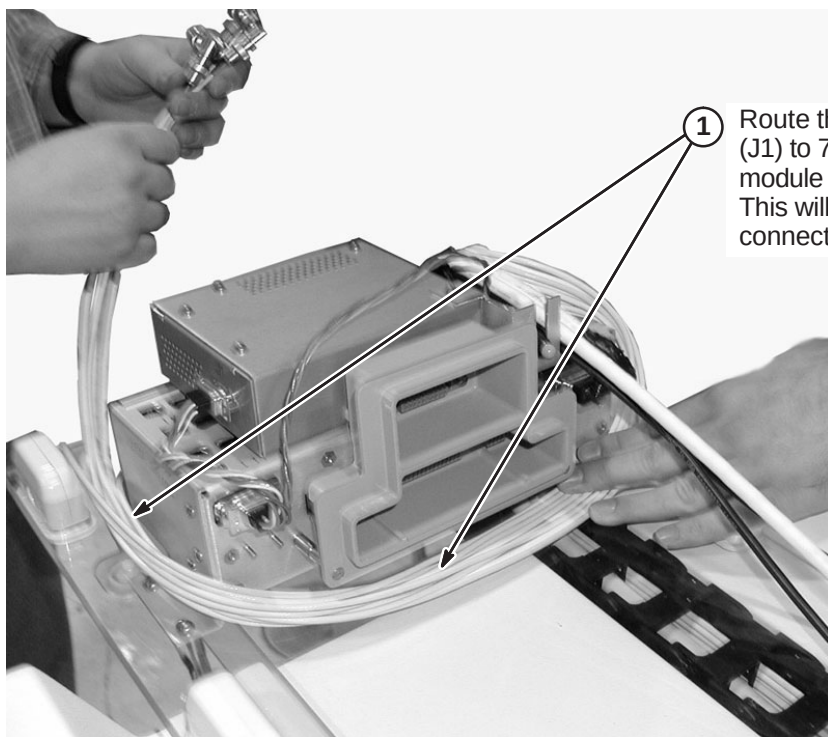


② Connect other end of cable track to Bridge using existing mounting hardware previously removed.

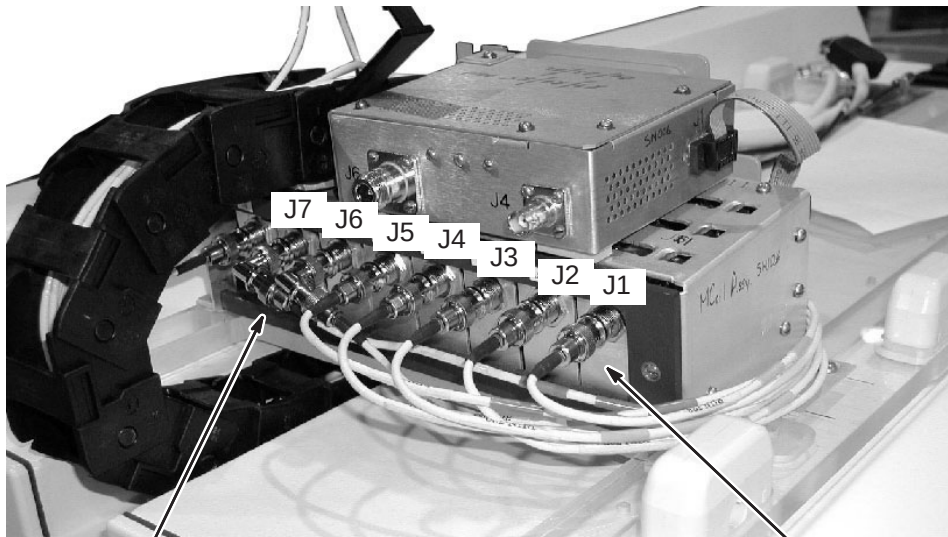
1-1-7 ROUTE/CONNECT RUN RUN 735 TO LPCA

- ① Before connecting other cables, route and connect BNC cable RUN 735 to MG2-A16OUT-J8.

Note that this cable is routed on opposite side of LPCA from other cables.

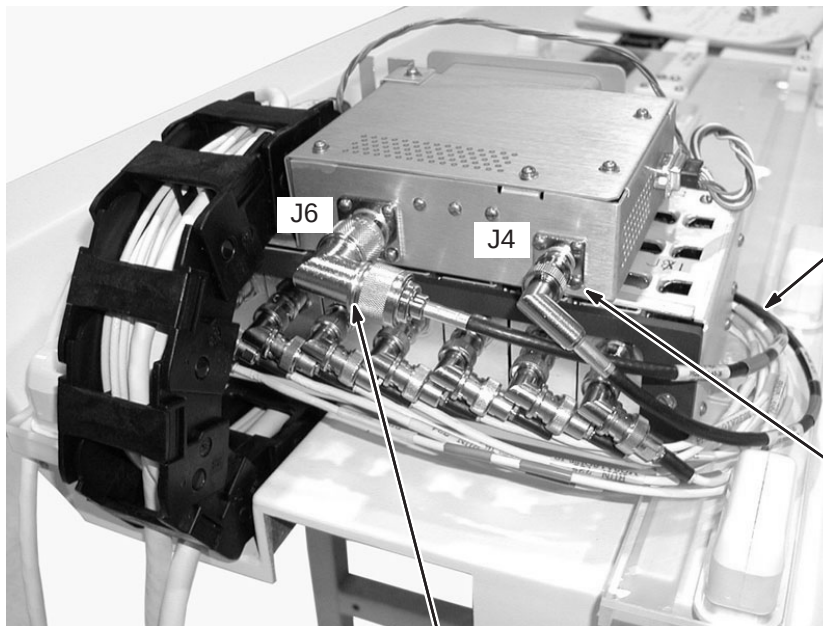
1-1-8 ROUTE RUNS 728 THRU 734

- ① Route the remaining 7 BNC cables, Runs 728 (J1) to 734 (J7), to the rear of the Multi-coil module as shown. Make sure they are snug. This will allow proper alignment when connecting to module.

1-1-9 CONNECT RUNS 728 THRU 734

② Two BNC Right Angle Adaptors (46-233422P1) are required for Runs 733 and 734 connecting to J6 and J7.

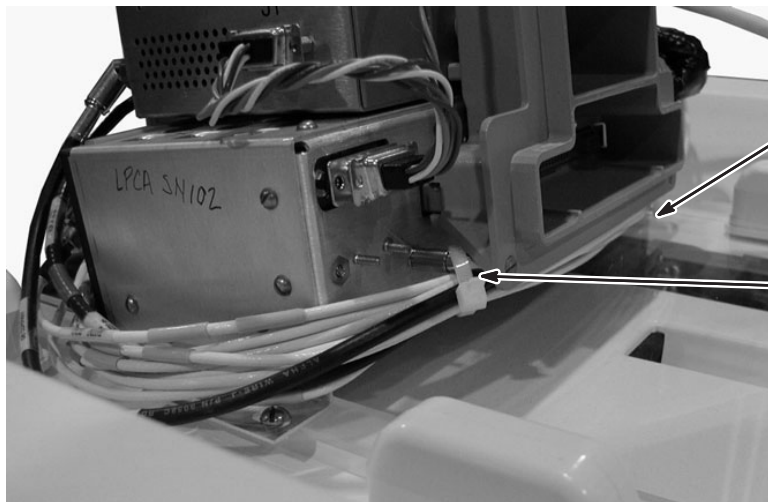
① Connect Runs 728 (J1) to 734 (J7), to the designated ports as shown.

1-1-10 ROUTE/CONNECT RUNS 404 AND 1035

① Route Runs 404 and 1035 along same path as BNC cables.

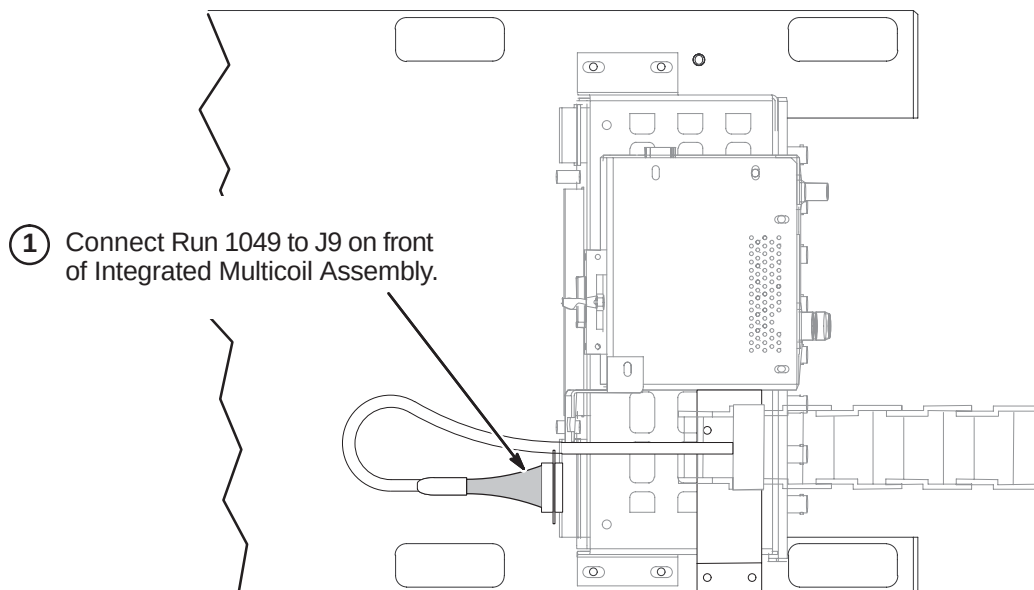
② Connect Run 1035 to J4. If cable does not have right angle connector, install BNC Right Angle Adaptor (46-233422P1) at J4.

③ Connect Run 404 to J6 on Head TR Module.

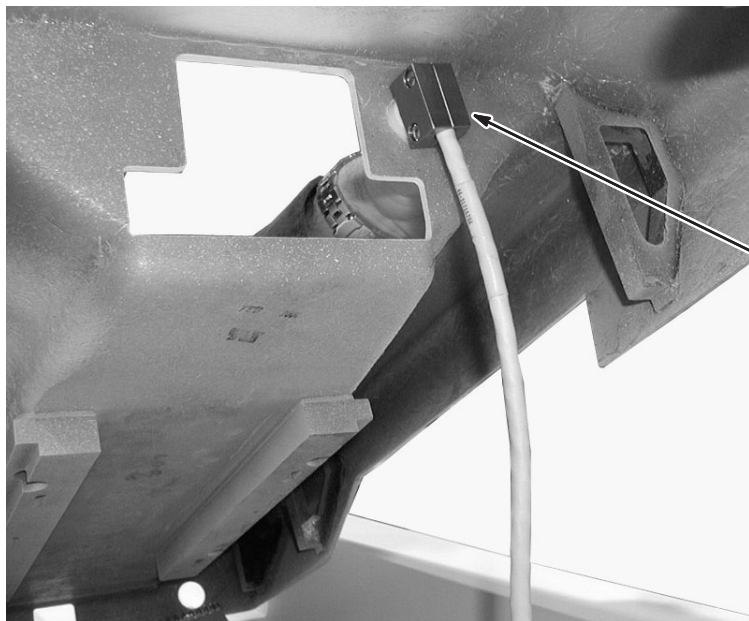
1-1-11 SECURE LPCA CABLES IN PLACE

① Dress cables to remove any slack. If necessary, carefully pull cables from other end of cable track to remove slack.

② Attach two Ty-wraps at this position.

1-1-12 CONNECT RUN 1049 TO INTEGRATED MULTICOIL ASSEMBLY

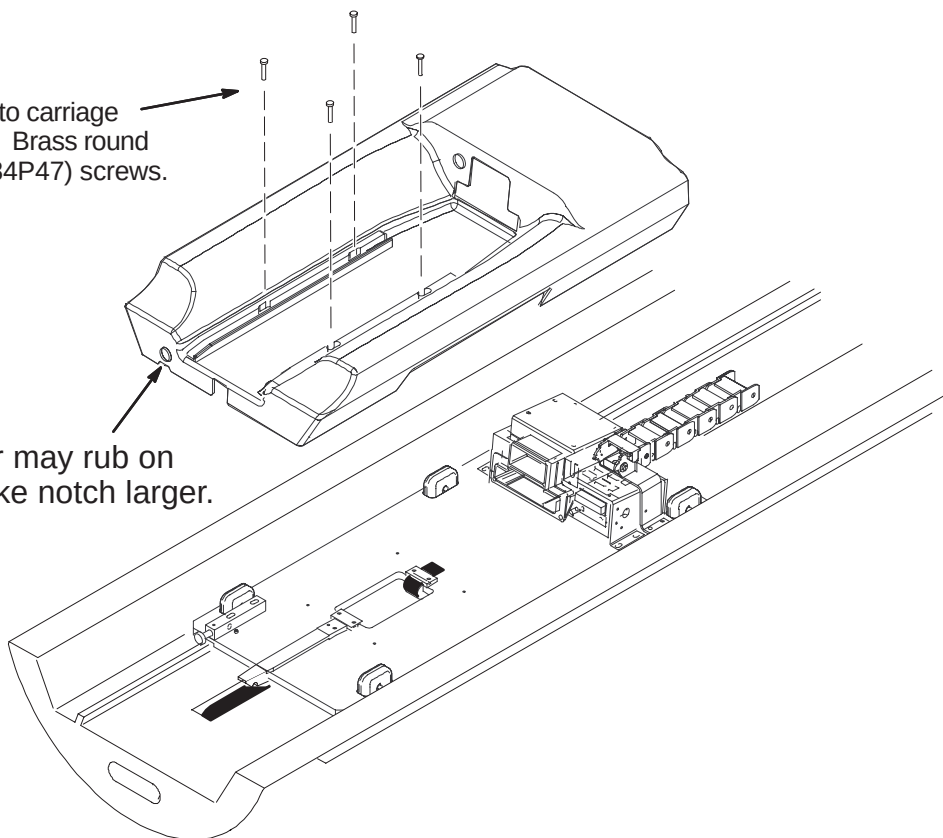
① Connect Run 1049 to J9 on front of Integrated Multicoil Assembly.

1-1-13 ROUTE/CONNECT MICROPHONE CABLE

- ① Route and connect Microphone cable to LPCA Cover using:
Cable Clamp (46-317135P1),
6-32 x 5/8 SS Screws (46-208560P111),
Microphone Housing (46-317134P1), and
O-Ring (46-136374P65).

1-1-14 INSTALL LPCA COVER

- ① Install new LPCA cover to carriage using four 10-32 x 1-1/2 Brass round head screws (46-220184P47) screws.

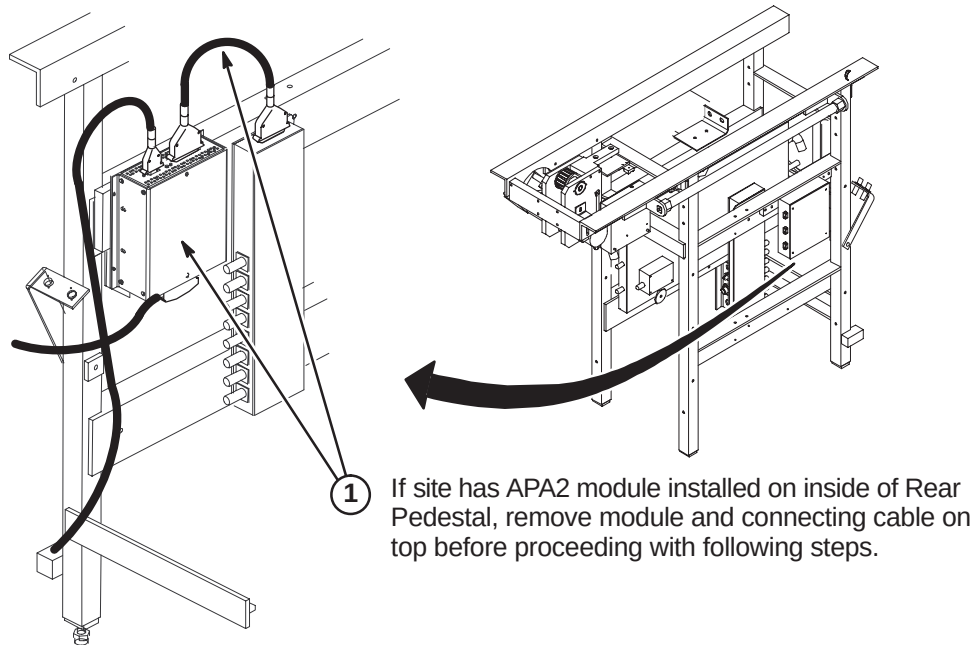


NOTE: Notch in cover may rub on bridge. If needed, make notch larger.

1-2 REAR PEDESTAL

The Rear Pedestal upgrade will include the 8-Channel Switch, Notch Filter, and the UTNS Antenna. The following upgrade instructions cover both the original Horizon and WideOpen style magnet enclosures.

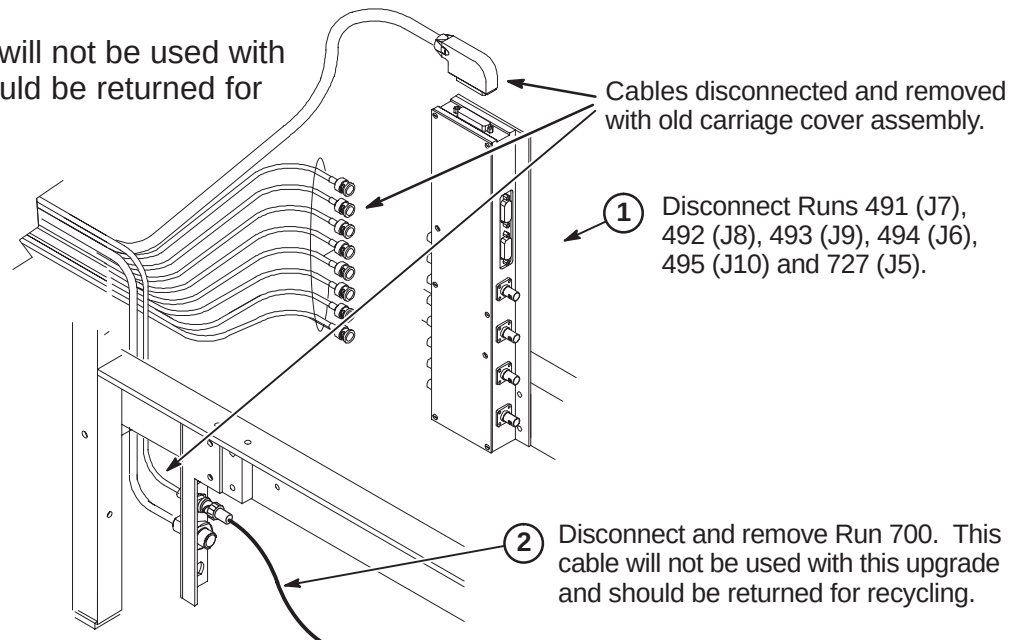
1-2-1 REMOVE APA2 MODULE (If installed)

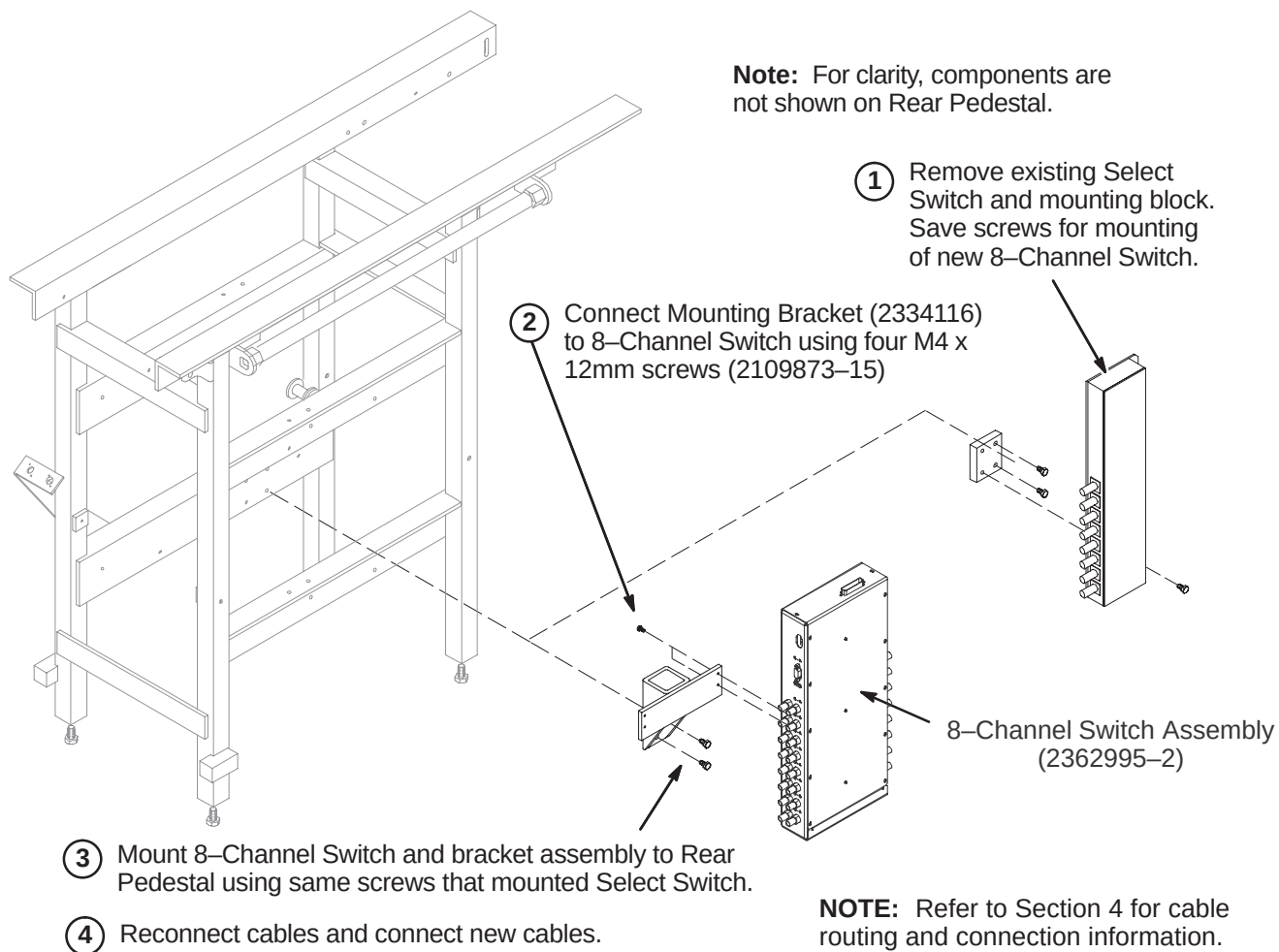


1-2-2 DISCONNECT CABLES IN REAR PEDESTAL

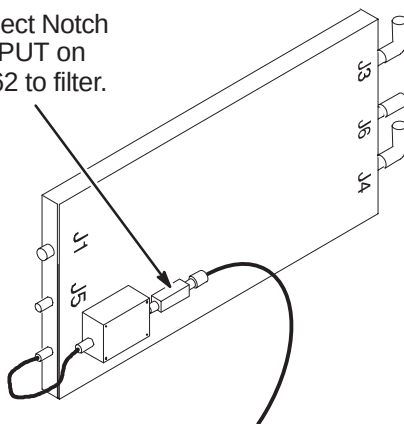
NOTE:

The following cables will not be used with this upgrade and should be returned for recycling.



1-2-3 INSTALL 8-CHANNEL SWITCH ASSEMBLY**1-2-4 INSTALL NOTCH FILTER ON HYBRID SPLITTER PREAMP**

- ① Disconnect Run 262, connect Notch Filter (2305086-2) to OUTPUT on Preamp, reconnect Run 262 to filter.

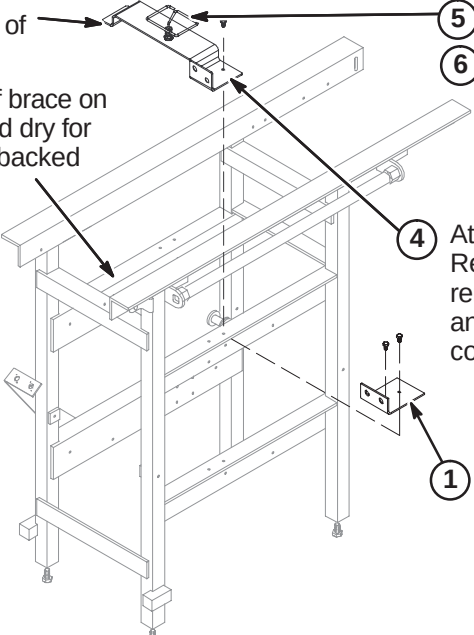


1-2-5 UTNS ANTENNA INSTALLATION

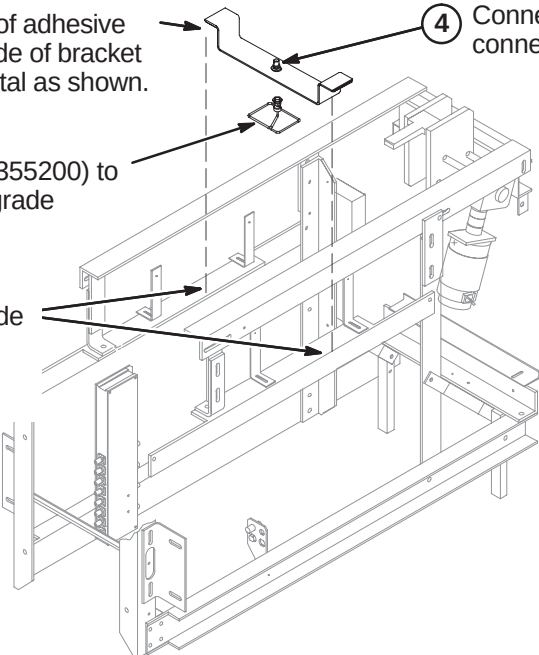
Two mounting brackets are provided for this upgrade. Only one will be installed depending on magnet enclosure.

- If site is a WideOpen style enclosure, proceed to section **1-2-5-1** and complete installation.
- If site is a Horizon style (non-WideOpen), proceed to section **1-2-5-2** to complete installation.

1-2-5-1 Install UTNS Antenna IN WideOpen Enclosure

- 
- The diagram shows a WideOpen magnet enclosure with a rear pedestal. Arrows point to various components and steps: (1) points to the existing mounting bracket on the rear pedestal; (2) points to the top surface of the rear pedestal brace; (3) points to the bottom of the rear pedestal brace; (4) points to the UTNS antenna bracket being attached to the rear pedestal; (5) points to the UTNS antenna being attached to the bracket; (6) points to the connection of Run 1057 to the antenna.
- At existing Mounting Bracket, disconnect Runs 743 and 744 and cables from RF Coil. Remove and save BNC cable connectors. Remove and discard bracket. Save mounting hardware for installation of new UTNS Antenna bracket.
 - Make sure top surface of brace on opposite side is clean and dry for attachment of adhesive backed Velcro.
 - Peel of liner from bottom of adhesive backed Velcro
 - Attach UTNS Antenna Bracket (2384445) to Rear Pedestal using mounting hardware removed in step 1 on this side of Rear Pedestal and Velcro on other side. Install BNC cable connectors.
 - Attach UTNS Antenna (2355200) to bracket.
 - Connect Run 1057 to UTNS Antenna. Reconnect Runs 743 and 744 and RF Coil cables

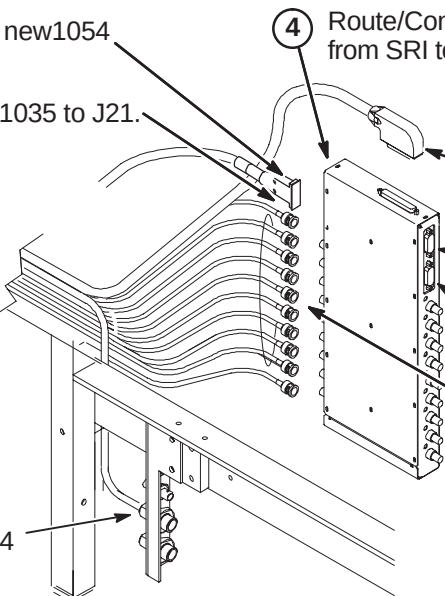
1-2-5-2 Install UTNS Antenna in Horizon Style Enclosure

- 
- The diagram shows a Horizon style magnet enclosure. Arrows point to various components and steps: (1) points to the top surfaces of the side braces; (2) points to the bottom side of the UTNS upgrade bracket; (3) points to the adhesive backed Velcro on the bracket; (4) points to the connection of Run 1057 to the antenna connector at the top of the bracket.
- Make sure top surfaces of side braces are clean and dry for attachment of adhesive backed Velcro.
 - Attach UTNS Antenna (2355200) to bottom side of UTNS Upgrade bracket (2384700).
 - Peel of liner from bottom of adhesive backed Velcro on each side of bracket and attach to Rear Pedestal as shown.
 - Connect Run 1057 to UTNS Antenna connector at top of bracket.

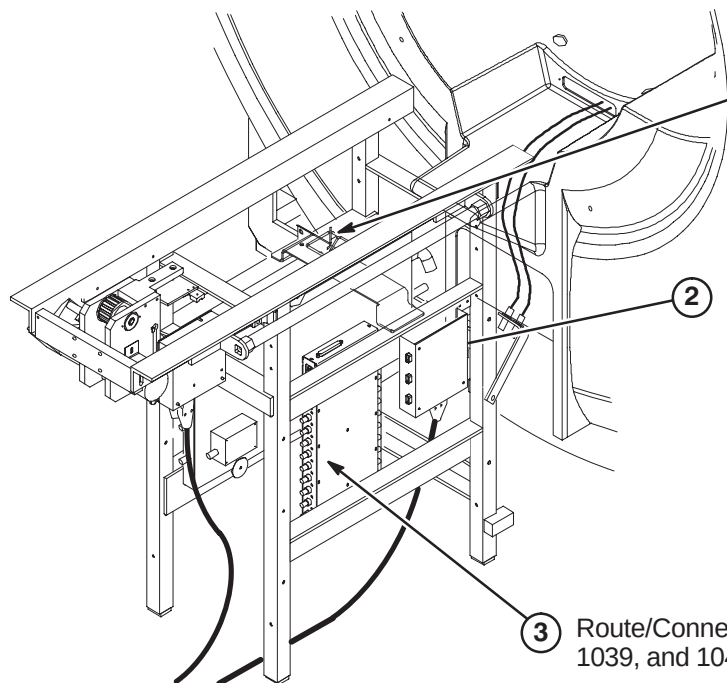
NOTE:

For clarity, Hybrid Splitter not shown in installed position.

1-2-6 CONNECT CABLES FROM LPCA AND SRI TO REAR PEDESTAL

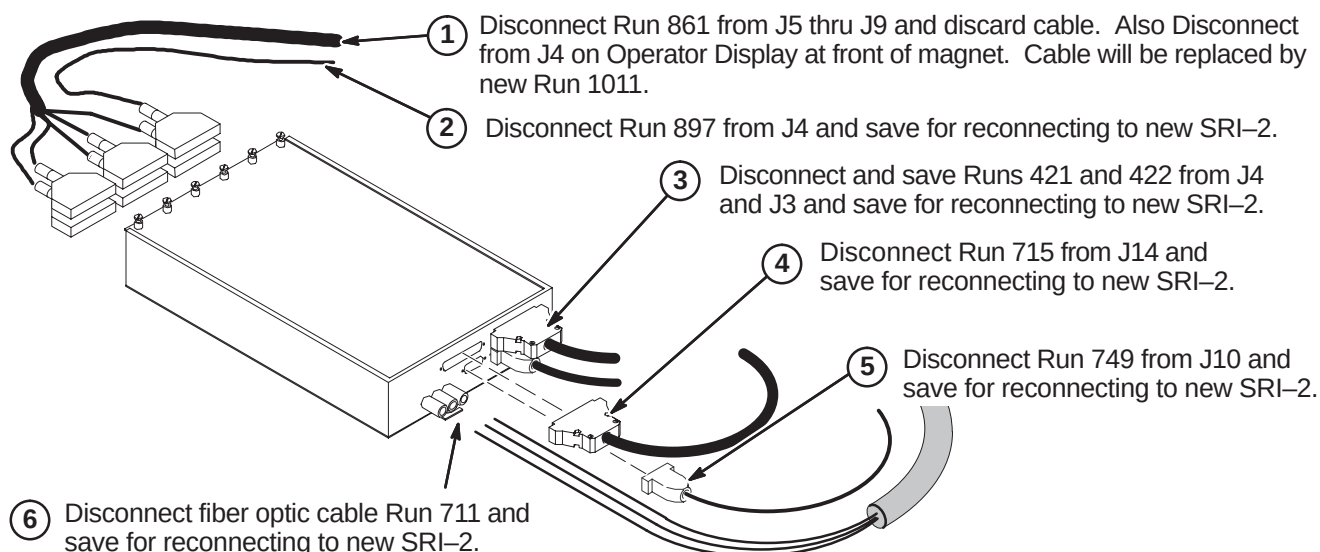
- 
- The diagram shows a rear pedestal with multiple cable ports. Arrows point from numbered callouts to specific cable runs and connection points.
- ① Route/Connect Run new1054 from SRI to J20.
 - ② Route/Connect Run 1035 to J21.
 - ③ Route/Connect Run 404 from LPCA to J1.
 - ④ Route/Connect new Run 1078 from SRI to J29 and J30.
 - ⑤ Route/Connect Run 1049 from LPCA to J19.
 - ⑥ Route/Connect Run 1041 from J78 on Penetration Panel to J10.
 - ⑦ Route/Connect Run 1042 from J77 on Penetration Panel to J5.
 - ⑧ Route/Connect Runs 728 thru 735 to J11 to J18 and Run 1049 to J19.

1-2-7 CONNECT REMAINING CABLES AT REAR PEDESTAL

- 
- The diagram shows the rear pedestal with a carriage cover and various cable ports. Arrows point from numbered callouts to specific cable runs and connection points.
- ① Route/Connect Run 1057 from J85 on Penetration panel to J1 on UTNS Antenna.
 - ② After Carriage cover is installed, connect Microphone cable from Carriage Cover to J7.
 - ③ Route/Connect Runs 491, 492, 493, 1036, 1037, 1038, 1039, and 1040 to J1 thru J9 on 8 Channel Switch.

1-3 SRI

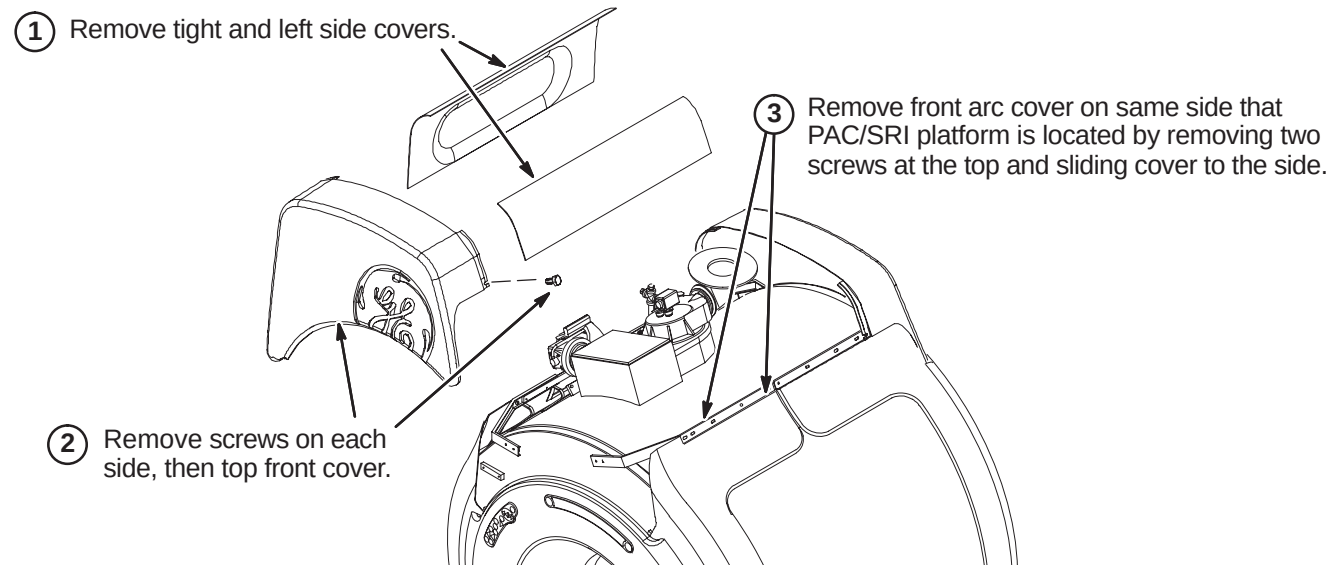
The EXCITE II upgrade provides a new SRI-2 for installation. Existing Run 861 will be replaced by Run 1011 and two new cables will be installed, Runs 1054 and 1078. Remaining cables will be reused. The mounting of the new SRI-2 within the WideOpen, Cx, or S-Series magnet enclosures is shown in the following procedures. It is different for each. The illustration below provides information for all enclosures on the disconnecting of existing cables. Verify that all cables are labeled before disconnecting.

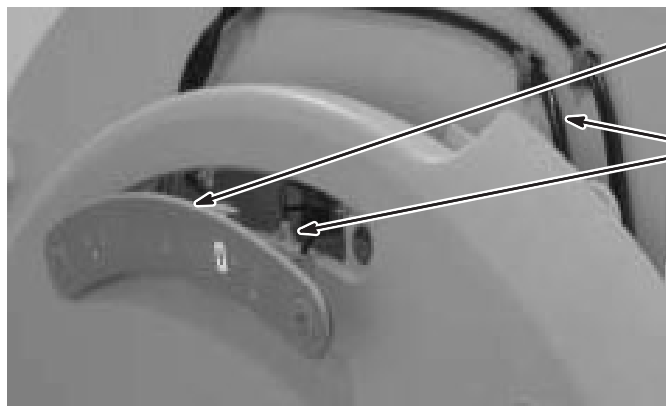


1-3-1 WideOpen ENCLOSURE UPGRADE FOR SRI-2

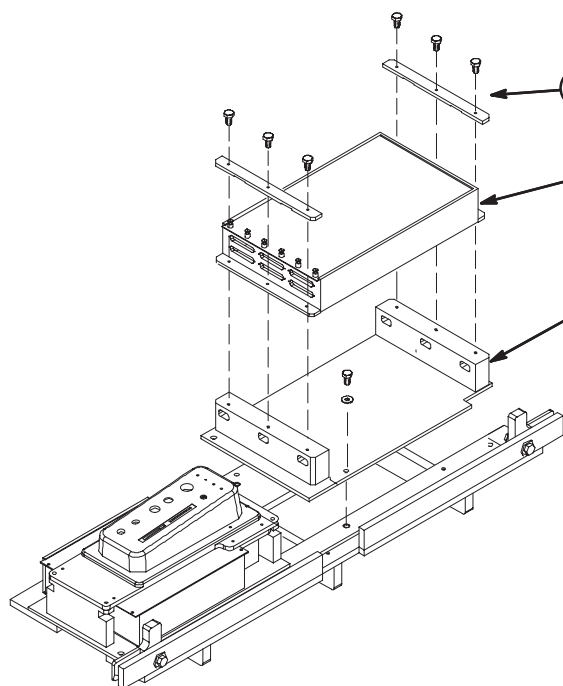
The following procedures are for upgrading the WideOpen enclosure to SRI-2, located on left or right side.

1-3-1-1 Remove WideOpen Enclosure Covers

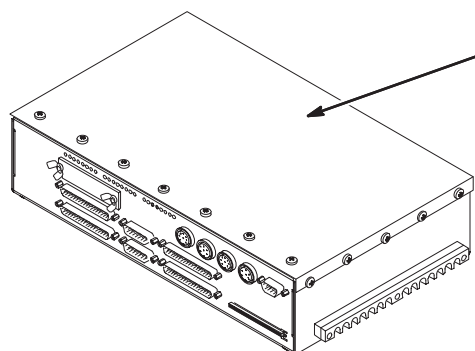


1-3-1-2 Replace Run 861 with Run 1011

- ① Carefully, using a small screwdriver, pry the Operator Display away from the front cover.
- ② Disconnect Run 861 from J4 on back of Operator Display and completely remove cable. There are five connections of this cable at the SRI.
- ③ Route new Run 1011 along same route as Run 861 and connect to J4 on back of Operator Display.
- ④ Re-install Operator Display, arc cover, top front cover, and top side covers.

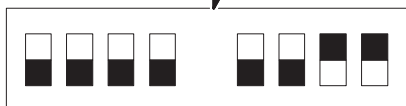
1-3-1-3 Remove Existing SRI

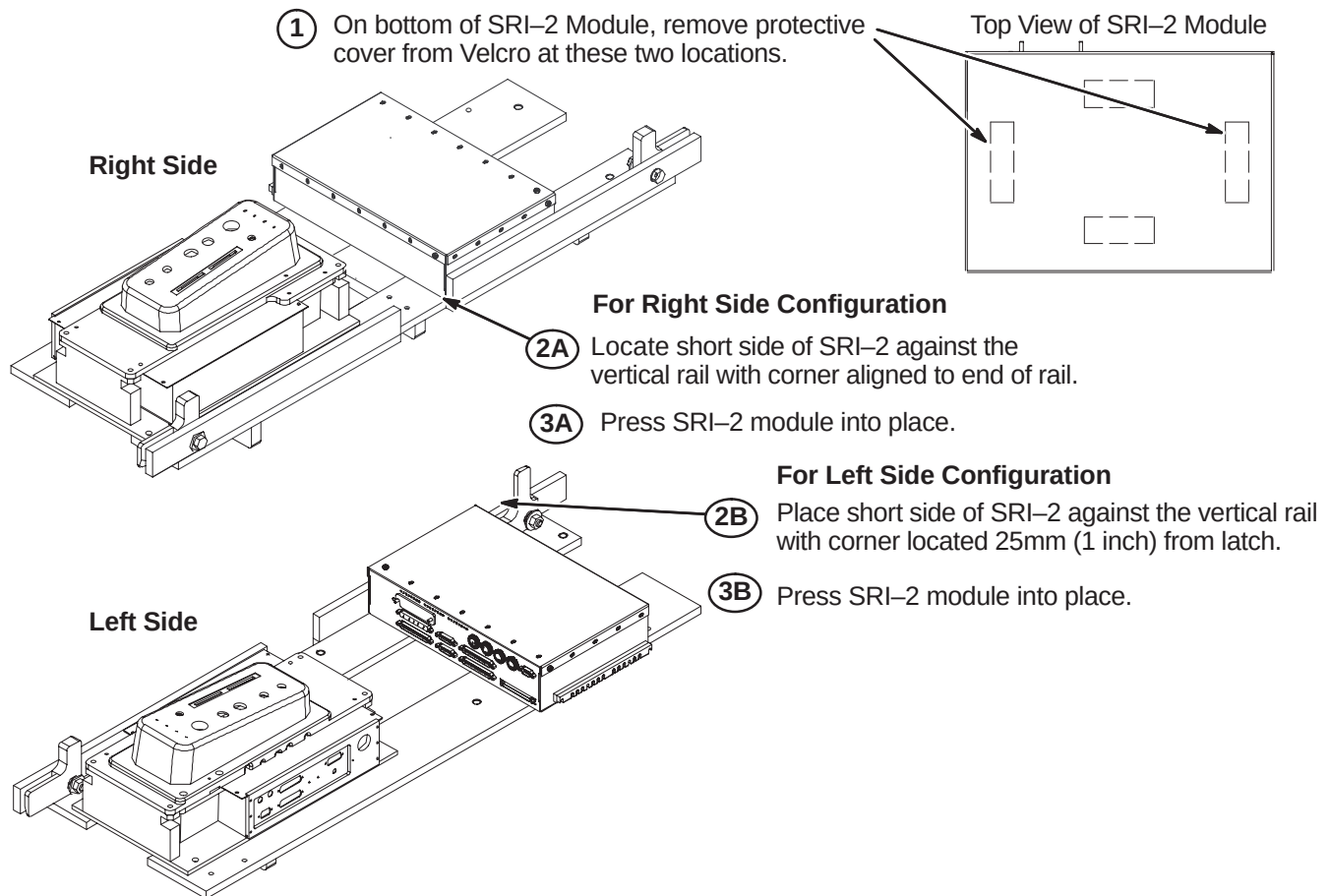
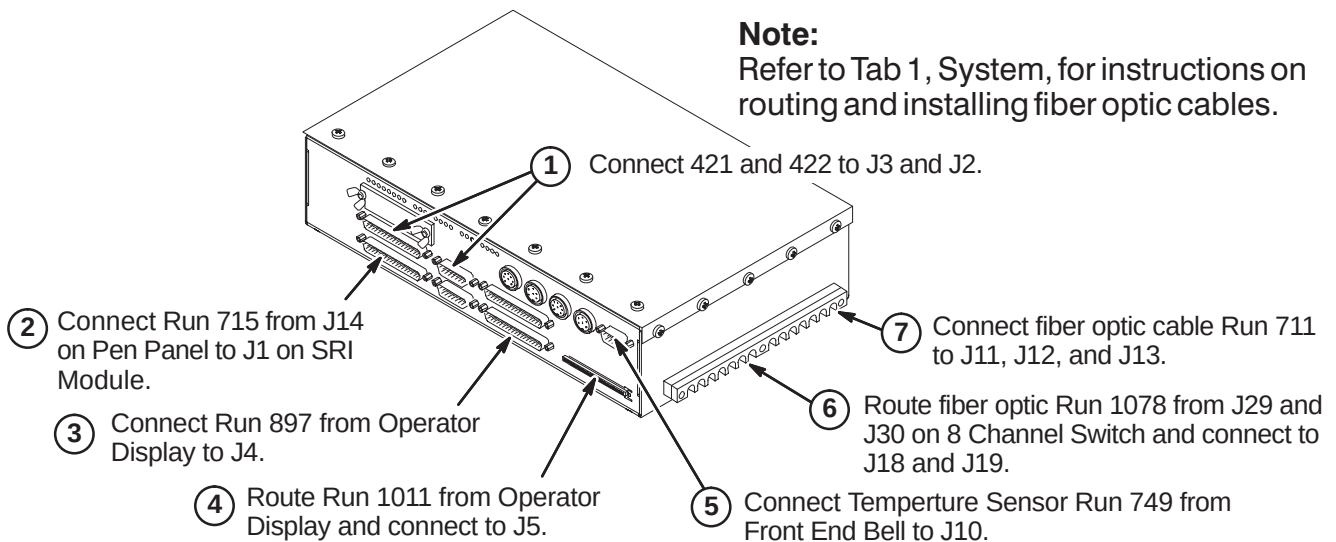
- ① Unfasten and remove mounting hardware and clamps.
- ② Remove existing SRI module.
- ③ Unfasten and remove mounting hardware and SRI platform.
- ④ Return SRI, mounting platform, hardware, and clamps for recycling.

1-3-1-4 Verify Jumper Positions for SRI-2

- ① Slide cover plate away on new SRI-2 and verify jumper settings for 1.5T with Coil ID.

- ② For the eight jumpers, the two jumpers on the right should be in the "ON" or located in the up position. All others should be "OFF" or down.

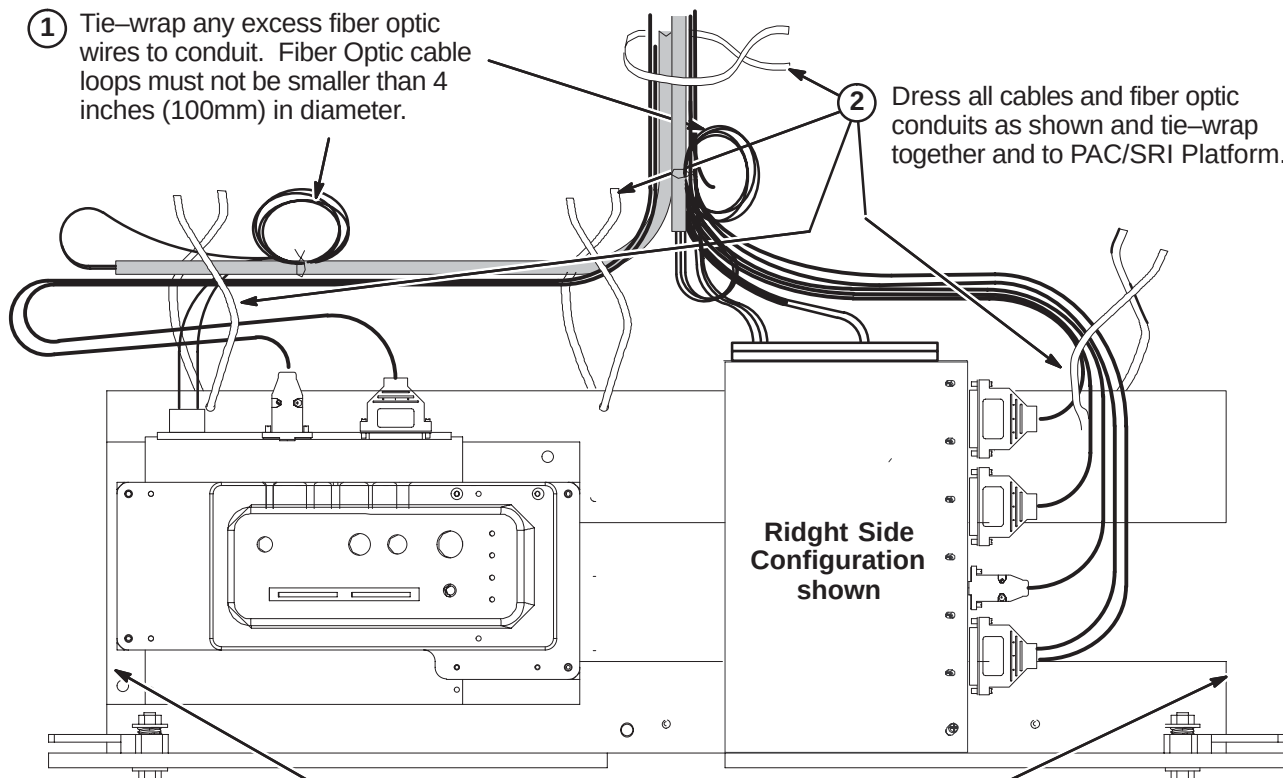


1-3-1-5 Install SRI-2 On Right or Left Side of Enclosure**1-3-1-6 Connect Cables to SRI-2**

1-3-1-7 Route and Tie-wrap Cables to SRI/PAC Platform

- ① Tie-wrap any excess fiber optic wires to conduit. Fiber Optic cable loops must not be smaller than 4 inches (100mm) in diameter.

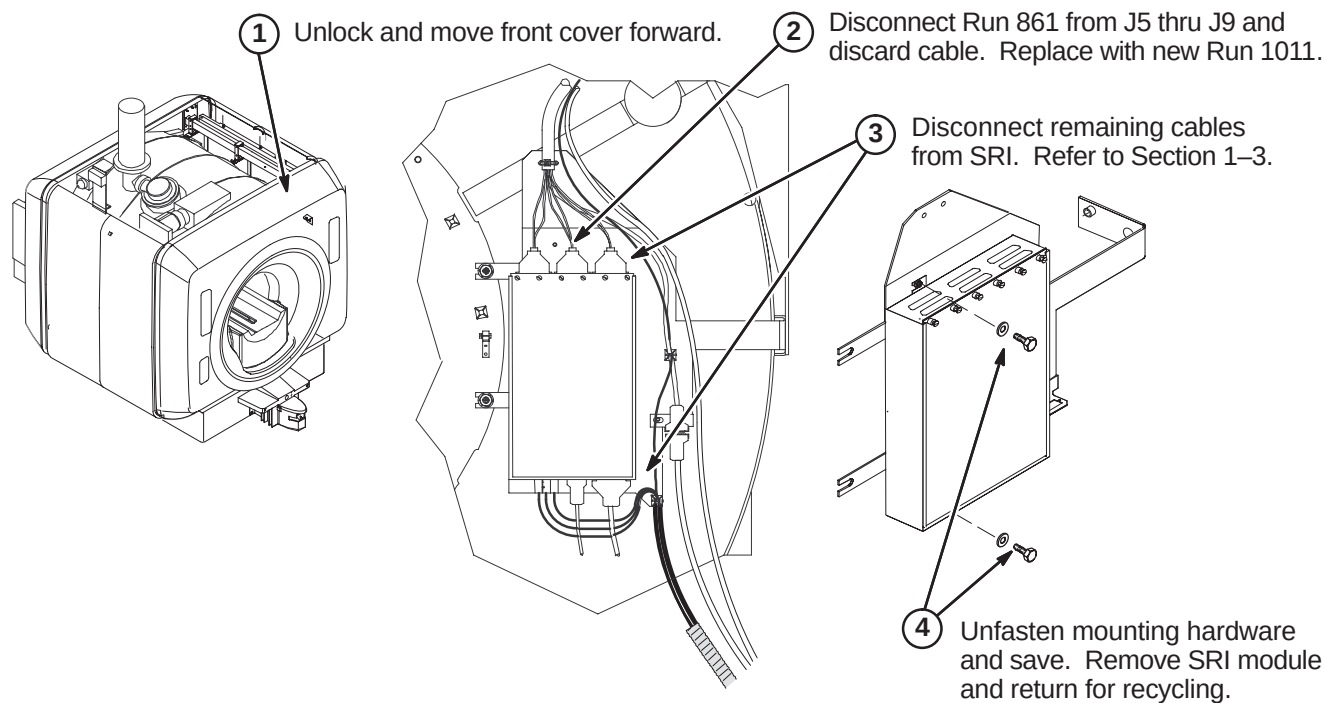
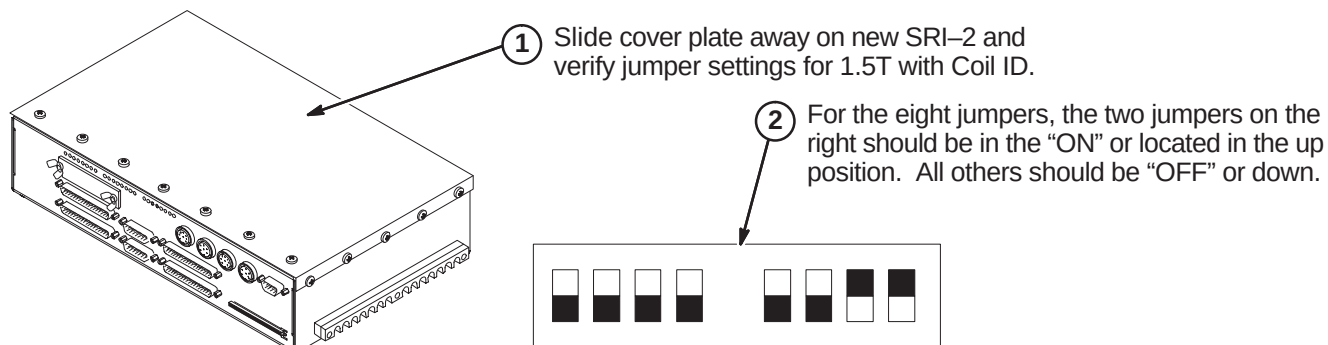
- ② Dress all cables and fiber optic conduits as shown and tie-wrap together and to PAC/SRI Platform.

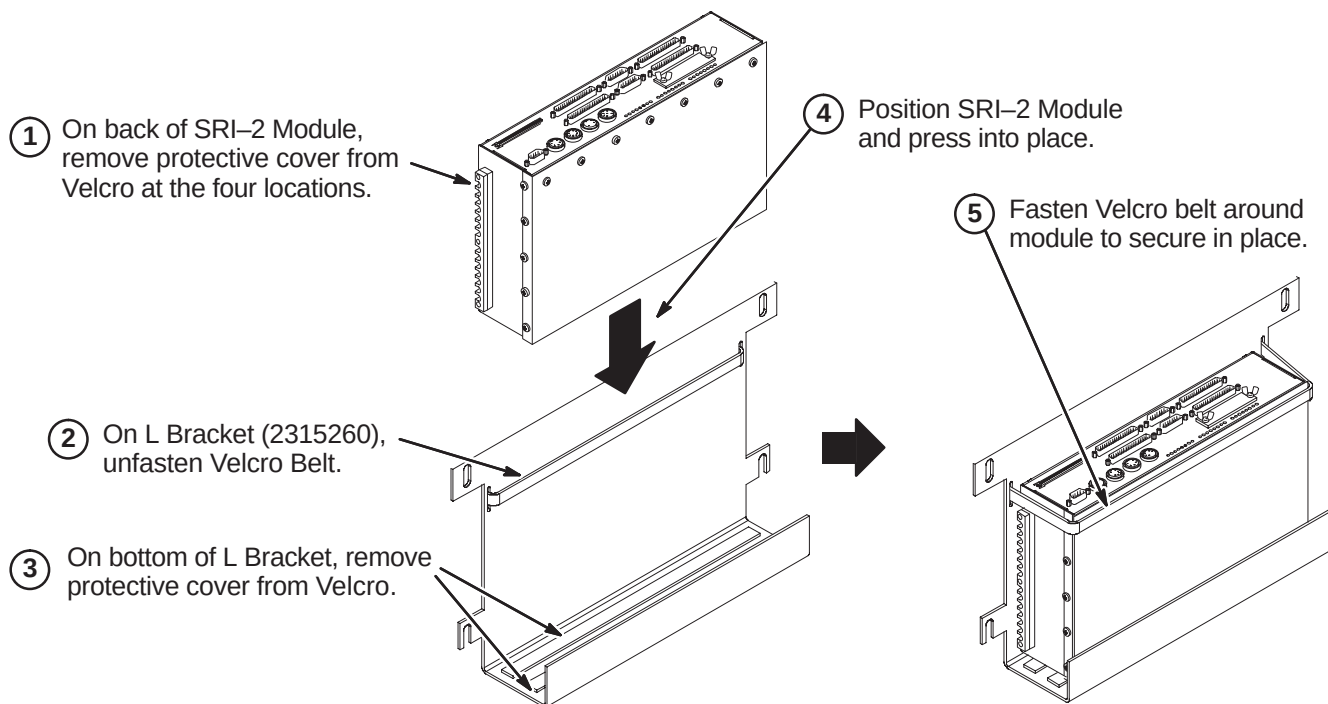
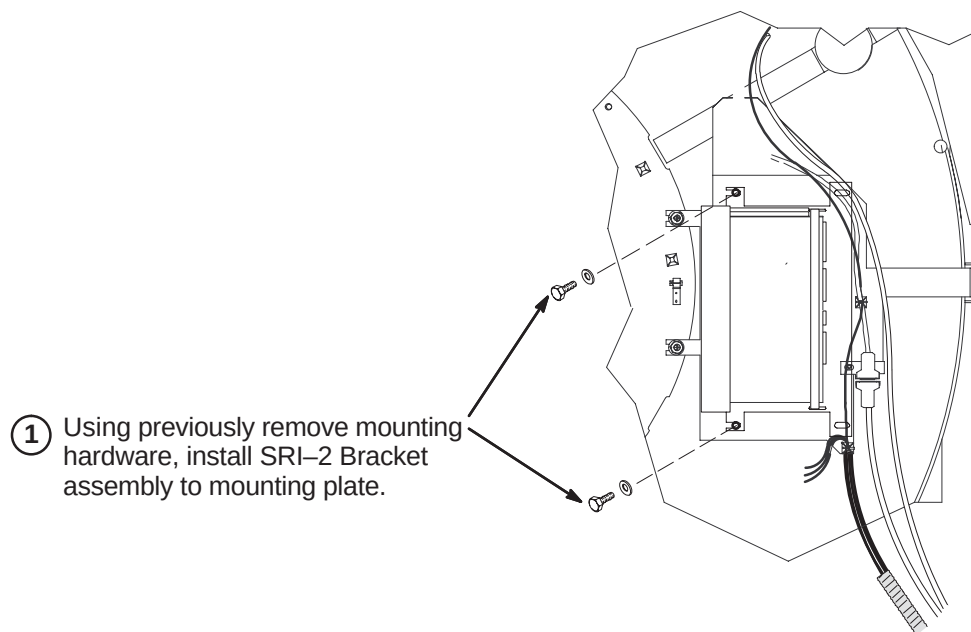


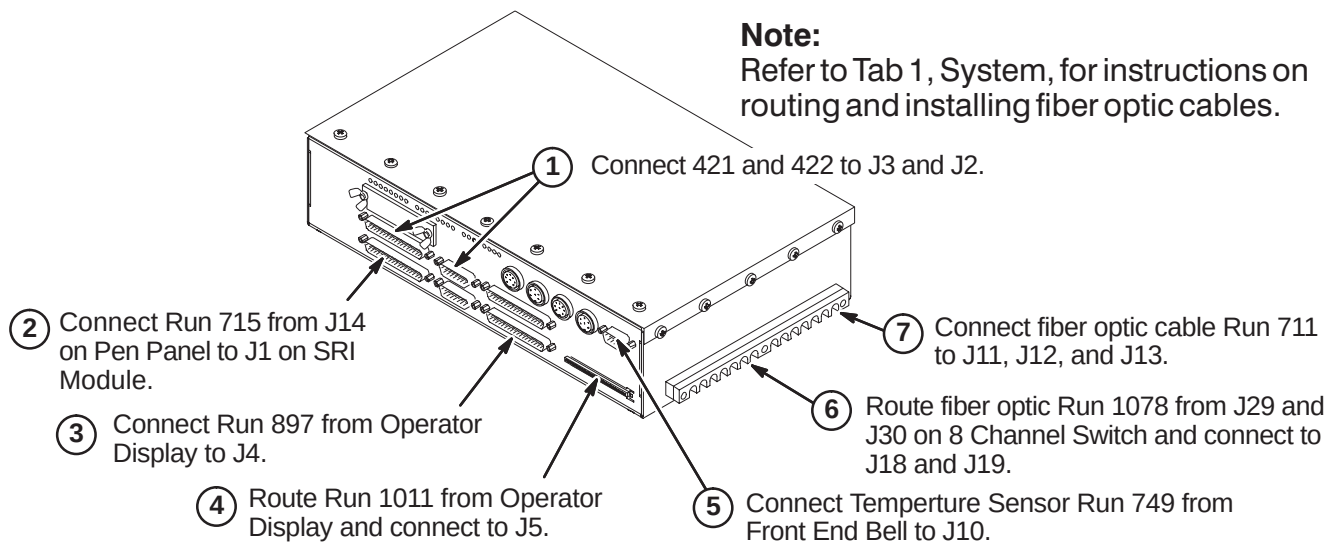
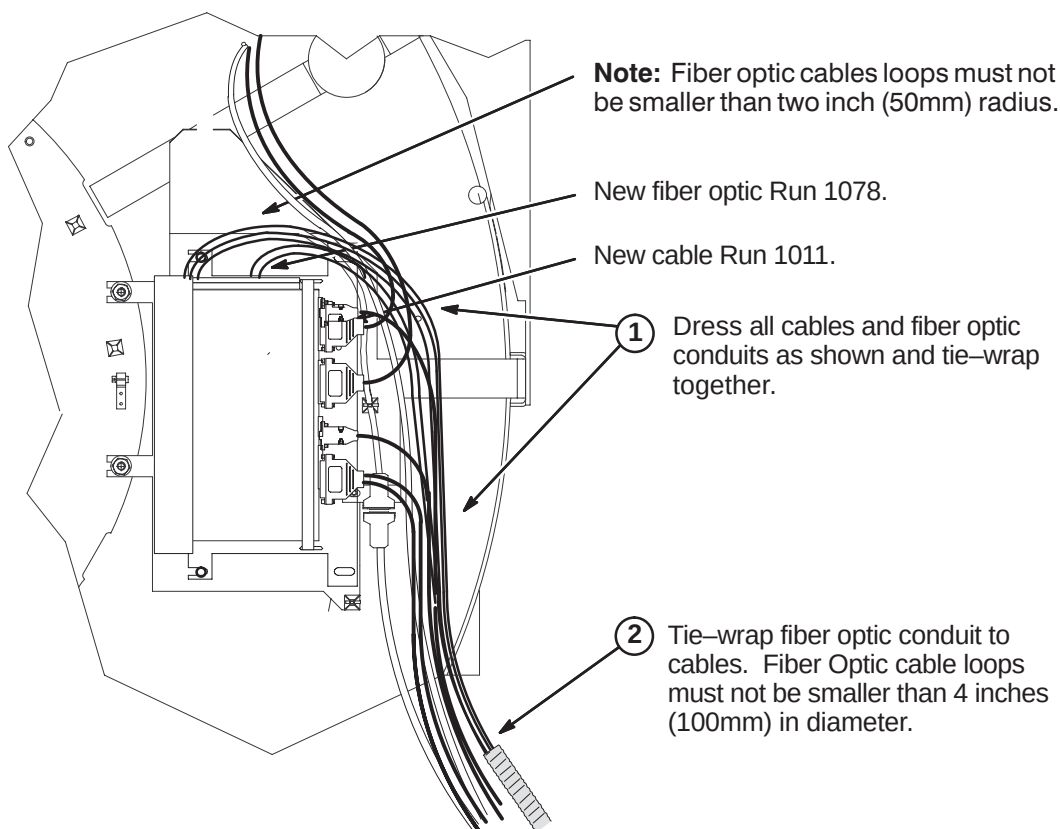
Note: Keep cables dressed inside edge so unit slides between legs easily for service.

1-3-2 Cx AND LCC HORIZON ENCLOSURE UPGRADE FOR SRI-2

The following procedures are for upgrading the Cx Horizon style enclosure to SRI-2.

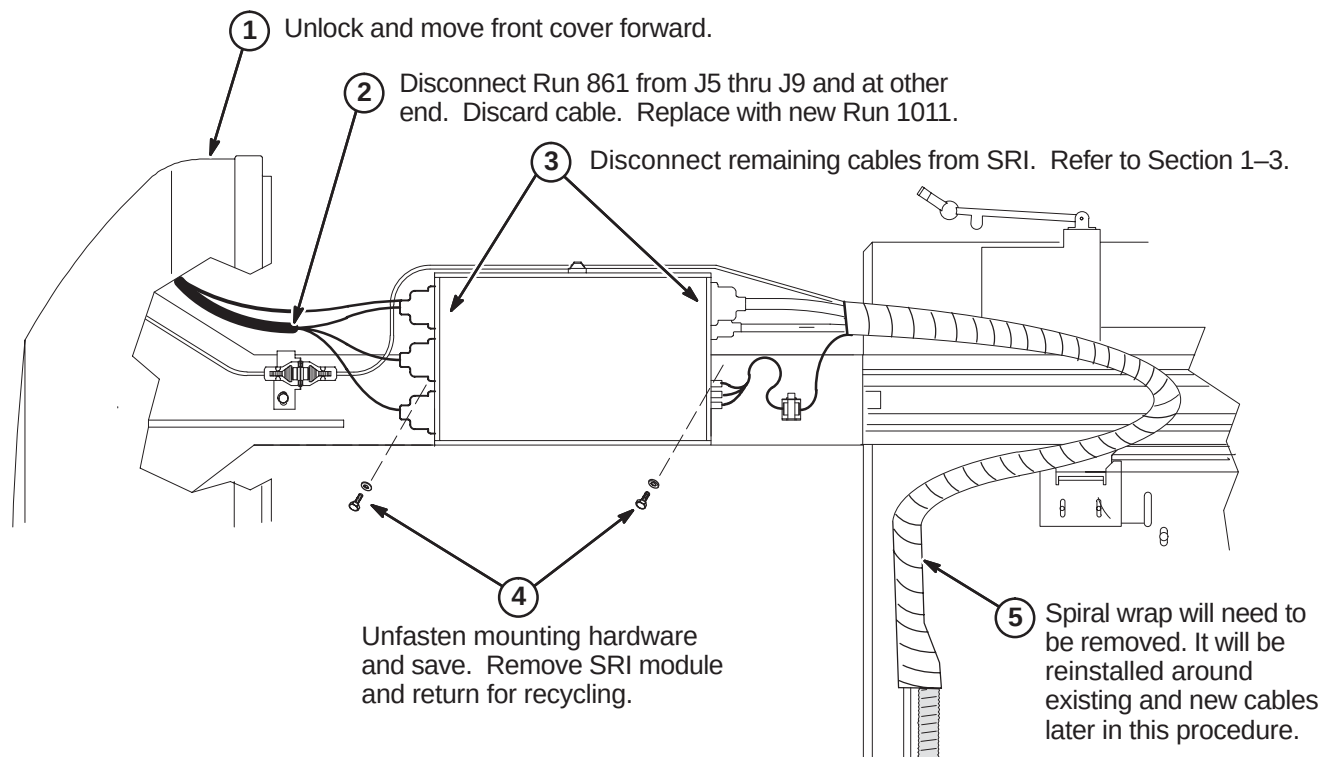
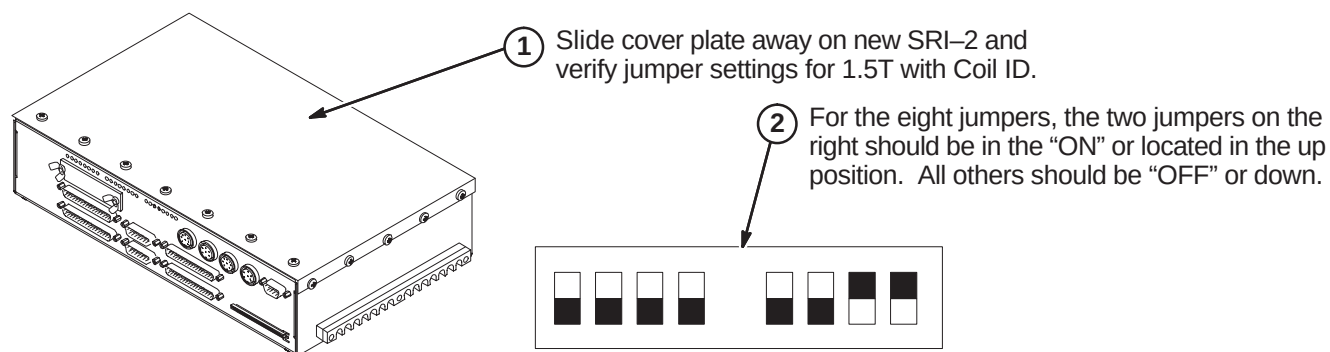
1-3-2-1 Remove Existing SRI Module**1-3-2-2 Verify Jumper Positions for SRI-2**

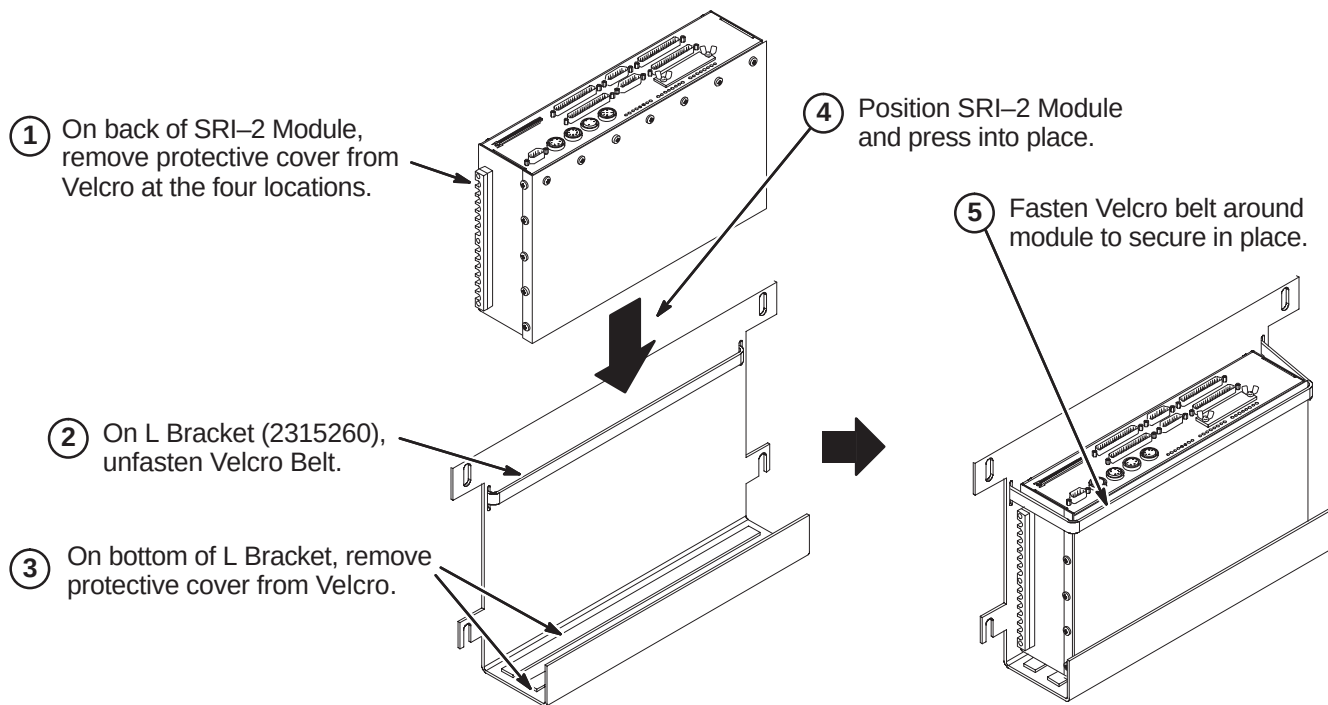
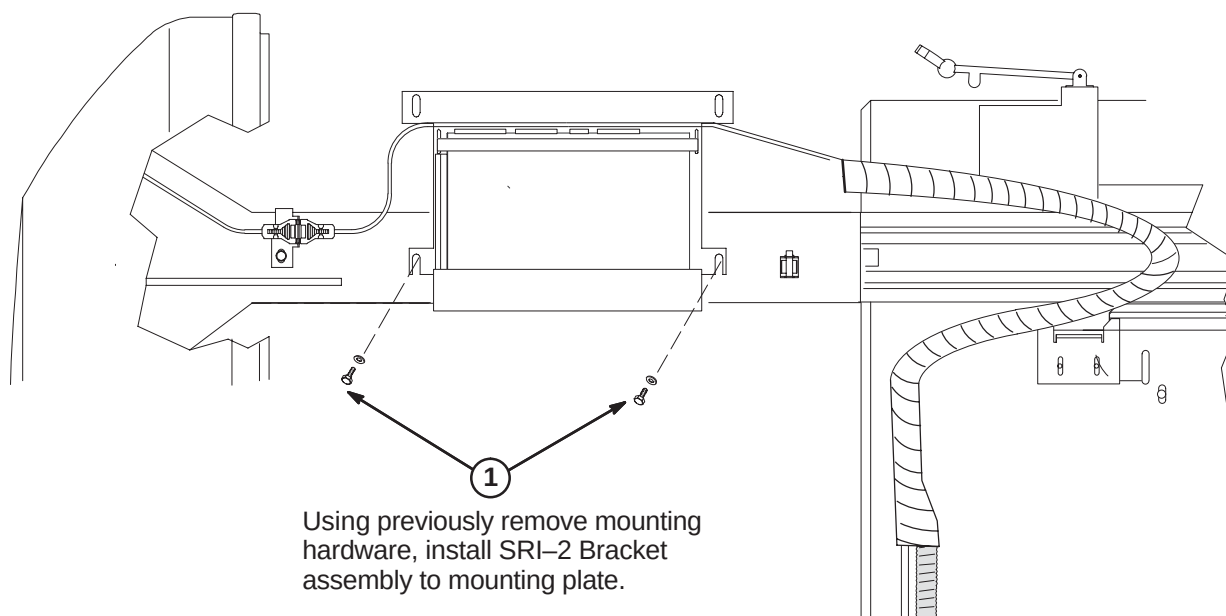
1-3-2-3 Position SRI-2 in L Bracket**1-3-2-4 Install SRI-2 on Magnet**

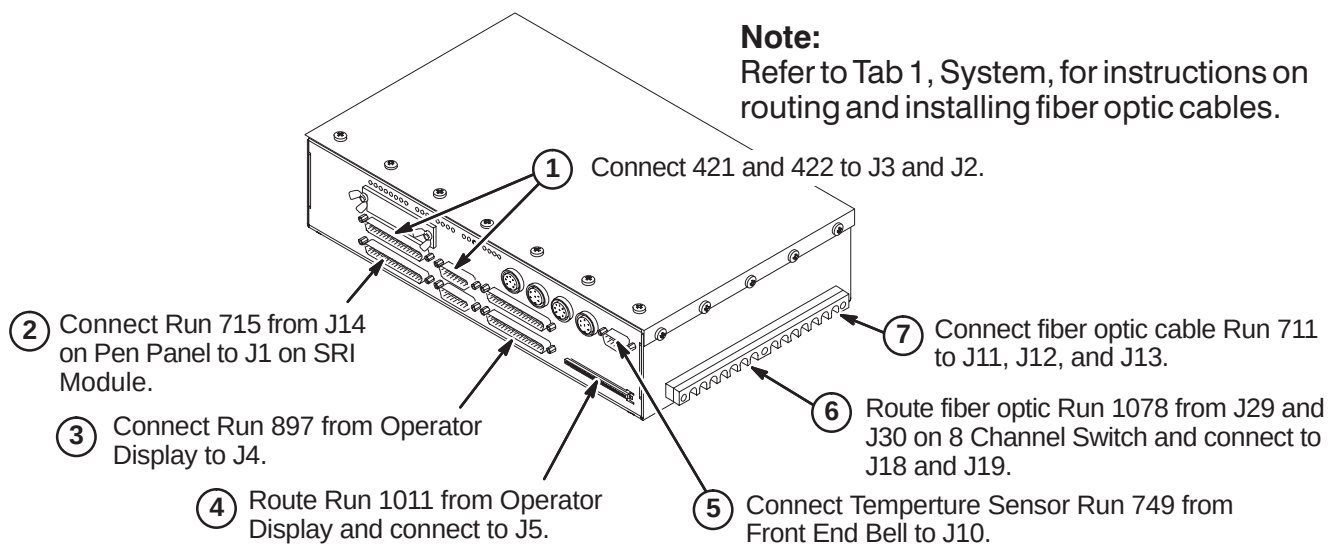
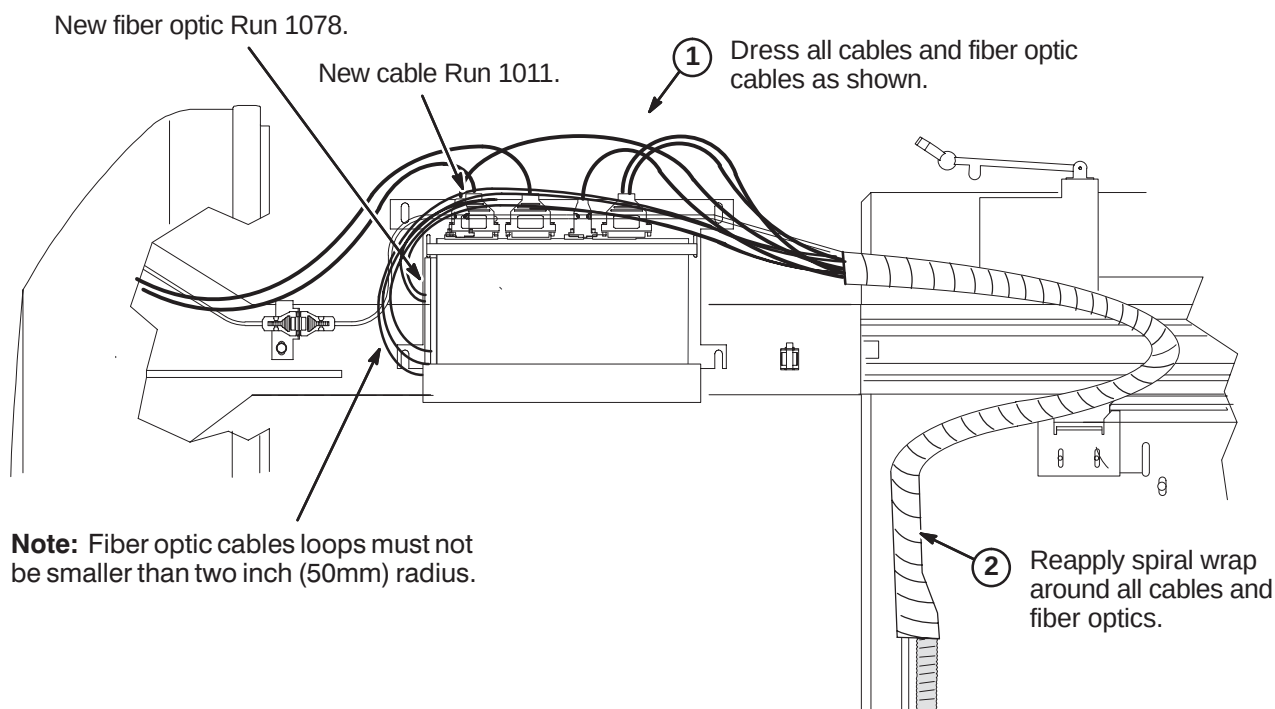
1-3-2-5 Connect Cables to SRI-2**1-3-2-6 Route and Tie-wrap Cables**

1-3-3 S-SERIES HORIZON ENCLOSURE UPGRADE FOR SRI-2

The following procedures are for upgrading the S-Series Horizon style enclosure to SRI-2.

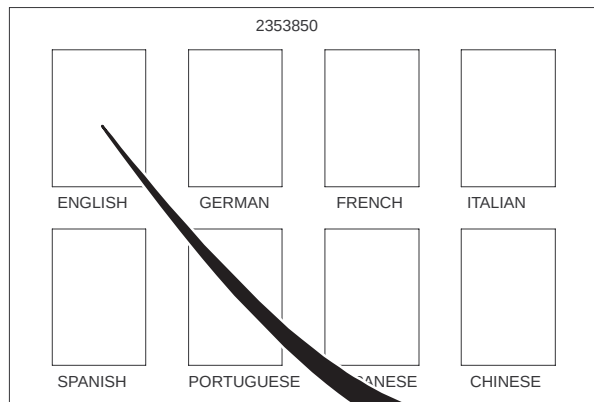
1-3-3-1 Remove Existing SRI Module**1-3-3-2 Verify Jumper Positions for SRI-2**

1-3-3-3 Position SRI-2 in L Bracket**1-3-3-4 Install SRI-2 on Magnet**

1-3-3-5 Connect Cables to SRI-2**1-3-3-6 Route and Spiral Wrap Cables**

1-4 INSTALLATION OF HEARING LOSS WARNING SIGN

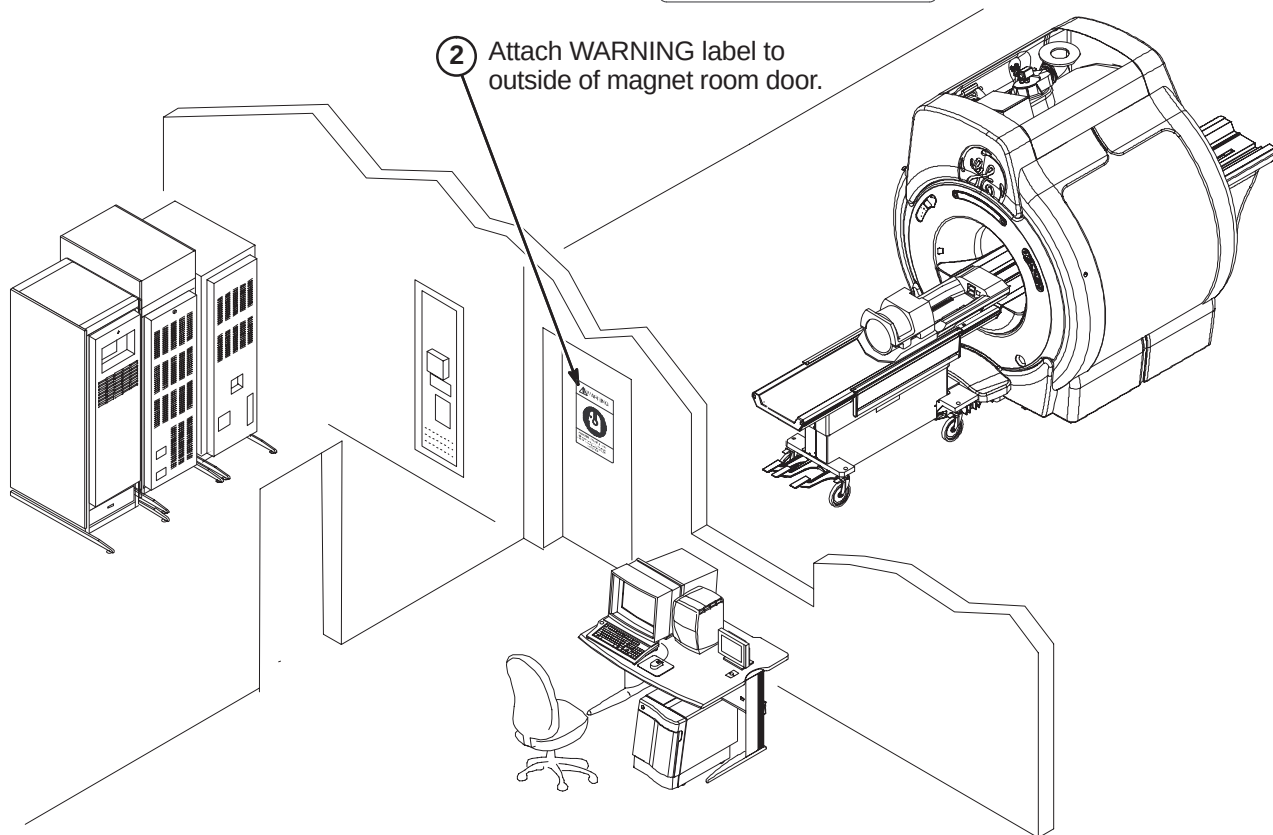
□ G1 - WARNING SIGN INSTALLATION



- ① Remove applicable language warning label from "HEARING PROTECTION REQUIRED" sheet of 8 labels (2353850).



- ② Attach WARNING label to outside of magnet room door.



BODY COIL INSTALLATION INSTRUCTIONS

1-1 Tab 3, Body Coil, contains installation instructions for the following:

- *DIRECTION 2223489* – BRM-D & CRM Body Coil in Magnet Upgrade Instructions
- *DIRECTION 2147882* – Body Coil in Magnet Upgrade Instructions
- *DIRECTION 2147883* – Heat Exchanger Cabinet Installation Instructions
- *DIRECTION 2231327* – Water Chiller Cabinet Installation Instructions

INSTALLATION STEPS SUMMARY

- ☐ A1 – Body Coil Installation Procedure
- ☐ A2 – Body Coil Cart and Cradle Components
- ☐ A3 – Loading Body Coil and Cradle or Pallet on Coil Cart
- ☐ A4 – Using Cart to transport Body Coil

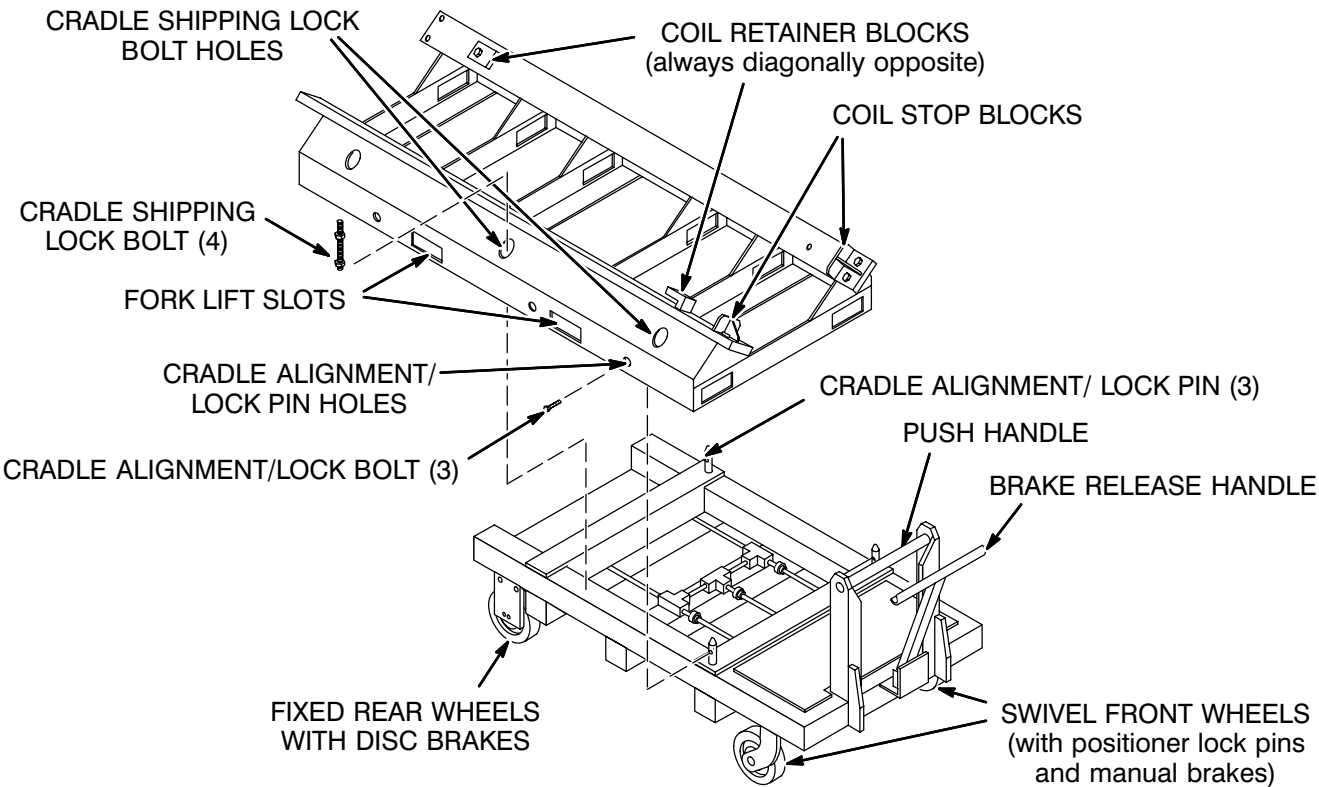
NOTE: For Transportable or Relocatable Upgrade on Site
Because of the weight of the coil cart/cradle, a lift truck may be required to lift the new body coil cart/cradle into the magnet room. Also, the floor may need to be reinforced with 3/4 inch plywood or other plate where ever the coil cart/cradle may roll. Contact van manufacturer for recommendation.

☐ A1 – BODY COIL INSTALLATION PROCEDURE

1. Remove existing pre-Horizon Body Coil. Refer to procedures in *MR Service Methods CD-ROM*, Replacement: Gradient Coil, Quad II/Roemer Coil 1.5T Replacement
2. This Upgrade Instruction contains information for installing Horizon Body Coils in all GE Magnets and the Oxford Magnet. Different installation requirements are found in the following procedure steps:
B4 - Different dimensions are given for positioning Body Coil in Oxford, GE S-I, and all other GE magnet bores.
C1a thru C2b - Instructions for centering Body Coil in all GE magnets.
D1 thru D2 - Instructions for centering Body Coil in Oxford magnet.
3. Horizon Upgrade body coils may be delivered on a cart/cradle or on a shipping pallet (cradle).
If body coil is delivered on a coil cart and cradle combination:
The body coil is transported to the magnet room. Review Step A2, skip Step A3, and proceed to Step A4.
If body coil is delivered on a pallet (or cradle) for use with a coil cart:
The body coil is loaded on the regional coil cart. Review Step A2 and proceed to Step A3.

☐ A2 – BODY COIL CART AND CRADLE COMPONENTS

The Coil Cart and Cradle combination, as shown below is used to install the Body Coil in the magnet bore. The body coil may also be shipped on a disposable wood and metal pallet which performs the functions of the metal cradle shown below.



☐ A3 – LOADING BODY COIL AND CRADLE OR PALLET ON COIL CART

- 1 Have a fork lift truck raise the body coil and cradle (or pallet) combination. The gradient leads are at the end of the cradle away from the handle, and the Coil Stop Blocks are near the handle.



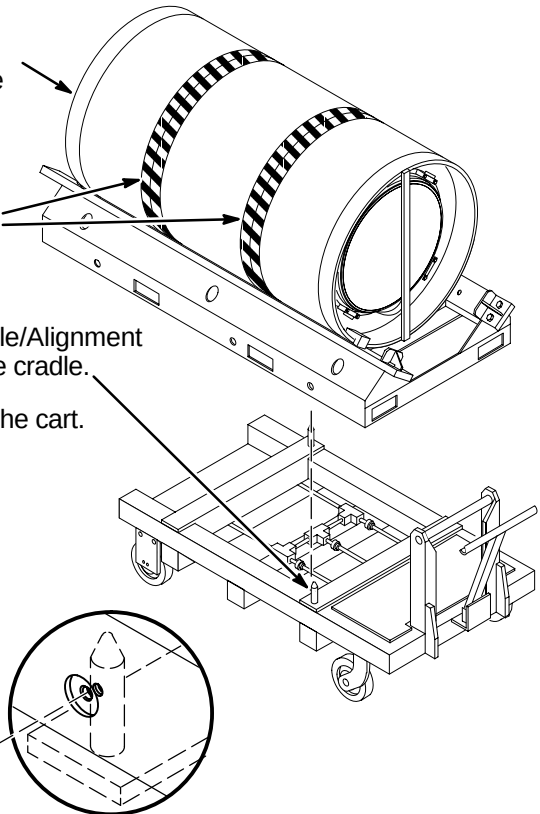
Do not lift or apply pressure to area of Body Coil marked with black/yellow tape labelled "NO LOAD ZONE". Doing so will damage the Body Coil.

- 2 Position cart beneath cradle with Cradle/Alignment Lock Pins aligned with their holes in the cradle.
- 3 Lower body coil cradle (or pallet) onto the cart.



Not having the cradle anchored to the cart could cause the coil/cradle to tip. The three Cradle Alignment/Lock Bolts MUST be in place before attempting to move the coil cart.

- 4 Insert the three Cradle Alignment/Lock Bolts.

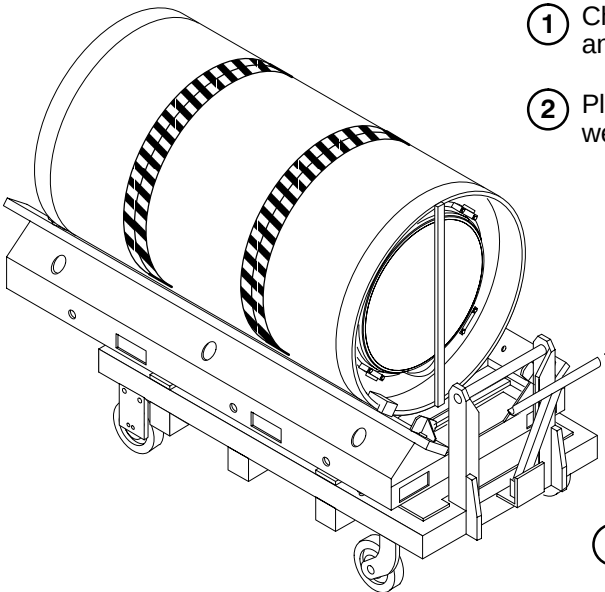


☐ A4 – USING CART TO TRANSPORT BODY COIL



Avoid possible bodily injury. The combined coil/cradle/cart weigh about 3300 pounds. To avoid physical injury, and to control this amount of weight as it moves, a minimum of four people must be used.

- 1 Check route to magnet room and verify route clearance and sufficient manpower to overcome inclines.
- 2 Place plywood sheets or metal plates along route to distribute weight and protect floor surface from damage.



Note: The Brake Release Handle controls the disc brakes on the fixed rear wheels. It must be pushed against the Push Handle in order to move the cart.

- 3 Roll coil cart to outside of magnet room for connection of 10 foot (305cm) Gradient extension cables.

Procedures B2 & B3 are for Non-Actively Shielded magnets only.

INSTALLATION STEPS SUMMARY

- ☐ B1 – Pre-Installation of PVC Clear Tubing
- ☐ B2 – Cable Terminations for 2119600 Series Gradient Extension Cables
- ☐ B3 – Connection of Body Coil Gradient Cables to Extension Cable

☐ B1 – PRE-INSTALLATION OF PVC CLEAR TUBING

1 Cut PVC Clear Tubing (46–221986P3) into at least 12 lengths.

2 Each length should be 2 to 3 inches (50 to 75mm).

3 Slide two of the above lengths over each Body Coil gradient cable prior to adding Extension cables.

NOTE:
Instructions for installing the short sections of PVC Clear Tubing are found in Procedure F1.

WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

IF MAGNET IS AT FULL FIELD – ATTACHING THE 2119600 SERIES EXTENSION CABLES TO THE BODY COIL GRADIENT CABLES MUST BE COMPLETED OUTSIDE THE MAGNET ROOM.

☐ B2 – CABLE TERMINATIONS FOR 2119600 SERIES GRADIENT EXTENSION CABLES

1 Cut off Butt terminal at this end only for all six extension cables.

2 Strip 0.75 inch (19mm) of insulation from end of wire for all six extension cables.

☐ B3 – CONNECTION OF BODY COIL GRADIENT CABLES TO EXTENSION CABLES

2119600 series Extension cables

Gradient Cables from Body Coil

1 Cable connection process shown below is for Run 762 (X1 & X2). Repeat process for Runs 763 (Y3 & Y4) and 764 (Z5 & Z6).

NOTE:
Red tape, X+ (RUN 762), Y+ (RUN 763) and Z+ (RUN 764) labels are attached to wire "ONE".

2 Position Cold Shrink tubing (2218148) on output cables prior to connecting.

3 Insert ends of Run 762 into correct Two-way Splicer (2218149) on Body Coil gradient cable.

4 Crimp butt splices using a T&B WT 3185 Gradient Cable Crimper (2144091) with #2 die.

5 Locate Cold Shrink tubing at desired starting point. Position tubing 1–1/2 to 2 inches (38 to 50mm) beyond end of connector.

6 Pull loose tab gently while unwinding in a counter-clockwise direction. Repeat process for other cable.

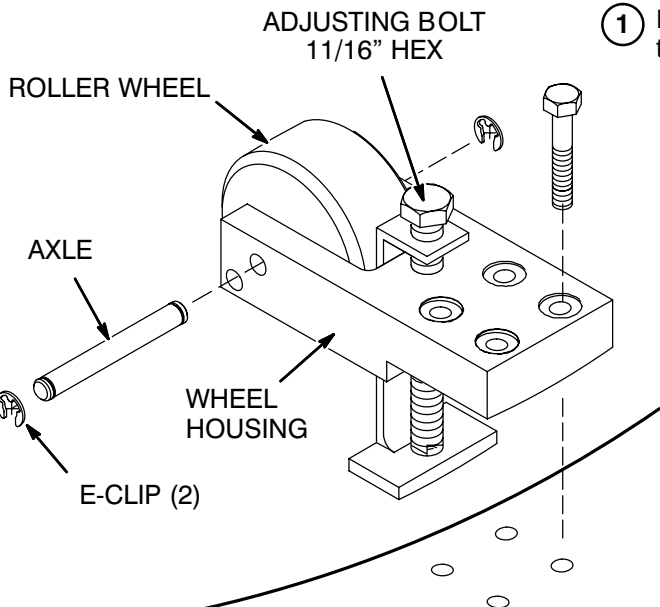
7 Connection of cables and installation of Cold Shrink tubing is now complete.

8 After completion of all connections above and prior to insertion of Body Coil into magnet, coil and place Gradient cables inside of Body Coil bore

INSTALLATION STEPS SUMMARY

- C1 – Installation of Roller Wheel Assemblies
- C2 – Remove Cradle Shipping Locking Bolts
- C3 – Position Body Coil at Front of Magnet Bore
- C4 – Locate Position of Body Coil in Magnet Bore

C1 – INSTALLATION OF ROLLER WHEEL ASSEMBLIES

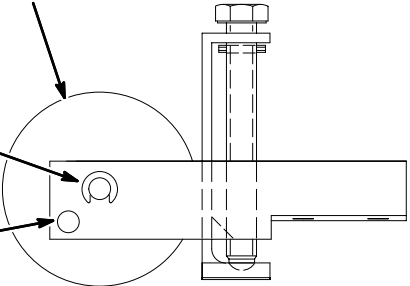


- Remove the two Coil Retainer Blocks (used to secure the coil to the cradle or pallet during shipment).
- Back the adjusting bolt all the way out by turning it counterclockwise.
- Remove one of the e-clips and remove wheel and axle.
- Place Wheel Housing in position over the four threaded holes on the coil and fasten with four bolts.
- Turn the adjusting bolt clockwise on all four Roller Wheel Assemblies to raise the coil to the point that the wheels and axles can be installed.
- Note which position to use and install wheels and axles.

Note: The following steps are used to mount the four Roller Wheel Assemblies to the coil, two at each end.

Use upper hole for GE magnets with passive or resistive shim and Oxford magnet.

Use lower hole for GE magnets without passive or resistive shim.

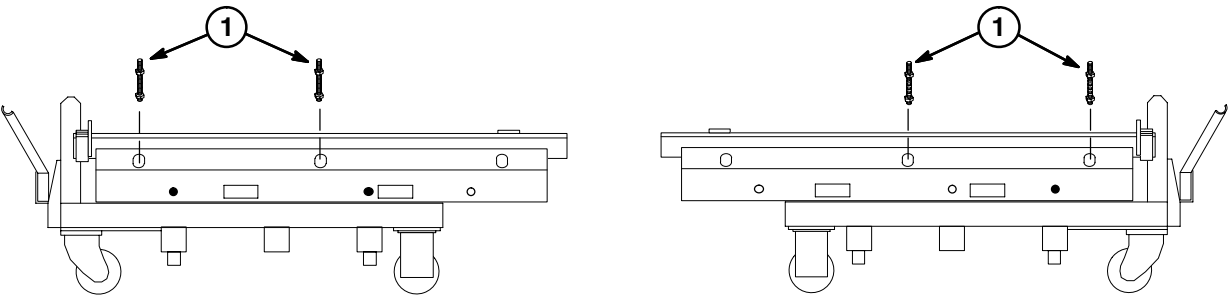


C2 – REMOVE CRADLE SHIPPING LOCKING BOLTS

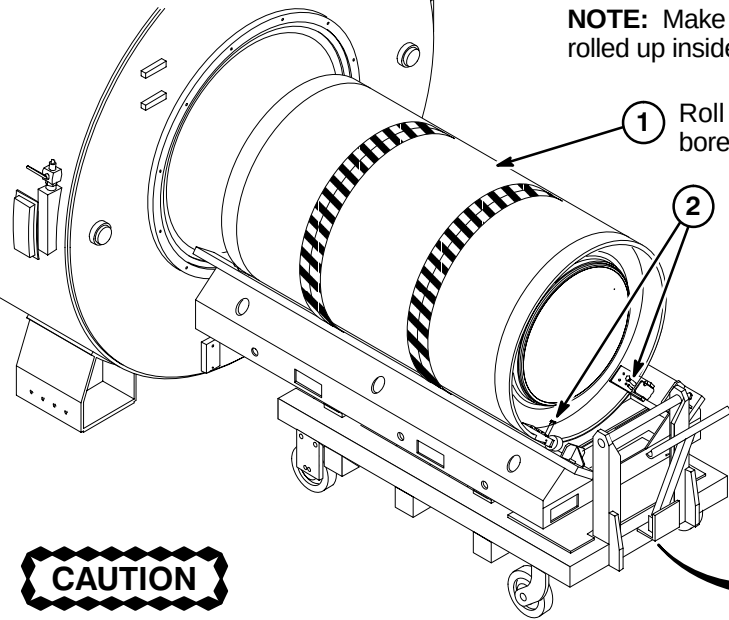
CAUTION

Attempting to raise the body coil with cradle shipping lock bolts installed will damage the elevating mechanism. The four Lock Bolts **MUST** be removed before attempting to raise the coil cart cradle.

- Remove the four Cradle Shipping Lock Bolts from the cart (two bolts on each side). This must be done before the Lift Assembly can be used to raise the cradle. These bolts belong with the cart; do not misplace them.



C3 – POSITION BODY COIL AT FRONT OF MAGNET BORE



NOTE: Make sure that all Body Coil gradient cables are rolled up inside the Body Coil prior to insertion into magnet.

- Roll the cart into position at the front of the bore for unloading the coil into the magnet.
- Lower the wheels on the Roller Wheel Assemblies by turning the adjusting screws counterclockwise.
- Use a 19-mm socket on the Lift Assembly Height Adjuster, turn clockwise (see caution) to raise the cradle even with the magnet bore.

CAUTION

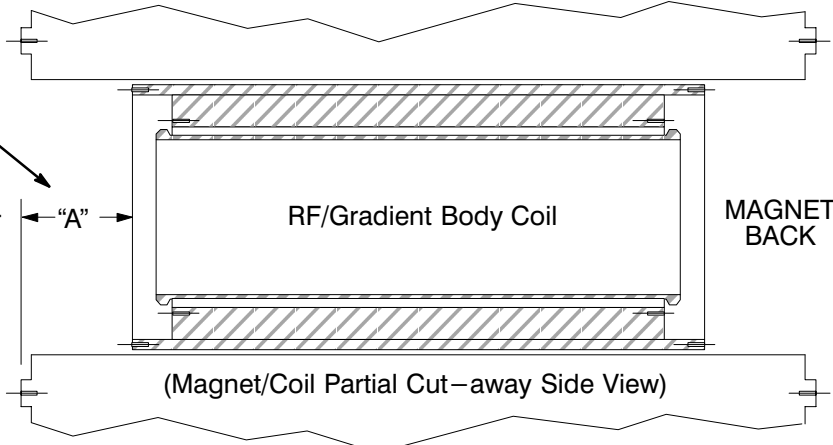
EQUIPMENT DAMAGE POSSIBILITY

Early carts had left-handed threads on the Lift Assembly Height Adjuster (Step A), i.e., clockwise rotation lowered the assembly, counterclockwise raised it. Gradually, these carts are being modified for right-handed threads. Be sure to check the cart that you are about to use. Damage to the mechanism can occur if the bolt is turned the wrong way with the full weight of a coil and cradle on it.

C4 – LOCATE POSITION OF BODY COIL IN MAGNET BORE

- Slowly and carefully roll the Body Coil into the magnet bore. **Do not roll Body Coil past rear end of magnet bore.**

- Measure distance from surface of front Magnet mounting flange to edge of outer gradient coil. (see Suggestion below)



- Refer to table below for distance "A". Select the correct distance according to Magnet type.

Body Coil	BRM-D		CRM	
Magnet	Oxford	GE S-I	All Other GE Magnets	
"A" Distance	13.18 Inch (335.0mm)	11.24 Inch (285.5mm)	12.00 Inch (304.5mm)	10.43 (264.5mm)
TOLERANCE: +0.0mm, -1.5mm				

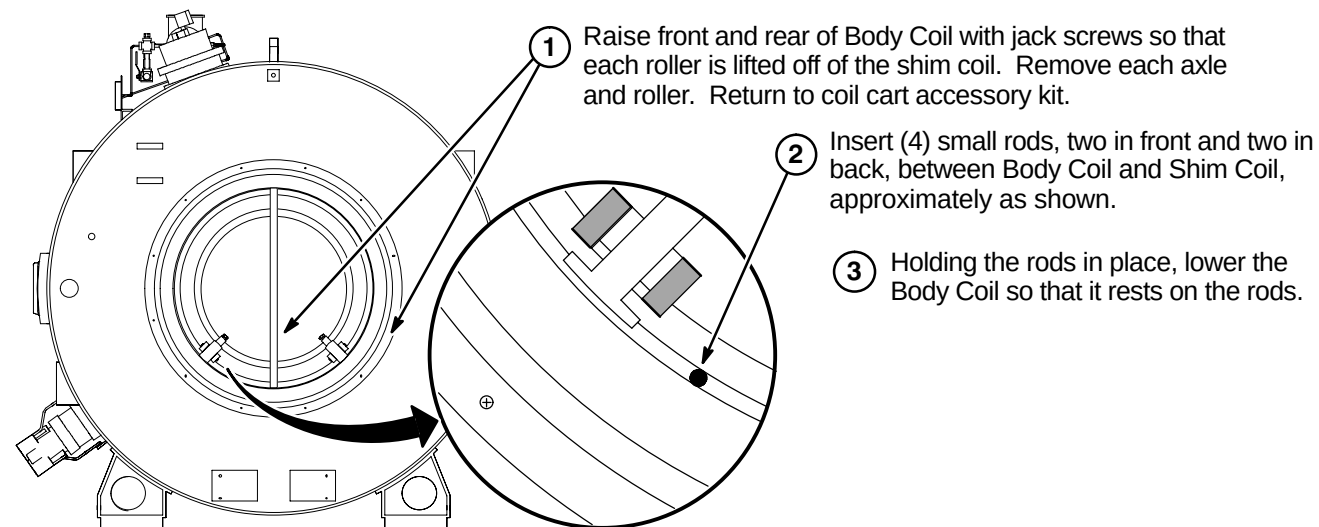
SUGGESTION:

A cabinet aluminum anti-tip leg, with ferrous leveling pad removed, can be placed against Magnet mounting flange and used as a straight edge to help measure distance from Magnet Interface Ring to Coil.

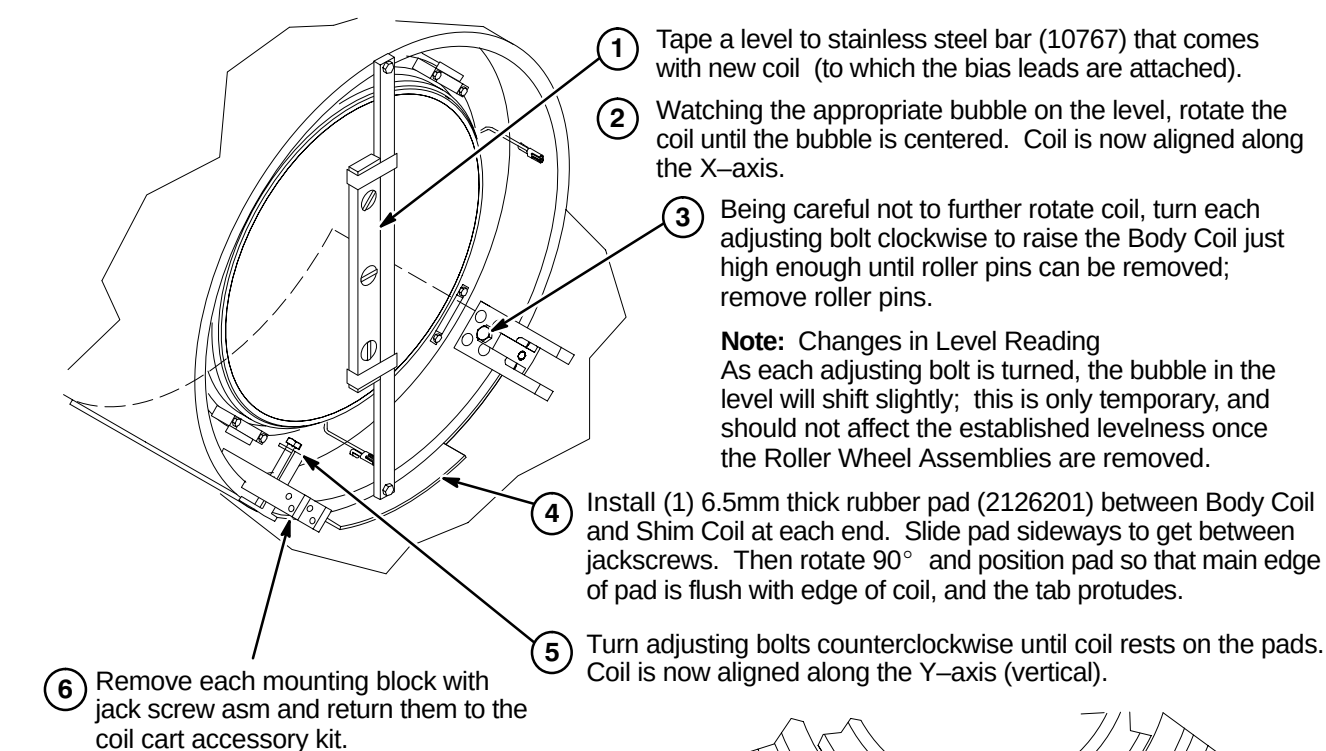
INSTALLATION STEPS SUMMARY

- ☐ D1a – Rotation Alignment of Body Coil in GE Magnet Bore with Shim Coil
☐ D2a – Centering Body Coil in GE Magnet Bore with Shim Coil
☐ D1b – Rotation Alignment of Body Coil in GE Magnet Bore (no Shim Coil)
☐ D2b – Centering Body Coil in GE Magnet Bore (no Shim Coil)
- } Perform either set

☐ D1a – ROTATION ALIGNMENT OF BODY COIL IN GE MAGNET WITH SHIM COIL



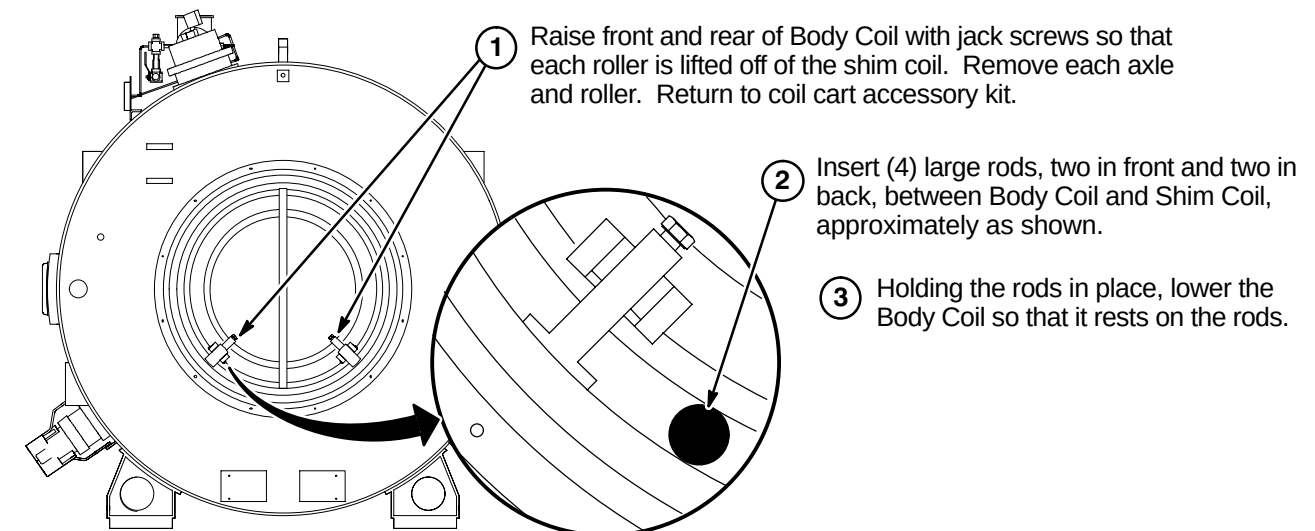
☐ **D2a – CENTERING BODY COIL IN GE MAGNET BORE WITH SHIM COIL**



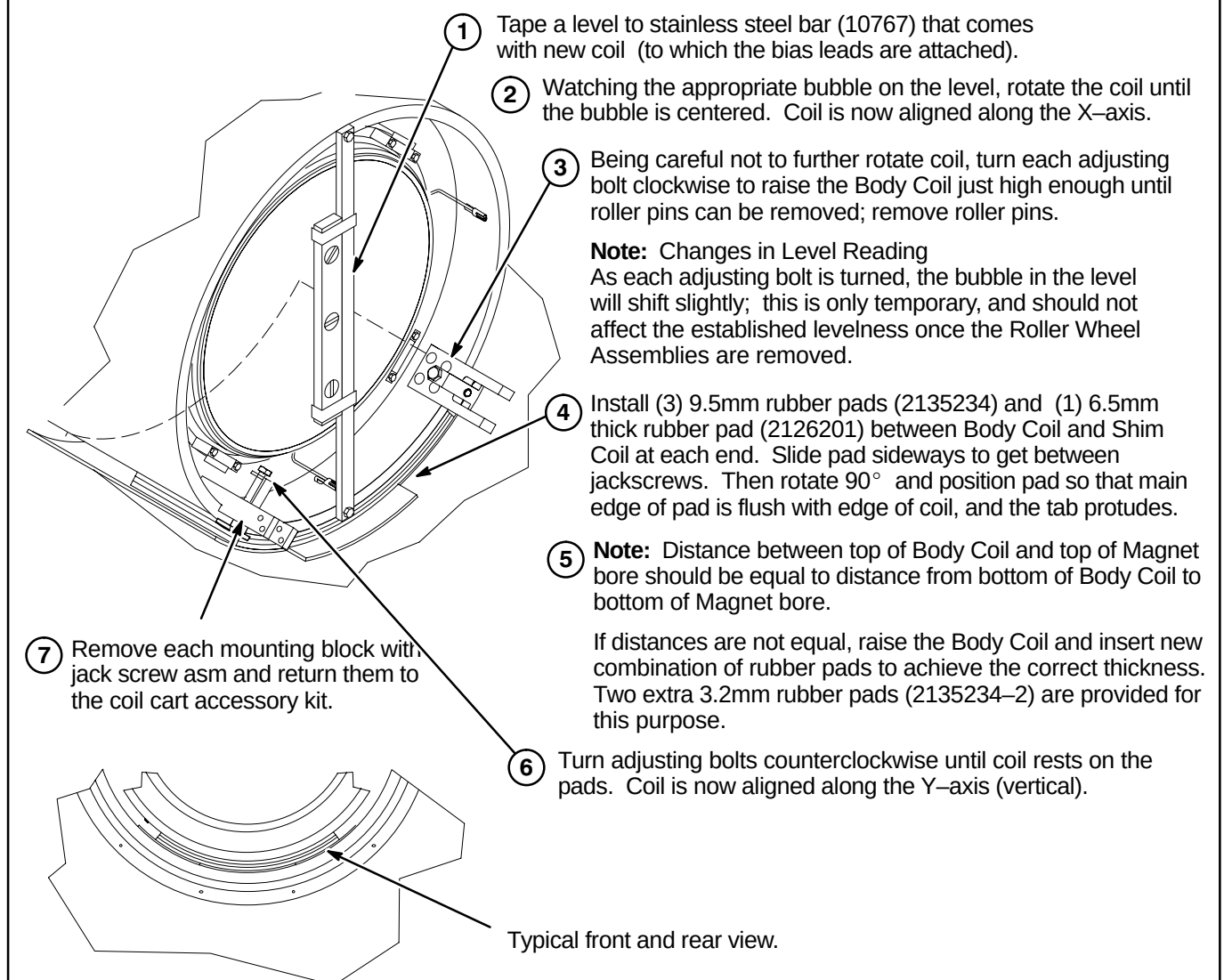
Note: Distance between top of Body Coil and top of Magnet bore should be equal to distance from bottom of Body Coil to bottom of Magnet bore.

Typical front and rear view.

☐ D1b – ROTATION ALIGNMENT OF BODY COIL IN GE MAGNET (NO SHIM COIL)



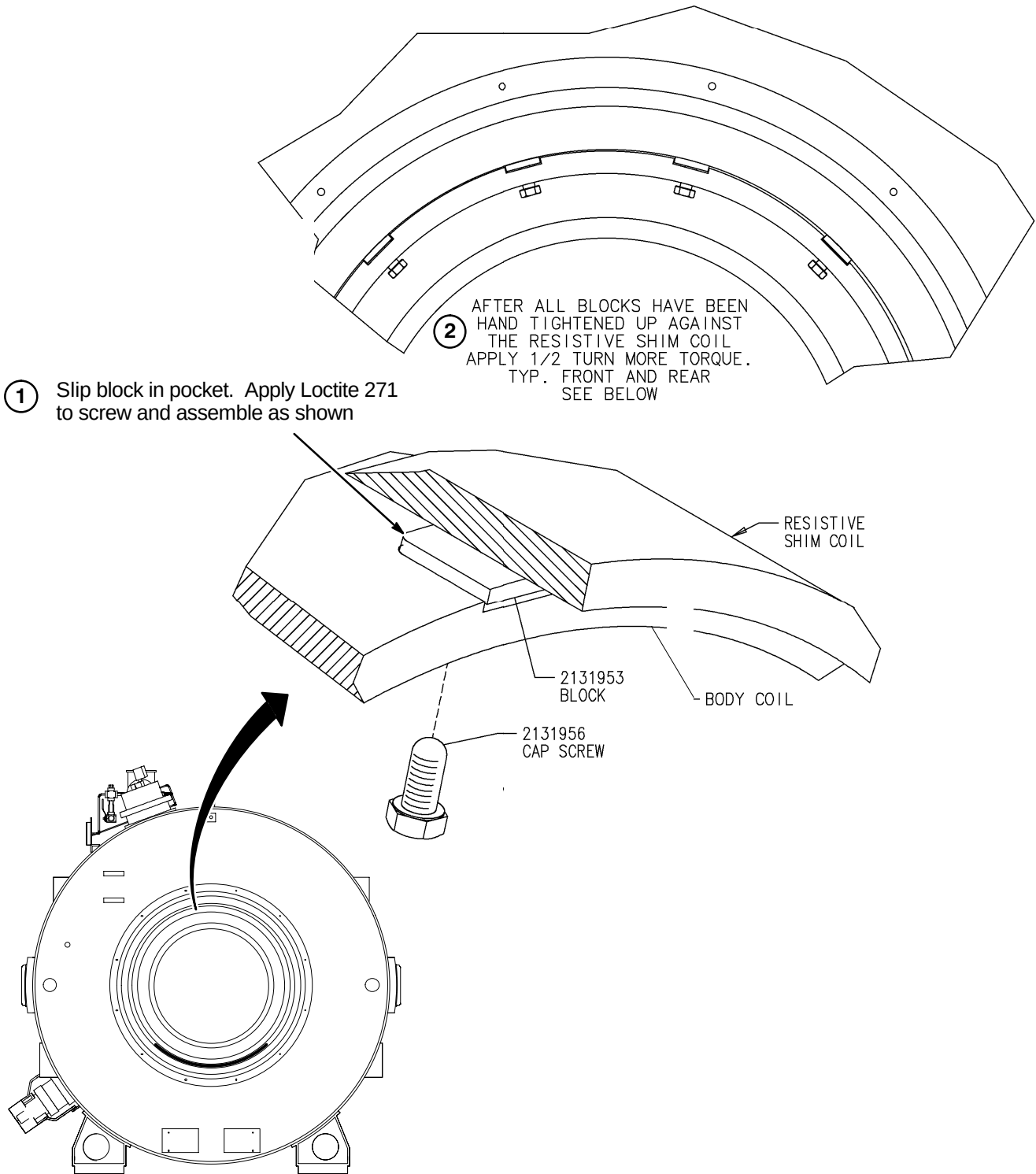
☐ **D2b – CENTERING BODY COIL IN GE MAGNET BORE (NO SHIM COIL)**



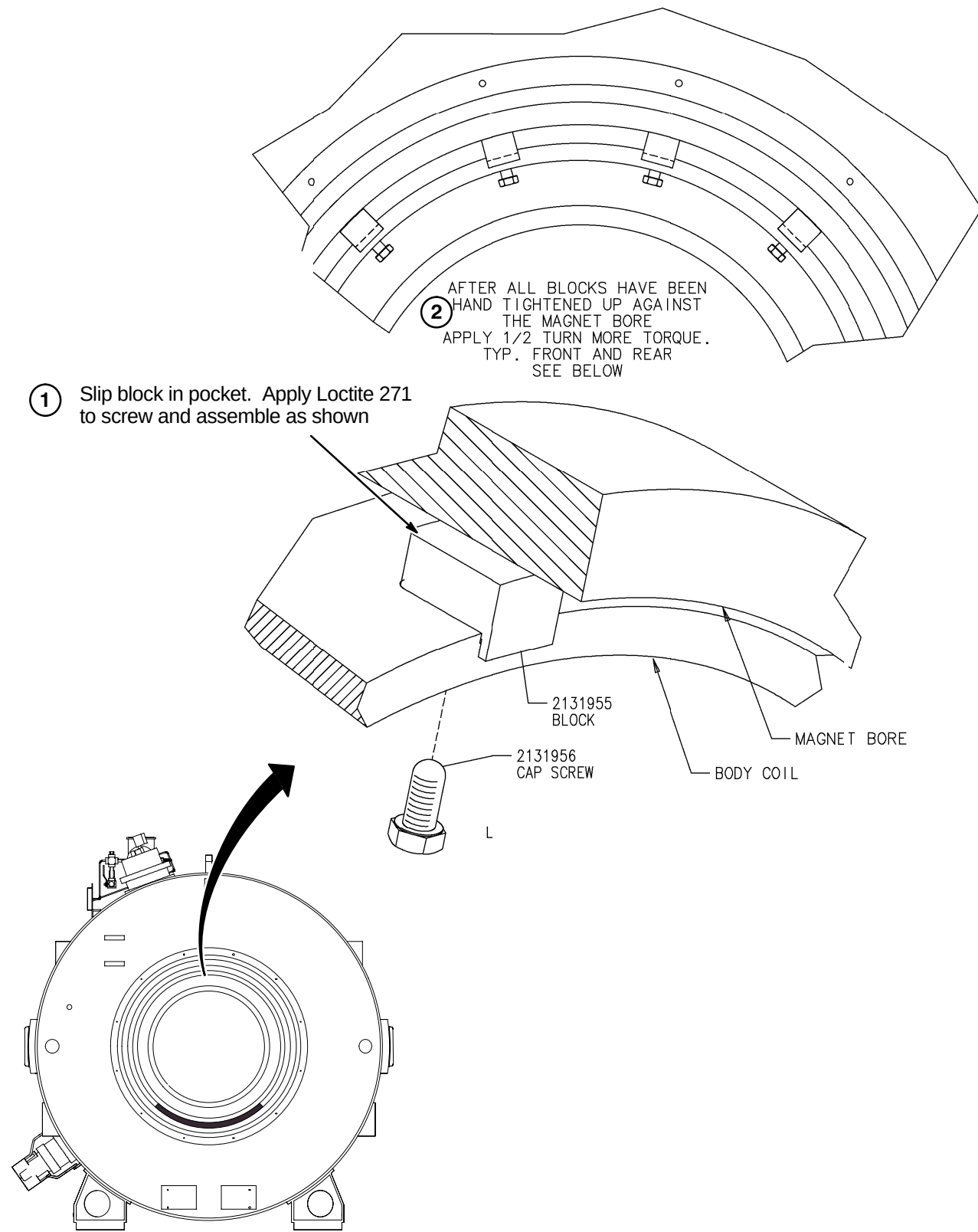
INSTALLATION STEPS SUMMARY

- ☐ E1 – Install Radial Support in GE Magnet with 900mm Diameter Bore
- OR
- ☐ E2 – Install Radial Support in GE Magnet with 950mm Diameter Bore

☐ E1 – INSTALL RADIAL SUPPORT IN GE MAGNET WITH 900mm DIAMETER BORE



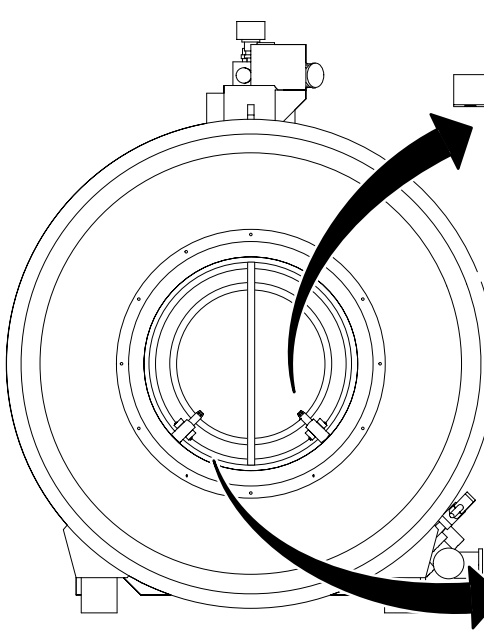
☐ E2 – INSTALL RADIAL SUPPORT IN GE MAGNET WITH 950mm DIAMETER BORE



INSTALLATION STEPS SUMMARY

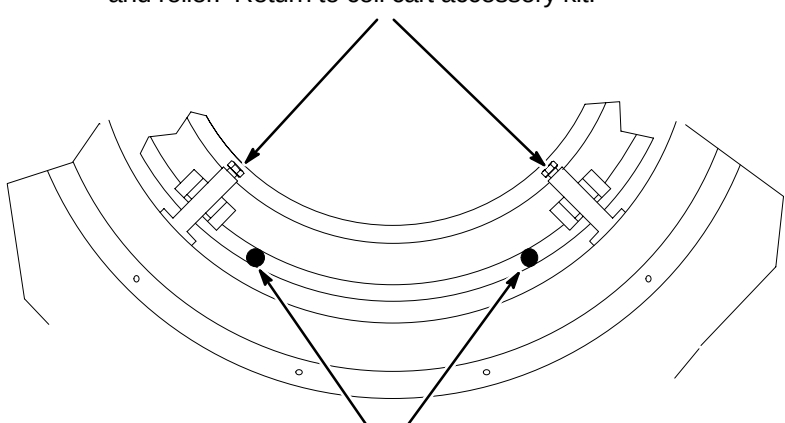
- F1 – Rotation Alignment of Body Coil in Oxford Magnet Bore
- F2 – Centering Body Coil in Oxford Magnet Bore

F1 – ROTATION ALIGNMENT OF BODY COIL IN OXFORD MAGNET BORE



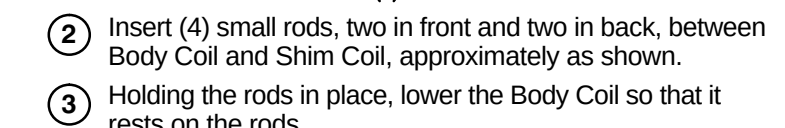
Note: For Oxford, make sure roller is mounted in this hole.

1 Raise front and rear of Body Coil with jack screws so that each roller is lifted off of the shim coil. Remove each axle and roller. Return to coil cart accessory kit.



2 Insert (4) small rods, two in front and two in back, between Body Coil and Shim Coil, approximately as shown.

3 Holding the rods in place, lower the Body Coil so that it rests on the rods.



F2 – CENTERING BODY COIL IN OXFORD MAGNET BORE

1 Tape a level to stainless steel bar (10767) that comes with new coil (to which the bias leads are attached).

2 Watching the appropriate bubble on the level, rotate the coil until the bubble is centered. Coil is now aligned along the X-axis.

3 Being careful not to further rotate coil, turn each adjusting bolt clockwise to raise the Body Coil just high enough until roller pins can be reoved; remove roller pins.

Note: Changes in Level Reading
As each adjusting bolt is turned, the bubble in the level will shift slightly; this is only temorary, and should not affect the established levelness once the Roller Wheel Assemblies are removed.

4 Install (2) 9.5mm thick rubber pad (2135234) between Body Coil and Shim Coil at each end.

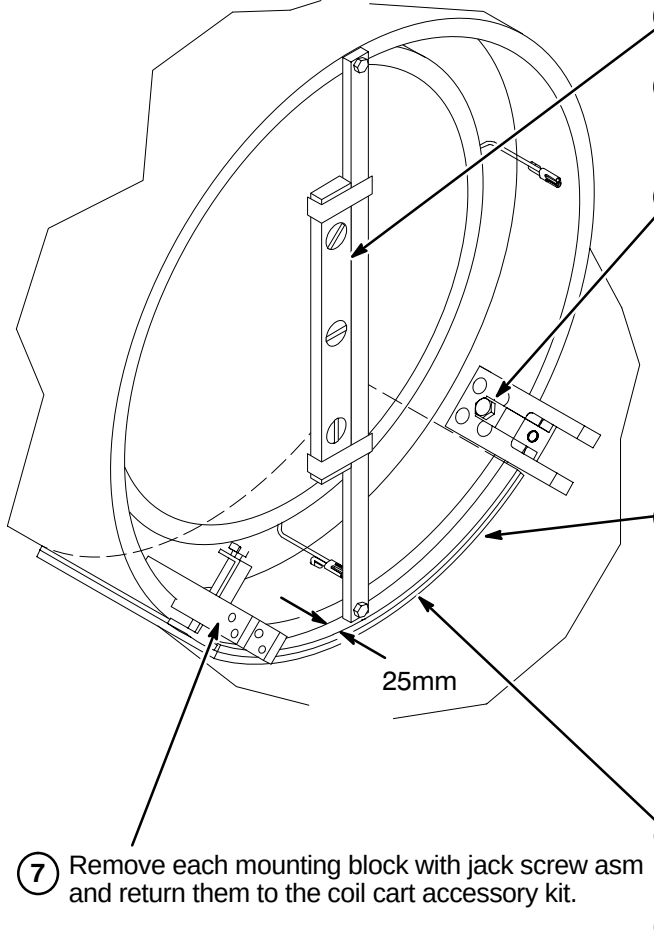
Note: Distance between top of Body Coil and top of Magnet bore should be equal to distance from bottom of Body Coil to bottom of Magnet bore.

If distances are not equal, raise the Body Coil and insert new combination of rubber pads to achieve the correct thickness. Two extra 6.5mm rubber pads (2126201) are provided for this purpose.

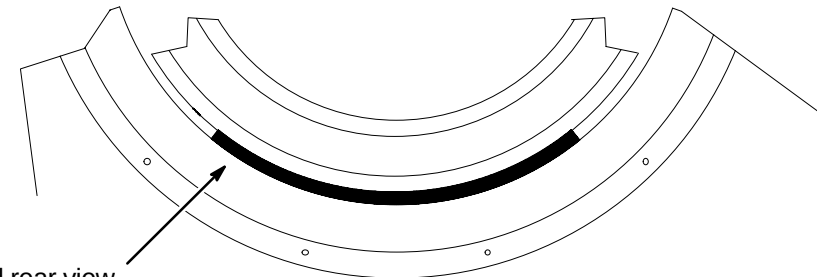
5 Center pads between mounting blocks and extend beyond gradient coil 25mm as shown.

6 Turn jackscrews and lower Body Coil down on pads.

7 Remove each mounting block with jack screw asm and return them to the coil cart accessory kit.



25mm

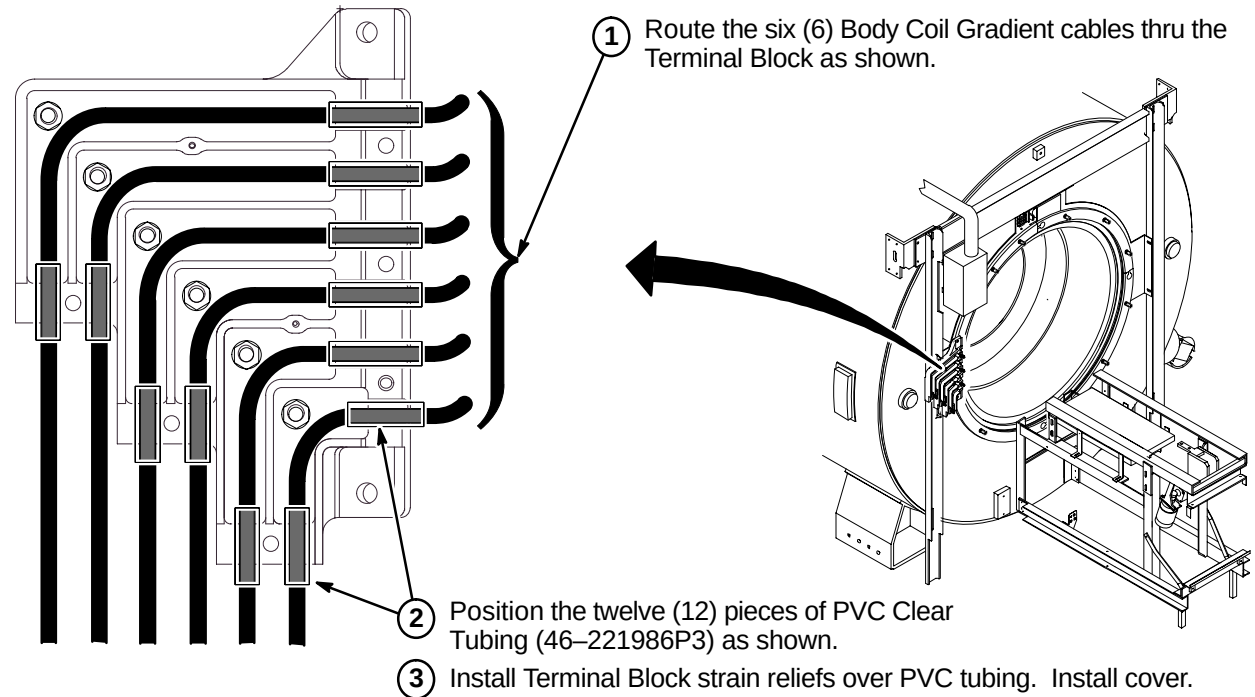


Typical front and rear view.

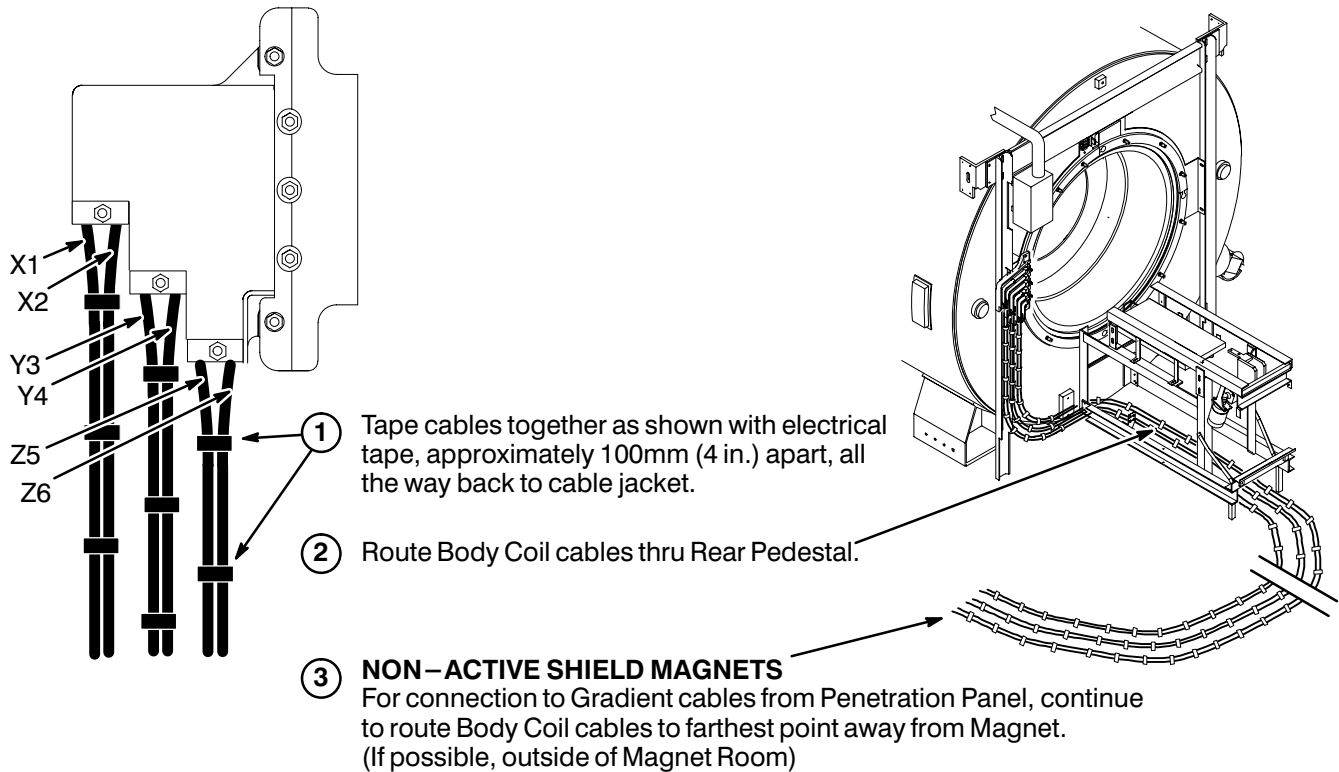
INSTALLATION STEPS SUMMARY

- G1 – Installation of PVC Clear Tubing
- G2 – Taping Together X, Y, & Z Gradient Cables from Body Coil
- G2 – Connection of Body Coil Gradient Cables to Runs 762, 763, and 764

□ G1 – INSTALLATION OF PVC CLEAR TUBING



□ G2 – TAPING TOGETHER X, Y, & Z GRADIENT CABLES FROM BODY COIL

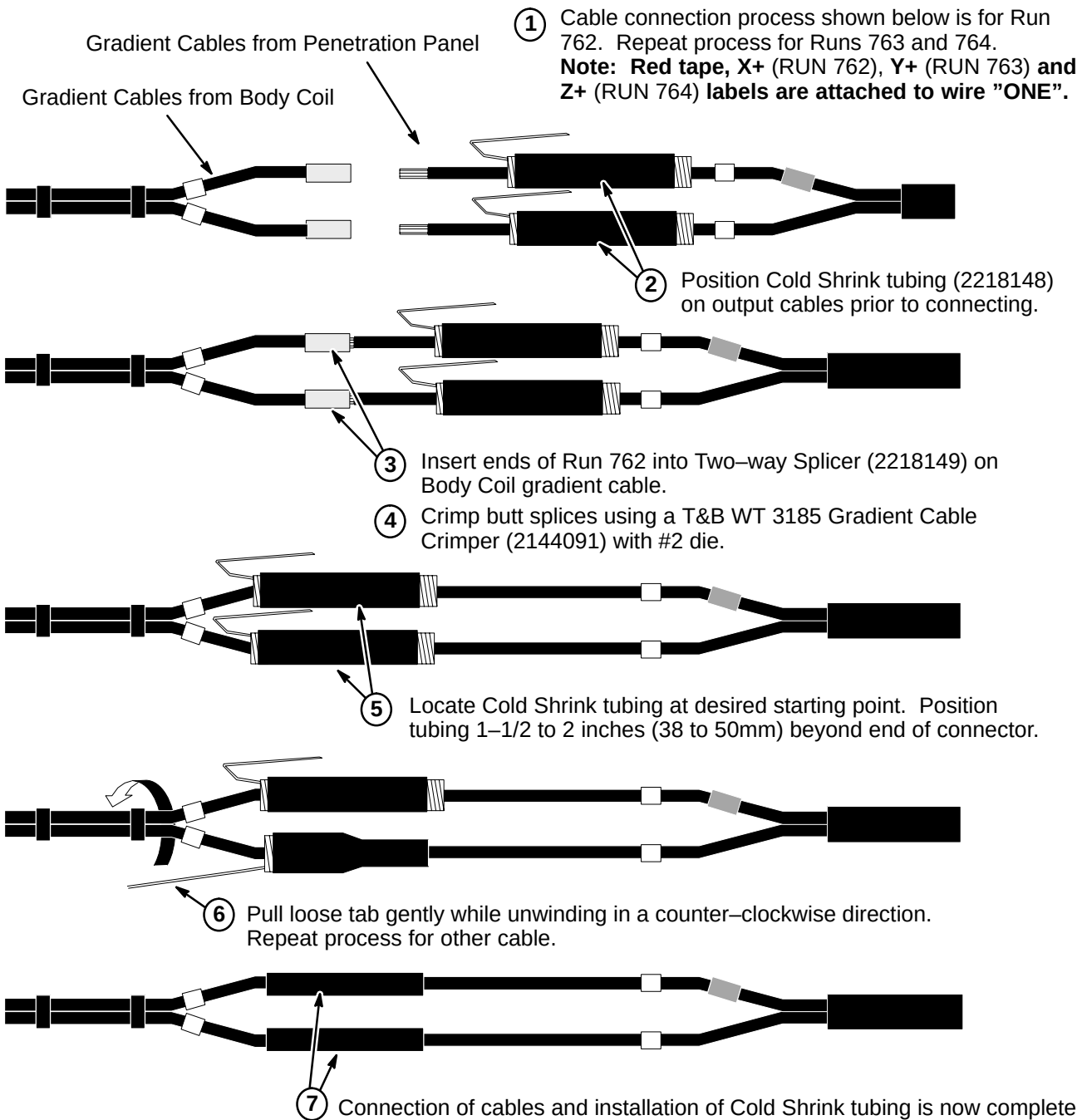


WARNING!

FERROUS MATERIAL HAZARD! THE RATCHETING CRIMP TOOL REQUIRED FOR THIS PROCEDURE CONTAINS FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME A DANGEROUS PROJECTILE. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

IF MAGNET IS NON-ACTIVE SHIELD AND IS AT FULL FIELD – MOVE END OF BODY COIL EXTENSION CABLES TO FARTHEST POINT AWAY FROM MAGNET TO PERFORM BUTT SPLICE CONNECTION TO RUNS 762, 763, AND 764 COMING FROM PENETRATION PANEL.

□ G3 – CONNECTION OF BODY COIL GRADIENT CABLES TO RUNS 762, 763, & 764



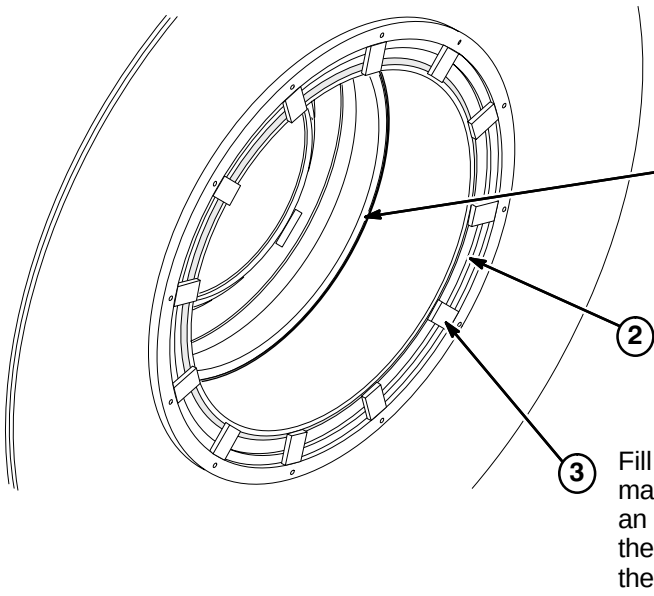
INSTALLATION STEPS SUMMARY

- ☐ H1 – Body Coil Installation Installation
- ☐ H2a – Installing Insulation in Magnet with Passive or Resistive Shim
- ☐ H2b – Installing Insulation in Magnet without Passive or Resistive Shim
- ☐ H3 – Installation of Velcro in Magnet Bore
- ☐ H4 – Acoustic Collar Installation

☐ H1 – BODY COIL INSULATION INSTALLATION

1. Locate accessories box that contains body coil soundproofing and air–sealing insulation.
2. Proceed to Step A2 or A3.

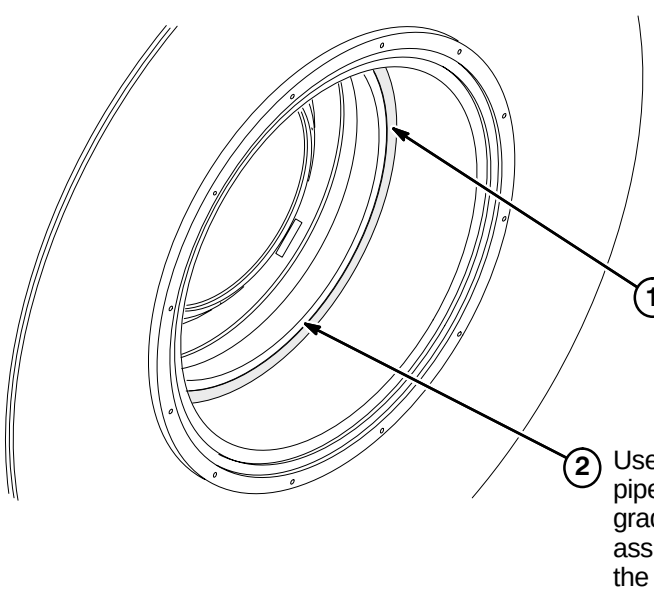
☐ H2a – INSTALLING INSULATION IN MAGNET WITH PASSIVE OR RESISTIVE SHIM



Note:
Rope caulk, pipe insulation and acoustical collar are installed after body coil centering.

- 1 Fill the gap between the Passive Shim Coil and outer Gradient Coil at each end. Use one, two, or three strips (rolled together) of rope caulk (46–287118P1) around entire perimeter.
- 2 Fill the gap between the Magnet bore and the Passive Shim Coil at each end. Use 8 in. (205 mm) sections of Pipe Insulation (46–320868P1) around entire perimeter.
- 3 Fill the gap between the Passive Shim drum stops and magnet bore. Use rope caulk (46–287118P1) to provide an air–tight seal around entire perimeter at the back of the magnet. This is to assure maximum air flow between the RF coil and the Gradient Coil.

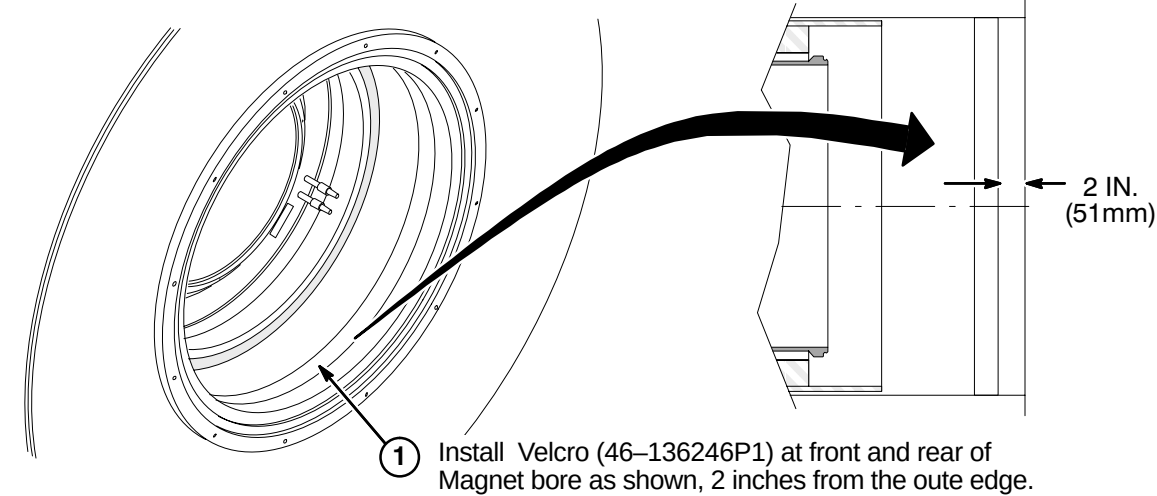
☐ H2b - INSTALLING INSULATION IN MAGNET WITHOUT PASSIVE OR RESISTIVE SHIM



Note:
Rope caulk, pipe insulation, and acoustical collars are installed after body coil centering.

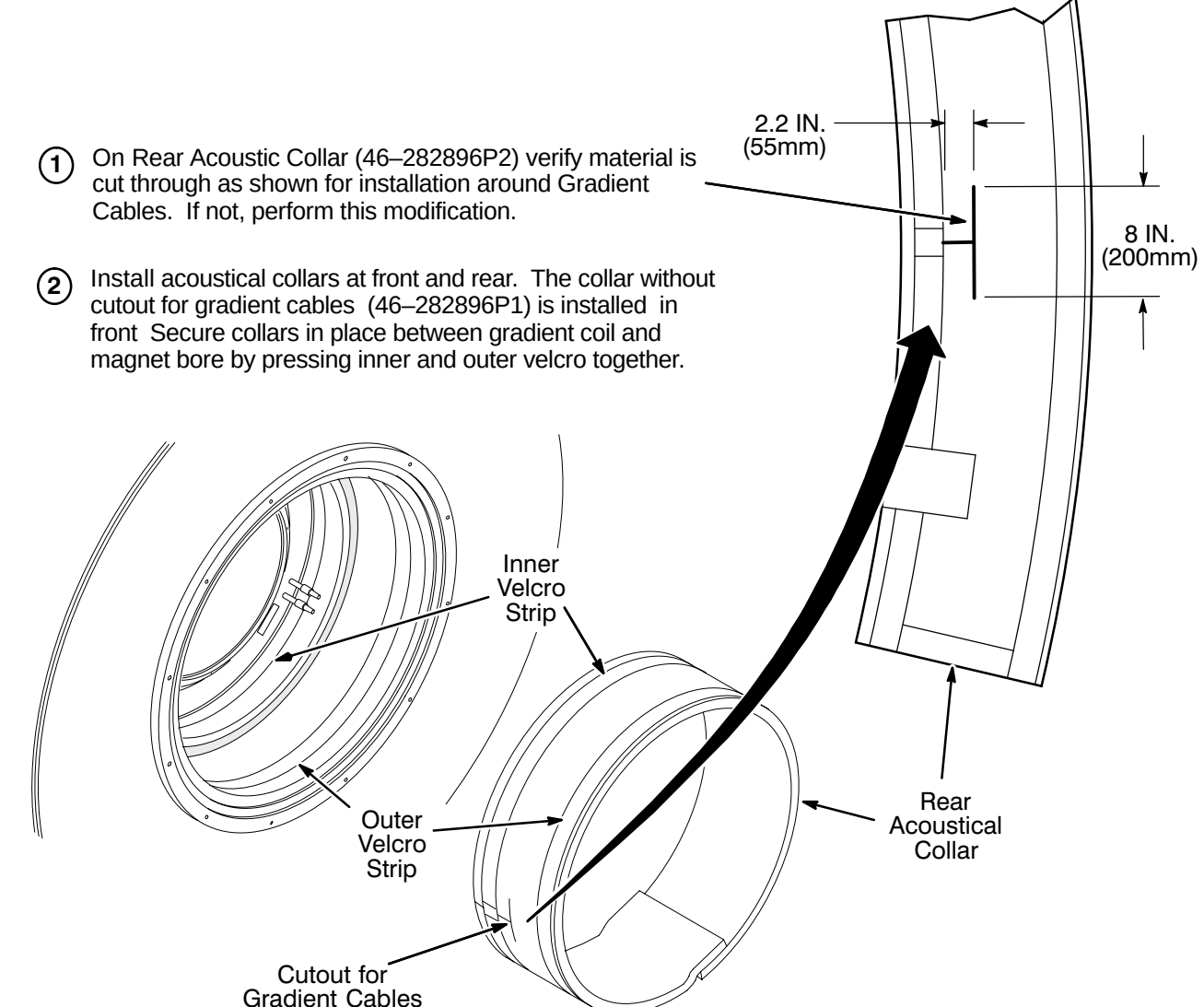
- 1 Fill the gap between the Magnet bore and outside of Gradient Coil at each end. Use Pipe Insulation (46–320868P1) and rope caulk (46–287118P1) around entire perimeter.
- 2 Use rope caulk (46–287118P1) as needed to supplement pipe insulation required for an air– tight seal between gradient coil and magnet bore at back of magnet. This is to assure maximum cooling air flow between the RF coil and the Gradient Coil.

☐ H3 – INSTALLATION OF VELCRO IN MAGNET BORE



1 Install Velcro (46–136246P1) at front and rear of Magnet bore as shown, 2 inches from the oute edge.

☐ H4 – ACOUSTIC COLLAR INSTALLATION



1 On Rear Acoustic Collar (46–282896P2) verify material is cut through as shown for installation around Gradient Cables. If not, perform this modification.

2 Install acoustical collars at front and rear. The collar without cutout for gradient cables (46–282896P1) is installed in front. Secure collars in place between gradient coil and magnet bore by pressing inner and outer velcro together.

Labels: Inner Velcro Strip, Outer Velcro Strip, Cutout for Gradient Cables, Rear Acoustical Collar

Dimensions: 2.2 IN. (55mm), 8 IN. (200mm)

INSTALLATION STEPS SUMMARY

- ☐ A1 – Body Coil Installation Procedure
- ☐ A2 – Body Coil Cart and Cradle Components
- ☐ A3 – Loading Body Coil and Cradle or Pallet on Coil Cart
- ☐ A4 – Using Cart to transport Body Coil

NOTE: For Transportable or Relocatable Upgraded on Site

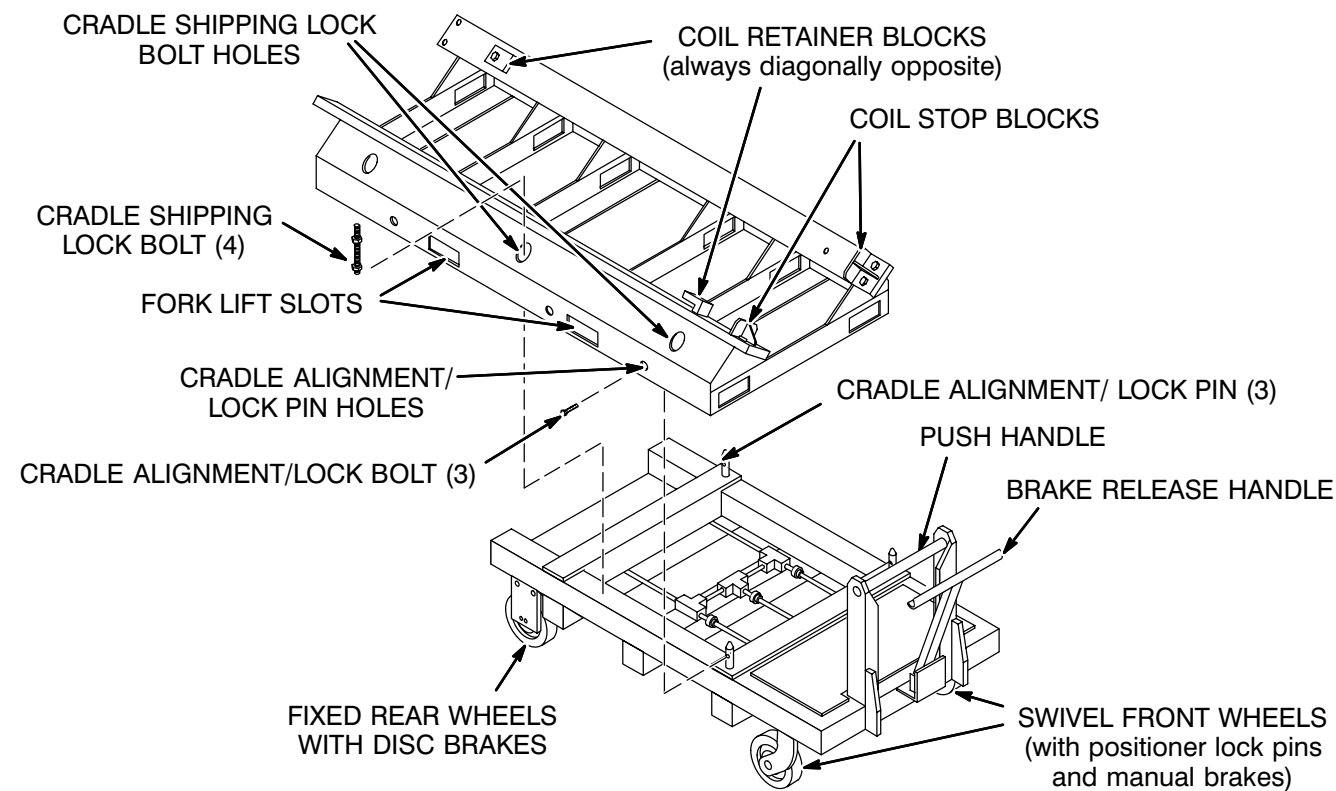
Because of the weight of the coil cart/cradle, a lift truck may be required to lift the new body coil cart/cradle into the magnet room. Also, the floor may need to be reinforced with 3/4 inch plywood or other plate where ever the coil cart/cradle may roll. Contact van manufacturer for recommendation.

☐ A1 – BODY COIL INSTALLATION PROCEDURE

1. Remove existing pre–Horizon Body Coil. Refer to procedures in *CD–ROM Direction 2124201, MR Release 5.x Signa Service Methods*, Replacement: Gradient Coil, Quad II/Roemer Coil 1.5T Replacement
2. This Upgrade Instruction contains information for installing Horizon Body Coils in all GE Magnets and the Oxford Magnet. Different installation requirements are found in the following procedure steps:
B4 - Different dimensions are given for positioning Body Coil in Oxford, GE S–I, and all other GE magnet bores.
C1a thru C2b - Instructions for centering Body Coil in all GE magnets.
D1 thru D2 - Instructions for centering Body Coil in Oxford magnet.
3. Horizon Upgrade body coils may be delivered on a cart/cradle or on a shipping pallet (cradle).
If body coil is delivered on a coil cart and cradle combination:
The body coil is transported to the magnet room. Review Step A2, skip Step A3, and proceed to Step A4.
If body coil is delivered on a pallet (or cradle) for use with a coil cart:
The body coil is loaded on the regional coil cart. Review Step A2 and proceed to Step A3.

☐ A2 – BODY COIL CART AND CRADLE COMPONENTS

The Coil Cart and Cradle combination, as shown below is used to install the Body Coil in the magnet bore. The body coil may also be shipped on a disposable wood and metal pallet which performs the functions of the metal cradle shown below.



☐ A3 – LOADING BODY COIL AND CRADLE OR PALLET ON COIL CART

- 1 Have a fork lift truck raise the body coil and cradle (or pallet) combination. The gradient leads are at the end of the cradle away from the handle, and the Coil Stop Blocks are near the handle.

CAUTION

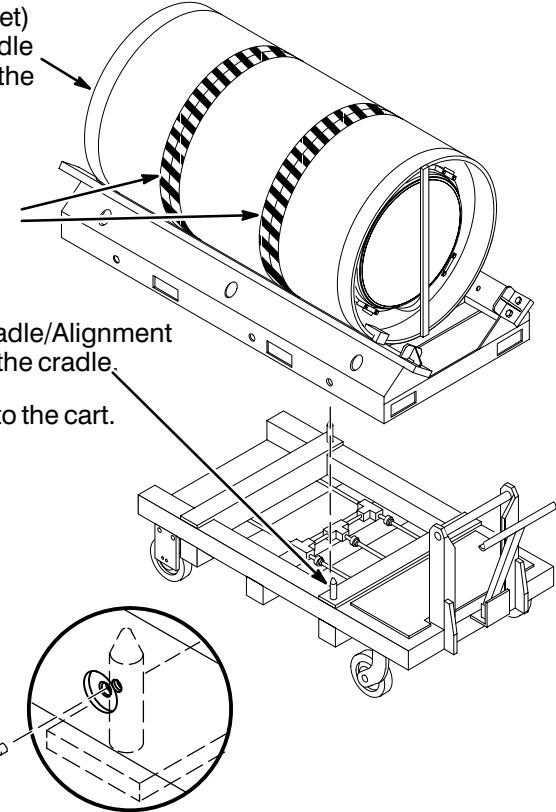
Do not lift or apply pressure to area of Body Coil marked with black/yellow tape labelled “NO LOAD ZONE”. Doing so will damage the Body Coil.

- 2 Position cart beneath cradle with Cradle/Alignment Lock Pins aligned with their holes in the cradle.
- 3 Lower body coil cradle (or pallet) onto the cart.

CAUTION

Not having the cradle anchored to the cart could cause the coil/cradle to tip. The three Cradle Alignment/Lock Bolts MUST be in place before attempting to move the coil cart.

- 4 Insert the three Cradle Alignment/Lock Bolts.

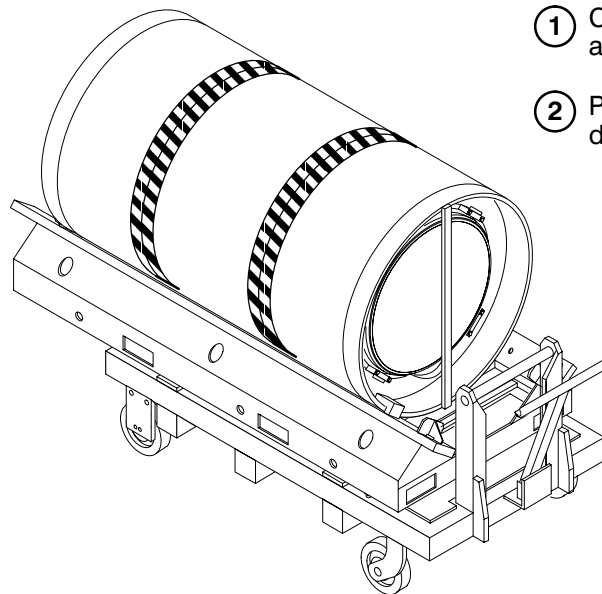


☐ A4 – USING CART TO TRANSPORT BODY COIL

CAUTION

Avoid possible bodily injury. The combined coil/cradle/cart weigh about 3300 pounds. To avoid physical injury, and to control this amount of weight as it moves, a minimum of four people must be used.

- 1 Check route to magnet room and verify route clearance and sufficient manpower to overcome inclines.
- 2 Place plywood sheets or metal plates along route to distribute weight and protect floor surface from damage.



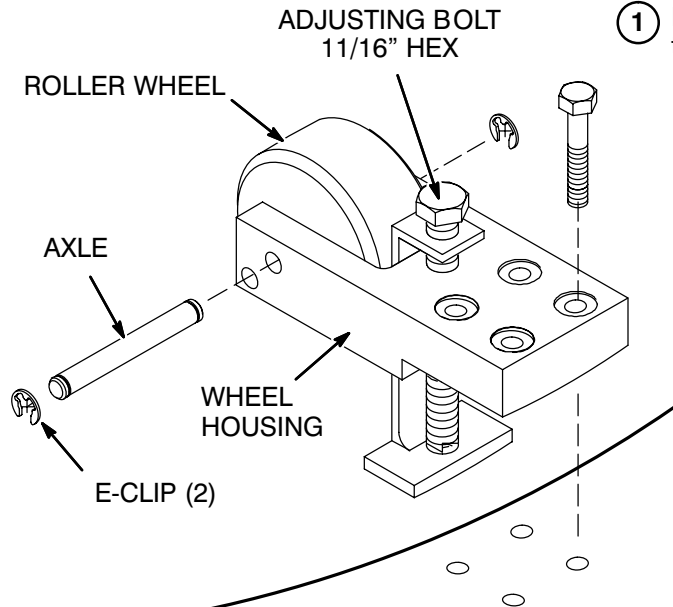
Note: The Brake Release Handle controls the disc brakes on the fixed rear wheels. It must be pushed against the Push Handle in order to move the cart.

- 3 Roll coil cart to magnet room and position at front center of magnet bore for unloading into the magnet.

INSTALLATION STEPS SUMMARY

- ❑ B1 – Installation of Roller Wheel Assemblies
- ❑ B2 – Remove Cradle Shipping Locking Bolts
- ❑ B3 – Position Body Coil at Front of Magnet Bore
- ❑ B4 – Locate Position of Body Coil in Magnet Bore

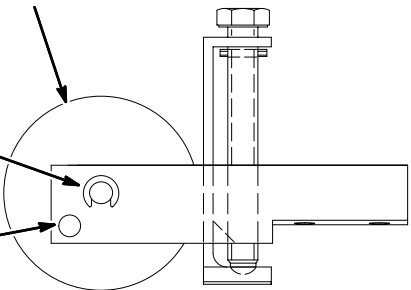
❑ B1 – INSTALLATION OF ROLLER WHEEL ASSEMBLIES



- ❶ Remove the two Coil Retainer Blocks (used to secure the coil to the cradle or pallet during shipment).
- Note:** The following steps are used to mount the four Roller Wheel Assemblies to the coil, two at each end.
- ❷ Back the adjusting bolt all the way out by turning it counterclockwise.
- ❸ Remove one of the e-clips and remove wheel and axle.
- ❹ Place Wheel Housing in position over the four threaded holes on the coil and fasten with four bolts.
- ❺ Turn the adjusting bolt clockwise on all four Roller Wheel Assemblies to raise the coil to the point that the wheels and axles can be installed.
- ❻ Note which position to use and install wheels and axles.

Use upper hole for **GE** magnets with passive or resistive shim and **Oxford** magnet.

Use lower hole for **GE** magnets without passive or resistive shim.

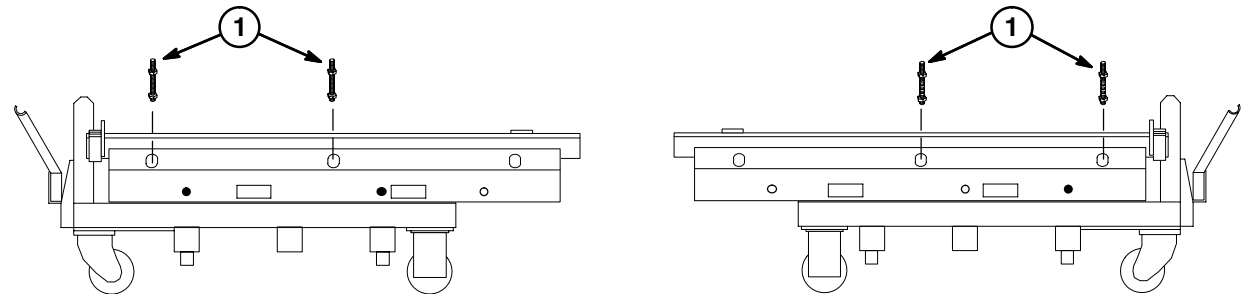


❑ B2 – REMOVE CRADLE SHIPPING LOCKING BOLTS

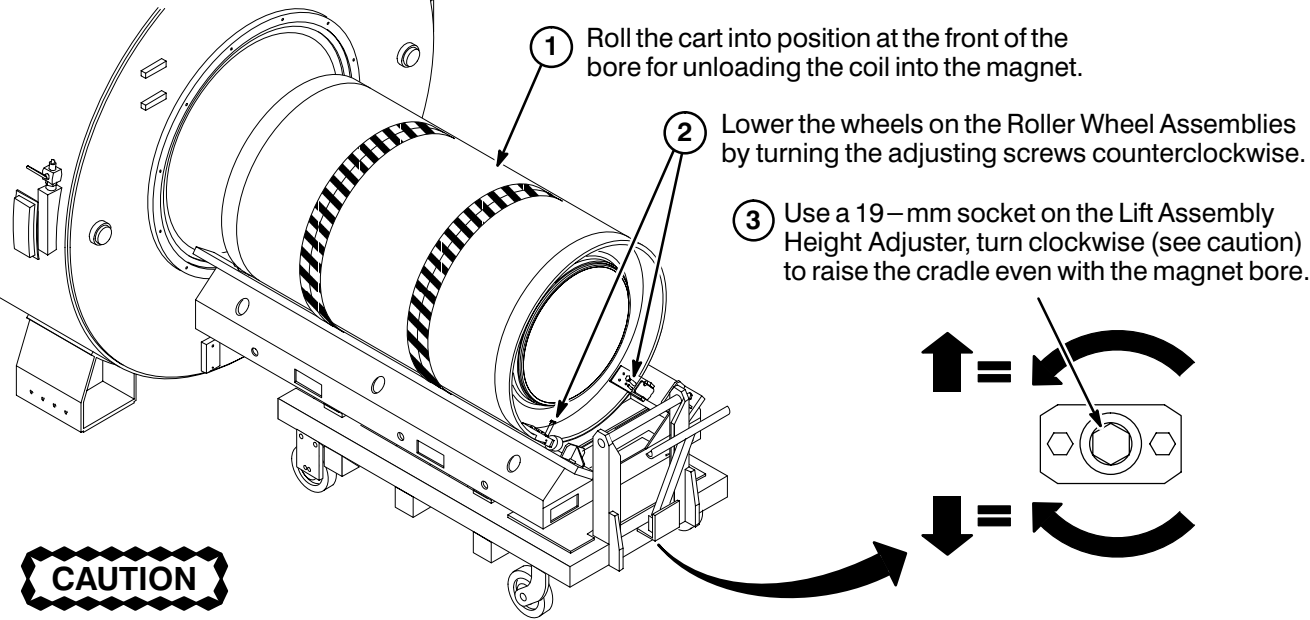
CAUTION

Attempting to raise the body coil with cradle shipping lock bolts installed will damage the elevating mechanism. The four Lock Bolts **MUST** be removed before attempting to raise the coil cart cradle.

- ❶ Remove the four Cradle Shipping Lock Bolts from the cart (two bolts on each side). This must be done before the Lift Assembly can be used to raise the cradle. These bolts belong with the cart; do not misplace them.



❑ B3 – POSITION BODY COIL AT FRONT OF MAGNET BORE



CAUTION

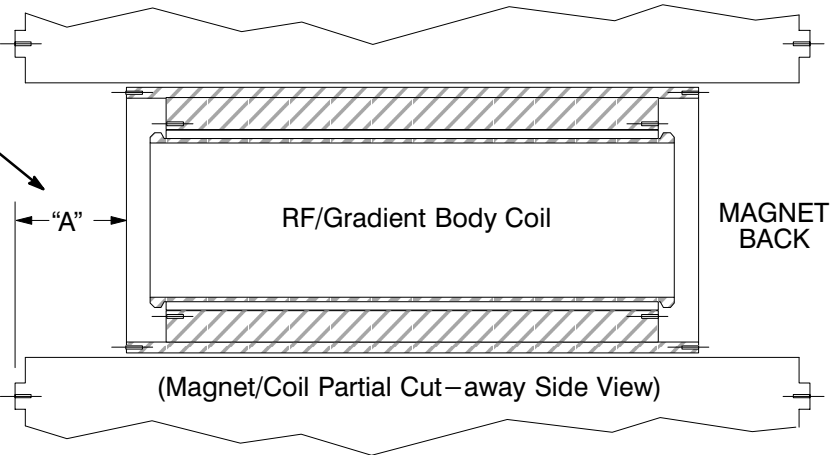
EQUIPMENT DAMAGE POSSIBILITY
Early carts had left-handed threads on the Lift Assembly Height Adjuster (Step ❸), i.e., clockwise rotation lowered the assembly, counterclockwise raised it. Gradually, these carts are being modified for right-handed threads. Be sure to check the cart that you are about to use. Damage to the mechanism can occur if the bolt is turned the wrong way with the full weight of a coil and cradle on it.

❑ B4 – LOCATE POSITION OF BODY COIL IN MAGNET BORE

- ❶ Slowly and carefully roll the Body Coil into the magnet bore. Do not roll Body Coil past rear end of magnet bore.

- ❷ Measure distance from surface of front Magnet mounting flange to edge of outer gradient coil. (see Suggestion below)

- ❸ Refer to table below for distance “A”. Select the correct distance according to Magnet type.



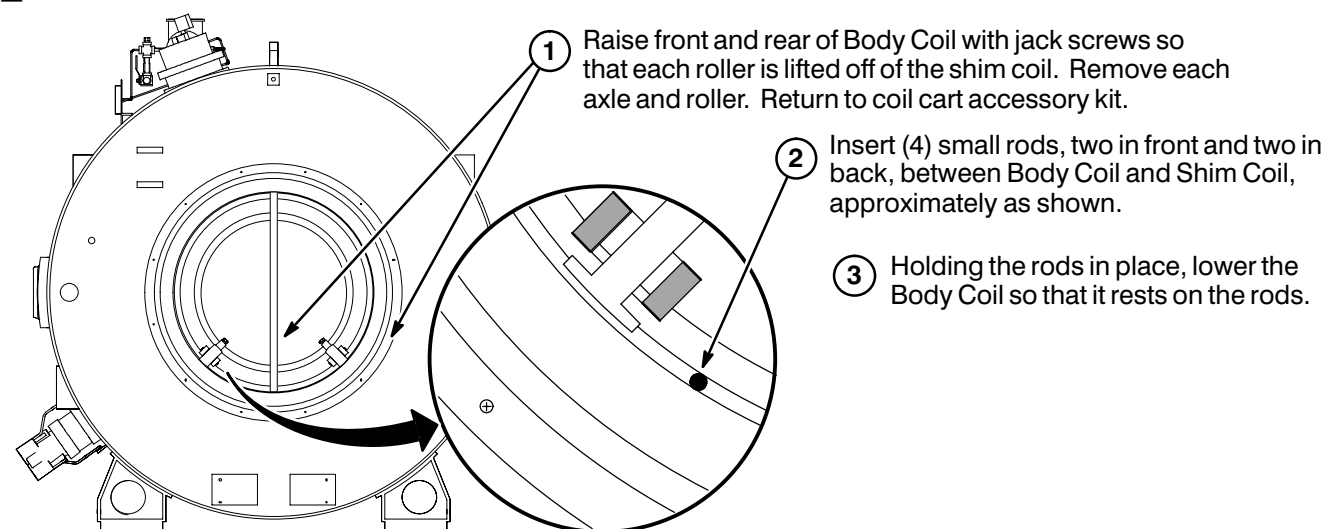
	Oxford	GE S-I	All Other GE Magnets
“A” Distance	13.18 Inch (335.0mm)	11.24 Inch (285.5mm)	12.00 Inch (304.5mm)
TOLERANCE: +0.0mm, -1.5mm			

SUGGESTION:
A cabinet aluminum anti-tip leg, with ferrous leveling pad removed, can be placed against Magnet mounting flange and used as a straight edge to help measure distance from Magnet Interface Ring to Coil.

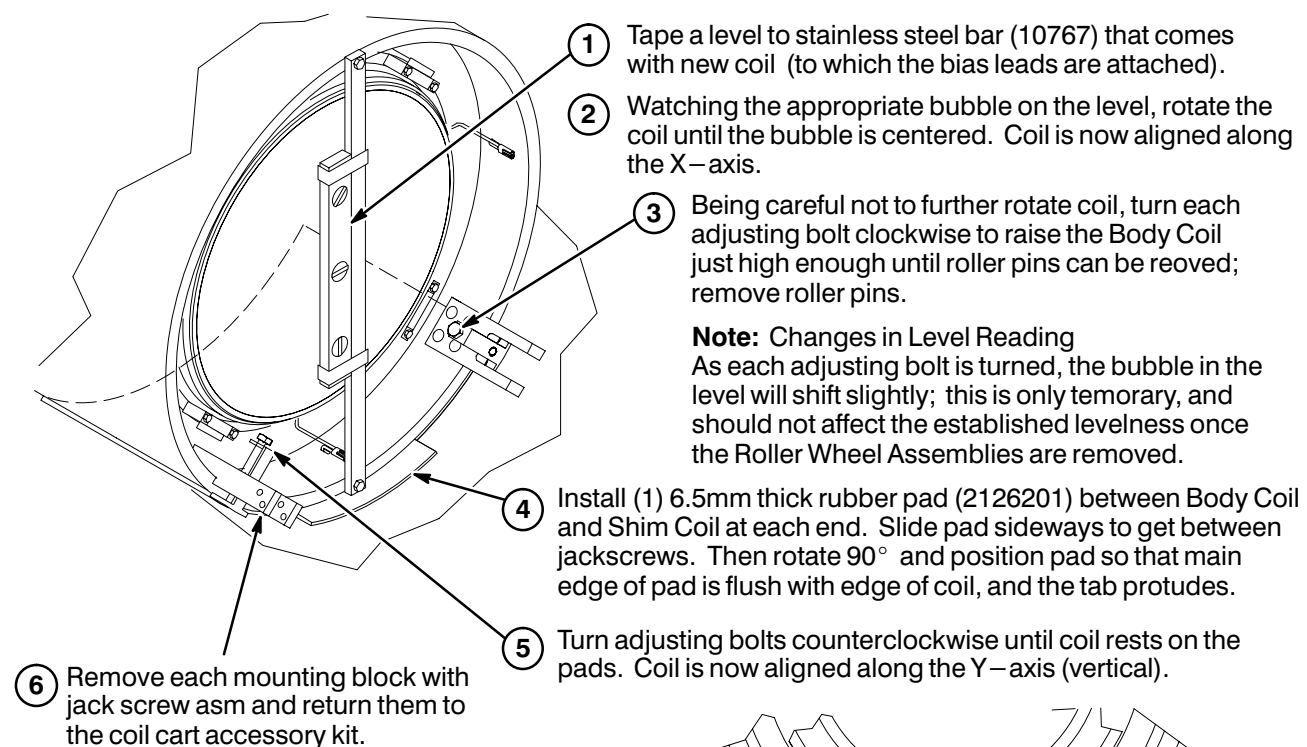
INSTALLATION STEPS SUMMARY

- ☐ C1a – Rotation Alignment of Body Coil in GE Magnet Bore with Shim Coil
☐ C2a – Centering Body Coil in GE Magnet Bore with Shim Coil
☐ C1b – Rotation Alignment of Body Coil in GE Magnet Bore (no Shim Coil)
☐ C2b – Centering Body Coil in GE Magnet Bore (no Shim Coil)
- } Perform either set

☐ C1a – ROTATION ALIGNMENT OF BODY COIL IN GE MAGNET WITH SHIM COIL



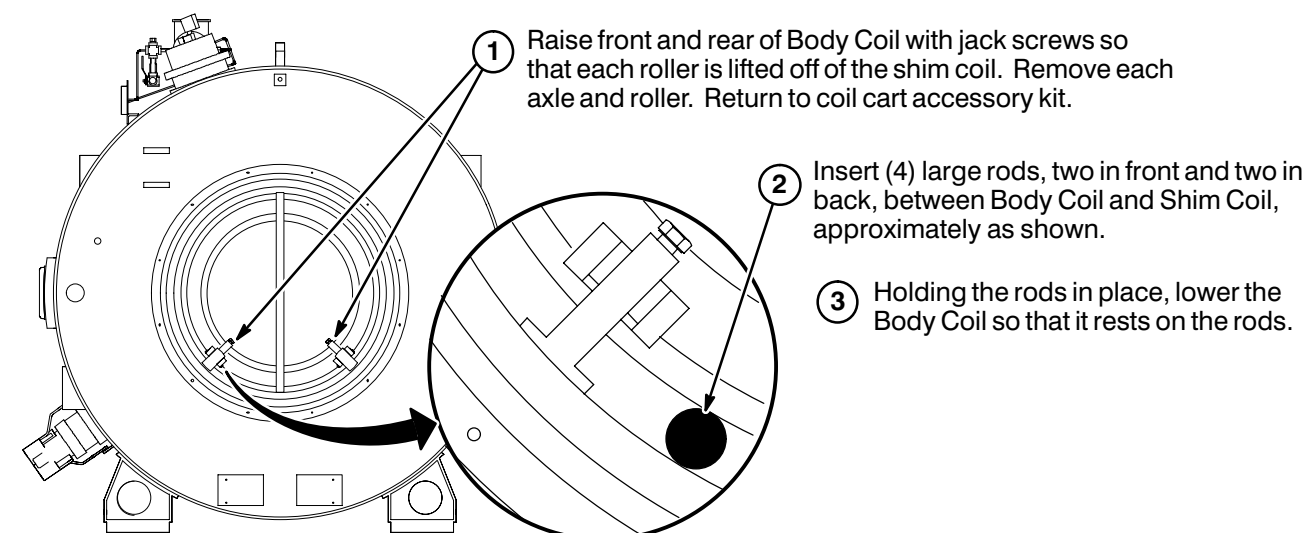
☐ **C2a – CENTERING BODY COIL IN GE MAGNET BORE WITH SHIM COIL**



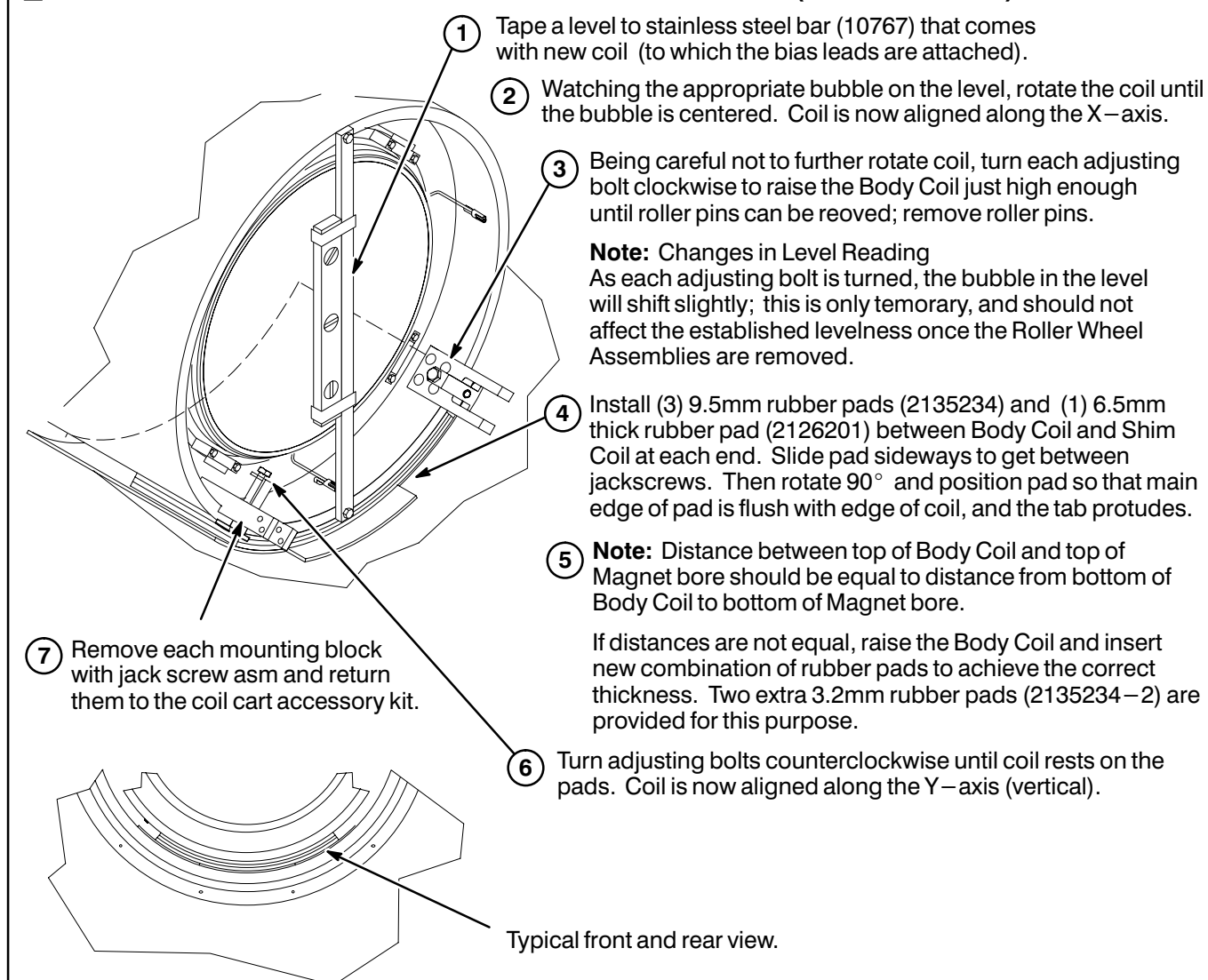
Note: Distance between top of Body Coil and top of Magnet bore should be equal to distance from bottom of Body Coil to bottom of Magnet bore.

Typical front and rear view.

☐ C1b – ROTATION ALIGNMENT OF BODY COIL IN GE MAGNET (NO SHIM COIL)



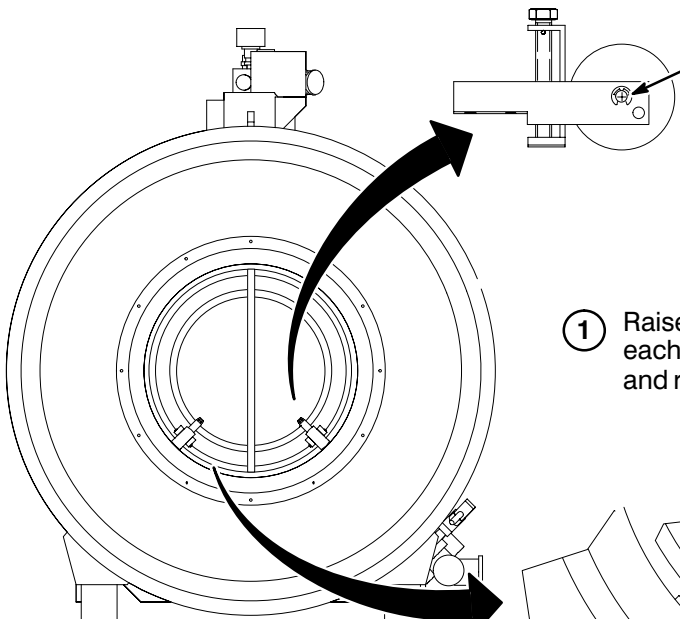
☐ C2b – CENTERING BODY COIL IN GE MAGNET BORE (NO SHIM COIL)



INSTALLATION STEPS SUMMARY

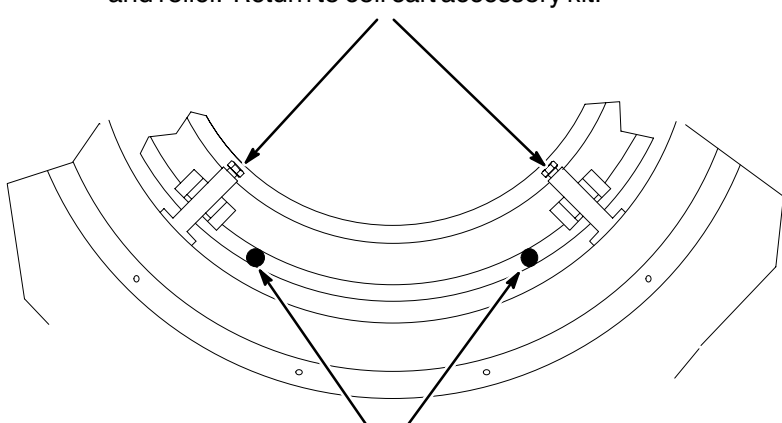
- D1 – Rotation Alignment of Body Coil in Oxford Magnet Bore
- D2 – Centering Body Coil in Oxford Magnet Bore

D1 – ROTATION ALIGNMENT OF BODY COIL IN OXFORD MAGNET BORE



Note: For Oxford, make sure roller is mounted in this hole.

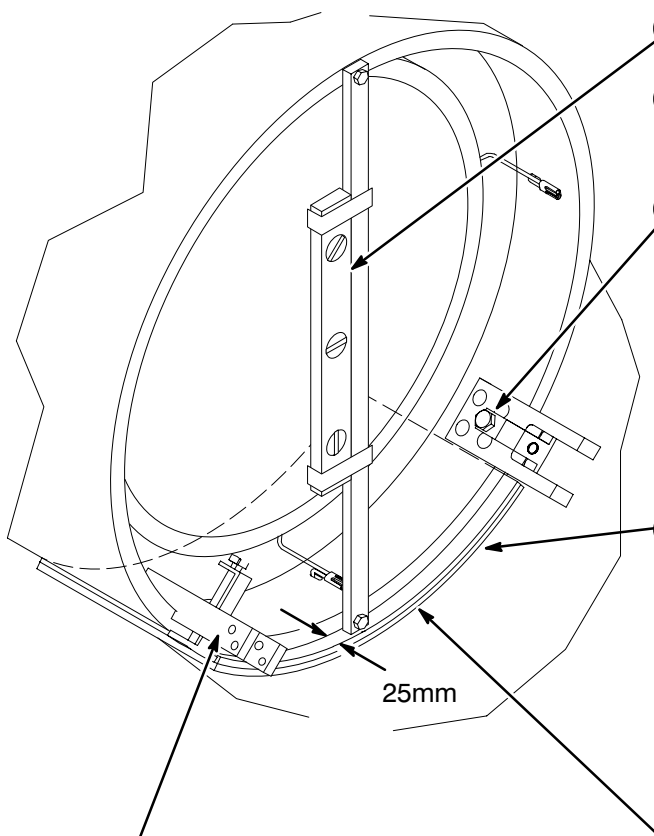
1 Raise front and rear of Body Coil with jack screws so that each roller is lifted off of the shim coil. Remove each axle and roller. Return to coil cart accessory kit.



2 Insert (4) small rods, two in front and two in back, between Body Coil and Shim Coil, approximately as shown.

3 Holding the rods in place, lower the Body Coil so that it rests on the rods.

D2 – CENTERING BODY COIL IN OXFORD MAGNET BORE



1 Tape a level to stainless steel bar (10767) that comes with new coil (to which the bias leads are attached).

2 Watching the appropriate bubble on the level, rotate the coil until the bubble is centered. Coil is now aligned along the X-axis.

3 Being careful not to further rotate coil, turn each adjusting bolt clockwise to raise the Body Coil just high enough until roller pins can be reoved; remove roller pins.

Note: Changes in Level Reading
As each adjusting bolt is turned, the bubble in the level will shift slightly; this is only temorary, and should not affect the established levelness once the Roller Wheel Assemblies are removed.

4 Install (2) 9.5mm thick rubber pad (2135234) between Body Coil and Shim Coil at each end.

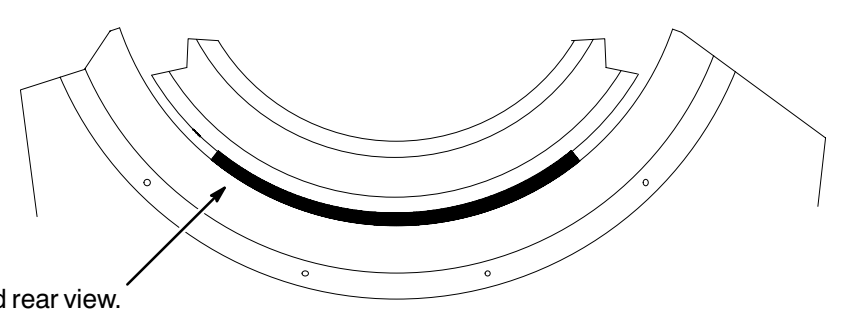
Note: Distance between top of Body Coil and top of Magnet bore should be equal to distance from bottom of Body Coil to bottom of Magnet bore.

If distances are not equal, raise the Body Coil and insert new combination of rubber pads to achieve the correct thickness. Two extra 6.5mm rubber pads (2126201) are provided for this purpose.

5 Center pads between mounting blocks and extend beyond gradient coil 25mm as shown.

6 Turn jackscrews and lower Body Coil down on pads.

7 Remove each mounting block with jack screw asm and return them to the coil cart accessory kit.

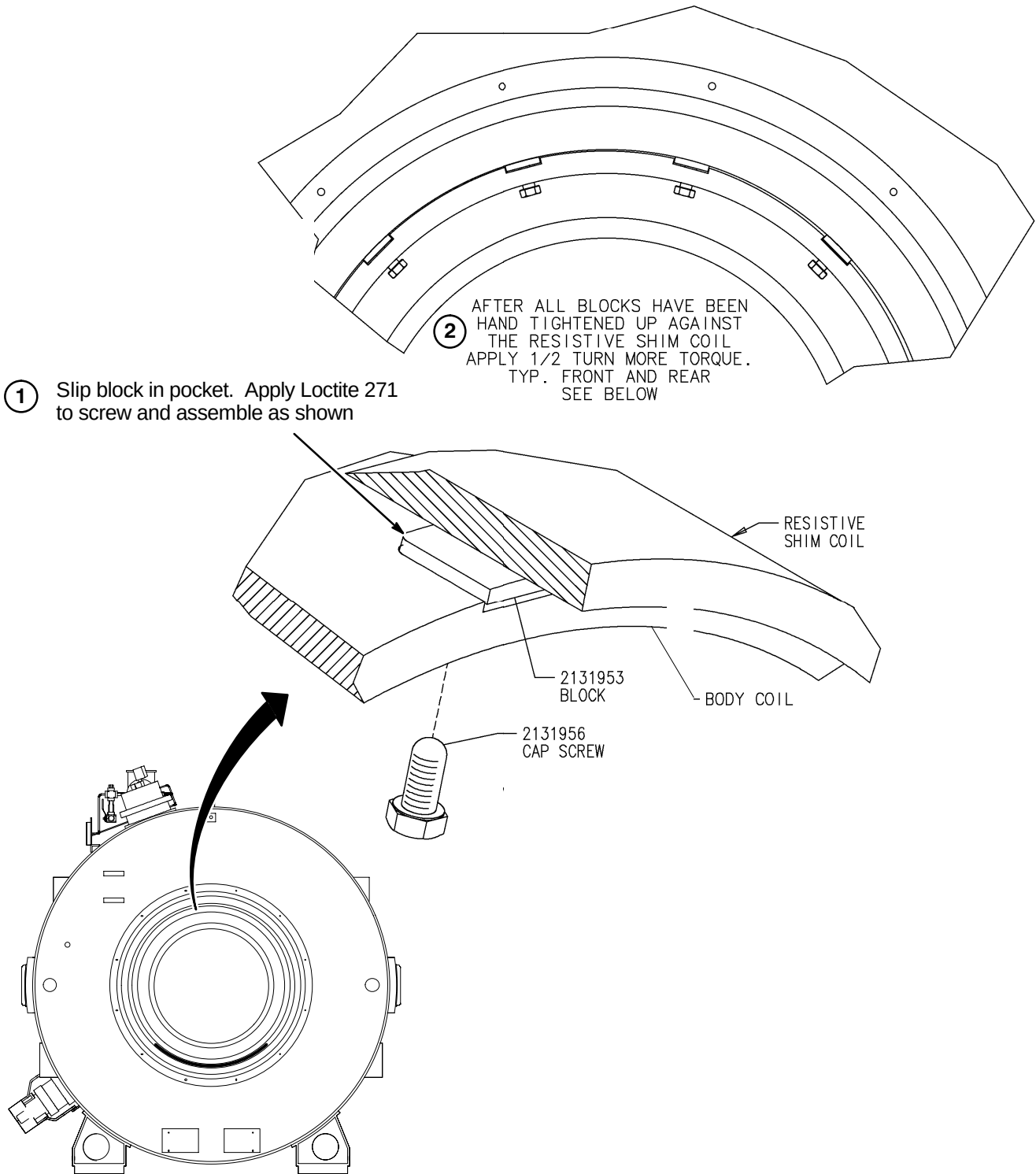


Typical front and rear view.

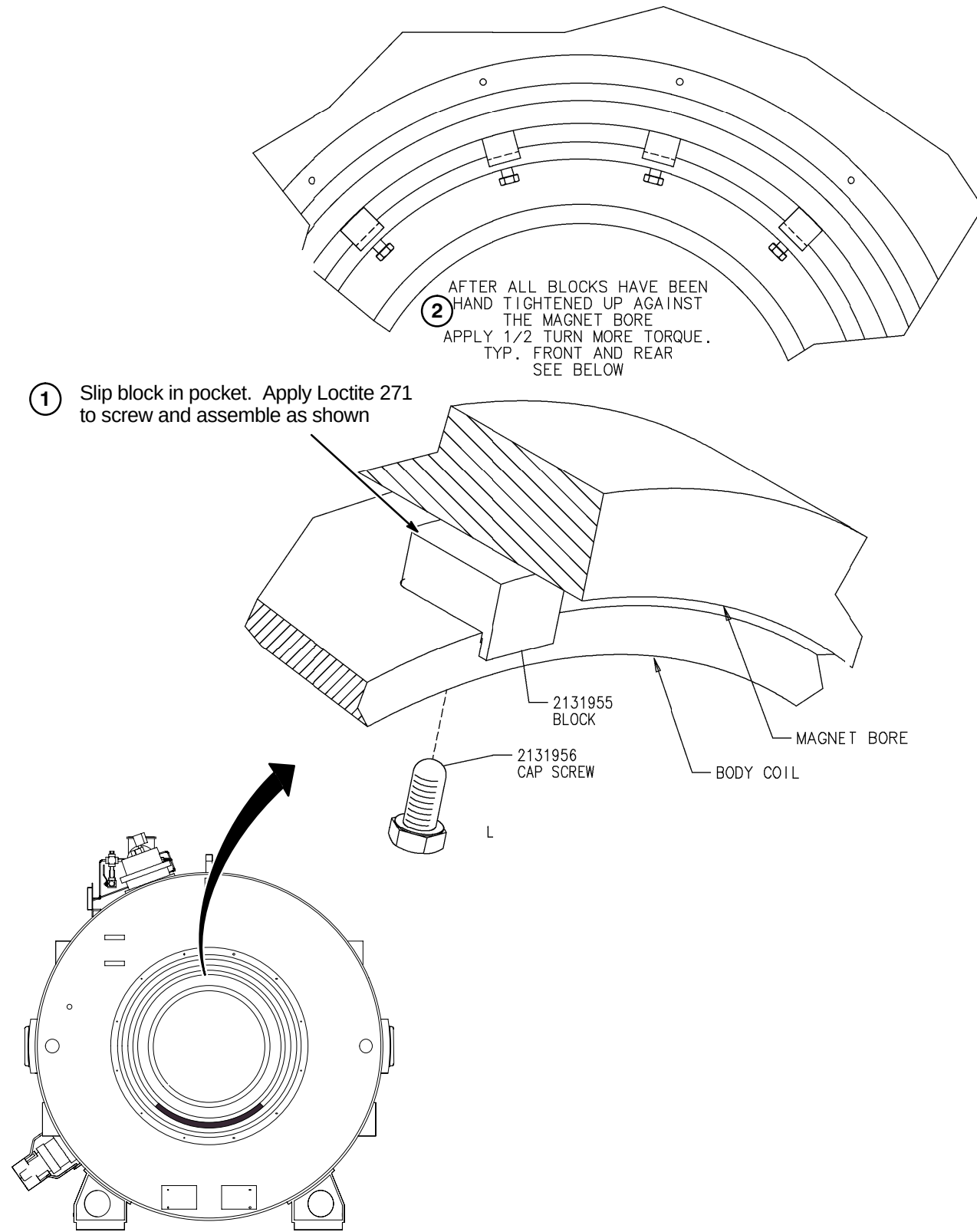
INSTALLATION STEPS SUMMARY

- ☐ E1 – Install Radial Support in GE Magnet with 900mm Diameter Bore
- OR
- ☐ E2 – Install Radial Support in GE Magnet with 950mm Diameter Bore

☐ E1 – INSTALL RADIAL SUPPORT IN GE MAGNET WITH 900mm DIAMETER BORE



☐ E2 – INSTALL RADIAL SUPPORT IN GE MAGNET WITH 950mm DIAMETER BORE



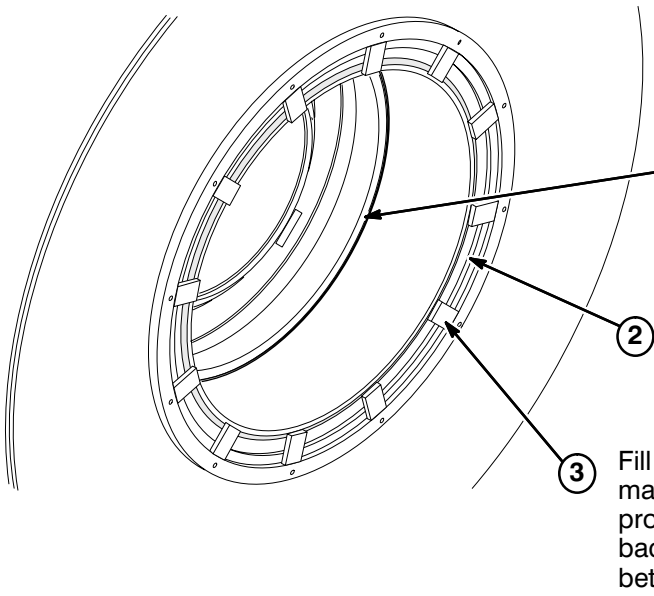
INSTALLATION STEPS SUMMARY

- ☐ F1 – Body Coil Installation Installation
- ☐ F1a – Installing Insulation in Magnet with Passive or Resistive Shim
- ☐ F2b – Installing Insulation in Magnet without Passive or Resistive Shim
- ☐ F3 – Installation of Velcro in Magnet Bore
- ☐ F4 – Acoustic Collar Installation

☐ F1 – BODY COIL INSULATION INSTALLATION

1. Locate accessories box that contains body coil soundproofing and air–sealing insulation.
2. Proceed to Step A2 or A3.

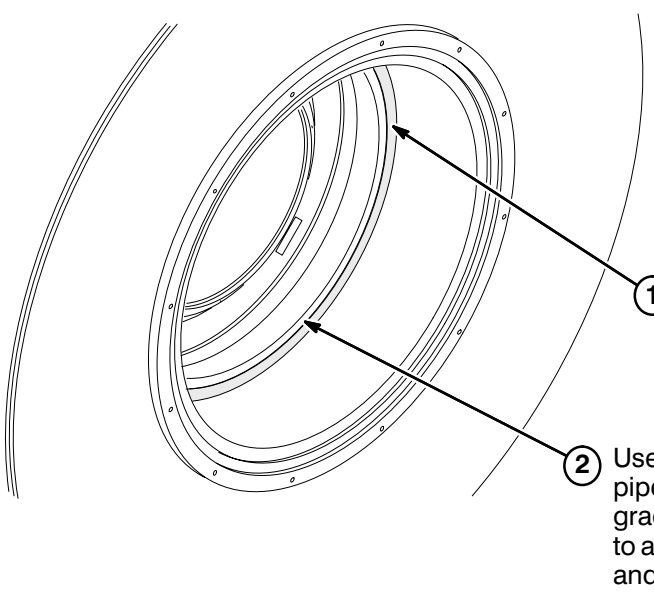
☐ F2a – INSTALLING INSULATION IN MAGNET WITH PASSIVE OR RESISTIVE SHIM



Note:
Rope caulk, pipe insulation and acoustical collar are installed after body coil centering.

- 1 Fill the gap between the Passive Shim Coil and outer Gradient Coil at each end. Use one, two, or three strips (rolled together) of rope caulk (46–287118P1) around entire perimeter.
- 2 Fill the gap between the Magnet bore and the Passive Shim Coil at each end. Use 8 in. (205 mm) sections of Pipe Insulation (46–320868P1) around entire perimeter.
- 3 Fill the gap between the Passive Shim drum stops and magnet bore. Use rope caulk (46–287118P1) to provide an air–tight seal around entire perimeter at the back of the magnet. This is to assure maximum air flow between the RF coil and the Gradient Coil.

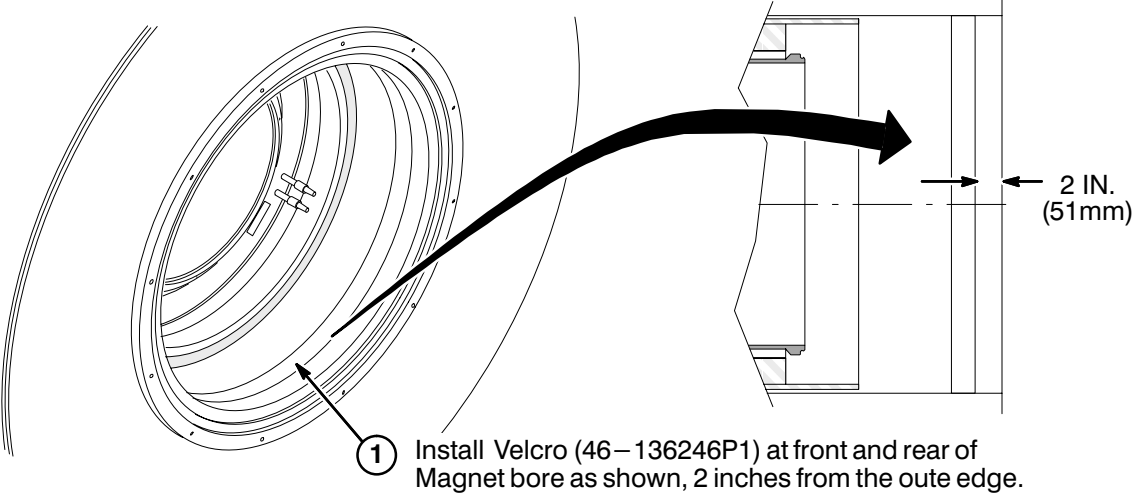
☐ F2b - INSTALLING INSULATION IN MAGNET WITHOUT PASSIVE OR RESISTIVE SHIM



Note:
Rope caulk, pipe insulation, and acoustical collars are installed after body coil centering.

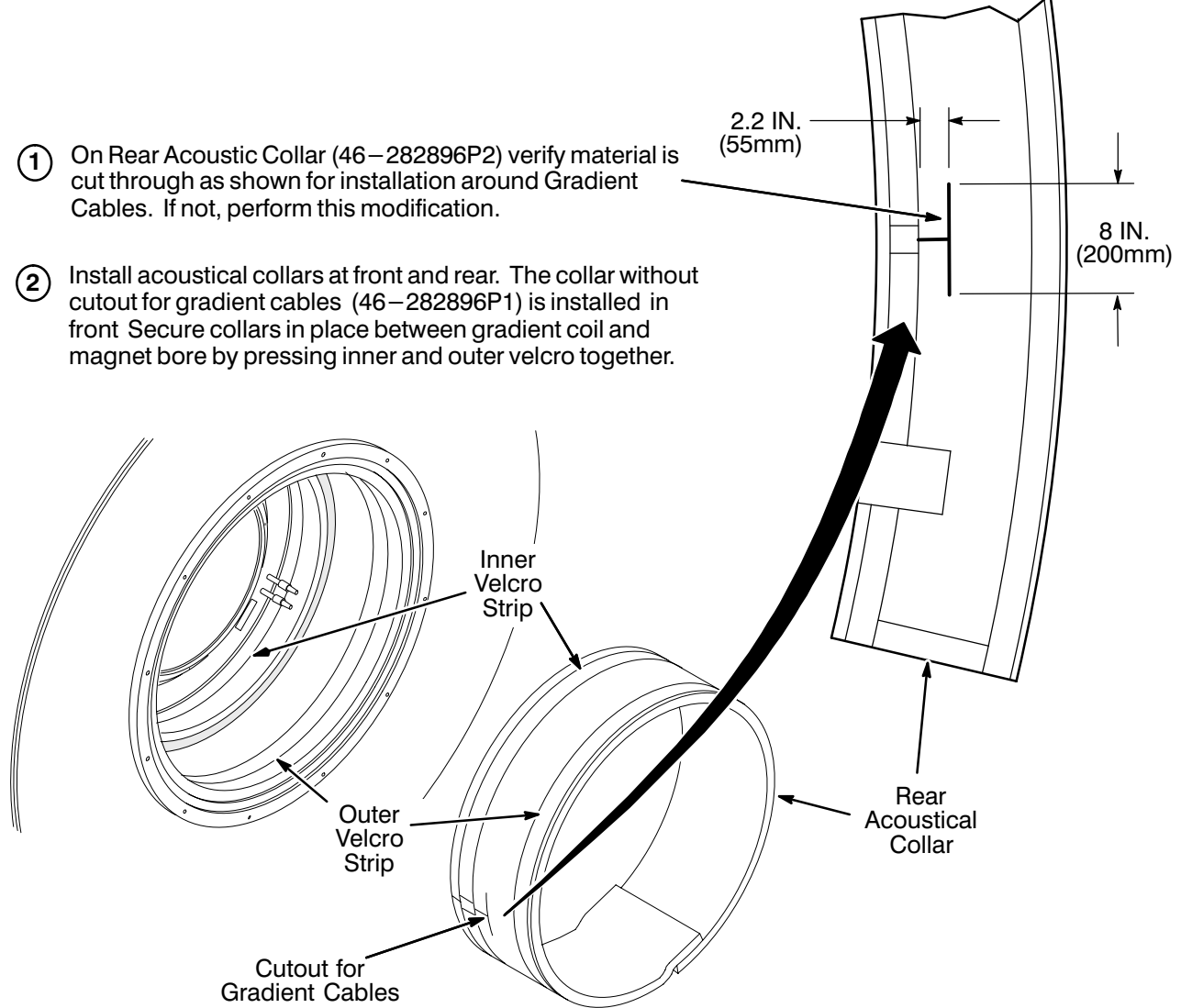
- 1 Fill the gap between the Magnet bore and outside of Gradient Coil at each end. Use Pipe Insulation (46–320868P1) and rope caulk (46–287118P1) around entire perimeter.
- 2 Use rope caulk (46–287118P1) as needed to supplement pipe insulation required for an air – tight seal between gradient coil and magnet bore at back of magnet. This is to assure maximum cooling air flow between the RF coil and the Gradient Coil.

☐ F3 – INSTALLATION OF VELCRO IN MAGNET BORE



1 Install Velcro (46–136246P1) at front and rear of Magnet bore as shown, 2 inches from the oute edge.

☐ F4 – ACOUSTIC COLLAR INSTALLATION



1 On Rear Acoustic Collar (46–282896P2) verify material is cut through as shown for installation around Gradient Cables. If not, perform this modification.

2 Install acoustical collars at front and rear. The collar without cutout for gradient cables (46–282896P1) is installed in front. Secure collars in place between gradient coil and magnet bore by pressing inner and outer velcro together.

Labels: Inner Velcro Strip, Outer Velcro Strip, Cutout for Gradient Cables, Rear Acoustical Collar

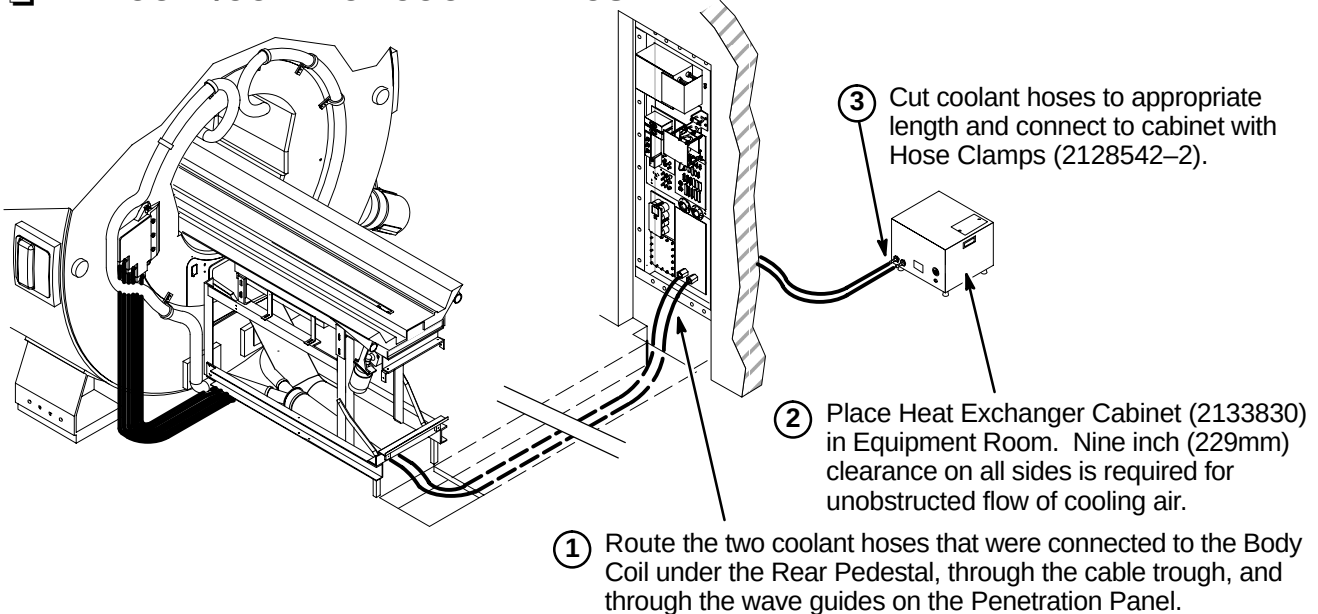
Dimensions: 2.2 IN. (55mm), 8 IN. (200mm)

HEAT EXCHANGER CABINET INSTALLATION

INSTALLATION STEPS SUMMARY

- ☐ A – Route/Connect Coolant Hose
- ☐ B – Install Power Cord
- ☐ C – Filling the System with Coolant

☐ A – ROUTE/CONNECT COOLANT HOSE

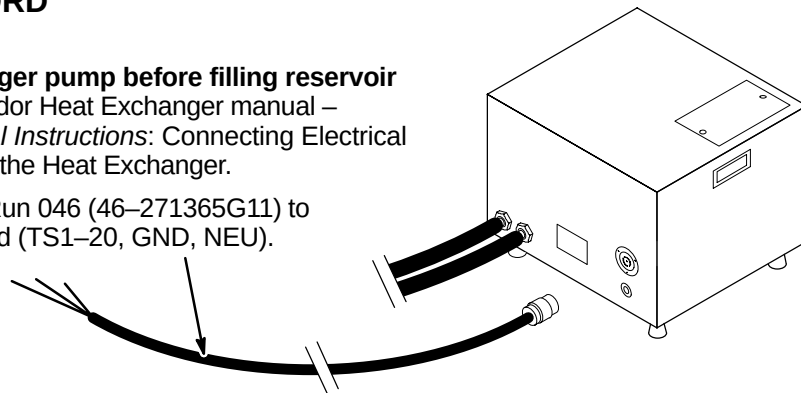


☐ B – INSTALL POWER CORD

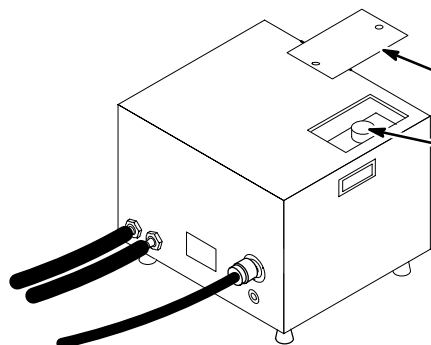
CAUTION:

Do not start Heat Exchanger pump before filling reservoir with coolant. Consult vendor Heat Exchanger manual – *Installation and Operational Instructions: Connecting Electrical Power*, that is shipped with the Heat Exchanger.

- ① Connect power cable Run 046 (46-271365G11) to existing PDU as marked (TS1-20, GND, NEU).



☐ C – FILLING THE SYSTEM WITH COOLANT



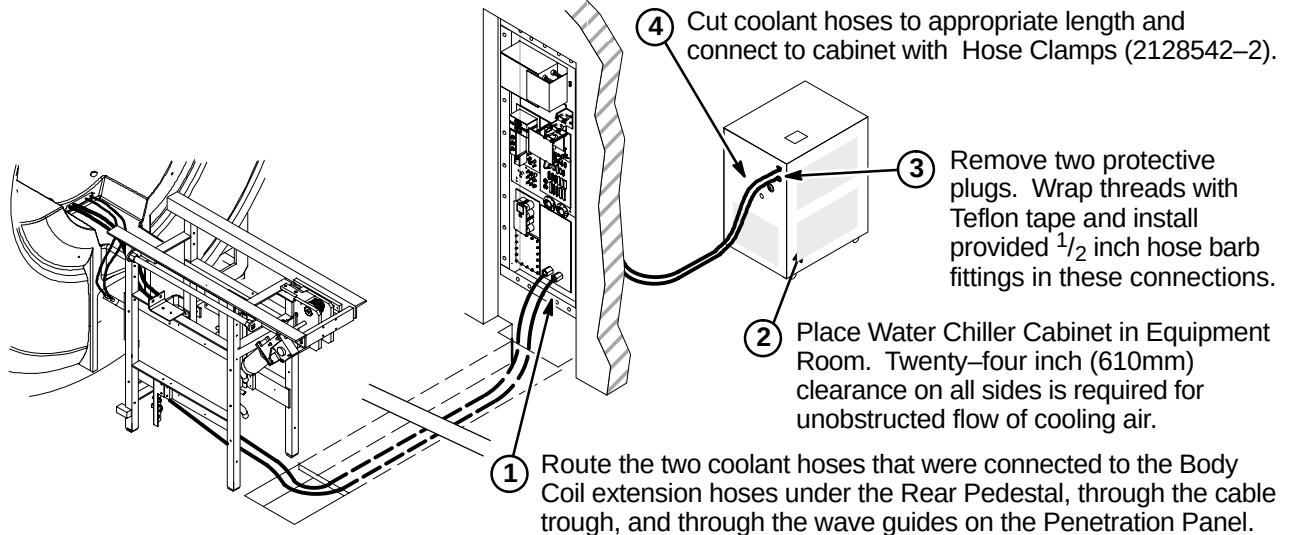
- ① Locate the six gallons of Body Coil Coolant (2138791) that was shipped with the Body Coil.
- ② Remove cover.
- ③ Consult vendor Heat Exchanger manual – *Installation and Operational Instructions: Initial Filling of Reservoir*, and the following steps for the proper procedure on priming and filling the system with coolant.
- ④ If supplied, attach plastic sleeve to side of Heat Exchanger and insert vendor manual. This will allow easy access to any future service/maintenance procedures. If plastic sleeve is not present, attach vendor manual to Heat Exchanger.

WATER CHILLER CABINET INSTALLATION

INSTALLATION STEPS SUMMARY

- ☐ A – Route/Connect Coolant Hose
- ☐ B – Install Power Cord
- ☐ C – Filling the System with Coolant

☐ A – ROUTE/CONNECT COOLANT HOSE



☐ B – INSTALL POWER CORD

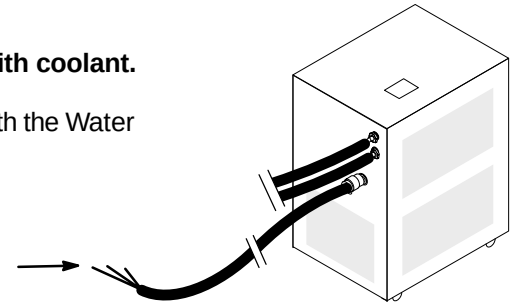
CAUTION:

Do not start Water Chiller pump before filling reservoir with coolant.

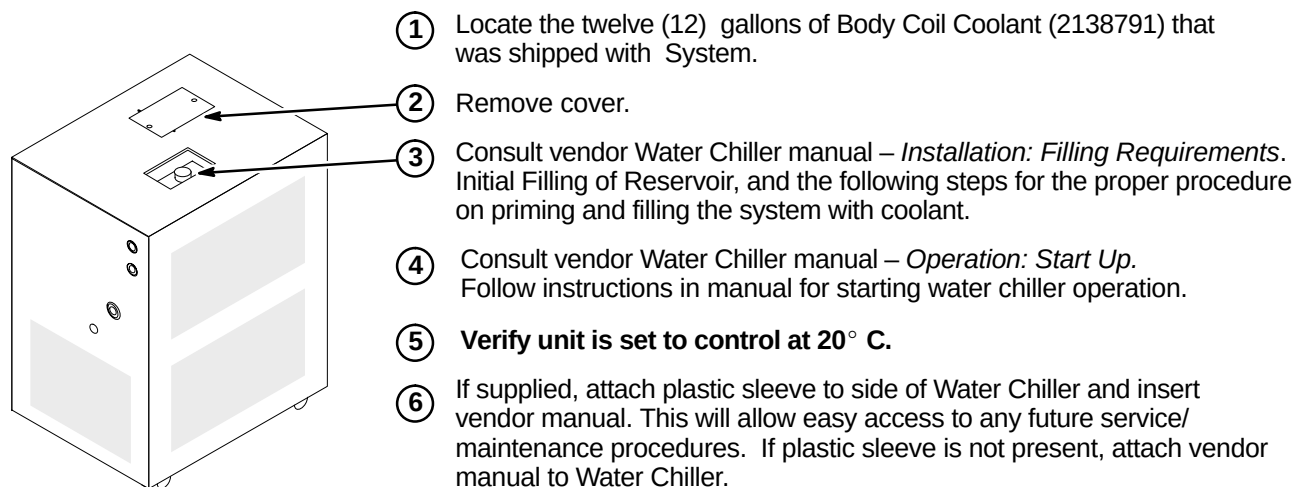
Consult vendor Water Chiller manual – *Installation, Operation, and Service Manual* that is shipped with the Water Chiller.

- ① **FOR RF/PDU:** Connect power cable Run 065 (2223355) to PDU as marked (TS1-26, 27, GND).

FOR ACGD/PDU: Connect power cable Run 970 (2280289) to PDU as marked (TS1-16,17,GND).



☐ C– FILLING THE SYSTEM WITH COOLANT AND START UP



PENETRATION PANEL UPGRADE INSTRUCTIONS

Tab 4, Penetration Panel, contains upgrade instructions for the following configurations:

- *DIRECTION 2147670* – Signa LX/EXCITE Penetration Panel Installation Instructions
- *DIRECTION 2147884* – Signa Horizon Penetration Panel Upgrade Instructions
- *DIRECTION 2333493* – Signa EXCITE Penetration Panel Upgrade Instructions
- *DIRECTION 2403003* – Signa EXCITE II Penetration Panel Upgrade Instructions

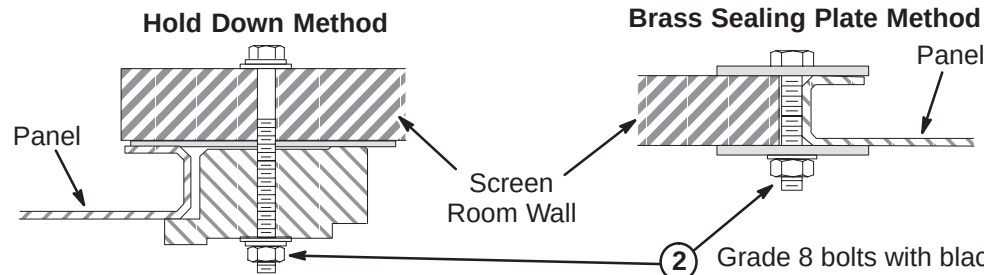
Use only the applicable Penetration Panel Instruction Direction for the site being upgraded

INSTALLATION STEPS SUMMARY

- A – Remove Blank Panel/Prepare Screen Room Opening Surface
- B – Install Penetration Panel
- C – Install Penetration Panel Stand-Off Hardware
- D – Install Penetration Panel Covers

A – REMOVE BLANK PANEL/PREPARE SCREEN ROOM OPENING SURFACE

- 1 If still in place, remove vendor supplied blank panel. Save mounting hardware of either method shown below for installation of Penetration Panel.

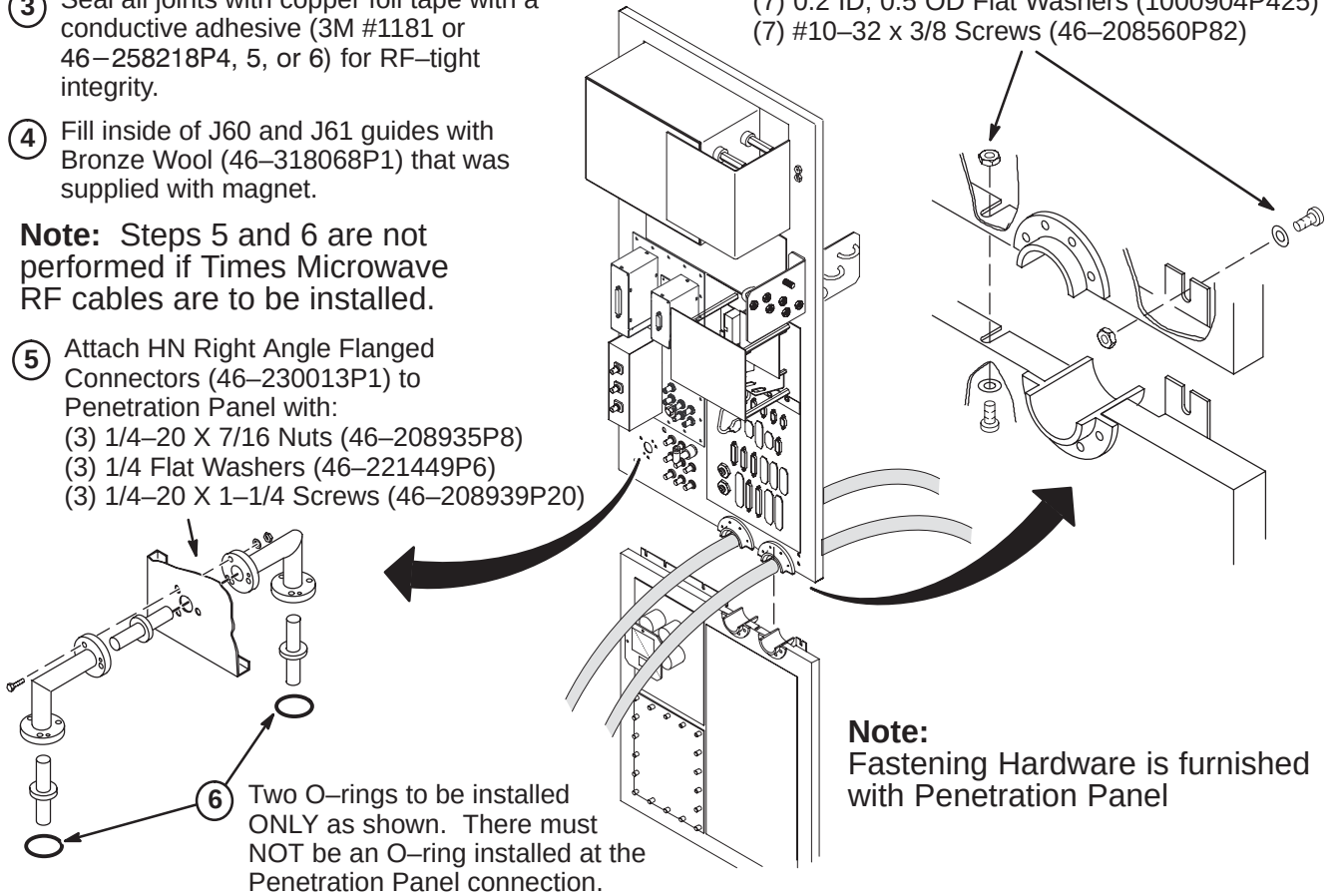


- 2 Grade 8 bolts with black oxide coating should be used as mounting hardware. Tighten to torque of 20 lb-ft (1.5 to 2 turns after lock washer flattens).
- 3 To insure good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**

B – INSTALL PENETRATION PANEL

- 2 Install Penetration Panel using same method and hardware for mounting of blank panel above.
- 3 Seal all joints with copper foil tape with a conductive adhesive (3M #1181 or 46-258218P4, 5, or 6) for RF-tight integrity.
- 4 Fill inside of J60 and J61 guides with Bronze Wool (46-318068P1) that was supplied with magnet.
- 5 Attach HN Right Angle Flanged Connectors (46-230013P1) to Penetration Panel with:
(3) 1/4-20 X 7/16 Nuts (46-208935P8)
(3) 1/4 Flat Washers (46-221449P6)
(3) 1/4-20 X 1-1/4 Screws (46-208939P20)

- 1 Assemble Penetration Panel around helium lines (Runs 621 and 622) using:
(7) #10-32 Nylock Lock Nuts (46-208937P6)
(7) 0.2 ID, 0.5 OD Flat Washers (1000904P425)
(7) #10-32 x 3/8 Screws (46-208560P82)

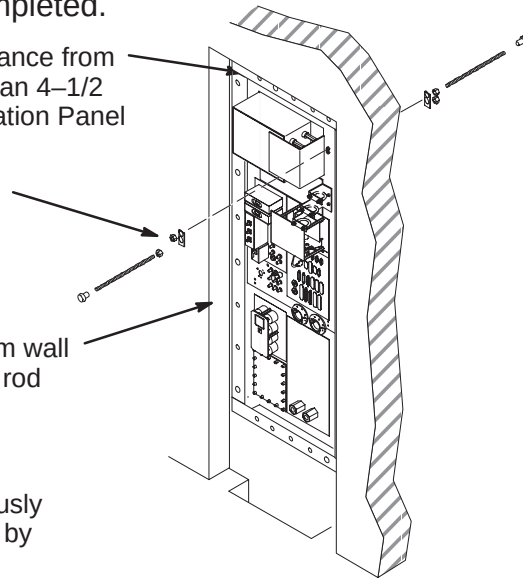


C – INSTALL PENETRATION PANEL STAND-OFF HARDWARE

Note: The Penetration Panel Floor plate that is supplied with the cover kit is to be installed prior to routing and connecting cables to Penetration Panel. Refer to Installation Drawing furnished with the Cover Kit.

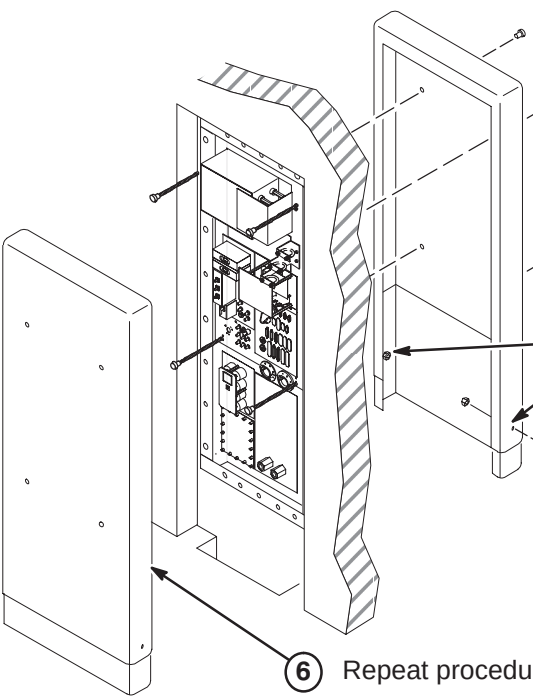
Note: The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

- 1 Verify that cover will enclose finished wall opening. The distance from the top of Penetration Panel to wall opening must be less than 4-1/2 inches (114mm). If not, new mounting stud holes in Penetration Panel may be required.
- 2 Place (8) Reinforcing Plates (46-301211P1) over each hole pair on both sides of Penetration Panel and attach using:
(16) 5/16-18 x 9/16 Brass Nut (46-208935P9)
(8) 5/16-18 x 18 inch Threaded Rod (46-301213P2)
(8) 5/16-18 x 1 inch Barrel Nut (46-301212P1)
- 3 Measure distance from face of Pen Panel to equipment room wall surface. Add six inches (152mm) to obtain length threaded rod should extend from panel.
- 4 Repeat step 3 procedure for exam room cover.
- 5 Adjust threaded rods with attached barrel nuts to the previously calculated dimensions from steps 3 and 4. Secure to panel by tightening brass nuts on both sides of Penetration Panel.



D – INSTALL PENETRATION PANEL COVERS

- 1 Remove all barrel nuts.
- 2 In the Magnet room, temporarily install Top (46-306936P1) and Bottom (46-306964P1) Cover assembly on threaded rods and secure with four barrel nuts.
- 3 Adjust position of bottom cover section and mark two lower fastener holes.
- 4 Remove covers and drill hole in lower cover at each mark.
- 5 Attach top and bottom covers with furnished 10-32 nuts and screws.
- 6 Repeat procedure for Equipment room cover.
- 7 After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.



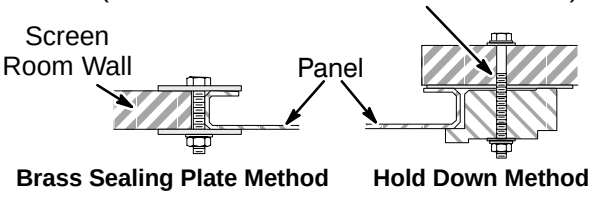
INSTALLATION STEPS SUMMARY

- ☐ A – Remove/Install Penetration Panel
- ☐ B – Transfer of Penetration Panel Components
- ☐ C – Install Penetration Panel Floor Plate
- ☐ D – Install Penetration Panel Stand–Off Hardware
- ☐ E – Install Penetration Panel Covers

☐ A – REMOVE/INSTALL PENETRATION PANEL

- ① If still in place, remove existing Penetration Panel. Install new Penetration Panel using the same mounting hardware of either method shown below. Apply copper tape (46–258218P4, 5, or 6) to seam between Panel and wall.

③ To insure good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**
- ② Grade 8 bolts with black oxide coating should be used as mounting hardware. Tighten to torque of 20 lb–ft (1.5 to 2 turns after lock washer flattens).



Screen Room Wall Panel Brass Sealing Plate Method Hold Down Method

☐ B – TRANSFER OF PENETRATION PANEL COMPONENTS

CAUTION: Sites with Oxford Magnets may have an Ion Pump installed on the magnet. The Ion Pump Power supply provides 4,000 volts to the Ion Pump via a cable connected through the penetration panel. To prevent electrical shock during servicing of the Penetration Panel, remove AC power to the Ion Pump Power Supply, located in the Equipment Room.

CAUTION: All cable connections **MUST BE LABELLED** before disconnect. Customer may have added connections to Penetration Panel which are not labelled but must be reconnected on new panel.

NOTE: If old system is 3.x or 3.2 Perf+

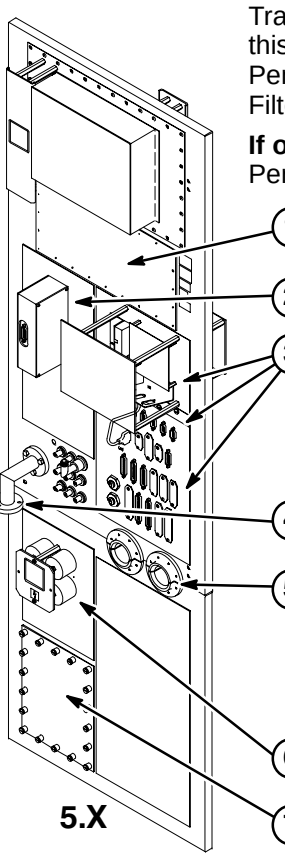
Transfer the Shim Filter, Step 1, only if it's a 46–271282G1, (if not this model, order one), and the heliax connector, Step 4, to new Penetration Panel. Discard the rest.

If old system is 4.x

Transfer the Shim Filter, Step 1, only if it's a 46–271282G1, (if not this model, order one), and the heliax connector, Step 4, to new Penetration Panel. Mount new Fiber Optic Repeater, Bar, and Filter Panel. Discard the rest.

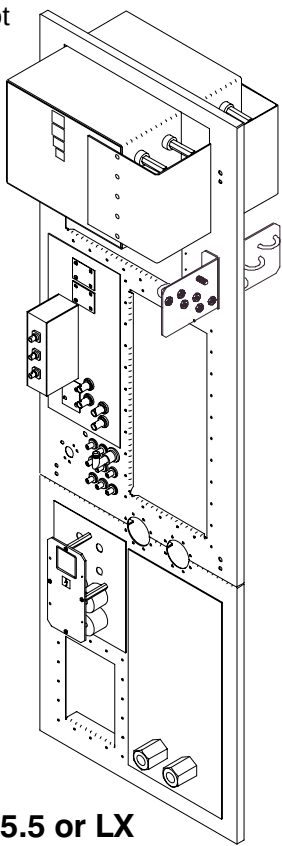
If old system is a 5.x

Perform the following steps.

- 
- Transfer blank panel or Shim Filter to new Penetration Panel re–using mounting hardware.
 - If part of existing system, transfer (3) Multi–coil boxes).
 - Transfer Fiber Optic Repeater, Bar, and Filter Panel to new Penetration Panel re–using mounting hardware.
 - Transfer Heliac Connector to new Penetration Panel re–using mounting hardware.
 - Transfer Wave Guides of blocking plate to holes on new Penetration Panel re–using mounting hardware.
 - Transfer Filters to holes on new Penetration Panel re–using mounting hardware.
 - Transfer Panel to new Penetration Panel re–using mounting hardware.

Note: If this is an early 5.X system without Repeater, Catalog M1090ZA should be part of this upgrade which includes this component.

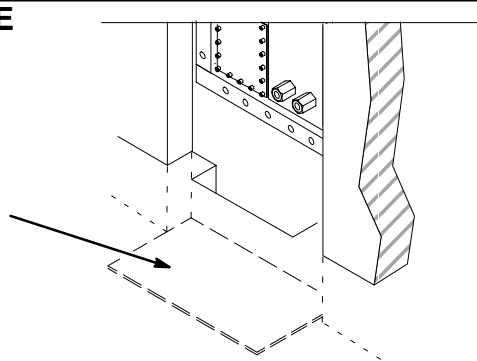
CAUTION! – OLDER SYSTEMS USED BERYLIUM PASTE ON WAVE GUIDES AND GAS LINES. WASH HANDS IF YOU COME IN CONTACT WITH THIS MATERIAL.



☐ C – INSTALL PENETRATION PANEL FLOOR PLATE

Note: The Penetration Panel Floor plate (2120824) that is supplied with the cover kit is to be installed prior to routing and connecting cables to Penetration Panel.

- ① Remove backing from tape and stick Floor Plate to floor beneath Penetration Panel.

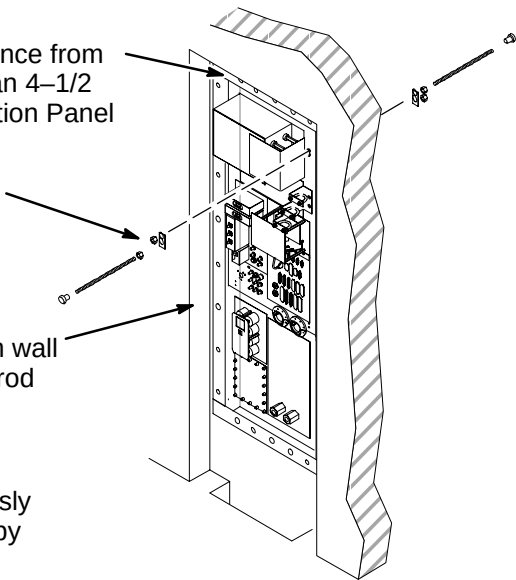


☐ D – INSTALL PENETRATION PANEL STAND–OFF HARDWARE

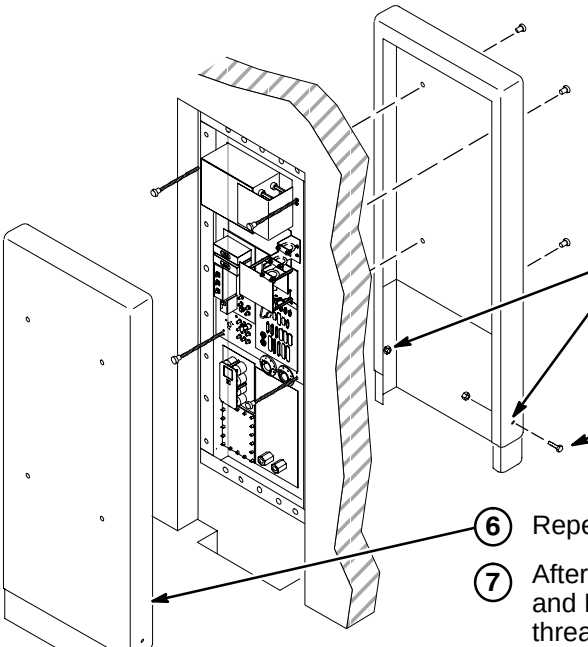
Note: The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

- Verify that cover will enclose finished wall opening. The distance from the top of Penetration Panel to wall opening must be less than 4–1/2 inches (114mm). If not, new mounting stud holes in Penetration Panel may be required.
- Place (8) Reinforcing Plates (46–301211P1) over each hole pair on both sides of Penetration Panel and attach using:
(16) 5/16–18 x 9/16 Brass Nut (46–208935P9)
(8) 5/16–18 x 18 inch Threaded Rod (46–301213P2)
(8) 5/16–18 x 1 inch Barrel Nut (46–301212P1)
- Measure distance from face of Pen Panel to equipment room wall surface. Add 7.5 inches (190mm) to obtain length threaded rod should extend from panel.
- Repeat step 3 procedure for exam room cover.

Adjust threaded rods with attached barrel nuts to the previously calculated dimensions from steps 3 and 4. Secure to panel by tightening brass nuts on both sides of Penetration Panel.



☐ E – INSTALL PENETRATION PANEL COVERS

- 
- Remove all barrel nuts.
 - In the Magnet room, temporarily install Top (46–306936P1) and Bottom (46–306964P1) Cover assembly on threaded rods and secure with four barrel nuts.
 - Adjust position of bottom cover section and mark two lower fastener holes.
 - Remove covers and drill hole in lower cover at each mark.
 - Attach top and bottom covers with furnished 10–32 nuts and screws.
 - Repeat procedure for Equipment room cover.
 - After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.

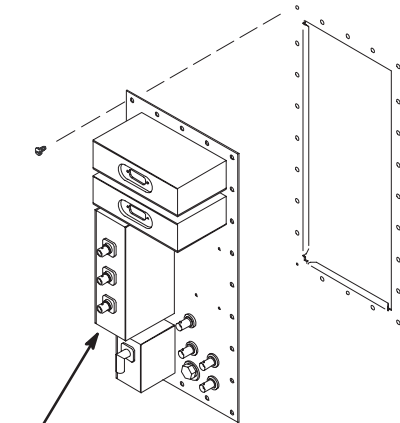
1-1 REMOVE OBSOLETE MULTICOIL PLATE FROM PENETRATION PANEL

CAUTION: All cable connections MUST BE LABELED before disconnect.

- Customer may have added connections for options to Penetration Panel which are not labelled but must be reconnected on new panel.

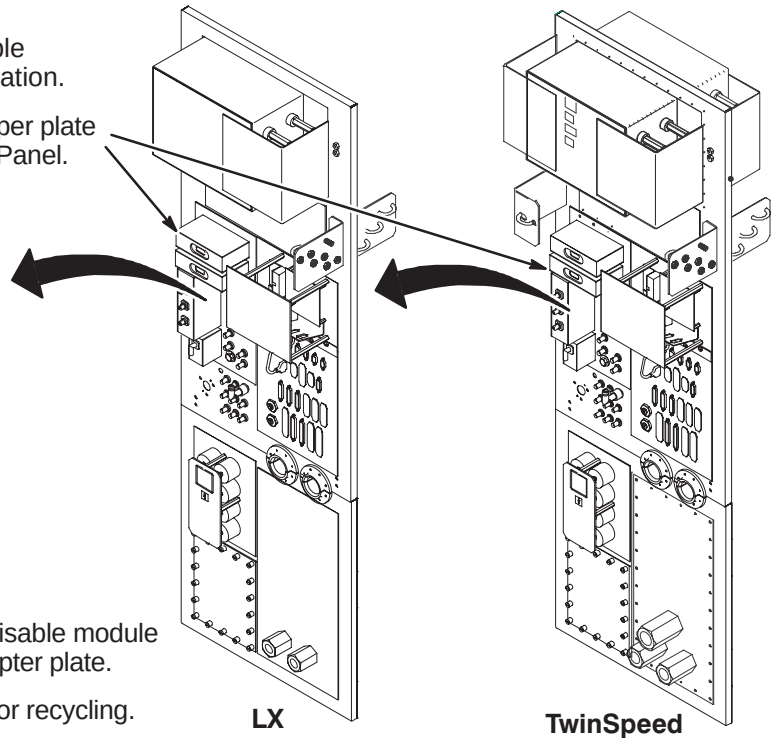
NOTE: Refer to Section 4 for cable disconnection and removal information.

- 1 Remove existing Multicoil Adapter plate at this location on Penetration Panel.



- 2 If needed, remove Dynamic Disable module for re-use on new EXCITE adapter plate.

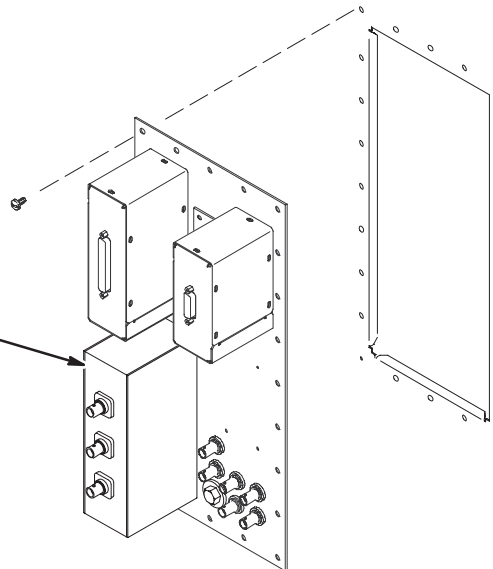
- 3 Return Multicoil Adaptor plate for recycling.



1-2 INSTALL NEW EXCITE ADAPTER PLATE ON PENETRATION PANEL

NOTE: Refer to Section 4 for cable routing and connection information.

- 1 If needed, install previously removed Dynamic Disable module on new EXCITE adapter plate.
- 2 Install new Excite Adapter plate using fasteners removed in previous step.
- 3 Reconnect existing cables and connect new cables to Penetration Panel.



INSTALLATION STEPS SUMMARY

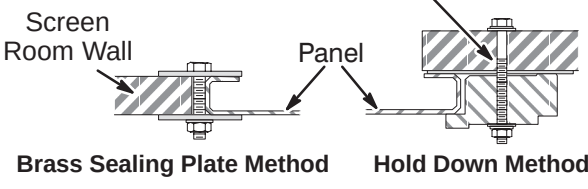
- ☐ A – Remove/Install Penetration Panel
- ☐ B – Transfer of Penetration Panel Components
- ☐ C – Install Penetration Panel Floor Plate
- ☐ D – Install Penetration Panel Stand–Off Hardware
- ☐ E – Install Penetration Panel Covers

A – REMOVE/INSTALL PENETRATION PANEL

WARNING:
Sites with Oxford Magnets may have an Ion Pump installed on the magnet. The Ion Pump Power supply provides 4,000 volts to the Ion Pump via a cable connected through the penetration panel. To prevent electrical shock during servicing of the Penetration Panel, remove AC power to the Ion Pump Power Supply, located in the Equipment Room.

- 1 If still in place, remove existing Penetration Panel. Install new Penetration Panel using the same mounting hardware of either method shown below. Apply copper tape (46–258218P4, 5, or 6) to seam between Panel and wall.

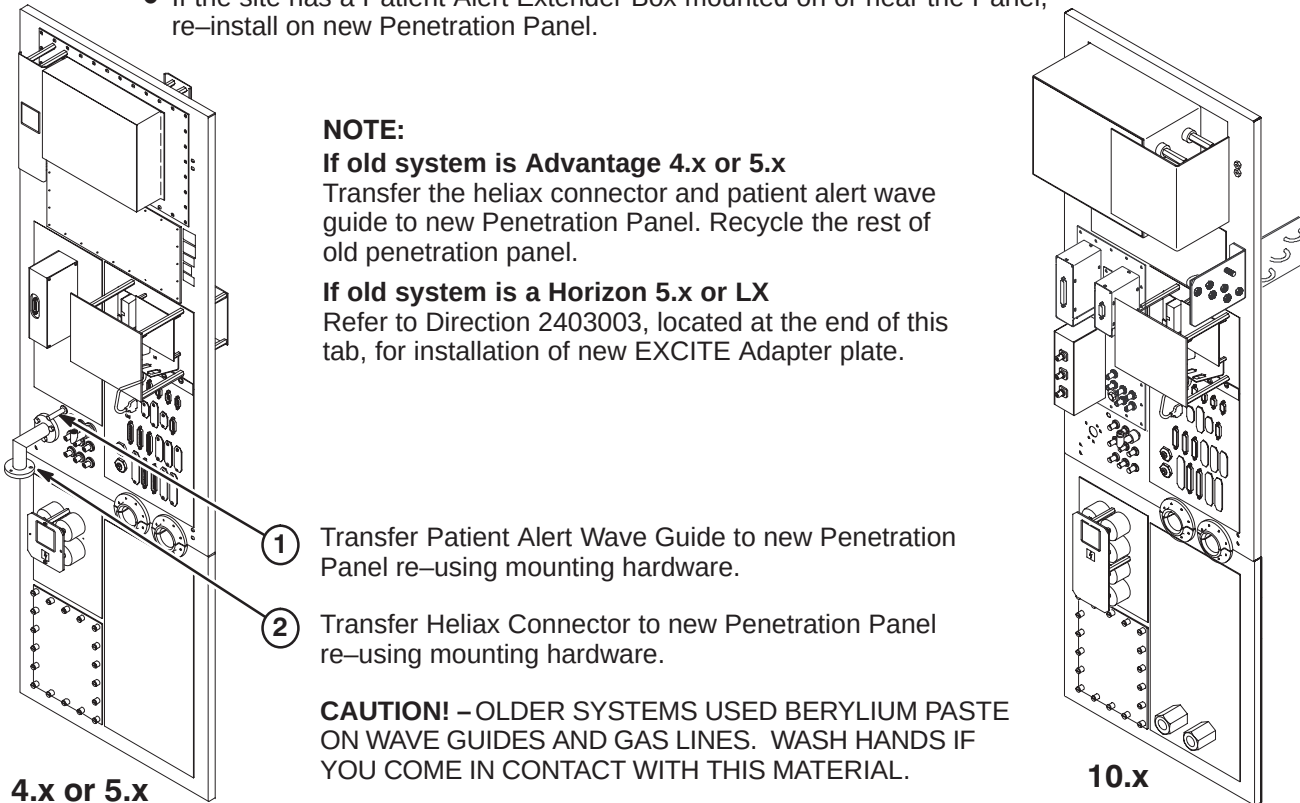
3 To insure good conductivity, use a fine sandpaper or scouring pad to clean surface of screen room wall which will be in contact with Penetration Panel. **Do not sand surface of Penetration Panel as it is coated.**
- 2 Grade 8 bolts with black oxide coating should be used as mounting hardware. Tighten to torque of 20 lb–ft (1.5 to 2 turns after lock washer flattens).



Brass Sealing Plate Method Hold Down Method

B – TRANSFER OF PENETRATION PANEL COMPONENTS

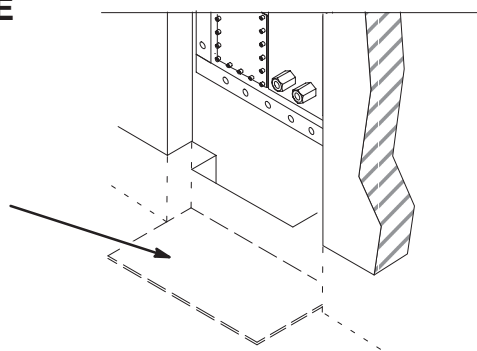
- CAUTION:** All cable connections **MUST BE LABELLED** before disconnect.
- Customer may have added connections for options to Penetration Panel which are not labelled but must be reconnected on new panel.
 - If the site has a Patient Alert Extender Box mounted on or near the Panel, re–install on new Penetration Panel.



C – INSTALL PENETRATION PANEL FLOOR PLATE

Note: The Penetration Panel Floor plate (2120824) that is supplied with the cover kit is to be installed prior to routing and connecting cables to Penetration Panel.

- 1 Remove backing from tape and stick Floor Plate to floor beneath Penetration Panel.

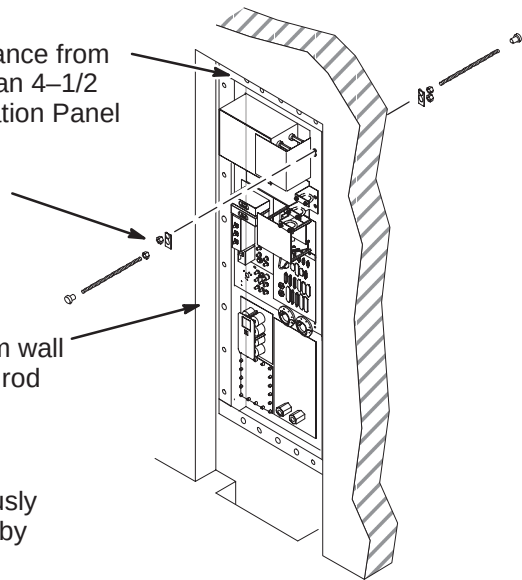


D – INSTALL PENETRATION PANEL STAND–OFF HARDWARE

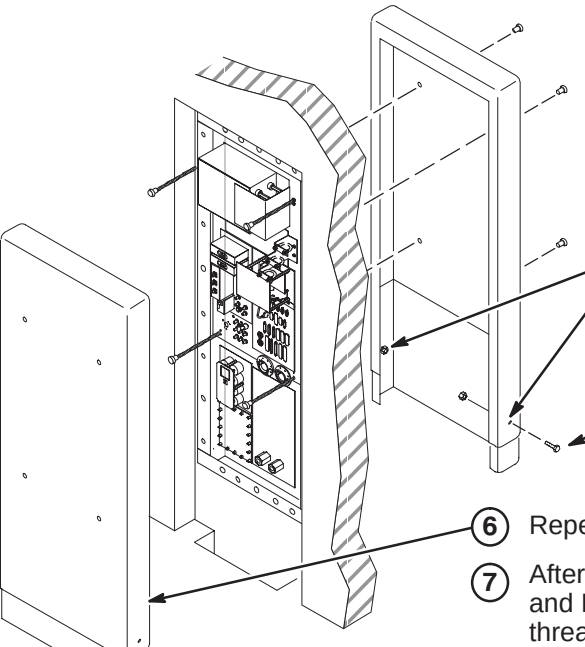
Note: The Penetration Panel Cover can be installed only after all connections and System installation has been completed.

- 1 Verify that cover will enclose finished wall opening. The distance from the top of Penetration Panel to wall opening must be less than 4–1/2 inches (114mm). If not, new mounting stud holes in Penetration Panel may be required.
- 2 Place (8) Reinforcing Plates (46–301211P1) over each hole pair on both sides of Penetration Panel and attach using:
(16) 5/16–18 x 9/16 Brass Nut (46–208935P9)
(8) 5/16–18 x 18 inch Threaded Rod (46–301213P2)
(8) 5/16–18 x 1 inch Barrel Nut (46–301212P1)
- 3 Measure distance from face of Pen Panel to equipment room wall surface. Add 7.5 inches (190mm) to obtain length threaded rod should extend from panel.
- 4 Repeat step 3 procedure for exam room cover.

Adjust threaded rods with attached barrel nuts to the previously calculated dimensions from steps 3 and 4. Secure to panel by tightening brass nuts on both sides of Penetration Panel.



E – INSTALL PENETRATION PANEL COVERS

- 
- 1 Remove all barrel nuts.
- 2 In the Magnet room, temporarily install Top (46–306936P1) and Bottom (46–306964P1) Cover assembly on threaded rods and secure with four barrel nuts.
- 3 Adjust position of bottom cover section and mark two lower fastener holes.
- 4 Remove covers and drill hole in lower cover at each mark.
- 5 Attach top and bottom covers with furnished 10–32 nuts and screws.
- 6 Repeat procedure for Equipment room cover.
- 7 After all installation procedures are completed in Magnet and Equipment rooms, install assembled covers on threaded rods and secure with four barrel nuts.

SYSTEM CABINET UPGRADE INSTRUCTIONS

Tab 8, System Cabinet, contains upgrade instructions for the following configurations:

- *DIRECTION 2170004* – Signa 8.x/9.x System Cabinet Installation Instructions
- *DIRECTION 2171796* – Signa LX System Cabinet Upgrade Instructions
- *DIRECTION 2338502* – Signa EXCITE I System Cabinet Installation Instructions
- *DIRECTION 2398344* – Signa EXCITE II System Cabinet Installation Instructions
- *DIRECTION 2402879* – Signa EXCITE II Upgrade System Cabinet Installation Instructions

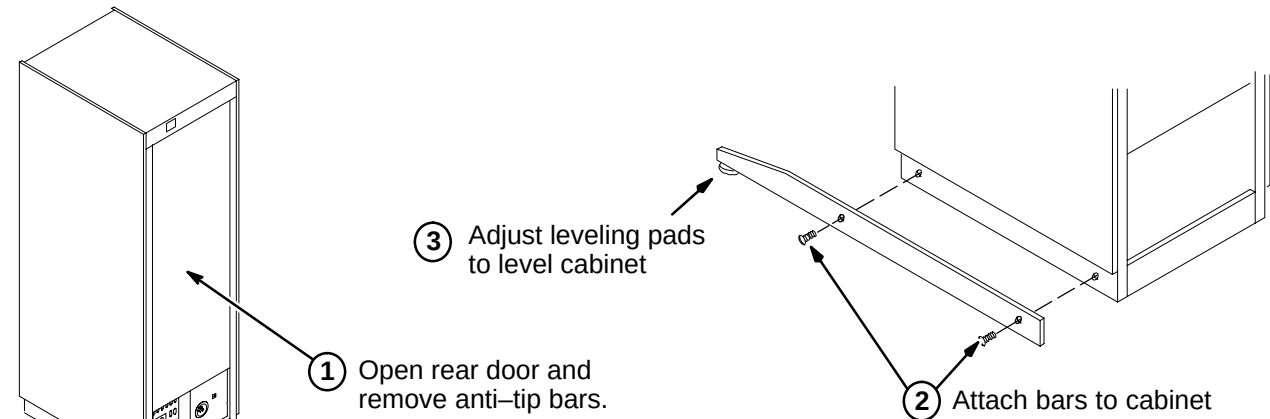
Use only the applicable System Cabinet Upgrade Instruction Direction for the site being upgraded

INSTALLATION STEPS SUMMARY

- ☐ A – Position/Secure System Cabinet
- ☐ B – Install Magnet Monitoring Cables
- ☐ C – Connect Fiber Optic Runs 708, 710, 711/712, and 836 to I/F Panel
- ☐ D – Route/Connect Fiber Optic Runs 836 to BIT3 Module
- ☐ E – Route/Connect Fiber Optic Runs 708, 710, and 711/712 to Tyme II
- ☐ F – Connect Power, Ground, and Data Cables
- ☐ G – Install InSite “SYSTEMS” Cabinet Magnet

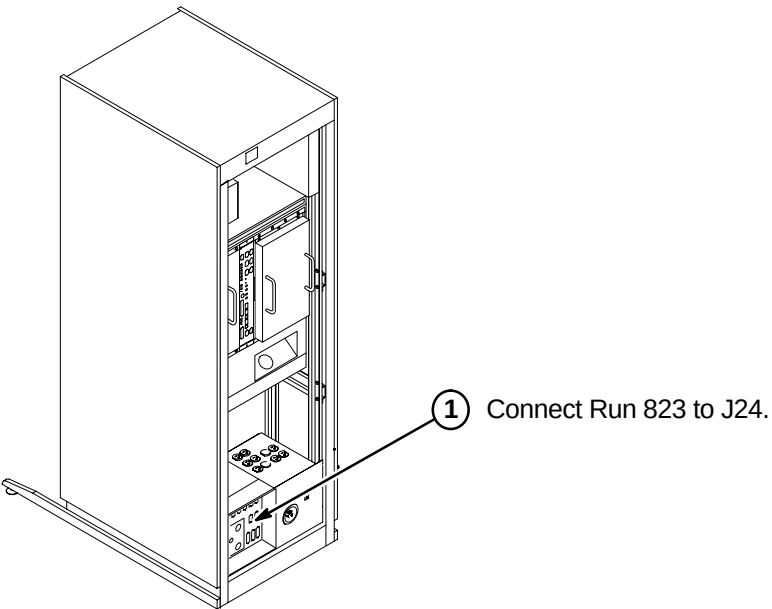
☐ A – POSITION/SECURE SYSTEM CABINET

Comply with UL requirements by either installing anti tip legs (see illustration below) or securing the cabinet to the floor with local supplied brackets and floor anchors.



☐ B – INSTALL MAGNET MONITORING CABLES

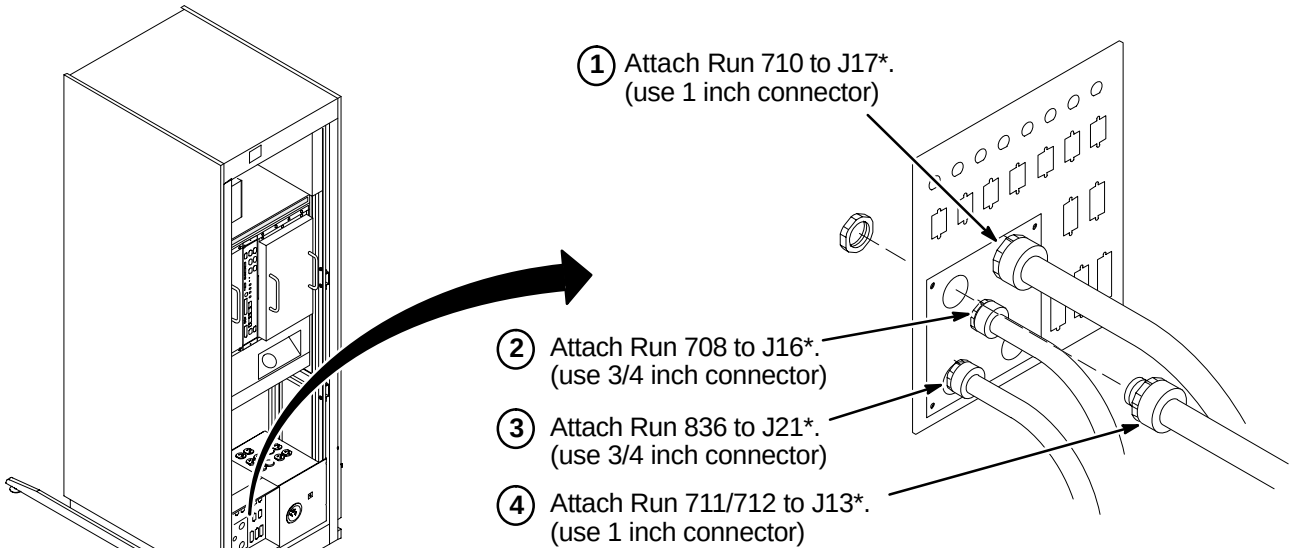
Note: Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual*, for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.



☐ C – CONNECT FIBER OPTIC RUNS 708, 710, 711/712, AND 836 TO I/F PANEL

CAUTION

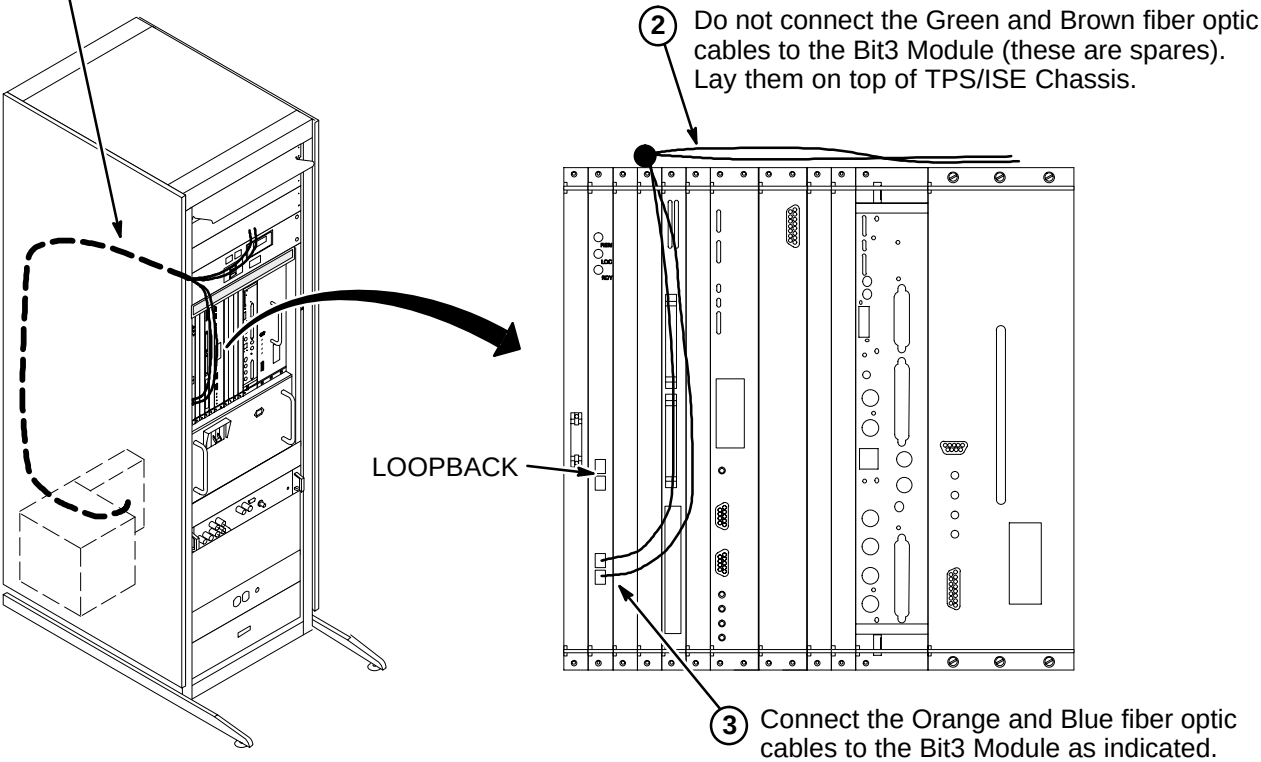
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



* Instructions provided with connectors.

☐ D – ROUTE/CONNECT FIBER OPTIC RUN 836 TO BIT3 MODULE

- 1 Route Run 836 fiber optic cables from cabinet I/F J21 to front of Bit3 module (MR2 A11 A30). Secure cable to side of cabinet. Do not bend fiber optic cables to radius smaller than two inches (50mm).

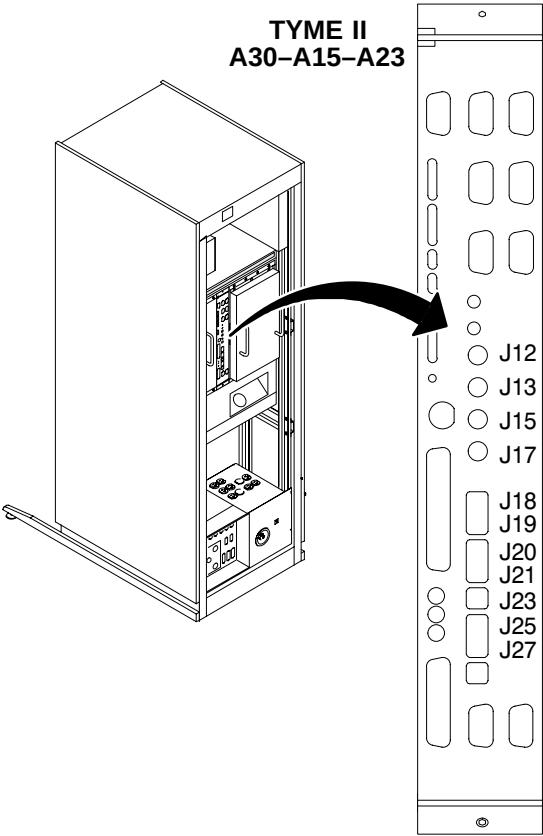


E – ROUTE/CONNECT FIBER OPTIC RUNS 708, 710, AND 711/712 TO TYME II

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.

Note:
Runs 708, 710, 711, and 712 are used for both Horizon 5.x and 8.x. The fiber optic connections to the TYME board are different for 5.x and 8.x. The labels on the cables include both the 5.x and 8.x connections.
The table below indicates the connection labels as they should be marked on this end of the fiber optic cables. **Use only the 8.x connections.**



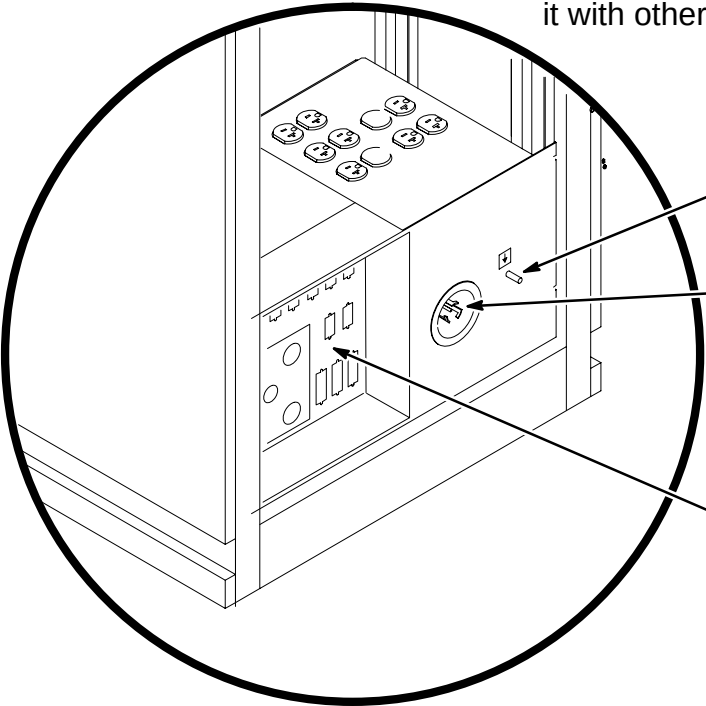
RUN #	5.x	8.x
708-1	J5	J19
710-1	J13	J12
710-2	J15	J13
710-3	J17	J15
710-4	J14	J17
710-5	J4	J18
711-1	J6,J7	J25,J27
711-2	J8	J23
712-1	J9,J10	J20,J21

- 1 Route/Connect as marked Run 710 fiber optic cables to J12, J13, J15, J17 and J18 on the TYME II Board.
- 2 Route/Connect as marked Run 708 fiber optic cable to J19 on the TYME II Board.
- 3 Route/Connect as marked Run 711/712 fiber optic cables to J20, J21, J23, J25, and J27 on the TYME II Board.

Note:
Use tie-wraps to secure fiber optic cables to ISE Chassis cables.

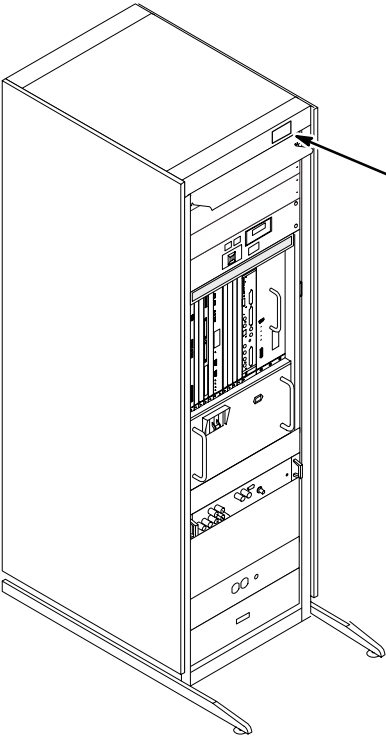
F – CONNECT POWER, GROUND, AND DATA CABLES

Note:
Locate CERD Test Cable, 2168505 (it is packed with the Operator Workspace cable kit, 2154392). Store it with other test cables or in the System Cabinet.



- 1 Connect Run 037 GND cable to ground stud.
- 2 Connect Run 030 or 968 cable plug to "J1". Be sure to push and turn plug for secure connection.
- 3 Perform Ground Resistance Checks before connecting remaining cabinet cables.
- 4 Connect remaining cabinet cables as marked.

G – INSTALL INSITE "SYSTEMS" CABINET MAGNET



- 1 From the 8.0 LX Insite Kit (46-301708G14), locate the "Systems" cabinet magnet (46-320095P5) and attach to front of cabinet as shown.

INSTALLATION STEPS SUMMARY

- ☐ A1 – System Cabinet Upgrade Guidelines
- ☐ A2 – Position/Secure System Cabinet

☐ A1 – SYSTEM CABINET UPGRADE GUIDELINES

The Signa Horizon 8.x System Cabinet Upgrade includes a new cabinet which will require items to be transferred from the old cabinet, such as the GE, GE S–I, Oxford Magnet Cryogen Monitors. This Upgrade also provides instructions for other components and options being transferred to the new System Cabinet. Note that various parts will be left over according to the type of system being upgraded. Discard these parts.

For any of the upgrades below, make sure to attach Rating Plate as shown in procedure **E4**.

UPGRADE FROM SIGNA ADVANTAGE 4.x SYSTEM CABINET

Any Signa Advantage 4.x System being upgraded to Signa 8.x SmartSpeed, HiSpeed Plus, or EchoSpeed Plus will receive a new System Cabinet without the TNF Module.

- To position cabinet and connect system cables, see prodedures **A2, D1, D2, D3, and D4**.
- Transfer Cryogen Meter with its cabling from 4.x System Cabinet to 8.x System Cabinet. Refer to procedure **B1** (GE) or **B2** (GE S–I) or **B3 & B4** (Oxford).
- The new Horizon 8.x System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old two–bay System Cabinet, it will need to be transferred. Refer to procedure **C1, C2, and C3 or C4**. If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade. If multi–coil is being installed, Catalogs M1090ZC or M1090ZG should have been ordered prior to upgrade.

UPGRADE FROM SIGNA ADVANTAGE 5.x SYSTEM CABINET

Any Signa Advantage 5.x System being upgraded to Signa 8.x SmartSpeed, HiSpeed Plus, or EchoSpeed Plus will receive a new System Cabinet without the TNF Module.

- To position cabinet and connect system cables, see prodedures **A2, D1, D2, D3, and D4**.
- Transfer Cryogen Meter with its cabling from 4.x System Cabinet to 8.x System Cabinet. Refer to procedure **B1** (GE) or **B2** (GE S–I) or **B3 & B4** (Oxford).
- The new Horizon 8.x System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old Advantage 5.x System Cabinet, it will need to be transferred. Refer to procedure **C1, C2, and C3 or C4**. If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade. If multi–coil is being installed, Catalogs M1090ZC or M1090ZG should have been ordered prior to upgrade.
- To position cabinet and connect system cables, see prodedures **A2, D1, D2, D3, and D4**.

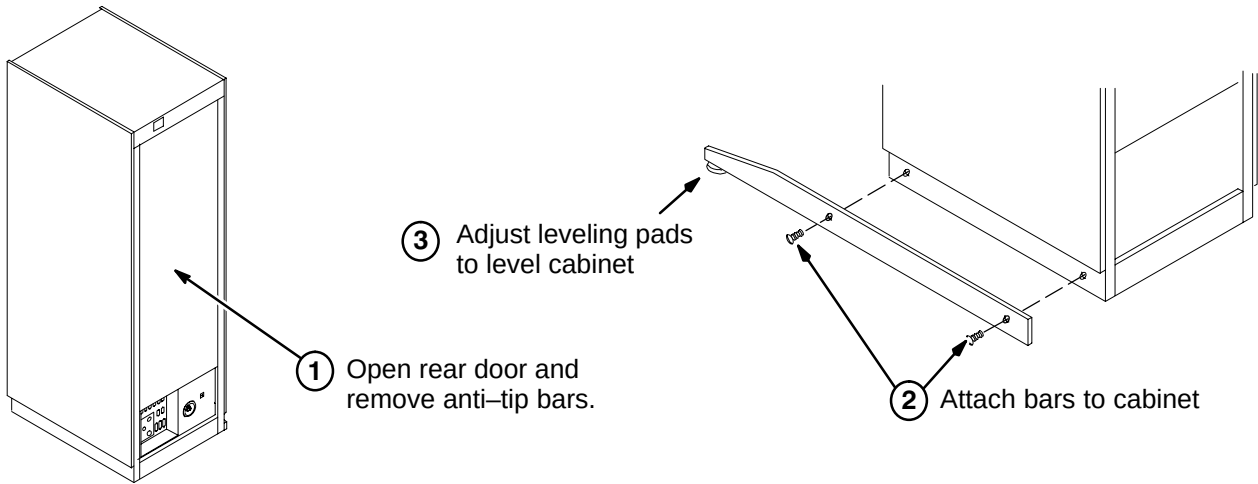
UPGRADE FROM SIGNA HORIZON 5.X SYSTEM CABINET

Any Signa Horizon 5.x System being upgraded to Signa 8.x SmartSpeed, HiSpeed Plus, or EchoSpeed Plus will receive a new System Cabinet without the TNF Module.

- To position cabinet and connect system cables, see prodedures **A2, D1, D2, D3, and D4**.
- Transfer Cryogen Meter with its cabling from 4.x System Cabinet to 8.x System Cabinet. Refer to procedure **B1** (GE) or **B2** (GE S–I) or **B3 & B4** (Oxford).
- The new Horizon 8.x System Cabinet does not have a TNF Assembly installed. If TNF Module assembly is present in the old Horizon 5.x System Cabinet, it will need to be transferred. Refer to procedure **C1, C2, and C3 or C4**. If not present Catalogs M1090ZB or M1090ZF should have been ordered prior to upgrade. If multi–coil is being installed, Catalogs M1090ZC or M1090ZG should have been ordered prior to upgrade.
- To position cabinet and connect system cables, see prodedures **A2, D1, D2, D3, and D4**.
- For upgrading from a Horizon 5.x EchoSpeed to a Horizon 8.x Echosped, the Fast Receiver and cable (2137306) will need to be transferred, refer to procedures **E1, E2, and E3**.

☐ A2 – POSITION/SECURE SYSTEM CABINET

Comply with UL requirements by either installing anti tip legs (see illustration below) or securing the cabinet to the floor with local supplied brackets and floor anchors.



INSTALLATION STEPS SUMMARY

- ☐ B1 – Transfer and Install GE Cryogen Meter
- ☐ B2 – Transfer and Install GE S–I Cryogen Meter
- ☐ B3 – Remove and Upgrade (46–317875G1) Oxford Cryogen Meter
- ☐ B4 – Install Oxford Cryogen Meter

☐ B1 – TRANSFER AND INSTALL GE CRYOGEN METER

1 Remove filler panel

2 If needed, move clip nuts to correct holes for mounting Cryogen Meter module.

3 Remove Cryogen Meter, attaching cables and mounting hardware from existing cabinet and install into new cabinet.

4 Connect Power Cord to Power Module J7.

5 Connect I/F cable from Interface Panel J11 to J403 at back of Monitor.

6 Connect cable from TYME II Board J5 to J1 at back of Monitor.

☐ B2 – TRANSFER AND INSTALL GE S–I CRYOGEN METER

1 Remove Cryogen Meter, attaching cables and mounting hardware from existing cabinet.

2 Remove (3) filler panels.

3 Remove and discard existing insert from Front Cover and replace with GE S–I Meter Access Insert (2155693) using existing hardware.

4 Move cover with new insert to front of cabinet to aid in aligning clip nuts to correct holes for mounting Cryogen Meter module.

5 Install GE S–I Cryogen Meter in cabinet using saved hardware.

6 Remove cover from J12 on I/F panel and install (2) Cable Mounts (46–265067P1) so Nitrogen Cable can be attached.

7 Connect power cords and signal cables as indicated above.

Connect Helium Cable to Interface Panel J11.

Connect Nitrogen Cable to Interface Panel J12.

Connect cable from TYME II Board J5 to J1 at back of Monitor.

Connect Power Cord to Power Module J7.

☐ B3 – REMOVE AND UPGRADE (46–317875G1) OXFORD CRYOGEN METER

Note: This procedure may have already been performed in previous upgrades.

1 Remove Cryogen Meter and its connecting cables from existing cabinet. Save hardware.

2 Remove Cryogen Meter top cover. Save hardware.

3 Drill .265 (17/64) hole in front cover at the indicated dimensions.

4 Mount Reed Switch/Harness (46–301387G1) in hole.

5 Route and secure brown twisted wires to existing cable with cable ties (46–208758P1).

6 Remove shrink tubing and unsolder white/red leads from terminal.

7 Splice white/red lead to one of the brown twisted wires. Solder other white/red lead and other brown twisted wire to terminal.

8 Replace top cover.

9 Attach “PRESS AND READ...” label to front cover.

☐ B4 – INSTALL OXFORD CRYOGEN METER

1 Remove filler panel.

2 Remove and discard existing insert from Front Cover and replace with Oxford Meter Access Insert (2155691–2) using existing hardware.

3 Move clip nuts to correct holes and install Cryogen Meter in space where filler panel was removed using saved hardware.

4 Remove cover from J12 on I/F panel and install (2) Cable Mounts (46–265067P1) so Nitrogen Cable can be attached.

5 Connect power cords and signal cables as indicated above.

Connect Helium Cable to Interface Panel J11.

Connect Nitrogen Cable to Interface Panel J12.

Connect Helium Power Cord to Power Module J7.

Connect Nitrogen Power Cord to Power Module J8.

Connect cable from TYME II Board J5 to J1 at back of Monitor.

INSTALLATION STEPS SUMMARY

- ☐ C1 – Remove TNF Module from Existing Cabinet
- ☐ C2 – Install TNF Module into New 8.x Cabinet
- ☐ C3 – TNF Module Wiring Diagram
- ☐ C4 – TNF Module with Expansion Assembly Wiring Diagram

C1 – REMOVE TNF MODULE FROM EXISTING CABINET

For Upgrading from Horizon 5.x: TNF Assembly is present in existing cabinet for transfer. The TNF Expansion Asm (2118107) should also have been installed if Multi-coil is being transferred.

For Upgrading from Advantage 5.x: If present, transfer TNF Assembly Module from existing cabinet. If TNF Assembly Module is not present, order 2118106 for non Multi-coil system or 2118106 and 2118107 for Multi-coil.

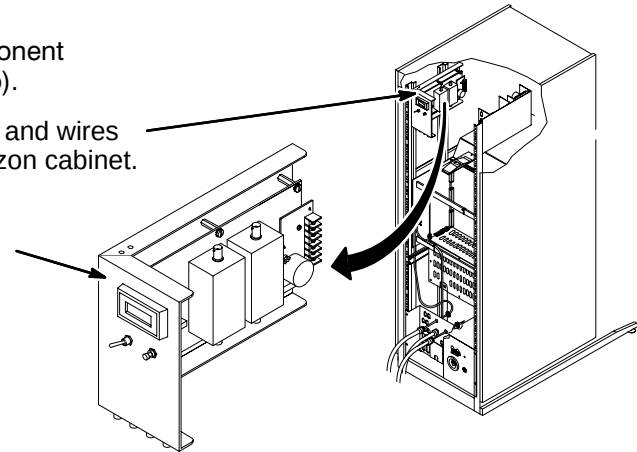
For Upgrading from Advantage 4.x: If present, transfer TNF Assembly Module from existing cabinet. If TNF Assembly Module is not present, order 2118106 for non Multi-coil system or 2118106 and 2118107 for Multi-coil.



Note: TNF Module is a static sensitive component and needs to be handled as such (wrist strap).

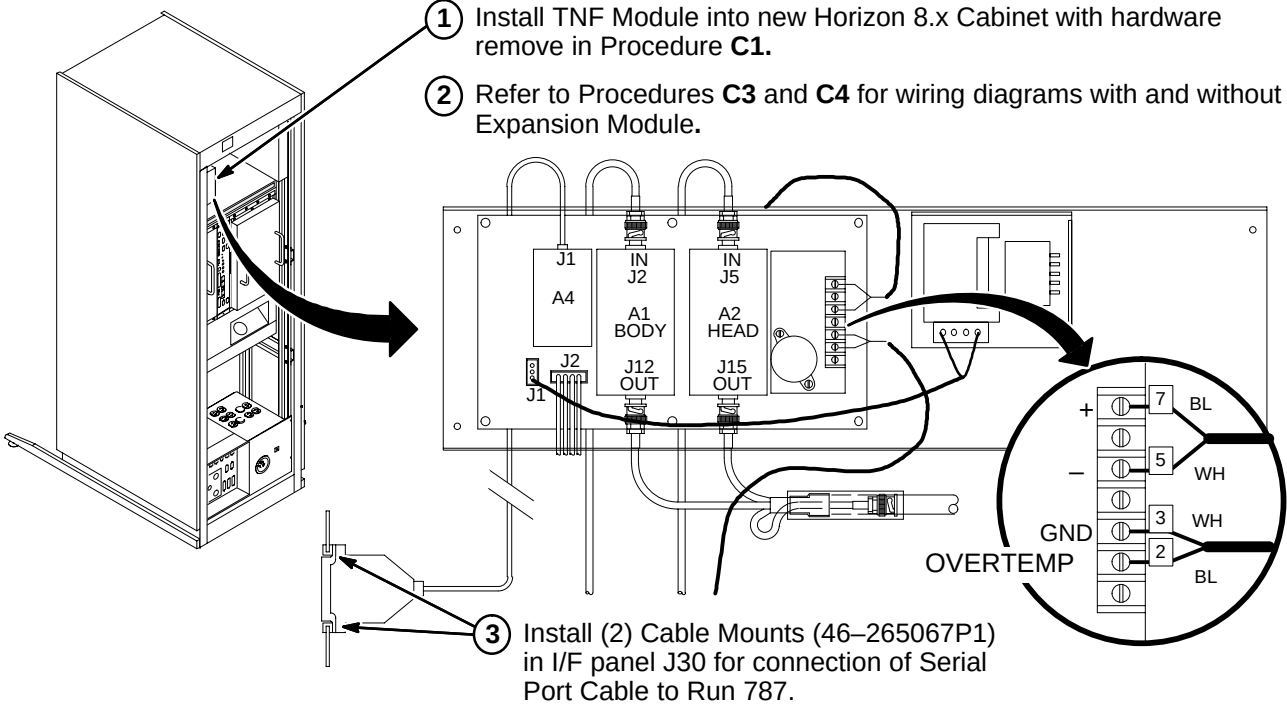
- 1 Disconnect and discard all cables and wires connected to TNF Module in Horizon cabinet.

- 2 Unfasten and remove TNF Assembly Module (with Expansion Asm if present). Save mounting hardware.



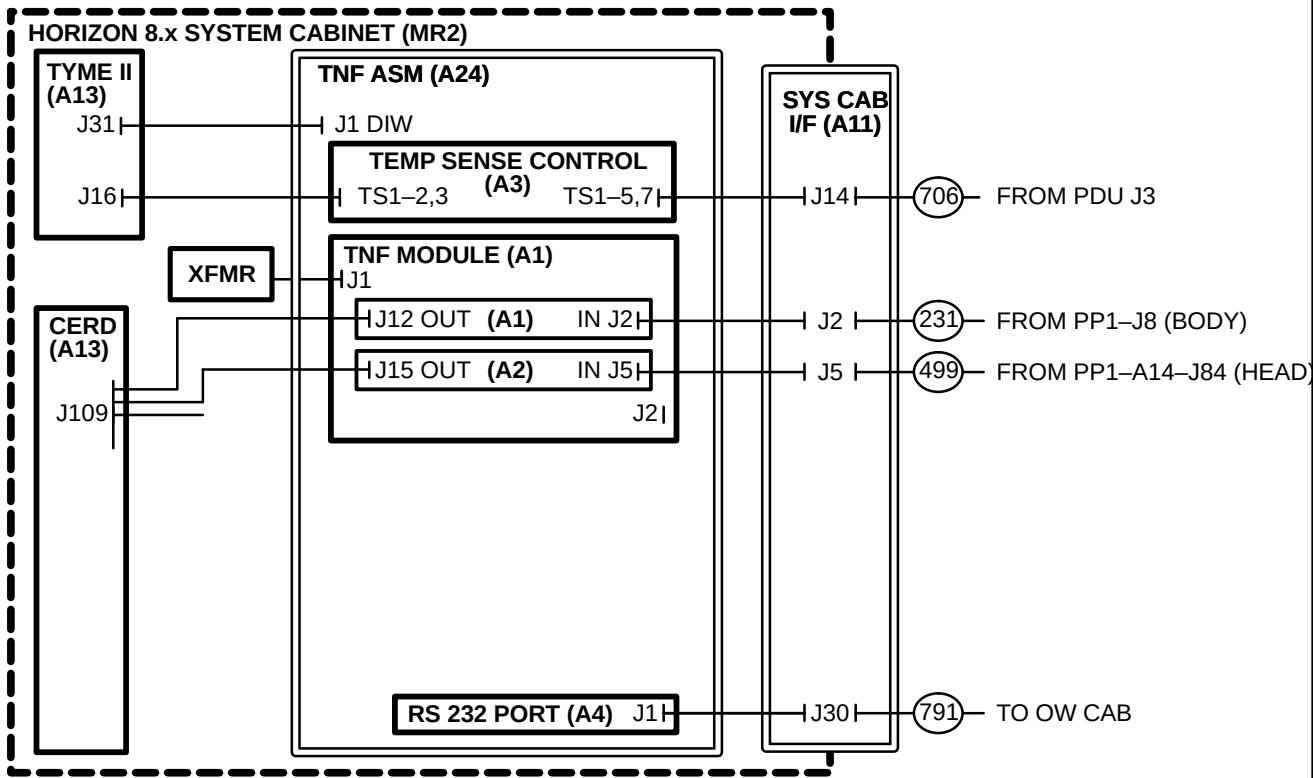
C2 – INSTALL TNF MODULE INTO NEW 8.x CABINET

- 1 Install TNF Module into new Horizon 8.x Cabinet with hardware remove in Procedure C1.
- 2 Refer to Procedures C3 and C4 for wiring diagrams with and without Expansion Module.



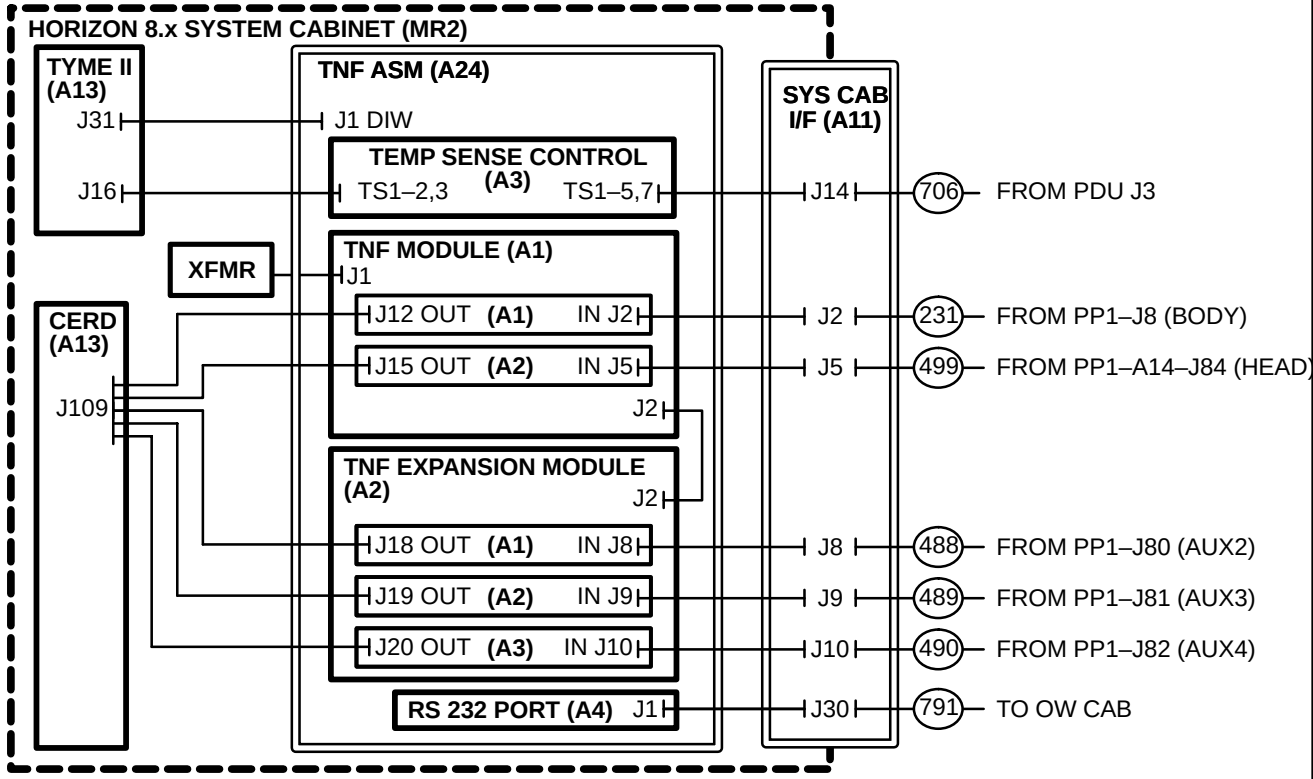
C3 – TNF MODULE WIRING DIAGRAM

- 1 Connect new System Cabinet cables to TNF Module as indicated below.



C4 – TNF MODULE WITH EXPANSION ASSEMBLY WIRING DIAGRAM

- 1 Connect new System Cabinet cables to TNF Module with Expansion Assembly as indicated below.



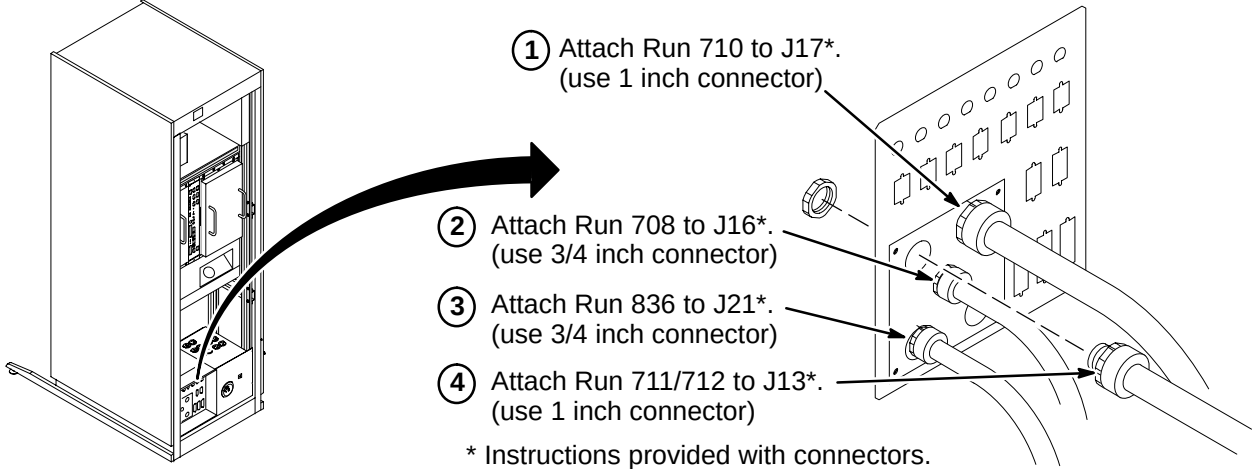
INSTALLATION STEPS SUMMARY

- ❑ D1 – Connect Fiber Optic Runs 708, 710, 711/712, and 836 to I/F Panel
- ❑ D2 – Route/Connect Fiber Optic Runs 790 to BIT3 Fiber Optic Interface
- ❑ D3 – Route/Connect Fiber Optic Runs 708, 710, and 711/712 to Tyme II
- ❑ D4 – Connect Power, Ground, and Data Cables

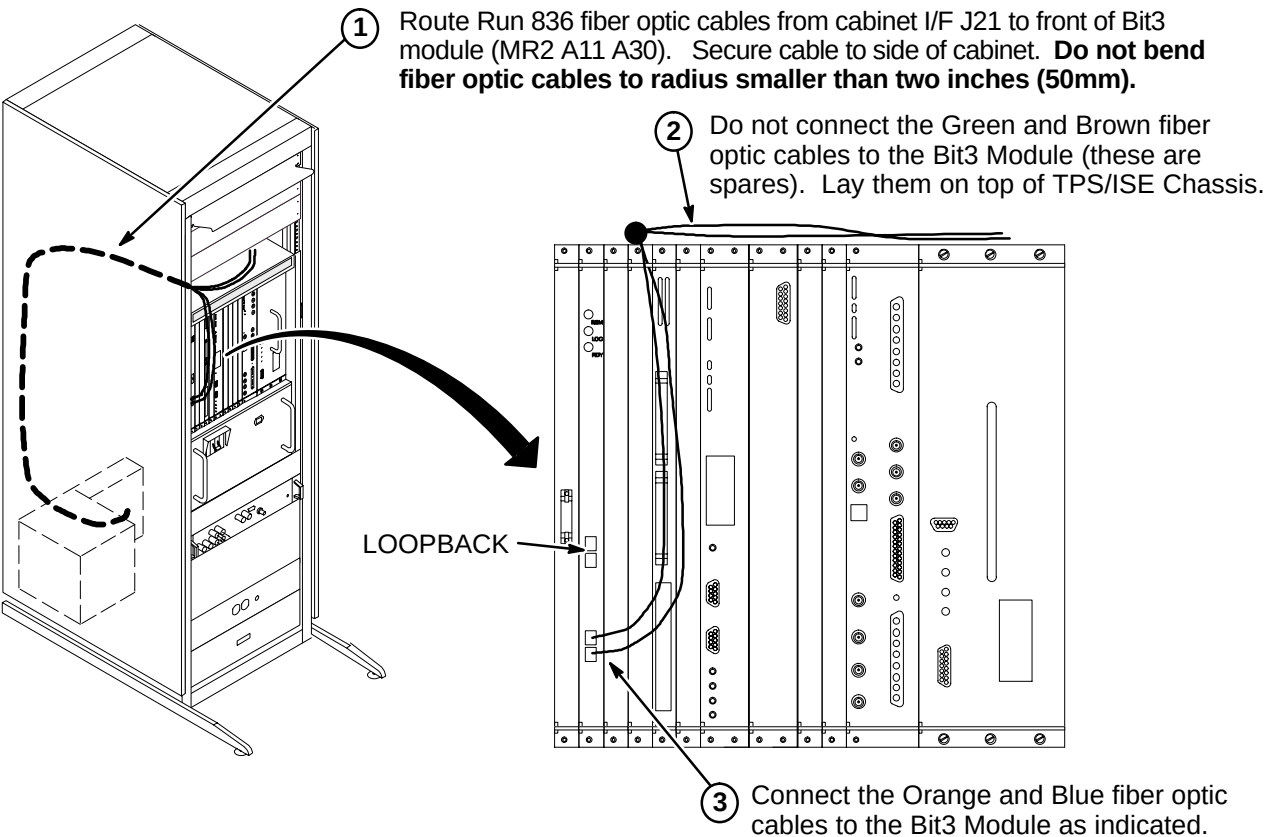
❑ D1 – CONNECT FIBER OPTIC RUNS 708, 710, 711/712, AND 836 TO I/F PANEL

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.



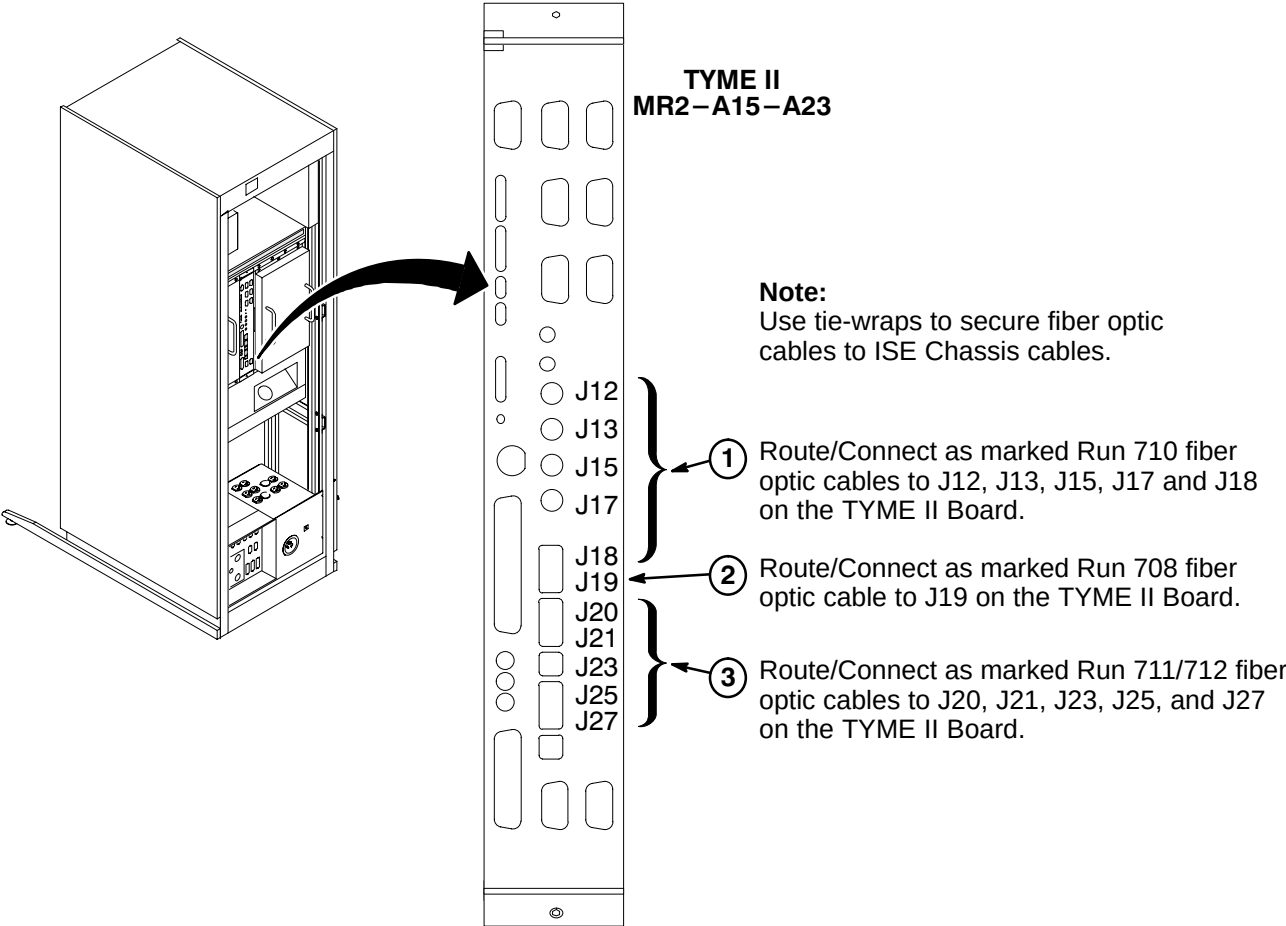
❑ D2 – ROUTE/CONNECT FIBER OPTIC RUN 836 TO BIT3 MODULE



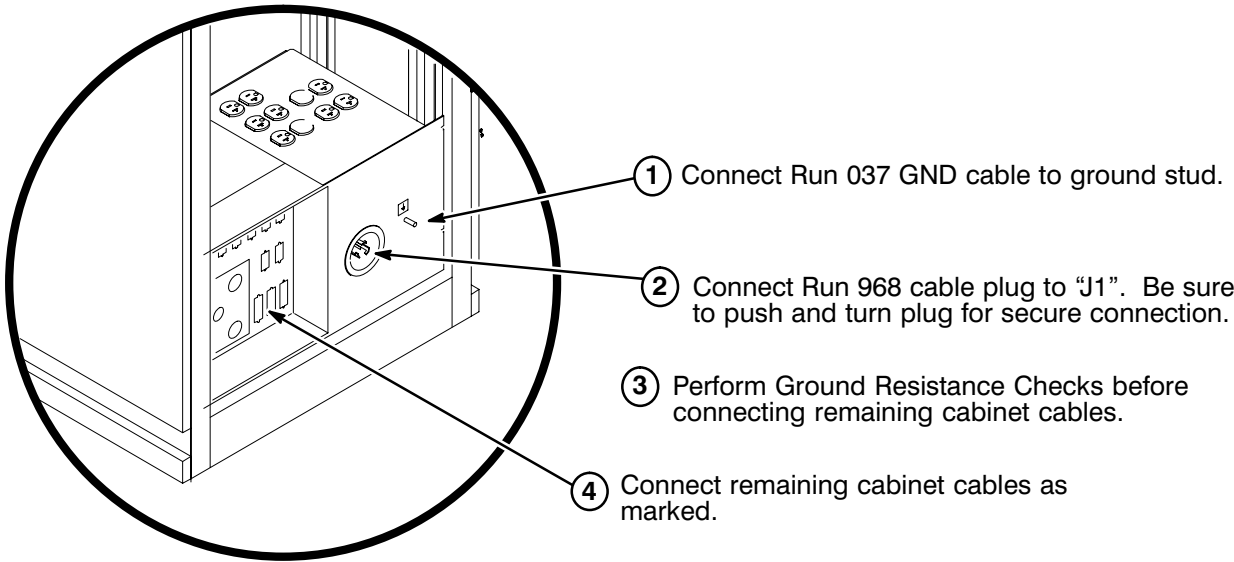
❑ D3 – ROUTE/CONNECT FIBER OPTIC RUNS 708, 710, AND 711/712 TO TYME II

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.



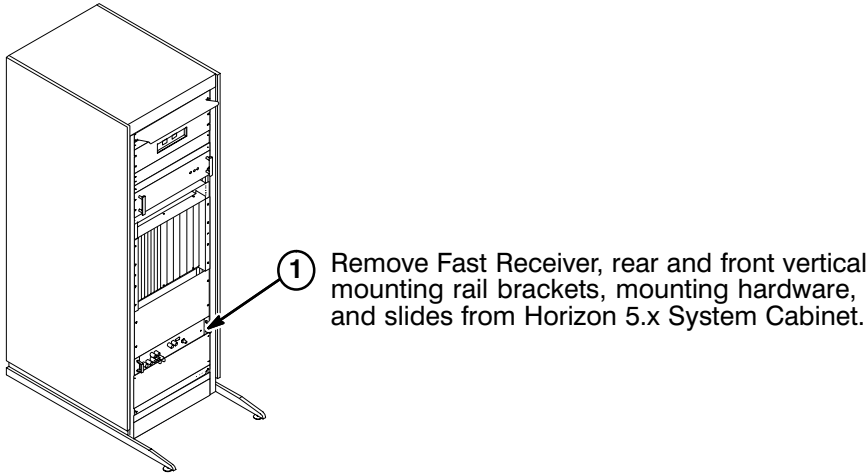
❑ D4 – CONNECT POWER, GROUND, AND DATA CABLES



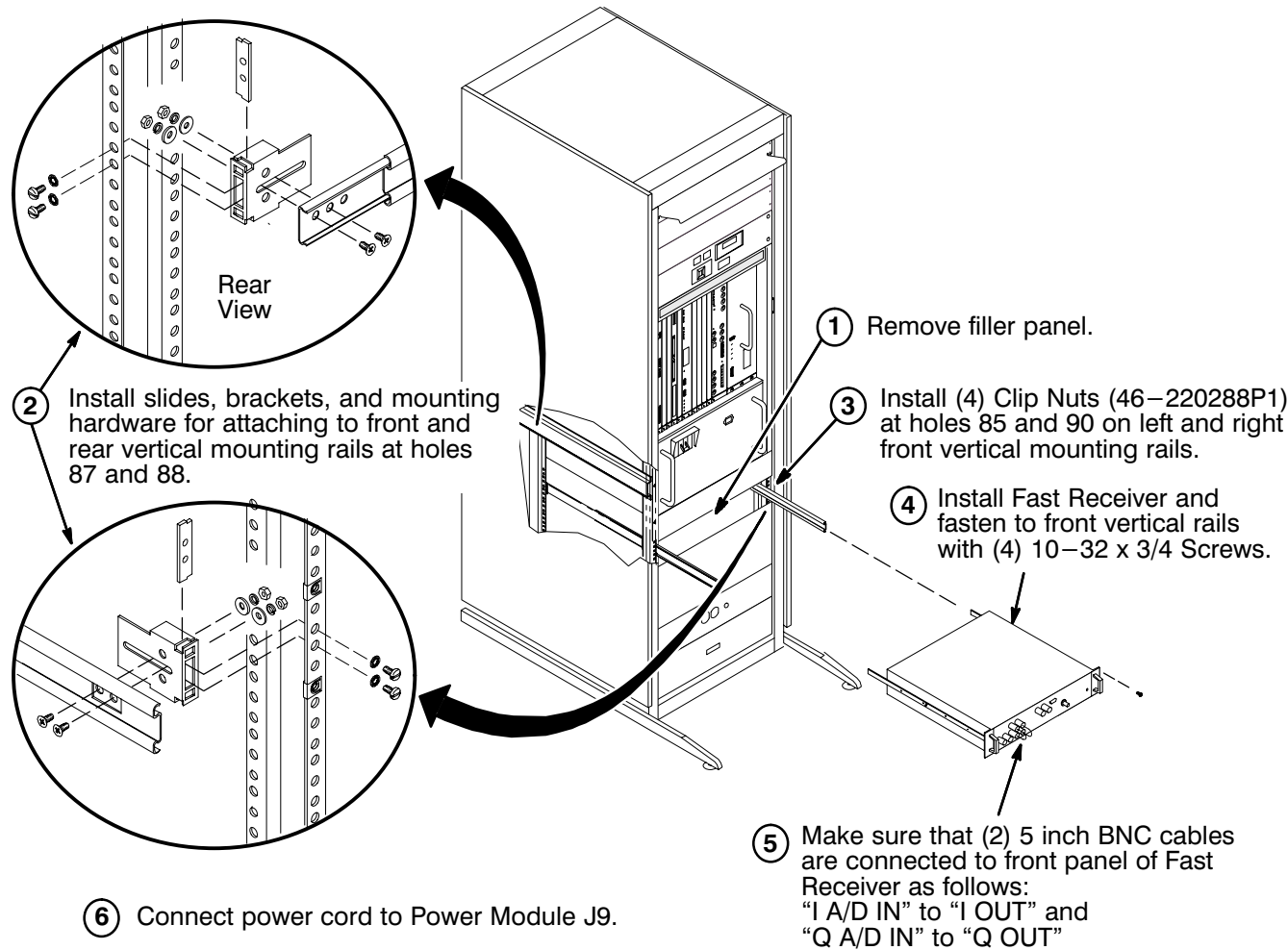
INSTALLATION STEPS SUMMARY

- ❑ E1 – Remove Fast Receiver Module form Horizon System Cabinet
- ❑ E2 – Transfer and Install Fast Receiver Module
- ❑ E3 – Relabel and Connect Fast Receiver Cables
- ❑ E4 – Install Rating Plate

E1 – REMOVE FAST RECEIVER MODULE FROM HORIZON SYSTEM CABINET

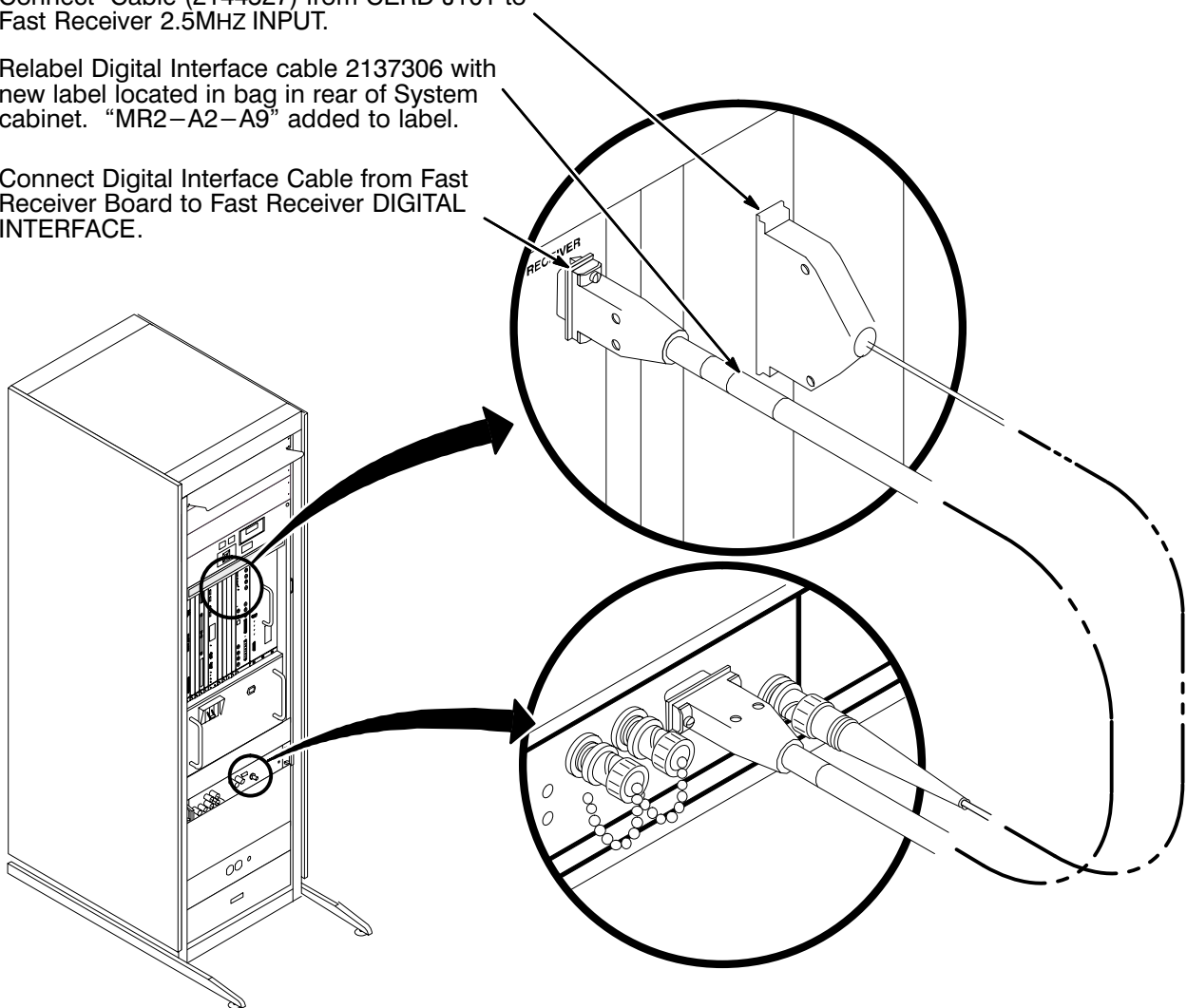


E2 – TRANSFER AND INSTALL FAST RECEIVER MODULE



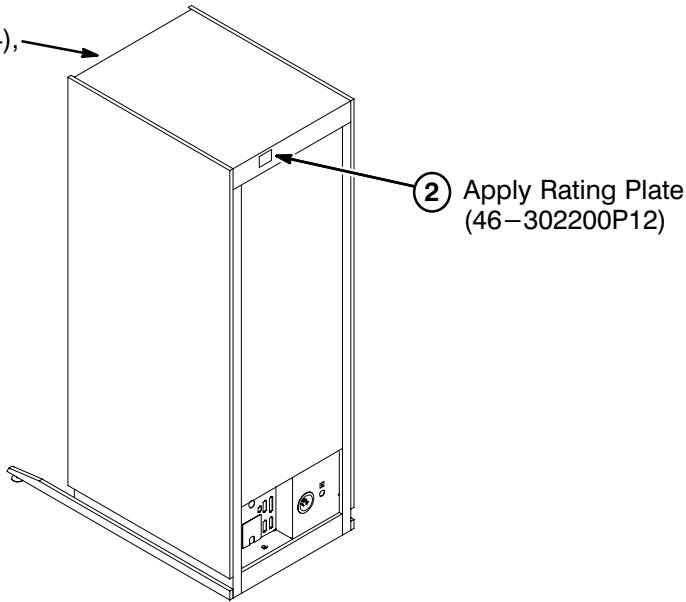
E3 – RELABEL AND CONNECT FAST RECEIVER CABLES

- 1 Connect Cable (2144527) from CERD J101 to Fast Receiver 2.5MHZ INPUT.
- 2 Relabel Digital Interface cable 2137306 with new label located in bag in rear of System cabinet. "MR2-A2-A9" added to label.
- 3 Connect Digital Interface Cable from Fast Receiver Board to Fast Receiver DIGITAL INTERFACE.



E4 – INSTALL RATING PLATE

- 1 From the 8.0 8.x Insite Kit (46-301708G14), locate the "Systems" cabinet magnet (46-320095P5) and attach to front of cabinet as shown.
- 2 Apply Rating Plate (46-302200P12)

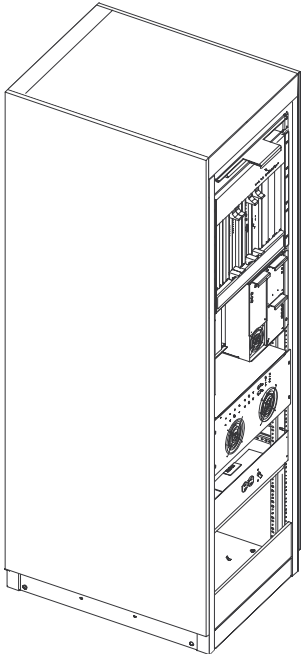


INSTALLATION STEPS SUMMARY

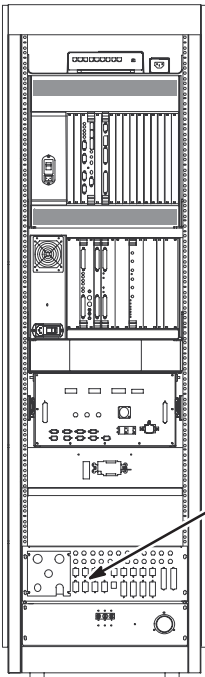
- ☐ A – Position/Secure System Cabinet
- ☐ B – Install Magnet Monitoring Cables
- ☐ C – Connect Fiber Optic Runs 1033, 1034, 1044, and 1045 to I/F Panel
- ☐ D – Route/Connect Fiber Optic Run 1044 to APS Board
- ☐ E – Route/Connect Fiber Optic Runs 1033, 1034, and 1045 to STIF Board
- ☐ F – Connect Power, Ground, and Data Cables
- ☐ G – Install InSite “SYSTEMS” Cabinet Magnet

A – POSITION/SECURE SYSTEM CABINET

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.



B – INSTALL MAGNET MONITORING CABLES



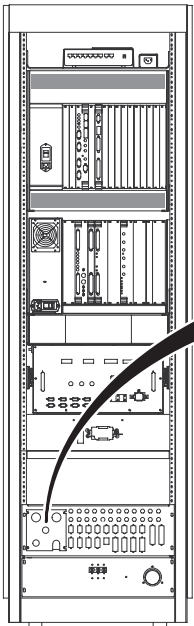
Note: Refer to Direction 2230681, *Magnet Monitor Hardware Installation Manual*, for instructions on connecting cables in the System Cabinet. This manual should be found in the box containing the parts for Magnet Monitoring.

- 1 Connect Run 823 to J24.

C – CONNECT FIBER OPTIC RUNS 1033, 1034, 1044, AND 1045 TO I/F PANEL

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

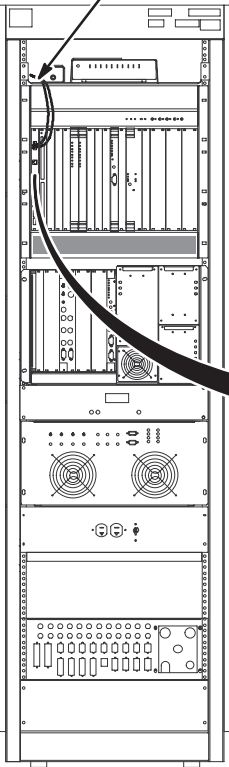


- 1 Attach Run 1034 to J17*.
(use 1 inch connector)
- 2 Attach Run 1045 to J13*.
(use 1 inch connector)
- 3 Attach Run 1033 to J16*.
(use 3/4 inch connector)
- 4 Attach Run 1044 to J22*.
(use 3/4 inch connector)

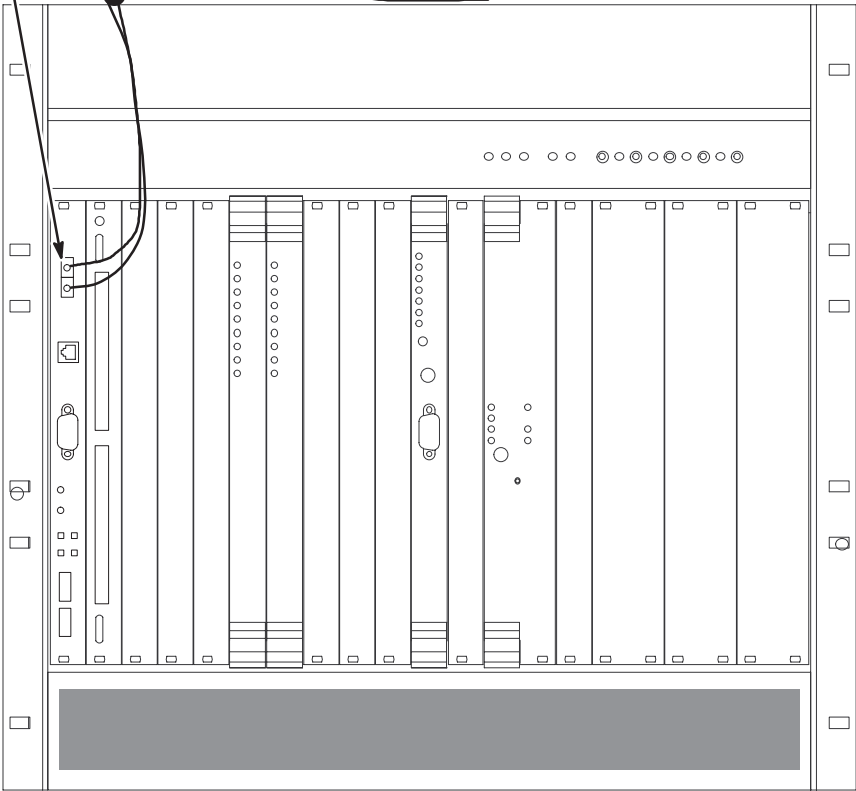
* Instructions provided with connectors.

D – ROUTE/CONNECT FIBER OPTIC RUN 1044 TO APS BOARD

- 1 Route Run 1044 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet. Do not bend fiber optic cables to radius smaller than two inches (50mm).



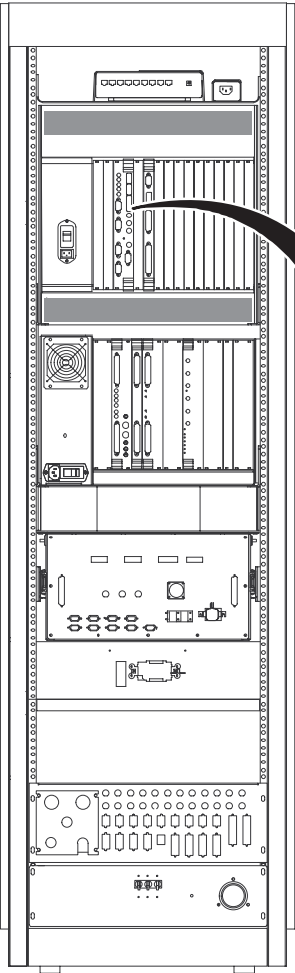
- 2 Connect the Orange and Blue fiber optic cables to the TI-MGD Daughter Board as indicated.
- 3 Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.



E – ROUTE/CONNECT FIBER OPTIC RUNS 1033, 1034, AND 1045 TO STIF BOARD

CAUTION

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.



NOTE:
Refer to table at right for relabel information if existing cables are to be reused.

New Cables		Existing Cables	
RUN #	10.x	RUN #	8.x/9.x
1033-1	J21	708-1	J19
1034-1	J14	710-1	J12
1034-2	J13	710-2	J13
1034-3	J12	710-3	J15
1034-4	J11	710-4	J17
1034-5	J20	710-5	J18
1045-1	J17	711-2	J23
1045-2	J18,J19	711-3	J20, J21
1045-3	J15,J16	711-1	J25, J27

- J21

J20

J19

J18

J17

J16

J15

J14

J13

J12

J11
- 1

2

3
- Relabel (if old cable re-used) and Route/Connect Run 1033 fiber optic cable to J21 on the STIF Board.

Part of Run 1034 below.

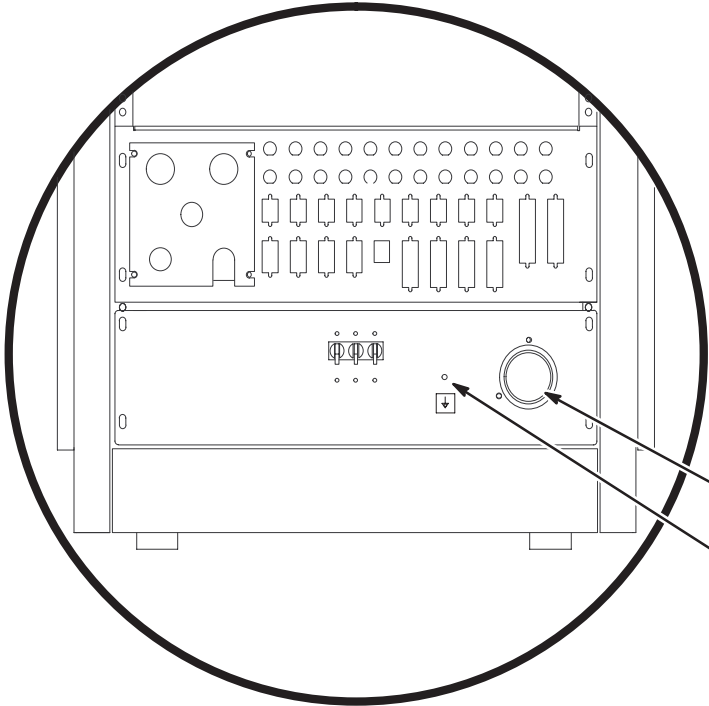
Relabel (if old cable re-used) and Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.

Relabel (if old cable re-used) and Route/Connect Run 1034 fiber optic cables to J11, J12, J13, J14, and J20 above on the STIF Board.

Note:
Use tie-wraps to secure fiber optic cables to MGD Chassis cables.

F – CONNECT POWER, GROUND, AND DATA CABLES

Note:
Locate CERD Test Cable, 2168505 (it is packed with the Operator Workspace cable kit, 2154392). Store it with other test cables or in the System Cabinet.



WARNING!

SYSTEM CABINET POWER AND GROUND CABLES MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.

- 1

2

3

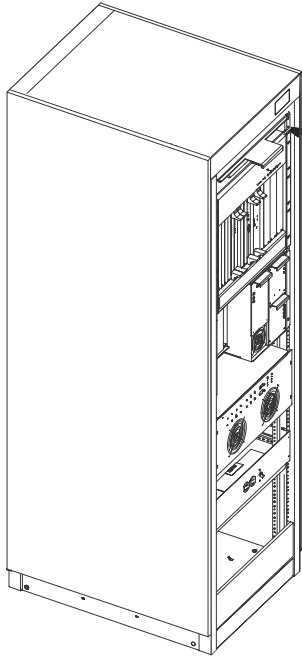
4
- Connect Run 968 cable plug to "J1". Be sure to push and turn plug for secure connection.

Connect Run 037 GND cable to ground stud.

Perform Ground Resistance Checks before connecting remaining cabinet cables.

Connect remaining cabinet cables as marked.

G – INSTALL INSITE "SYSTEMS" CABINET MAGNET



- 1
- From the LX Insite Kit (46-301708G14), locate the "Systems" cabinet magnet (46-320095P5) and attach to front of cabinet as shown.

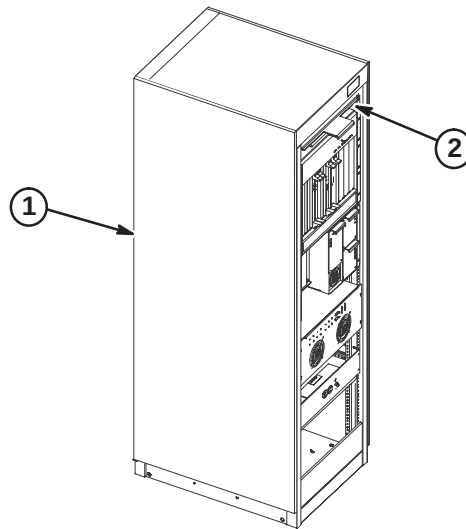
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1-5 ROUTE/CONNECT FIBER OPTIC RUNS 1033, 1034, AND 1045 TO STIF BOARD	1-5
1-6 CONNECT COMMUNICATION CABLES TO SYSTEM CABINET I/F	1-6

1-1 POSITION AND SECURE SYSTEM CABINET

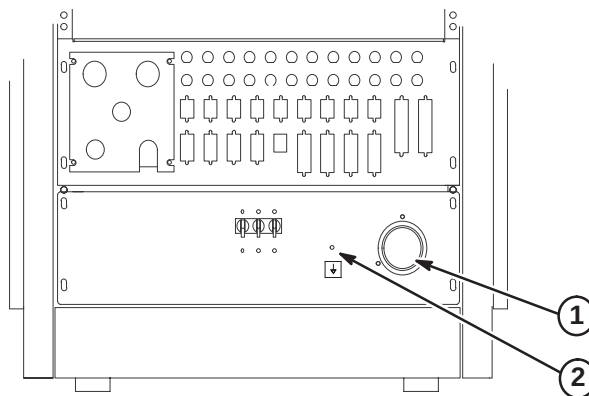
Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

1. Position System cabinet to final position as indicated by site layout drawings.
2. From the LX Insite Kit (46-301708G14), locate the "Systems" cabinet magnet (46-320095P5) and attach to front of cabinet as shown.



1-2 SYSTEM CABINET POWER AND GROUND CABLE CONNECTIONS

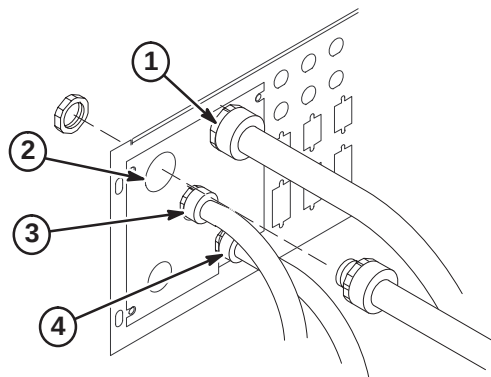
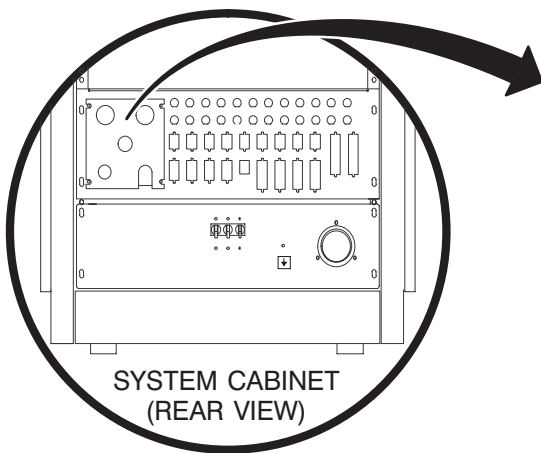
1. Connect Run 968 cable plug to "J1". Be sure to push and turn plug for secure connection.
2. Connect Run 037 GND cable to ground stud.
3. Perform Ground Resistance Checks before connecting remaining cabinet cables.



1-3 CONNECT FIBER OPTIC RUNS 1033, 1034, 1045, AND 1083 TO SYSTEM CABINET I/F PANEL**CAUTION**

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Attach Run 1034 to J17*. (use 1 inch connector)
2. Attach Run 1045 to J13*. (use 1 inch connector)
3. Attach Run 1033 to J16*. (use 3/4 inch connector)
4. Attach Run 1083 to J22*. (use 3/4 inch connector)

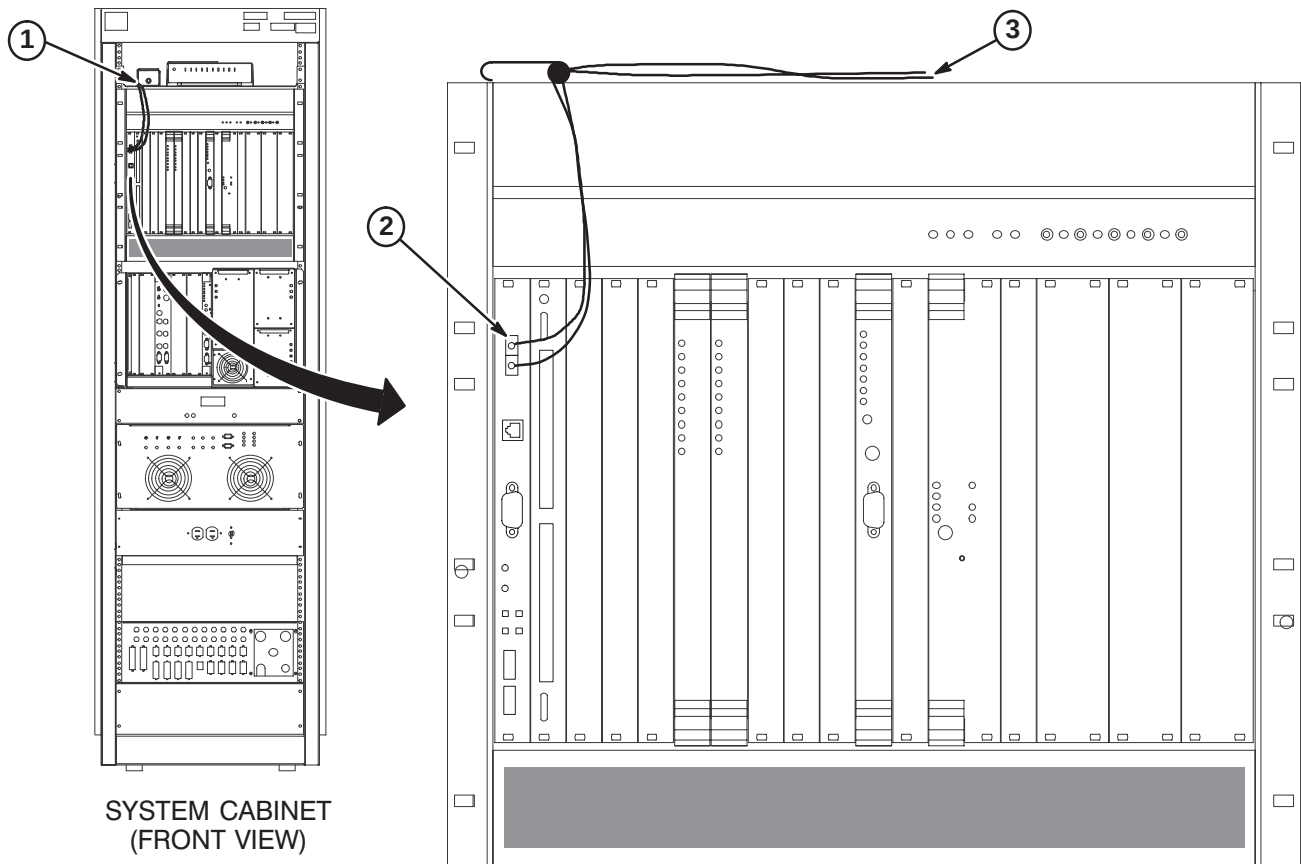


* Instructions provided with connectors.

1-4 ROUTE/CONNECT FIBER OPTIC RUN 1083 TO APS BOARD**CAUTION**

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route Run 1083 fiber optic cables from cabinet I/F J22 to front of APS module. Secure cable to side of cabinet. **Do not bend fiber optic cables to radius smaller than two inches (50mm).**
2. Connect the Orange and Blue fiber optic cables to the TI-MGD Daughter Board as indicated.
3. Do not connect the Green and Brown fiber optic cables to the APS Module (these are spares). Lay them on top of MGD Chassis.

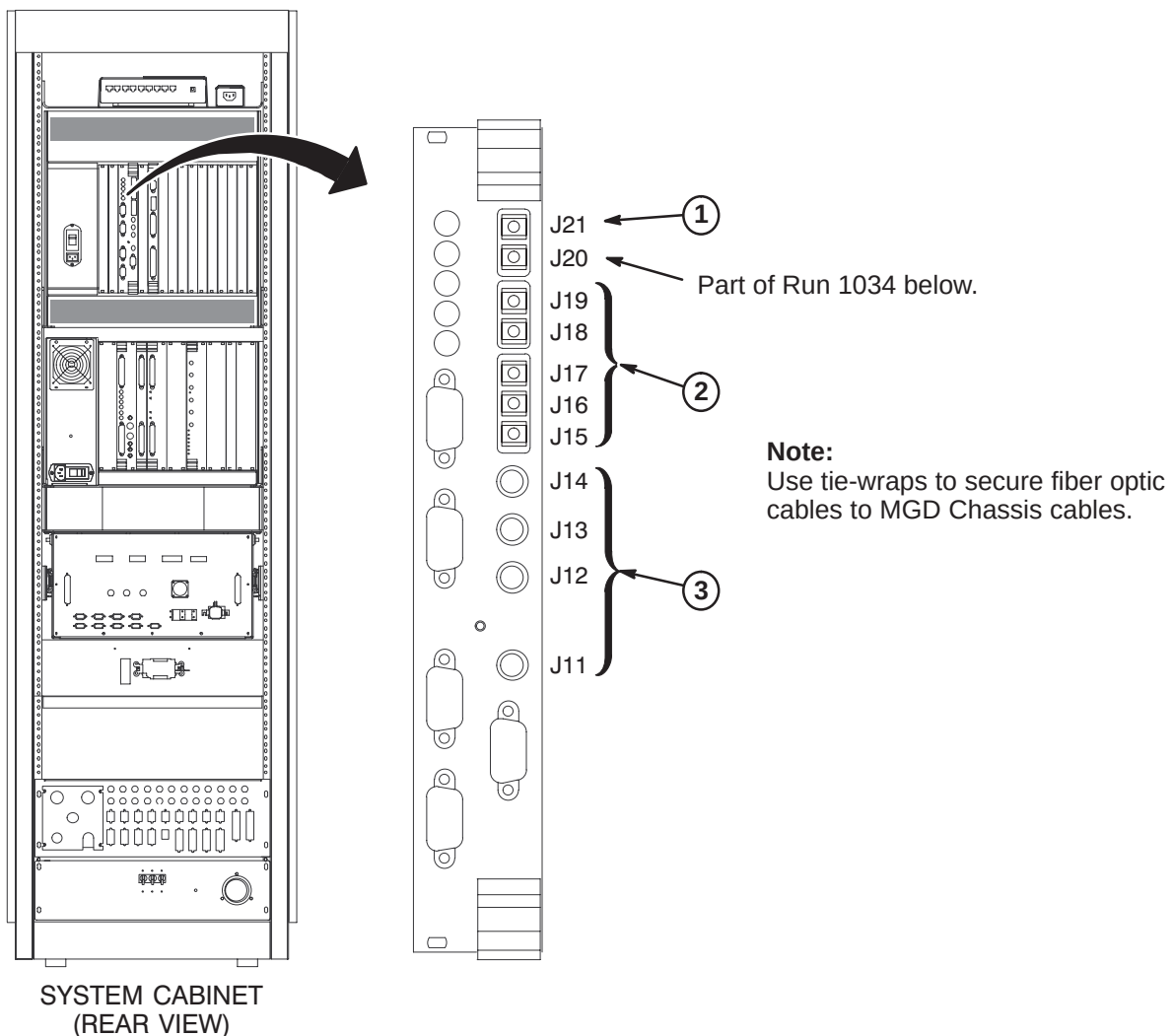


SYSTEM CABINET
(FRONT VIEW)

1-5 ROUTE/CONNECT FIBER OPTIC RUNS 1033, 1034, AND 1045 TO STIF BOARD**CAUTION**

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

1. Route/Connect Run 1033 fiber optic cable to J21 on the STIF Board.
2. Route/Connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
3. Route/Connect Run 1034 fiber optic cables to J11, J12, J13, J14, and J20 above on the STIF Board.



1-6 CONNECT COMMUNICATION CABLES TO SYSTEM CABINET I/F

1. Connect RF Door Switch cable Run 701 to J15
2. Connect Magnet Monitor cable Run 823 to J24
3. Connect remaining cables as marked. Refer to cable map in Cable section of Overview tab.

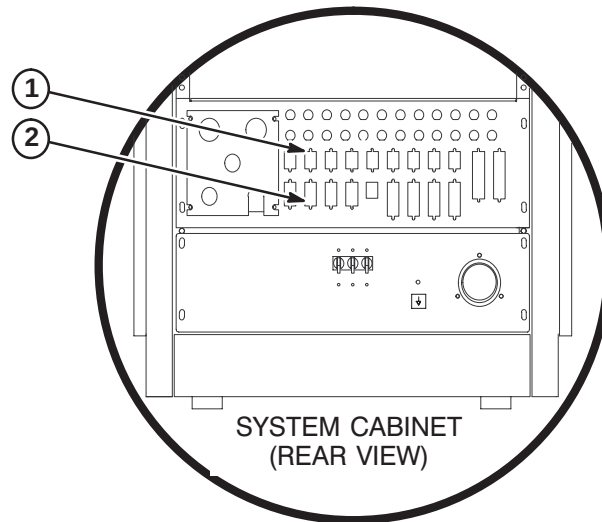
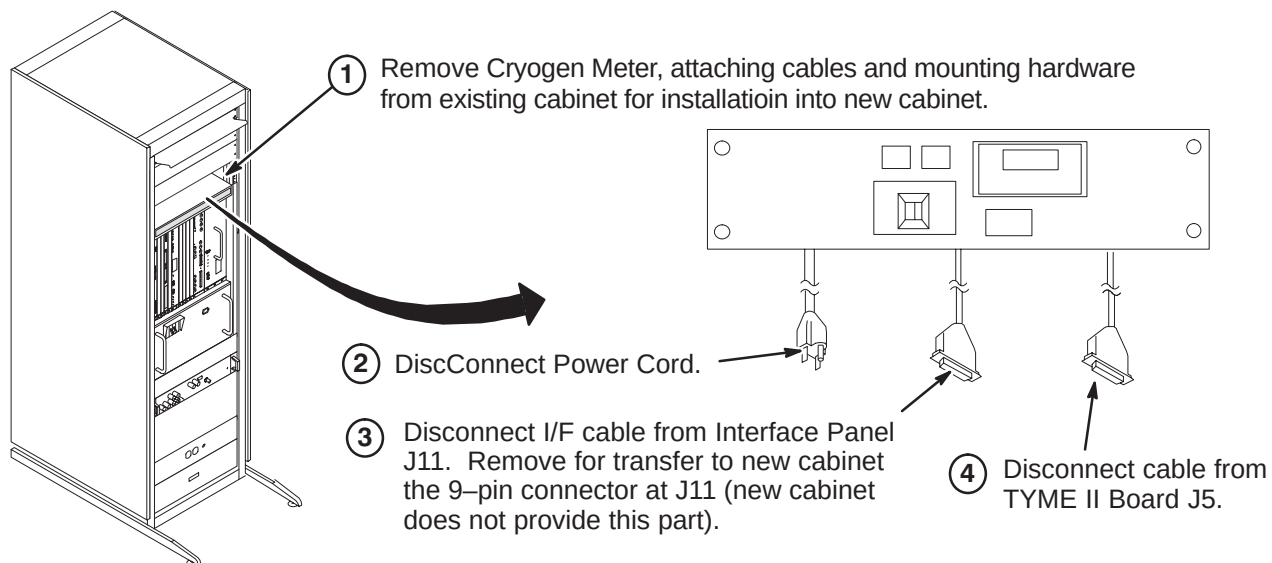
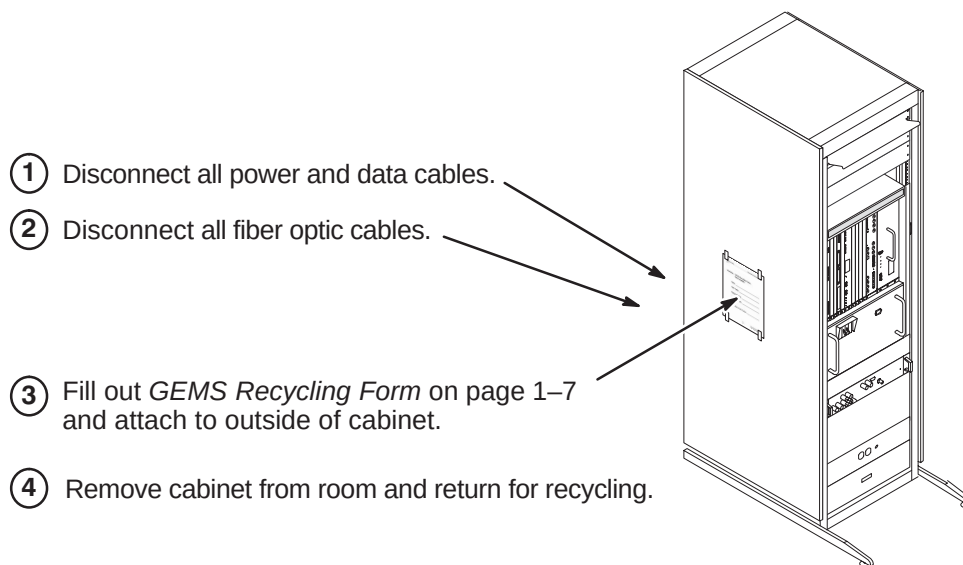


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1-1 REMOVE CRYOGEN METER (for non-Magnet Monitor systems)

The new EXCITE System Cabinet does not contain a cryogen meter. For those non-Magnet Monitor sites being upgraded to EXCITE, the cryogen meter will have to be transferred to the new System cabinet. This section refers to removal of cryogen meter from existing cabinet and Section 1-5 refers to installing it in new EXCITE System cabinet.

**1-2 DISCONNECT CABLES AND PREPARE CABINET FOR RETURN**

1-3 POSITION AND SECURE EXCITE SYSTEM CABINET

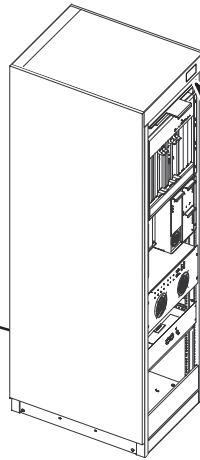
Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

1. Position System cabinet to final position as indicated by site layout drawings.
2. From the LX Insite Kit (46-301708G14), locate the "Systems" cabinet magnet (46-320095P5) and attach to front of cabinet as shown.

Comply with UL requirements or local codes for securing the cabinet to the floor with locally supplied brackets and floor anchors.

NOTE: Refer to Section 4 for cable routing and connection information.

- ① Position System cabinet to final position as indicated by site layout drawings.

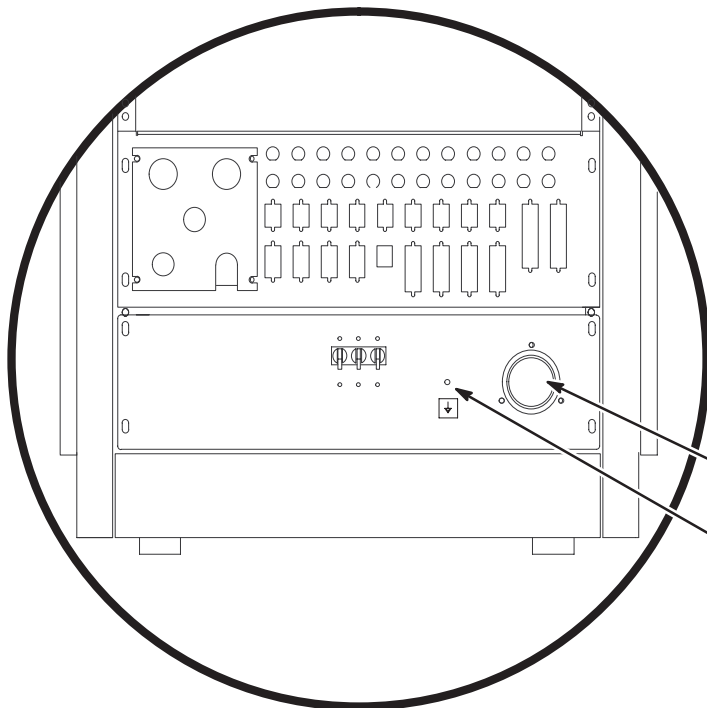


- ② From the LX Insite Kit (46-301708G14), locate the "Systems" cabinet magnet (46-320095P5) and attach to front of cabinet as shown.

1-4 SYSTEM CABINET POWER AND GROUND CABLE CONNECTIONS

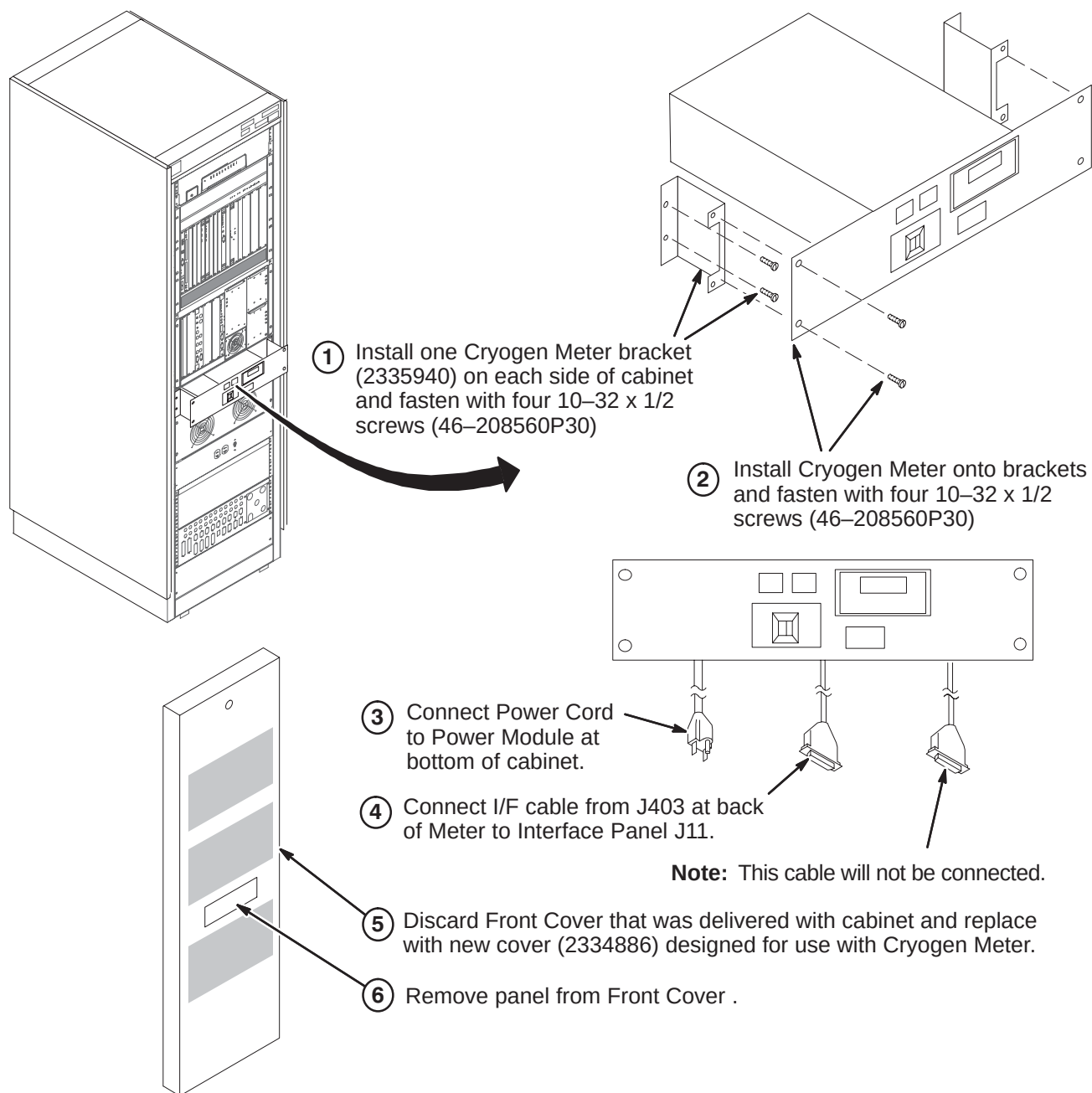
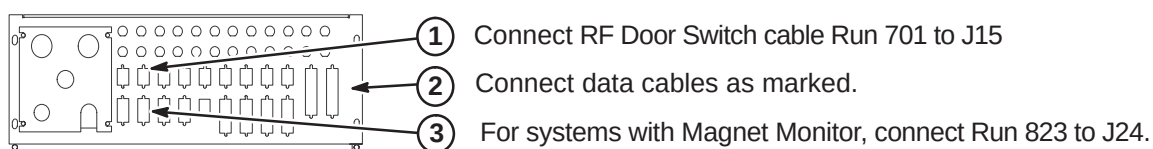
Note:

Transfer CERD Test Cable, 2168505 from removed System cabinet. Store it with other test cables or in the new Excite System Cabinet.



**SYSTEM CABINET POWER
AND GROUND CABLES
MUST BE CONNECTED TO
PDU MODULE LOCATED
WITHIN ACGD/PDU CABINET.**

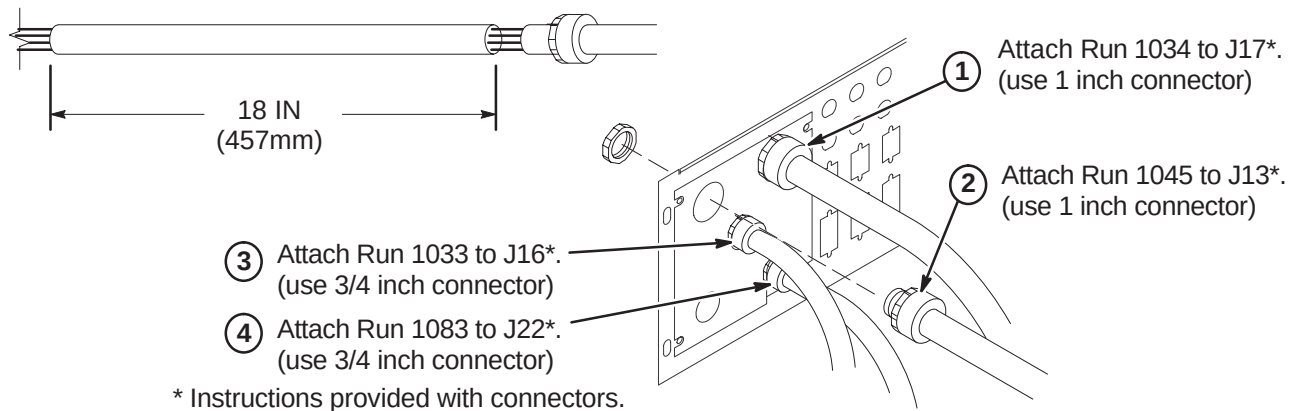
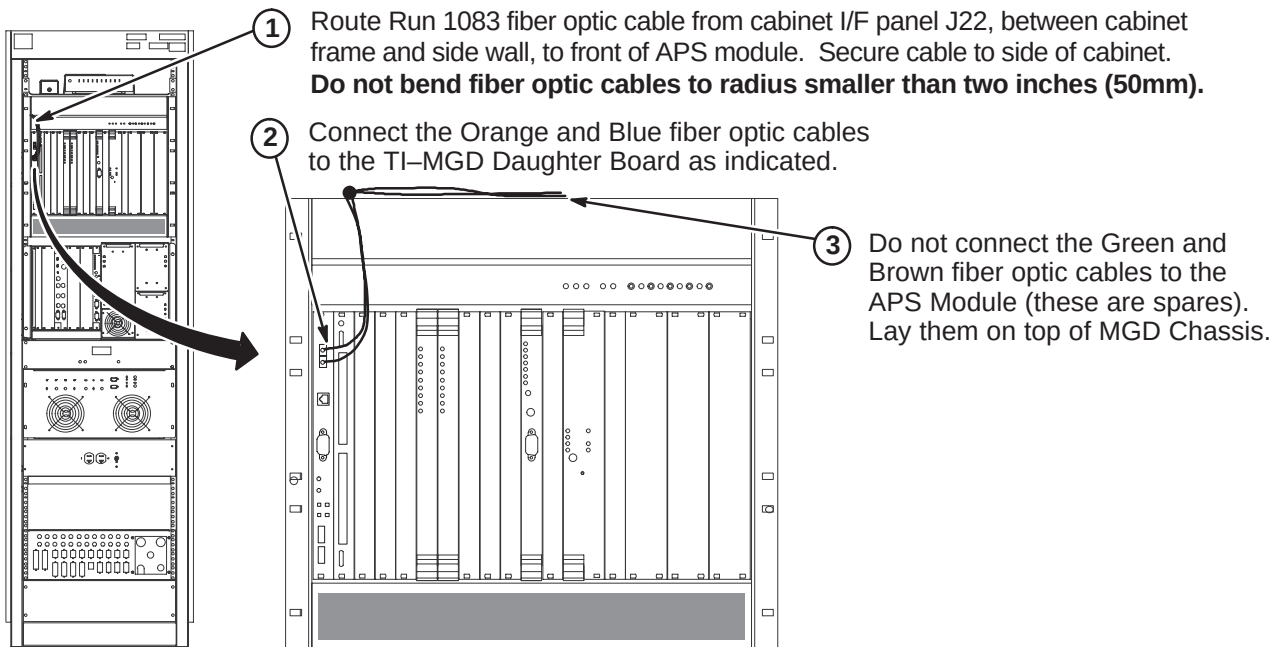
- ① Connect Run 968 cable plug to "J1". Be sure to push and turn plug for secure connection.
- ② Connect Run 037 GND cable to ground stud.
- ③ Perform Ground Resistance Checks before connecting remaining cabinet cables.

1-5 INSTALL CRYOGEN METER (for non-Magnet Monitor systems)**1-6 CONNECT COMMUNICATION CABLES TO SYSTEM CABINET I/F**

1-7 CONNECT FIBER OPTIC RUNS 1033, 1034, 1045, AND 1083 TO SYSTEM CABINET I/F PANEL**CAUTION**

Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches (50mm). Avoid scratching connector ends. Keep connectors protected until ready to connect.

NOTE: Runs 710, and 711/712 may be relabeled as Runs 1034 and 1045 and reused rather than installing new Runs 1034 and 1045. To reuse these Runs, the conduit will need to be shortened about 18 inches (457mm) at the System Cabinet end (see below). This is required because longer fiber optic wires are needed for routing within the cabinet. **If conduit is to be shortened, care must be taken to not damage the fiber optic wires contained within the conduit when cutting.** If damaged, remove existing Run and replace with new run 1034 or 1045 delivered with this Upgrade.

**1-8 ROUTE/CONNECT FIBER OPTIC RUN 1083 TO APS BOARD**

1-9 ROUTE/CONNECT FIBER OPTIC RUNS1033, 1034, AND 1045 TO STIF BOARD

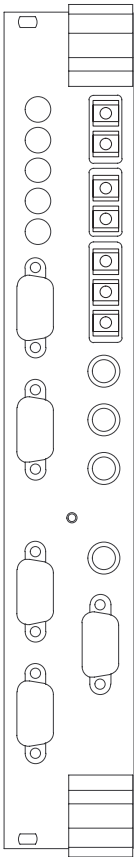
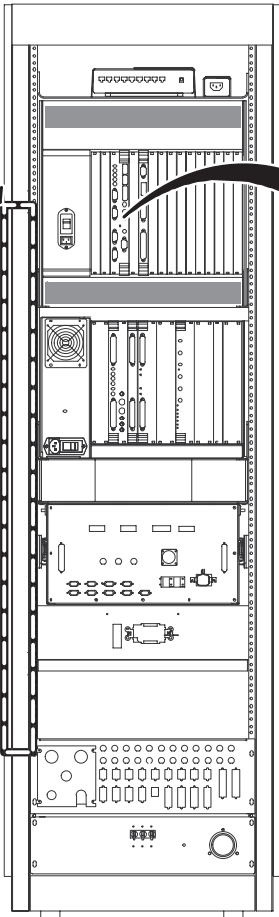


Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Do not coil naked outside of cabinet, or store fiber optic cables in bottom of cabinet.

NOTE:
Refer to table at right for relabel information if existing cables are to be reused.

New Cables		Existing Cables	
RUN #	10.x/11.x	RUN #	8.x/9.x
1033-1	J21	708-1	J19
1034-1	J14	710-1	J12
1034-2	J13	710-2	J13
1034-3	J12	710-3	J15
1034-4	J11	710-4	J17
1034-5	J20	710-5	J18
1045-1	J17	711-2	J23
1045-2	J18,J19	711-3	J20, J21
1045-3	J15,J16	711-1	J25, J27

1 Route fiber optic cables between cabinet frame and side cover to STIF module. Ty-wrap to existing cable bundle routed in this area.



- 2 Relabel (if old cable re-used) and connect Run 1033 fiber optic cable to J21 on the STIF Board.
- Part of Run 1034 below.
- 3 Relabel (if old cable re-used) and connect Run 1045 fiber optic cables to J15, J16, J17, J18 and J19 on the STIF Board.
- 4 Relabel (if old cable re-used) and connect Run 1034 fiber optic cables to J11, J12, J13, J14, and J20 above on the STIF Board.

Note:
Use ty-wraps to secure fiber optic cables to MGD Chassis cables.

1-10 GEMS GOLDSEAL RECYCLING FORM

Attention: GEMS GoldSeal

FDO: _____

Site Name: _____

Removal Date: _____

Field Engineer: _____

SYSTEM Cabinet Serial Number: _____

INSTALLATION STEPS SUMMARY

- ☐ A1 – Preinstallation Planning
- ☐ A2 – Squeeze Bulb Hanger Installation
- ☐ A3 – Squeeze Bulb Installation
- ☐ A4 – Magnet Room Routing of Tubing & Wave Guide Installation

☐ A1 – PREINSTALLATION PLANNING

Since the Squeeze bulb tubing will connect to the front panel of the PAC-II on the new front cover, additional tubing is supplied to permit re-routing of tubing from control box, through Penetration Panel and to rear of PAC-II.

The customer and users should be involved in the following decisions:

Location of Squeeze Bulb Hanger: Procedure A2

Re-location of Control Box for Lightning Upgrade: Procedures B1 thru B5

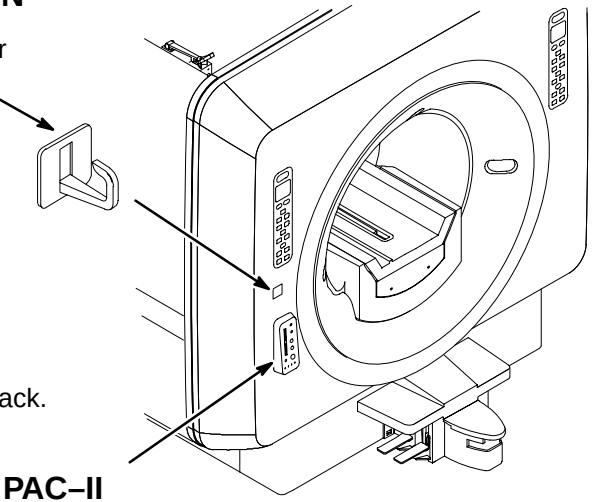
The Control Box may need to be re-located if it was fastened to the old Operator Console or in a place that is now unreachable with the installation of the Operator Workspace. Mounting instructions are included for attaching the Control Box to the new location.

☐ A2 – SQUEEZE BULB HANGER INSTALLATION

- ① Consult with customer and choose mounting location for the Squeeze Bulb Hanger. Verify squeeze bulb hanger will not be in the way during patient transportation, or servicing. Mount high enough above the floor to allow adequate room to hang the tubing and squeeze bulb.

NOTE: A typical mounting location for the squeeze bulb hanger would be on Magnet Enclosure front cover above the PAC-II.

- ② Clean mounting surface.
- ③ Remove protective paper from Squeeze Bulb Hanger back.
- ④ Stick the Squeeze Bulb hanger to mounting surface.

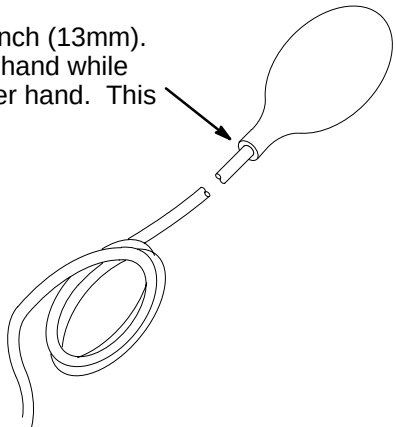


☐ A3 – SQUEEZE BULB INSTALLATION

- ① Insert end of Pneumatic Tubing into Squeeze Bulb approximately 1/2 inch (13mm). For ease of pneumatic tubing insertion, pinch pneumatic tubing in one hand while squeezing squeeze bulb and pushing pneumatic tubing in with the other hand. This will create a positive pressure and expand squeeze bulb opening.
- ② Before cutting pneumatic tubing from roll, verify that when the squeeze bulb is held by a patient being scanned, the tubing length can be routed to the "CALL" connector on the PAC-II front panel without getting in the way of operator or patient during use. .

NOTE: It may be necessary to cut off additional tubing if the squeeze bulb hanger is attached to an Enclosure side cover.

- ③ Connect end of tubing to "CALL" connection on front of PAC-II.



☐ A4 – MAGNET ROOM ROUTING OF TUBING & WAVE GUIDE INSTALLATION

CAUTION

Make sure that pneumatic tubing is not routed where it can be stepped on, pinched, or have an object set on it. The control box alarm will not work if pneumatic tubing is pinched.

NOTE: Also, make sure that the pneumatic tubing can be routed from the connector, on the back of the PAC-II, through the front cover, into the cable take-up, through the Rear Pedestal, and on to the Penetration panel without being pinched along the route by other cables.

- ① Push end of pneumatic tubing onto PAC-II tubing connector.

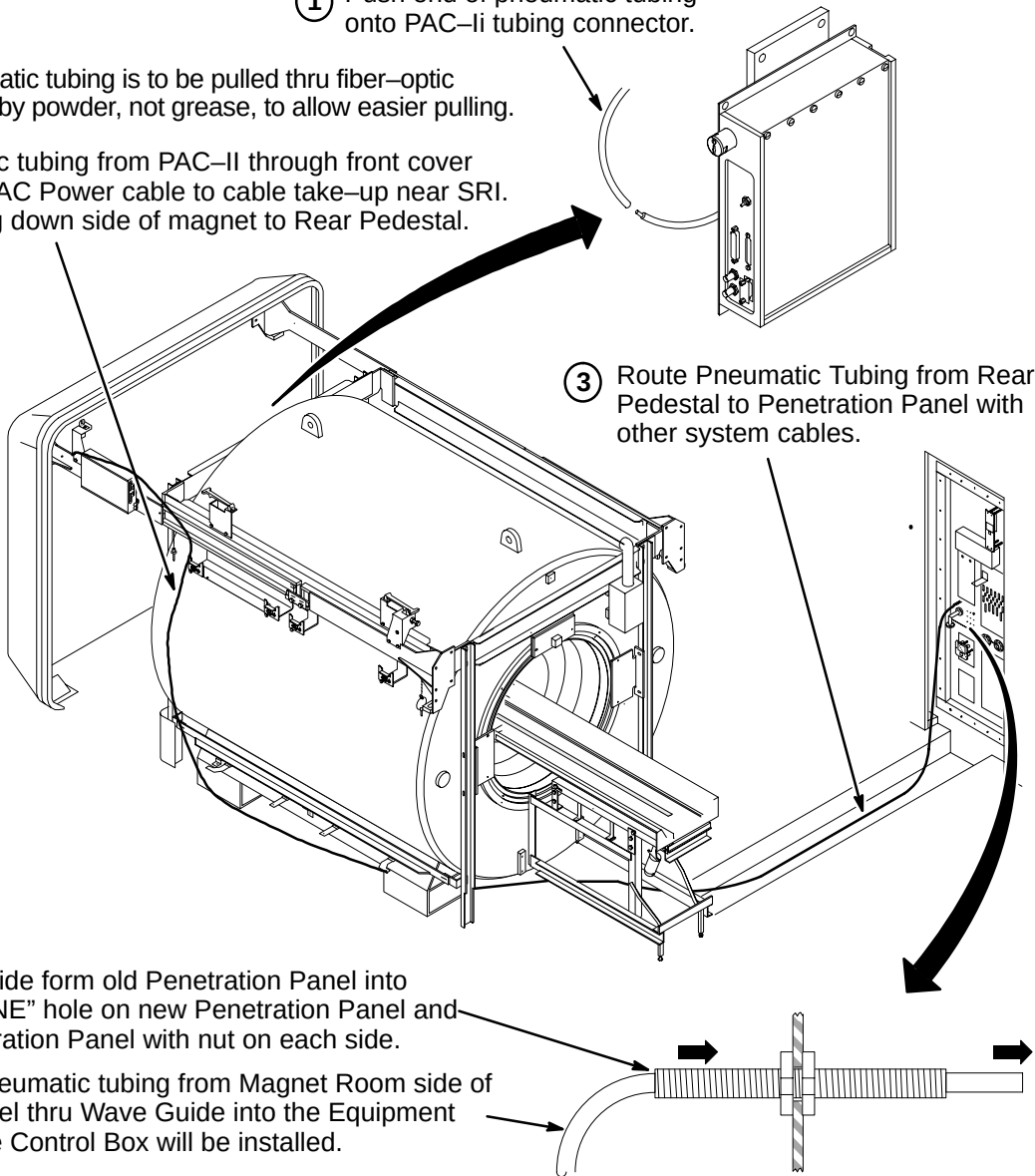
NOTE: If the pneumatic tubing is to be pulled thru fiber-optic conduit, coat with baby powder, not grease, to allow easier pulling.

- ② Route pneumatic tubing from PAC-II through front cover along route of PAC Power cable to cable take-up near SRI. Continue routing down side of magnet to Rear Pedestal.

- ③ Route Pneumatic Tubing from Rear Pedestal to Penetration Panel with other system cables.

- ④ Insert Wave Guide form old Penetration Panel into "ALERT AIR LINE" hole on new Penetration Panel and fasten to Penetration Panel with nut on each side.

- ⑤ Route end of pneumatic tubing from Magnet Room side of Penetration Panel thru Wave Guide into the Equipment Room where the Control Box will be installed.

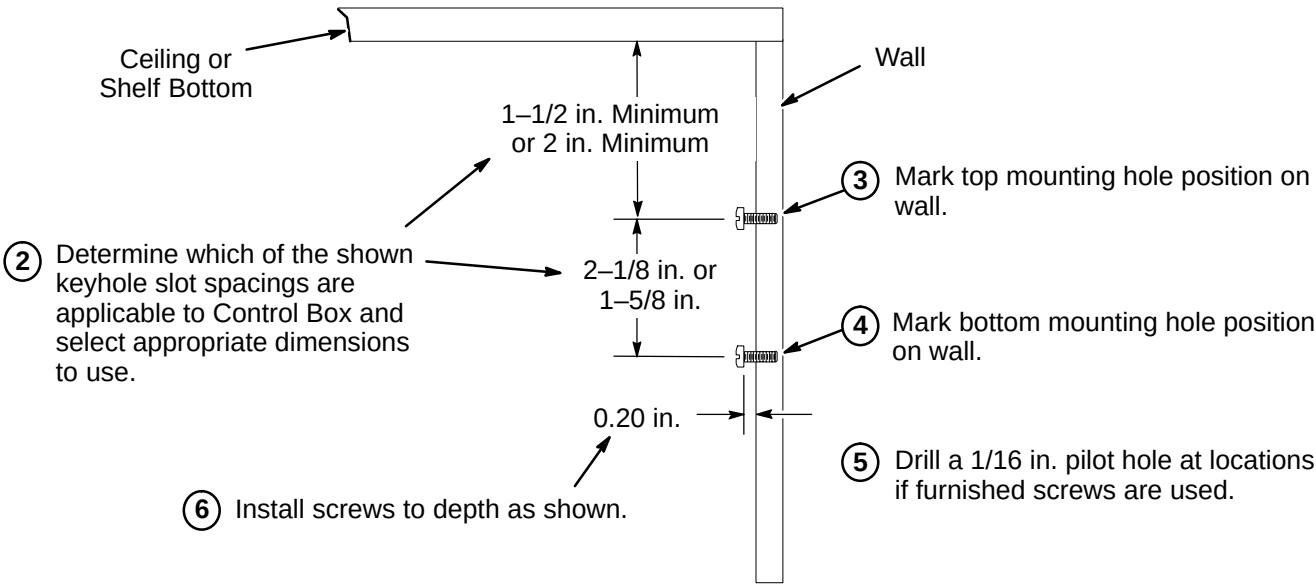


INSTALLATION STEPS SUMMARY

- ☐ B1 – Installing Mounting Screws in Vertical Surface
- ☐ B2 – Installing Control Box on Vertical Surface
- ☐ B3 – Installing Mounting Screws Under Shelf
- ☐ B4 – Installing Control Box Under Shelf
- ☐ B5 – Final Connections to Control Box

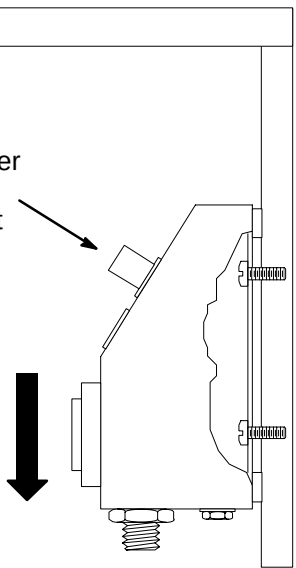
B1 – INSTALLING MOUNTING SCREWS IN VERTICAL SURFACE

- 1 Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.



B2 – INSTALLING CONTROL BOX ON VERTICAL SURFACE

- 1 Center box keyhole slots over screws, push in on box, and then push downward to seat screws within slots.

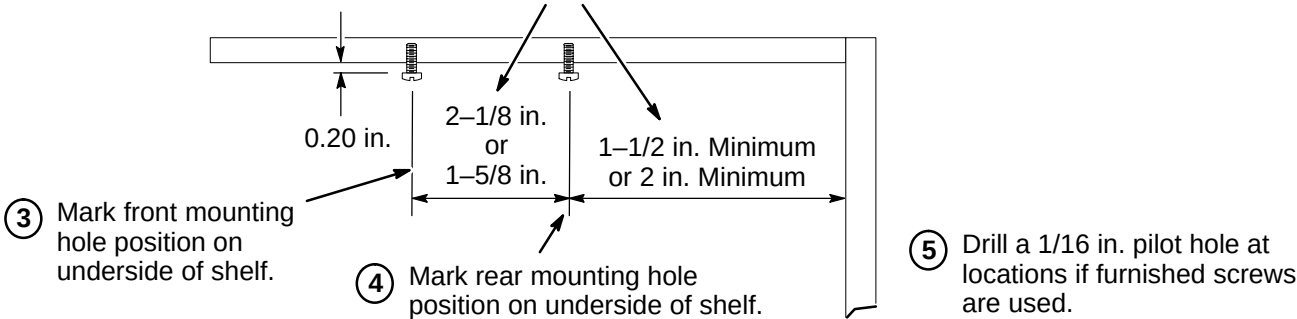


- 2 Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
- 3 Proceed to Step B5, FINAL CONNECTIONS TO CONTROL BOX.

B3 – INSTALLING MOUNTING SCREWS UNDER SHELF

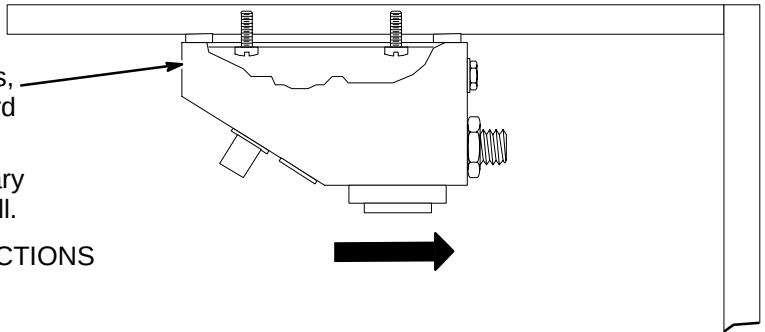
- 1 Choose the mounting area near the Operator Console and install mounting screws (item 9). Control box should be mounted within sight of operator.

- 2 Determine which of the shown keyhole slot spacings are applicable to Control Box and select appropriate dimensions to use.



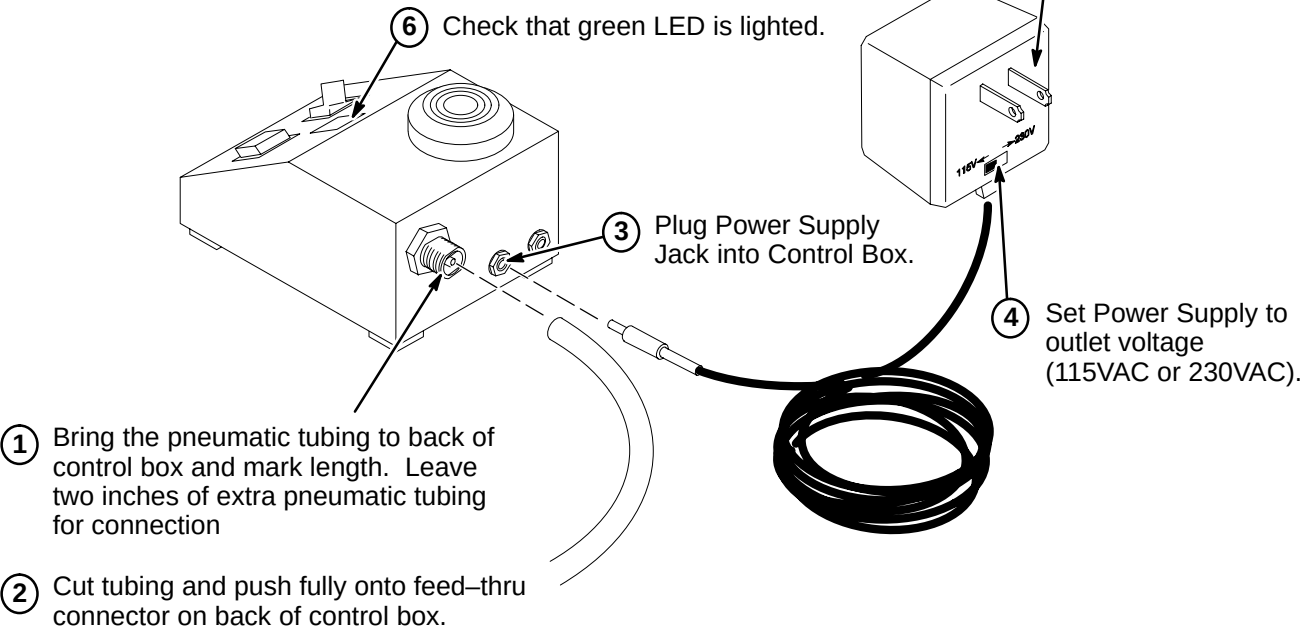
B4 – INSTALLING CONTROL BOX UNDER SHELF

- 1 Center box keyhole slots over screws, push up on box, and then push inward to seat screws within slots.
- 2 Tighten or loosen screws as necessary to obtain a snug fit of box against wall.
- 3 Proceed to Step B5, FINAL CONNECTIONS TO CONTROL BOX.



B5 – FINAL CONNECTIONS TO CONTROL BOX

Note: Steps 1 and 2 are not needed if Extender Box (46-317758P2) has been installed.



OPERATOR WORKSPACE UPGRADE INSTRUCTIONS

Tab 13, Operator Workspace, contains installation instructions for the following:

- *DIRECTION 2210982* – Signa 8.x Octane Operator Workspace Installation Instructions
- *DIRECTION 2386129* – Signa Linux Operator Workspace Installation Instructions
- *DIRECTION 2403543* – Operator Workspace Upgrade to EXCITE II Installation Instructions

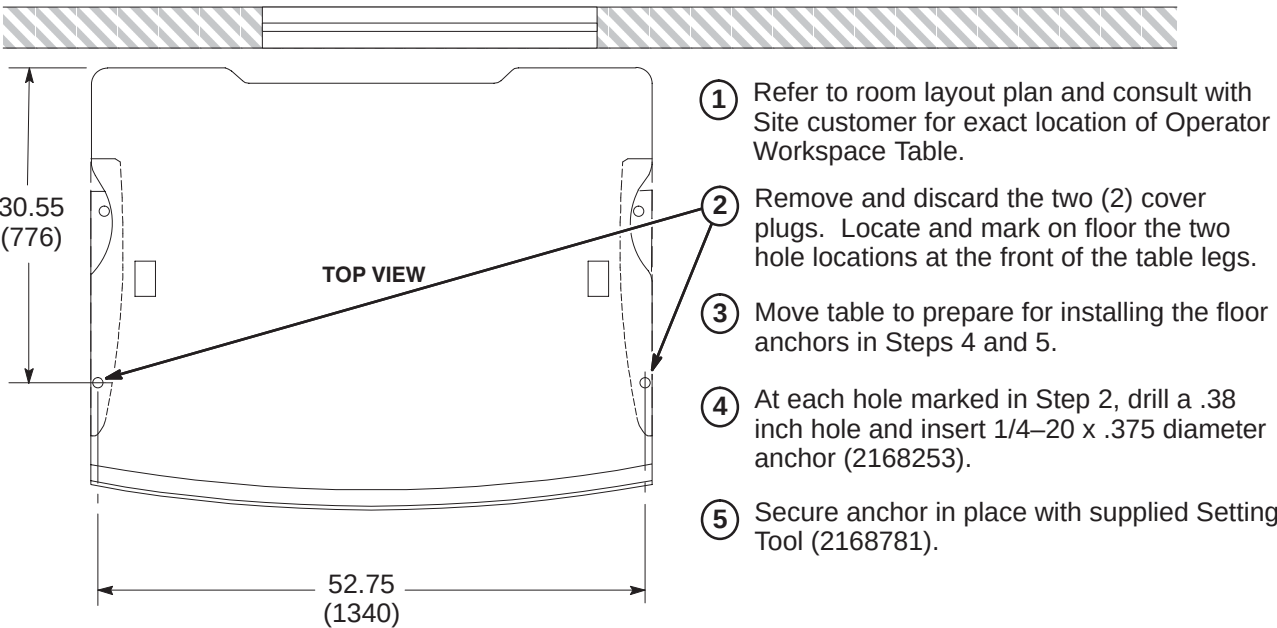
Use only the applicable System Upgrade Instruction Directions for the site being upgraded

INSTALLATION STEPS SUMMARY

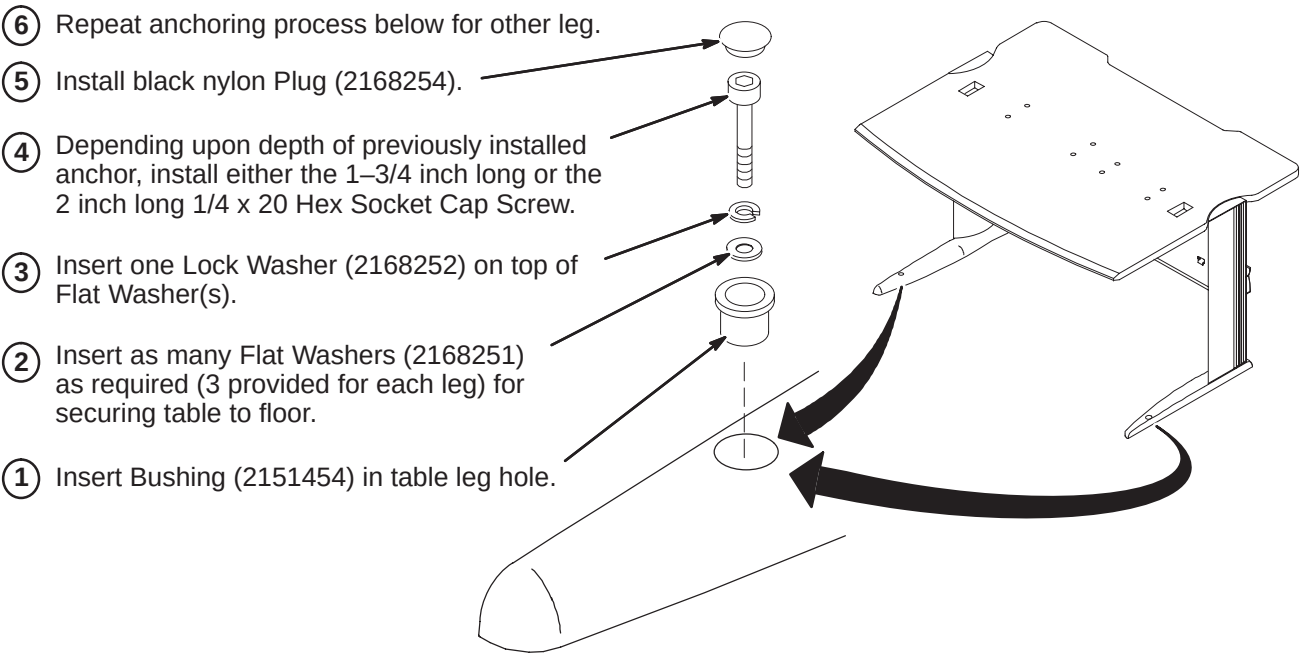
- A1 – Operator Workspace Table Anchor Installation
- A2 – Securing Operator Workspace Table to Floor
- A3 – Attach Workspace Interface Module (2141978 series) to Table
- A4 – Attach Power Distribution Box (46–307830G1) to Table

Note: The CERD Spike Noise Test Cable (2168505) is part of the 8.x Octane Operator Workspace Cable collector (2154392–2). Remove this cable from the collector and place it in the TPS RF Service Interface Kit (46–301927G1).

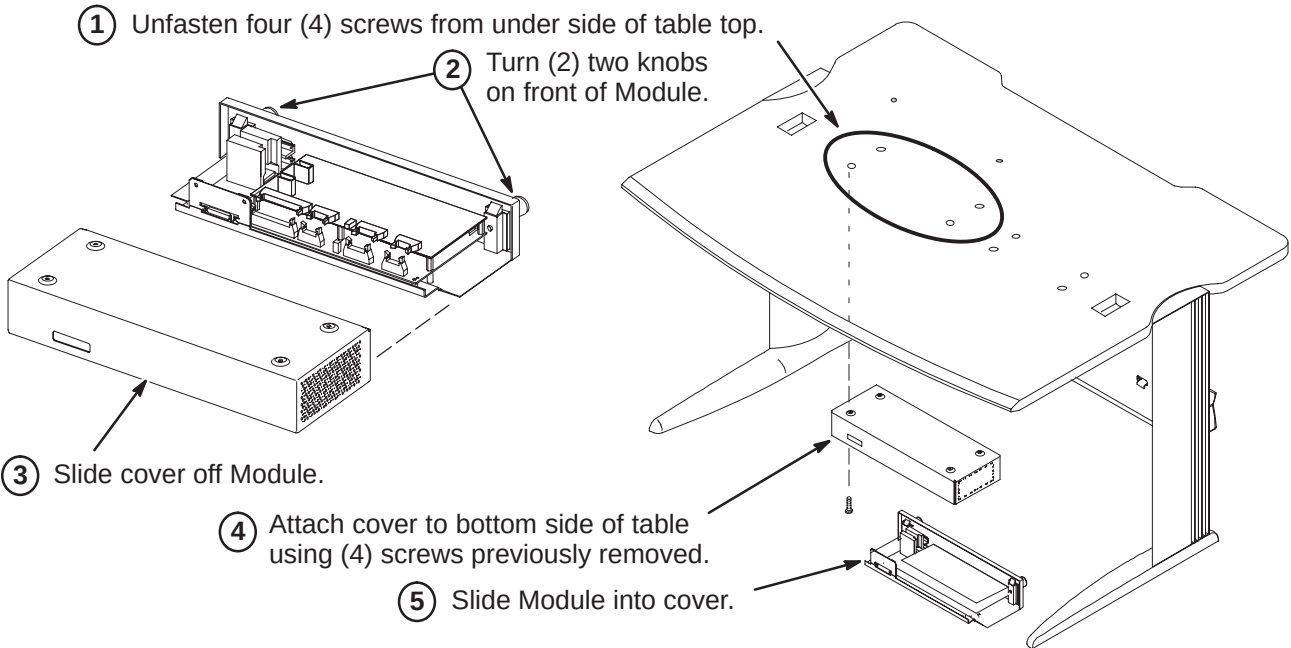
A1 – OPERATOR WORKSPACE TABLE ANCHOR INSTALLATION



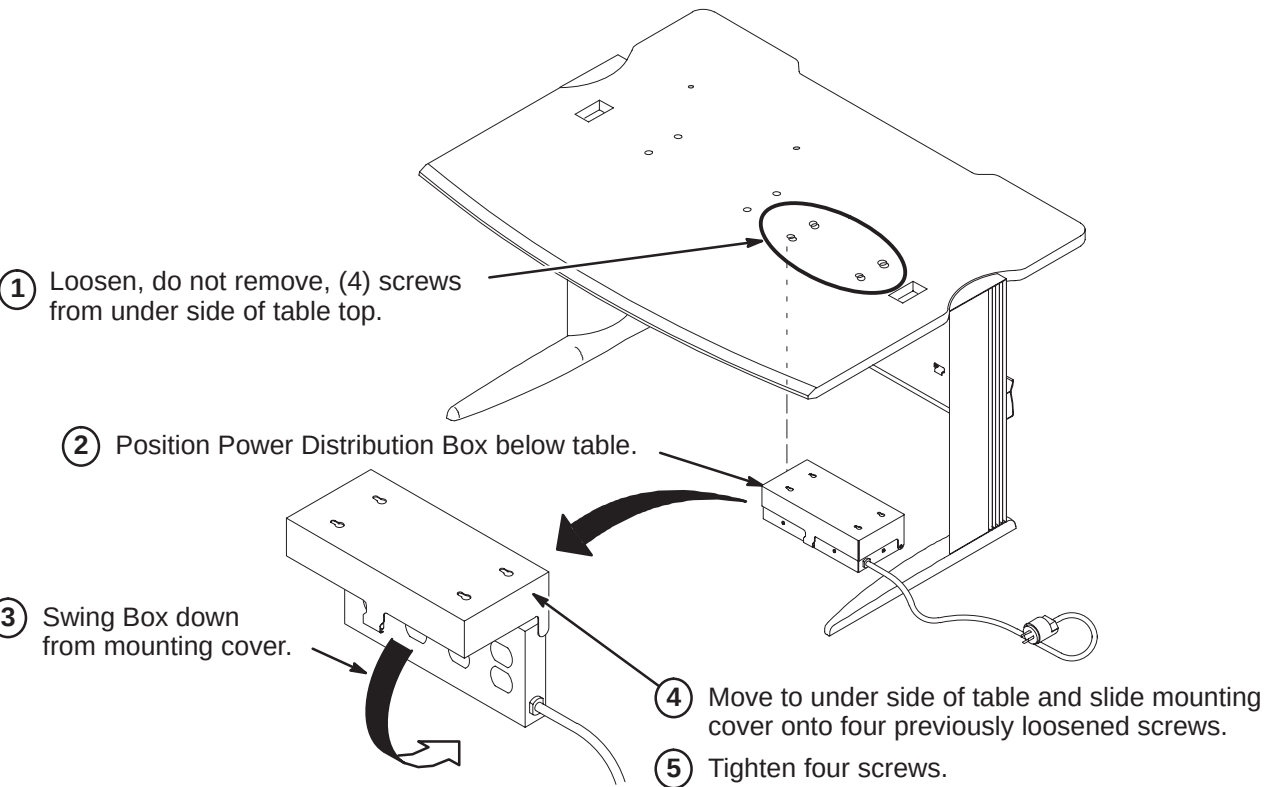
A2 – SECURING OPERATOR WORKSPACE TABLE TO FLOOR



A3 – ATTACH WORKSPACE INTERFACE MODULE (2141978 series) TO TABLE



A4 – ATTACH POWER DISTRIBUTION BOX (46–307830G1) TO TABLE

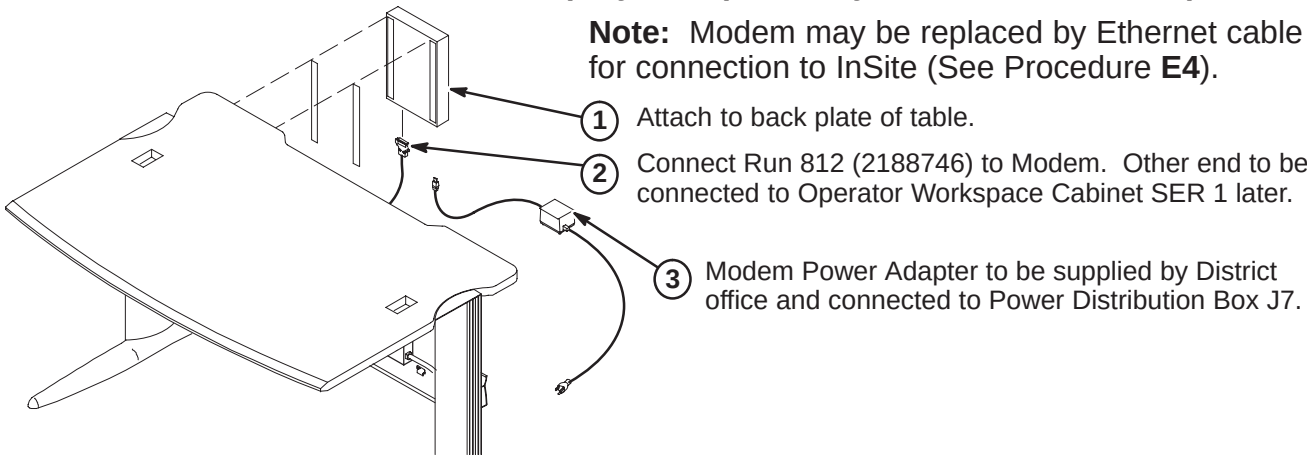


INSTALLATION STEPS SUMMARY

- ☐ B1 – Attach Modem to Table (May be replaced by ethernet connection)
- ☐ B2 – Attach DASM to Table
- ☐ B3 – Attach Serial Expansion Box Asm to Underside of Table Top
- ☐ B4 – Attach Serial Coverter to Underside of Table Top (MGD & OpenSpeed only)

☐ B1 – ATTACH MODEM TO TABLE (May be replaced by ethernet connection)

Note: Modem may be replaced by Ethernet cable for connection to InSite (See Procedure E4).



☐ B2 – ATTACH DASM TO TABLE

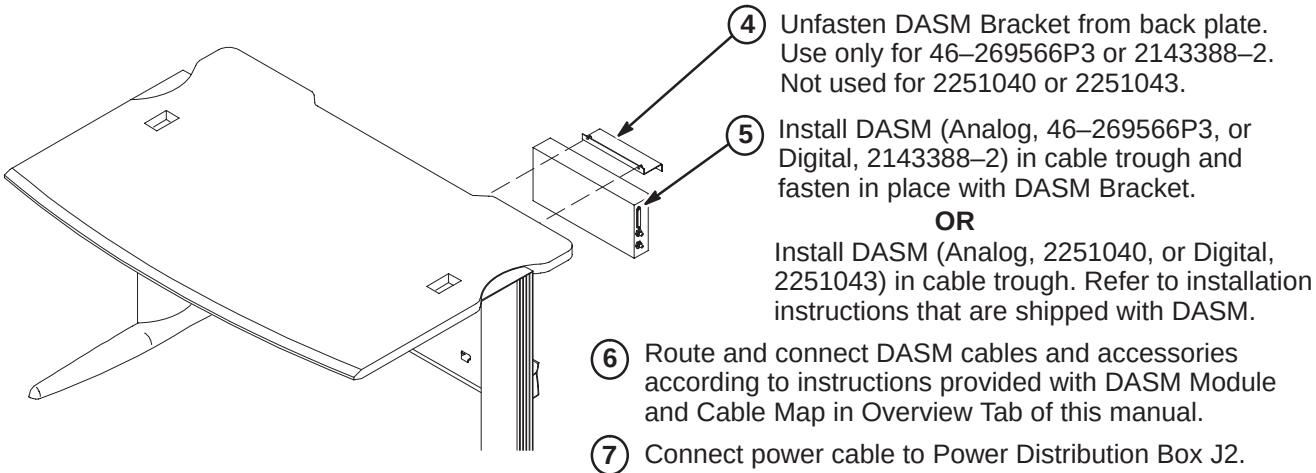
Note: The Analog and Digital DASMs are shipped with the jumper set to SCSI ID=0. This is the correct ID for an AW, but incorrect for Signa Horizon LX (SCSI ID=3).

- 1 Unfasten DASM cover screw and remove cover.
- 2 For SCSI ID = 3, set Configuration Jumper J2 as follows:
5=OFF, 6=OFF, 7=ON
Note: A jumper in the OFF position enables the SCSI ID bit.
- 3 Re-attach DASM cover.

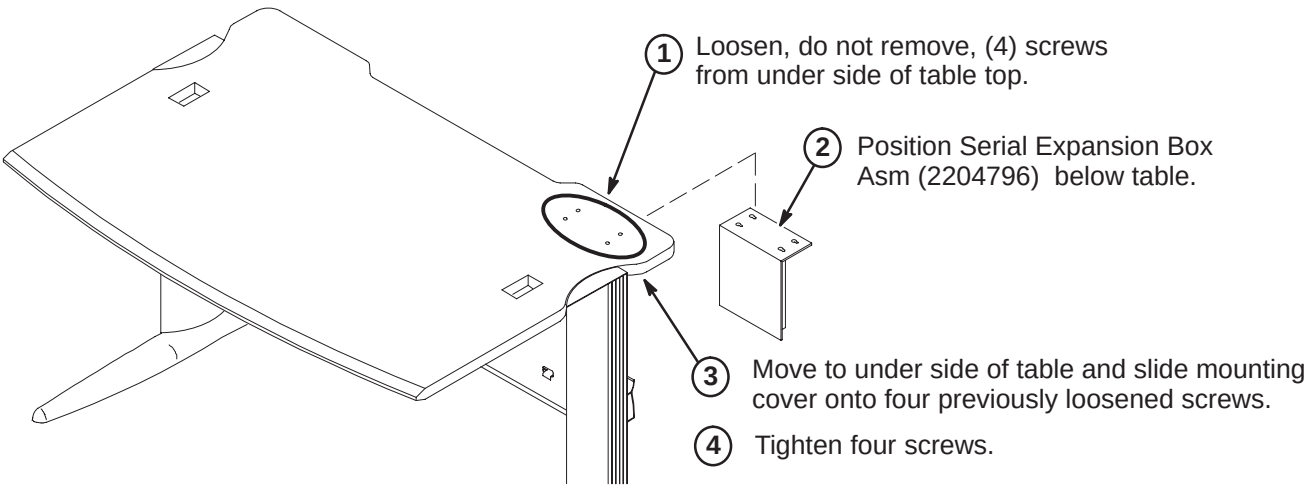


Note: Consult with Site customer for preference of DASM location. This procedure installs DASM in the cable trough. Customer may prefer DASM on top of table.

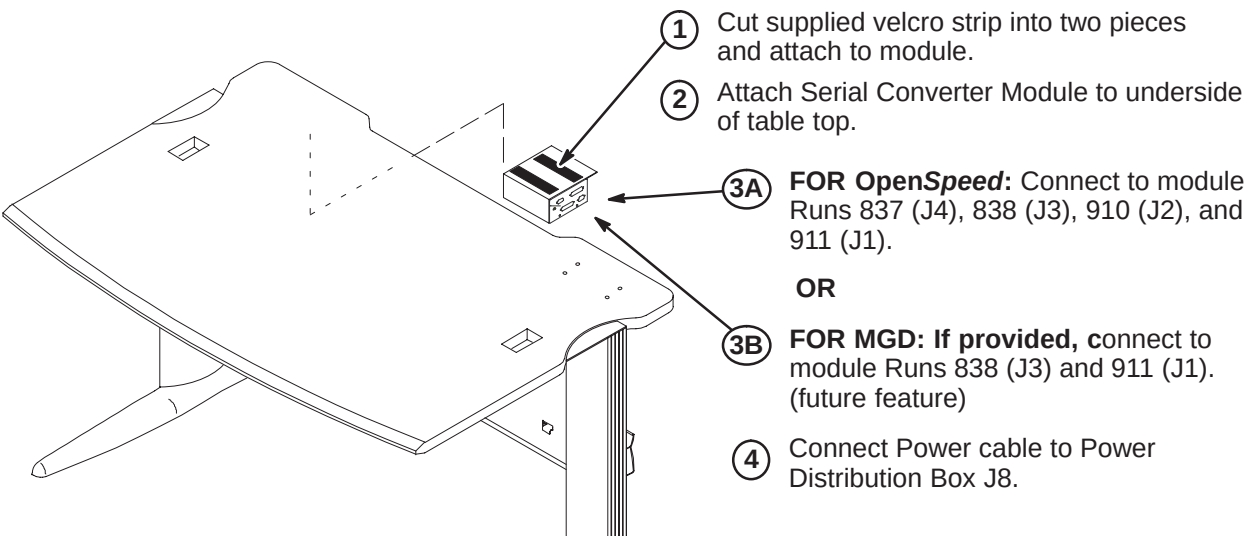
CAUTION: Longer SCSI cables cannot be substituted.



☐ B3 – ATTACH SERIAL EXPANSION BOX ASM TO UNDERSIDE OF TABLE TOP



☐ B4 – ATTACH SERIAL CONVERTER TO BOTTOM OF TABLE (MGD & OpenSpeed)



INSTALLATION STEPS SUMMARY

- ☐ C1 – Install Keyboard on Table
- OR
- ☐ C2 – Install SCIM/Keyboard Data Cable on Table
- ☐ C3 – Install SCIM/Keyboard on Table
- ☐ C4 – Install Flat Screen Color Monitor on Table

NOTE:
Consult with Site customer for exact location of Color Monitor on Operator Workspace Table.

☐ C1 – INSTALL KEYBOARD ON TABLE

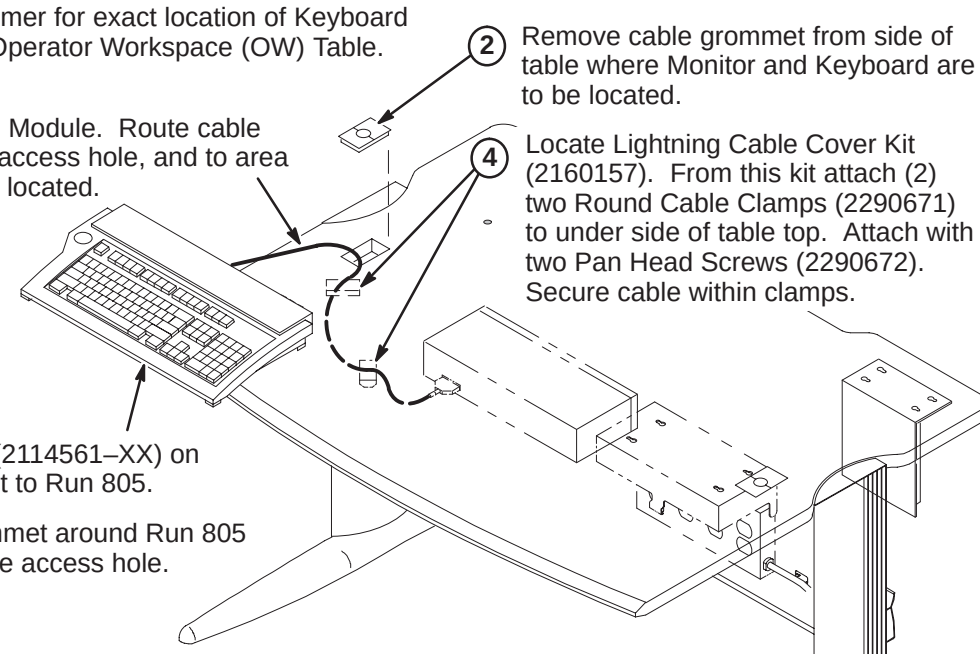
- 1 Consult with Site customer for exact location of Keyboard and Color Monitor on Operator Workspace (OW) Table.

3 Connect Run 805 to I/F Module. Route cable under table, thru cable access hole, and to area where Keyboard will be located.

5 Place Keyboard (2114561-XX) on table and connect to Run 805.

6 Place cable grommet around Run 805 and insert in cable access hole.

2 Remove cable grommet from side of table where Monitor and Keyboard are to be located.

4 Locate Lightning Cable Cover Kit (2160157). From this kit attach (2) two Round Cable Clamps (2290671) to under side of table top. Attach with two Pan Head Screws (2290672). Secure cable within clamps.
- 
- The diagram shows a side view of the operator workspace table. A keyboard is being placed on the table surface. A cable (Run 805) is shown being routed under the table through a cable access hole. The cable is then secured under the table top using two round cable clamps attached with pan head screws. A cable grommet is shown being placed around the cable where it enters the table.

OR

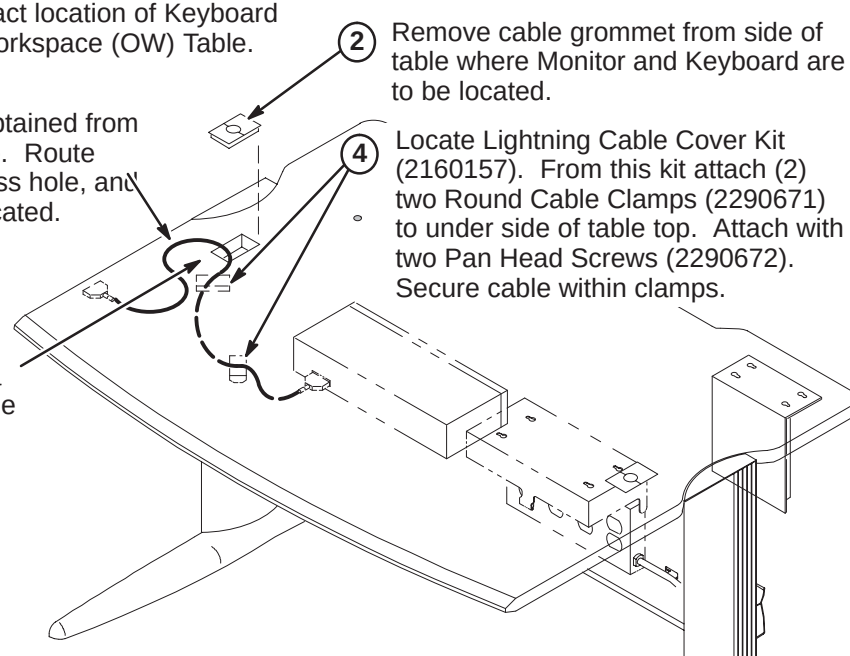
☐ C2 – INSTALL SCIM/KEYBOARD DATA CABLE ON TABLE

- 1 Consult with Site customer for exact location of Keyboard and Color Monitor on Operator Workspace (OW) Table.

3 Connect Data Cable (Run 805), obtained from SCIM Kit (2307175), to I/F Module. Route cable under table, thru cable access hole, and to area where Keyboard will be located.

5 Place cable grommet around Data Cable (Run 805) and insert in cable access hole.

2 Remove cable grommet from side of table where Monitor and Keyboard are to be located.

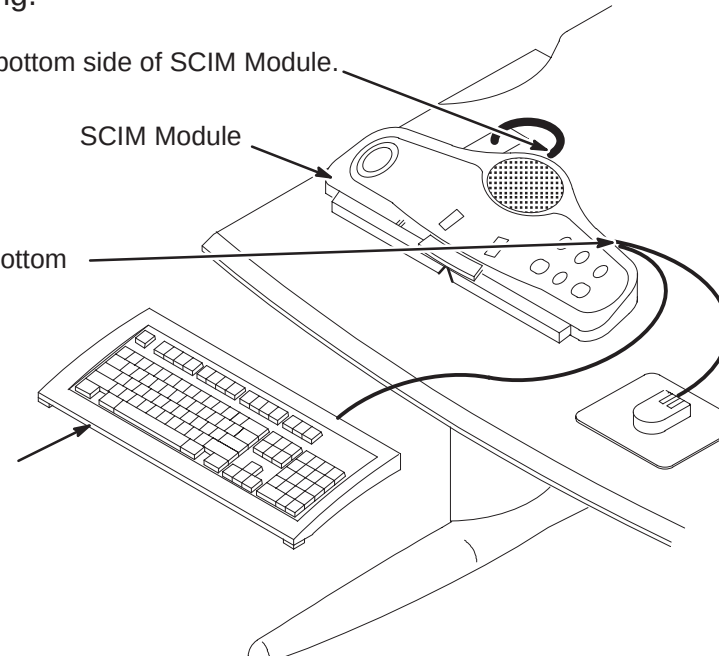
4 Locate Lightning Cable Cover Kit (2160157). From this kit attach (2) two Round Cable Clamps (2290671) to under side of table top. Attach with two Pan Head Screws (2290672). Secure cable within clamps.
- 
- The diagram shows a side view of the operator workspace table. A data cable (Run 805) is shown being routed under the table through a cable access hole. The cable is then secured under the table top using two round cable clamps attached with pan head screws. A cable grommet is shown being placed around the cable where it enters the table.

☐ C3 – INSTALL SCIM/KEYBOARD ON TABLE

NOTE: Refer to Installation document OW1INA3.DOC on Service Methods CD-ROM or MR Service Engineering web site for installation information on language labels and keyboard mounting.

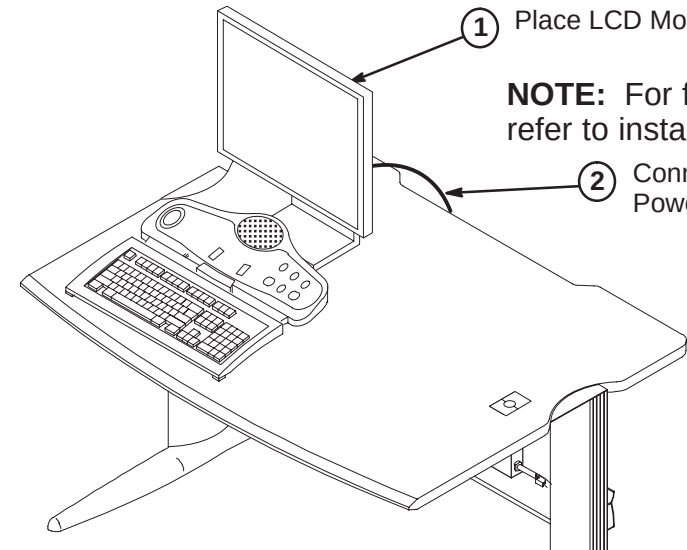
- 1 Connect Data Cable to bottom side of SCIM Module.

2 Connect Keyboard and Mouse cables to bottom side of SCIM Module as marked.

3 Place Keyboard on table in front of SCIM Module.
- 
- The diagram shows a top-down view of the SCIM module on the table. A data cable is connected to the bottom of the module. Keyboard and mouse cables are also connected to the bottom of the module. The keyboard is placed on the table in front of the module.

☐ C4 – INSTALL FLAT SCREEN COLOR MONITOR ON TABLE

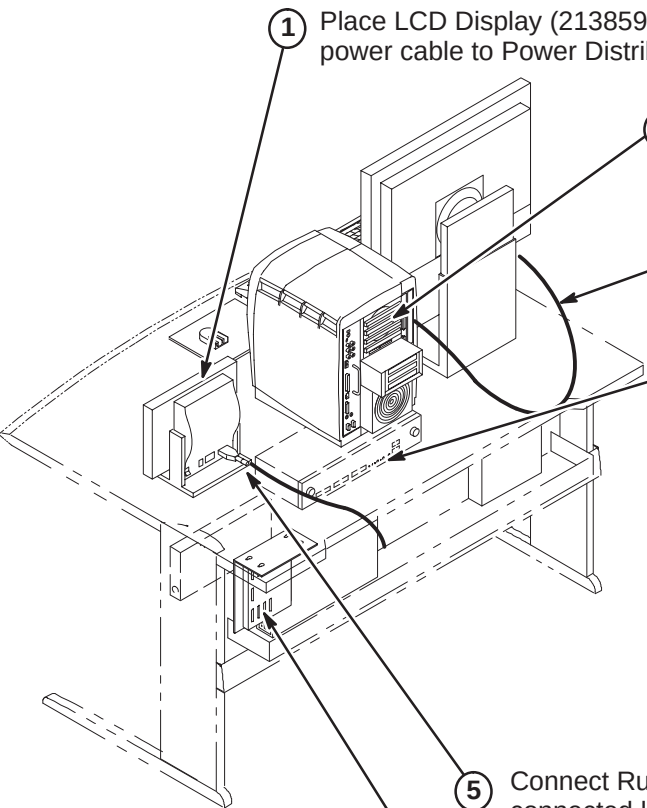
- 1 Place LCD Monitor on table in back of SCIM.

2 Connect power cable to Power Distribution Box J1.
- NOTE:** For fastening monitor to surface, refer to installation drawing 2204493DDW.
- 
- The diagram shows a top-down view of the operator workspace table. A flat screen color monitor is placed on the table in front of the SCIM module. A power cable is shown being connected to the Power Distribution Box J1.

INSTALLATION STEPS SUMMARY

- D1 – Install LCD Display on Table and Position Octane Computer
- D2 – Connect Table Component Cables
- D3 – Serial Expansion Box Set Up
- D4 – SGI Octane Computer Cables Connections

D2 – CONNECT TABLE COMPONENT CABLES



1 Place LCD Display (2138599) on table. Connect power cable to Power Distribution Box J5.

2 Place Octane Computer on table.
NOTE: Refer to Procedure D4 for Run connections to SGI Octane Computer.

3 Connect Run 798/821 (2150533–2) to R IN, B IN, G IN on Monitor and route/connect to MONITOR on the Octane Computer module.

4 Connect the following cables from the OW I/F Module (OW A1 A4) for later connection to Operator Workspace Cabinet:

RUN #	CONNECTION	PART #
805	J6	2114561–38
810	J8	2141980–2
814	J9	2196990
817	J17	46–328000G958

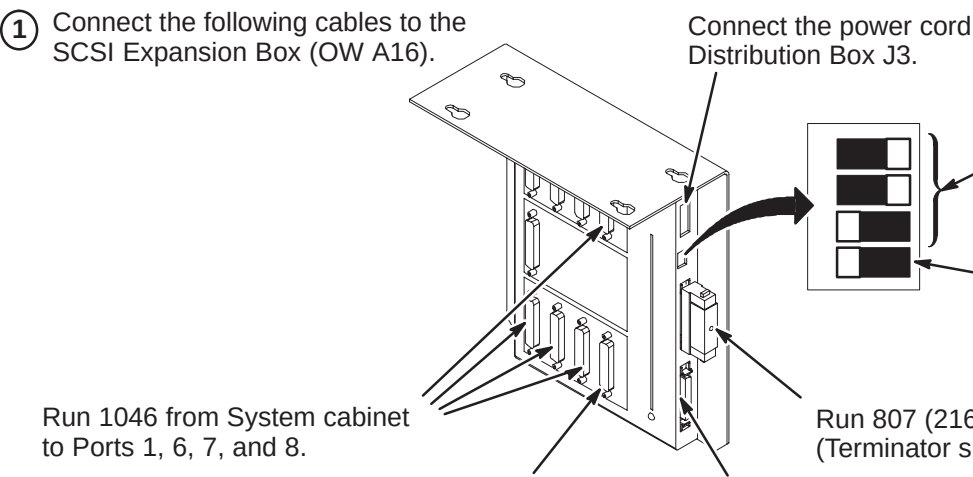
Connect the following cables to OW I/F Module. These cables will be routed to the System Cabinet and Penetration Panel:

RUN #	CONNECTION	PART #
789	J18	46–317068G938
788	J7	46–328000G937

5 Connect Run 797 (2138599–2) to LCD Display. Other end is connected later to PC in the Operator Workspace Cabinet.

NOTE: Refer to Procedure D3 for Run connections and DIP switch settings.

D3 – SERIAL EXPANSION BOX SET UP FOR MGD (10.x)



1 Connect the following cables to the SCSI Expansion Box (OW A16).

Connect the power cord to Power Distribution Box J3.

2 Set SCSI ID switches on side of module as shown. (Addr = 4)

Termination Enabled
NOTE: If DASM not installed, move switch setting to disabled

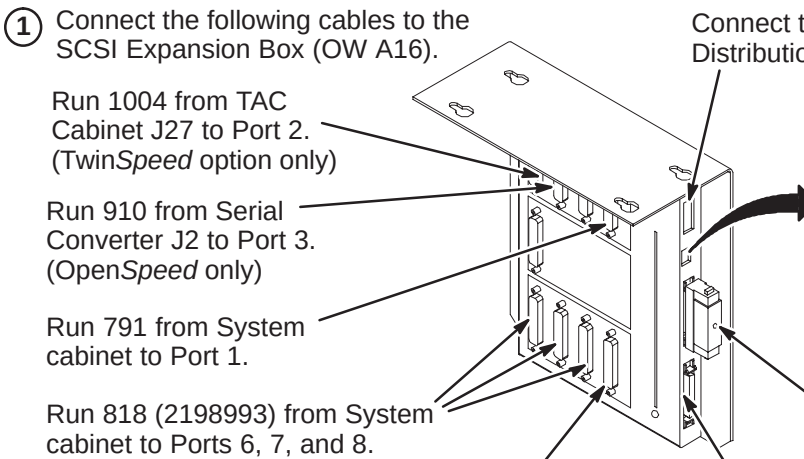
Run 1046 from System cabinet to Ports 1, 6, 7, and 8.

Run 807 (2163886) from DASM. (Terminator shown)

Run 911 from Serial Converter J1 to Port 5. (Future)

Run 815 (2198983) from Octane computer

D3 – SERIAL EXPANSION BOX SET UP (9.x)



1 Connect the following cables to the SCSI Expansion Box (OW A16).

Run 1004 from TAC Cabinet J27 to Port 2. (TwinSpeed option only)

Run 910 from Serial Converter J2 to Port 3. (OpenSpeed only)

Run 791 from System cabinet to Port 1.

Run 818 (2198993) from System cabinet to Ports 6, 7, and 8.

Run 911 from Serial Converter J1 to Port 5. (OpenSpeed only)

Connect the power cord to Power Distribution Box J3.

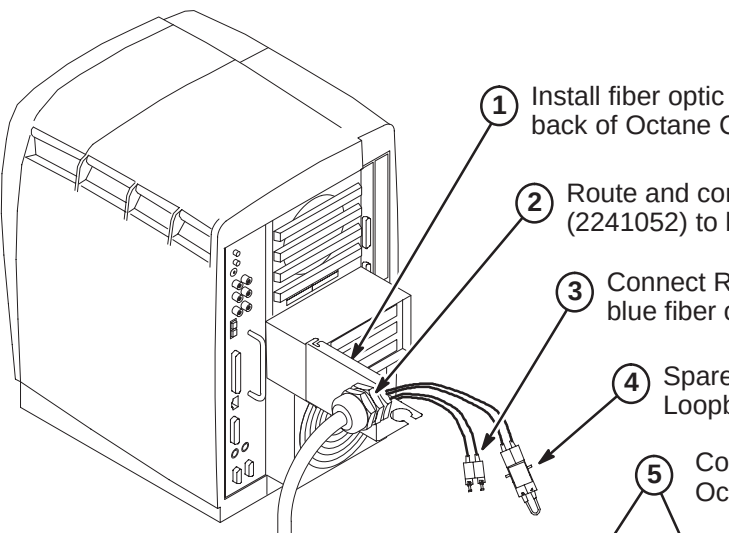
2 Set SCSI ID switches on side of module as shown. (Addr = 4)

Termination Enabled
NOTE: If DASM not installed, move switch setting to disabled

Run 807 (2163886) from DASM. (Terminator shown)

Run 815 (2198983) from Octane computer

D4 – SGI OCTANE COMPUTER CABLES CONNECTIONS



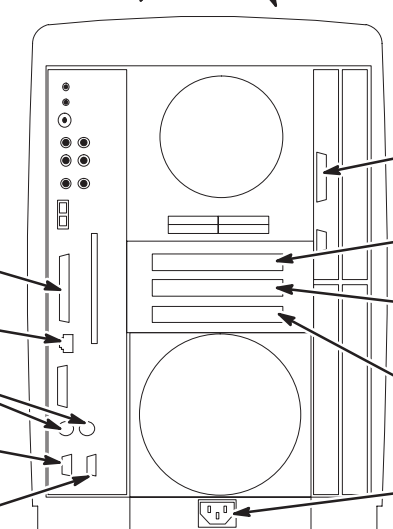
1 Install fiber optic Strain Relief Bracket (2245994) to back of Octane Computer using existing screws.

2 Route and connect Bit3 Fiber Optic cable Run 836 (2241052) to bracket.

3 Connect Run 836 (or Run 1044, MGD) orange and blue fiber optic cables as indicated in Step 5 below.

4 Spare Fiber Optic cable and Loopback Assembly (2241052–6).

5 Connect the following cables to the SGI Octane Computer (OW A15) as marked.



Run 798 or 821 (2150533–2) OR Run 799 or 820 (2150533)

Run 1031 (MGD)

Run 836 (2241052) OR Run 1044 (MGD)

Run 815 (2198983)

Connect power cable to Power distribution Box J4.

Run 819

Run 811 (2153396–3)

Run 810 (2141980–2)

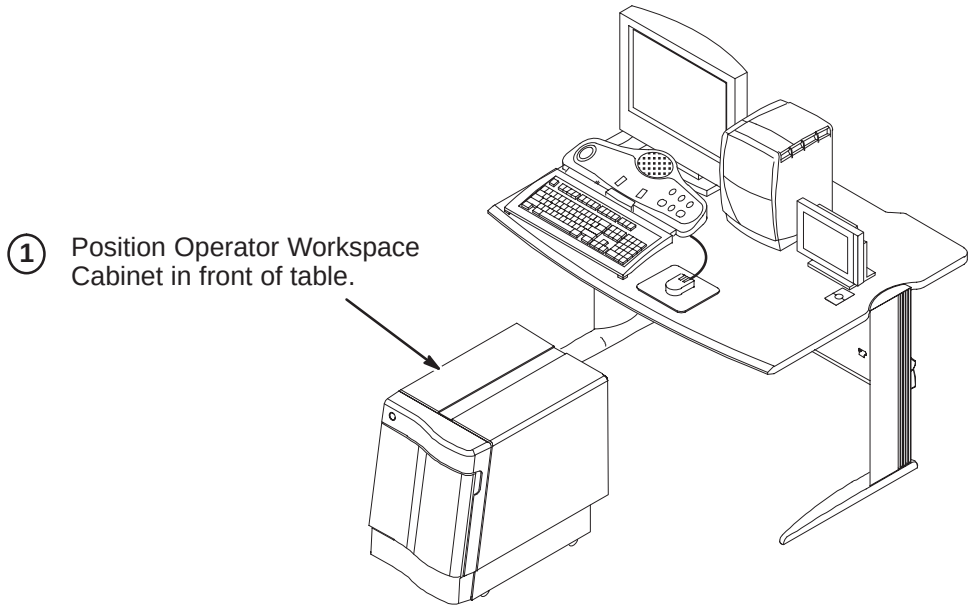
Run 813 (2188747)

Run 812 (2188746)

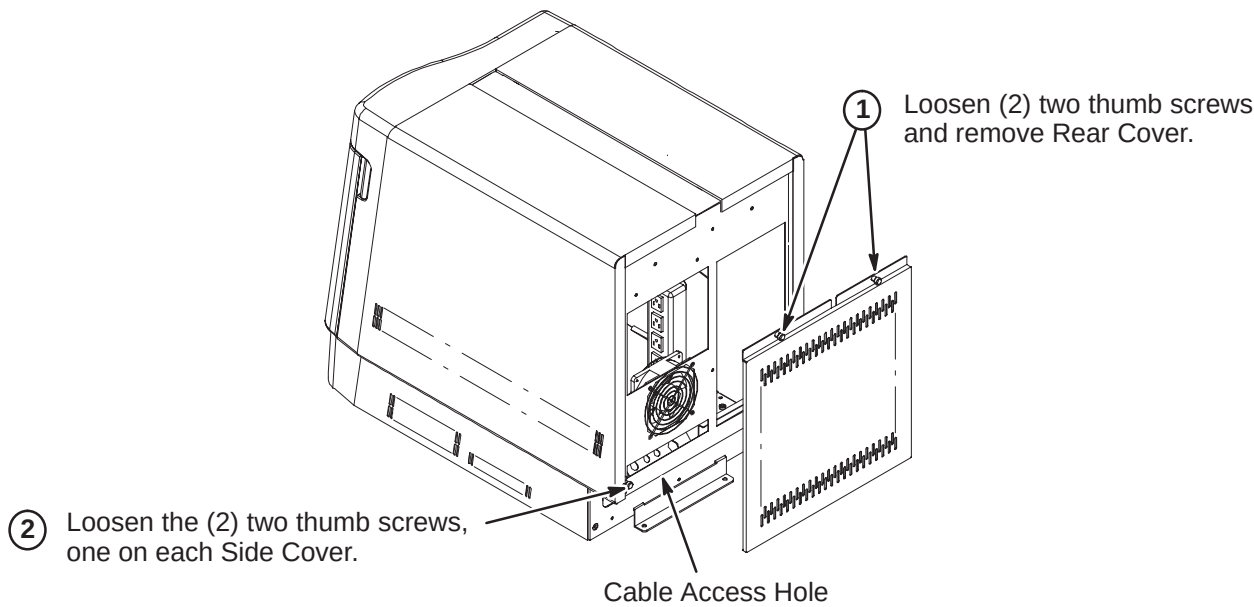
INSTALLATION STEPS SUMMARY

- E1 – Locate Operator Workspace Cabinet in Front of Table
- E2 – Remove Operator Workspace Cabinet Rear Cover
- E3 – Remove Operator Workspace Cabinet Front And Side Covers
- E4 – Connect Ethernet Cables
- E5 – Route/Connect Runs 792, 797, 813, and 814 to PC Module

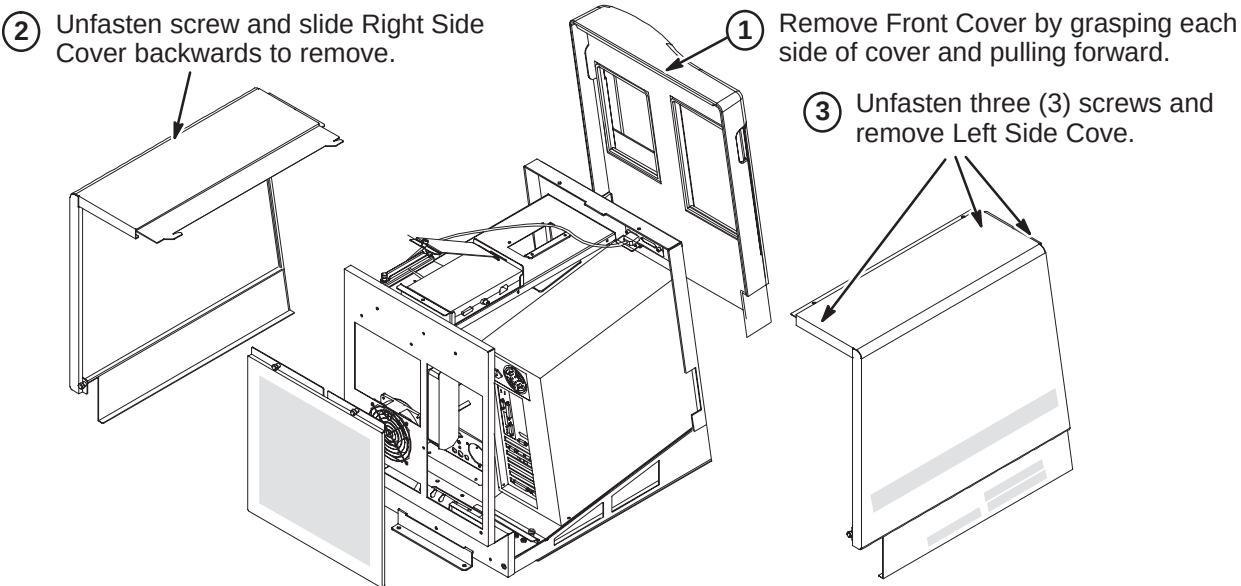
E1 – LOCATE OPERATOR WORKSPACE CABINET IN FRONT OF TABLE



E2 – REMOVE OPERATOR WORKSPACE CABINET REAR COVER

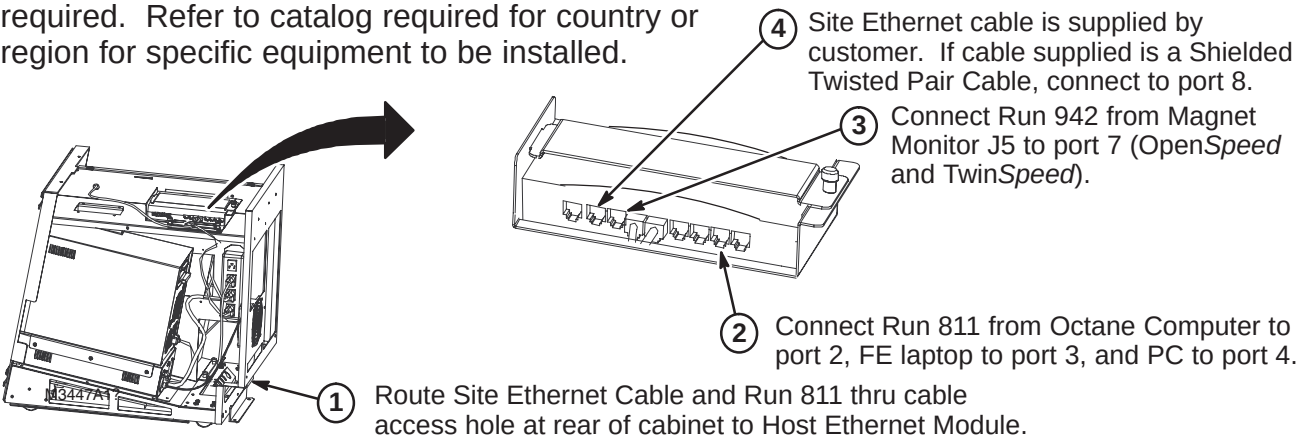


E3 – REMOVE OPERATOR WORKSPACE CABINET FRONT AND SIDE COVERS



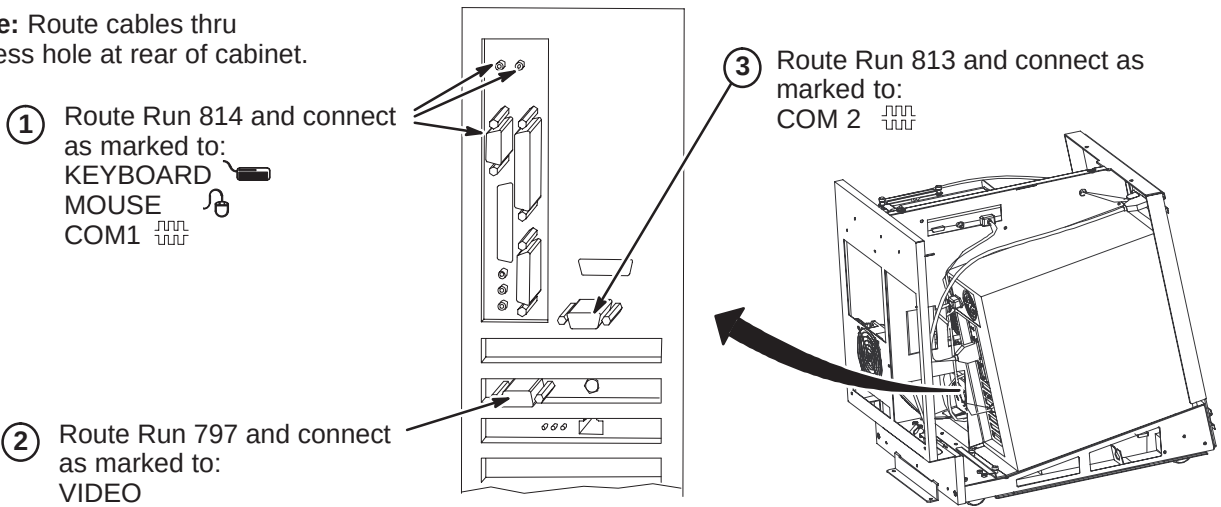
E4 – CONNECT ETHERNET CABLES

Note: Site Ethernet cable for InSite connection is supplied by customer. For Europe, connection to ISDN router is required. Refer to catalog required for country or region for specific equipment to be installed.



E5 – ROUTE/CONNECT RUNS 792, 797, 813, AND 814 TO PC MODULE

Note: Route cables thru access hole at rear of cabinet.



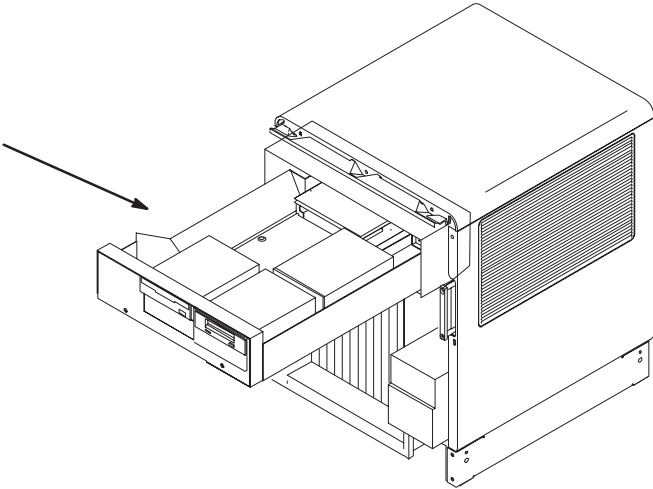
PROCEDURES F1 THRU F5 ARE USED ONLY WHEN UPGRADING FROM 5.x PLATFORMS, **DO NOT PERFORM PROCEDURES F1 THRU F5 FOR NEW OCTANE INSTALLATIONS**

INSTALLATION STEPS SUMMARY

- ☐ F1 – Removal of DAT and MOD Drives from Genesis Computer
- ☐ F2 – Remove SCSI Expansion Box
- ☐ F3 – DAT Drive Settings
- ☐ F4 – MOD Drive Settings
- ☐ F5 – Install DAT and MOD Drives (Legacy Media) in SCSI Expansion Box

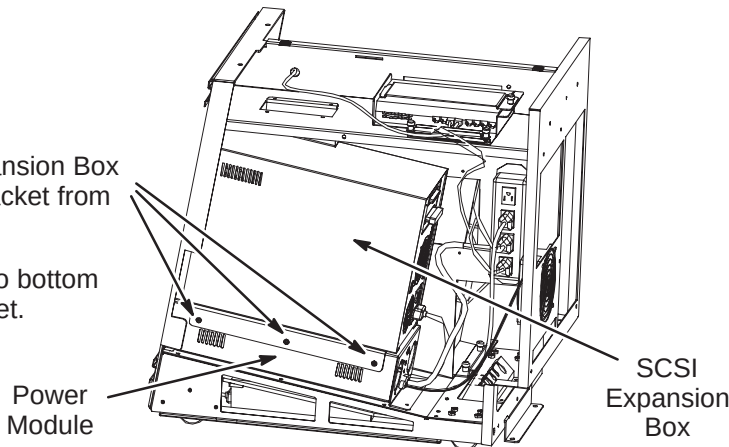
☐ F1 – REMOVAL OF DAT AND MOD DRIVES FROM GENESIS COMPUTER

UPGRADE FROM 5.x OPTIONS:
For transferring Legacy Media DAT and MOD Drives from Genesis Computer – refer to Tab 13, Directions 2174955 and 2174956, steps **A1** thru **A4** only for removal of existing drives.



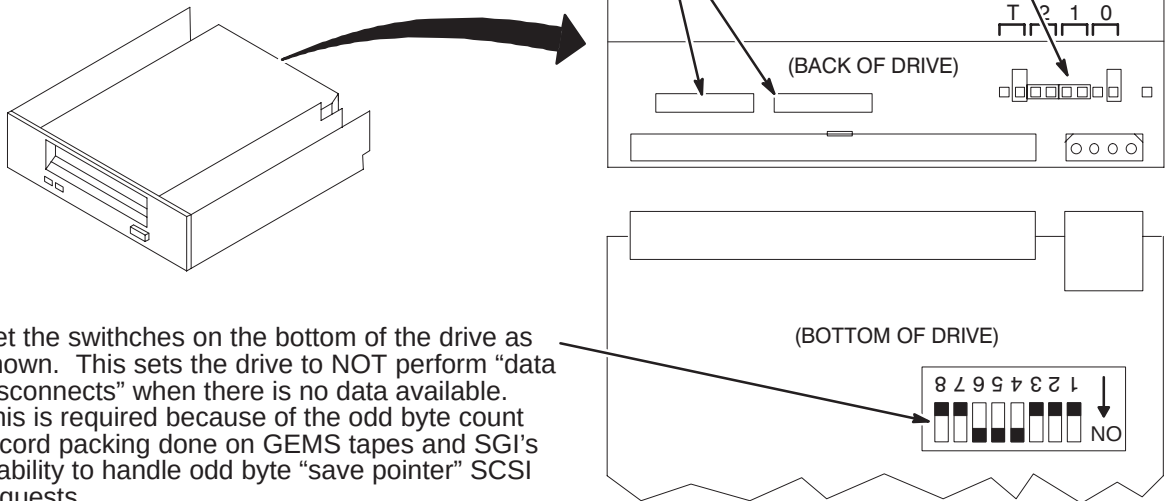
☐ F2 – REMOVE SCSI EXPANSION BOX

- ① Unfasten three (3) screws from SCSI Expansion Box mounting bracket and remove Box and bracket from power module.
- ② Unfasten six (6) screws that hold bracket to bottom of SCSI Expansion Box and remove bracket.
- ③ Remove SCSI Expansion Box cover.



☐ F3 – DAT DRIVE SETTINGS

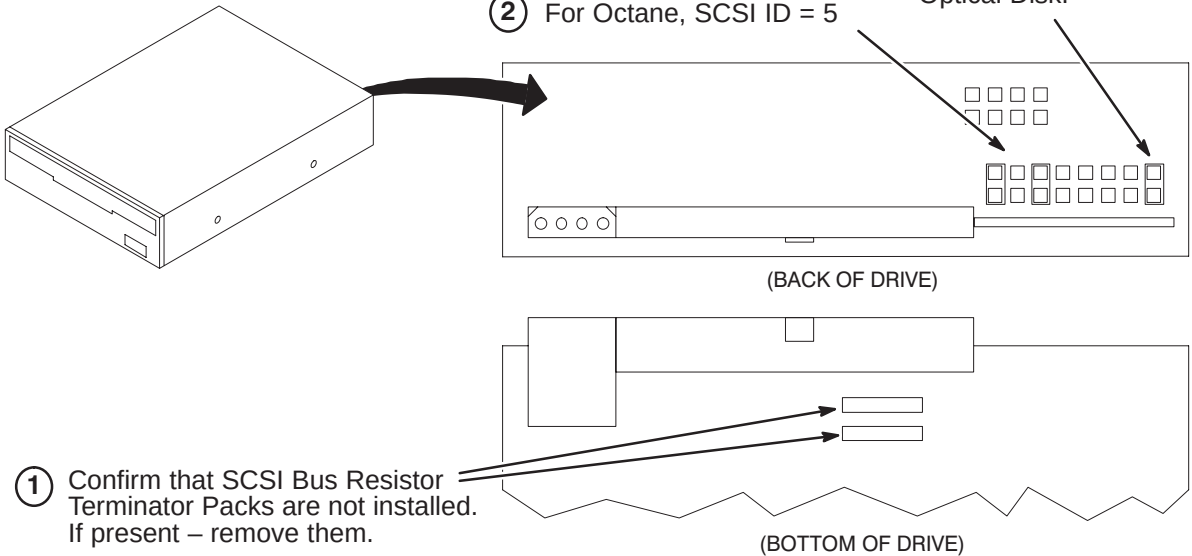
- ① Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.
- ② For Octane, set SCSI ID = 6



- ③ Set the switches on the bottom of the drive as shown. This sets the drive to NOT perform “data disconnects” when there is no data available. This is required because of the odd byte count record packing done on GEMS tapes and SGI's inability to handle odd byte “save pointer” SCSI requests.

☐ F4 – MOD DRIVE SETTINGS

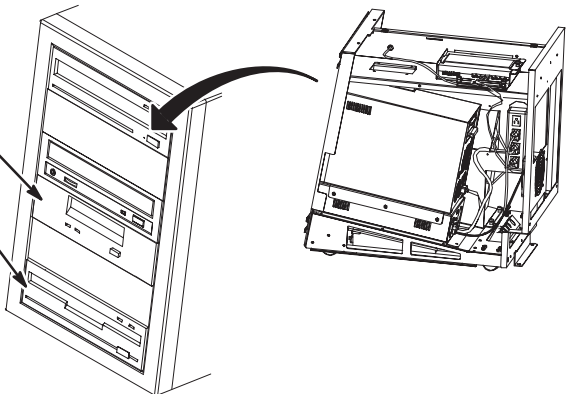
- ② For Octane, SCSI ID = 5
- ③ Make sure jumper is installed on pins at position 8 for SCSI Optical Disk.



- ① Confirm that SCSI Bus Resistor Terminator Packs are not installed. If present – remove them.

☐ F5 – INSTALL DAT AND MOD DRIVES (LEGACY MEDIA) IN SCSI EXPANSION BOX

- ① Install DAT Drive in slot 4 with existing hardware.
- ② Install MOD Drive in slot 6 with existing hardware.
- ③ Connect power cords to DAT and MOD Drives.
- ④ Connect ribbon cable to DAT and MOD Drives.
- ⑤ Install SCSI Expansion Box cover. Reattach bracket to bottom of Box. Reattach bracket and Box assembly to power module.

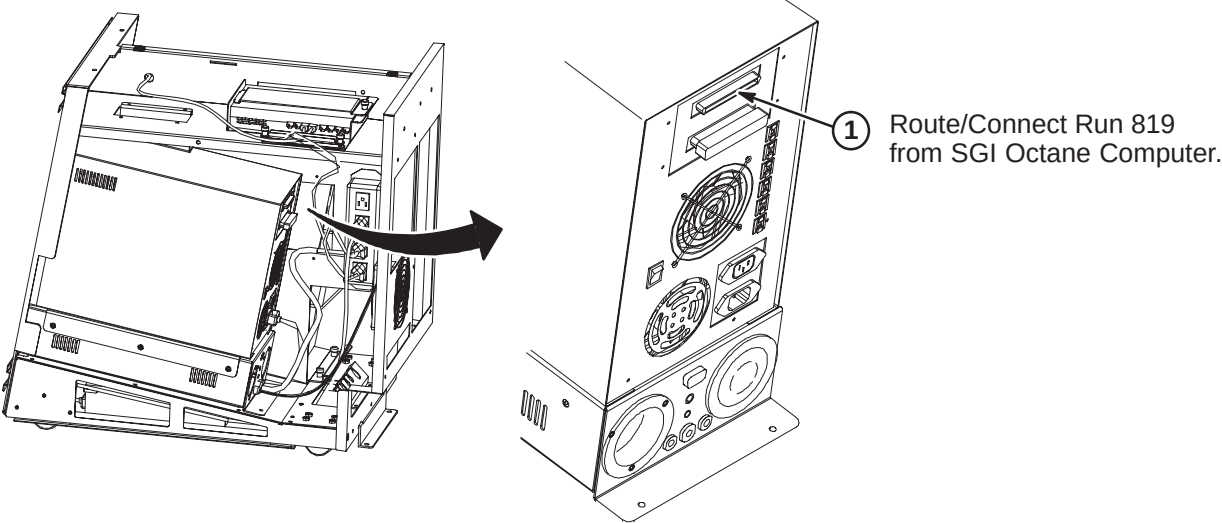


INSTALLATION STEPS SUMMARY

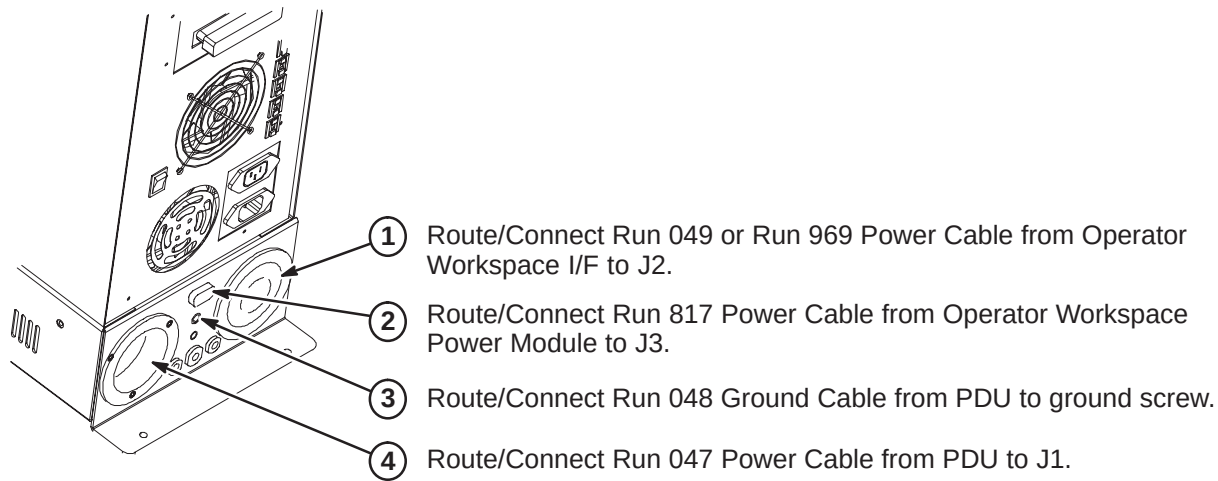
- G1 – Route/Connect Run 819 to SCSI Expansion Box
- G2 – Route/Connect Runs 047, 048, 049/969, and 817 to Power Module
- G3 – Align Operator Workspace Cabinet Cables & Attach InSite “Computer” Magnet
- G4 – Bundle Operator Workspace Cabinet Cables

G1 – ROUTE/CONNECT RUN 819 TO SCSI EXPANSION BOX

Note: Route all cables from SGI Octane Computer thru cable access hole at rear of cabinet.

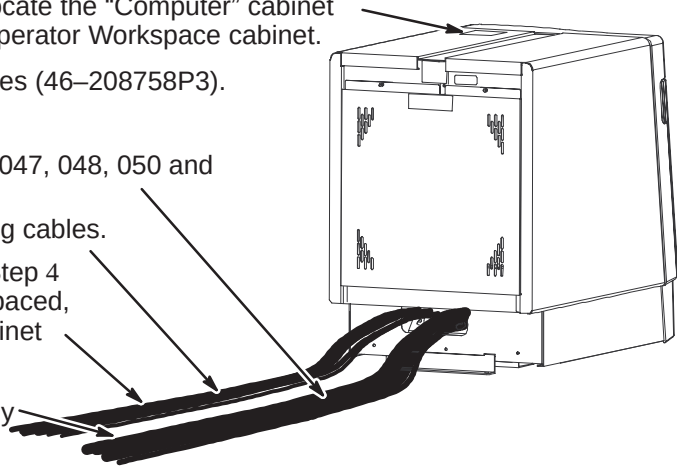


G2 – ROUTE/CONNECT RUNS 047, 048, 049/969, AND 817 TO POWER MODULE



G3 – ALIGN CABINET CABLES & ATTACH InSite “COMPUTER” MAGNET

- 1 Install Operator Workspace Cabinet right and left Side covers, Front cover and Rear cover.
- 2 From the 8.0 LX InSite Kit (46–301708G14), locate the “Computer” cabinet magnet (46–320095P1) and attach to top of Operator Workspace cabinet.
- 3 Locate the twenty (20) seven inch long cable ties (46–208758P3).
- 4 Separate cables into two bundles;
The first bundle will include power cable Runs 047, 048, 050 and fiber optic cable Run 790.
The second bundle will include all the remaining cables.
- 5 Align the cables of the first bundle created in Step 4 and attach ten (10) of the cable ties, equally spaced, around the cable bundle between the OW Cabinet and the OW Table.
- 6 Attach the remaining ten (10) cable ties, equally spaced, around the second cable bundle.



G4 – BUNDLE OPERATOR WORKSPACE CABINET CABLES

- 1 For each tie–wrapped bundle, place Cable Cover (46–307782P1) underneath.
- 2 Attach mating VelcroT to outside surface of Operator Workspace cabinet.
- 3 Wrap Cable Cover around bundle attaching the VelcroT hooks and loops.
- 4 Attach cable bundle Clamp (46–307873P1) to center of Cable Cover.
- 5 Carefully roll OW Cabinet into final position. Make sure cable bundles are unobstructed and slide easily. Attach to mating Velcro on cabinet.
- 6 Repeat Step 5 as often as necessary.

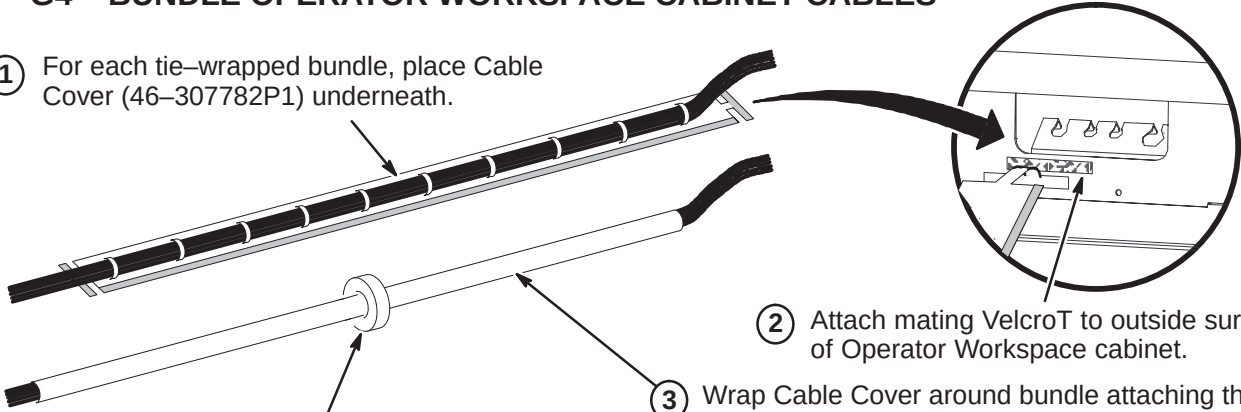


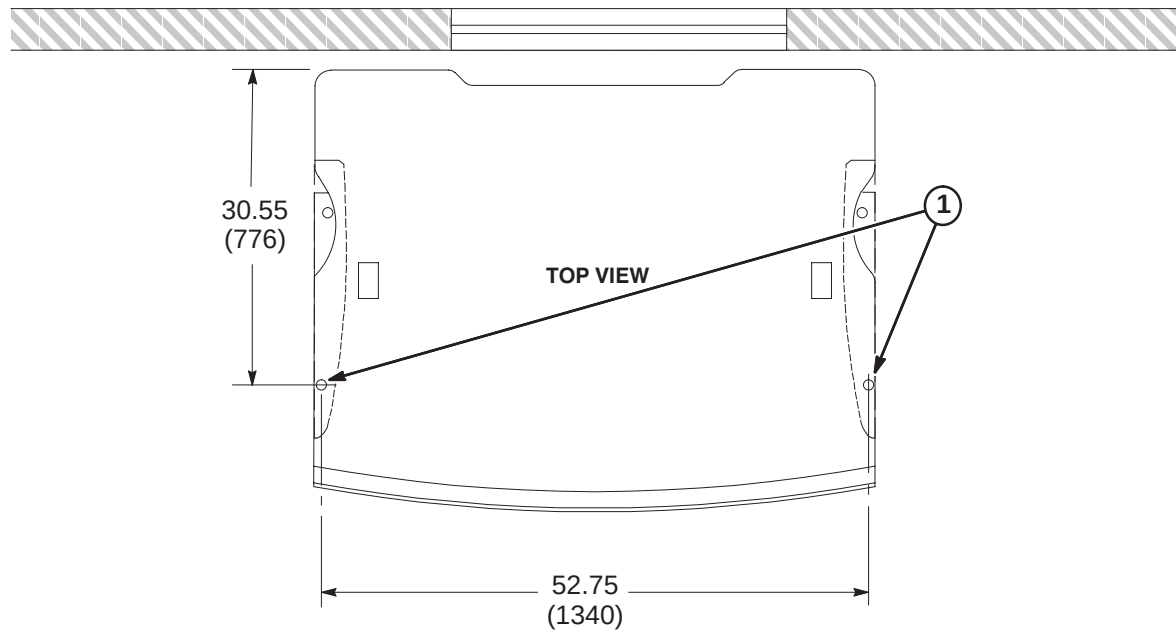
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1-1 OPERATOR WORKSPACE TABLE ANCHOR INSTALLATION

Refer to room layout plan and consult with Site customer for exact location of Operator Workspace Table. Items installed are part of Color Monitor Table Tie Down Kit, 2154928.

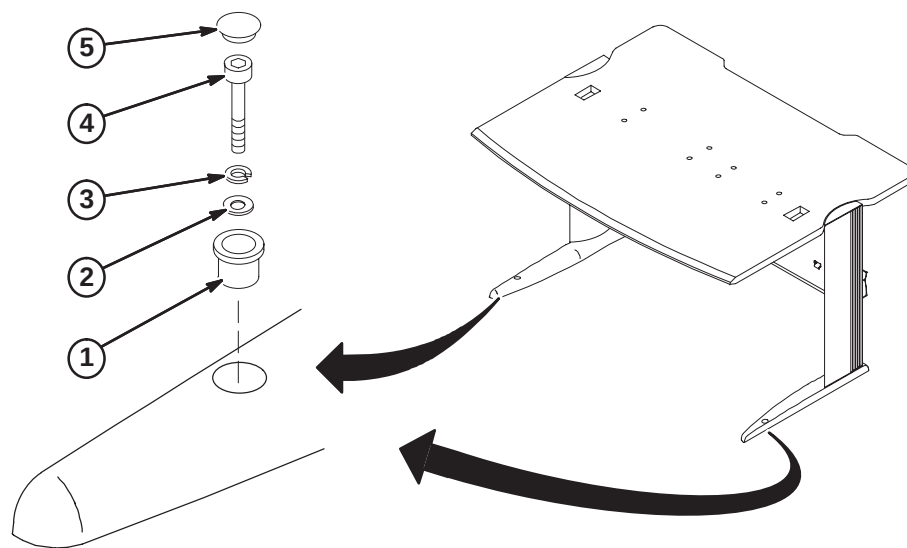
1. Remove and discard the two (2) cover plugs. Locate and mark on floor the two hole locations at the front of the table legs.
2. Move table to prepare for installing the floor anchors in Steps 3 and 4.
3. At each hole marked in Step 2, drill a .38 inch hole and insert 1/4-20 x .375 diameter anchor (2168253).
4. Secure anchor in place with supplied Setting Tool (2168781).



1-2 SECURE OPERATOR WORKSPACE TABLE TO FLOOR

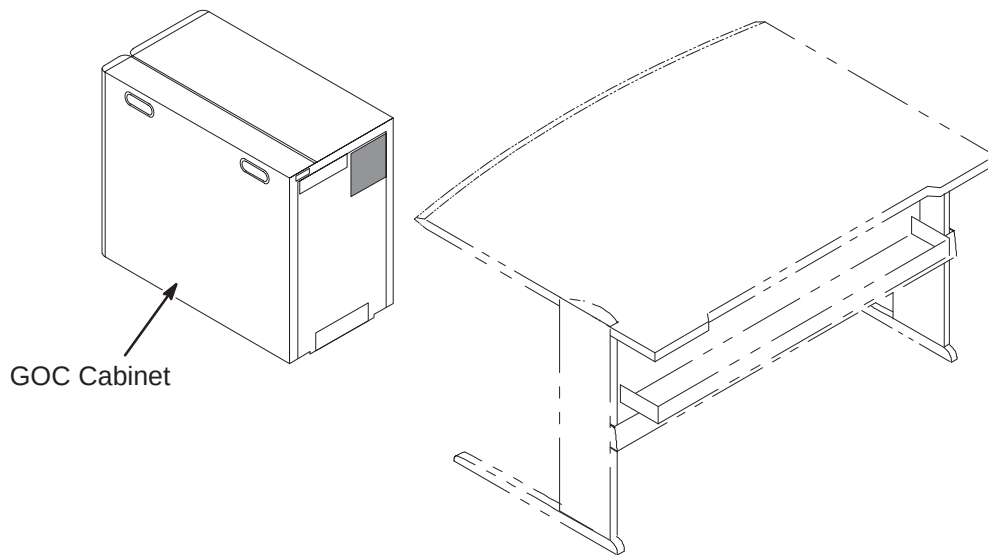
Items installed are part of Color Monitor Table Tie Down Kit, 2154928.

1. Insert Bushing (2151454) in table leg hole.
2. Insert as many Flat Washers (2168251) as required (3 provided for each leg) for securing table to floor.
3. Insert one Lock Washer (2168252) on top of Flat Washer(s).
4. Depending upon depth of previously installed anchor, install either the 1-3/4 inch long or the 2 inch long 1/4 x 20 Hex Socket Cap Screw.
5. Install black nylon Plug (2168254).
6. Repeat anchoring process below for other leg.

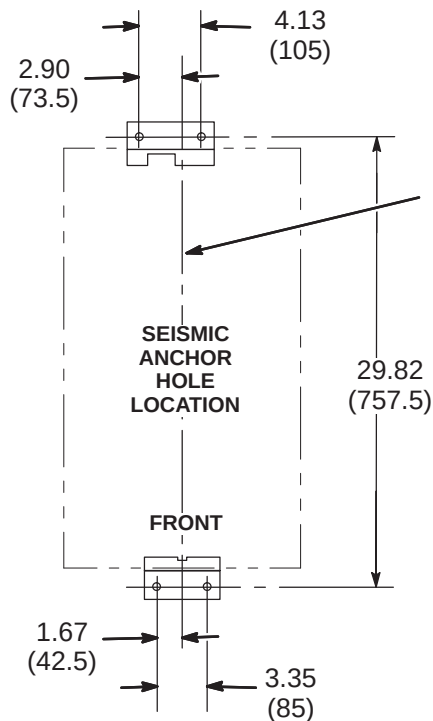


1-3 POSITION COMPUTER CABINET

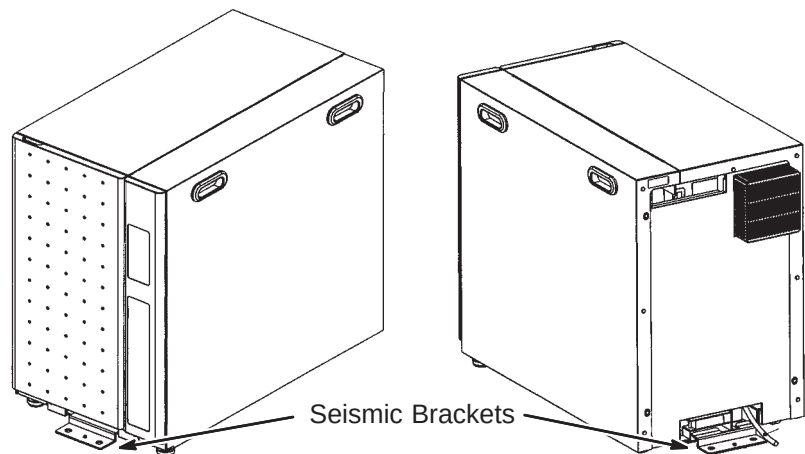
Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.

**1-4 SEISMIC ANCHORING (IF REQUIRED)**

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets.

**NOTE:**

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



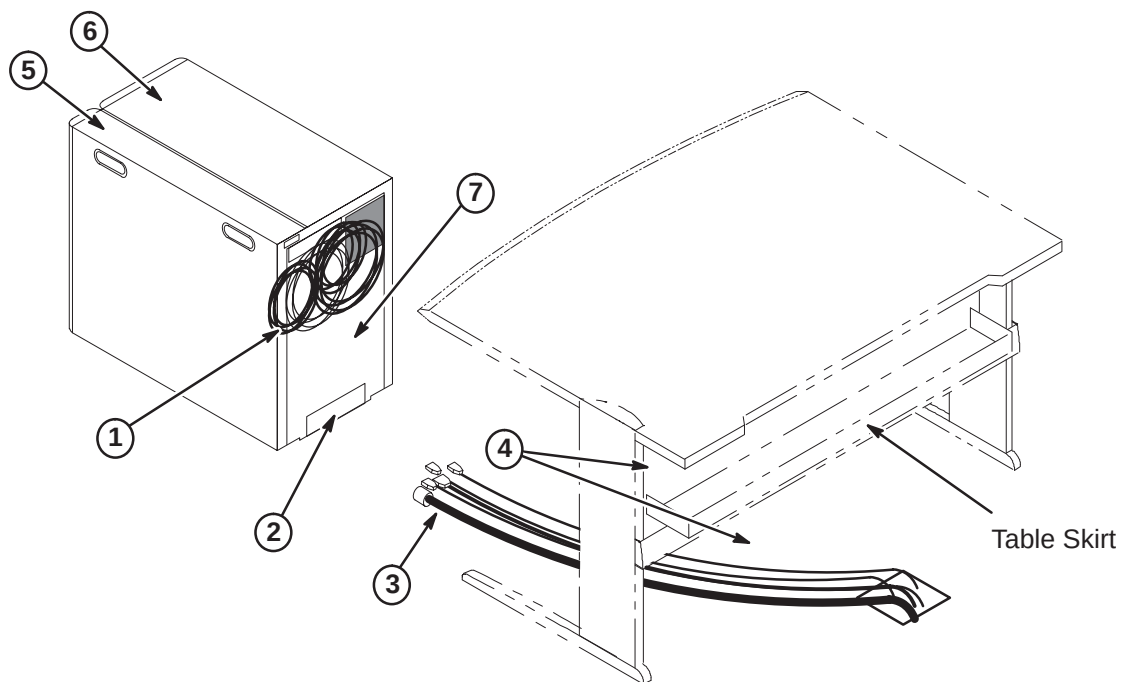
1-5 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with seven cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Three power cords
- Two video output cables
- Keyboard output cable
- SCSI output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

1. Runs 1062, 1063, 1064, 1065, 1068, 1069, and 1070 are pre-connected to computer and are routed thru top opening.
2. Runs 1077, 1081, 1082, 1083, 1084, 1085, power cord for the Patient Alert system, and any other option cables will be routed thru the lower opening.
3. Route all cables before connecting any cables. Refer to cable map for connection information.
4. If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the table skirt. The cables coming out of the lower opening will be routed below the table skirt.



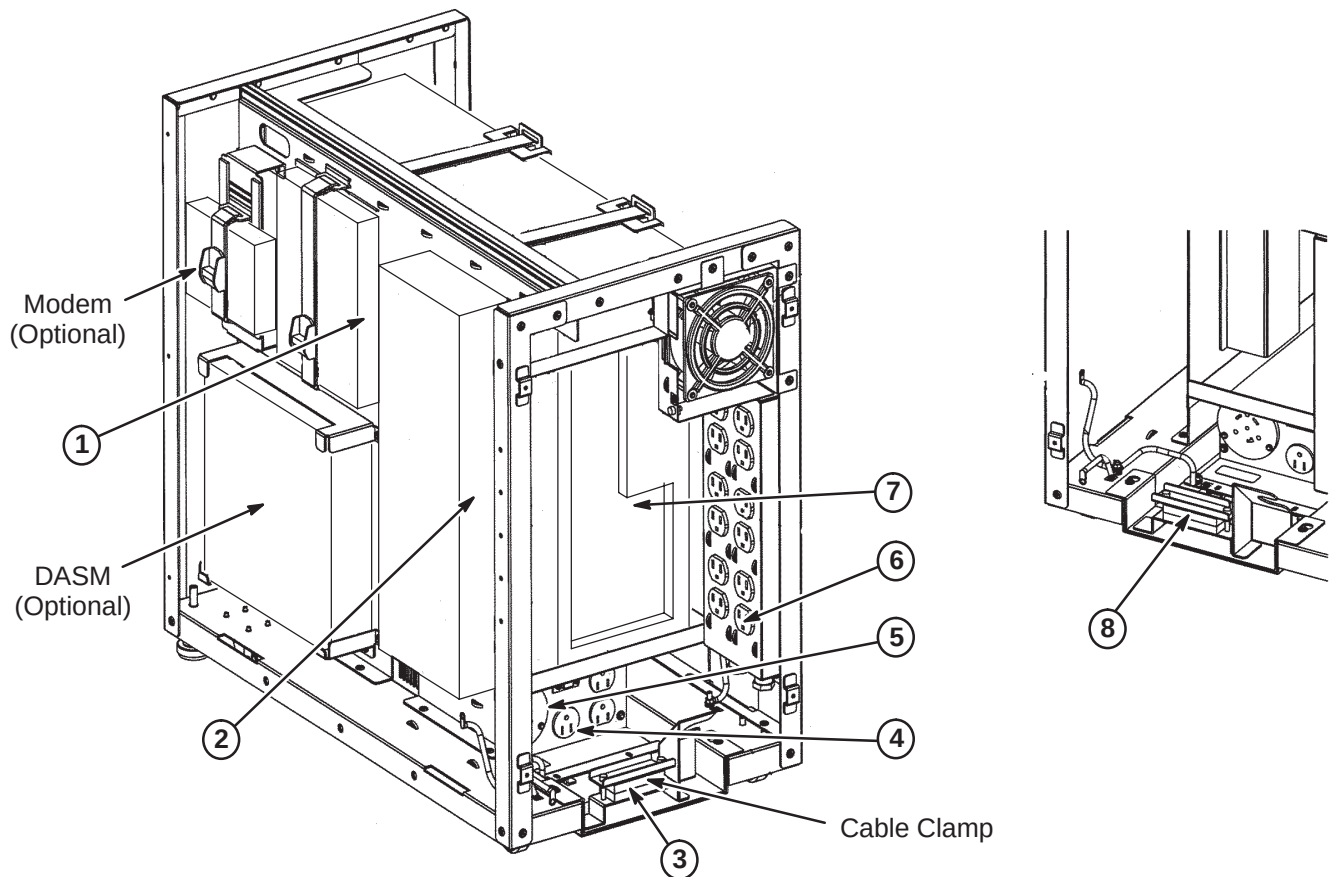
The Computer Cabinet covers need to be removed to access the cable connection points.

5. For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
6. For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
7. Loosen four screws holding the back cover. Lift and remove cover.

1-6 ROUTE CABLES FOR LOWER OPENING

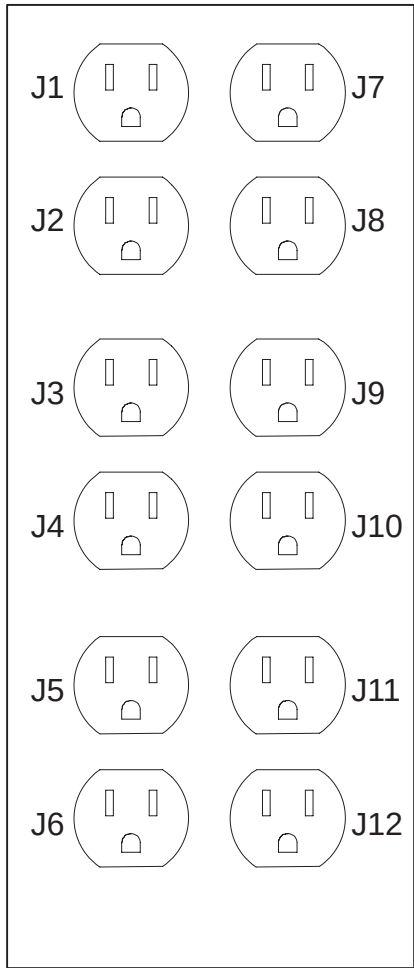
Refer to cable map for information on cables to be connected. Route cables thru clamp at lower rear of cabinet and connect cable runs as shown in illustration below.

1. Run 1077 to Lan Hub.
2. Run 1085 to WIM.
3. Run 1084 routes from GOC Cabinet thru clamp to J26 on System Cabinet.
4. Make sure Modem power cable is connected to J5 on Computer Cabinet PDU.
5. Run 1081 to PDU J1.
6. Patient Alert Power cable to power strip at J12. Refer to Section 1-7 for other power cable connections to power strip.
7. Runs 1082 and 1083 to Linux PC computer. Refer to Section 1-8 for all cables connected to Linux PC computer.
8. Tighten clamp to secure cables in place.



1-7 POWER STRIP CONNECTIONS

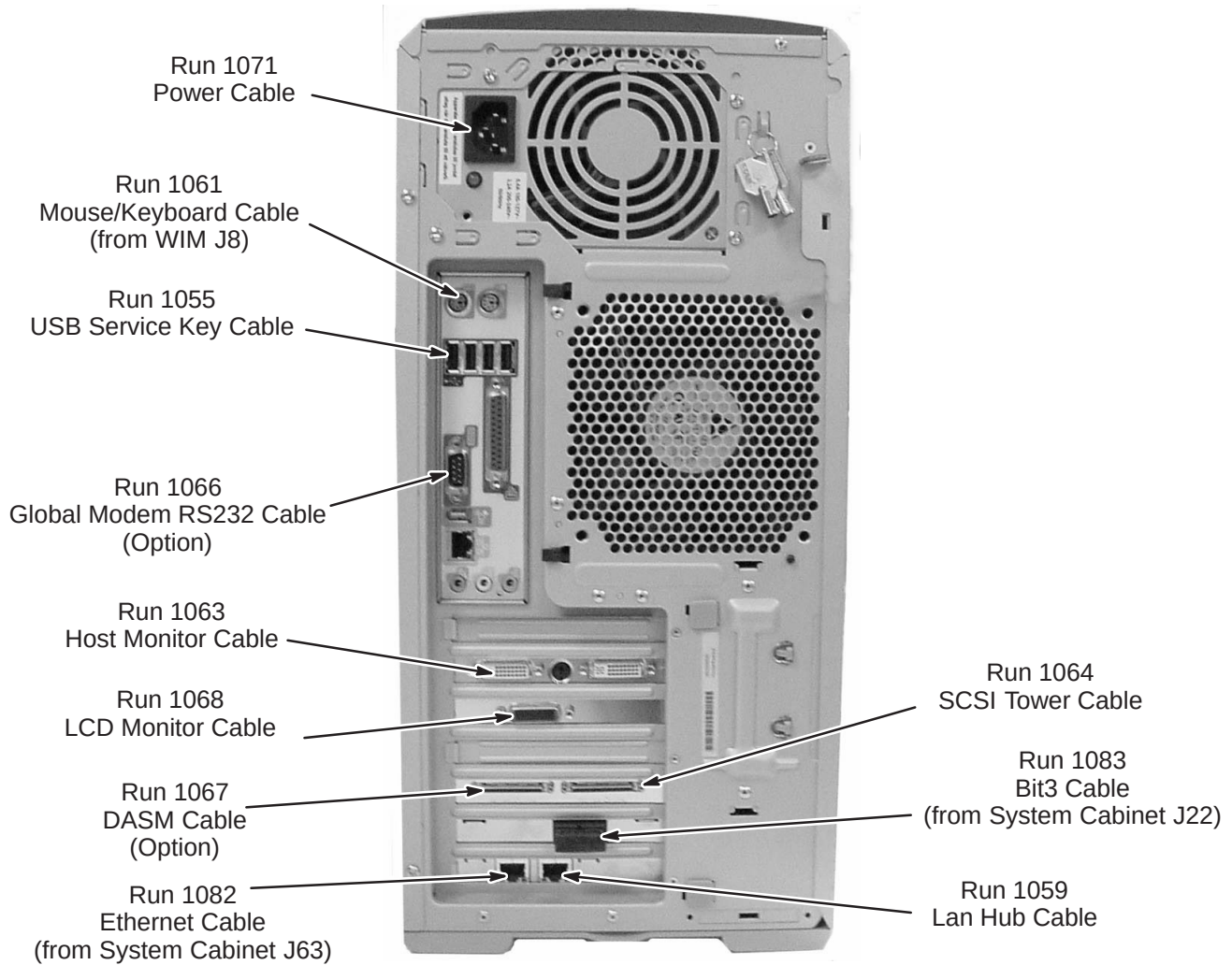
The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.



POSITIONNAME		MAX AMP	
		CIRCUIT 1	CIRCUIT 2
J1	Host	6.4	
J2	Host LCD	0.8	
J3	Fan	0.2	
J4			
J5			
J6			
J7	Hub		0.2
J8	2nd LCD		0.8
J9			
J10	DASM		0.6
J11	SCSI Tower		7.0
J12	Patient Alert		0.2
TOTAL		7.4	8.8

1-8 LINUX PC CABLE CONNECTIONS

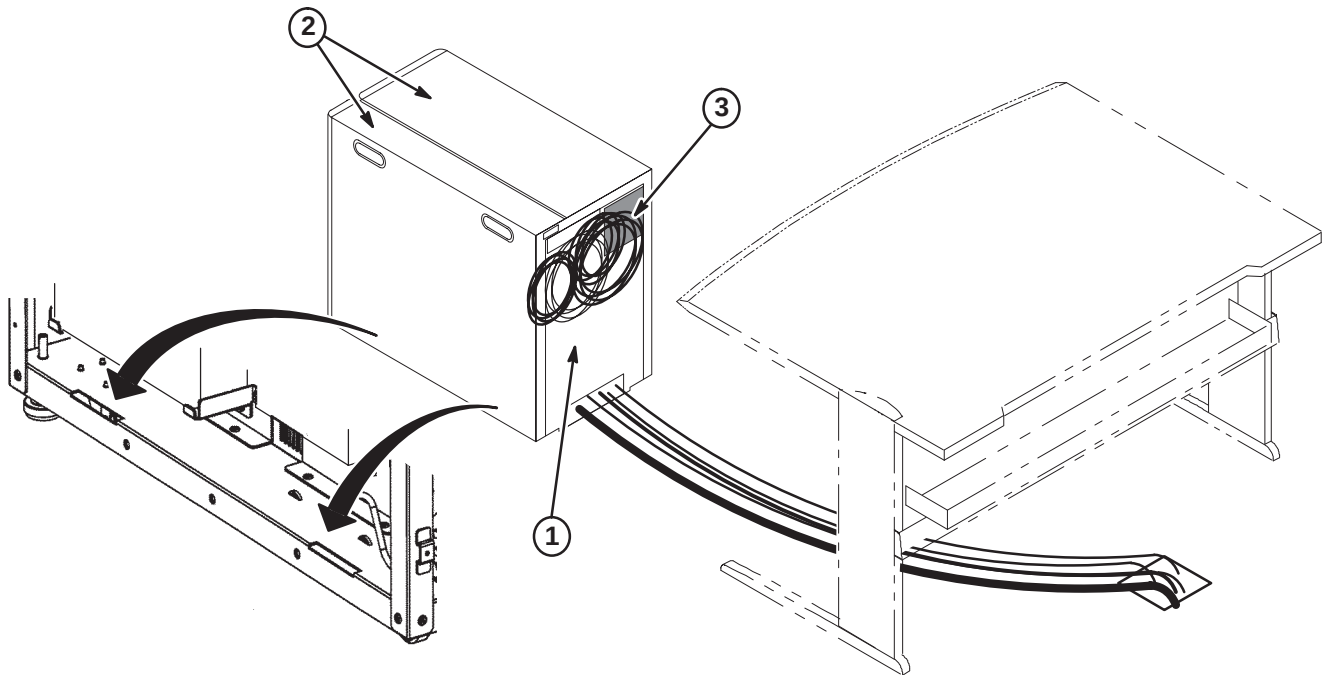
The illustration below shows the connection points of cables routed to the Linux PC. All cables shown are preconnected except for Runs 1082 and 1083, which have to be routed and connected at site.



1-9 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.

1. Re-attach rear cover and tighten screws.
2. Install both side covers by inserting tabs at bottom of covers into slots (as indicated in illustration below) and slide forward. Install screws at rear of cover that were removed in earlier procedure.
3. Route remaining cables from top opening at rear of GOC Cabinet thru opening of table skirt.
4. Place cabinet in final position as determined by customer.



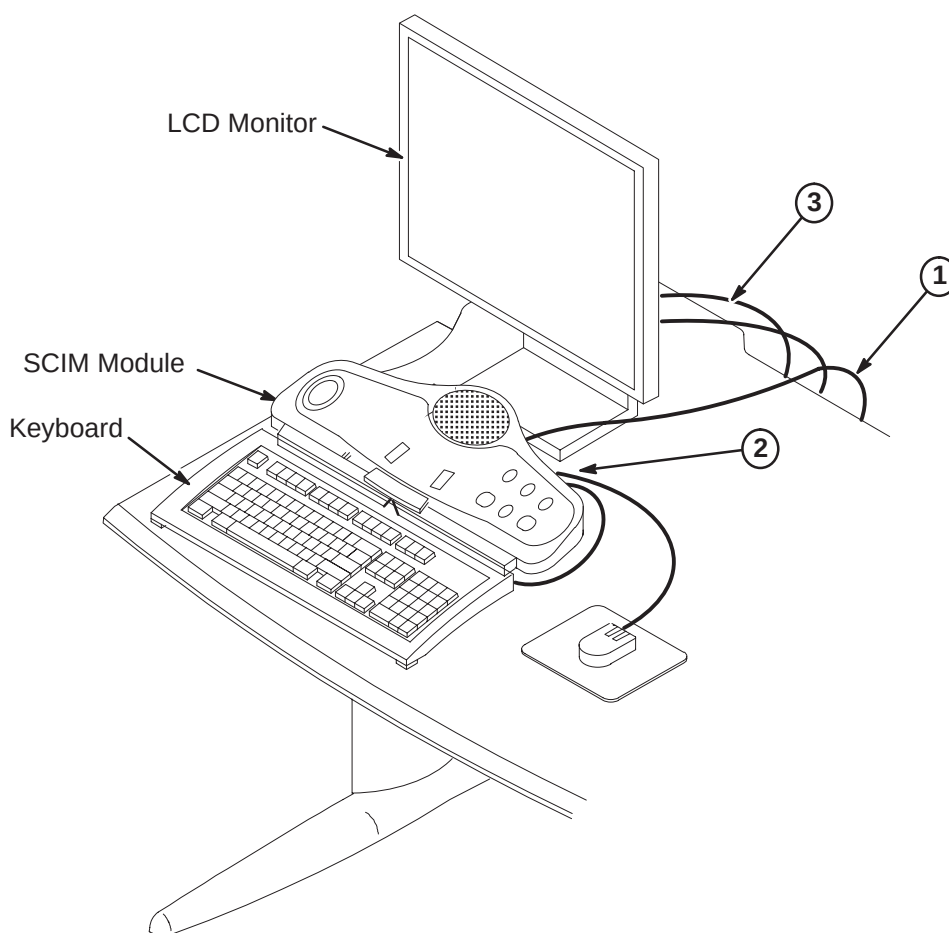
1-10 INSTALL SCIM/KEYBOARD AND MONITOR

Consult with Site customer for exact location of Keyboard and Monitor on Operator Workspace (OW) Table.

Note

Refer to Installation document OW1INA3.DOC on Service Methods CD-ROM or MR Service Engineering web site for installation information on language labels and keyboard mounting.

1. Route data cable Run 1062 from Computer Cabinet to bottom side of SCIM Module and connect as marked. (Run 805, supplied with SCIM Kit (2307175) is not used for EXCITE) Run 1062 is preconnected to the Computer Cabinet and is located on back side of cabinet.
2. Place Keyboard and Mouse on table and connect cables to bottom side of SCIM Module as marked.
3. Place LCD Host Monitor on table in back of SCIM. Route Runs 1063 and 1065 from Computer Cabinet and connect to monitor.



1-11 INSTALL SCSI EXPANSION BOX AND LCD DISPLAY

1. Place the SCSI Tower on the OW table and connect Runs 1064 and 1070 from GOC Cabinet. Install terminator.
2. Place LCD Display on OW table and connect Runs 1068 and 1069 from GOC Cabinet.

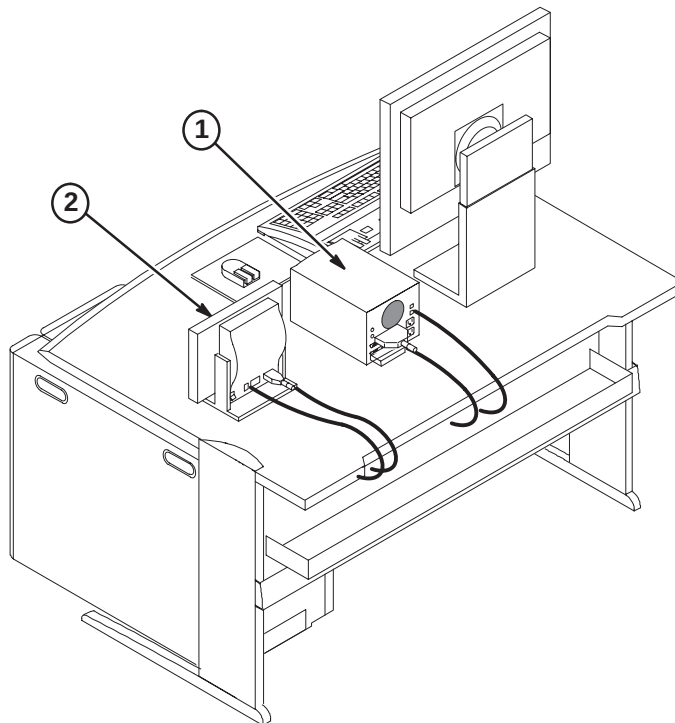


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1-1 INTRODUCTION

This upgrade procedure is to be performed on LX sites that have an Indigo or Octane computer and are upgrading to EXCITE II with a Linux computer.

1-2 DEINSTALLATION OF DUPLEX CABINET

-
- ② Loosen thumb screw, unfasten top screw and slide Right Side Cover backwards to remove.
- ④ Disconnect and save all customer supplied site specific ethernet cables. These will be reconnected in new GOC Cabinet.
- ③ Loosen thumb screw, unfasten three (3) top screws and remove Left Side Cove.
- ① Loosen (2) two thumb screws and remove Rear Cover.
- ⑤ Disconnect and save Run 969 (Octane) or Run 047 (Indigo) Power Cable from J1. The cable will be relabeled as Run 1081 and reused in the new GOC Cabinet.
- ⑥ The Duplex must be returned for recycling to Gold Seal. Duplex can be inserted onto skid that was used to ship the GOC by removing the white foam on the bottom that supported the GOC.
- ⑦ The remaining Duplex cables will not be used on the new GOC Cabinet and must be returned for recycling to Gold Seal.

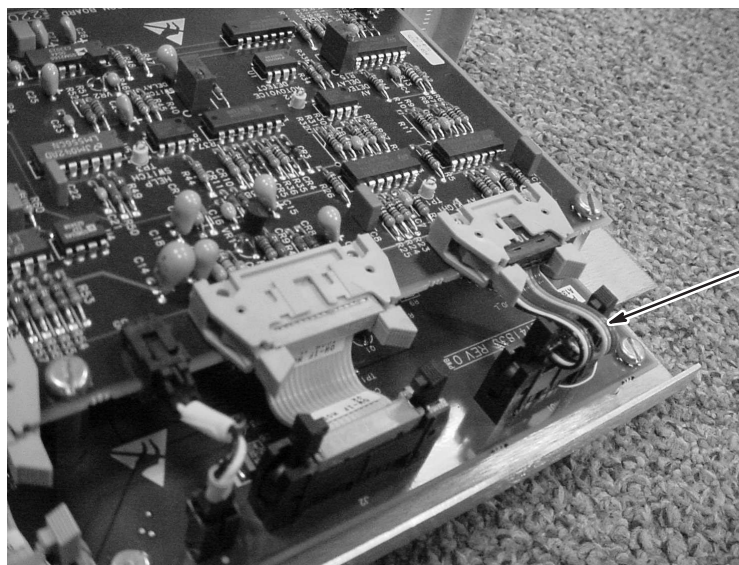
1-3 DEINSTALLATION OF OPERATOR WORKSPACE TABLE AREA

The WIM (Workspace Interface Module) will be transferred to the new GOC but needs a modification. If present, other modules (DASM, Modem) will be retained for use with new EXCITE II Operator Workspace.

1-3-1 WIM REMOVAL

- ① Disconnect keyboard cable from front of WIM.
- ② Turn (2) two knobs on front of module counter-clockwise and unseat WIM from the housing by moving it to the rear of the table.
- ③ Disconnect all cable at rear of WIM.
- ④ Unfasten four screws holding WIM enclosure bracket to bottom of table and transfer enclosure bracket to GOC Cabinet.
- ⑤ Runs 788 and 789 will be used on EXCITE II and relabeled as Runs 1085 and 1084. Refer to table in Section 4 for relabeling information.
- ⑥ The remaining cables will not be used on the new EXCITE II and must be returned for recycling to Gold Seal.

1-3-2 WIM MODIFICATION

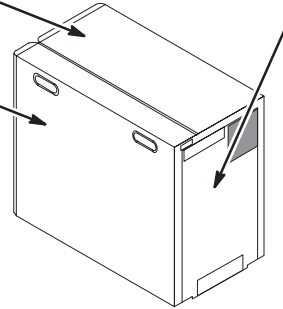
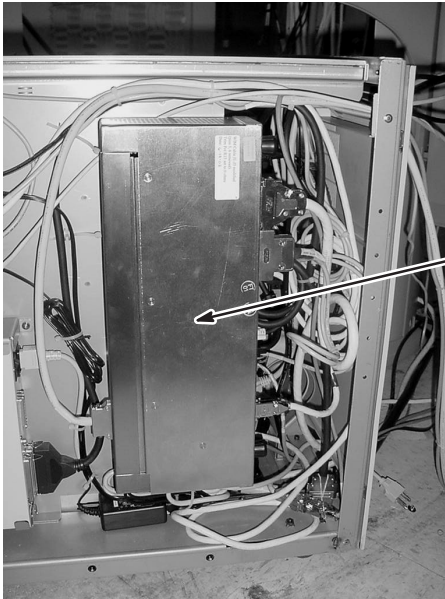


- ① Remove the short jumper cable on the WIM, between J1 on motherboard to J3 on the console intercom board.
- ② Install the replacement cable that came inside of the new GOC.

1-3-3 WIM INSTALLATION INTO GOC

The Computer Cabinet covers need to be removed for installation of WIM and other cables.

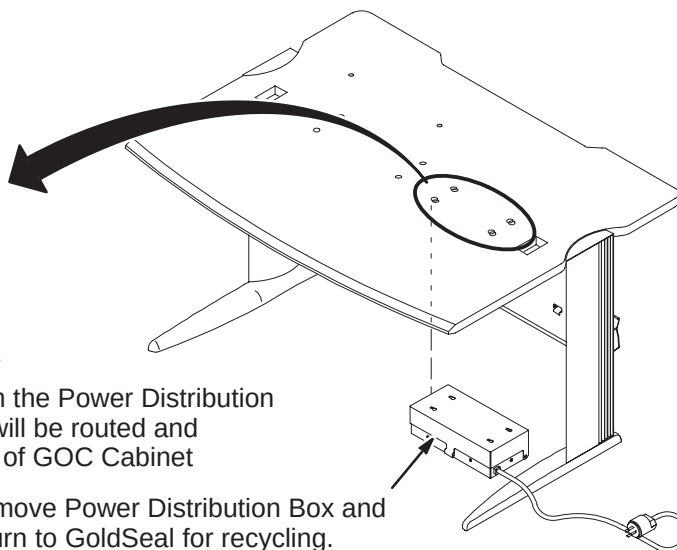
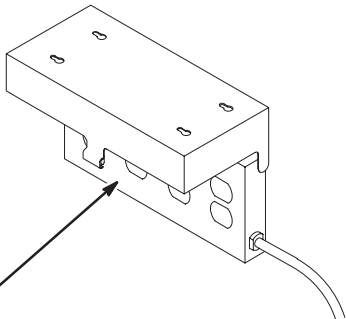
- ② For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
- ③ Loosen four screws holding the back cover. Lift and remove cover.
- ① For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.



- ④ Install WIM enclosure bracket into the GOC.
- ⑤ Install the WIM into the enclosure bracket.
- ⑥ Connect Run 1062 (supplied with GOC) to J6.
- ⑦ Connect Run 1061 (supplied with GOC) to J8.
- ⑧ Connect Run 1060 (supplied with GOC) to J17.

1-3-4 REMOVAL OF POWER DISTRIBUTION BOX

- ① Disconnect all power cables from the Power Distribution Box. All required power cables will be routed and connected to PDU module inside of GOC Cabinet
- ② Remove Power Distribution Box and return to GoldSeal for recycling.



1-3-5 DEINSTALL OW TABLE MODULES (If Applicable)

① SCIM is required for EXCITE II. If older style keyboard is part of existing site, it will have to be replaced.

② Color Monitor can be used with EXCITE II. Black and white monitor has to be replaced by new LCD Monitor.

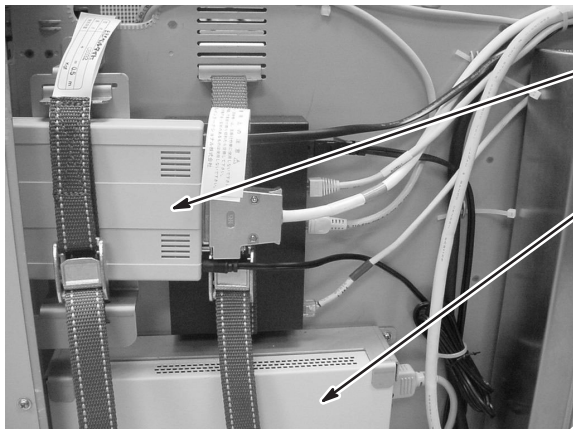
③ If Modem is part of existing site, remove Modem and power cord and transfer to GOC Cabinet.

NOTE: DASM-1 is not compatible with the Linux PC, must use DASM-2.

④ Remove DASM and cable to filming device and transfer to GOC Cabinet.

⑤ Remove SCSI Expansion Box and return to GoldSeal for recycling.

NOTE: If site is TwinSpeed, and has HOS option, disconnect Run 1004 from SCSI Expansion Box and save. This cable will be reconnected to GOC.

1-3-6 INSTALL MODEM AND DASM IN GOC CABINET (If Applicable)

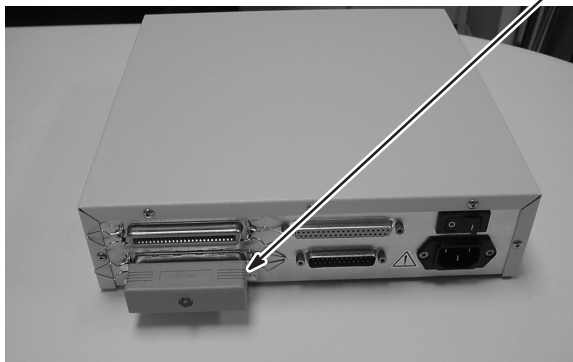
① Mount Modem to GOC so that the underside is facing out to cover. Route customer provided phone line and connect to Modem.

② Mount DASM at this location. Route and connect cable from filming device.

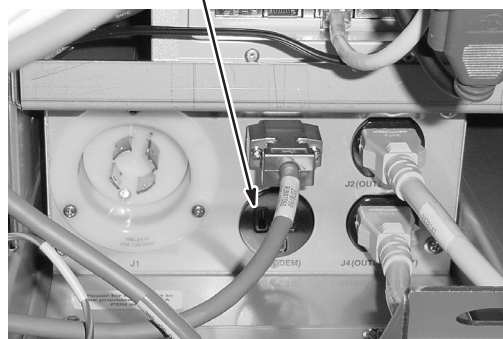
③ Cable routed between DASM or Modem to the Linux PC are provided. Connect Run 1066 to Modem and Run 1067 to DASM.

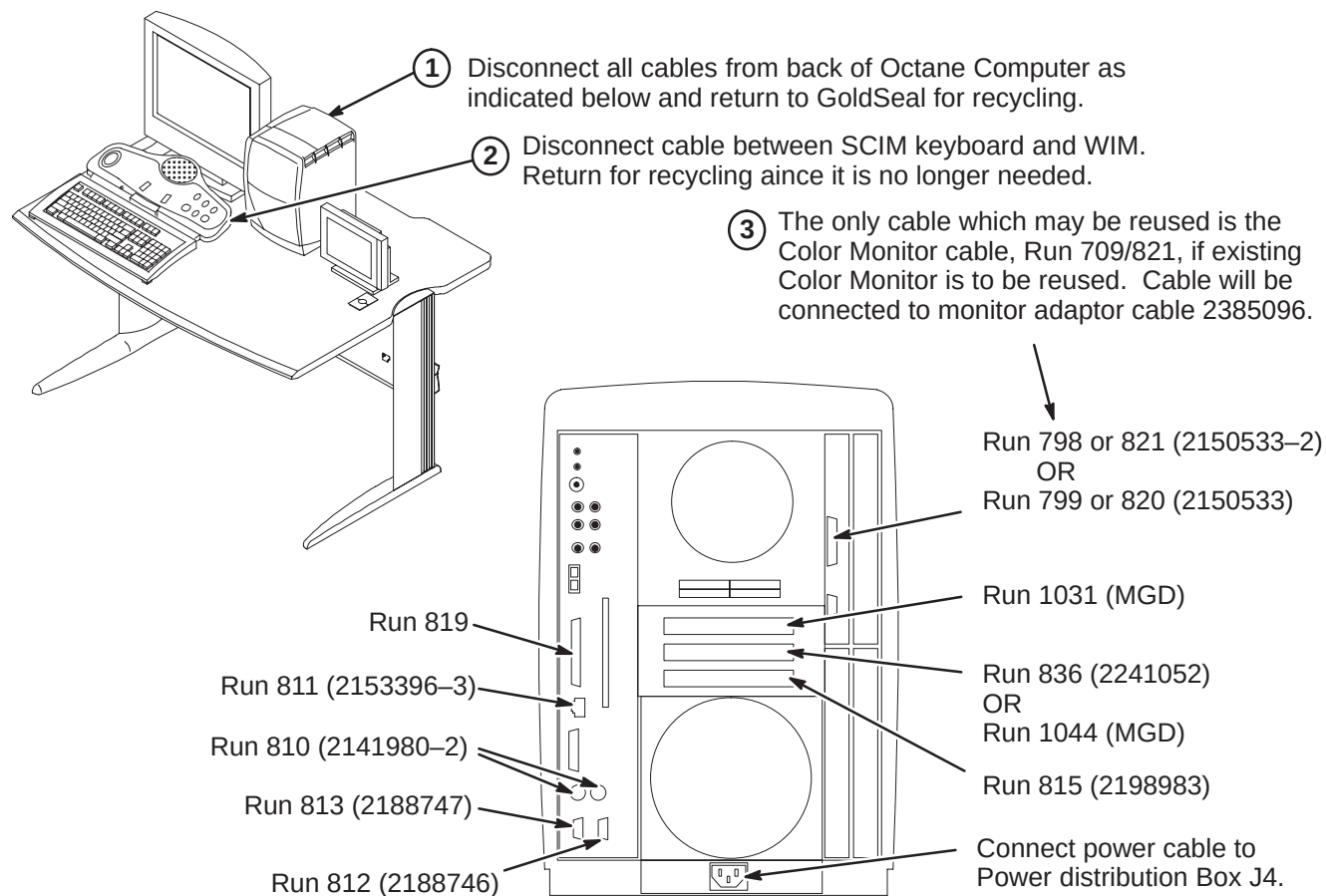
④ If applicable, install the SCSI Terminator onto the bottom port of the DASM.

⑤ Verify that the **Active SCSI Terminator** located on the bottom of the DASM is in the **OFF** position.



⑥ Modem power cord must be connected to J5 on GOC PDU Module

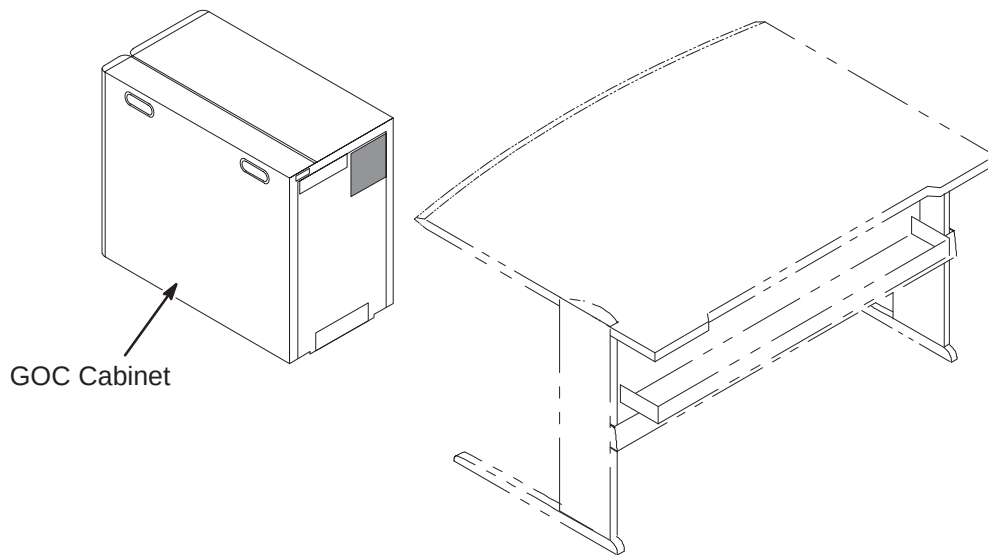


1-3-7 DEINSTALL OCTANE COMPUTER

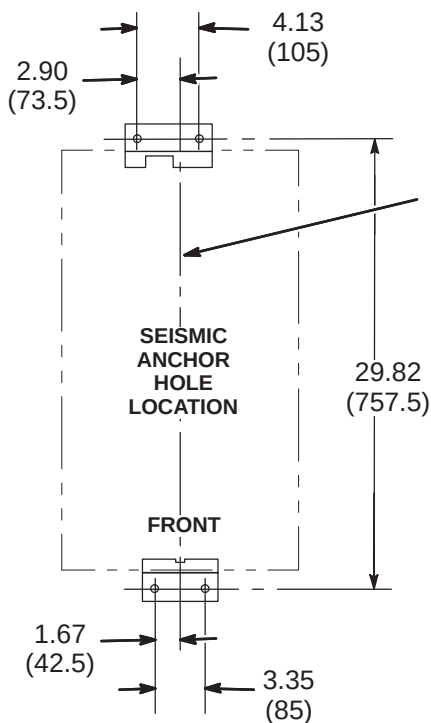
④ Octane Computer MUST be returned to GoldSeal for recycling.

1-4 POSITION NEW GOC COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Cabinet (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.

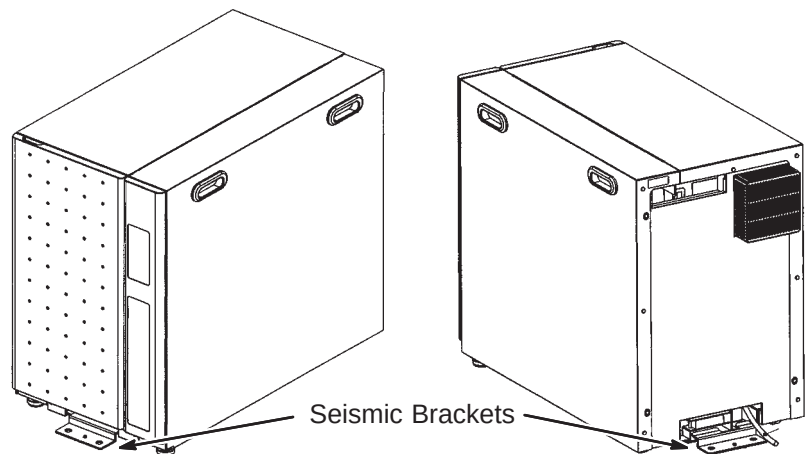
**1-5 SEISMIC ANCHORING (IF REQUIRED)**

If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets.

**NOTE:**

- ALL DIMENSIONS ARE IN INCHES.
ALL BRACKETED () DIMENSIONS
ARE IN MILLIMETERS.

CENTERLINE OF FRAME

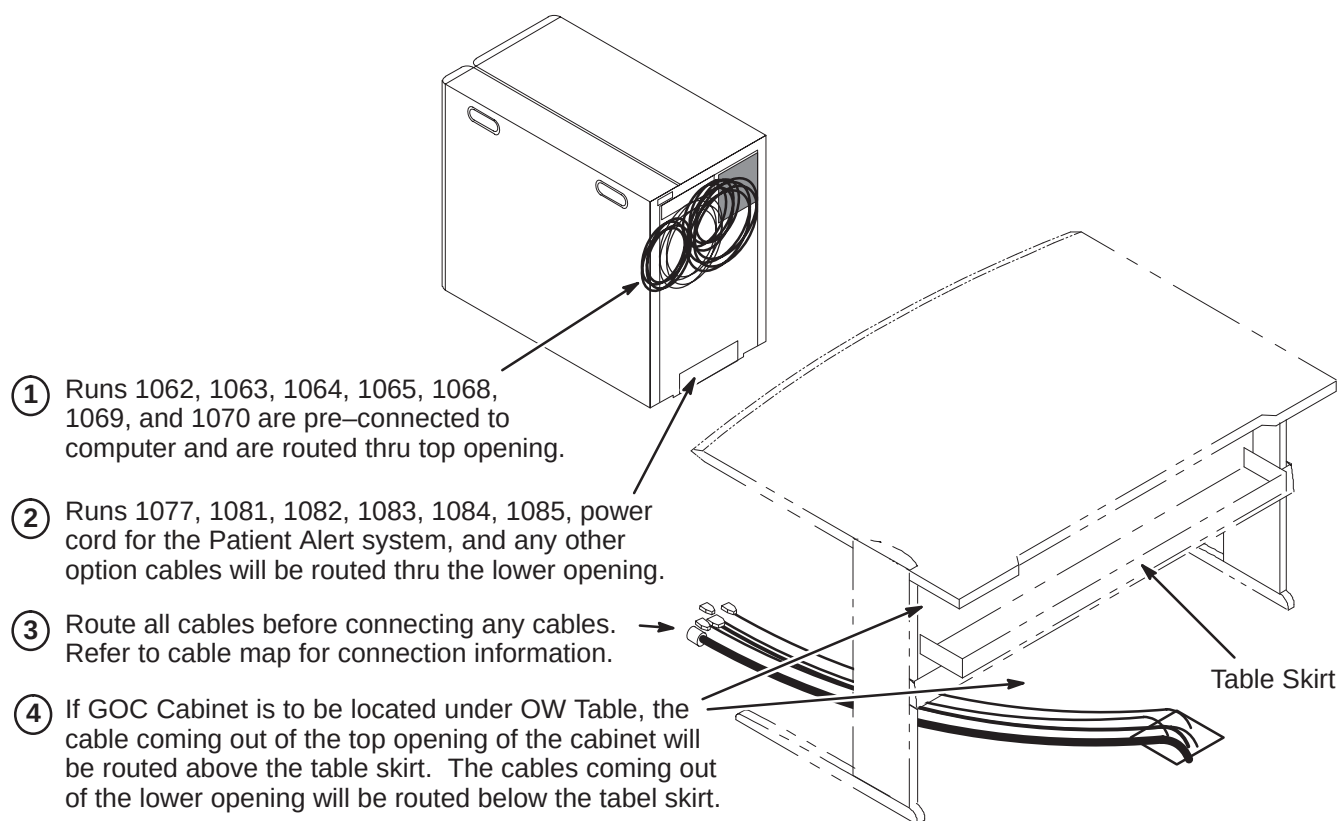


1-6 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with seven cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

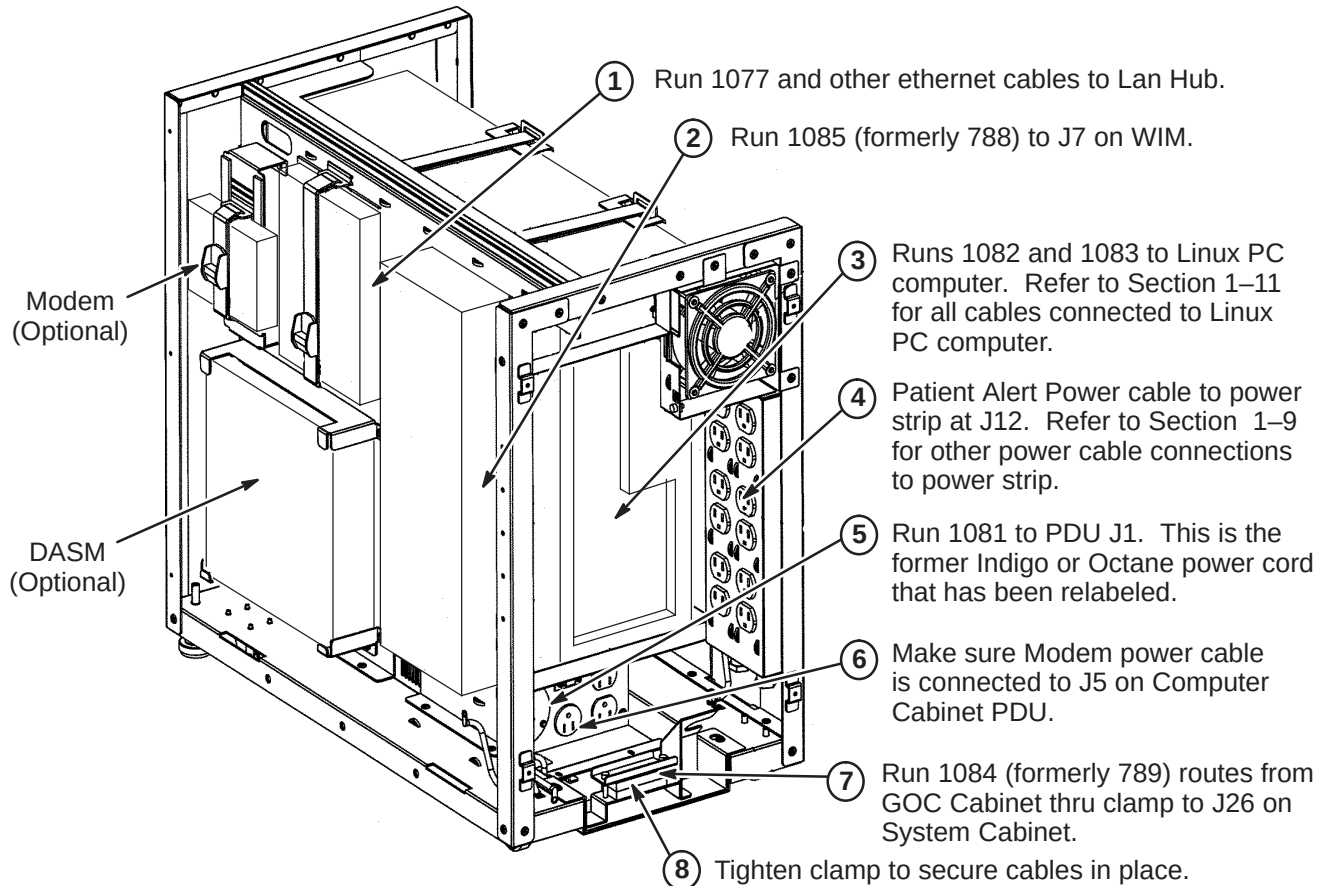
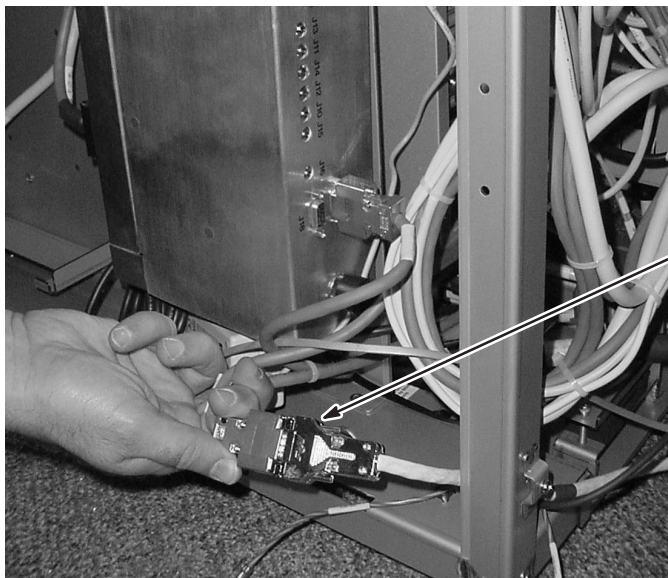
- Three power cords
- Two video output cables
- Keyboard output cable
- SCSI output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.



1-7 ROUTE CABLES FOR LOWER OPENING

Refer to cable map for information on cables to be connected thru lower opening.

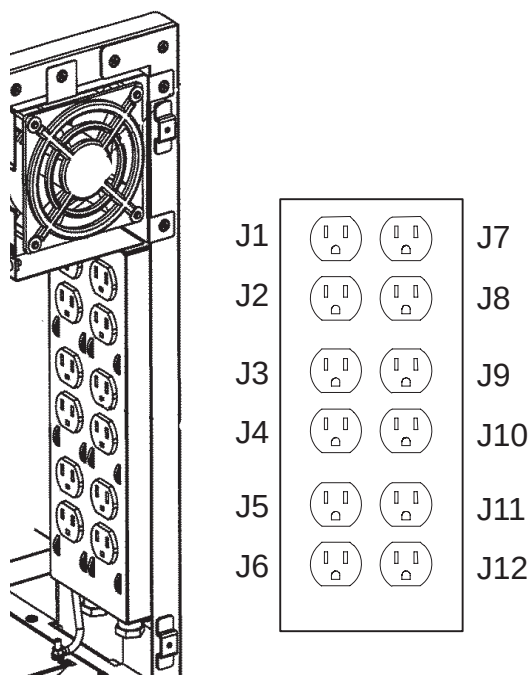
**1-8 CONNECT RUN 1084****IMPORTANT**

Do not connect Run 1084 to J18 on the WIM. The E-stop circuit will be open.

- 1 Connect Run 1084 to Run 1062 "Y" cable on the bottom of the GOC cabinet.

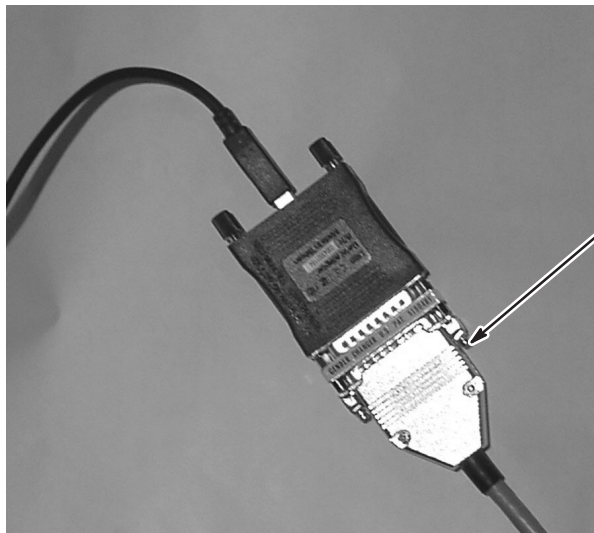
1-9 POWER STRIP CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.



POSITION	NAME	MAX AMP	
		CIRCUIT 1	CIRCUIT 2
J1	Host Comp	6.4	
J2	Host LCD	0.8	
J3	Fan	0.2	
J4			
J5			
J6			
J7	LAN Hub		0.2
J8	2nd LCD		0.8
J9			
J10	DASM		0.6
J11	SCSI Tower		7.0
J12	Patient Alert		0.2
TOTAL		7.4	8.8

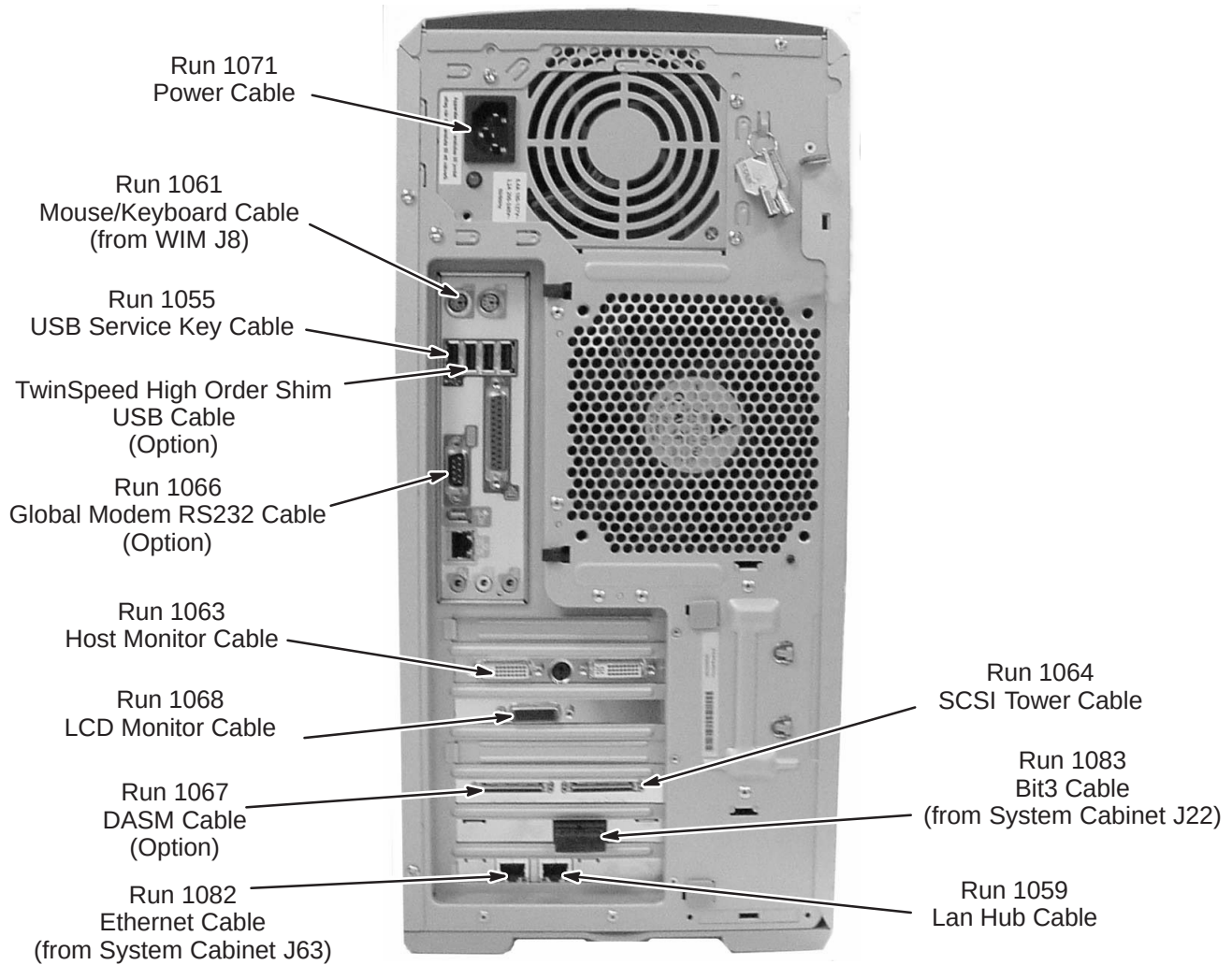
1-10 HIGH ORDER SHIM OPTION CABLE INSTALLATION (TwinSpeed Only)



- ① If the site is TwinSpeed with the High Order Shim option, connect cable Run 1076 (formerly 1004) to the cable located in the rear of the GOC cabinet. The other end of this cable routes to a USB port on the Linux PC.

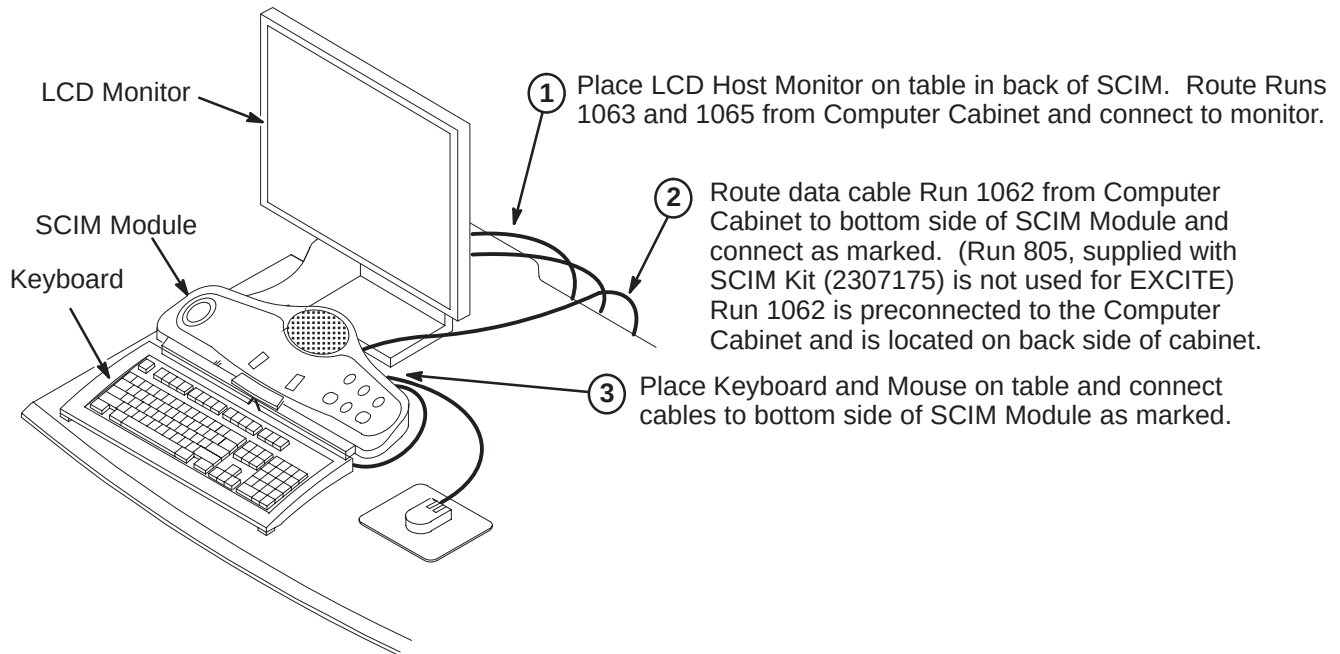
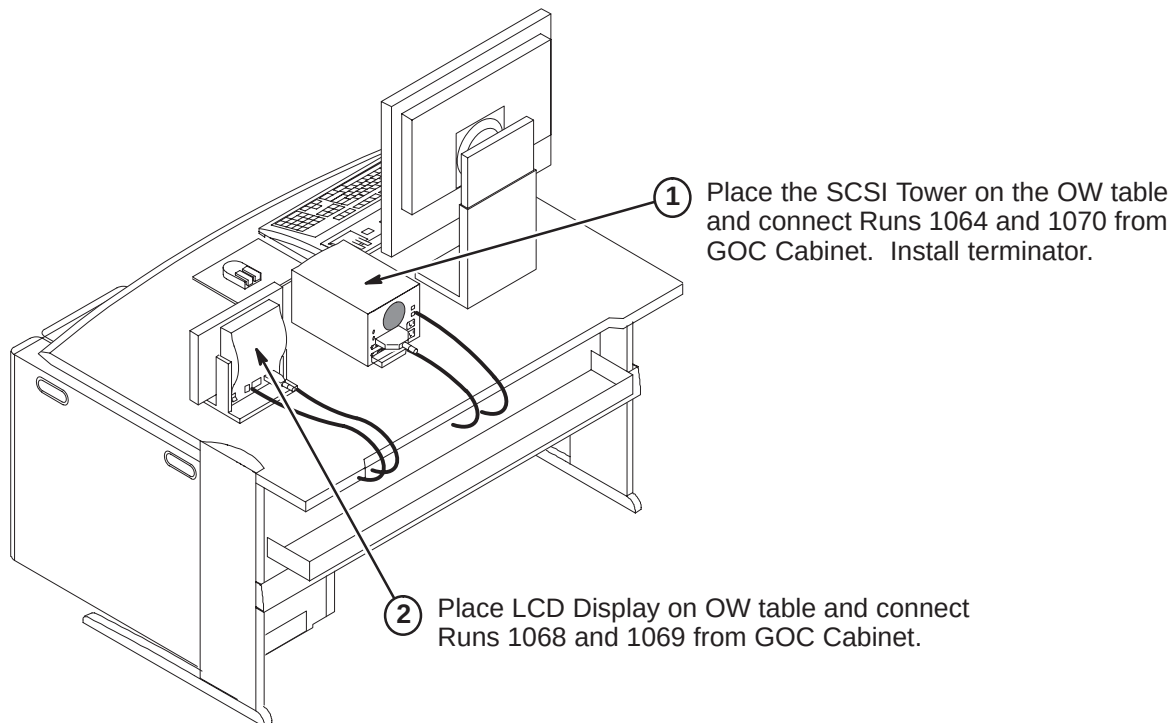
1-11 LINUX PC CABLE CONNECTIONS

The illustration below shows the connection points of cables routed to the Linux PC. All cables shown are preconnected except for Runs 1082 and 1083, which have to be routed and connected at site.



1-12 INSTALL SCIM/KEYBOARD AND MONITOR**Note**

Refer to Installation document OW1INA3.DOC on Service Methods CD-ROM or MR Service Engineering web site for installation information on language labels and keyboard mounting.

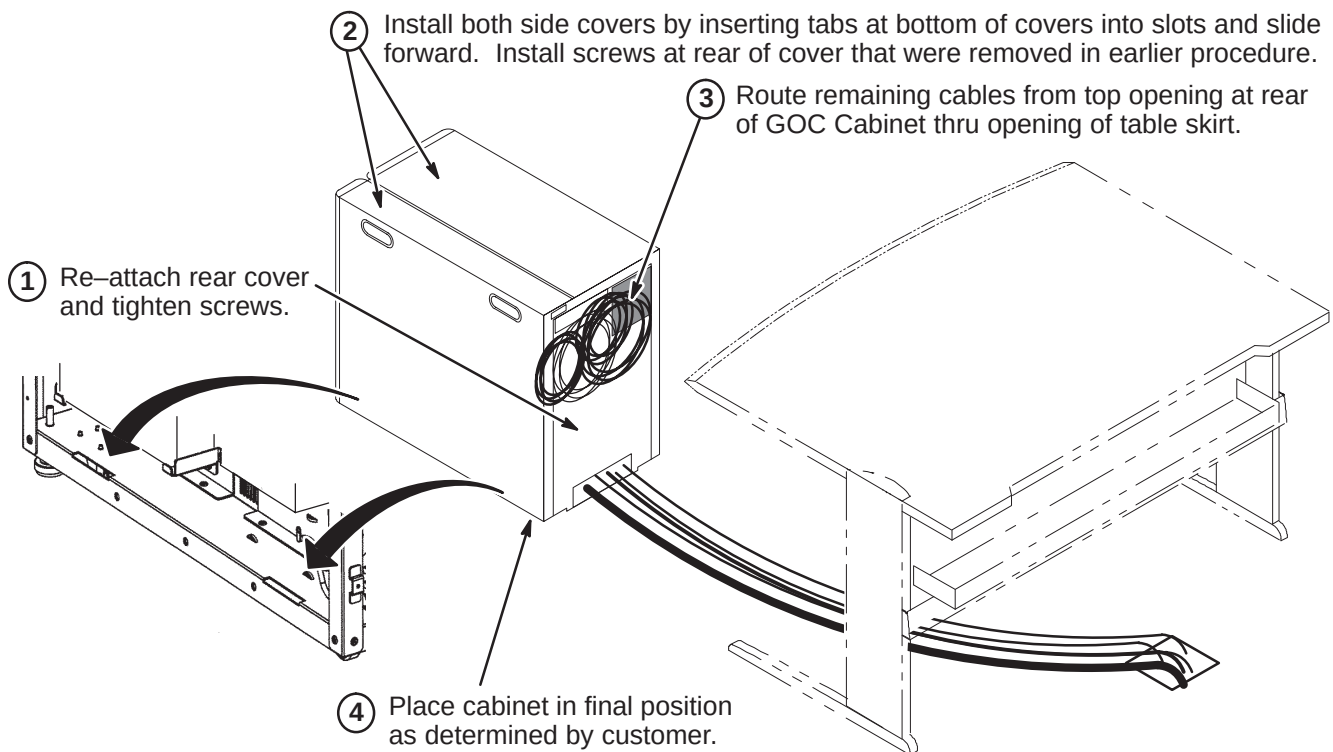
**1-13 INSTALL SCSI EXPANSION BOX AND LCD DISPLAY**

1-14 POWER ON INFORMATION**IMPORTANT**

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the WIM and Modem power.

1-15 FINAL POSITIONING OF GOC CABINET**1-16 RETURN OF REPLACED PARTS**

Refer to Section 1-4 in Overview Tab for instructions on returning discarded parts.

1. Remove the white Styrofoam from the bottom of the GOC skid. The Duplex will now fit onto the bottom of the GOC skid. Place the removed Duplex onto the GOC Skid. Properly secure the top cardboard box onto the skid using straps. Return to GoldSeal.
2. Return the Octane or Indigo computer to GoldSeal.
3. Return all cables that have been replaced by this upgrade to GoldSeal.

ACGD/PDU UPGRADE INSTRUCTIONS

Tab 15, ACGD/PDU Cabinet, contains installation instructions for the following:

- *DIRECTION 2286811* – Signa ACGD/PDU Cabinet Installation Instructions
- *DIRECTION 2403139* – Signa ACGD/PDU GP Module Upgrade for EXCITE

Use only the applicable System Upgrade Instruction Directions for the site being upgraded

INSTALLATION STEPS SUMMARY

- A1 – Position ACGD/PDU Cabinet
- A2 – Install Fan Assembly and Attach InSite Cabinet Magnet
- A3 – Input Voltage Selection
- A4 – Circuit Breaker DIP Switch Settings

A1 – POSITION ACGD/PDU CABINET

- 1 Move cabinet to final position.

WARNING!

DUE TO WEIGHT OF CABINET, AT LEAST TWO PEOPLE ARE REQUIRED TO MOVE CABINET

- 2 Anti-tip bars for the ACGD/PDU Cabinet are shipped in the SRF Cabinet. Remove from SRF Cabinet for this procedure.
- 3 Attach bars to cabinet.
- 4 Adjust leveling pads to level cabinet.

A2 – INSTALL FAN ASSEMBLY AND ATTACH INSITE MAGNET

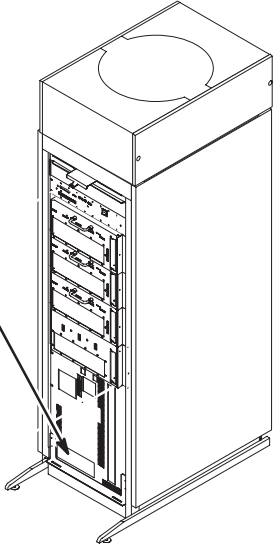
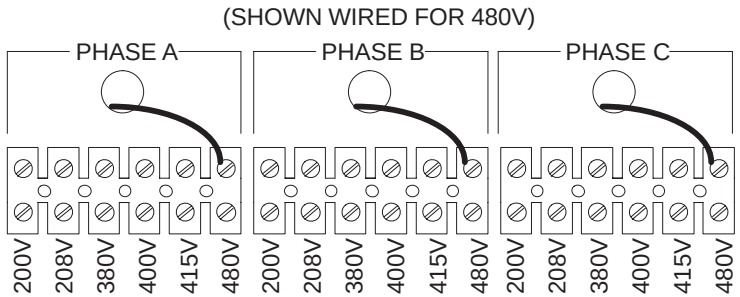
- 1 Remove Fan Assembly from shipping container.
- 2 Remove two side trim panels (two screws from each).
- 3 Requiring two people, place the Fan Assembly on top of cabinet with the cable connector facing the rear of the cabinet.
- 4 Secure Fan Assembly in place with (4) four screws and washers, two on each side of cabinet.
- Note:** The fasteners are provided in a plastic bag located in PDU Module. Rear PDU covers will need to be removed for access to plastic bag with fasteners.
- 5 Install two side trim panels with previously removed screws.
- 6 Connect cable from SGA Power Supply to the fan connector.

DANGER!!!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU. CHECK THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

A3 – INPUT VOLTAGE SELECTION

- 1 Using a voltmeter or other voltage indicating device, check to be sure that no voltage is being applied to the input of the PDU.
- 2 Remove the plate from the front of the PDU marked INPUT CIRCUIT BREAKER CURRENT SELECT / INPUT VOLTAGE SELECT.
- 3 For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



A4 – CIRCUIT BREAKER DIP SWITCH SETTINGS

- 1 Remove small cover plate under the main circuit breaker to access DIP switches controlling the circuit breaker operation.
- 2 Table below shows the position of the DIP switches for each input voltage selection. The switches immediately under the letter "L" and "I" are the only ones that should be changed.

INPUT VOLTAGE	DIP SWITCH "L" SETTING	L	I	DIP SWITCH "I" SETTING	INPUT VOLTAGE
200V					200V
208V					208V
380V					380V
400V					400V
415V					415V
480V					480V

Set these switches according to the input voltage shown at the left.

Set these switches according to the input voltage shown at the right.

If these switches are present, set both of them in the **DOWN** position as shown.

INSTALLATION STEPS SUMMARY

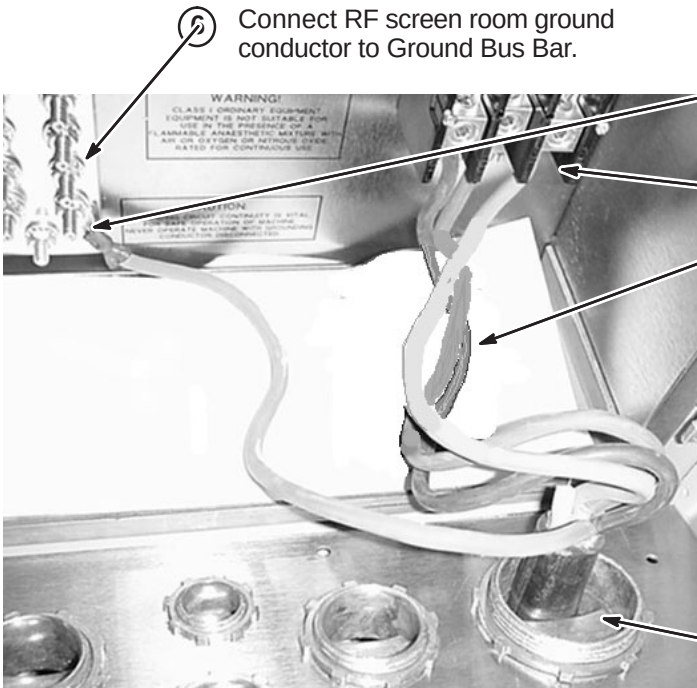
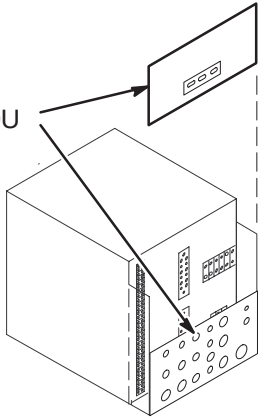
- ☐ B1 – Facility Power and Ground Connention
- ☐ B2 – Route Power Cables Thru Interface Panel
- ☐ B3 – Run 967 Power Cable Alteration
- ☐ B4 – Connect Power Cables to PDU Module

B1 – FACILITY POWER AND GROUND CONNECTION

WARNING!

IF 3 PHASE WYE WITH NEUTRAL AND GROUND (5 WIRE SYSTEM) INPUT IS USED THE NEUTRAL MUST BE TERMINATED INSIDE THE MAIN DISCONNECT CONTROL AND NOT BROUGHT TO THE ACGD/PDU CABINET.

- 1 Remove rear PDU module covers.



- Connect RF screen room ground conductor to Ground Bus Bar.

- 5 Connect facility input dedicated ground conductor to Ground (GRN/YEL) Bus Bar.

- 4 Connect L1–L3 conductors to terminal block.

- 3 NOTE: Air flow thru cabinet must exit thru top of cabinet. If not, fan at top of cabinet is rotating in wrong direction. To reverse rotation of fan, switch any two of the three incoming L1–L3 power wires.

DANGER!!!

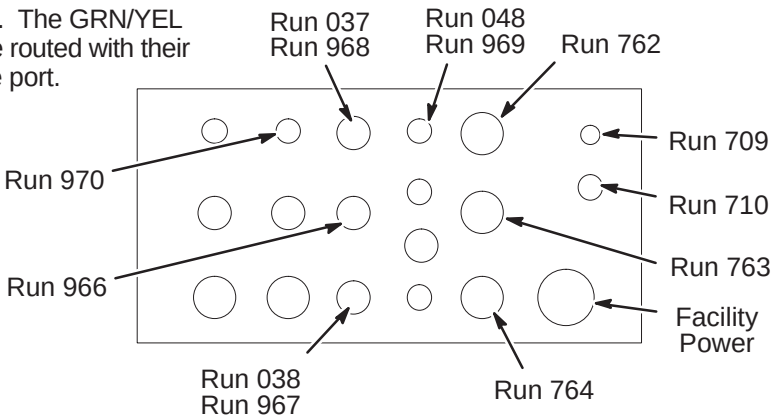
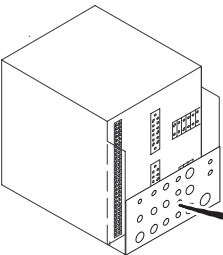
MAKE SURE ALL POWER TO CABINET IS OFF, LOCKED, & TAGGED BEFORE SWITCHING ANY ABOVE WIRES.

- 2 Route facility input power cable and ground wire thru interface panel.

B2 – ROUTE POWER CABLES THRU INTERFACE PANEL

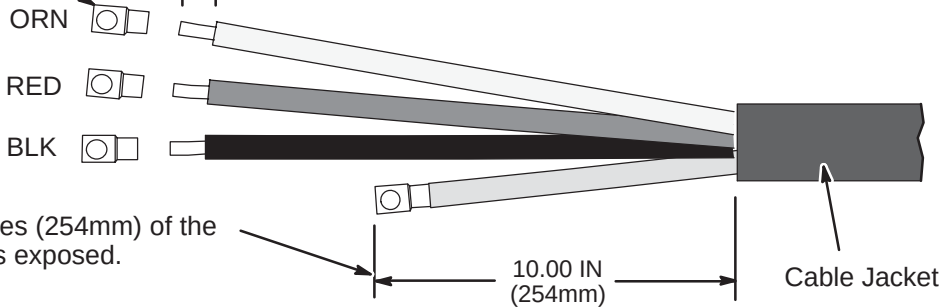
- 1 Route Runs thru interface panel as marked. The GRN/YEL ground wires (Runs 037 and 048) MUST be routed with their respective power cable Runs thru the same port.

- 2 Route any option Runs as marked.



B3 – RUN 967 POWER CABLE ALTERATION

- 1 Cut off ring terminal from ORN, RED, and BLK wires.
- 2 Strip 0.75 IN (19mm) of insulation from end of each wire.

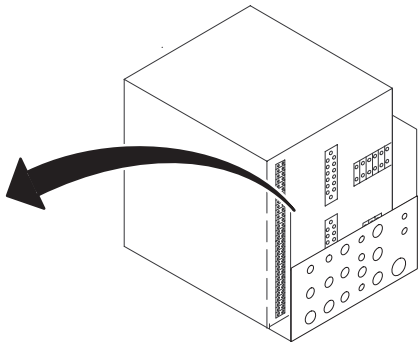


- 3 Strip jacket until 10 Inches (254mm) of the GRN/YEL ground wire is exposed.

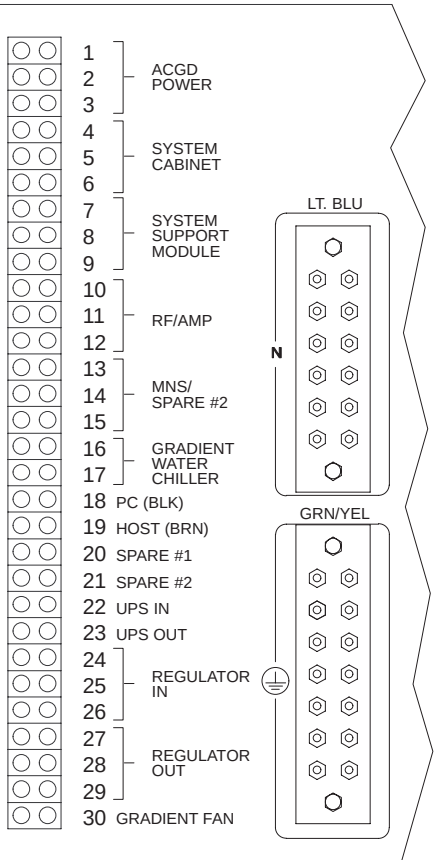
B4 – CONNECT POWER CABLES TO PDU MODULE

WARNING!

ALL POWER CABLES LISTED BELOW MUST BE CONNECTED TO PDU MODULE LOCATED WITHIN ACGD/PDU CABINET.



- 1 Route/connect power cables to pressure connectors and studs on back of PDU module as shown in Table below.
- 2 Trim and connect ground wire cables (Runs 037, 038, and 048) to ground bus connectors.
- 3 Tighten clamps of interface panel to secure cables in place.



CONNECT TO	RUN	PHASE	DESTINATION
4,5,6	968	φ A, B, C	System Cabient 3φ, Neutral, Ground
7,8,9	966	φ A, B, C	SSM 3φ Neutral, Ground
10,11,12	967	φ A, B, C	RFI/AMP 3φ, Ground
16,17	970	φ A,B	Water Chiller Single φ, Ground
18,19	969	φ C,A	Operator Workspace Single φ, Neutral, GND
20	973	φ A	Heat Exchanger Single φ, Neutral, GND
GRN/YEL	037		System Cabinet Ground Wire
GRN/YEL	038		SRF Cabinet Ground Wire
GRN/YEL	048		Operator Workspace Ground Wire

COLOR CODE: for remainder of cables:
3φ= Black, Red, Orange Neutral = Light blue Gnd = Grn/Yel
Single φ = Brown, Black

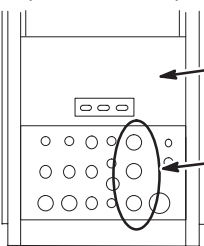
INSTALLATION STEPS SUMMARY

- C1 – Gradient Output Cable Interface Panel Access
- C2 – Gradient Output Cable Routing
- C3 – Route/Connect Fiber Optics Cables to GP Module
- C4 – Connect Runs 971 and 972 to PDU Module
- C5 – Attach InSite Cabinet Magnets

C1 – GRADIENT OUTPUT CABLE INTERFACE PANEL ACCESS

ACGD CABINET
(REAR VIEW)

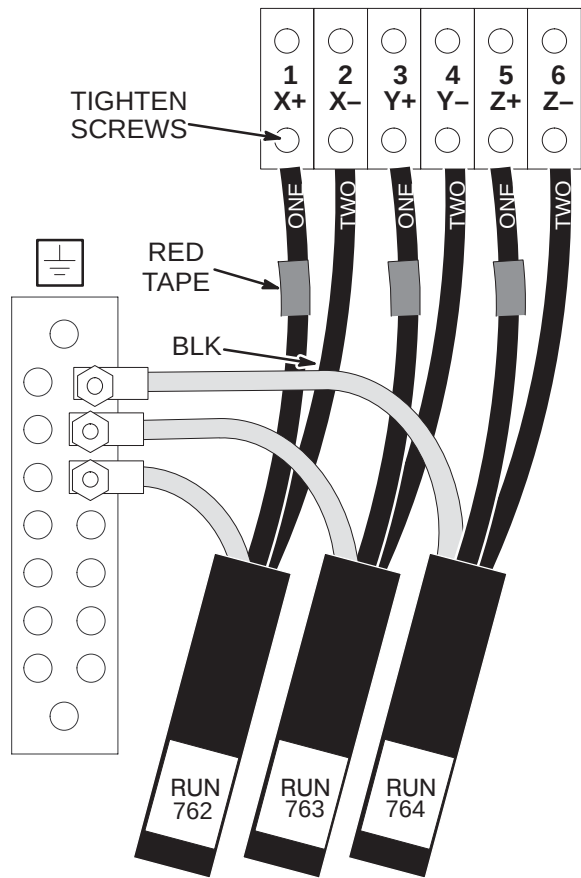
NOTE: Output cables (Runs 762, 763, and 764) were previously routed, cut to length, and terminated.



- Remove Rear Cover Plate by removing seven screws that hold it in place.
- Loosen screws on clamps for gradient output cables. Insert ends of gradient output cables into Cabinet Interface as marked. (Runs 762 into J3, Run 763 into J4, and Run 764 into J5)

C2 – GRADIENT OUTPUT CABLE ROUTING

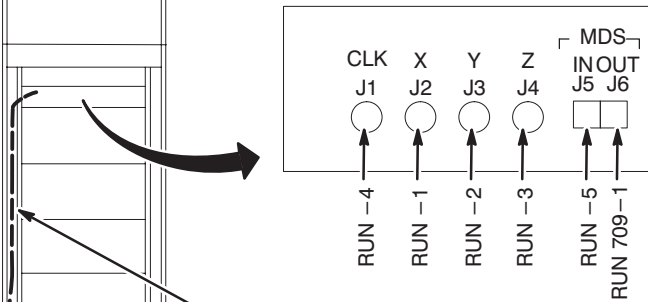
GRADIENT CABLE CONNECTIONS



- Connect gradient output cables with red-taped (+) #1 wires to + output terminals and black (-) #2 wires to - output terminals.
NOTE: OVER TIGHTENING OF SCREWS IN STEP 2 CAN DAMAGE TERMINAL BLOCK.
- Tighten screws to 52–65 inch-lbs (5.88–7.34 N-M). This is about what can be applied using only your thumb and little finger on wrench. Loosen 1/2 turn and re-tighten to the same torque. Repeat approximately 6 times or until wrench returns to the same spot after torquing.
- Connect the GND leads of each run to the Ground Bus Bar studs.
- Tighten clamps on Interface Panel to secure gradient output cables.

C3 – ROUTE/CONNECT FIBER OPTICS TO GP MODULE

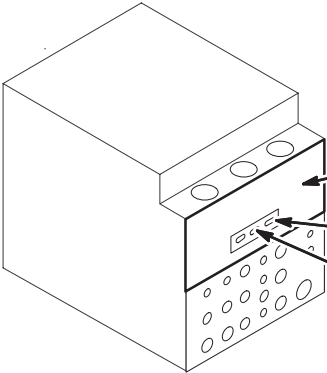
GP INTERFACE PANEL



SYSTEM		GP J#
10.x	LX	
709-1	709-1	J6
1034-5	710-5	J5
1034-3	710-3	J4
1034-2	710-2	J3
1034-1	710-1	J2
1034-4	710-4	J1

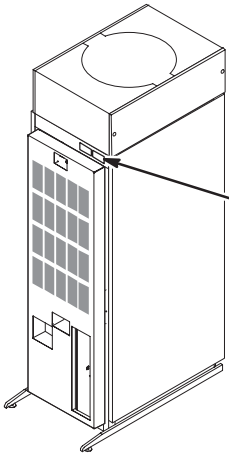
- Route fiber optic cables to GP Module. Leave enough slack for extending GP module forward.
- Attach Run 709 to I/F J1
- Attach Run 710 or Run 1034 to I/F J2
- Connect fiber optics to GIP module according to table.

C4 – CONNECT RUNS 971 AND 972 TO PDU MODULE



- Attach rear PDU module covers removed in Procedure B1.
- Connect Run 972 to J3.
- Connect Run 971 to J2.

C5 – ATTACH INSITE CABINET MAGNETS



- Attach front and rear covers.
- From the 8.0 LX InSite Kit (46-301708G14), locate the "Gradient" cabinet magnet (46-320095P7) and the "PDU" cabinet magnet (46-320095P4). Attach to front of cabinet as shown.

1-1 INTRODUCTION

Specific GP modules may need to be upgraded because of intermittent current distortion faults, specifically on oblique imaging prescriptions.

Perform the EEPROM firmware upgrade in Section 1-3 if your GP is one of following:

- All GP's (2248992) without Deviation M-179-02 applied
- GP2's (2248992-3) Revision 0 only

Perform the Capacitor/Jumper upgrade in Section 1-4 if your GP is one of following:

- All GP's (2248992)
- GP2's (2248992-3) with Revision 0, 1 or 2

NOTE

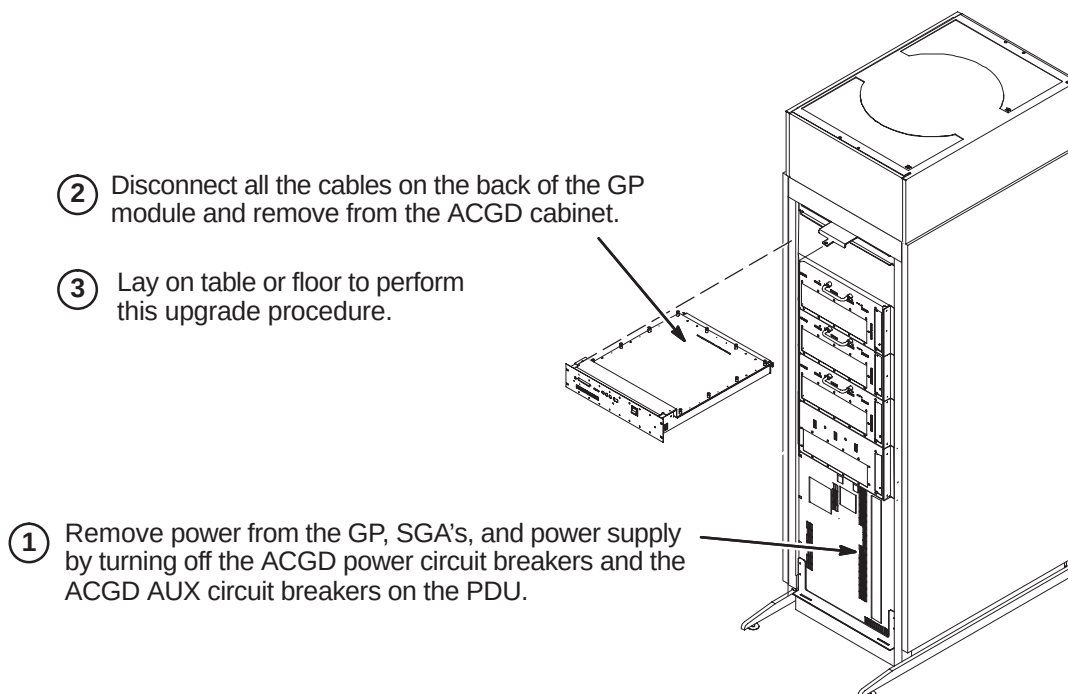
The GP Module upgrade must be completed before EPI calibrations are performed.



CAUTION:

GP Module is a static sensitive component and needs to be handled as such (wrist strap).

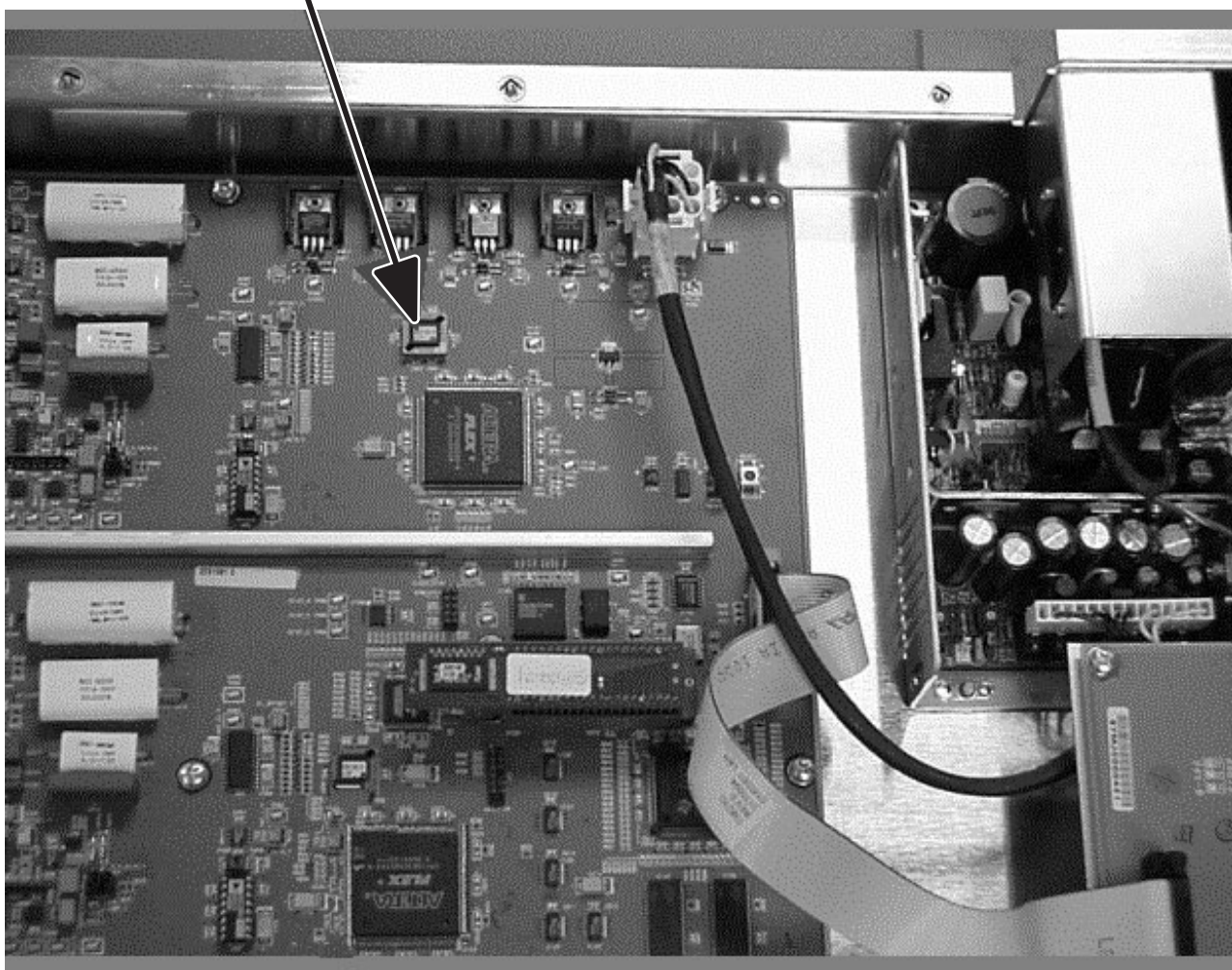
1-2 REMOVE GP MODULE FROM ACGD/PDU CABINET



1-3 REPLACE EEPROM AT U108

NOTE: These devices are extremely ESD sensitive. Use of grounding strap is required.

- ① Remove top cover.
- ② Carefully pull U108 from its socket.
- ③ Place the new EEPROM (2261124-4) at U108. Make sure that the notched corner of the EEPROM lines up with the notch on the socket. (top left corner on the image below)

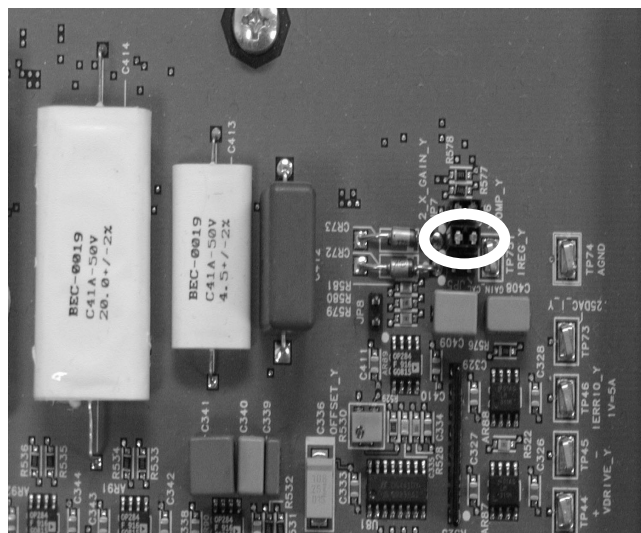
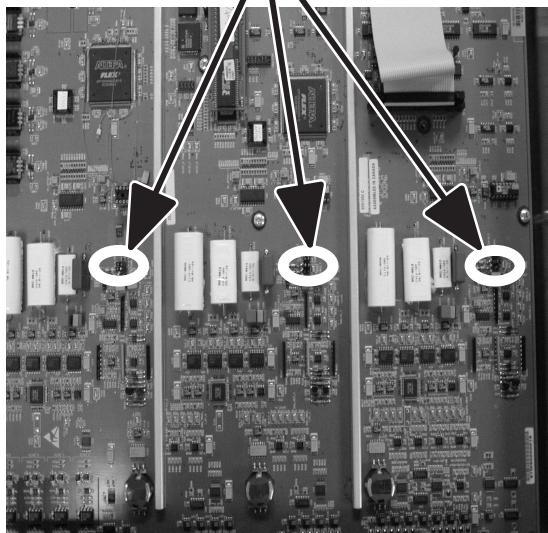
U108**2261124-x
GGxxxx**

1-4 GP CAPACITOR/JUMPER INSTALLATION

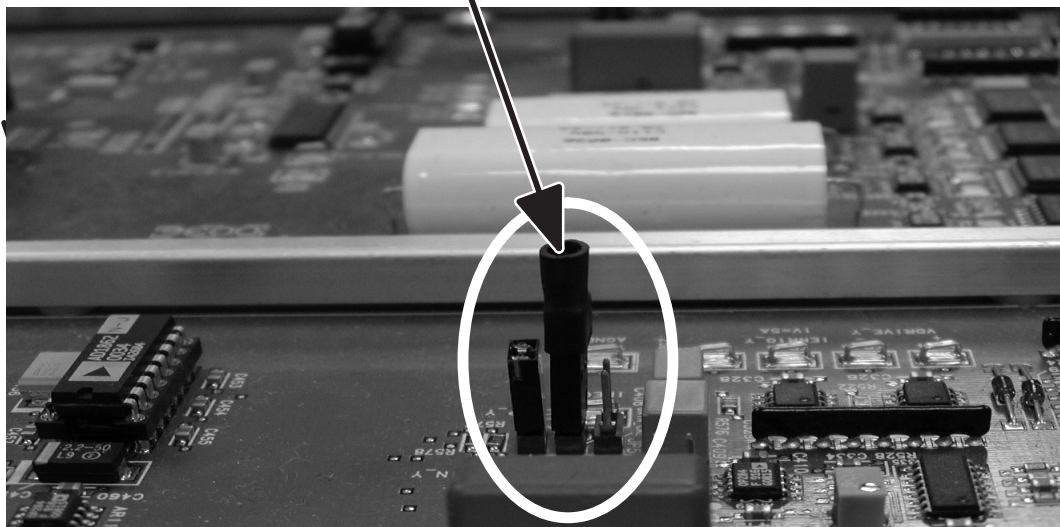


NOTE: These devices are extremely ESD sensitive. Use of grounding strap is required.

- ① Locate jumpers JP10, JP6 and JP2 on the board.



- ② For each of the jumpers remove the shunt and replace with the Capacitor/Jumper Adaptor assembly (2354359) as shown.



- ③ Replace the GP cover (be sure to torque down the captive screws).
④ Reinstall GP Module into the cabinet and reconnect all cables.
⑤ Turn power on.

SRF CABINET UPGRADE INSTRUCTIONS

Tab 16, SRF Cabinet, contains installation instructions for the following:

- *DIRECTION 2286812* – Signa SRF Cabinet Installation Instructions
- *DIRECTION 2346942* – Signa SRFD2 Cabinet Installation Instructions

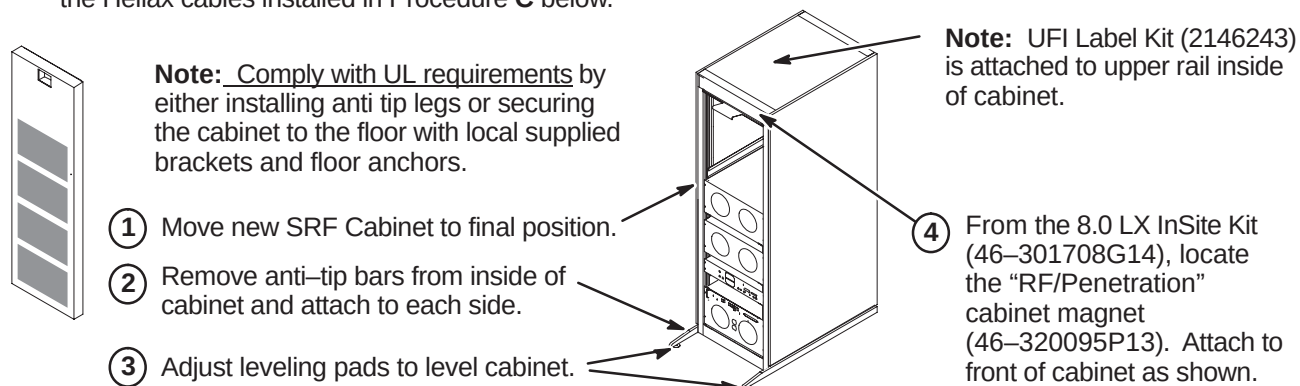
Use only the applicable SRF Cabinet Installation Instruction Direction for the site being upgraded

INSTALLATION STEPS SUMMARY

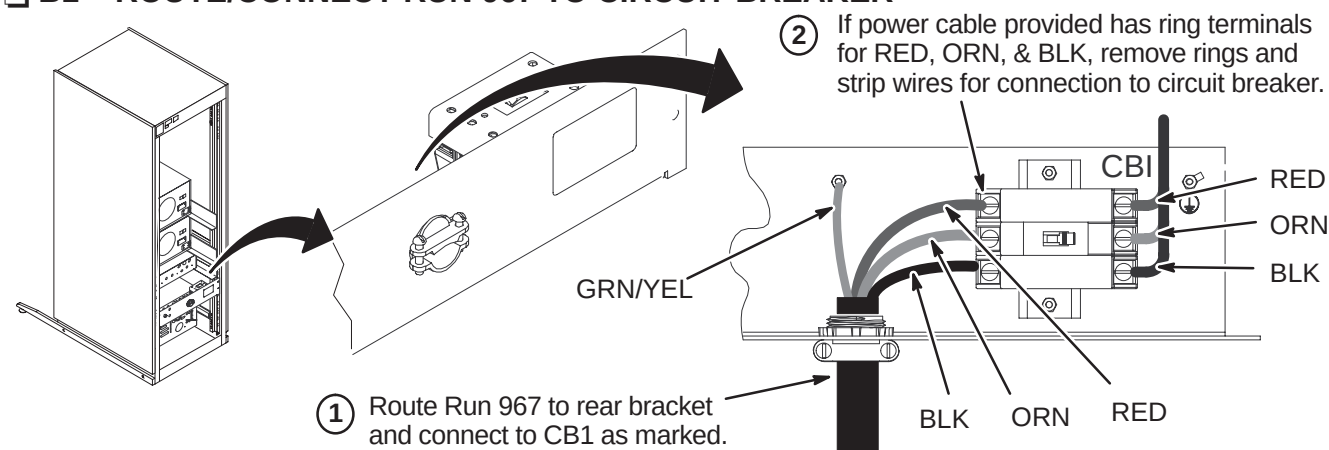
- ☐ A – Install New SRF Cabinet
 - ☐ B1 – Route/Connect Run 967 to Circuit Breaker
 - OR
 - ☐ B2 – Route/Connect Run 967 to Terminal Strip
 - ☐ C – Connect Runs 235/745 and 236/748 to RF Interface (RFI) Module
 - ☐ D – Connect Fiber Optics and Run 229
 - ☐ E – System Support Module Cable Connections
- } Use the applicable procedure.

A – INSTALL NEW SRF CABINET

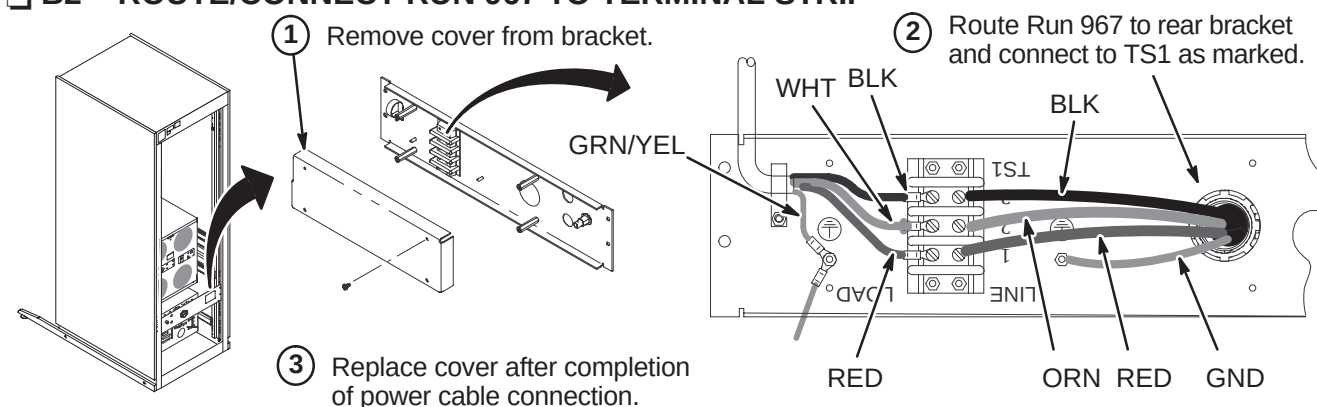
Note: The 1.5T SRF (Scaleable RF) cabinet is shown in these procedures. The 1.0T is the same except for the Helix cables installed in Procedure **C** below.



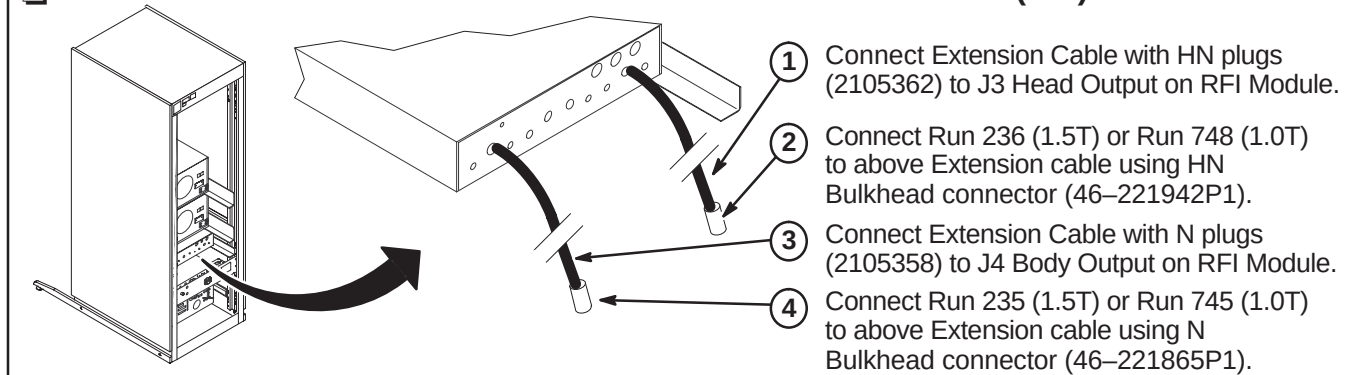
☐ **B1 – ROUTE/CONNECT RUN 967 TO CIRCUIT BREAKER**



☐ B2 – ROUTE/CONNECT RUN 967 TO TERMINAL STRIP



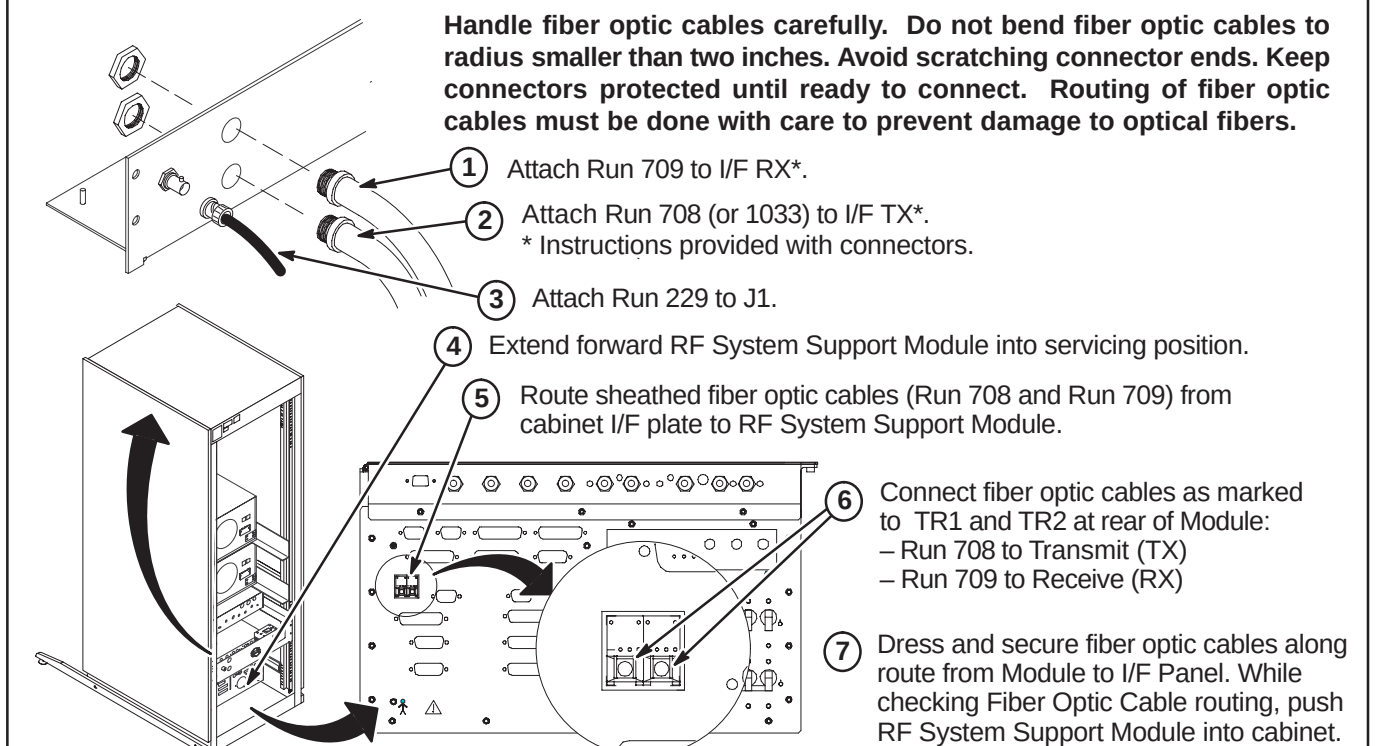
❑ C – CONNECT RUNS 235/745 AND 236/748 TO RF INTERFACE (RFI) MODULE



D – ROUTE/CONNECT FIBER OPTICS AND RUN 229

CAUTION

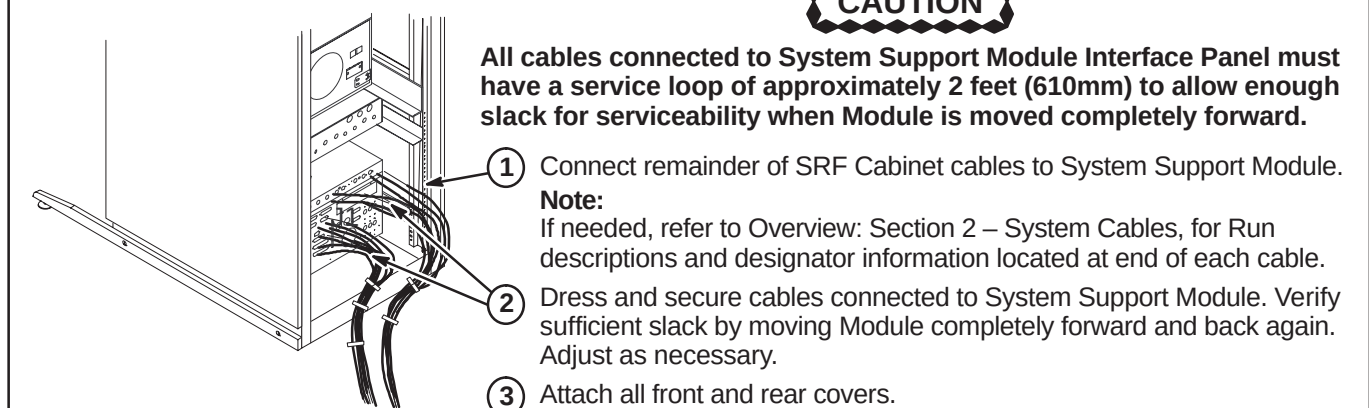
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.



☐ E – SYSTEM SUPPORT MODULE CABLE CONNECTIONS

CAUTION

All cables connected to System Support Module Interface Panel must have a service loop of approximately 2 feet (610mm) to allow enough slack for serviceability when Module is moved completely forward.



INSTALLATION STEPS SUMMARY

- A – Install New SRFD2 Cabinet
- B – Route/Connect Run 967
- C – Connect Runs 235 and 236 to RF Amp & Interface Module
- D – Connect Fiber Optics and Run 229
- E – System Support Module Cable Connections
- F – Attach InSite Cabinet Magnets

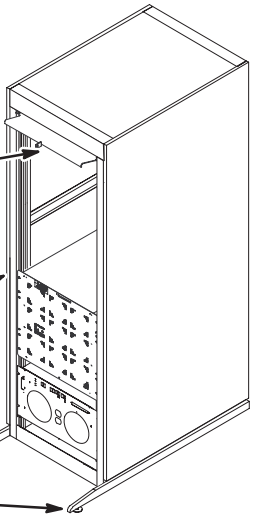
A – INSTALL NEW SRFD2 CABINET

Note: Comply with UL requirements by either installing anti tip legs or securing the cabinet to the floor with local supplied brackets and floor anchors.

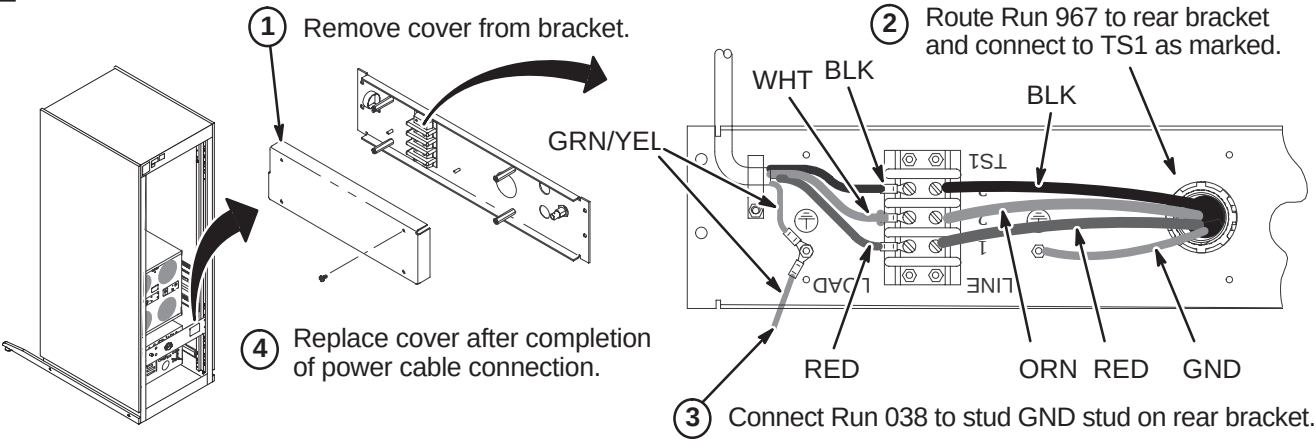


Note: UFI Label Kit (2146243) is attached to upper rail inside of cabinet.

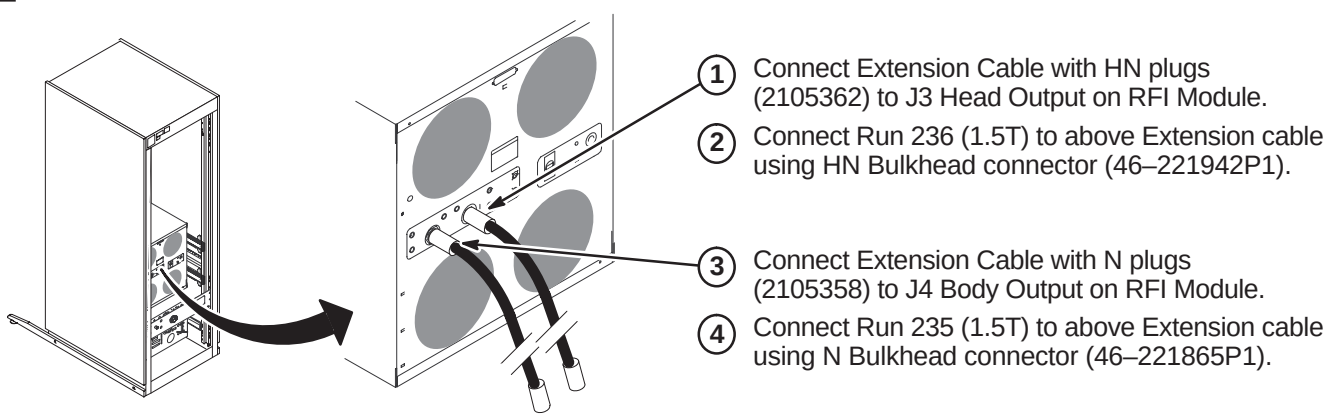
- 1 Move new SRFD2 Cabinet to final position.
- 2 Remove anti-tip bars from inside of cabinet and attach to each side.
- 3 Adjust leveling pads to level cabinet.



B – ROUTE/CONNECT RUN 967



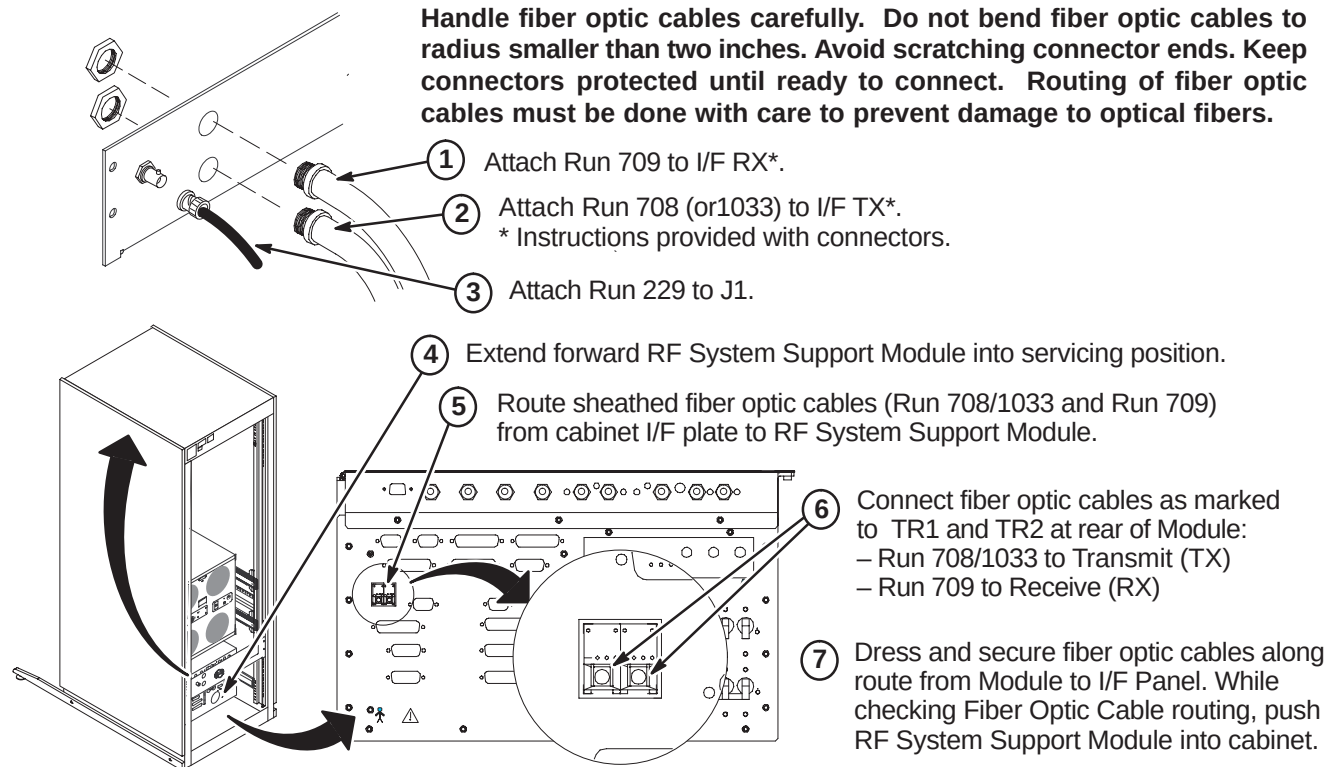
C – CONNECT RUNS 235 AND 236 TO RF AMP & INTERFACE MODULE



D – ROUTE/CONNECT FIBER OPTICS AND RUN 229

CAUTION

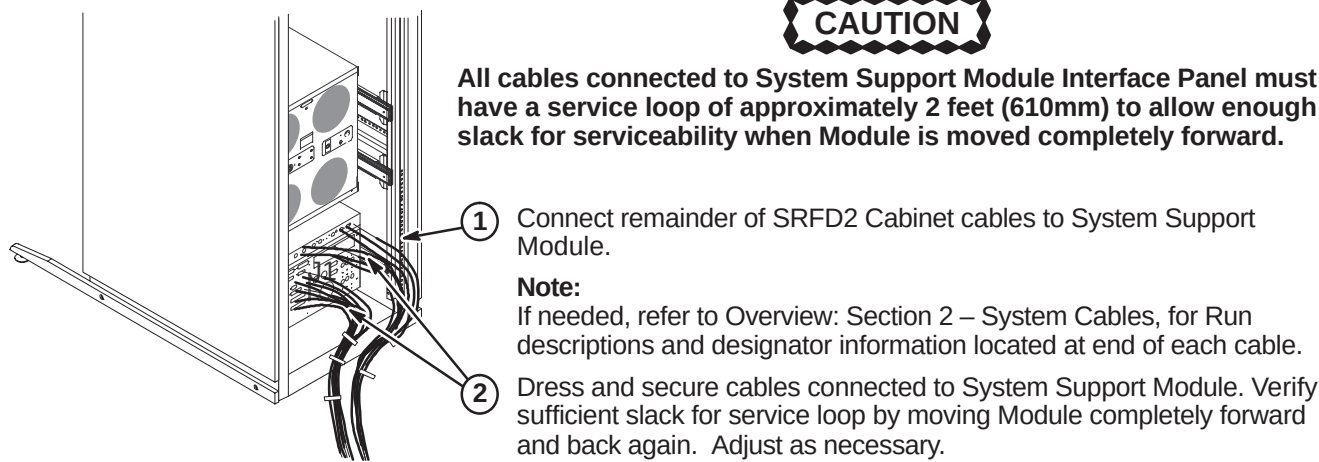
Handle fiber optic cables carefully. Do not bend fiber optic cables to radius smaller than two inches. Avoid scratching connector ends. Keep connectors protected until ready to connect. Routing of fiber optic cables must be done with care to prevent damage to optical fibers.



E – SYSTEM SUPPORT MODULE CABLE CONNECTIONS

CAUTION

All cables connected to System Support Module Interface Panel must have a service loop of approximately 2 feet (610mm) to allow enough slack for serviceability when Module is moved completely forward.



F – ATTACH INSITE CABINET MAGNET

